

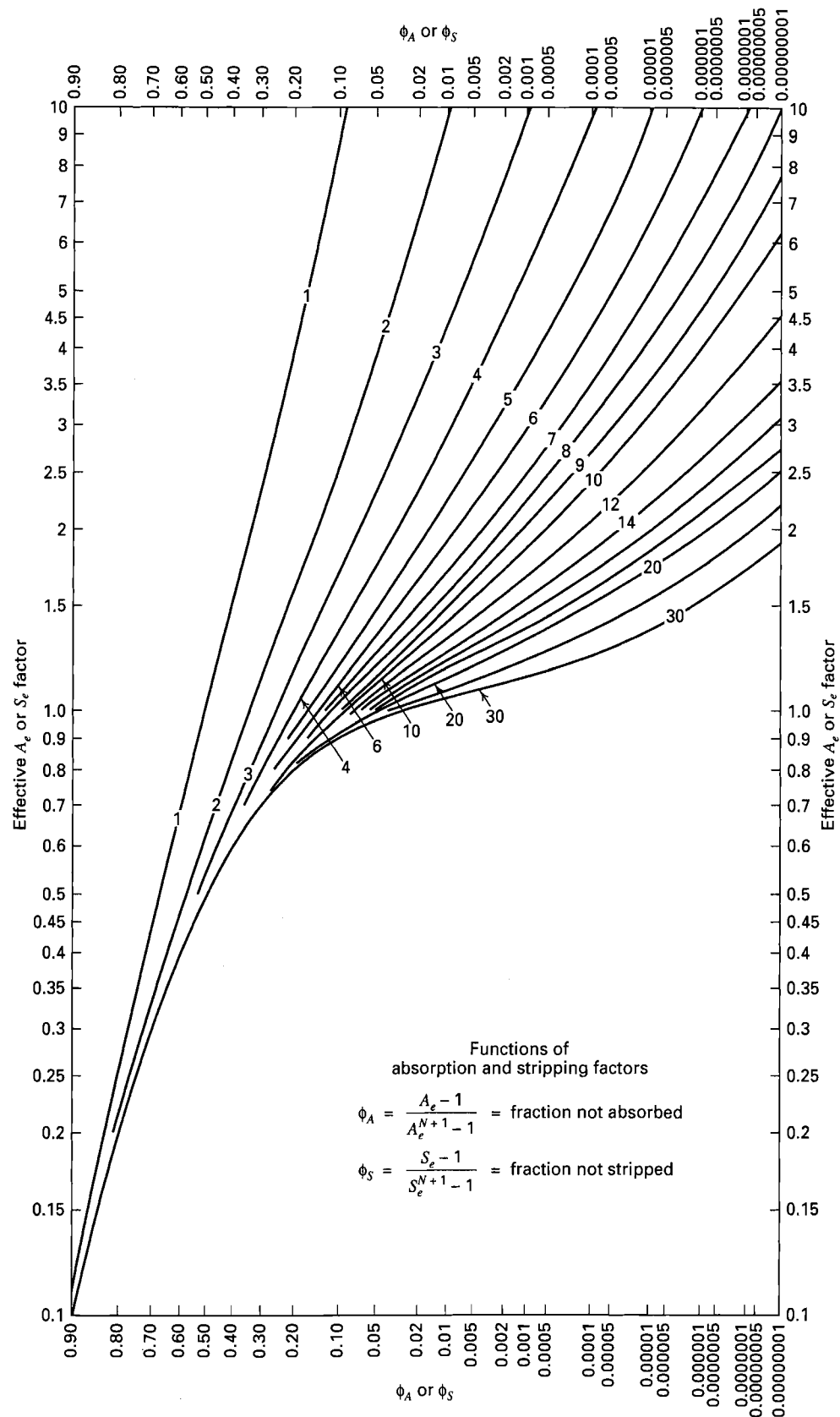


Figure 5.9 Plot of Kremser equation for a single-section countercurrent cascade.

[From W. C. Edmister, *AIChE J.*, **3**, 165-171 (1957).]

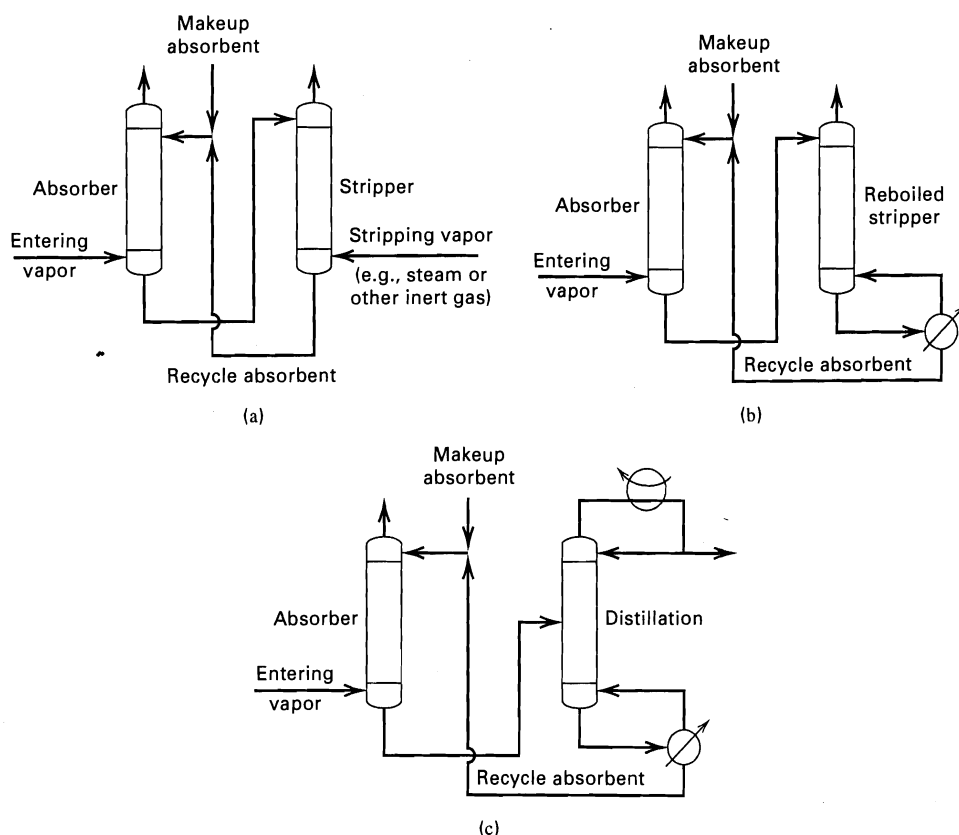


Figure 5.10 Various coupling schemes for absorbent recovery: (a) use of steam or inert gas stripper; (b) use of reboiled stripper; (c) use of distillation.

which is generally applied to each component in the vapor entering the absorber. Equation (5-52) is used for species that appear only in the entering liquid. The analogous equation for a stripper in Figure 5.8b is

$$l_1 = l_{N+1}\phi_S + v_0(1 - \phi_A) \quad (5-55)$$

EXAMPLE 5.3

In Figure 5.11, the heavier components in a slightly superheated hydrocarbon gas are to be removed by absorption at 400 psia (2,760 kPa) with a high-molecular-weight oil. Estimate exit vapor and exit liquid flow rates and compositions by the approximate group method of Kremser. Assume that effective absorption and stripping factors for each component can be estimated from the entering values of L , V , and the component K -values, as listed below based on an average entering temperature of $(90 + 105)/2 = 97.5^\circ\text{F}$.

Component	$K@97.5^\circ\text{F},$ 400 psia	A	S	ϕ_A	ϕ_S	v_1	l_6
C_1	6.65	0.0310	—	0.969	—	155.0	5.0
C_2	1.64	0.126	—	0.874	—	323.5	46.5
C_3	0.584	0.353	—	0.647	—	155.4	84.6
nC_4	0.195	1.06	0.946	0.119	0.168	3.02	22.03
nC_5	0.0713	2.89	0.346	0.00112	0.654	0.28	5.5
Oil	0.0001	—	0.0005	—	0.9995	0.075	164.095
						637.275	327.725

SOLUTION

From (5-38) and (5-51),

$$A_i = L/K_i V = 165/[K_i(800)] = 0.206/K_i$$

$$S_i = 1/A_i = 4.85K_i$$

$$N = 6 \text{ stages}$$

Values of ϕ_A and ϕ_S are obtained from (5-48) and (5-50) or Figure 5.9. Values of $(v_i)_1$, the component flow rates in the exit vapor, are computed from (5-54). Values of $(l_i)_6$, the component flow rates in the exit liquid, are computed from an overall component material balance using Figure 5.8a:

$$(l_i)_6 = (l_i)_0 + (v_i)_7 - (v_i)_1 \quad (1)$$

The computations, which are best made in tabular fashion with a spreadsheet computer program, give the following results:

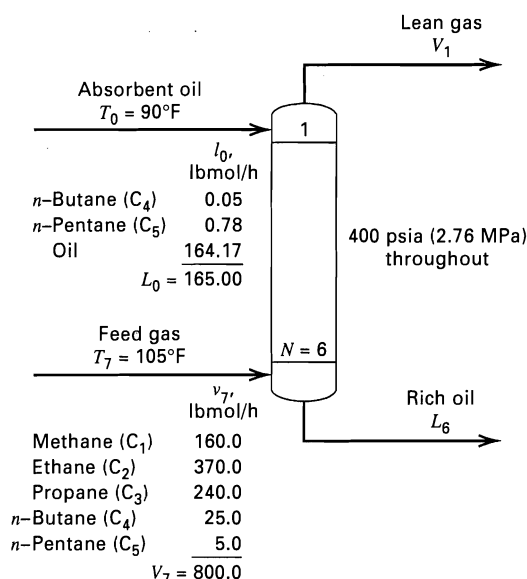


Figure 5.11 Specifications for absorber of Example 5.3.

The above results indicate that approximately 20% of the gas is absorbed. Less than 0.1% of the absorbent oil is stripped.

Two-Section Cascades

A single-stage flash distillation produces a vapor that is somewhat richer in the lower-boiling constituents than the feed. Further enrichment can be achieved by a series of flash distillations in which the vapor from each stage is condensed, then reflash. In principle, any desired product purity can be obtained by a multistage flash technique, provided a suitable volatility difference exists and a suitable number of stages is employed. In practice, however, the recovery of product is small, heating and cooling requirements are high, and relatively large quantities of various liquid products are produced.

As an example, consider Figure 5.12a, where *n*-hexane (H) is separated from *n*-octane by a series of three flashes at 1 atm (pressure drop and pump needs are ignored). The feed to the first flash stage is an equimolar bubble-point liquid at a flow rate of 100 lbmol/h. A bubble-point temperature calculation yields 192.3°F. If the vapor rate leaving stage 1 is set equal to the amount of *n*-hexane in the feed to stage 1, the calculated equilibrium exit phases are as shown. The vapor V_1 is enriched to a hexane mole fraction of 0.690. The heating requirement is 751,000 Btu/h. Equilibrium vapor from stage 1 is condensed to bubble-point liquid with a cooling duty of 734,000 Btu/h. Repeated flash calculations for stages 2 and 3 give the results shown. For each stage, the leaving molar vapor rate is set equal to the moles of hexane in the feed to the stage. The purity of *n*-hexane is increased from 50 mol% in the feed to 86.6 mol% in the final condensed vapor product, but the recovery of hexane is only 27.7(0.866)/50 or 48%. Total heating requirement is 1,614,000 Btu/h and liquid products total 72.3 lbmol/h.

In comparing feed and liquid products from two contiguous stages, we note that liquid from the later stage and the feed to the earlier stage are both leaner in hexane, the more volatile species, than the feed to the later stage. Thus, if intermediate streams are recycled, intermediate recovery of hexane is improved. This processing scheme is depicted in Figure 5.12b, where again the molar fraction vaporized in each stage equals the mole fraction of hexane in the combined feeds to the stage. The mole fraction of hexane in the final condensed vapor product is 0.853, just slightly less than that achieved by successive flashes without recycle. However, the use of recycle increases recovery of hexane from 48% to 61.6%. As shown in Figure 5.12b, increased recovery of hexane is accompanied by approximately 28% increased heating and cooling requirements. If the same degree of heating and cooling is used for the no-recycle scheme in Figure 5.12a as in Figure 5.12b, the final hexane

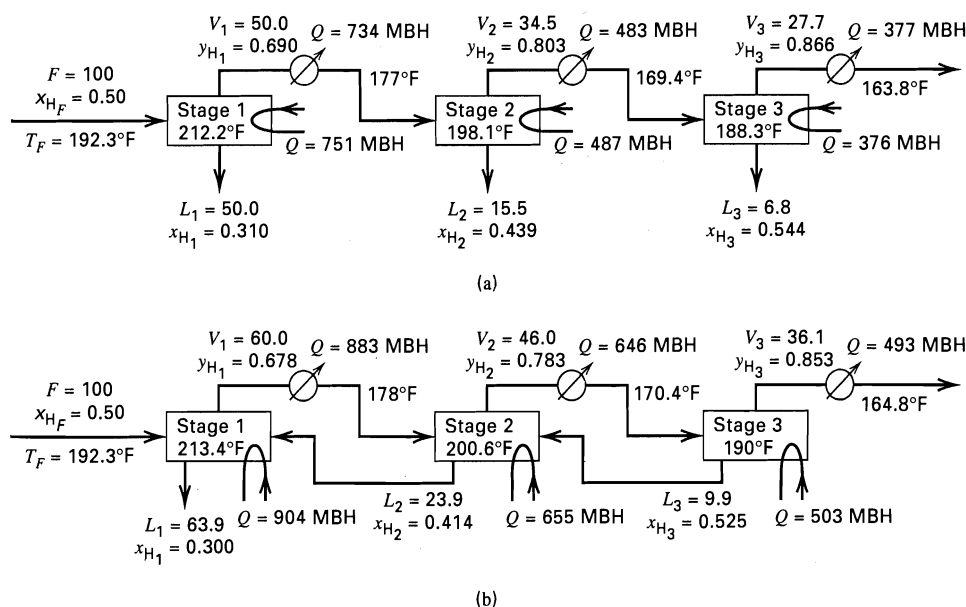


Figure 5.12 Successive flashes for recovering hexane from octane: (a) no recycle; (b) with recycle. Flow rates in lbmol/h. MBH = 1,000 Btu/h.

mole fraction is reduced from 0.866 to 0.815, but hexane recovery is increased to $36.1(0.815)/50$ or 58.8%.

Both of the successive flash arrangements in Figure 5.12 involve a considerable number of heat exchangers and pumps. Except for stage 1, the heaters in Figure 5.12a can be eliminated if the two intermediate total condensers are converted to partial condensers with duties of $734 - 487 = 247$ MBH (MBH = 1,000 Btu/h) and $483 - 376 = 107$ MBH. Total heating duty is now only 751,000 Btu/h, and total cooling duty is 731,000 Btu/h. Similarly, if heaters for stages 2 and 3 in Figure 5.12b are removed by converting the two total condensers to partial condensers, total heating duty is 904,000 Btu/h (20% greater than the no-recycle case), and cooling duty is 864,000 Btu/h (18% greater than the no-recycle case).

A considerable simplification of the successive flash technique with recycle is shown in Figure 5.13a. The total heating duty is provided by a feed boiler ahead of stage 1. The total cooling duty is utilized at the opposite end to condense totally the vapor leaving stage 3. Condensate in excess of distillate is returned as reflux to the top stage, from which it passes successively from stage to stage countercurrently to vapor flow. Vertically arranged adiabatic stages eliminate the need for interstage pumps, and all stages are contained within a single piece of less expensive equipment. The set of stages is called a *rectifying section*. As discussed in Chapter 2, such an arrangement is thermodynamically inefficient, however, because heat is added at the highest temperature level and removed at the lowest temperature level.

The number of degrees of freedom for the arrangement in Figure 5.13a is determined by the method of Chapter 4 to be $(C + 2N + 10)$. If all independent feed conditions, number of stages (3), and all stage pressures (1 atm), bubble-point liquid leaving the condenser, and adiabatic stages are specified, two degrees of freedom remain. These are specified to be a heating duty for the boiler and a distillate rate equal to that of Figure 5.12b. Calculations result in a mole fraction of 0.872 for hexane in the distillate. This is somewhat greater than that shown in Figure 5.12b.

The same principles by which we have concluded that the adiabatic, multistage, countercurrent-flow arrangement is advantageous for concentrating a light component in an overhead product can be applied to the concentration of a heavy component in a bottoms product, as in Figure 5.13b. Such a set of stages is called a *stripping section*.

Figure 5.13c, a combination of Figures 5.13a and 5.13b with a liquid feed, is a complete column for rectifying and stripping a feed to effect a sharper separation between a selected more volatile component, called the *light key*, and a less volatile component, called the *heavy key component*, than is possible with either a stripping or an enriching section alone. Adiabatic flash stages are placed above and below the feed. Recycled liquid reflux, L_R , is produced in the condenser and vapor boilup, V_1 , in the reboiler. The reflux ratios are L_R/V_N and L_2/V_1 at the top and bottom of the apparatus, respectively. All interstage flows are countercurrent. Two-section cascades are widely used in industry for multistage distillation.

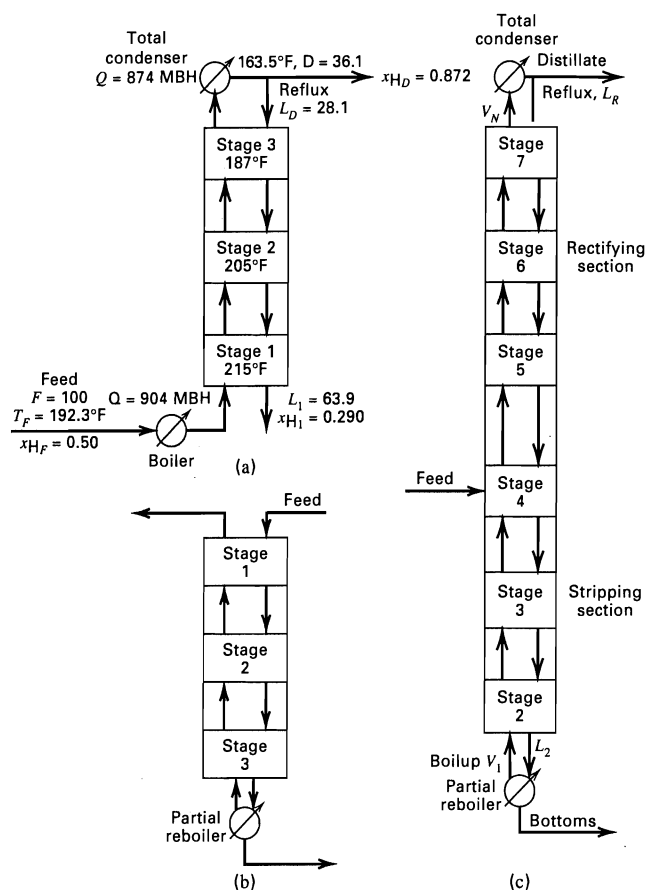


Figure 5.13 Successive adiabatic flash arrangements: (a) rectifying section; (b) stripping section; (c) multistage distillation.

The rectifying stages above the point of feed introduction purify the light product by contacting upward flowing vapor with successively richer liquid reflux. Stripping stages below the feed increase light-product recovery because vapor relatively low in volatile constituents strips light components out of the liquid. For the heavy product, the functions are reversed: The stripping section increases purity; the enriching section increases recovery.

Edmister [3] applied the Kremser group method for absorbers and strippers to distillation where two cascades are coupled to a condenser, a reboiler, and a feed stage. In Figure 5.14, five separation zones are shown: (1) partial condenser, C; (2) absorption or rectifying cascade (enriching section), E; (3) feed-flash stage, F; (4) stripping cascade (exhausting section), X; and (5) partial reboiler, B.

In Figure 5.14, N stages for the enricher are numbered from the top down and the overhead product is distillate; whereas for the exhauster, M stages are numbered from the bottom up. Component feeds to the enricher section are vapor, v_F , from the feed stage and liquid, l_C , from the condenser. Component feeds to the exhauster are liquid, l_F , from the feed stage and vapor, v_B , from the reboiler. Component flows leaving the enricher cascade are vapor, v_{TE} , from the top stage, 1, and liquid, l_{BE} , from the bottom stage, N . Component flows leaving the exhauster cascade are vapor, v_{TX} , from the top stage, M , and liquid, l_{BX} , from the bottom stage, 1. The recovery equations for the enricher are obtained from (5-54)

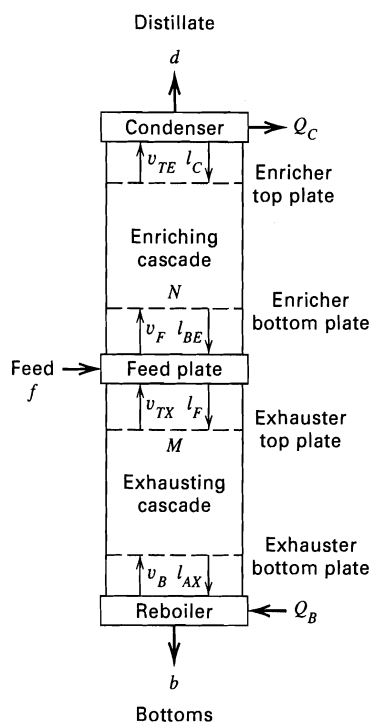


Figure 5.14 Countercurrent distillation cascade.

by making the following substitutions, which are obtained from material balance and equilibrium considerations. For each component in the feed,

$$v_{TE} = l_C + d \quad (5-56)$$

$$v_F = l_{BE} + d \quad (5-57)$$

and

$$l_C = dA_C \quad (5-58)$$

where

$$A_C = \frac{L_C}{DK_C} \quad (\text{for a partial condenser}) \quad (5-59)$$

$$A_C = \frac{L_C}{D} \quad (\text{for a total condenser}) \quad (5-60)$$

The resulting enricher recovery equations for each species are

$$\frac{l_{BE}}{d} = \frac{A_C \phi_{SE} + 1}{\phi_{AE}} - 1 \quad (5-61)$$

or

$$\frac{v_F}{d} = \frac{A_C \phi_{SE} + 1}{\phi_{AE}} \quad (5-62)$$

where the additional subscript *E* on ϕ refers to the enricher.

The recovery equations for the exhauster are obtained in a similar manner, as

$$\frac{v_{TX}}{b} = \frac{S_B \phi_{AX} + 1}{\phi_{SX}} - 1 \quad (5-63)$$

$$\frac{l_F}{b} = \frac{S_B \phi_{AX} + 1}{\phi_{SX}} \quad (5-64)$$

where

$$S_B = K_B V_B / B$$

for a partial reboiler, and additional subscript *X* on ϕ denotes an exhauster.

For either an enricher or exhauster, ϕ_A and ϕ_E are given, from above, by (5-48) and (5-50), respectively, or from Figure 5.9.

To couple the enriching and exhausting cascades, a feed stage is employed for which the absorption factor is related to the streams leaving the feed stage by

$$A_F = \frac{L_F}{K_F V_F} = \frac{L_F}{(v_F L_F / V_F l_F) V_F} = \frac{l_F}{v_F} \quad (5-65)$$

For the distillation column of Figure 5.14, (5-62), (5-64), and (5-65) are combined to eliminate l_F and v_F . The result is

$$\frac{b}{d} = A_F \frac{[(A_C \phi_{SE} + 1)/\phi_{AE}]}{[(S_B \phi_{AX} + 1)/\phi_{SX}]} \quad (5-66)$$

To apply (5-66) for the calculation of component split ratios, b/d , it is necessary to establish values of absorption factors A_F and A_C , and the stripping factor S_B . Average values for factors A_E , A_X , S_E , and S_X for each component are also required for the two cascades to determine the corresponding ϕ values. To establish these values, it is necessary to estimate temperatures and molar vapor and liquid, V and L , flow rates. An approximate method for making these estimates is given in the following example.

EXAMPLE 5.4

The hydrocarbon gas of Example 5.3 is distilled at 400 psia (2.76 MPa), to separate ethane from propane, for the conditions shown in Figure 5.15. Estimate the distillate and bottoms compositions using (5-66). This example is best solved by using a spreadsheet computer program.

SOLUTION

Assume a feed stage temperature equal to the feed temperature, 105°F. To estimate the condenser and reboiler temperatures, assume a perfect split for the specified distillate rate of 530 lbmol/h, with all methane and ethane going to the distillate and all propane and heavier going to the bottoms. Thus the preliminary material balance is

Component	lbmol/h		
	Feed, <i>f</i>	Assumed Distillate, <i>d</i>	Assumed Bottoms, <i>b</i>
C ₁	160	160	0
C ₂	370	370	0
C ₃	240	0	240
<i>n</i> C ₄	25	0	25
<i>n</i> C ₅	5	0	5
	800	530	270

For these assumed products, applying procedures in Section 4.4, a distillate temperature of 12°F is obtained from a dew-point calculation and a bottoms temperature of 165°F from a bubble-point calculation. Average temperatures of $(12 + 105)/2 = 59^\circ\text{F}$ and $(105 + 165)/2 = 135^\circ\text{F}$ are estimated for the enriching and exhausting cascades, respectively. Assuming that total molar flow rates are constant in each cascade, the following vapor and liquid flow-rate estimates are obtained by working down from the top of

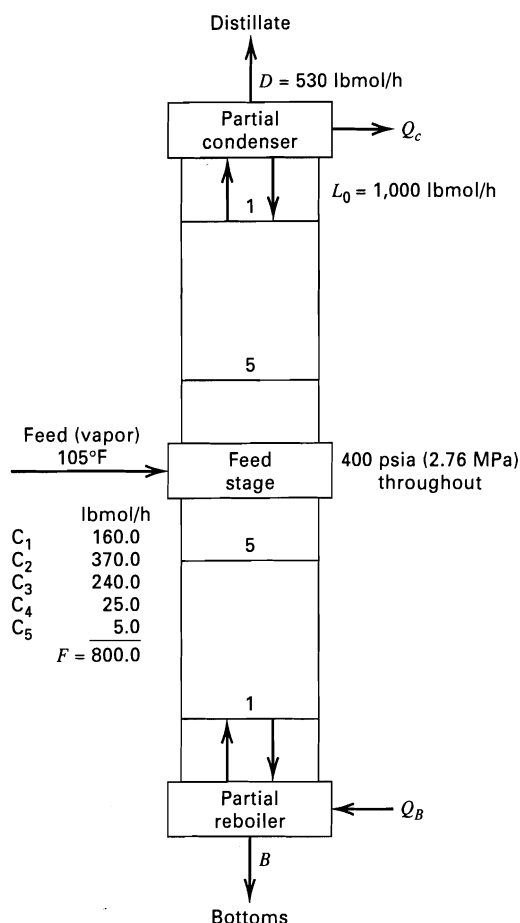


Figure 5.15 Specifications for fractionator of Example 5.4.

the column, where the liquid reflux is specified:

Stage or Section	Average Temperature, °F	Average Flow Rates, lbmol/h	
		Vapor	Liquid
Condenser	12	530	1,000
Enricher	59	1,530	1,000
Feed	105	1,530	1,000
Exhauster	135	730	1,000
Reboiler	165	730	270

From the column pressure and the estimated temperature values, K -values are read from Figure 2.8. These values are then used to estimate absorption and stripping factors for the five sections, with the following results:

	Component				
	C_1	C_2	C_3	nC_4	nC_5
A_C	0.356	2.45	10.48	43.9	172
A_E	0.131	0.511	1.72	5.78	18.7
$S_E = 1/A_E$	7.65	1.96	0.581	0.173	0.0536
A_F	0.111	0.363	1.108	3.19	9.08
A_X	0.205	0.652	1.73	4.72	12.5
$S_X = 1/A_X$	4.89	2.26	0.577	0.212	0.080
S_B	19.5	7.03	2.78	1.08	0.433

From the values of A_E , A_X , S_E , and S_X , and the numbers of theoretical stages specified in Figure 5.15, the following values of ϕ are computed from (5-48) and (5-50) or read from Figure 5.9:

	Component				
	C_1	C_2	C_3	nC_4	nC_5
ϕ_{AE}	0.869	0.498	0.0289	0.0001	0.000
ϕ_{AX}	0.796	0.377	0.028	0.00034	0.000
ϕ_{SE}	0.0000	0.0173	0.435	0.827	0.946
ϕ_{SX}	0.0003	0.0444	0.439	0.788	0.920

From the values in the above two tables, values of (b/d) are computed for each component from (5-66). Since an overall balance for each component is given by $f = d + b$, values of d and b can then be computed from

$$d = \frac{f}{1 + (b/d)} \quad (1)$$

$$b = f - d \quad (2)$$

The following results are obtained:

	(b/d)	lbmol/h	
		d	b
C_1	0.000002	160	0
C_2	0.00924	366.6	3.4
C_3	86.8	2.7	237.3
nC_4	937,000	0	25
nC_5	Very large	0	5
Totals		529.3	270.7

Total distillate rate is somewhat less than the 530.0 lbmol/h specified. Values of d_i and b_i can be corrected to force the total to 530.0 by the method of Lyster et al. [5], which involves finding the positive root of θ in the relation

$$D = \sum_i \frac{f_i}{1 + \theta(b_i/d_i)}$$

followed by recalculation of d_i from

$$d_i = \frac{f_i}{1 + \theta(b_i/d_i)}$$

and b_i from $f_i - d_i$. The resulting value of θ is 0.8973, which gives $d_{C_2} = 367$, $b_{C_2} = 3$, $d_{C_3} = 3$, and $b_{C_3} = 237$, with no changes for other components.

The separation achieved by distillation is considerably improved over the separation achieved by absorption in Example 5.3. Although overhead vapor flow rates are approximately the same (530 lbmol/h) in this example and in Example 5.3, a reasonably sharp split between ethane and propane occurs for distillation because of the two-section cascade, while the absorber, with only a one-section cascade, allows appreciable quantities of both ethane and propane to exit in the overhead vapor and bottoms liquid. Even if the absorbent rate in Example 5.3 is doubled so that the recovery of propane in the bottoms exit liquid approaches 100%, more than 50% of the ethane also appears in the bottoms.

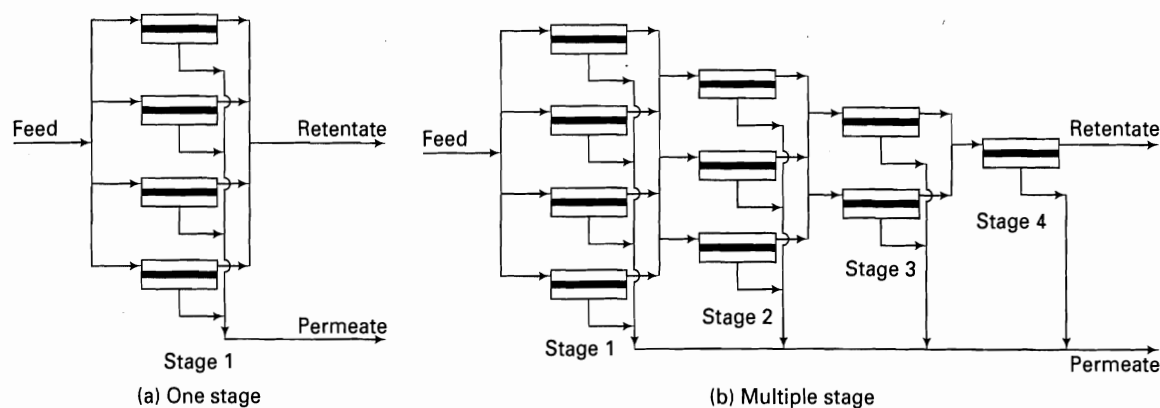


Figure 5.16 Parallel units of membrane separators.

5.5 MEMBRANE CASCADES

Membrane separation systems frequently consist of multiple-membrane units or modules. One reason for this is that a single module of the maximum size available may not be large enough to handle the required feed rate. In that case, it is necessary to use a number of modules of identical size in parallel as shown in Figure 5.16a, with retentates and permeates from each module combined, respectively, to obtain the final retentate and final permeate. For example, a membrane-separation system for separating hydrogen from methane might require a membrane area of 9,800 ft². If the largest membrane module available has 3,300 ft² of membrane surface, three modules in parallel are required. The parallel units in Figure 5.16a constitute a single stage of membrane separation. If, in addition, a large fraction of the feed is to become permeate, it may be necessary to carry out the membrane separation in two or more stages, as shown in Figure 5.16b for four stages, with the number of modules reduced for each successive stage as the flow rate on the feed-retentate side of the membrane decreases. The combined retentate from each stage becomes the feed for the next stage. The combined permeates for each stage differ in composition. They can be further combined to give an overall permeate, as shown in Figure 5.16b, or not, to give two or more permeate products of different composition.

A second reason for using multiple-membrane modules is that a single-membrane stage is often limited in the degree of separation achievable. In some cases, a high purity can be obtained, but only at the expense of a low recovery. In other cases, neither a high purity nor a high recovery can be obtained. The following table gives two examples of the degree of separation achieved for a single stage of gas permeation using a commercially available membrane.

Feed Molar Composition	More Permeable Component	Product Molar Composition	Percent Recovery
85% H ₂ 15% CH ₄	H ₂	99% H ₂ 1% N ₂ in the permeate	60% of H ₂ in the feed
80% CH ₄ 20% N ₂	N ₂	97% CH ₄ 3% N ₂ in the retentate	57% of CH ₄ in the feed

In the first example, the component of highest percentage in the feed is the most permeable component. The permeate purity is quite high, but the recovery is not. In the second example, the component of highest purity in the feed is not the most permeable component. The purity of the retentate is reasonably high, but, again, the recovery is not. To further increase the purity of one product and the recovery of the main component in that product, membrane stages are cascaded with recycle. Consider the separation of air to produce a high-purity nitrogen retentate and an oxygen-enriched permeate. Shown in Figure 5.17 are three membrane-separation systems, studied by Prasad et al. [6] for the production of high-purity nitrogen from air, using a membrane material that is more permeable to oxygen. The first system is just a single stage. The second system is a cascade of two stages, with recycle of permeate from the second stage to the first stage. The third system is a cascade of three stages with permeate recycles from stage 3 to stage 2 and stage 2 to stage 1. The two cascades are similar to the single-section, counter-current stripping cascade shown in Figure 5.8b, with the membrane feed, permeate, and retentate corresponding, respectively, to the stripper entering liquid, exiting vapor, and exiting liquid. However, the membrane cascades do not include a stream corresponding to the stripper entering

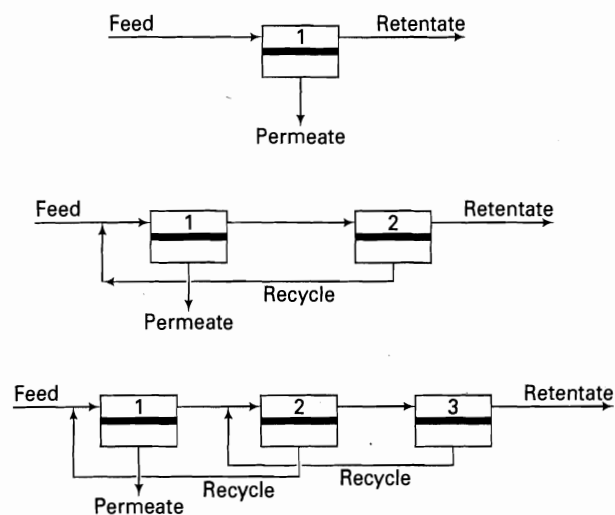


Figure 5.17 Membrane cascades.

vapor. Not shown in Figure 5.17 are recycle gas compressors. Typical calculations of Prasad et al. [6] give the following results:

Membrane System	Mol% N ₂ in Retentate	% Recovery of N ₂
Single stage	98	45
Two-stage cascade	99.5	48
Three-stage cascade	99.9	50

These results show that a high purity can be obtained with a single-section membrane cascade, but without major improvement in the recovery. To obtain both high purity and high recovery, a two-section membrane cascade is necessary, as discussed in Section 14.3.

5.6 HYBRID SYSTEMS

To reduce costs, particularly energy cost, make possible a difficult separation, and/or improve the degree of separation, *hybrid systems*, consisting of two or more separation operations of different types in series are used. Although combinations of membrane separators with other separation operations are the most common, other combinations have found favor. Table 5.1 is a partial list of hybrid systems that are used commercially or have received considerable attention. Examples of applications are included for some hybrid systems. Not included in Table 5.1 are hybrid systems consisting of distillation combined with extractive distillation, azeotropic distillation, and/or liquid-liquid extraction, which are very common and are considered in detail in Chapter 11.

The first example in Table 5.1 is a hybrid system that combines pressure-swing adsorption (PSA), to preferentially

Table 5.1 Hybrid Systems

Hybrid System	Separation Example
Adsorption—gas permeation	Nitrogen—Methane
Simulated moving bed adsorption—distillation	Metaxylene-paraxylene with ethylbenzene eluent
Chromatography—crystallization	—
Crystallization—distillation	—
Crystallization—pervaporation	—
Crystallization—liquid-liquid extraction	Sodium carbonate—water
Distillation—adsorption	Ethanol—water
Distillation—crystallization	—
Distillation—gas permeation	Propylene—propane
Distillation—pervaporation	Ethanol—water
Gas permeation—absorption	Dehydration of natural gas
Reverse osmosis—distillation	Carboxylic acids—water
Reverse osmosis—evaporation	Concentration of wastewater
Stripper—gas permeation	Recovery of ammonia and hydrogen sulfide from sour water

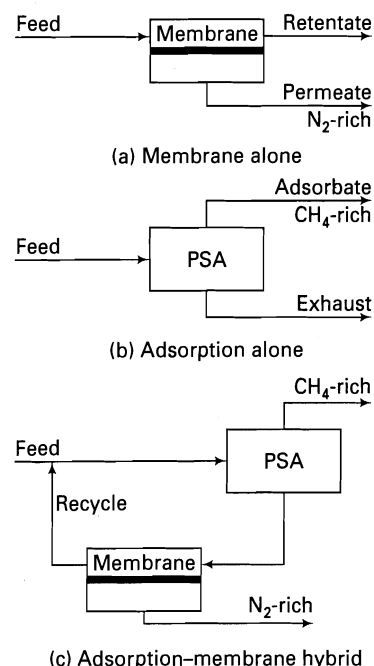
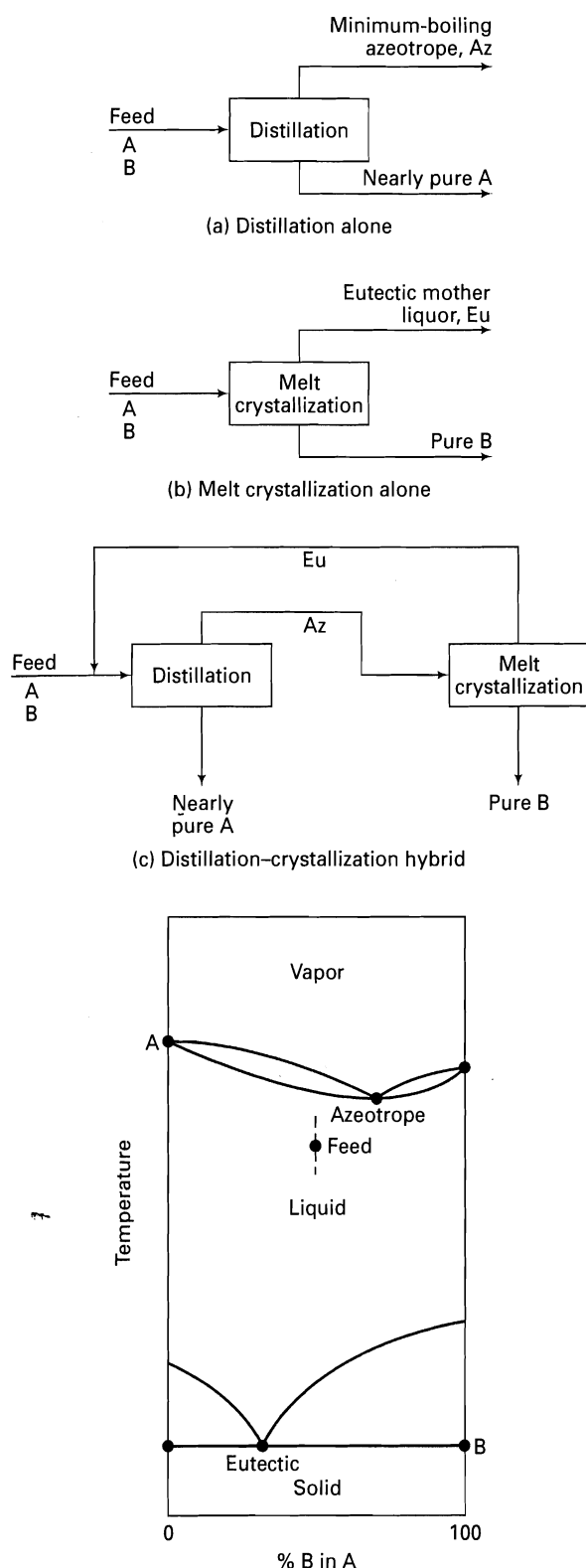


Figure 5.18 Separation of methane from nitrogen.

remove methane, with a gas-permeation membrane operation to preferentially remove nitrogen. The permeate is recycled to the adsorption step. Figure 5.18 shows this hybrid system compared to the use of just a single-stage gas-permeation membrane operation and a single-stage pressure-swing adsorption operation. Only the hybrid system is capable of making a relatively sharp separation between methane and nitrogen. Typical products obtained from these three processes are compared in Table 5.2 for 100,000 scfh of a feed containing 80 mol% methane and 20 mol% nitrogen. For all three processes, the methane-rich product contains 97 mol% methane. However, only the hybrid system gives a nitrogen-rich product containing a nitrogen composition greater than 90 mol%, and a high recovery of methane (98%). The methane recovery for a membrane alone is

Table 5.2 Typical Products for Processes in Figure 5.18

	Flow Rate, Mscfh	Mol% CH ₄	Mol% N ₂
Feed gas	100	80	20
Membrane only:			
Retentate	47.1	97	3
Permeate	52.9	65	35
PSA only:			
Adsorbate	70.6	97	3
Exhaust	29.4	39	61
Hybrid system:			
CH ₄ -rich	81.0	97	3
N ₂ -rich	19.0	8	92



(d) Phase diagram for distillation-crystallization hybrid system.

Figure 5.19 Separation of an azeotropic- and eutectic-forming mixture.

only 57%, while the adsorber alone gives 86%. The hybrid system is clearly superior to a single membrane or adsorber.

No application is shown in Table 5.1 for hybrid systems of crystallization and distillation. However, there is much interest because Berry and Ng [7] show such systems can

overcome the limitations of eutectics in crystallization and azeotropes in distillation. Furthermore, although streams containing solids are more difficult to process than fluids, crystallization requires just a single stage to obtain high-purity crystals. Figure 5.19 includes one of the many distillation and crystallization hybrid configurations discussed by Berry and Ng [7]. The feed is a mixture of A and B, which, as shown in the accompanying phase diagram, form both an azeotrope in the vapor-liquid region and a eutectic in the liquid-solid region at a lower temperature. With respect to component B, the feed composition in Figure 5.19 lies between the eutectic and azeotropic compositions. If distillation alone is used with a sufficient number of stages, the distillate composition will approach that of the minimum-boiling azeotrope, Az, and the bottoms will approach pure A. If melt crystallization alone is used, the two products will be crystals of pure B and a mother liquor approaching the eutectic composition, Eu. The hybrid system in Figure 5.19 combines distillation with melt crystallization to produce both pure B and nearly pure A. The feed enters the distillation column, where the distillate of near-azeotropic composition is sent to the melt crystallizer. Here, the mother liquor of near-eutectic composition is recovered and recycled to the distillation column. The net result is a separation, with nearly pure A obtained as bottoms from the distillation column and pure B obtained from the crystallizer.

Another hybrid system receiving considerable attention is the combination of distillation and pervaporation for separation of azeotropic mixtures, particularly ethanol-water. As discussed in Section 14.7, distillation produces a bottoms of nearly pure water and a distillate of the azeotrope, which is sent to the pervaporation step, producing a nearly pure ethanol retentate and a water-rich permeate that is recycled to the distillation step.

5.7 DEGREES OF FREEDOM AND SPECIFICATIONS FOR COUNTERCURRENT CASCADES

The solution to a multicomponent, multiphase, multistage separation problem is found in the simultaneous solution of the material balance, energy balance, and phase equilibria equations. This implies that a sufficient number of design variables is specified so that the number of remaining unknown (output) variables exactly equals the number of independent equations. In this section, the degrees-of-freedom analysis discussed in Section 4.1 for a single equilibrium stage is extended to one- and multiple-section countercurrent cascades.

An intuitively simple, but operationally complex, method of finding N_D , the number of independent design variables, *degrees of freedom*, or *variance* in the process, is to enumerate all pertinent variables, N_V , and to subtract from these the total number of independent equations or relationships, N_E , relating the variables:

$$N_D = N_V - N_E \quad (5-67)$$

This approach to separation process design was developed by Kwauk [8], and a modification of his methodology forms the basis for this discussion.

Typically, the variables in a separation process are intensive variables such as composition, temperature, and pressure; extensive variables such as flow rate or the heat-transfer rate; and equipment parameters such as the number of equilibrium stages. Physical properties such as enthalpy or K -values are not counted because they are functions of the intensive variables. The variables are relatively easy to enumerate, but to achieve an unambiguous count of N_E it is necessary to carefully seek out all independent relationships due to material and energy conservations, phase-equilibria restrictions, process specifications, and equipment configurations.

Separation equipment consists of physically identifiable elements (equilibrium stages, condensers, reboilers, etc.) as well as stream dividers and stream mixers. It is helpful to examine each element separately, before synthesizing the complete system.

Stream Variables

For each single-phase stream containing C components, a complete specification of intensive variables consists of C mole fractions (or other concentration variables) plus temperature and pressure, or $C + 2$ variables. However, only $C - 1$ of the feed mole fractions are independent, because the other mole fraction must satisfy the mole-fraction constraint:

$$\sum_{i=1}^C \text{mole fractions} = 1.0$$

Thus, only $C + 1$ intensive stream variables can be specified. This is in agreement with the phase rule, which states that, for a single-phase system, the intensive variables are specified by $C - \mathcal{P} + 2 = C + 1$ variables. To this number can be added the total flow rate of the stream, an extensive variable. Although the missing mole fraction is often treated implicitly, it is preferable for completeness to include the missing mole fraction in the list of stream variables and then to include in the list of equations the above mole-fraction constraint. Thus, associated with each stream are $C + 3$ variables. For example, for a liquid-phase stream, the variables are liquid mole fractions $x_1, x_2, \dots, x_C$; total molar flow rate L ; temperature T ; and pressure P .

Adiabatic or Nonadiabatic Equilibrium Stage

For a single adiabatic or nonadiabatic equilibrium stage with two entering streams and two exit streams, as shown in Figure 5.20, the variables are those associated with the four streams plus the heat transfer rate to or from the stage. Thus:

$$N_V = 4(C + 3) + 1 = 4C + 13$$

The exiting streams V_{OUT} and L_{OUT} are in equilibrium, so there are equilibrium restrictions as well as component material balances, a total material balance, an energy balance, and mole fraction constraints. Thus, the equations relating

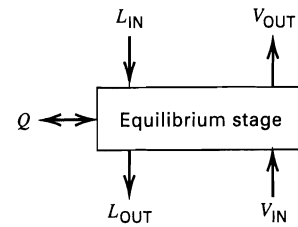


Figure 5.20 Equilibrium stage with heat addition.

these variables and N_E are

Equations	Number of Equations
Pressure equality $P_{VOUT} = P_{LOUT}$	1
Temperature equality, $T_{VOUT} = T_{LOUT}$	1
Phase equilibrium relationships, $(y_i)_{VOUT} = K_i(x_i)_{LOUT}$	C
Component material balances, $L_{IN}(x_i)_{L_{IN}} + V_{IN}(y_i)_{V_{IN}} = L_{OUT}(x_i)_{L_{OUT}} + V_{OUT}(y_i)_{V_{OUT}}$	$C - 1$
Total material balance, $L_{IN} + V_{IN} = L_{OUT} + V_{OUT}$	1
Energy balance, $Q + h_{L_{IN}}L_{IN} = h_{V_{IN}}V_{IN} = h_{L_{OUT}}L_{OUT} + h_{V_{OUT}}V_{OUT}$	1
Mole fraction constraints in entering and exiting streams e.g., $\sum_{i=1}^C (x_i)_{L_{IN}} = 1$	4
	$N_E = 2C + 7$

Alternatively, C , instead of $C - 1$, component material balances can be written. The total material balance is then a dependent equation obtained by summing the component material balances and applying the mole-fraction constraints to eliminate the mole fractions. From (5-67),

$$N_D = (4C + 13) - (2C + 7) = 2C + 6$$

Notice that the coefficient of C is equal to 2, the number of streams entering the stage.

Several different sets of design variables can be specified. A typical set includes complete specification of the two entering streams as well as the stage pressure and heat transfer rate.

Variable Specification	Number of Variables
Component mole fractions, $(x_i)_{L_{IN}}$	$C - 1$
Total flow rate, L_{IN}	1
Component mole fractions, $(y_i)_{V_{IN}}$	$C - 1$
Total flow rate, V_{IN}	1
Temperature and pressure of L_{IN}	2
Temperature and pressure of V_{IN}	2
Stage pressure, $(P_{VOUT}$ or $P_{LOUT})$	1
Heat transfer rate, Q	1
	$N_D = 2C + 6$

Specification of these $(2C + 6)$ variables permits calculation of the unknown variables L_{OUT} , V_{OUT} , $(x_C)_{L_{IN}}$, $(y_C)_{V_{IN}}$, all $(x_i)_{L_{OUT}}$, T_{OUT} , and all $(y_i)_{V_{OUT}}$, where C denotes the missing mole fractions in the two entering streams.

Single-Section, Countercurrent Cascade

Consider the N -stage, single-section, countercurrent cascade unit shown in Figure 5.21. This cascade consists of N adiabatic or nonadiabatic equilibrium-stage elements of the type shown in Figure 5.20. An algorithm is easily developed for enumerating variables, equations, and degrees of freedom for combinations of such elements to form a unit. The number of design variables for the unit is obtained by summing the variables associated with each element and then subtracting from the total variables the $C + 3$ variables for each of the N_R redundant interconnecting streams that arise when the output of one element becomes the input to another. Also, if an unspecified number of repetitions of any element occurs within the unit, an additional variable is added, one for each group of repetitions, giving a total of N_A additional variables. In a similar manner, the number of independent equations for the unit is obtained by summing the values of N_E for the units and then subtracting the N_R redundant mole-fraction constraints. The number of degrees of freedom is obtained as before, from (5-67). Thus,

$$(N_V)_{\text{unit}} = \sum_{\text{all elements, } e} (N_V)_e - N_R(C + 3) + N_A \quad (5-68)$$

$$(N_E)_{\text{unit}} = \sum_{\text{all elements, } e} (N_E)_e - N_R \quad (5-69)$$

Combining (5-67), (5-68), and (5-69), we have

$$(N_D)_{\text{unit}} = \sum_{\text{all elements, } e} (N_D)_e - N_R(C + 2) + N_A \quad (5-70)$$

or

$$(N_D)_{\text{unit}} = (N_V)_{\text{unit}} - (N_E)_{\text{unit}} \quad (5-71)$$

For the N -stage cascade unit of Figure 5.21, with reference to the above degrees-of-freedom analysis for the single

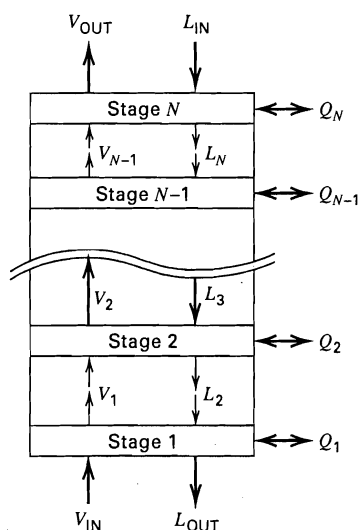


Figure 5.21 An N -stage cascade.

adiabatic or nonadiabatic equilibrium-stage element, the total number of variables from (5-68) is

$$\begin{aligned} (N_V)_{\text{unit}} &= N(4C + 13) - [2(N - 1)](C + 3) + 1 \\ &= 7N + 2NC + 2C + 7 \end{aligned}$$

since $2(N - 1)$ interconnecting streams exist. The additional variable is the total number of stages (i.e., $N_A = 1$).

The number of independent relationships from (5-69) is

$$(N_E)_{\text{unit}} = N(2C + 7) - 2(N - 1) = 5N + 2NC + 2$$

since $2(N - 1)$ redundant mole-fraction constraints exist.

The number of degrees of freedom from (5-71) is

$$(N_D)_{\text{unit}} = N_V - N_E = 2N + 2C + 5$$

Note, again, that the coefficient of C is 2, the number of streams entering the cascade. For a cascade, the coefficient of N is always 2 (corresponding to stage P and Q).

One possible set of design variables is

Variable Specification	Number of Variables
Heat transfer rate for each stage (or adiabaticity)	N
Stage pressures	N
Stream V_{IN} variables	$C + 2$
Stream L_{IN} variables	$C + 2$
Number of stages	1
	$2N + 2C + 5$

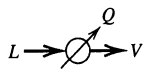
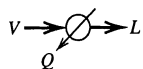
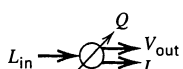
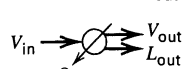
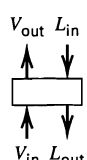
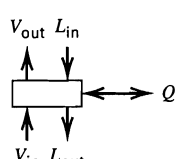
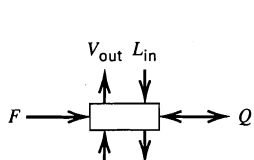
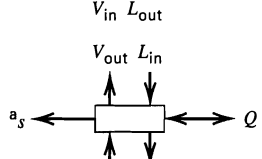
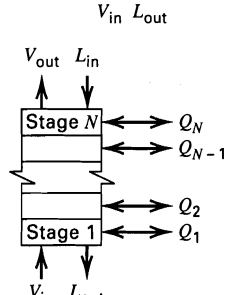
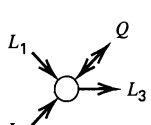
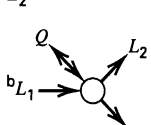
Output variables for this specification include missing mole fractions for V_{IN} and L_{IN} , stage temperatures, and the variables associated with the V_{OUT} stream, L_{OUT} stream, and interstage streams. This N -stage cascade unit can represent simple absorbers, strippers, or liquid-liquid extractors.

Two-Section, Countercurrent Cascades

Two-section, countercurrent cascades can consist not only of adiabatic or nonadiabatic equilibrium-stage elements, but also of other elements of the type shown in Table 5.3, including total and partial reboilers; total and partial condensers; equilibrium stages with a feed, F , or a sidestream S ; and stream mixers and dividers. These different elements can be combined into any of a number of complex cascades by applying to (5-68) to (5-71) the values of N_V , N_E , and N_D given in Table 5.3 for the different elements.

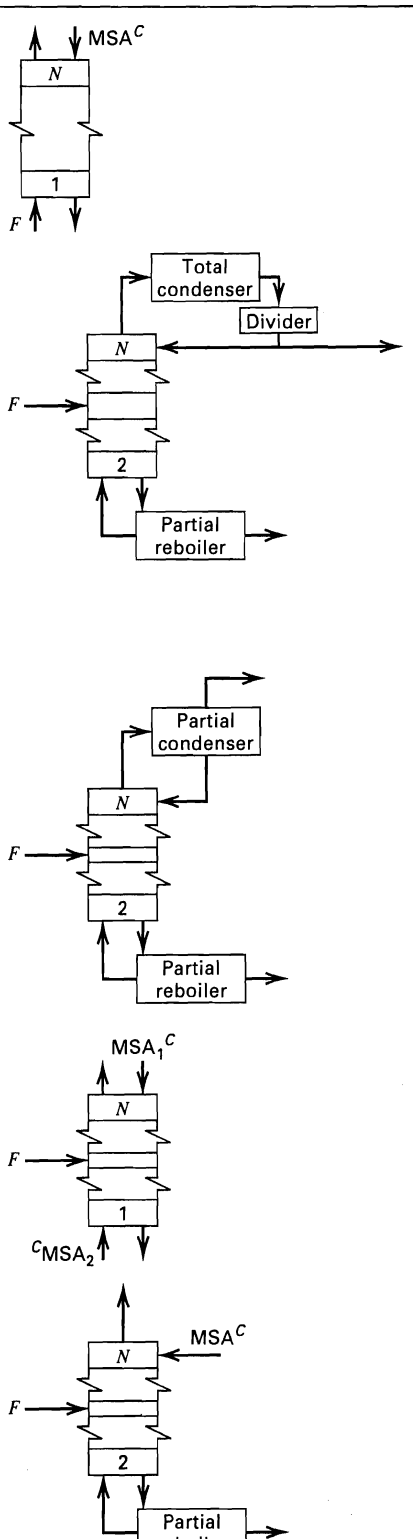
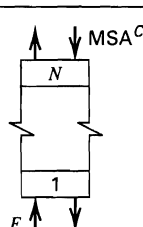
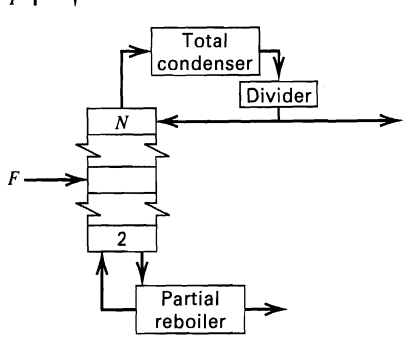
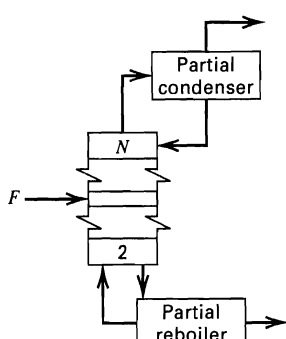
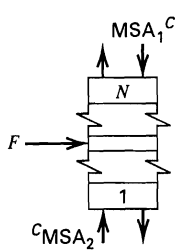
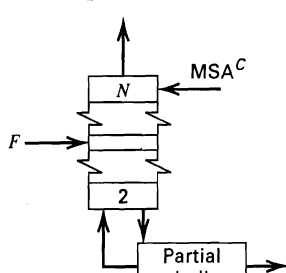
The design or simulation of multistage separation operations involves solving the variable relationships for output variables after selecting values of design variables to satisfy the degrees of freedom. Two cases are commonly encountered. In case I, the design case, recovery specifications are made for one or two key components and the number of required equilibrium stages is determined. In case II, the simulation case, the number of equilibrium stages is specified and component separations are computed. For rigorous calculations involving multicomponent feeds, the second case is more widely applied because less computational complexity is involved with the number of stages fixed. Table 5.4 is a

Table 5.3 Degrees of Freedom for Separation Operation Elements and Units

Schematic	Element or Unit Name	N_V , Total Number of Variables	N_E , Independent Relationships	N_D , Degrees of Freedom
(a) 	Total boiler (reboiler)	$(2C + 7)$	$(C + 3)$	$(C + 4)$
(b) 	Total condenser	$(2C + 7)$	$(C + 3)$	$(C + 4)$
(c) 	Partial (equilibrium) boiler (reboiler)	$(3C + 10)$	$(2C + 6)$	$(C + 4)$
(d) 	Partial (equilibrium) condenser	$(3C + 10)$	$(2C + 6)$	$(C + 4)$
(e) 	Adiabatic equilibrium stage	$(4C + 12)$	$(2C + 7)$	$(2C + 5)$
(f) 	Equilibrium stage with heat transfer	$(4C + 13)$	$(2C + 7)$	$(2C + 6)$
(g) 	Equilibrium feed stage with heat transfer and feed	$(5C + 16)$	$(2C + 8)$	$(3C + 8)$
(h) 	Equilibrium stage with heat transfer and sidestream	$(5C + 16)$	$(3C + 9)$	$(2C + 7)$
(i) 	N -connected equilibrium stages with heat transfer	$(7N + 2NC + 2C + 7)$	$(5N + 2NC + 2)$	$(2N + 2C + 5)$
(j) 	Stream mixer	$(3C + 10)$	$(C + 4)$	$(2C + 6)$
(k) 	Stream divider	$(3C + 10)$	$(2C + 5)$	$(C + 5)$

^aSidestream can be vapor or liquid.^bAlternatively, all streams can be vapor.

Table 5.4 Typical Variable Specifications for Design Cases

Unit Operation		N_D	Variable Specification ^a	
			Case I, Component Recoveries Specified	Case II, Number of Equilibrium Stages Specified
(a) Absorption (two inlet streams)		$2N + 2C + 5$	1. Recovery of one key component	1. Number of stages
(b) Distillation (one inlet stream, total condenser, partial reboiler)		$2N + C + 9$	1. Condensate at saturation temperature 2. Recovery of light key component 3. Recovery of heavy key component 4. Reflux ratio (> minimum) 5. Optimal feed stage ^b	1. Condensate at saturation temperature 2. Number of stages above feed stage 3. Number of stages below feed stage 4. Reflux ratio 5. Distillate flow rate
(c) Distillation (one inlet stream, partial condenser, partial reboiler, vapor distillate only)		$(2N + C + 6)$	1. Recovery of light key component 2. Recovery of heavy key component 3. Reflux ratio (> minimum) 4. Optimal feed stage ^b	1. Number of stages above feed stage 2. Number of stages below feed stage 3. Reflux ratio 4. Distillate flow rate
(d) Liquid-liquid extraction with two solvents (three inlet streams)		$2N + 3C + 8$	1. Recovery of key component 1 2. Recovery of key component 2	1. Number of stages above feed 2. Number of stages below feed
(e) Reboiled absorption (two inlet streams)		$2N + 2C + 6$	1. Recovery of light key component 2. Recovery of heavy key component 3. Optimal feed stage ^b	1. Number of stages above feed 2. Number of stages below feed 3. Bottoms flow rate

(continued)

Table 5.4 (Continued)

Unit Operation	N_D	Variable Specification ^a	
		Case I, Component Recoveries Specified	Case II, Number of Equilibrium Stages Specified
(f) Reboiled stripping (one inlet stream)	$2N + C + 3$	1. Recovery of one key component 2. Reboiler heat duty ^d	1. Number of stages 2. Bottoms flow rate
(g) Distillation (one inlet stream, partial condenser, partial reboiler, both liquid and vapor distillates)	$2N + C + 9$	1. Ratio of vapor distillate to liquid distillate 2. Recovery of light key component 3. Recovery of heavy key component 4. Reflux ratio (> minimum) 5. Optimal feed stage ^b	1. Ratio of vapor distillate to liquid distillate 2. Number of stages above feed stage 3. Number of stages below feed stage 4. Reflux ratio 5. Liquid distillate flow rate
(h) Extractive distillation (two inlet streams, total condenser, partial reboiler, single-phase condensate)	$2N + 2C + 12$	1. Condensate at saturation temperature 2. Recovery of light key component 3. Recovery of heavy key component 4. Reflux ratio (> minimum) 5. Optimal feed stage ^b 6. Optimal MSA stage ^b	1. Condensate at saturation temperature 2. Number of stages above MSA stage 3. Number of stages between MSA and feed stages 4. Number of stages below feed stage 5. Reflux ratio 6. Distillate flow rate
(i) Liquid-liquid extraction (two inlet streams)	$2N + 2C + 5$	1. Recovery of one key component	1. Number of stages
(j) Stripping (two inlet streams)	$2N + 2C + 5$	1. Recovery of one key component	1. Number of stages

^aDoes not include the following variables, which are also assumed specified: all inlet stream variables ($C + 2$ for each stream); all element and unit pressures; all element and unit heat transfer rates except for condensers and reboilers.

^bOptimal stage for introduction of inlet stream corresponds to minimization of total stages.

^cFor case I variable specifications, MSA flow rates must be greater than minimum values for specified recoveries.

^dFor case I variable specifications, reboiler heat duty must be greater than minimum value for specified recovery.

summary of possible variable specifications for each of these two cases for a number of separator types discussed in Chapter 1 and shown in Table 1.1. For all separators in Table 5.4, it is assumed that all inlet streams are completely specified (i.e., $C - 1$ mole fractions, total flow rate, temperature, and

pressure) and all element and unit pressures and heat transfer rates (except for condensers and reboilers) are specified. Thus, only variable specifications for satisfying the remaining degrees of freedom are listed.

EXAMPLE 5.5

Consider a multistage distillation column with one feed, one sidestream, a total condenser, a partial reboiler, and provisions for heat transfer to or from any stage. Determine the number of degrees of freedom and a reasonable set of specifications.

SOLUTION

This separator is assembled as shown in Figure 5.22, from the circled elements and units, which are all found in Table 5.3. The total variables are determined by summing the variables $(N_V)_e$ for each

element from Table 5.3 and then subtracting the redundant variables due to interconnecting flows. As before, redundant mole-fraction constraints are subtracted from the summation of independent relationships for each element $(N_E)_e$. This problem was first treated by Gilliland and Reed [9] and more recently by Kwauk [8]. Differences in N_D obtained by various authors are due, in part, to their method of numbering stages. Here, the partial reboiler is the first equilibrium stage. From Table 5.3, element variables and relationships are as follows:

Element or Unit	$(N_V)_e$	$(N_E)_e$
Total condenser	$(2C + 7)$	$(C + 3)$
Reflux divider	$(3C + 10)$	$(2C + 5)$
$(N - S)$ stages	$[7(N - S) + 2(N - S)C + 2C + 7]$	$[5(N - S) + 2(N - S)C + 2]$
Sidestream stage	$(5C + 16)$	$(3C + 9)$
$(S - 1) - F$ stages	$[7(S - 1 - F) + 2(S - 1 - F)C + 2C + 7]$	$[5(S - 1 - F) + 2(S - 1 - F)C + 2]$
Feed stage	$(5C + 16)$	$(2C + 8)$
$(F - 1) - 1$ stages	$[7(F - 2) + 2(F - 2)C + 2C + 7]$	$[5(F - 2) + 2(F - 2)C + 2]$
Partial reboiler	$(3C + 10)$	$(2C + 6)$
	$\sum(N_V)_e = 7N + 2NC + 18C + 59$	$\sum(N_E)_e = 5N + 2NC + 4C + 22$

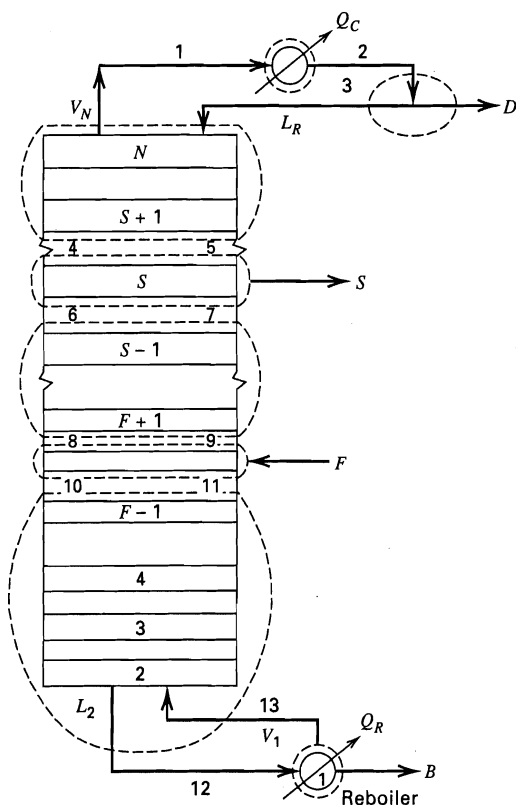


Figure 5.22 Complex distillation unit.

Subtracting $(C + 3)$ redundant variables for 13 interconnecting streams, according to (5-68), with $N_A = 0$ (no unspecified repetitions), gives

$$(N_V)_{\text{unit}} = \sum(N_V)_e - 13(C + 3) = 7N + 2NC + 5C + 20$$

Subtracting the corresponding 13 redundant mole-fraction constraints, according to (5-69),

$$(N_E)_{\text{unit}} = \sum(N_E)_e - 13 = 5N + 2NC + 4C + 9$$

Therefore, from (5-71),

$$N_D = (7N + 2NC + 5C + 20) - (5N + 2NC + 4C + 9) = 2N + C + 11$$

Note that the coefficient of C is only 1, because there is only one feed, and, again, the coefficient of N is 2.

A set of feasible design variable specifications is

Variable Specification	Number of Variables
1. Pressure at each stage (including partial reboiler)	N
2. Pressure at reflux divider outlet	1
3. Pressure at total condenser outlet	1
4. Heat transfer rate for each stage (excluding partial reboiler)	$(N - 1)$
5. Heat transfer rate for divider	1
6. Feed mole fractions and total feed rate	C

Variable Specification	Number of Variables
7. Feed temperature	1
8. Feed pressure	1
9. Condensate temperature (e.g., saturated liquid)	1
10. Total number of stages, N	1
11. Feed stage location	1
12. Sidestream stage location	1
13. Sidestream total flow rate, S	1
14. Total distillate flow rate, D or D/F	1
15. Reflux flow rate, L_R , or reflux ratio, L_R/D	1

$$N_D = (2N + C + 11)$$

In most separation operations, variables related to feed conditions, stage heat-transfer rates, and stage pressure are known or set. Remaining specifications have proxies, provided that the variables are mathematically independent of each other and of those already known. Thus, in the above list, the first nine entries are almost always known or specified. Variables 10 to 15, however, have surrogates. Some of these are

- 16. Condenser heat duty, Q_C
- 17. Reboiler heat duty, Q_R
- 18. Recovery or mole fraction of one component in bottoms
- 19. Recovery or mole fraction of one component in distillate

Heat duties Q_C and Q_R are not good design variables because they are difficult to specify. Condenser duty Q_C , for example, must be specified so that the condensate temperature lies between that corresponding to a saturated liquid and the freezing point of the condensate. Otherwise, a physically unrealizable (or no) solution to the problem is obtained. Similarly, it is much easier to calculate Q_R knowing the total flow rate and enthalpy of the bottom streams than vice versa. In general, Q_R and Q_C are so closely related that it is not advisable to specify both.

Other proxies are possible, such as a stage temperature, a flow rate leaving a stage, or any independent variable that characterizes the process. The problem of independence of variables requires careful consideration. Distillate product rate, Q_C , and L_R/D , for example, are not independent. It should also be noted that, for the design case, recoveries of no more than two species (items 18 and 19) are specified. These species are referred to as key components. Attempts to specify recoveries of three or four species will usually result in an unsuccessful solution of the equations.

The degrees of freedom for the complex distillation unit of Figure 5.22 can be determined quickly by modifying a

similar unit operation in Table 5.4. The closest unit is (b), which differs from the unit in Figure 5.22 by only a sidestream. From Table 5.3, we see that an equilibrium stage with heat transfer but without a sidestream [element (f)] has $N_D = (2C + 6)$, while an equilibrium stage with heat transfer and with a sidestream [element (h)] has $N_D = (2C + 7)$ or one additional degree of freedom. In addition, when this sidestream stage is placed in a cascade, an additional degree of freedom is added for the location of the sidestream stage. Thus, two degrees of freedom are added to $N_D = 2N + C + 9$ for unit operation (b) in Table 5.4. The result is $N_D = 2N + C + 11$, which is identical to that determined in the above example.

In a similar manner, the above example can be readily modified to include a second feed stage. By comparing values of N_D for elements (f) and (g) in Table 5.3, it is seen that a feed adds $C + 2$ degrees of freedom. In addition, one more degree of freedom must be added for the location of this feed stage in a cascade. Thus, a total of $C + 3$ degrees of freedom are added, giving $N_D = 2N + 2C + 14$.

SUMMARY

1. A cascade is a collection of contacting stages arranged to: (a) accomplish a separation that cannot be achieved in a single stage, and/or (b) reduce the amount of mass- or energy-separating agent.
2. Cascades are single- or multiple-sectioned and may be configured in cocurrent, crosscurrent, or countercurrent arrangements. Cascades are readily computed when governing equations are linear in component split ratios.
3. Stage requirements for a countercurrent solid-liquid leaching and/or washing cascade, involving constant underflow and mass transfer of one component, are given by (5-10).
4. Stage requirements for a single-section, liquid-liquid extraction cascade assuming a constant distribution coefficient and immiscible solvent and carrier are given by (5-19), (5-22), and (5-29) for cocurrent, crosscurrent, and countercurrent flow arrangements, respectively. The countercurrent cascade is the most efficient.

5. Single-section stage requirements for a countercurrent cascade for absorption and stripping can be estimated with the Kremser equations, (5-48), (5-50), (5-54), and (5-55). A single-section, countercurrent cascade is limited in its ability to achieve a separation between two components.
6. The Kremser equations can be combined for a two-section cascade to give (5-66), which is suitable for making approximate calculations of component splits for distillation. A two-section, countercurrent cascade can achieve a sharp split between two key components. The rectifying section purifies the light components and increases recovery of heavy components. The stripping section provides the opposite function.
7. Equilibrium cascade equations involve parameters referred to as washing W , extraction E , absorption A , and stripping S , factors that involve distribution coefficients, such as K , K_D , and R , and phase flow ratios, such as S/F and L/V .

8. Single-section membrane cascades increase purity of one product and recovery of the main component in that product.
9. Hybrid systems of different types reduce energy expenditures, make possible separations that are otherwise difficult, and/or improve the degree of separation.
10. The number of degrees of freedom (number of specifications) for a mathematical model of a cascade is the difference between

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2. KREMSER, A., *Natl. Petroleum News*, **22**(21), 43–49 (May 21, 1930).
3. EDMISTER, W.C., *AIChE J.*, **3**, 165–171 (1957).
4. SMITH, B.D., and W.K. BRINKLEY, *AIChE J.*, **6**, 446–450 (1960).
5. LYSTER, W.N., S.L. SULLIVAN, JR., D.S. BILLINGSLEY, and C.D. HOLLAND, *Petroleum Refiner*, **38**(6), 221–230 (1959).

EXERCISES

Section 5.1

5.1 Devise an interlinked cascade of the type shown in Figure 5.2e, but consisting of three columns for the separation of a four-component feed into four products.

5.2 A liquid–liquid extraction process is conducted batchwise as shown in Figure 5.23. The process begins in vessel 1 (original),

the number of unique variables and the number of independent equations that relate the variables. For a single-section, counter-current cascade, the recovery of one component can be specified. For a two-section countercurrent cascade, the recoveries of two components can be specified.

6. PRASAD, R., F. NOTARO, and D.R. THOMPSON, *J. Membrane Science*, **94**, Issue 1, 225–248 (1994).
7. BERRY, D.A., and K.M. NG, *AIChE J.*, **43**, 1751–1762 (1997).
8. KWAUK, M., *AIChE J.*, **2**, 240–248 (1956).
9. GILLILAND, E.R., and C.E. REED, *Ind. Eng. Chem.*, **34**, 551–557 (1942).

where 100 mg each of solutes A and B are dissolved in 100 ml of water. After adding 100 ml of an organic solvent that is more selective for A than B, the distribution of A and B becomes that shown for equilibration 1 with vessel 1. The organic-rich phase is transferred to vessel 2 (transfer), leaving the water-rich phase in vessel 1 (transfer). Assume that water and the organic solvent are immiscible. Next, 100 ml of water is added to vessel 2, resulting in the phase distribution shown for vessel 2 (equilibration 2). Also, 100 ml of organic solvent is added to vessel 1 to give the phase distribution shown for vessel 1 (equilibration 2). The batch process is continued by adding vessel 3 and then 4 to obtain the results shown.

(a) Carefully study the process in Figure 5.23 and then draw a corresponding cascade diagram, labeled in a manner similar to Figure 5.2(b).

(b) Is the process of the cocurrent, countercurrent, or crosscurrent type?

(c) Compare the separation achieved with that for a single-batch equilibrium step.

(d) How could the process be modified to make it a countercurrent cascade [see O. Post and L.C. Craig, *Anal. Chem.*, **35**, 641 (1963)].

5.3 Nitrogen is to be removed from a gas mixture with methane by gas permeation (see Table 1.2) using a glassy polymer membrane that is selective for nitrogen. However, the desired degree of separation cannot be achieved in one stage. Draw sketches of two different two-stage membrane cascades that might be considered to perform the desired separation.

Section 5.2

5.4 In Example 4.9, 83.25% of the oil in soybeans is leached by benzene using a single equilibrium stage. Calculate the percent extraction of oil if:

(a) Two countercurrent equilibrium stages are used to process 5,000 kg/h of soybean meal with 5,000 kg/h of benzene.

(b) Three countercurrent equilibrium stages are used to process the same flows as in part (a).

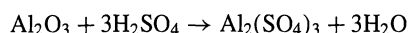
(c) Also, determine the number of countercurrent equilibrium stages required to extract 98% of the oil if a solvent rate of twice the minimum value is used.

Vessel 1				Original
Organic				
Aqueous	100 A 100 B			
Vessel 1				Equilibration 1
Organic	66.7 A 33.3 B			
Aqueous	33.3 A 66.7 B			
Vessel 2				Transfer
Organic		66.7 A 33.3 B		
Aqueous	33.3 A 66.7 B			
Vessel 2				Equilibration 2
Organic	22.2 A 22.2 B	44.4 A 11.1 B		
Aqueous	11.1 A 44.4 B	22.2 A 22.2 B		
Vessel 3				Transfer
Organic		22.2 A 22.2 B	44.4 A 11.1 B	
Aqueous	11.1 A 44.4 B	22.2 A 22.2 B		
Vessel 3				Equilibration 3
Organic	7.4 A 14.8 B	29.6 A 14.8 B	29.6 A 3.7 B	
Aqueous	3.7 A 29.6 B	14.8 A 29.6 B	14.8 A 7.4 B	
Vessel 4				Transfer
Organic		7.4 A 14.8 B	29.6 A 3.7 B	
Aqueous	3.7 A 29.6 B	14.8 A 29.6 B	14.8 A 7.4 B	
Vessel 4				Equilibration 4
Organic	2.5 A 9.9 B	14.8 A 14.8 B	29.6 A 7.4 B	
Aqueous	1.2 A 19.7 B	7.4 A 29.6 B	14.8 A 14.8 B	

Figure 5.23 Liquid–liquid extraction process for Exercise 5.2.

5.5 For Example 5.1, involving the separation of sodium carbonate from an insoluble oxide, compute the minimum solvent feed rate in pounds per hour. What is the ratio of actual solvent rate to the minimum solvent rate? Determine and plot the percent recovery of soluble solids with a cascade of five countercurrent equilibrium stages for solvent flow rates from 1.5 to 7.5 times the minimum value.

5.6 Aluminum sulfate, commonly called alum, is produced as a concentrated aqueous solution from bauxite ore by reaction with aqueous sulfuric acid, followed by a three-stage, countercurrent washing operation to separate soluble aluminum sulfate from the insoluble content of the bauxite ore, followed by evaporation. In a typical process, 40,000 kg/day of solid bauxite ore containing 50 wt% Al_2O_3 and 50% inert is crushed and fed together with the stoichiometric amount of 50 wt% aqueous sulfuric acid to a reactor, where the Al_2O_3 is reacted completely to alum by the reaction



The slurry effluent from the reactor (digester), consisting of solid inert material from the ore and an aqueous solution of aluminum sulfate is then fed to a three-stage, countercurrent washing unit to separate the aqueous aluminum sulfate from the inert material. If the solvent is 240,000 kg/day of water and the underflow from each washing stage is 50 wt% water on a solute-free basis, compute the flow rates in kilograms per day of aluminum sulfate, water, and inert solid in each of the two product streams leaving the cascade. What is the percent recovery of the aluminum sulfate? Would the addition of one more stage be worthwhile?

5.7 (a) When rinsing clothes with a given amount of water, would one find it more efficient to divide the water and rinse several times; or should one use all the water in one rinse? Explain. (b) Devise a clothes-washing machine that gives the most efficient rinse cycle for a fixed amount of water.

Section 5.3

5.8 An aqueous acetic-acid solution containing 6.0 moles of acid per liter is to be extracted in the laboratory with chloroform at 25°C to recover the acid (B) from chloroform-insoluble impurities present in the water. The water (A) and chloroform (C) are essentially immiscible. If 10 liters of solution are to be extracted at 25°C , calculate the percent extraction of acid obtained with 10 liters of chloroform under the following conditions:

- Using the entire quantity of solvent in a single batch extraction
- Using three batch extractions with one-third of the total solvent used in each batch
- Using three batch extractions with 5 liters of solvent in the first, 3 liters in the second, and 2 liters in the third batch

Assume that the volumetric amounts of the feed and solvent do not change during extraction. Also, assume the distribution coefficient for the acid, $K''_{DB} = (c_B)_C / (c_B)_A = 2.8$, where $(c_B)_C$ = concentration of acid in chloroform and $(c_B)_A$ = concentration of acid in water, both in moles per liter.

5.9 A 20 wt% solution of uranyl nitrate (UN) in water is to be treated with tributyl phosphate (TBP) to remove 90% of the uranyl nitrate. All operations are to be batchwise equilibrium contacts. Assuming that water and TBP are mutually insoluble, how much TBP is required for 100 g of solution if at equilibrium (g UN/g TBP) = 5.5(g UN/g H_2O) and:

- All the TBP is used at once in one stage?
- Half is used in each of two consecutive stages?

- Two countercurrent stages are used?
- An infinite number of crosscurrent stages is used?
- An infinite number of countercurrent stages is used?

5.10 The uranyl nitrate (UN) in 2 kg of a 20 wt% aqueous solution is to be extracted with 500 g of tributyl phosphate. Using the equilibrium data in Exercise 5.9, calculate and compare the percentage recoveries for the following alternative procedures:

- A single-stage batch extraction
- Three batch extractions with one-third of the total solvent used in each batch (the solvent is withdrawn after contacting the entire UN phase)
- A two-stage cocurrent extraction
- A three-stage countercurrent extraction
- An infinite-stage countercurrent extraction
- An infinite-stage crosscurrent extraction

5.11 One thousand kilograms of a 30 wt% dioxane in water solution is to be treated with benzene at 25°C to remove 95% of the dioxane. The benzene is dioxane-free, and the equilibrium data of Example 5.2 can be used. Calculate the solvent requirements for:

- A single batch extraction
- Two crosscurrent stages using equal amounts of benzene
- Two countercurrent stages
- An infinite number of crosscurrent stages
- An infinite number of countercurrent stages

5.12 Chloroform is to be used to extract benzoic acid from wastewater effluent. The benzoic acid is present at a concentration of 0.05 mol/liter in the effluent, which is discharged at a rate of 1,000 liter/h. The distribution coefficient for benzoic acid at process conditions is given by

$$c^I = K_D^{II} c^{II}$$

where $K_D^{II} = 4.2$, c^I = molar concentration of solute in solvent, and c^{II} = molar concentration of solute in water. Chloroform and water may be assumed immiscible. If 500 liters/h of chloroform is to be used, compare the fraction benzoic acid removed in

- A single equilibrium contact
- Three crosscurrent contacts with equal portions of chloroform
- Three countercurrent contacts

5.13 Repeat Example 5.2 with a solvent for which $E = 0.90$. Display your results in a plot like Figure 5.7. Does countercurrent flow still have a marked advantage over crosscurrent flow? Is it desirable to choose the solvent and solvent rate so that $E > 1$? Explain.

Section 5.4

5.14 Repeat Example 5.3 for $N = 1, 3, 10$, and 30 stages. Plot the percent absorption of each of the five hydrocarbons and the total feed gas, as well as the percent stripping of the oil versus the number of stages, N . What can you conclude about the effect of the number of stages on each component?

5.15 Solve Example 5.3 for an absorbent flow rate of 330 lbmol/h and three theoretical stages. Compare your results to the results of Example 5.3 and discuss the effect of trading stages for absorbent flow.

5.16 Estimate the minimum absorbent flow rate required for the separation calculated in Example 5.3 assuming that the key component is propane, whose flow rate in the exit vapor is to be 155.4 lbmol/h.

5.17 Solve Example 5.3 with the addition of a heat exchanger at each stage so as to maintain isothermal operation of the absorber at

- (a) 125°F
(b) 150°F

What is the effect of temperature on absorption in the range of 100 to 150°F?

5.18 One million pound-moles per day of a gas of the following composition is to be absorbed by *n*-heptane at -30°F and 550 psia in an absorber having 10 theoretical stages so as to absorb 50% of the ethane. Calculate the required flow rate of absorbent and the distribution, in lbmol/h, of all the components between the exiting gas and liquid streams.

Component	Mole Percent in Feed Gas	<i>K</i> -value @ -30°F and 550 psia
C ₁	94.9	2.85
C ₂	4.2	0.36
C ₃	0.7	0.066
<i>n</i> C ₄	0.1	0.017
<i>n</i> C ₅	0.1	0.004

5.19 A stripper operating at 50 psia with three equilibrium stages is used to strip 1,000 kmol/h of liquid at 300°F having the following molar composition: 0.03% C₁, 0.22% C₂, 1.82% C₃, 4.47% *n*C₄, 8.59% *n*C₅, 84.87% *n*C₁₀. The stripping agent is 1,000 kmol/h of superheated steam at 300°F and 50 psia. Use the Kremser equation to estimate the compositions and flow rates of the stripped liquid and exiting rich gas.

Assume a *K*-value for C₁₀ of 0.20 and assume that no steam is absorbed. However, calculate the dew-point temperature of the exiting rich gas at 50 psia. If that temperature is above 300°F, what would you suggest be done?

5.20 In Figure 5.12, is anything gained by totally condensing the vapor leaving each stage? Alter the processes in Figure 5.12a and 5.12b so as to eliminate the addition of heat to stages 2 and 3 and still achieve the same separations.

5.21 Repeat Example 5.4 for external reflux flow rates *L*₀ of

- (a) 1,500 lbmol/h
(b) 2,000 lbmol/h
(c) 2,500 lbmol/h

Plot *d*_{C₃}/*b*_{C₃} as a function of *L*₀ from 1,000 to 2,500 lbmol/h. In making the calculations, assume that stage temperatures do not change from the results of Example 5.4. Discuss the effect of reflux ratio on the separation.

5.22 Repeat Example 5.4 for the following numbers of equilibrium stages (see Figure 5.15):

- (a) *M* = 10, *N* = 10
(b) *M* = 15, *N* = 15

Plot *d*_{C₃}/*b*_{C₃} as a function of *M* + *N* from 10 to 30 stages. In making the calculations, assume that state temperatures and total flow rates do not change from the results of Example 5.4. Discuss the effect of the number of stages on the separation.

5.23 Use the Edmister group method to determine the compositions of the distillate and bottoms for the distillation operation shown in Figure 5.24. At column conditions, the feed is approximately 23 mol% vapor.

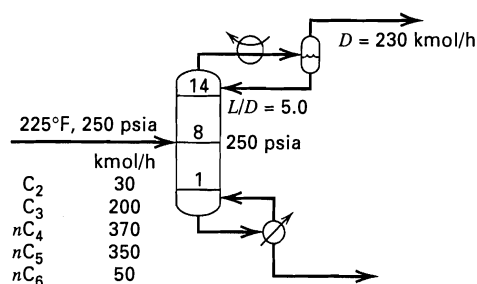


Figure 5.24 Conditions for Exercise 5.23.

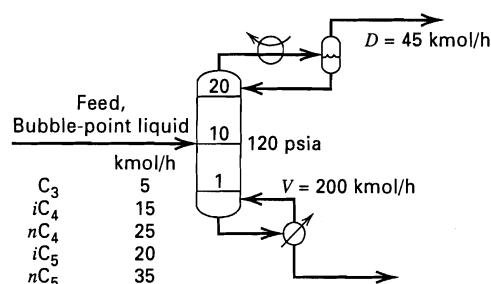


Figure 5.25 Conditions for Exercise 5.24.

5.24 A bubble-point liquid feed is to be distilled as shown in Figure 5.25. Use the Edmister group method to estimate the mole-fraction compositions of the distillate and bottoms. Assume initial overhead and bottoms temperatures are 150 and 250°F, respectively.

Section 5.7

5.25 Verify the values given in Table 5.3 for *N_V*, *N_E*, and *N_D* for a partial reboiler and a total condenser.

5.26 Verify the values given in Table 5.3 for *N_V*, *N_E*, and *N_D* for a stream mixer and a stream divider.

5.27 A mixture of maleic anhydride and benzoic acid containing 10 mol% acid is a product of the manufacture of phthalic anhydride. The mixture is to be distilled continuously in a column with a total condenser and a partial reboiler at a pressure of 13.2 kPa (100 torr) with a reflux ratio of 1.2 times the minimum value to give a product of 99.5 mol% maleic anhydride and a bottoms of 0.5 mol% anhydride. Is this problem completely specified?

5.28 Verify *N_D* for the following unit operations in Table 5.4: (b), (c), and (g). How would *N_D* change if two feeds were used instead of one?

5.29 Verify *N_D* for unit operations (e) and (f) in Table 5.4. How would *N_D* change if a vapor side stream was pulled off some stage located between the feed stage and the bottom stage?

5.30 Verify *N_D* for unit operation (h) in Table 5.4. How would *N_D* change if a liquid side stream was added to a stage that was located between the feed stage and stage 2?

5.31 The following are not listed as design variables for the distillation unit operations in Table 5.4:

- (a) Condenser heat duty
(b) Stage temperature
(c) Intermediate-stage vapor rate
(d) Reboiler heat load

Under what conditions might these become design variables? If so, which variables listed in Table 5.4 would you eliminate?

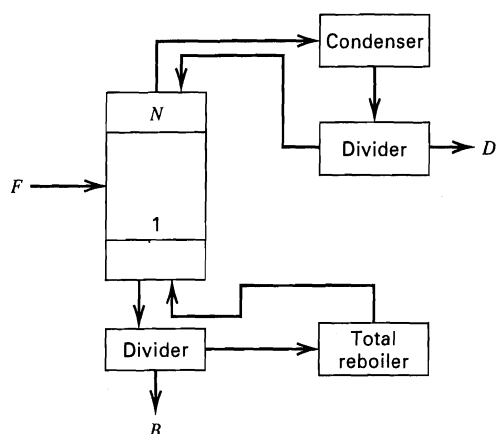


Figure 5.26 Conditions for Exercise 5.34.

5.32 Show for distillation that, if a total condenser is replaced by a partial condenser, the number of degrees of freedom is reduced by 3, provided that the distillate is removed solely as a vapor.

5.33 Unit operation (b) in Table 5.4 is to be heated by injecting live steam directly into the bottom plate of the column instead of by using a reboiler, for a separation involving ethanol and water. Assuming a fixed feed, an adiabatic operation, atmospheric pressure throughout, and a top alcohol concentration specification:

- What is the total number of design variables for the general configuration?
- How many design variables are needed to complete the design? Which variables do you recommend?

5.34 (a) For the distillation column shown in Figure 5.26, determine the number of independent design variables.

(b) It is suggested that a feed consisting of 30% A, 20% B, and 50% C, all in moles, at 37.8°C and 689 kPa, be processed in the unit of Figure 5.26, consisting of a 15-plate, 3-m-diameter column that is designed to operate at vapor velocities of 0.3 m/s and an L/V of 1.2. The pressure drop per plate is 373 Pa at these conditions, and the condenser is cooled by plant water at 15.6°C.

The product specifications in terms of the concentration of A in the distillate and C in the bottoms have been set by the process department, and the plant manager has asked you to specify a feed rate for the column. Write a memorandum to the plant manager pointing out why you can't do this, and suggest some alternatives.

5.35 Calculate the number of degrees of freedom for the mixed-feed, triple-effect evaporator system shown in Figure 5.27. Assume that the steam and all drain streams are at saturated conditions and the feed is an aqueous solution of a dissolved organic solid. Also,

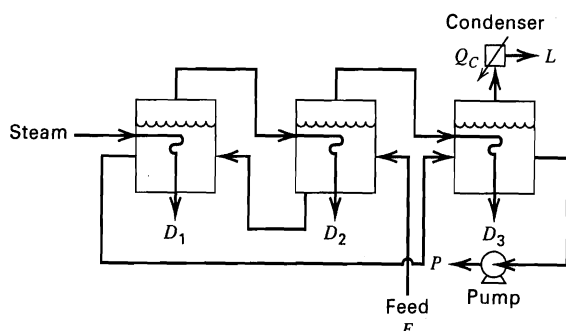


Figure 5.27 Conditions for Exercise 5.35.

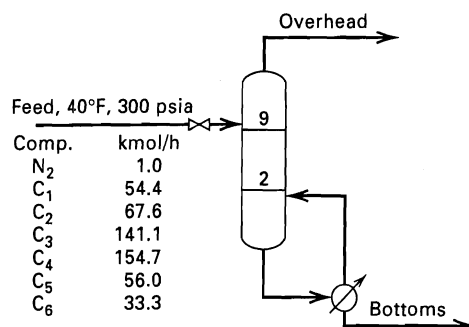


Figure 5.28 Conditions for Exercise 5.36.

assume that all overhead streams are pure water vapor, with no entrainment.

If this evaporator system is used to concentrate a feed containing 2 wt% dissolved organic to a product with 25 wt% dissolved organic, using 689-kPa saturated steam, calculate the number of unspecified design variables and suggest likely candidates. Assume perfect insulation against heat loss for each effect.

5.36 A reboiled stripper as shown in Figure 5.28 is to be designed for the task shown. Determine

- The number of variables.
- The number of equations relating the variables.
- The number of degrees of freedom and indicate.
- Which additional variables, if any, need to be specified.

5.37 The thermally coupled distillation system shown in Figure 5.29 is to be used to separate a mixture of three components into three products. Determine for the system

- The number of variables.
- The number of equations relating the variables.
- The number of degrees of freedom and propose.
- A reasonable set of design variables.

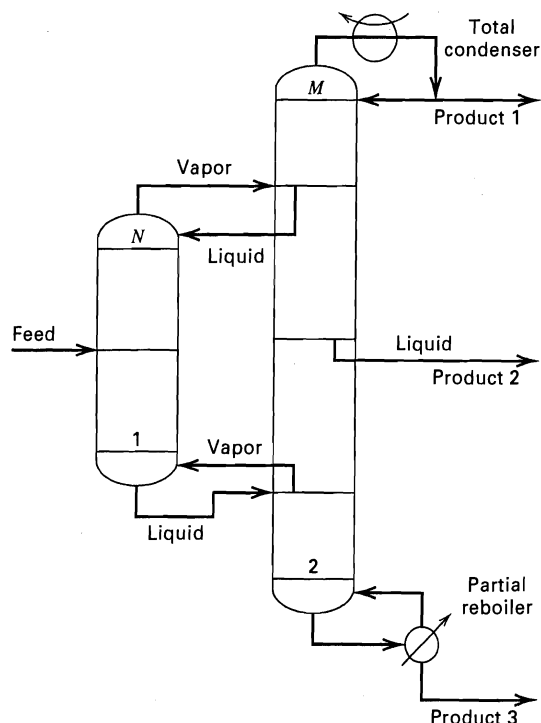


Figure 5.29 Conditions for Exercise 5.37.

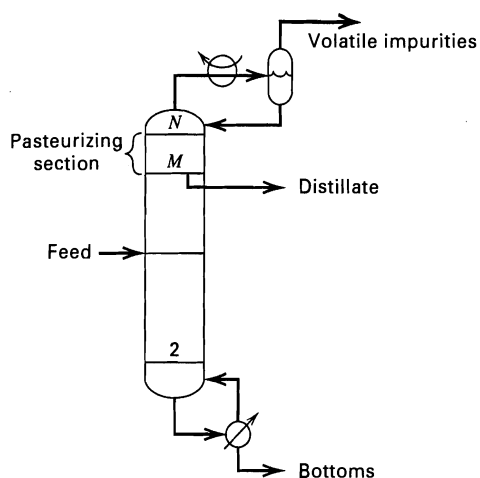


Figure 5.30 Conditions for Exercise 5.38.

5.38 When the feed to a distillation column contains a small amount of impurities that are much more volatile than the desired distillate, it is possible to separate the volatile impurities from the distillate by removing the distillate as a liquid sidestream from a stage located several stages below the top stage. As shown in Figure 5.30, this additional top section of stages is referred to as a pasteurizing section.

- Determine the number of degrees of freedom for the unit
- Determine a reasonable set of design variables

5.39 A system for separating a mixture into three products is shown in Figure 5.31. For it, determine

- The number of variables.
- The number of equations relating the variables.
- The number of degrees of freedom and propose.
- A reasonable set of design variables.

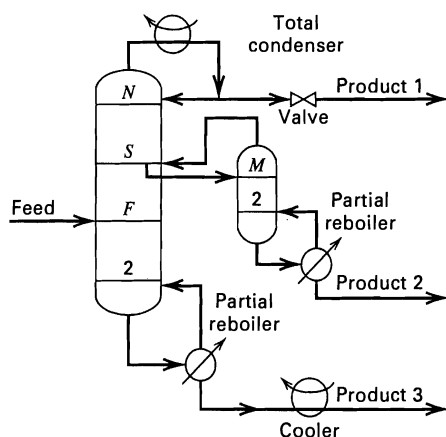


Figure 5.31 Conditions for Exercise 5.39.

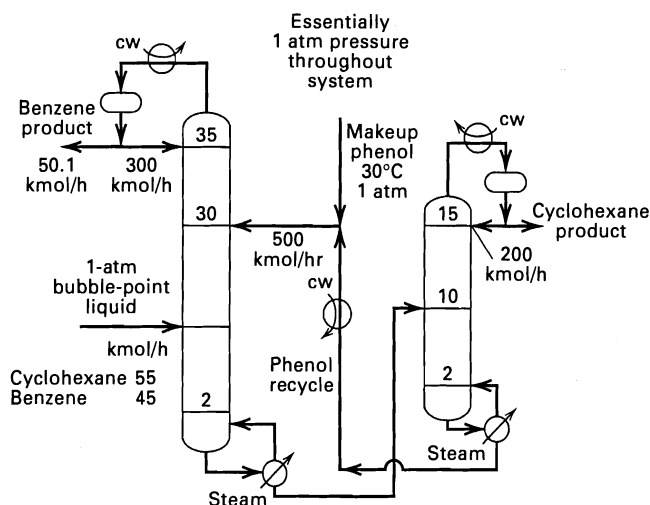


Figure 5.32 Conditions for Exercise 5.40.

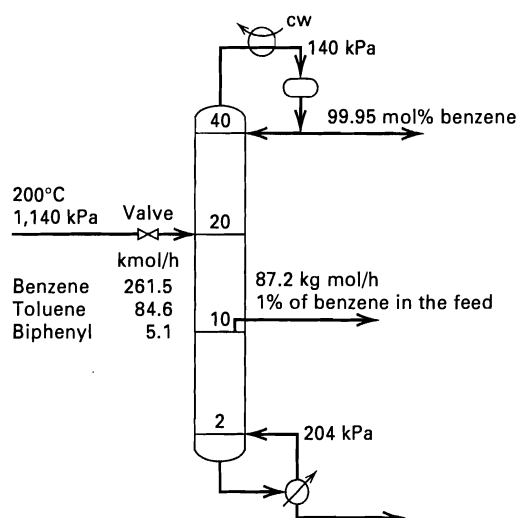


Figure 5.33 Conditions for Exercise 5.41.

5.40 A system for separating a binary mixture by extractive distillation, followed by ordinary distillation for recovery and recycle of the solvent, is shown in Figure 5.32. Are the design variables shown sufficient to specify the problem completely? If not, what additional design variable(s) would you select?

5.41 A single distillation column for separating a three-component mixture into three products is shown in Figure 5.33. Are the design variables shown sufficient to specify the problem completely? If not, what additional design variable(s) would you select?

Part 2

Separations by Phase Addition or Creation

Among the most widely used industrial separation operations are absorption, stripping, various types of distillation, and liquid–liquid extraction, all of which involve separations by selective mass transfer of components from one fluid phase to another. The other phase is created by thermal energy input (energy-separating agent) or by addition (mass-separating agent). In most cases, these operations are based on the use of countercurrent cascades of multiple stages. Detailed descriptions of, and design and analysis calculations for, these vapor–liquid and liquid–liquid operations are presented in Chapters 6 through 13. Two types of mathematical models are considered: (1) stages that attain thermodynamic phase equilibrium and (2) stages that do not reach phase equilibrium but are governed by rates of mass transfer. The less-complex equilibrium-stage models are more widely used, with a stage efficiency, but the availability of fast and inexpensive digital computations is encouraging an increase in the application of more tedious, but more accurate, mass-transfer models.

Absorption (vapor-phase feed) and stripping (liquid-phase feed) are covered in Chapter 6. These two operations usually rely on the addition of a mass-separating agent (liquid absorbent and vapor-stripping agent, respectively), but may also use heat transfer to produce the other phase. In general, these operations are not used to make a sharp separation, but can achieve a high recovery of a key component in the feed by its transfer to the other phase. Absorption and stripping equipment most often consists of columns containing trays or packing for contacting the two phases, with continuous flow of the fluid phases; calculation methods, graphical and algebraic, are presented for both types of contacting. Methods for estimating tray efficiency, column height, and diameter are also presented.

The continuous distillation of binary mixtures in multiple-stage, trayed or packed columns is covered in Chapter 7, with emphasis on the classical McCabe–Thiele graphical, equilibrium-stage model. Typically, this separation operation utilizes energy to achieve the separation and two sections (rectifying and stripping), which make possible, with non-azeotropic-forming mixtures, the separation of a binary mixture into two nearly pure products. Equipment-sizing methods of Chapter 6 generally apply to distillation in Chapter 7.

When the separation of a liquid binary mixture by distillation is infeasible or too expensive, liquid–liquid extraction using a selective solvent is considered, as presented in Chapter 8. Although many equipment configurations are available, columns or vessels with mechanically assisted agitation are the most useful when multiple stages are needed to achieve the desired recovery. This chapter emphasizes graphical, equilibrium-stage methods using triangular diagrams for treating ternary systems.

Equilibrium-stage models and calculations for multicomponent mixtures are considerably more complex than those for binary mixtures, as in distillation, and for ternary mixtures, as in absorption, stripping, and liquid–liquid extraction. Approximate algebraic methods are presented in Chapter 9, while rigorous algebraic methods are developed in Chapter 10. These methods are implemented in process simulators and widely used.

Chapter 11 considers equilibrium-stage calculation methods for so-called enhanced distillation of mixtures that are difficult to separate by conventional distillation or liquid–liquid extraction. An important aspect of enhanced distillation is the determination of feasible products, which uses residue-curve maps. Extractive, azeotropic, and salt distillation use mass-addition as well as thermal energy input to achieve separation.

Pressure-swing distillation involves use of two columns operating at different pressures. Reactive distillation strives to couple a chemical reaction with separation of the products. Included in Chapter 11 is supercritical-fluid extraction, which makes use of the favorable properties in the vicinity of the critical point to achieve a separation.

Mass-transfer models for multicomponent, multi-stage, vapor-liquid separation operations are available

in several process simulators. These models are particularly useful in cases where stage efficiency is low or uncertain and are described in Chapter 12.

Batch distillation has become increasingly popular with the trend toward production of specialty products. Calculation methods, widely used in process simulators, for both binary and multicomponent mixtures are presented in Chapter 13, with an introduction to methods for determining an optimal set of operation steps.

Absorption and Stripping of Dilute Mixtures

In *absorption* (also called *gas absorption*, *gas scrubbing*, and *gas washing*), a gas mixture is contacted with a liquid (the *absorbent* or *solvent*) to selectively dissolve one or more components by mass transfer from the gas to the liquid. The components transferred to the liquid are referred to as *solutes* or *absorbate*. Absorption is used to separate gas mixtures; remove impurities, contaminants, pollutants, or catalyst poisons from a gas; or recover valuable chemicals. Thus, the species of interest in the gas mixture may be all components, only the component(s) not transferred, or only the component(s) transferred.

The opposite of absorption is *stripping* (also called *desorption*), wherein a liquid mixture is contacted with a gas to selectively remove components by mass transfer from the liquid to the gas phase. As discussed in Chapter 5, absorbers are frequently coupled with strippers to permit regeneration (or recovery) and recycling of the absorbent. Because stripping is not perfect, absorbent recycled to the absorber contains species present in the vapor entering the absorber. When water is used as the absorbent, it is more common to separate the absorbent from the solute by distillation rather than stripping.

6.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Explain the difference between absorption and stripping.
- Explain the difference between physical and chemical absorption.
- Explain why absorbers are best operated at high pressure and low temperature, while strippers are best operated at low pressure and high temperature.
- Enumerate different types of industrial equipment for absorption and stripping and explain which are most popular.
- Explain how vapor and liquid streams flow from one tray to another in a trayed tower.
- Compare three different types of trays.
- Explain the difference between random and structured packings and cite examples of each.
- Explain the importance of the liquid distributor and redistributors in a packed column with respect to liquid flow.
- Derive the “operating-line equation,” used in graphical methods, starting with a component material balance.
- Calculate the minimum MSA flow rate to achieve a specified recovery of a key component in a single-section, countercurrent cascade.
- Determine graphically, by stepping off stages, or algebraically, the required number of equilibrium stages in a countercurrent cascade to achieve a specified recovery of a key component, given an MSA flow rate greater than the minimum value.
- Define the overall stage efficiency and explain why efficiency values are relatively low for absorbers and at a moderate level for strippers.
- Make preliminary estimates of overall stage efficiency of absorbers and strippers.
- Explain why multiple liquid-flow passes are necessary in trayed columns of moderate to large column diameter.
- Define Murphree point and tray vapor efficiencies and their relationship to overall stage efficiency.
- Explain how experimental stage-efficiency data from a small laboratory Oldershaw column can be scaled up to a large-diameter column.
- Explain two mechanisms by which a trayed column can flood.
- Enumerate the contributions to pressure drop in a trayed column.
- Estimate column diameter and tray pressure drop for a trayed column.
- Estimate tray efficiency from correlations of mass-transfer coefficients using two-film theory.
- Estimate weeping, entrainment, and downcomer backup in a trayed column.

- For a packed column, define the “height equivalent to a theoretical (equilibrium) stage (plate),” HETP, and explain how it and the number of equilibrium stages differ from “height of a transfer unit,” HTU, and “number of transfer units,” NTU, respectively.
- Explain differences between “loading point” and “flooding point” in a packed column.
- Estimate packed height, packed-column diameter, and pressure drop across the packing.
- Estimate HTU from correlations of mass-transfer coefficients.
- Explain how the number of theoretical stages is computed for concentrated solutions in which equilibrium and operating lines are curved.

Industrial Example

A typical absorption operation is shown in Figure 6.1. The feed, which contains air (21% O₂, 78% N₂, and 1% Ar), water vapor, and acetone vapor, is the gas leaving a dryer where solid cellulose acetate fibers, wet with water and acetone, are dried. The purpose of the 30-tray (equivalent to 10 equilibrium stages) absorber is to remove the acetone by contacting the gas with a suitable absorbent, water. By using countercurrent flow of gas and liquid in a multiple-stage device, the material balance, shown in Figure 6.1, indicates that 99.5% of the acetone is absorbed. The gas leaving the absorber contains only 143 ppm (parts per million) by weight of acetone vapor and can be recycled to the dryer or exhausted to the atmosphere. Although the major component transferred between phases is acetone, the material balance indicates that small amounts of oxygen and nitrogen are also absorbed by the water solvent. Because water is present in both the feed gas and the absorbent, it can be both absorbed and stripped. As seen in Figure 6.1, the net effect is that water is stripped because more water appears in the exit gas than in the feed gas. The exit gas is almost saturated with water vapor and the exit liquid is almost saturated with air. The temperature of the absorbent decreases by 3°C to supply the energy of vaporization needed to strip the water, which

in this example is greater than the energy of condensation liberated from the absorption of acetone.

As was shown in Figure 5.9, the fraction of a component absorbed in a countercurrent cascade depends on the number of equilibrium stages and the absorption factor, $A = L/(KV)$, for that component. For the conditions of Figure 6.1, using $L = 1943$ kmol/h and $V = 703$ kmol/h, estimated K -values and absorption factors, which range over many orders of magnitude, are

Component	$A = L/(KV)$	K -value
Water	89.2	0.031
Acetone	1.38	2.0
Oxygen	0.00006	45,000
Nitrogen	0.00003	90,000
Argon	0.00008	35,000

For acetone, the K -value is based on Eq. (4) of Table 2.3, the modified Raoult's law, $K = \gamma P^s/P$, with $\gamma = 6.7$ for a dilute solution of acetone in water at 25°C and 101.3 kPa. For oxygen and nitrogen, K -values are based on the use of Eq. (6) of Table 2.3, Henry's law, $K = H/P$, using constants from Figure 4.27 at 25°C. For water, the K -value is obtained from Eq. (3) of Table 2.3, Raoult's law, $K = P^s/P$, which applies because the mole fraction of water in the liquid phase is close to 1. For argon, the Henry's law constant at 25°C was obtained from the International Critical Tables [1].

Figure 5.9 shows that if the value of A is greater than 1, any degree of absorption can be achieved: the larger the value of A , the fewer the number of stages required to absorb a desired fraction of the solute. However, very large values of A can correspond to absorbent flow rates that are larger than necessary. From an economic standpoint, the value of A , for the main (key) species to be absorbed, should be in the range of 1.25 to 2.0, with 1.4 being a frequently recommended value. Thus, the above value of 1.38 for acetone is favorable.

For a given feed-gas flow rate and choice of absorbent, factors that influence the value of A are absorbent flow rate, temperature, and pressure. Because $A = L/(KV)$, the larger the absorbent flow rate is, the larger will be the value of A . The required absorbent flow rate can be reduced by reducing the K -value of the solute. Because the K -value for many solutes varies exponentially with temperature and is inversely proportional to pressure, this reduction can be achieved by

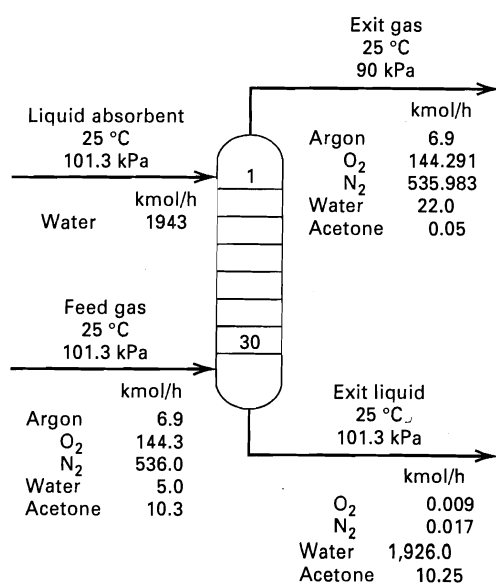


Figure 6.1 Typical absorption process.

reducing the temperature and/or increasing the pressure. Increasing the pressure also serves to reduce the diameter of the equipment for a given gas throughput. However, temperature adjustment by feed-gas refrigeration and/or absorbent refrigeration, and/or adjustment of the feed-gas pressure by gas compression can be expensive. For these reasons, the absorber in Figure 6.1 operates at near-ambient conditions.

For a stripper, the stripping factor, $S = 1/A = KV/L$, is crucial. To reduce the required flow rate of stripping agent, operation of the stripper at a high temperature and/or a low pressure is desirable, with an optimum stripping factor in the vicinity of 1.4.

Absorption and stripping are technically mature separation operations. Design procedures are well developed and commercial processes are common. Table 6.1 lists representative, commercial absorption applications. In most cases, the solutes are contained in gaseous effluents from chemical reactors. Passage of strict environmental standards with respect to pollution by emission of noxious gases has greatly increased the use of gas absorbers in the past decade.

When water and hydrocarbon oils are used as absorbents, no significant chemical reactions occur between the absorbent and the solute, and the process is commonly referred to as *physical absorption*. When aqueous sodium hydroxide

(a strong base) is used as the absorbent to dissolve an acid gas, absorption is accompanied by a rapid and irreversible neutralization reaction in the liquid phase and the process is referred to as *chemical absorption* or *reactive absorption*. More complex examples of chemical absorption are processes for absorbing CO_2 and H_2S with aqueous solutions of monoethanolamine (MEA) and diethanolamine (DEA), where a reversible chemical reaction takes place in the liquid phase. Chemical reactions can increase the rate of absorption, increase the absorption capacity of the solvent, increase selectivity to preferentially dissolve only certain components of the gas, and convert a hazardous chemical to a safe compound.

In this chapter, trayed and packed-column equipment for conducting absorption and stripping operations is discussed and fundamental *equilibrium-based* and *rate-based* (mass-transfer) models and calculation procedures, both graphical and algebraic, are presented for physical absorption and stripping of mainly dilute mixtures. The methods also apply to reactive absorption with irreversible and complete chemical reactions of the solute in the liquid phase. Calculations for concentrated mixtures and reactive absorption with reversible chemical reactions are best handled by computer-aided calculations, which are discussed in Chapters 10 and 11. An introduction to calculations for concentrated mixtures in packed columns is given in the last section of this chapter.

Table 6.1 Representative, Commercial Applications of Absorption

Solute	Absorbent	Type of Absorption
Acetone	Water	Physical
Acrylonitrile	Water	Physical
Ammonia	Water	Physical
Ethanol	Water	Physical
Formaldehyde	Water	Physical
Hydrochloric acid	Water	Physical
Hydrofluoric acid	Water	Physical
Sulfur dioxide	Water	Physical
Sulfur trioxide	Water	Physical
Benzene and toluene	Hydrocarbon oil	Physical
Butadiene	Hydrocarbon oil	Physical
Butanes and propane	Hydrocarbon oil	Physical
Naphthalene	Hydrocarbon oil	Physical
Carbon dioxide	Aq. NaOH	Irreversible chemical
Hydrochloric acid	Aq. NaOH	Irreversible chemical
Hydrocyanic acid	Aq. NaOH	Irreversible chemical
Hydrofluoric acid	Aq. NaOH	Irreversible chemical
Hydrogen sulfide	Aq. NaOH	Irreversible chemical
Chlorine	Water	Reversible chemical
Carbon monoxide	Aq. cuprous ammonium salts	Reversible chemical
CO_2 and H_2S	Aq. monoethanolamine (MEA) or diethanolamine (DEA)	Reversible chemical
CO_2 and H_2S	Diethyleneglycol (DEG) or triethyleneglycol (TEG)	Reversible chemical
Nitrogen oxides	Water	Reversible chemical

6.1 EQUIPMENT

Absorption and stripping are conducted mainly in trayed towers (plate columns) and packed columns, and less often in spray towers, bubble columns, and centrifugal contactors, as shown schematically in Figure 6.2. A trayed tower is a vertical, cylindrical pressure vessel in which vapor and liquid, which flow countercurrently, are contacted on a series of trays or plates, an example of which is shown in Figure 6.3. Liquid flows across each tray, over an outlet weir, and into a downcomer, which takes the liquid by gravity to the tray below. Gas flows upward through openings in each tray, bubbling through the liquid on the tray. When the openings are holes, any of the five two-phase-flow regimes shown in

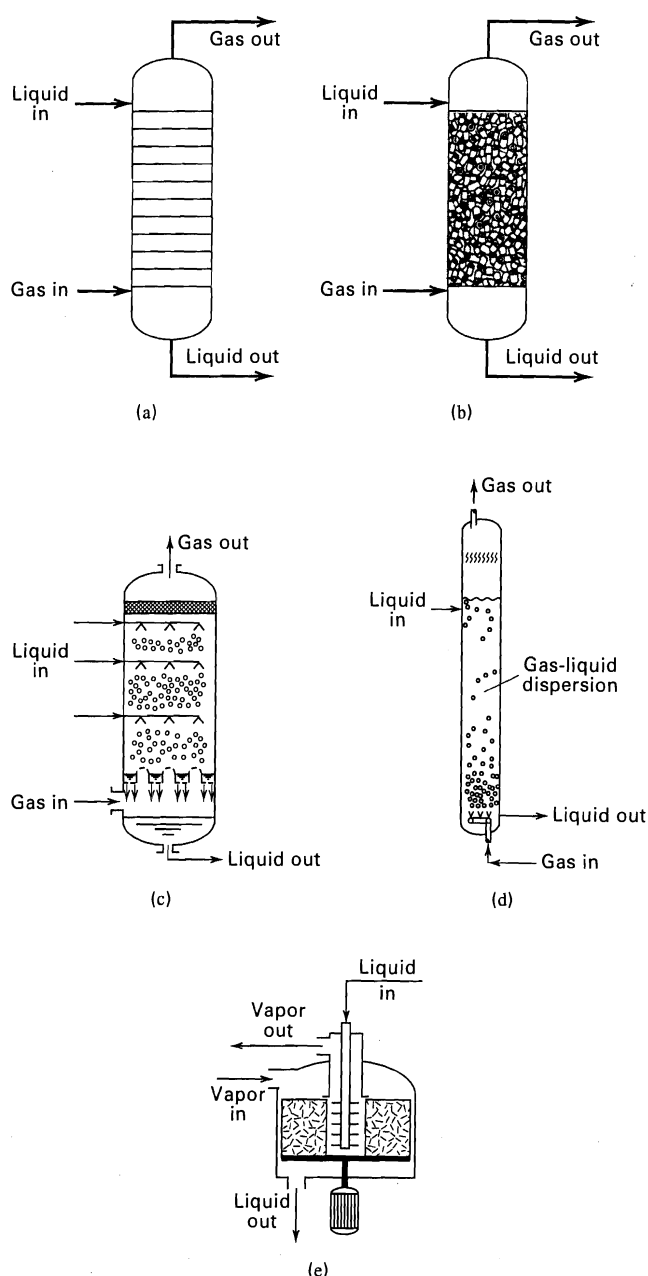


Figure 6.2 Industrial equipment for absorption and stripping: (a) trayed tower; (b) packed column; (c) spray tower; (d) bubble column; (e) centrifugal contactor.

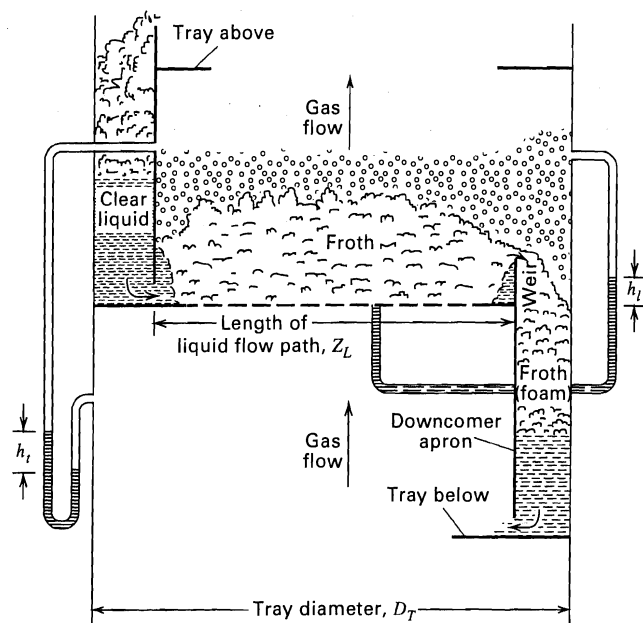


Figure 6.3 Details of a contacting tray in a trayed tower.

[Adapted from B.F. Smith, *Design of Equilibrium Stage Processes*, McGraw-Hill, New York (1963).]

Figure 6.4, and considered in detail by Lockett [2], may occur. The most common and favored regime is the *froth regime*, in which the liquid phase is continuous and the gas passes through in the form of jets or a series of bubbles. The *spray regime*, in which the gas phase is continuous, occurs for low weir heights (low liquid depths) at high gas rates. For low gas rates, the *bubble regime* can occur, in which the liquid is fairly quiescent and bubbles rise in swarms. At high liquid rates, small gas bubbles may be undesirably emulsified. If bubble coalescence is hindered, an undesirable foam forms. Ideally, the liquid carries no vapor bubbles (*occlusion*) to the tray below, the vapor carries no liquid droplets (*entrainment*) to the tray above, and there is no *weeping* of liquid through the holes in the tray. With good contacting, equilibrium between the exiting vapor and liquid phases is approached on each tray.

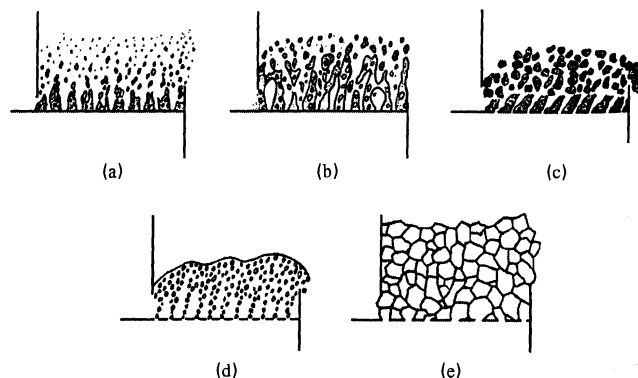


Figure 6.4 Possible vapor-liquid flow regimes for a contacting tray: (a) spray; (b) froth; (c) emulsion; (d) bubble; (e) cellular foam.

[Reproduced by permission from M.J. Lockett, *Distillation Tray Fundamentals*, Cambridge University Press, London (1986).]

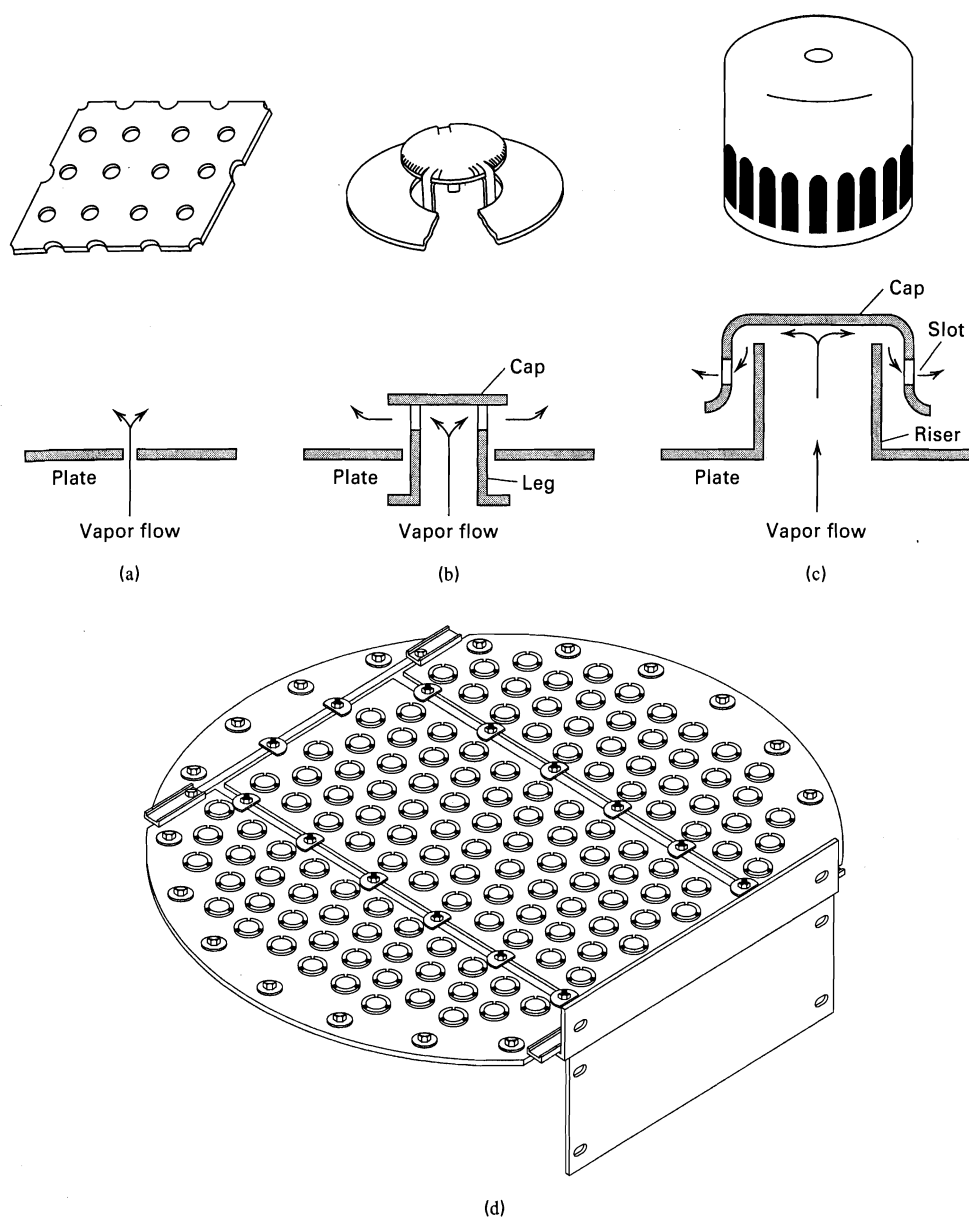


Figure 6.5 Three types of tray openings for passage of vapor up into liquid: (a) perforation; (b) valve cap; (c) bubble cap; (d) tray with valve caps.

As shown in Figure 6.5, openings in the tray for the passage of vapor are most commonly perforations, valves, and/or bubble caps. The simplest is perforations, usually $\frac{1}{8}$ to $\frac{1}{2}$ in. in diameter, used in a so-called *sieve tray* (also called a *perforated tray*). A *valve tray* has much larger openings, commonly from 1 to 2 in. in diameter. Each hole is fitted with a valve that consists of a cap, which overlaps the hole, with legs or a cage to limit the vertical rise while maintaining the horizontal location of the valve. With no vapor flow, each valve sits on the tray, over a hole. As the vapor rate is increased, the valve rises, providing a larger and larger peripheral opening for vapor to flow into the liquid to create a froth. A bubble-cap tray has bubble caps that consist of a fixed cap, 3 to 6 in. in diameter, mounted over and above a concentric riser of 2 to 3 in. in diameter. The cap has rectangular or triangular slots cut around its side. The vapor flows up through the tray opening into the riser, turns around, and passes out through the slots of the cap, into the liquid to form

a froth. An 11-ft-diameter column might have trays with 50,000 $\frac{3}{16}$ -in.-diameter perforations, or 1,000 2-in.-diameter valve caps, or 500 4-in.-diameter bubble caps.

As listed in Table 6.2, tray types are compared on the basis of cost, pressure drop, mass-transfer efficiency, vapor capacity, and flexibility in terms of turndown ratio (ratio of

Table 6.2 Comparison of Types of Trays

	Sieve Trays	Valve Trays	Bubble-Cap Trays
Relative cost	1.0	1.2	2.0
Pressure drop	Lowest	Intermediate	Highest
Efficiency	Lowest	Highest	Highest
Vapor capacity	Highest	Highest	Lowest
Typical turndown ratio	2	4	5

maximum to minimum vapor capacity). At the limiting vapor capacity, *flooding* of the column occurs because of excessive entrainment of liquid droplets in the vapor causing the liquid flow rate to exceed the capacity of the downcomer and, thus, go back up the column. At low vapor rates, weeping of liquid through the tray openings or vapor pulsation becomes excessive. Because of their low relative cost, sieve trays are preferred unless flexibility is required, in which case valve trays are best. Bubble-cap trays, which exist in many pre-1950 installations, are rarely specified for new installations, but may be preferred when the amount of liquid holdup on a tray must be controlled to provide adequate residence time for a chemical reaction or when weeping must be prevented.

A *packed column*, shown in detail in Figure 6.6, is a vertical, cylindrical pressure vessel containing one or more sections of a packing material over whose surface the liquid flows downward by gravity, as a film or as droplets between packing elements. Vapor flows upward through the wetted packing, contacting the liquid. The sections of packing are contained between a lower gas-injection support plate, which holds the packing, and an upper grid or mesh hold-down plate, which prevents packing movement. A *liquid distributor*, placed above the hold-down plate, ensures uniform distribution of liquid over the cross-sectional area of the column as it enters the packed section. If the depth of packing is more than about 20 ft, liquid channeling may occur, causing the liquid to flow down the column mainly near the wall, and

gas to flow mainly up the center of the column, thus greatly reducing the extent of vapor-liquid contact. In that case, a liquid *redistributor* should be installed.

Commercial packing materials include *random* (dumped) packings, some of which are shown in Figure 6.7a, and *structured* (also called arranged, ordered, or stacked packings), some of which are shown in Figure 6.7b. Among the random packings, which are poured into the column, are the old (1895–1950) ceramic Raschig rings and Berl saddles, which are seldom specified for new installations. They have been largely replaced by metal and plastic Pall rings, metal Bialecki rings, and ceramic Intalox saddles, which provide more surface area for mass transfer, a higher flow capacity, and a lower pressure drop. More recently, through-flow packings of a lattice-work design have been developed. These packings, which include metal Intalox IMTP; metal, plastic, and ceramic Cascade Mini-Rings; metal Levapak; metal, plastic, and ceramic Hiflow rings; metal tri-packs; and plastic Nor Pac rings, exhibit even lower pressure drop per unit height of packing and even higher mass-transfer rates per unit volume of packing. Accordingly, they are called “high-efficiency” random packings. Most random packings are available in nominal diameters, ranging from 1 in. to 3.5 in. As packing size increases, mass-transfer efficiency and pressure drop may decrease. Therefore, for a given column diameter an optimal packing size exists that represents a compromise between these two factors, since low pressure drop and high mass-transfer rates are both desirable. However, to minimize channeling of liquid, the nominal diameter of the packing should be less than one-eighth of the column diameter. Most recently, a “fourth generation” of random packings, including VSP rings, Fleximax, and Raschig super-rings, has been developed, which features a very open undulating geometry that promotes even wetting, but with recurrent turbulence promotion. The result is lower pressure drop, but sustained mass-transfer efficiency that may not decrease noticeably with increasing column diameter and may permit a larger depth of packing before a liquid redistributor is necessary. Metal packings are usually preferred because of their superior strength and good wettability. Ceramic packings, which have superior wettability but inferior strength, are used only to resist corrosion at elevated temperatures, where plastics would fail. Plastic packings, usually of polypropylene, are inexpensive and have sufficient strength, but may experience poor wettability, particularly at low liquid rates.

Representative structured packings include the older corrugated sheets of metal gauze, such as Sulzer BX, Montz A, Gempak 4BG, and Intalox High-Performance Wire Gauze Packing. Newer and less-expensive structured packings, which are fabricated from sheet metal and plastics and may or may not be perforated, embossed, or surface roughened, include metal and plastic Mellapak 250Y, metal Flexipac, metal and plastic Gempak 4A, metal Montz B1, and metal Intalox High-Performance Structured Packing. Structured

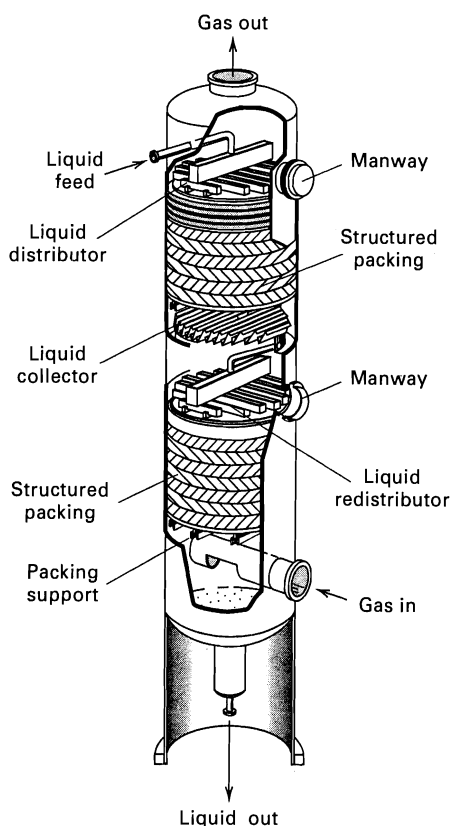


Figure 6.6 Details of internals used in a packed column.

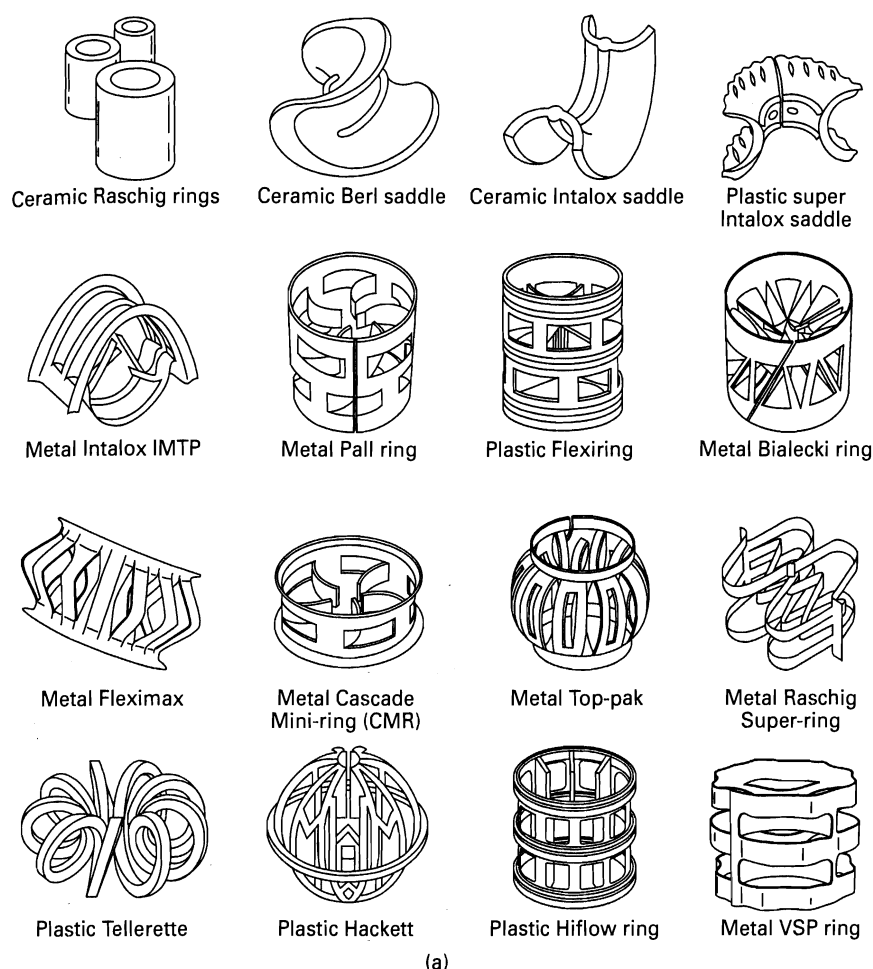


Figure 6.7 Typical materials used in a packed column: (a) random packing materials; (continued)

packings come with different size openings between adjacent corrugated layers and are stacked in the column. Although structured packings are considerably more expensive per unit volume than random packings, structured packings exhibit far less pressure drop per theoretical stage and have higher efficiency and capacity.

As shown in Table 6.3, packings are usually compared on the basis of the same factors used to compare tray types. However, the differences between random and structured packings are much greater than the differences among the three types of trays listed in Table 6.2.

Table 6.3 Comparison of Types of Packing

	Random		Structured
	Raschig Rings and Saddles	"Through Flow"	
Relative cost	Low	Moderate	High
Pressure drop	Moderate	Low	Very low
Efficiency	Moderate	High	Very high
Vapor capacity	Fairly high	High	High
Typical turndown ratio	2	2	2

If only one or two theoretical stages are required, only a very low pressure drop is allowed, and the solute is very soluble in the liquid phase, the use of a *spray tower* may be advantageous. As shown in Figure 6.2, a spray tower consists of a vertical, cylindrical vessel filled with gas into which liquid is sprayed. A *bubble column*, also shown in Figure 6.2, consists of a vertical, cylindrical vessel partially filled with liquid into which the vapor is bubbled. Vapor pressure drop is high, and only one or two theoretical stages can be achieved. Such a device has a low vapor throughput and should not be considered unless the solute has a very low solubility in the liquid and/or a slow chemical reaction takes place in the liquid phase, thus requiring an appreciable residence time. A novel device is the *centrifugal contactor*, one example of which, as shown in Figure 6.2, consists of a stationary, ringed housing, intermeshed with a ringed rotating section. The liquid phase is fed near the center of the packing, from which it is caused to flow outward by centrifugal force. The vapor phase flows inward by a pressure driving force. Very high mass-transfer rates can be achieved with only moderately high rotation rates. It is possible to obtain the equivalent of several equilibrium stages in a very compact unit. This type of contact is favored when headroom for a trayed tower or packed column is not available or when a short residence time is desired.

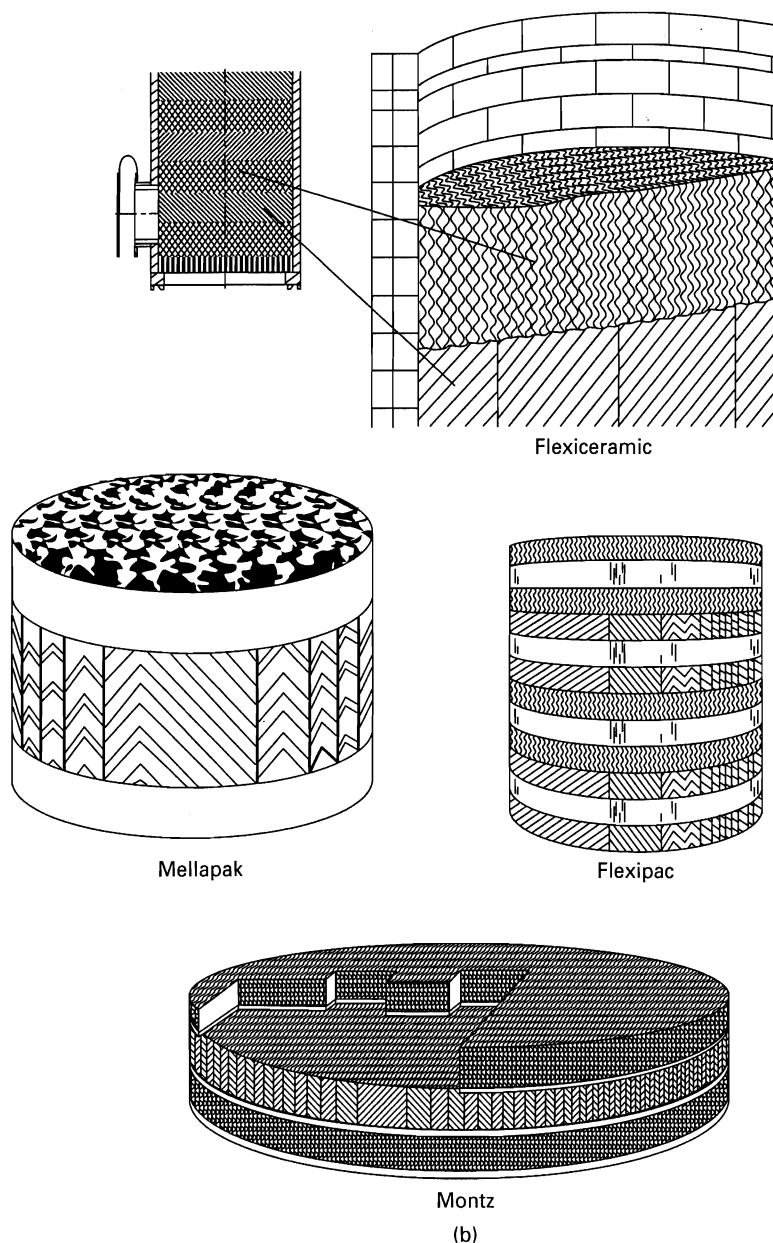


Figure 6.7 (Continued) (b) structured packing materials.

In most applications, the choice of contacting device is between a trayed tower and a packed column. The latter, using dumped packings, is almost always favored when a column diameter of less than 2 ft and a packed height of not more than 20 ft are sufficient. In addition, packed columns should be considered for corrosive services where ceramic or plastic materials are preferred over metals, in services where foaming may be severe if trays are used, and when pressure drop must be low, as in vacuum or near-ambient-pressure operations. Otherwise, trayed towers, which can be designed and scaled up more reliably, are preferred. Although structured packings are quite expensive, they may be the best choice for a new installation when pressure drop must be very low or for replacing existing trays (retrofitting) when a higher capacity or degree of separation is required in an existing column. Trayed towers are preferred when liquid velocities are low, while columns with random packings are best for high-liquid velocities. The use of structured

packings should be avoided at high-pressures (> 200 psia) and high-liquid flow rates (> 10 gpm/ft²), Kister [33]. In general, a continuous, turbulent liquid flow is desirable if mass transfer is limiting in the liquid phase, while a continuous, turbulent gas flow is desirable if mass transfer is limiting in the gas phase.

6.2 GENERAL DESIGN CONSIDERATIONS

Design or analysis of an absorber (or stripper) requires consideration of a number of factors, including:

1. Entering gas (liquid) flow rate, composition, temperature, and pressure
2. Desired degree of recovery of one or more solutes
3. Choice of absorbent (stripping agent)
4. Operating pressure and temperature, and allowable gas pressure drop

5. Minimum absorbent (stripping agent) flow rate and actual absorbent (stripping agent) flow rate as a multiple of the minimum rate needed to make the separation
6. Number of equilibrium stages and stage efficiency
7. Heat effects and need for cooling (heating)
8. Type of absorber (stripper) equipment
9. Height of absorber (stripper)
10. Diameter of absorber (stripper)

The ideal absorbent should (a) have a high solubility for the solute(s) to minimize the need for absorbent, (b) have a low volatility to reduce the loss of absorbent and facilitate separation of absorbent from solute(s), (c) be stable to maximize absorbent life and reduce absorbent makeup requirement, (d) be noncorrosive to permit use of common materials of construction, (e) have a low viscosity to provide low pressure drop and high mass- and heat-transfer rates, (f) be nonfoaming when contacted with the gas so as to make it unnecessary to increase absorber dimensions, (g) be nontoxic and nonflammable to facilitate its safe use, and (h) be available, if possible, within the process, to make it unnecessary to provide an absorbent from external sources, or be inexpensive. As already indicated at the beginning of this chapter, the most widely used absorbents are water, hydrocarbon oils, and aqueous solutions of acids and bases. The most common stripping agents are steam, air, inert gases, and hydrocarbon gases.

In general, operating pressure should be high and temperature low for an absorber, to minimize stage requirements and/or absorbent flow rate and to lower the equipment volume required to accommodate the gas flow. Unfortunately, both compression and refrigeration of a gas are expensive. Therefore, most absorbers are operated at feed-gas pressure, which may be greater than ambient pressure, and ambient temperature, which can be achieved by cooling the feed gas and absorbent with cooling water, unless one or both streams already exist at a subambient temperature. Operating pressure should be low and temperature high for a stripper to minimize stage requirements or stripping agent flow rate. However, because maintenance of a vacuum is expensive, strippers are commonly operated at a pressure just above ambient. A high temperature can be used, but it should not be so high as to cause undesirable chemical reactions. Of course, operating temperature and pressure must be compatible with the necessary phase conditions of the streams being contacted. For example, an absorber should not be operated at a pressure and/or temperature that would condense the feed gas, and a stripper should not be operated at a pressure and/or temperature that would vaporize the feed liquid. The possibility of such conditions occurring can be checked by bubble-point and dew-point calculations, discussed in Chapter 4.

For given feed-gas (liquid) flow rate, extent of solute absorption (stripping), operating pressure and temperature, and absorbent (stripping agent) composition, a minimum

absorbent (stripping agent) flow rate exists that corresponds to an infinite number of countercurrent equilibrium contacts between the gas and liquid phases. In every design problem involving flow rates of the absorbent (stripping agent) and number of stages, a trade-off exists between the number of equilibrium stages and the absorbent (stripping agent) flow rate at rates greater than the minimum value. Graphical and analytical methods for computing the minimum flow rate and this trade-off are developed in the following sections for a mixture that is dilute in the solute(s). For this essentially isothermal case, the energy balance can be ignored. As discussed in Chapters 10 and 11, computer-aided methods are best used for concentrated mixtures, where multicomponent phase-equilibrium and mass-transfer effects can become complicated and it is necessary to consider the energy balance.

6.3 GRAPHICAL EQUILIBRIUM-STAGE METHOD FOR TRAYED TOWERS

Consider the countercurrent-flow, trayed tower for absorption (or stripping) operating under isobaric, isothermal, continuous, steady-state flow conditions shown in Figure 6.8. For convenience, the stages are numbered from top to bottom for the absorber and from bottom to top for the stripper. Phase equilibrium is assumed to be achieved at each of the N trays between the vapor and liquid streams leaving the tray. That is, each tray is treated as an equilibrium stage. Assume that the only component transferred from one phase to the

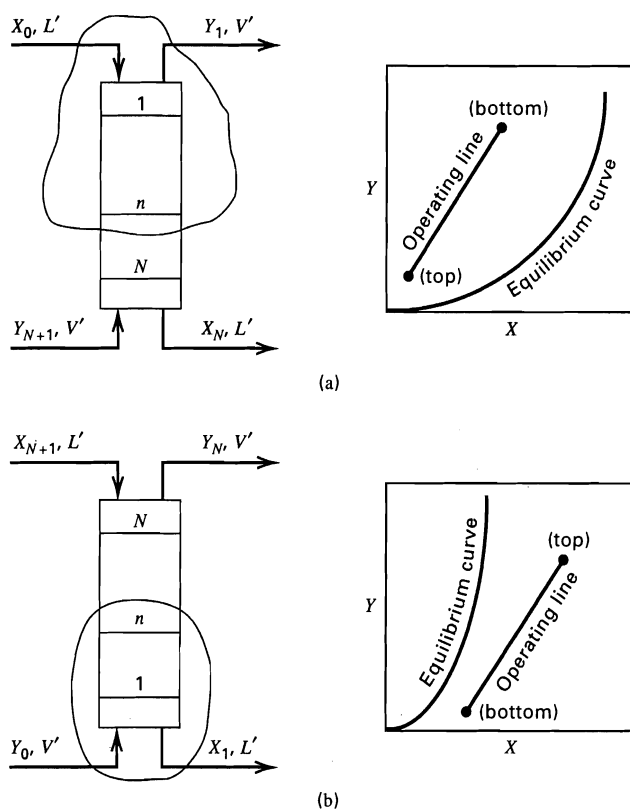


Figure 6.8 Continuous, steady-state operation in a countercurrent cascade with equilibrium stages: (a) absorber; (b) stripper.

other is the solute. For application to an absorber, let:

L' = molar flow rate of solute-free absorbent

V' = molar flow rate of solute-free gas (carrier gas)

X = mole ratio of solute to solute-free absorbent in the liquid

Y = mole ratio of solute to solute-free gas in the vapor

Note that with these definitions, values of L' and V' remain constant through the tower, assuming no vaporization of absorbent into carrier gas or absorption of carrier gas by liquid. For the solute at any equilibrium stage, n , the K -value is given in terms of X and Y as:

$$K_n = \frac{y_n}{x_n} = \frac{Y_n/(1 + Y_n)}{X_n/(1 + X_n)} \quad (6-1)$$

where $Y = y/(1 - y)$ and $X = x/(1 - x)$.

For the fixed temperature and pressure and a series of values of x , equilibrium values of y in the presence of the solute-free absorbent and solute-free gas are estimated by methods discussed in Chapter 2. From these values, an equilibrium curve of Y as a function of X is calculated and plotted, as shown in Figure 6.8. In general, this curve will not be a straight line, but it will pass through the origin. If the solute undergoes, in the liquid phase, a complete irreversible conversion by chemical reaction, to a nonvolatile solute, the equilibrium curve will be a straight line of zero slope passing through the origin.

At either end of the towers shown in Figure 6.8, entering and leaving streams and solute mole ratios are paired. For the absorber, the pairs are $(X_0, L'$ and $Y_1, V')$ at the top and $(X_N, L'$ and $Y_{N+1}, V')$ at the bottom; for the stripper, $(X_{N+1}, L'$ and $Y_N, V')$ at the top and $(X_1, L'$ and $Y_0, V')$ at the bottom. These terminal pairs can be related to intermediate pairs of passing streams by the following solute material balances for the envelopes shown in Figure 6.8. The balances are written around one end of the tower and an arbitrary intermediate equilibrium stage, n .

For the absorber,

$$X_0 L' + Y_{N+1} V' = X_N L' + Y_1 V' \quad (6-2)$$

or, solving for Y_{N+1} ,

$$Y_{N+1} = X_N (L'/V') + Y_1 - X_0 (L'/V') \quad (6-3)$$

For the stripper,

$$X_{N+1} L' + Y_0 V' = X_1 L' + Y_N V' \quad (6-4)$$

or, solving for Y_N ,

$$Y_N = X_{N+1} (L'/V') + Y_0 - X_1 (L'/V') \quad (6-5)$$

Equations (6-3) and (6-5), which are called *operating-line equations*, are plotted in Figure 6.8. The terminal points of these lines represent the conditions at the top and bottom of the towers. For the absorber, the operating line is above the equilibrium line because, for a given solute concentration in

the liquid, the solute concentration in the gas is always greater than the equilibrium value, thus providing the driving force for mass transfer of solute from the gas to the liquid. For the stripper, the operating line lies below the equilibrium line for the opposite reason. For the coordinate systems in Figure 6.8, the operating lines are straight with a slope of L'/V' .

For an absorber, the terminal point of the operating line at the top of the tower is fixed at X_0 by the amount of solute, if any, in the entering absorbent, and the specified degree of absorption of the solute, which fixes the value of Y_1 in the leaving gas. The terminal point of the operating line at the bottom of the tower depends on Y_{N+1} and the slope of the operating line and, thus, the flow rate, L' , of solute-free absorbent.

Minimum Absorbent Flow Rate

Operating lines for four different absorbent flow rates are shown in Figure 6.9, where each operating line passes through the terminal point, (Y_1, X_0) , at the top of the column, and corresponds to a different liquid absorbent rate and corresponding slope, L'/V' . To achieve the desired value of Y_1 for given Y_{N+1} , X_0 , and V' , the solute-free absorbent flow rate L' , must lie in the range of ∞ (operating line 1) to $L'_{\min}$ (operating line 4). The value of the solute concentration in the outlet liquid, X_N , depends on L' by a material balance on

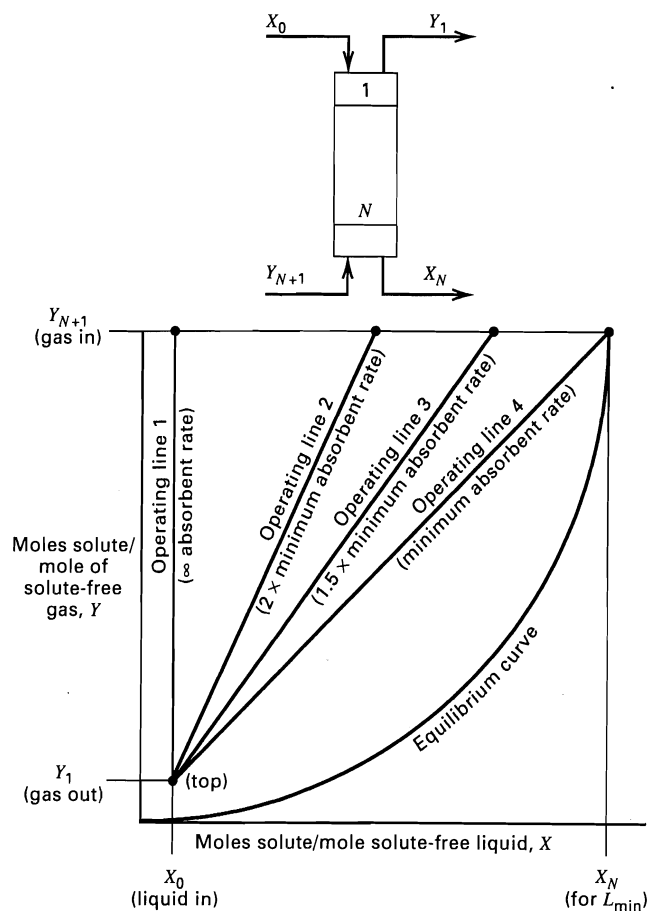


Figure 6.9 Operating lines for an absorber.

the solute for the entire absorber. From (6-2), for $n = N$,

$$X_0 L' + Y_{N+1} V' = X_N L' + Y_1 V' \quad (6-6)$$

or

$$L' = \frac{V'(Y_{N+1} - Y_1)}{(X_N - X_0)} \quad (6-7)$$

Note that the operating line can terminate at the equilibrium line, as for operating line 4, but cannot cross it because that would be a violation of the second law of thermodynamics.

The value of $L'_{\min}$ corresponds to a value of X_N (leaving the bottom of the tower) in equilibrium with Y_{N+1} , the solute concentration in the feed gas. It takes an infinite number of stages for this equilibrium to be achieved. An expression for $L'_{\min}$ of an absorber can be derived from (6-7) as follows.

For stage N , (6-1) becomes, for the minimum absorbent rate,

$$K_N = \frac{Y_{N+1}/(1 + Y_{N+1})}{X_N/(1 + X_N)} \quad (6-8)$$

Solving (6-8) for X_N and substituting the result into (6-7) gives

$$L'_{\min} = \frac{V'(Y_{N+1} - Y_1)}{\{Y_{N+1}/[Y_{N+1}(K_N - 1) + K_N]\} - X_0} \quad (6-9)$$

For dilute-solute conditions, where $Y \approx y$ and $X \approx x$, (6-9) approaches

$$L'_{\min} = V' \left(\frac{\frac{y_{N+1} - y_1}{\frac{y_{N+1}}{K_N} - x_0}} \right) \quad (6-10)$$

Furthermore, if the entering liquid contains no solute, that is, $X_0 \approx 0$, (6-10) approaches

$$L'_{\min} = V' K_N \text{ (fraction of solute absorbed)} \quad (6-11)$$

This equation is reasonable because it would be expected that $L'_{\min}$ would increase with increasing V' , K -value, and fraction of solute absorbed.

The selection of the actual operating absorbent flow rate is based on some multiple of $L'_{\min}$, typically from 1.1 to 2. A value of 1.5 corresponds closely to the value of 1.4 for the optimal absorption factor mentioned earlier. In Figure 6.9, operating lines 2 and 3 correspond to 2.0 and 1.5 times $L'_{\min}$, respectively. As the operating line moves from 1 to 4, the number of required equilibrium stages, N , increases from zero to infinity. Thus, a trade-off exists between L' and N , and an optimal value of L' exists.

A similar derivation of $V'_{\min}$, for the stripper of Figure 6.8, results in an expression analogous to (6-11):

$$V'_{\min} = \frac{L'}{K_N} \text{ (fraction of solute stripped)} \quad (6-12)$$

Number of Equilibrium Stages

As shown in Figure 6.10a, the operating line relates the solute concentration in the vapor passing upward between

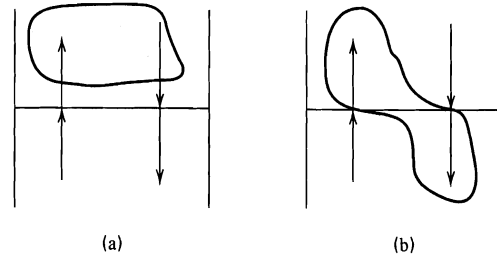


Figure 6.10 Vapor-liquid stream relationships: (a) operating line (passing streams); (b) equilibrium curve (leaving streams).

two stages to the solute concentration in the liquid passing downward between the same two stages. Figure 6.10b illustrates that the equilibrium curve relates the solute concentration in the vapor leaving an equilibrium stage to the solute concentration in the liquid leaving the same stage. This makes it possible, in the case of an absorber, to start from the top of the tower (at the bottom of the Y - X diagram) and move to the bottom of the tower (at the top of the Y - X diagram) by constructing a staircase alternating between the operating line and the equilibrium curve, as shown in Figure 6.11a. The number of equilibrium stages required for a particular absorbent flow rate corresponding to the slope of the operating line, which in Figure 6.11a is for $(L'/V') = 1.5(L'_{\min}/V')$, is stepped off by moving up the staircase, starting from the point (Y_1, X_0) , on the operating line and moving horizontally to the right to the point (Y_1, X_1) on the equilibrium curve. From there, a vertical move is made to the point (Y_2, X_1) on the operating line. Proceeding in this manner, the staircase is climbed until the terminal point (Y_{N+1}, X_N) on the operating line is reached. As shown in Figure 6.11a, the stages are counted at the points of the staircase on the equilibrium curve. As the slope (L'/V') is increased, fewer equilibrium stages are required. As (L'/V') is decreased, more stages are required until $(L'_{\min}/V')$ is reached, at which the operating line and equilibrium curve intersect at a so-called *pinch point*, for which an infinite number of stages is required. Operating line 4 in Figure 6.9 has a pinch point at Y_{N+1}, X_N . If (L'/V') is reduced below $(L'_{\min}/V')$, the specified extent of absorption of the solute cannot be achieved.

The number of equilibrium stages required for stripping a solute is determined in a manner similar to that for absorption. An illustration is shown in Figure 6.11b, which refers to Figure 6.8b. For given specifications of Y_0, X_{N+1} , and the extent of stripping of the solute, which corresponds to a value of X_1 , $V'_{\min}$ is determined from the slope of the operating line that passes through the points (Y_0, X_1) , and (Y_N, X_{N+1}) on the equilibrium curve. The operating line in Figure 6.11b is for $V' = 1.5V'_{\min}$ or a slope of $(L'/V') = (L'/V'_{\min})/1.5$.

In Figure 6.11, the number of equilibrium stages for the absorber and stripper is exactly three each. These integer results are coincidental. Ordinarily, the result is some fraction above an integer number of stages, as is the case in the following example. In practice, the result is usually rounded to the next highest integer.

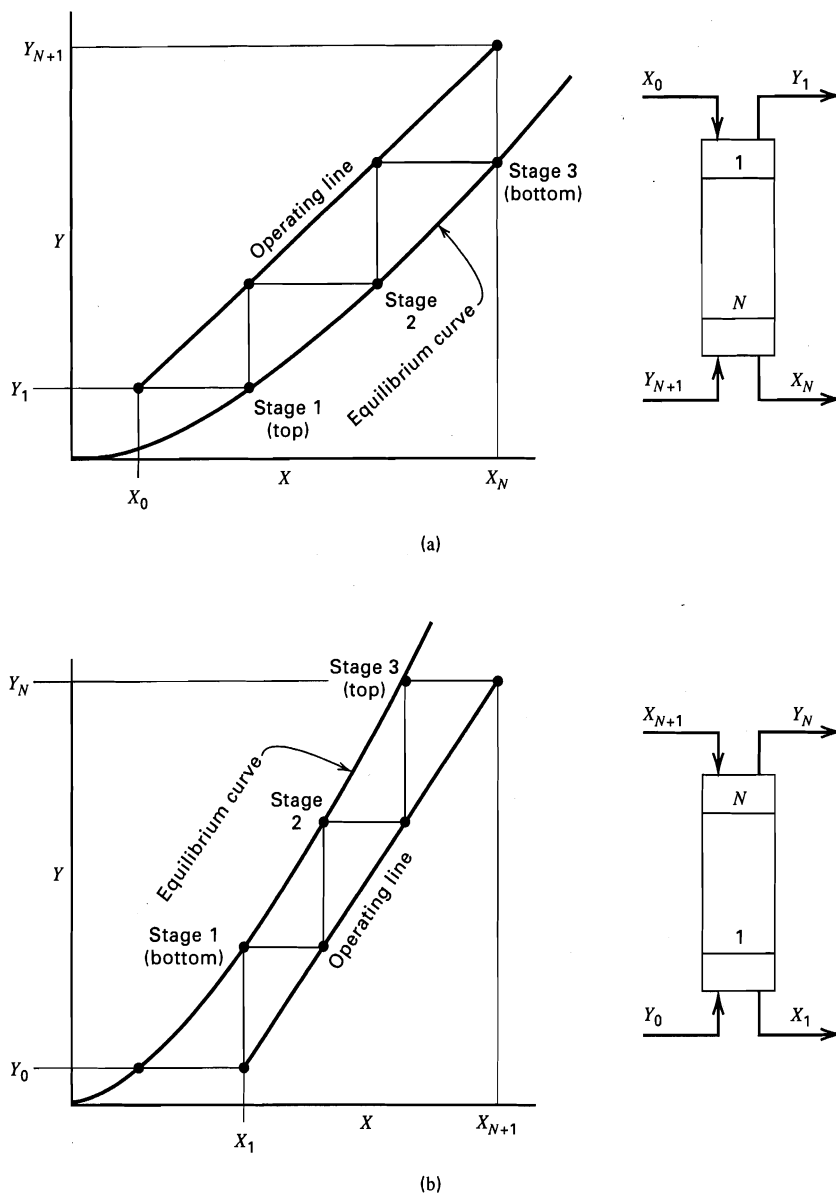


Figure 6.11 Graphical determination of the number of equilibrium stages for (a) absorber and (b) stripper.

EXAMPLE 6.1

When molasses is fermented to produce a liquor containing ethyl alcohol, a CO_2 -rich vapor containing a small amount of ethyl alcohol is evolved. The alcohol can be recovered by absorption with water in a sieve-tray tower. For the following conditions, determine the number of equilibrium stages required for countercurrent flow of liquid and gas, assuming isothermal, isobaric conditions in the tower and neglecting mass transfer of all components except ethyl alcohol.

Entering gas:

180 kmol/h; 98% CO_2 , 2% ethyl alcohol; 30°C , 110 kPa

Entering liquid absorbent:

100% water; 30°C , 110 kPa

Required recovery (absorption) of ethyl alcohol: 97%

SOLUTION

From Section 5.7 for a single-section, countercurrent cascade, the number of degrees of freedom is $2N + 2C + 5$. All stages operate adiabatically at a pressure of approximately 1 atm, taking $2N$ degrees of freedom. The entering gas is completely specified, tak-

ing $C + 2$ degrees of freedom. The entering liquid flow rate is not specified; thus, only $C + 1$ degrees of freedom are taken by the entering liquid. The recovery of ethyl alcohol takes one additional degree of freedom. Thus, the total number of degrees of freedom taken by the problem specification is $2N + 2C + 4$. This leaves one additional specification to be made, which in this example can be the entering liquid flow rate at, say, 1.5 times the minimum value.

The above application of the degrees of freedom analysis from Chapter 5 has assumed the use of an energy balance for each stage. The energy balances are assumed to result in the assumed isothermal operation at 30°C .

Assume that the exiting absorbent will be dilute in ethyl alcohol, whose K -value is determined from a modified Raoult's law, $K = \gamma P^s / P$. The vapor pressure of ethyl alcohol at 30°C is 10.5 kPa. At infinite dilution in water at 30°C , the liquid-phase activity coefficient of ethyl alcohol is taken as 6. Therefore, $K = (6)(10.5)/110 = 0.57$. The minimum solute-free absorbent rate is given by (6-11), where the solute-free gas rate, V' , is $(0.98)(180) = 176.4$ kmol/h. Thus,

$$L'_{\min} = (176.4)(0.57)(0.97) = 97.5 \text{ kmol/h}$$

The actual solute-free absorbent rate, at 50% above the minimum rate, is

$$L' = 1.5(97.5) = 146.2 \text{ kmol/h}$$

The amount of ethyl alcohol transferred from the gas to the liquid is 97% of the amount of alcohol in the entering gas or

$$(0.97)(0.02)(180) = 3.49 \text{ kmol/h}$$

The amount of ethyl alcohol remaining in the exiting gas is

$$(1.00 - 0.97)(0.02)(180) = 0.11 \text{ kmol/h}$$

We now compute the alcohol mole ratios at both ends of the operating line as follows, referring to Figure 6.8a:

$$\text{top } \begin{cases} X_0 = 0, & Y_1 = \frac{0.11}{176.4} = 0.0006 \end{cases}$$

$$\text{bottom } \begin{cases} Y_{N+1} = \frac{0.11 + 3.49}{176.4} = 0.0204, & X_N = \frac{3.49}{146.2} = 0.0239 \end{cases}$$

The equation for the operating line from (6-3) with $X_0 = 0$ is

$$Y_{N+1} = \left(\frac{146.2}{176.4} \right) X_N + 0.0006 = 0.829X_N + 0.0006 \quad (1)$$

It is clear that we are dealing with a dilute system. The equilibrium curve for ethyl alcohol can be determined from (6-1) using the value of $K = 0.57$ computed above. From (6-1),

$$0.57 = \frac{Y/(1+Y)}{X/(1+X)}$$

Solving for Y , we obtain

$$Y = \frac{0.57X}{1 + 0.43X} \quad (2)$$

To cover the entire column, the necessary range of X for a plot of Y vs X is 0 to almost 0.025. From the Y - X equation, (2),

Y	X
0.00000	0.000
0.00284	0.005
0.00569	0.010
0.00850	0.015
0.01130	0.020
0.01410	0.025

For this dilute system in ethyl alcohol, the maximum error in Y is 1.0% if Y is taken simply as $Y = KX = 0.57X$.

The equilibrium curve, which is almost straight in this example, and a straight operating line drawn through the terminal points (Y_1, X_0) and (Y_{N+1}, X_N) is given in Figure 6.12. The determination of points for the operating line and the equilibrium curve, as well as the plot of the points, is conveniently done with a spreadsheet program on a computer using Eqs. (1) and (2). The theoretical stages are stepped off as shown starting from the top stage (Y_1, X_0) located near the lower left corner of Figure 6.12. The required number of theoretical stages for 97% absorption of ethyl alcohol is just slightly more than six. Accordingly, it is best to provide seven theoretical stages.

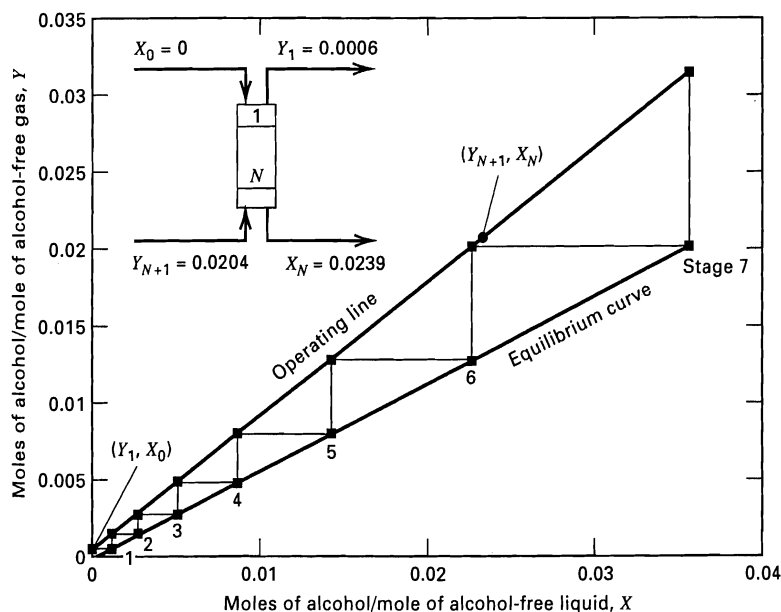


Figure 6.12 Graphical determination of number of equilibrium stages for an absorber.

6.4 ALGEBRAIC METHOD FOR DETERMINING THE NUMBER OF EQUILIBRIUM STAGES

Graphical methods for determining equilibrium stages have great educational value because a fairly complex multistage problem can be readily followed and understood. Furthermore, one can quickly gain visual insight into the phenomena involved. However, the application of a graphical

method can become very tedious when (1) the problem specification fixes the number of stages rather than the percent recovery of solute, (2) when more than one solute is being absorbed or stripped, (3) when the best operating conditions of temperature and pressure are to be determined so that the location of the equilibrium curve is unknown, and/or (4) if very low or very high concentrations force the graphical construction to the corners of the diagram so that multiple y - x diagrams of varying sizes and dimensions are needed.

Then, the application of an algebraic method may be preferred.

The Kremser method for single-section cascades, as developed in Section 5.4, is ideal for absorption and stripping of dilute mixtures. For example, (5-48) and (5-50) can be written in terms of the fraction of solute absorbed or stripped as

$$\text{Fraction of a solute, } i, \text{ absorbed} = \frac{A_i^{N+1} - A_i}{A_i^{N+1} - 1} \quad (6-13)$$

and

$$\text{Fraction of a solute, } i, \text{ stripped} = \frac{S_i^{N+1} - S_i}{S_i^{N+1} - 1} \quad (6-14)$$

where the solute absorption and stripping factors are, respectively,

$$A_i = L/(K_i V) \quad (6-15)$$

$$S_i = K_i V/L \quad (6-16)$$

Values of L and V in moles per unit time may be taken as entering values. Values of K_i depend mainly on temperature, pressure, and liquid-phase composition. Methods for estimating K -values are discussed in detail in Chapter 2. At near-ambient pressure, for dilute mixtures, some common expressions are

$$K_i = P_i^s/P \quad (\text{Raoult's law}) \quad (6-17)$$

$$K_i = \gamma_{iL}^\infty P_i^s/P \quad (\text{modified Raoult's law}) \quad (6-18)$$

$$K_i = H_i/P \quad (\text{Henry's law}) \quad (6-19)$$

$$K_i = P_i^s/x_i^* P \quad (\text{solubility}) \quad (6-20)$$

The first expression applies for ideal solutions involving solutes at subcritical temperatures. The second expression is useful for moderately nonideal solutions when activity coefficients are known at infinite dilution. For solutes at supercritical temperatures, the use of Henry's law may be preferable. For sparingly soluble solutes at subcritical temperatures, the fourth expression is preferred when solubility data in mole fractions, x_i^* , are available. This expression is derived by considering a three-phase system consisting of an ideal-vapor-containing solute, carrier vapor, and solvent; a pure or near-pure solute as liquid (1); and the solvent liquid (2) with dissolved solute. In that case, for solute, i , at equilibrium between the two liquid phases,

$$x_i^{(1)} \gamma_{iL}^{(1)} = x_i^{(2)} \gamma_{iL}^{(2)}$$

But,

$$x_i^{(1)} \approx 1, \quad \gamma_{iL}^{(1)} \approx 1, \quad x_i^{(2)} = x_i^*$$

Therefore,

$$\gamma_{iL}^{(2)} \approx 1/x_i^*$$

and from (6-18),

$$K_i^{(2)} = \gamma_{iL}^{(2)} P_i^s/P = P_i^s/(x_i^* P)$$

The advantage of (6-13) and (6-14) is that they can be solved directly for the percent absorption or stripping of a

solute when the number of theoretical stages, N , and the absorption or stripping factor are known.

EXAMPLE 6.2

As discussed by Okoniewski [3], volatile organic compounds (VOCs) can be stripped from wastewater by air. Such compounds are to be stripped at 70°F and 15 psia from 500 gpm of wastewater with 3,400 scfm of air (standard conditions of 60°F and 1 atm) in an existing tower containing 20 plates. A chemical analysis of the wastewater shows three organic chemicals in the amounts shown in the following table. Included are necessary thermodynamic properties from the 1966 *Technical Data Book—Petroleum Refining* of the American Petroleum Institute. For all three organic compounds, the wastewater concentrations can be shown to be below the solubility values.

Organic Compound	Concentration in the Wastewater, mg/L	Solubility in Water at 70°F, mole fraction	Vapor Pressure at 70°F, psia
Benzene	150	0.00040	1.53
Toluene	50	0.00012	0.449
Ethylbenzene	20	0.000035	0.149

It is desirable that 99.9% of the total VOCs be stripped, but the plate efficiency of the tower is uncertain, with an estimated range of 5% to 20%, corresponding to one to four theoretical stages for the 20-plate tower. Calculate and plot the percent stripping of each of the three organic compounds for one, two, three, and four theoretical stages. Under what conditions can we expect to achieve the desired degree of stripping? What should be done with the exiting air?

SOLUTION

Because the wastewater is dilute in the VOCs, the Kremser equation may be applied independently to each of the three organic chemicals. We will ignore the absorption of air by the water and the stripping of water by the air. The stripping factor for each compound is given by $S_i = K_i V/L$, where V and L will be taken at entering conditions. The K -value may be computed from a modified Raoult's law, $K_i = \gamma_{iL} P_i^s/P$, where for a compound that is only slightly soluble, take $\gamma_{iL} = 1/x_i^*$, where x_i^* is the solubility in mole fraction. Thus, from (6-20), $K_i = P_i^s/x_i^* P$

$$V = 3,400(60)/(379 \text{ scf/lbmol}) \quad \text{or} \quad 538 \text{ lbmol/h}$$

$$L = 500(60)(8.33 \text{ lb/gal})/(18.02 \text{ lb/lbmol}) \quad \text{or} \quad 13,870 \text{ lbmol/h}$$

The corresponding K -values and stripping factors are

Component	K at 70°F, 15 psia	S
Benzene	255	9.89
Toluene	249	9.66
Ethylbenzene	284	11.02

From (6-14),

$$\text{Fraction stripped} = \frac{S^{N+1} - S}{S^{N+1} - 1}$$

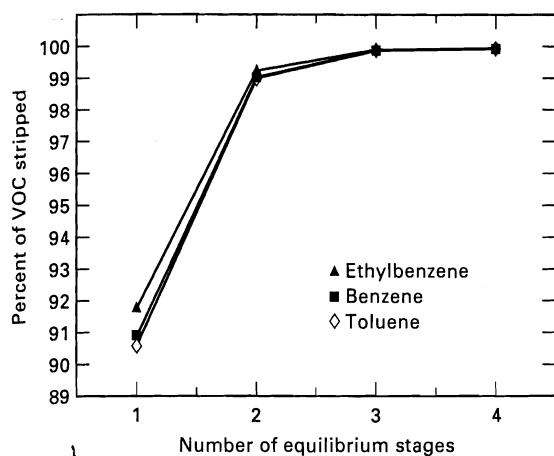


Figure 6.13 Results of Example 6.2 for stripping of VOCs from water with air.

The calculations when carried out with a spreadsheet computer program give the following results:

Component	Percent Stripped			
	1 Stage	2 Stages	3 Stages	4 Stages
Benzene	90.82	99.08	99.91	99.99
Toluene	90.62	99.04	99.90	99.99
Ethylbenzene	91.68	99.25	99.93	99.99

The results are quite sensitive to the number of theoretical stages as shown in Figure 6.13. To achieve 99.9% removal of the total VOCs, three theoretical stages are needed, corresponding to the necessity for a 15% stage efficiency in the existing 20-tray tower.

It is best to process the exiting air to remove or destroy the VOCs, particularly the benzene, which is a carcinogen [4]. The amount of benzene stripped is

$$(500 \text{ gpm})(60 \text{ min/h})(3.785 \text{ liters/gal})(150 \text{ mg/liters}) \\ = 17,030,000 \text{ mg/h or } 37.5 \text{ lb/h}$$

If benzene is valued at \$0.30/lb, the annual value is approximately \$100,000. It is doubtful that this would justify a recovery technique, such as carbon adsorption. It is perhaps preferable to destroy the VOCs by incineration. For example, the air can be sent to a utility boiler, a waste-heat boiler, or a catalytic incinerator. It is also to be noted that the amount of air was arbitrarily given as 3,400 scfm. To complete the design procedure, various air rates should be investigated. It will also be necessary to verify by methods given later in this chapter that, at the chosen air flow rates, no flooding or weeping will occur in the column.

6.5 STAGE EFFICIENCY

Graphical and algebraic methods for determining stage requirements for absorption and stripping assume equilibrium with respect to both heat and mass transfer at each stage. Thus, the number of *equilibrium stages* (*theoretical stages*, *ideal stages*, or *ideal plates*) is determined or specified when

using those methods. Except when temperature changes significantly from stage to stage, the assumption that vapor and liquid phases leaving a stage are at the same temperature is often reasonable. The assumption of equilibrium with respect to mass transfer, however, is not often reasonable and, for streams leaving a stage, vapor-phase mole fractions are not related to liquid-phase mole fractions simply by thermodynamic K -values. To determine the actual number of plates, the number of equilibrium stages must be adjusted with a *stage efficiency* (*plate efficiency* or *tray efficiency*).

Stage efficiency concepts are applicable to devices in which the phases are contacted and then separated, that is, when discrete stages can be identified. This is not the case for packed columns or continuous-contact devices. For these, the efficiency is imbedded into an equipment- and system-dependent parameter, an example of which is the HETP (height of packing equivalent to a theoretical plate).

The simplest approach for staged columns, in preliminary design studies and in the evaluation of the performance of an existing column, is to apply an overall stage (or column) efficiency, defined by Lewis [5] as

$$E_o = N_t/N_a \quad (6-21)$$

where E_o is the fractional overall stage efficiency, usually less than 1.0; N_t is the calculated number of equilibrium (theoretical) stages; and N_a is the actual number of contacting trays or plates (usually greater than N_t) required. Based on the results of extensive research conducted over a period of more than 60 years, the overall stage efficiency has been found to be a complex function of the

1. Geometry and design of the contacting trays
2. Flow rates and flow paths of vapor and liquid streams
3. Compositions and properties of vapor and liquid streams

For well-designed trays and for flow rates near the capacity limit, E_o depends mainly on the physical properties of the vapor and liquid streams.

Values of E_o can be predicted by any of the following four methods:

1. Comparison with performance data from industrial columns for the same or similar systems
2. Use of empirical efficiency models derived from data on industrial columns
3. Use of semitheoretical models based on mass- and heat-transfer rates
4. Scale-up from data obtained with laboratory or pilot-plant columns

These methods, which are discussed in some detail in the following four subsections, are applied to other vapor-liquid separation operations, such as distillation, as well as to absorption and stripping. Suggested correlations of mass-transfer coefficients for trayed towers are deferred to Section 6.6, following the discussion of tray capacity.

Table 6.4 Performance Data for Absorbers and Strippers in Hydrocarbon Service

Service	Type of Tray	Column Diameter, ft	No. of Trays	Tray Spacing, in.	Average Pressure, psia	Average Temp., °F	Molar Average Liquid Viscosity, cP	Overall Stage Efficiency, %
Absorption of butane	Bubble cap	4	24	18	260	120	0.48	36
Absorption of butane	Bubble cap	5	16	30	254	132	0.31	50
Absorption of butane	Bubble cap	4	16	24	94	117	1.41	10.4
Steam stripping of kerosene	Bubble cap	5	4	30	68	448	0.205	57
Steam stripping of gas oil	Bubble cap	5	6	30	60	507	0.250	49

Source: H.G. Drickamer and J.R. Bradford [6].

Performance Data

Performance data obtained from industrial absorption and stripping columns equipped with trays generally include gas- and liquid-feed and product flow rates and compositions, average column pressure and temperature or pressures and temperatures at the bottom and top of the column, number of actual trays, N_a , column diameter, and type of tray with, perhaps, some details of the tray design. From these data, particularly if the system is dilute with respect to the solute(s), the graphical or algebraic methods, described in Sections 6.3 and 6.4, respectively, can be used to estimate the number of equilibrium stages, N_e , required. Then (6-21) can be applied to determine the overall stage efficiency, E_o . Values of E_o for absorbers and strippers are typically low, often less than 50%.

Table 6.4 presents performance data, from a study by Drickamer and Bradford [6], for five industrial hydrocarbon absorption and stripping operations using columns with bubble-cap trays. For the three absorbers, the stage efficiencies are based on the absorption of *n*-butane as the key component. For the two strippers, both of which use steam as the stripping agent, the key component is not given, but is probably *n*-heptane. Although the data cover a wide range of average pressure and temperature, the overall stage efficiencies, which cover a wide range of 10.4% to 57%, appear to depend primarily on the molar average liquid viscosity, a key factor for the rate of mass transfer in the liquid phase.

The gas feed to a hydrocarbon absorber contains a range of light hydrocarbons, each of which is absorbed to a different extent based on its *K*-value, as illustrated in Example 5.3. The data of Jackson and Sherwood [7] for a 9-ft-diameter hydrocarbon absorber equipped with 19 bubble-cap trays on 30-in. tray spacing and operating at 92 psia and 60°F, as analyzed by O'Connell [8] and summarized in Table 6.5, show that each component being absorbed has a different overall stage efficiency, which appears to increase with decreasing *K*-value (increasing solubility in the liquid absorbent). For

Table 6.5 Effect of Species on Overall Stage Efficiency in a 9-ft-Diameter Industrial Absorber Using Bubble-Cap Trays

Component	Overall Stage Efficiency, %
Ethylene	10.3
Ethane	14.9
Propylene	25.5
Propane	26.8
Butylene	33.8

Source: H.E. O'Connell [8].

the same molar-average liquid viscosity (1.90 cps), the overall stage efficiency is seen to vary from as low as 10.3% for ethylene, the most-volatile species considered, to 33.8% for butylene (presumably *n*-butene), the least-volatile species considered.

An even more dramatic effect of the species solubility in the absorbent on the overall stage efficiency is seen in Table 6.6, from a study by Walter and Sherwood [9] using small laboratory, bubble-cap tray columns ranging in size from 2 to 18 in. in diameter. Stage efficiencies vary over a very wide range from 0.65% to 69%. Comparing the data for the water absorption of ammonia (a very soluble gas) and carbon dioxide (a slightly soluble gas), it is clear that the solubility of the gas (i.e., the *K*-value) has a large effect on stage efficiency. Thus, low stage efficiency can occur when the liquid viscosity is high and/or the gas solubility is low (high *K*-value); high stage efficiency can occur when the liquid viscosity is low and the gas solubility is high (low *K*-value).

Empirical Correlations

Using 20 sets of performance data from industrial hydrocarbon absorbers and strippers, including the data in Table 6.4, Drickamer and Bradford [6] correlated the overall stage efficiency of the key component absorbed or stripped with just

Table 6.6 Performance Data for Absorption in Laboratory Bubble-Cap Tray Columns

Service	Column Diameter, in.	No. of Trays	Tray Spacing, in.	Average Pressure, psia	Average Temp., °F	Overall Stage Efficiency, %
Absorption of ammonia in water	18	1	—	14.7	57	69
Absorption of isobutylene in heavy naphtha	2	1	—	66	78.8	36.4
Absorption of propylene in gas oil	2	1	—	66	118.4	13.1
Absorption of propylene in gas lube oil	2	1	—	66	105.8	4.7
Absorption of carbon dioxide in water	18	1	—	14.7	50.4	2.0
Desorption of carbon dioxide from 43.7 wt% aqueous glycerol	5	4	11	14.7	77	0.65

Source: J.F. Walter and T.K. Sherwood [9].

the molar-average viscosity of the rich oil (liquid leaving an absorber or liquid entering a stripper) at the average tower temperature over a viscosity range of 0.19 to 1.58 cP. The empirical equation,

$$E_o = 19.2 - 57.8 \log \mu_L, \quad 0.2 < \mu_L < 1.6 \text{ cP} \quad (6-22)$$

where E_o is in percent and μ is in centipoise, fits the data with average- and maximum-percent deviations of 10.3% and 41%, respectively. A plot of the Drickamer and Bradford correlation, compared to performance data, is given in Figure 6.14. Equation (6-22) should not be used for absorption into nonhydrocarbon liquids and is restricted to the listed range of the liquid viscosity data used to develop the correlation.

Mass-transfer theory indicates that when the volatility of species being absorbed or stripped covers a wide range, the

relative importance of liquid-phase and gas-phase mass-transfer resistances can shift. Thus, O'Connell [8] found that the Drickamer-Bradford correlation, (6-22), was inadequate for absorbers and strippers when applied to species covering a wide range of volatility or K -values. This additional effect is indicated clearly in the performance data of Tables 6.5 and 6.6, where liquid viscosity alone cannot correlate the data. O'Connell obtained a more general correlation by using a parameter that included not only the liquid viscosity but also the liquid density and the Henry's law constant of the species being absorbed or stripped. Edmister [10] and Lockhart and Leggett [11] suggested slight modifications to the O'Connell correlation to permit its use with K -values (instead of Henry's-law constants). An O'Connell-type plot of overall stage efficiency for absorption or stripping in bubble-cap tray columns is given in Figure 6.15.

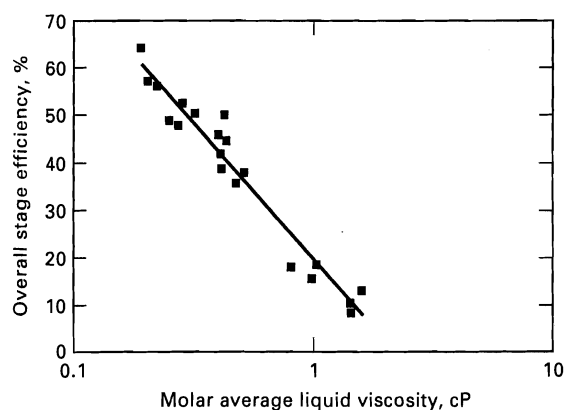


Figure 6.14 Drickamer and Bradford correlation for plate efficiency of hydrocarbon absorbers and strippers.

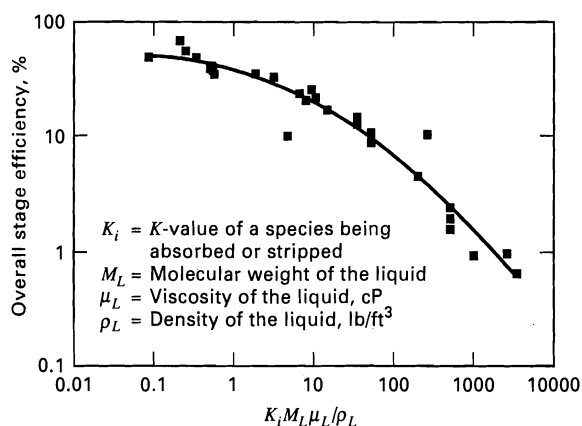


Figure 6.15 O'Connell correlation for plate efficiency of absorbers and strippers.

The correlating parameter, suggested by Edmister, is $K_i M_L \mu_L / \rho_L$, where:

K_i = K -value of species being absorbed or stripped

M_L = molecular weight of the liquid, lb/lbmol

μ_L = viscosity of the liquid, cP

ρ_L = density of the liquid, lb/ft³

Thus, the correlating parameter has the units of cP-ft³/lbmol. A reasonable fit to the 33 data points used by O'Connell is given by the empirical equation

$$\log E_o = 1.597 - 0.199 \log \left(\frac{K M_L \mu_L}{\rho_L} \right) - 0.0896 \left[\log \left(\frac{K M_L \mu_L}{\rho_L} \right) \right]^2 \quad (6-23)$$

The average and maximum deviations of (6-23) for the 33 data points of Figure 6.15 are 16.3% and 157%, respectively. More than 50% of the data points, including points for the highest- and lowest-observed efficiencies, are predicted to within 10%.

The 33 data points in Figure 6.15 cover a wide range of conditions:

Column diameter:	2 in. to 9 ft
Average pressure:	14.7 to 485 psia
Average temperature:	60 to 138°F
Liquid viscosity:	0.22 to 21.5 cP
Overall stage efficiency:	0.65 to 69%

Absorbents include both hydrocarbons and water. For the absorption or stripping of more than one species, because of the effect of species K -value, different stage efficiencies are predicted, as observed from performance data of the type shown in Table 6.5. The inclusion of the K -value also permits the correlation to be used for aqueous systems where the solute may exhibit a very wide range of solubility (e.g., ammonia versus carbon dioxide) as included in Table 6.6. In using Figure 6.15 or Eq. (6-23), the K -value and absorbent properties are best evaluated at the end of the tower where the liquid phase is richest in solute(s). Prudent designs use the lowest predicted efficiency.

Most of the data used to develop the correlation of Figure 6.15 are for columns having a liquid flow path across the active tray area of from 2 to 3 ft. Theory and experimental data show that higher efficiencies are achieved for longer flow paths. For short liquid flow paths, the liquid flowing across the tray is usually completely mixed. For longer flow paths, the equivalent of two or more completely mixed, successive liquid zones may be present. The result is a greater average driving force for mass transfer and, thus, a higher efficiency—perhaps greater than 100%. For example, a column with a 10-ft liquid flow path may have an efficiency as much as 25% greater than that predicted by (6-23). However, at high liquid rates, long liquid-path lengths are

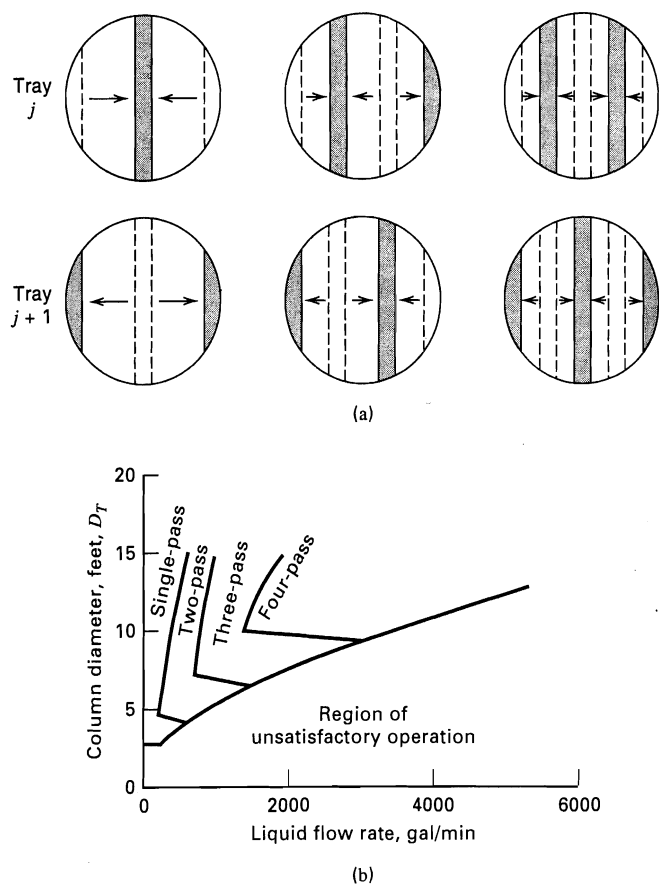


Figure 6.16 Estimation of number of required liquid flow passes. (a) Multipass trays: (1) two-pass; (2) three-pass; (3) four-pass. (b) Flow pass selection.

(Derived from *Koch Flexitray Design Manual, Bulletin 960*, Koch Engineering Co., Inc., Wichita, KA, 1960.)

undesirable because they lead to excessive hydraulic gradients. When the effective height of a liquid on a tray is appreciably higher on the inflow side than at the overflow weir, vapor may prefer to enter the tray in the latter region, leading to nonuniform bubbling action. Multipass trays, as shown in Figure 6.16a, are used to prevent excessive liquid gradients. Estimation of the desired number of flow paths can be made with Figure 6.16b, where, e.g., a 10-foot-diameter column with a liquid flow rate of 1000 gpm should use a three-pass tray.

Based on estimates of the number of actual trays and tray spacing, the height of a column between the top tray and the bottom tray is computed. By adding another 4 ft above the top tray for removal of entrained liquid and 10 ft below the bottom tray for bottoms surge capacity, the total column height is estimated. If the height is greater than 212 ft (equivalent to 100 trays on 24-in. spacing), two or more columns arranged in series may be preferable to a single column. Perhaps the tallest column in the world, located at the Shell Chemical Company complex in Deer Park, Texas, stands 338 ft tall [*Chem. Eng.*, 84 (26), 84 (1977)].

EXAMPLE 6.3

Performance data, given below, for a bubble-cap tray absorber located in a Texas petroleum refinery, were reported by Drickamer and Bradford [6]. Based on these data, back-calculate the overall stage efficiency for *n*-butane and compare the result with both the Drickamer-Bradford and O'Connell correlations. Lean oil and rich gas enter the tower; rich oil and lean gas leave the tower.

Performance Data

Number of plates	16
Plate spacing, in.	24
Tower diameter, ft	4
Tower pressure, psig	79
Lean oil temperature, °F	102
Rich oil temperature, °F	126
Rich gas temperature, °F	108
Lean gas temperature, °F	108
Lean oil rate, lbmol/h	368
Rich oil rate, lbmol/h	525.4
Rich gas rate, lbmol/h	946
Lean gas rate, lbmol/h	786.9
Lean oil molecular weight	250
Lean oil viscosity at 116°F, cP	1.4
Lean oil gravity, °API	21

Stream Compositions, Mol%

Component	Rich Gas	Lean Gas	Rich Oil	Lean Oil
C ₁	47.30	55.90	1.33	
C ₂	8.80	9.80	1.16	
C ₃	5.20	5.14	1.66	
C ₃	22.60	21.65	8.19	
C ₄	3.80	2.34	3.33	
<i>n</i> C ₄	7.40	4.45	6.66	
<i>n</i> C ₅	3.00	0.72	4.01	
<i>n</i> C ₆	1.90		3.42	
Oil absorbent			70.24	100
Totals	100.00	100.00	100.00	100

SOLUTION

Before computing the overall stage efficiency for *n*-butane, it is worthwhile to check the consistency of the plant data by examining the overall material balance and the material balance for each component. From the above stream compositions, it is apparent that the compositions have been normalized to total 100%.

The overall material balance is

$$\text{Total flow into tower} = 368 + 946 = 1,314 \text{ lbmol/h}$$

$$\text{Total flow from tower} = 525.4 + 786.9 = 1,312.3 \text{ lbmol/h}$$

These two totals agree to within 0.13%. This is excellent agreement.

The component material balance for the oil absorbent is

$$\text{Total oil in} = 368 \text{ lbmol/h}$$

$$\text{Total oil out} = (0.7024)(525.4) = 369 \text{ lbmol/h}$$

These two totals agree to within 0.3%. Again, this is excellent agreement.

Component material balances for other hydrocarbons from spreadsheet calculations are as follows.

Component	lbmol/h			
	Lean Gas	Rich Oil	Total Out	Total In
C ₁	439.9	7.0	446.9	447.5
C ₂	77.1	6.1	83.2	83.2
C ₃	40.4	8.7	49.1	49.2
C ₃	170.4	43.0	213.4	213.8
C ₄	18.4	17.5	35.9	35.9
<i>n</i> C ₄	35.0	35.0	70.0	70.0
<i>n</i> C ₅	5.7	21.1	26.8	28.4
<i>n</i> C ₆	0.0	18.0	18.0	18.0
	786.9	156.4	943.3	946.0

Again, we see excellent agreement. The largest difference is 6% for pentanes. Plant data are not always so consistent.

For the back-calculation of stage efficiency from the performance data, the Kremser equation is applied to compute the number of equilibrium stages required for the measured absorption of *n*-butane.

$$\text{Fraction of } nC_4 \text{ absorbed} = \frac{35}{70} = 0.50$$

$$\text{From (6-13), } 0.50 = \frac{A^{N+1} - A}{A^{N+1} - 1}$$

$$\text{where } A = \text{absorption factor} = \frac{L}{KV}$$

Because *L* and *V* vary greatly through the column, let

$$L = \text{average liquid rate} = \frac{368 + 525.4}{2} = 446.7 \text{ lbmol/h}$$

and let

$$V = \text{average vapor rate} = \frac{946 + 786.9}{2} = 866.5 \text{ lbmol/h}$$

Assume average tower temperature = the average of inlet and outlet temperatures = (102 + 126 + 108 + 108)/4 = 111°F. Also assume that the viscosity of the lean oil at 116°F equals the viscosity of the rich oil at 111°F. Therefore, $\mu = 1.4 \text{ cP}$.

Assume the ambient pressure is 14.7 psia. Then

$$\text{Tower pressure} = 79 + 14.7 = 93.7 \text{ psia}$$

From Figure 2.8, at 93.7 psia and 111°F, $K_{nC_4} = 0.7$. Thus,

$$A = \frac{446.7}{(0.7)(866.5)} = 0.736$$

$$\text{Therefore, } 0.50 = \frac{0.736^{N+1} - 0.736}{0.736^{N+1} - 1}$$

$$\text{Solving, } N = N_t = 1.45$$

From the performance data, $N_a = 16$

$$\text{From (6-21), } E_o = \frac{1.45}{16} = 0.091 \text{ or } 9.1\%$$

Equation (6-22) is applicable to *n*-butane, because that component is absorbed to the extent of about 50% and thus can be considered one of the key components. Other possible key components are butenes and *n*-pentane.

$$\text{From (6-22), } E_o = 19.2 - 57.8 \log(1.4) = 10.8\%$$

To estimate the stage efficiency from the O'Connell correlation, use the following properties for the rich oil at 126°F, 93.7 psia, and 30 mol% light hydrocarbons/70 mol% of 250-MW oil, as obtained from a simulation program.

$$K = 0.77 \text{ for } n\text{-butane}$$

$$M_L = 195$$

$$\mu_L = 0.9 \text{ cP}$$

$$\rho_L = 44.1 \text{ lb/ft}^3$$

$$\text{Therefore, } \frac{K M_L \mu_L}{\rho_L} = \frac{0.77(195)(0.9)}{(44.1)} = 3.1$$

From (6-23),

$$\log E_o = 1.597 - 0.199 \log(3.1) - 0.0896[\log(3.1)]^2 = 1.48$$

$$E_o = 10^{1.48} = 30.2$$

For this hydrocarbon absorber, the Drickamer and Bradford correlation (10.8%) gives better agreement than the O'Connell correlation (30.2%) with the plant performance data (9.1%).

Semitheoretical Models

A third method for predicting the overall stage efficiency involves the application of a semitheoretical tray model based on mass- and heat-transfer rates. With this model, the fractional approach to equilibrium, called the *plate* or *tray efficiency*, is estimated for each component in the mixture for each tray in the column. These efficiency values are then utilized to determine conditions for each tray, or averaged for the column to obtain the overall plate efficiency.

Tray efficiency models, in order of increasing complexity, have been proposed by Holland [12], Murphree [13], Hausen [14], and Standart [15]. All four models are based on the assumption that vapor and liquid streams entering each tray are of uniform compositions. The *Murphree vapor efficiency*, which is the oldest and most widely used, is derived with the additional assumptions of (1) complete mixing of the liquid flowing across the tray such that the liquid is of a uniform concentration, equal to the composition of the liquid leaving the tray and entering the next tray below, and (2) plug flow of the vapor passing up through the liquid, as indicated in Figure 6.17 for tray n . Considering species i , let

n = rate of mass transfer for absorption from the gas to the liquid

K_G = overall gas mass-transfer coefficient based on a partial-pressure driving force

a = vapor-liquid interfacial area per volume of combined gas and liquid holdup (froth or dispersion) on the tray,

A_b = active bubbling area of the tray (total cross-sectional area minus liquid down-comer areas)

Z_f = height of combined gas and liquid holdup on the tray

P = total absolute pressure

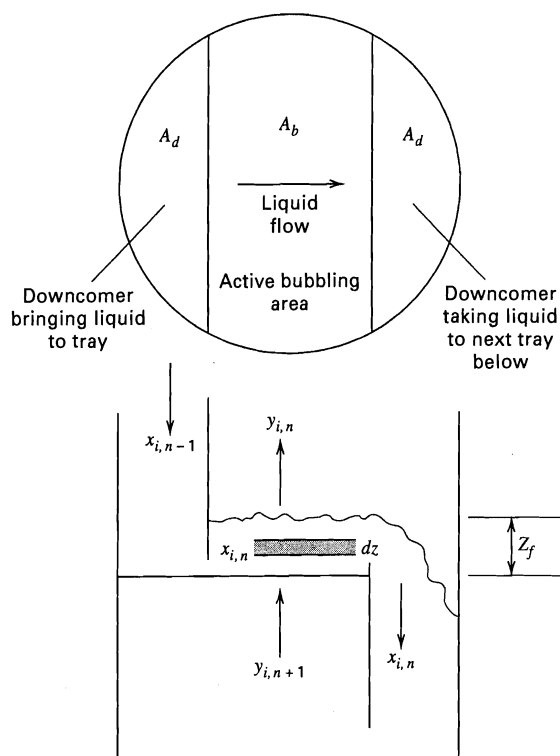


Figure 6.17 Schematic top and side views of tray for derivation of Murphree vapor-tray efficiency.

y_i = mole fraction of i in the vapor rising up through the liquid

y_i^* = vapor mole fraction of i in equilibrium with the completely mixed liquid on the tray

Then the differential rate of mass transfer for a differential height of holdup on tray n , numbered down from the top, is

$$dn_i = K_G a (y_i - y_i^*) P A_b dZ \quad (6-24)$$

where K_G takes into account both gas- and liquid-phase resistances to mass transfer. By material balance, assuming a negligible change in V across the stage,

$$dn_i = -V dy_i \quad (6-25)$$

where V = molar gas flow rate up through the liquid on the tray.

Combining (6-24) and (6-25) to eliminate dn_i , separating variables, and converting to integral form,

$$A_b \int_0^{Z_f} \frac{K_G a P}{V} dZ = \int_{y_{i,n+1}}^{y_{i,n}} \frac{dy_i}{y_{i,n+1}^* - y_i} = N_{OG} \quad (6-26)$$

where a second subscript involving the tray number, n , has been added to the mole fraction of the vapor phase. The vapor enters tray n at $y_{i,n+1}$ and exits at $y_{i,n}$. This equation defines

N_{OG} = number of overall gas-phase mass-transfer units

Values of K_G , a , P , and V may vary somewhat as the gas flows up through the liquid on the tray, but if they as well as

y_i^* are taken to be constant, (6-26) can be integrated to give

$$N_{OG} = \frac{K_G a P Z_f}{(V/A_b)} = \ln \left(\frac{y_{i,n+1} - y_{i,n}^*}{y_{i,n} - y_{i,n}^*} \right) \quad (6-27)$$

A rearrangement of (6-27) in terms of the fractional approach of y_i to equilibrium defines the Murphree vapor efficiency as

$$E_{MV} = \frac{y_{i,n+1} - y_{i,n}}{y_{i,n+1} - y_{i,n}^*} = 1 - e^{-N_{OG}} \quad (6-28)$$

or

$$N_{OG} = -\ln(1 - E_{MV}) \quad (6-29)$$

Suppose that measurements give

$$y_i \text{ entering tray } n = y_{i,n+1} = 0.64$$

$$y_i \text{ leaving tray } n = y_{i,n} = 0.61$$

and, from thermodynamics or phase equilibrium data, y_i^* in equilibrium with x_i on and leaving tray $n = 0.60$.

Then, from (6-28),

$$E_{MV} = (0.64 - 0.61)/(0.64 - 0.60) = 0.75$$

or a 75% approach to equilibrium. From (6-29),

$$N_{OG} = -\ln(1 - 0.75) = 1.386$$

When $N_{OG} = 1$, $E_{MV} = 1 - e^{-1} = 0.632$.

The derivation of the Murphree vapor efficiency does not consider the exiting stream temperatures. However, it is implied that the completely mixed liquid phase is at its bubble-point temperature so that the equilibrium vapor phase mole fraction, $y_{i,n}^*$, can be computed.

For multicomponent mixtures, values of E_{MV} are component-dependent and can vary from tray to tray; but at each tray it can be shown that the number of independent values of E_{MV} is one less than the number of components. The dependent value of E_{MV} is determined by forcing $\sum y_i = 1$. It is thus possible that a negative value of E_{MV} can result for a component in a multicomponent mixture. Such negative efficiencies are possible because of mass-transfer coupling among concentration gradients in a multicomponent mixture, which is discussed in Chapter 12. However, for a binary mixture, values of E_{MV} are always positive and identical for the two components.

Only if liquid travel distance across a tray is small will the liquid on a tray approach the complete-mixing assumption used to derive (6-27). To handle the more general case of incomplete liquid mixing, a *Murphree vapor-point efficiency* is defined by assuming that liquid composition varies with distance of travel across a tray, but is uniform in the vertical direction. Thus, for species i on tray n , at any horizontal distance from the downcomer that directs liquid onto tray n , as shown in Figure 6.18,

$$E_{OV} = \frac{y_{i,n+1} - y_i}{y_{i,n+1} - y_i^*} \quad (6-30)$$

Because x_i varies across a tray, y_i^* and y_i also vary. However, the exiting vapor is then assumed to mix completely to give a uniform $y_{i,n}$ before entering the tray above. Because E_{OV} is

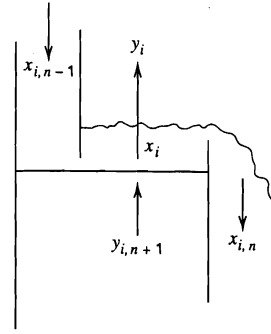


Figure 6.18 Schematic of tray for Murphree vapor-point efficiency.

a more fundamental quantity than E_{MV} , E_{OV} serves as the basis for semitheoretical estimates of tray efficiency and overall column efficiency.

Lewis [16] integrated E_{OV} over a tray for several cases. For complete mixing of liquid on a tray to give a uniform composition, $x_{i,n}$, it is obvious that

$$E_{OV} = E_{MV} \quad (6-31)$$

For plug flow of liquid across a tray with no longitudinal diffusion (no mixing of liquid in the horizontal direction), Lewis derived

$$E_{MV} = \frac{1}{\lambda} (e^{\lambda E_{OV}} - 1) \quad (6-32)$$

with

$$\lambda = mV/L \quad (6-33)$$

where V and L are gas and liquid molar flow rates, respectively, and $m = dy/dx$ = slope of the equilibrium line for a species, using the expression $y = mx + b$. If b is taken as zero, then m is the K -value, and for the key component, k , being absorbed,

$$\lambda = K_k V/L = 1/A_k$$

If A_k , the key-component absorption factor, is given the typical value of 1.4, $\lambda = 0.71$. Suppose the measured or predicted point efficiency is $E_{OV} = 0.25$. From (6-32),

$$E_{MV} = 1.4(e^{0.71(0.25)} - 1) = 0.27$$

which is only 9% higher than E_{OV} . However, if $E_{OV} = 0.9$, E_{MV} is 1.25, which is significantly higher and equivalent to more than a theoretical stage. This surprising result is due to the concentration gradient in the liquid across the length of travel on the tray, which allows the vapor to contact a liquid having an average concentration of species k that can be appreciably lower than that in the liquid leaving the tray.

Equations (6-31) and (6-32) represent extremes between complete mixing and no mixing of the liquid phase, respectively. A more realistic, but considerably more complex model that accounts for partial liquid mixing on the tray, as developed by Gerster et al. [17], is

$$\frac{E_{MV}}{E_{OV}} = \frac{1 - e^{-(\eta + N_{Pe})}}{(\eta + N_{Pe})\{1 + [(\eta + N_{Pe})/\eta]\}} + \frac{e^\eta - 1}{\eta\{1 + [\eta/(\eta + N_{Pe})]\}} \quad (6-34)$$

where

$$\eta = \frac{N_{Pe}}{2} \left[\left(1 + \frac{4\lambda E_{OV}}{N_{Pe}} \right)^{1/2} - 1 \right] \quad (6-35)$$

The dimensionless Peclet number, N_{Pe} , which serves as a partial-mixing parameter, is defined by

$$N_{Pe} = Z_L^2 / D_E \theta_L = Z_L u / D_E \quad (6-36)$$

where Z_L is the length of liquid flow path across the tray as shown in Figure 6.3, D_E is the eddy diffusion coefficient in the direction of liquid flow, θ_L is the average liquid residence time on the tray, and $u = Z_L / \theta_L$ is the mean liquid velocity across the tray. Equation (6-34) is plotted in Figure 6.19 for wide ranges of N_{Pe} and λE_{OV} . When $N_{Pe} = 0$, (6-31) holds; when $N_{Pe} = \infty$, (6-32) holds.

From (6-36), the Peclet number can be viewed as the ratio of the mean liquid bulk velocity to the eddy-diffusion velocity. When N_{Pe} is small, eddy diffusion is important and the liquid approaches a well-mixed condition. When N_{Pe} is large, bulk flow predominates and the liquid approaches plug flow. Experimental measurements of D_E in bubble-cap and sieve-plate columns [18–21] cover a range of 0.02 to 0.20 ft²/s. Values of u/D_E typically range from 3 to 15 ft⁻¹. Based on the second form of (6-36), N_{Pe} increases directly with increasing Z_L and, therefore, column diameter. A typical value of N_{Pe} for a 2-ft-diameter column is 10; for a 6-ft-diameter column, N_{Pe} might be 30. For N_{Pe} values of this

magnitude, Figure 6.19 shows that values of E_{MV} can be significantly larger than E_{OV} for large values of λ .

Lewis [16] showed that when the equilibrium and operating lines are straight, but not necessarily parallel, the overall stage efficiency, defined by (6-21), is related to the Murphree vapor efficiency by

$$E_o = \frac{\log[1 + E_{MV}(\lambda - 1)]}{\log \lambda} \quad (6-37)$$

When the two lines are not only straight but parallel, such that $\lambda = 1$, (6-37) becomes $E_o = E_{MV}$. Also, when $E_{MV} = 1$, then $E_o = 1$ regardless of the value of λ .

Scale-up from Laboratory Data

When vapor–liquid equilibrium data for a system are unavailable or not well known, and particularly if the system forms a highly nonideal liquid solution with possible formation of azeotropes, tray requirements are best estimated, and the feasibility of achieving the desired degree of separation verified, by conducting laboratory tests. A particularly useful apparatus is a small glass or metal sieve-plate column with center-to-side downcomers developed by Oldershaw [22]

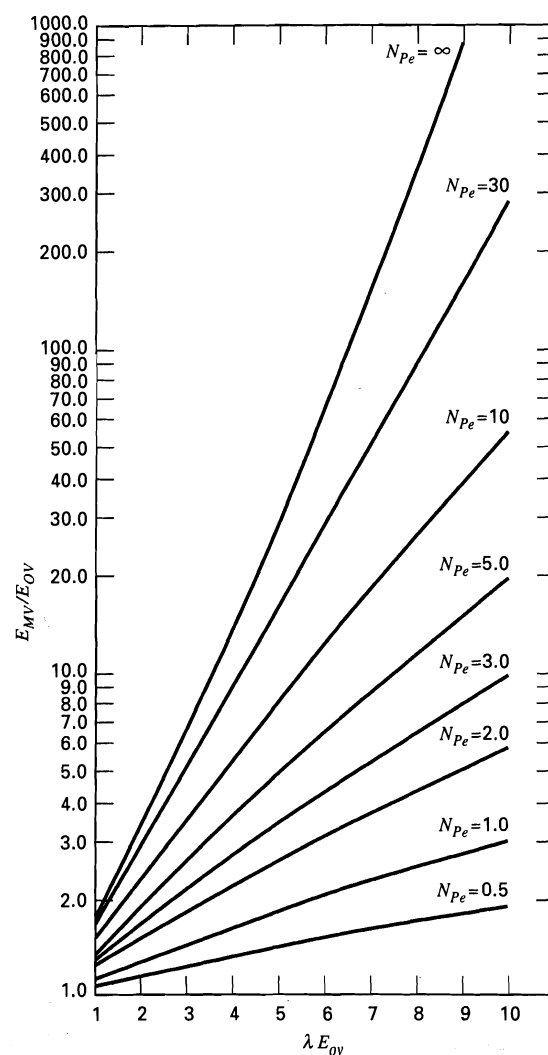
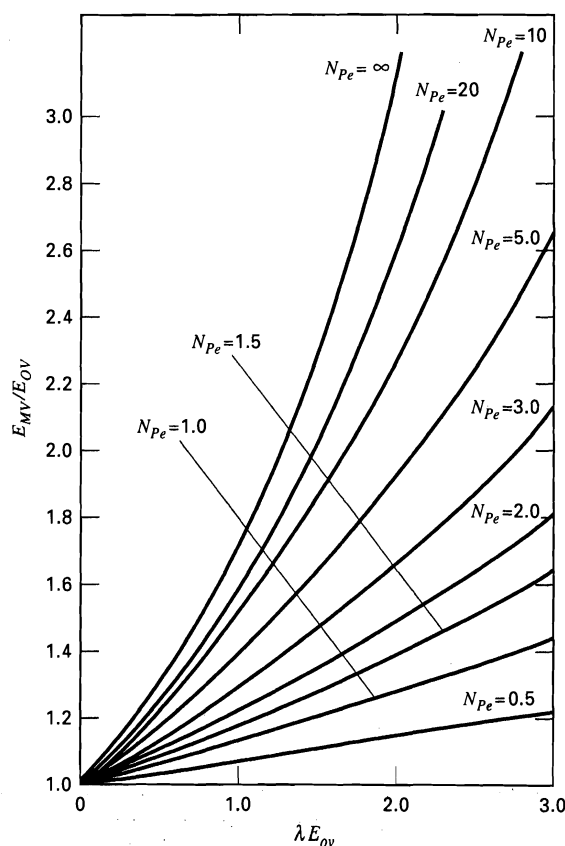


Figure 6.19 Effect of longitudinal mixing on Murphree vapor tray efficiency.

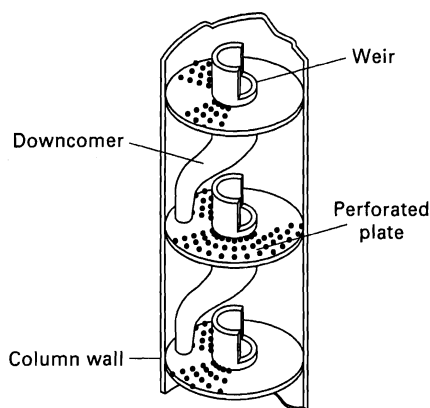


Figure 6.20 Oldershaw column.

and shown schematically in Figure 6.20. Oldershaw columns are typically 1 to 2 in. in diameter and can be assembled with almost any number of sieve plates, usually containing 0.035- to 0.043-in. holes with a hole area of approximately 10%. A detailed study by Fair, Null, and Bolles [23] showed that overall plate efficiencies of Oldershaw columns operated over a pressure range of 3 to 165 psia are in conservative agreement with distillation data obtained from sieve-tray, pilot-plant and industrial-size columns ranging in size from 18 in. to 4 ft in diameter when operated in the range of 40% to 90% of flooding. It may be assumed that similar agreement might be realized for absorption and stripping.

It is believed that the small-diameter Oldershaw column achieves essentially complete mixing of liquid on each tray, thus permitting the measurement of a point efficiency. As discussed above, somewhat larger efficiencies may be observed in much-larger-diameter columns due to incomplete liquid mixing, which results in a higher Murphree tray efficiency and, therefore, higher overall plate efficiency.

Fair et al. [23] recommend the following conservative scale-up procedure for the Oldershaw column:

1. Determine the flooding point.
2. Establish operation at about 60% of flooding (but 40 to 90% seems acceptable).
3. Run the system to find a combination of plates and flow rates that gives the desired degree of separation.
4. Assume that the commercial column will require the same number of plates for the same ratio of liquid to vapor molar flow rates.

If reliable vapor-liquid equilibrium data are available, they can be used with the Oldershaw data to determine the overall column efficiency, E_o . Then (6-37) and (6-34) can be used to estimate the average point efficiency. For the commercial-size column, the Murphree vapor efficiency can be determined from the Oldershaw column point efficiency using (6-34), which takes into account incomplete liquid mixing. In general, the tray efficiency of the commercial column, depending on the length of the liquid flow path, will be higher than for the Oldershaw column at the same percentage of flooding.

EXAMPLE 6.4

Assume that the column diameter for the absorption operation of Example 6.1 is 3 ft. If the overall stage efficiency, E_o , is 30% for the absorption of ethyl alcohol, estimate the average Murphree vapor efficiency, E_{MV} , and the possible range of the Murphree vapor-point efficiency, E_{OV} .

SOLUTION

For Example 6.1, the system is dilute in ethyl alcohol, the main component undergoing mass transfer. Therefore, the equilibrium and operating lines are essentially straight, and (6-37) can be applied. From the data of Example 6.1, $\lambda = KV/L = 0.57(180)/151.5 = 0.68$.

Solving (6-37) for E_{MV} , using $E_o = 0.30$,

$$E_{MV} = (\lambda^{E_o} - 1)/(\lambda - 1) = (0.68^{0.30} - 1)/(0.68 - 1) = 0.34$$

For a 3-ft-diameter column, the degree of liquid mixing probably lies intermediate between complete mixing and plug flow. From (6-31) for the former case, $E_{OV} = E_{MV} = 0.34$. From a rearrangement of (6-32) for the latter case, $E_{OV} = \ln(1 + \lambda E_{MV})/\lambda = \ln[1 + 0.68(0.34)]/0.68 = 0.31$. Therefore, E_{OV} lies in the range of 31% to 34%, probably closer to 34% for complete mixing. However, the differences between E_o , E_{MV} , and E_{OV} for this example are almost negligible.

6.6 TRAY DIAMETER, PRESSURE DROP, AND MASS TRANSFER

In the trayed tower shown in Figure 6.21, vapor flows vertically upward, contacting liquid in crossflow on each tray. When trays are designed properly, a stable operation is achieved wherein (1) vapor flows only through the perforations or open regions of the tray between the downcomers, (2) liquid flows from tray to tray only by means of the downcomers, (3) liquid neither weeps through the tray perforations nor is carried by the vapor as entrainment to the tray above, and (4) vapor is neither carried (occluded) down by the liquid in the downcomer to the tray below nor allowed to bubble up through the liquid in the downcomer. Tray design includes the determination of tray diameter and the division of the tray cross-sectional area, A , as shown in Figure 6.21, into active vapor bubbling area, A_a , and liquid downcomer area, A_d . With the tray diameter fixed, vapor pressure drop and mass-transfer coefficients can be estimated.

Tray Diameter

For a given liquid flow rate, as shown in Figure 6.22 for a sieve-tray column, a maximum vapor flow rate exists beyond which incipient column flooding occurs because of backup of liquid in the downcomer. This condition, if sustained, leads to carryout of liquid with the overhead vapor leaving the column. *Downcomer flooding* takes place when, in the absence of entrainment, liquid backup is caused by downcomers of inadequate cross-sectional area, A_d , to carry

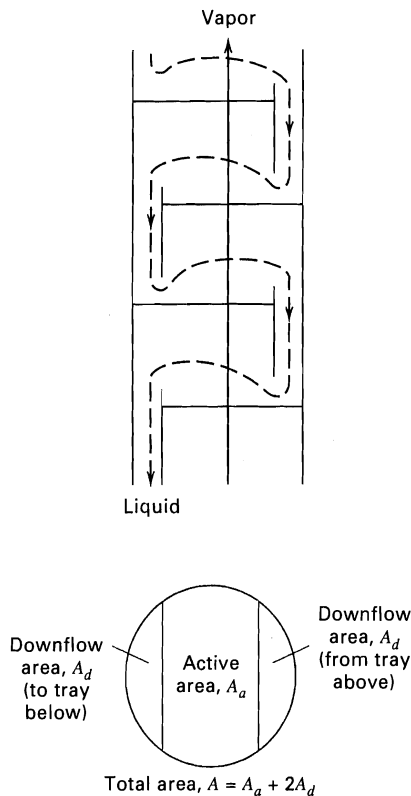


Figure 6.21 Vapor and liquid flow through a trayed tower.

the liquid flow. It rarely occurs if downcomer cross-sectional area is at least 10% of total column cross-sectional area and if tray spacing is at least 24 in. The usual design limit is *entrainment flooding*, which is caused by excessive carry-up of liquid, at the rate e , by vapor entrainment to the tray above. At incipient flooding, $(e + L) \gg L$ and downcomer cross-sectional area is inadequate for the excessive liquid load $(e + L)$. Tray diameter is determined as follows to avoid entrainment flooding.

Entrainment of liquid is due to carry-up of suspended droplets by rising vapor or to throw-up of liquid particles by vapor jets formed at tray perforations, valves, or bubble-cap slots. Souders and Brown [24] successfully correlated

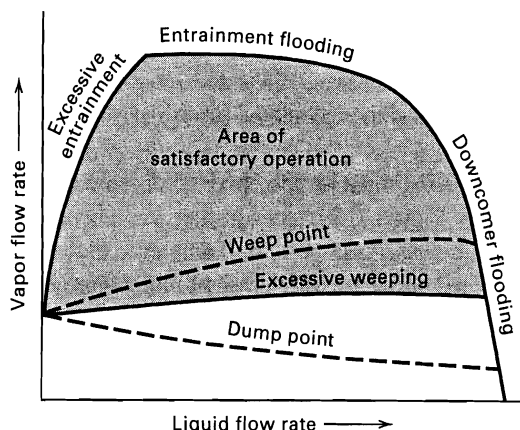


Figure 6.22 Limits of stable operation in a trayed tower. [Reproduced by permission from H.Z. Kister, *Distillation Design*, McGraw-Hill, New York (1992).]

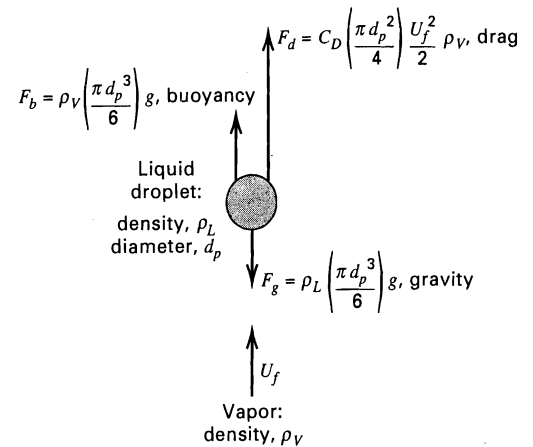


Figure 6.23 Forces acting on a suspended liquid droplet.

entrainment flooding data for 10 commercial trayed columns by assuming that carry-up of suspended droplets controls entrainment. At low vapor velocity, a droplet settles out; at high vapor velocity, it is entrained. At flooding or incipient entrainment velocity, U_f , the droplet is suspended such that the vector sum of the gravitational, buoyant, and drag forces acting on the droplet, as shown in Figure 6.23, are zero. Thus,

$$\sum F = 0 = F_g - F_b - F_d \quad (6-38)$$

In terms of droplet diameter, d_p , terms on the right-hand side of (6-38) become, respectively,

$$\rho_L \left(\frac{\pi d_p^3}{6} \right) g - \rho_v \left(\frac{\pi d_p^3}{6} \right) g - C_D \left(\frac{\pi d_p^2}{4} \right) \frac{U_f^2}{2} \rho_v = 0 \quad (6-39)$$

where C_D is the drag coefficient. Solving for flooding velocity,

$$U_f = C \left(\frac{\rho_L - \rho_v}{\rho_v} \right)^{1/2} \quad (6-40)$$

where C = capacity parameter of Souders and Brown. According to the above theory,

$$C = \left(\frac{4d_p g}{3C_D} \right)^{1/2} \quad (6-41)$$

Parameter C can be calculated from (6-41) if the droplet diameter d_p is known. In practice, however, d_p is distributed over a wide range and C is treated as an empirical parameter determined using experimental data obtained from operating equipment. Souders and Brown considered all the important variables that could influence the value of C and obtained a correlation for commercial-size columns with bubble-cap trays. Data covered column pressures from 10 mmHg to 465 psia, plate spacings from 12 to 30 in., and liquid surface tensions from 9 to 60 dyne/cm. In accordance with (6-41), the value of C increases with increasing surface tension, which increases d_p . Also, C increases with increasing tray spacing, since this allows more time for agglomeration to a larger d_p .

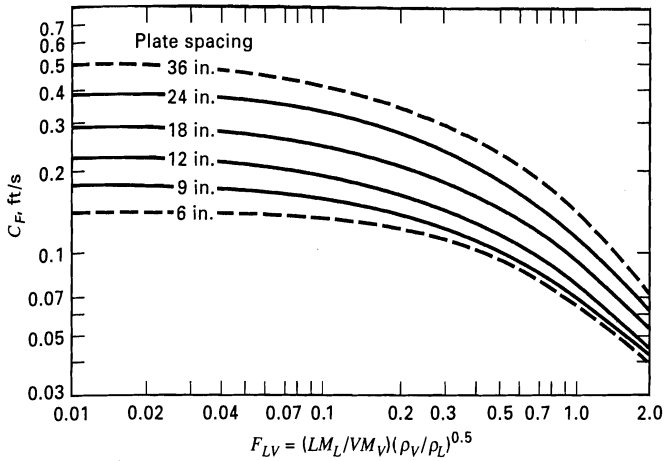


Figure 6.24 Entrainment flooding capacity in a trayed tower.

Using additional commercial operating data, Fair [25] produced the more general correlation of Figure 6.24, which is applicable to columns with bubble cap and sieve trays. Whereas Souders and Brown base the vapor velocity on the entire column cross-sectional area, Fair utilizes a net vapor flow area equal to the total inside column cross-sectional area minus the area blocked off by the downcomer, that is, $A - A_d$ in Figure 6.21. The value of C_F in Figure 6.24 depends on tray spacing and the abscissa ratio $F_{LV} = (LM_L/VM_V)(\rho_V/\rho_L)^{0.5}$ (where flow rates are in molar units), which is a kinetic energy ratio first used by Sherwood, Shipley, and Holloway [26] to correlate packed-column flooding data. The value of C in (6-41) is obtained from Figure 6.24 by correcting C_F for surface tension, foaming tendency, and the ratio of vapor hole area A_h to tray active area A_a , according to the empirical relationship

$$C = F_{ST} F_F F_{HA} C_F \quad (6-42)$$

where

$$F_{ST} = \text{surface tension factor} = (\sigma/20)^{0.2}$$

$$F_F = \text{foaming factor}$$

$$F_{HA} = 1.0 \text{ for } A_h/A_a \geq 0.10 \text{ and } 5(A_h/A_a) + 0.5 \text{ for } 0.06 \leq A_h/A_a \leq 0.1$$

$$\sigma = \text{liquid surface tension, dyne/cm}$$

For nonfoaming systems, $F_F = 1.0$; for many absorbers, F_F may be 0.75 or even less. The quantity A_h is the area open to the vapor as it penetrates into the liquid on a tray. It is the total cap slot area for bubble-cap trays and the perforated area for sieve trays.

Figure 6.24 appears to be conservative for valve trays. This is shown in Figure 6.25, where entrainment-flooding data of Fractionation Research, Inc. (FRI) [27,28], for a 4-ft-diameter column equipped with Glitsch type A-1 and V-1 valve trays on 24-in. spacing are compared to the correlation in Figure 6.24. For valve trays, the slot area A_h is taken as the full valve opening through which vapor enters the frothy liquid on the tray at a 90° angle with the axis of the column.

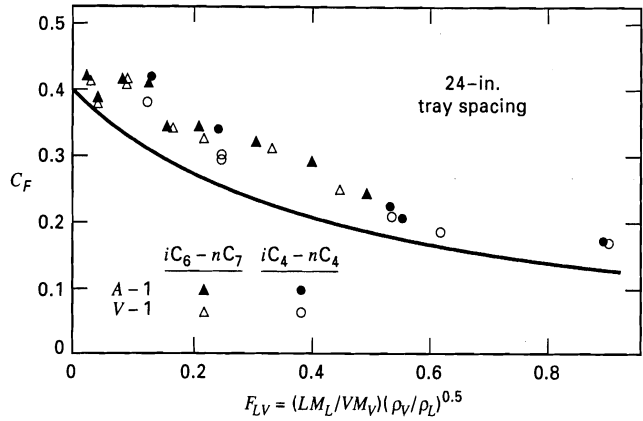


Figure 6.25 Comparison of flooding correlation with data for valve trays.

Column diameter D_T is based on a fraction, f , of flooding velocity U_f , which is calculated from (6-40), using C from (6-42), based on C_F from Figure 6.24. By the continuity equation from fluid mechanics (flow rate = velocity $\times$ flow area $\times$ density), the molar vapor flow rate is related to the flooding velocity by

$$V = (fU_f)(A - A_d) \frac{\rho_V}{M_V} \quad (6-43)$$

where A = total column cross-sectional area $= \pi D_T^2/4$. Thus,

$$D_T = \left[\frac{4VM_V}{fU_f\pi(1 - A_d/A)\rho_V} \right]^{0.5} \quad (6-44)$$

Typically, the fraction of flooding, f , is taken as 0.80.

Oliver [29] suggests that A_d/A be estimated from F_{LV} in Figure 6.24 by

$$\frac{A_d}{A} = \begin{cases} 0.1, & F_{LV} \leq 0.1 \\ 0.1 + \frac{(F_{LV} - 0.1)}{9}, & 0.1 \leq F_{LV} \leq 1.0 \\ 0.2, & F_{LV} \geq 1.0 \end{cases}$$

Column diameter is calculated at both the top and bottom of the column, with the larger of the two diameters used for the entire column unless the two diameters differ appreciably. Because of the need for internal access to columns with trays, a packed column, discussed later in this chapter, is generally used if the calculated diameter from (6-44) is less than 2 ft.

Tray spacing must be specified to compute column diameter using Figure 6.24. As spacing is increased, column height is increased but column diameter is reduced. A spacing of 24 in., which provides ease of maintenance, is optimal for a wide range of conditions; however, a smaller spacing may be desirable for small-diameter columns with a large number of stages; and larger spacing is frequently used for large-diameter columns with a small number of stages.

As shown in Figure 6.22, a minimum vapor rate exists below which liquid weeps (dumps) through tray perforations or risers instead of flowing completely across the active area

and into the downcomer. Below this minimum, the degree of contacting of liquid with vapor is reduced, causing tray efficiency to decline. The ratio of the vapor rate at flooding to the minimum vapor rate is the *turn-down ratio*, which is approximately 8 for bubble-cap trays, 5 for valve trays, but only about 2 for sieve trays.

When vapor and liquid flow rates change appreciably from tray to tray, column diameter, tray spacing, or hole area can be varied to reduce column cost and ensure stable operation at high tray efficiency. Variation of tray spacing is particularly applicable to columns with sieve trays because of their low turn-down ratio.

High-Capacity Trays

Since the 1990s, a number of high-capacity trays have been introduced and installed in hundreds of columns. By various changes to the conventional tray design shown in Figure 6.3, capacity increases of more than 20% of that predicted by Figure 6.24 have been achieved with both perforated trays and valve trays. These changes, which are discussed in some detail by Sloley [71], have included:

1. Sloping or stepping of the downcomer to make the downcomer area smaller at the bottom than at the top so as to increase the active flow area.
2. Vapor flow through that portion of the tray located beneath the downcomer, in addition to the normal active flow area.
3. Use of staggered, louvered downcomer floor plates to impart a horizontal velocity to the liquid exiting the downcomer, thus enhancing the ability to allow vapor flow beneath the downcomer.
4. Elimination of vapor impingement from adjacent valves, in valve trays, by using bi-directional fixed valves.
5. Use of multiple-downcomer trays that provide very long outlet weirs leading to low crest heights and lower froth heights. The downcomers terminate in the active vapor space of the tray below.
6. Directional slotting of sieve trays to impart a horizontal component to the up-flowing vapor, enhance plug flow of liquid across the tray, and eliminate dead areas.

Regardless of the tray design, as shown by Stupin and Kister [72], an ultimate capacity, independent of tray spacing, exists for a countercurrent-flow, vapor-liquid contactor, in which the superficial vapor velocity in the column exceeds the settling velocity of large liquid droplets. Their correlation, based on FRI data, uses the following form of (6-40):

$$U_{S,ult} = C_{S,ult} \left(\frac{\rho_L - \rho_V}{\rho_V} \right)^{1/2} \quad (6-45)$$

where $U_{S,ult}$ is the superficial vapor velocity in m/s based on the column cross-sectional area. The ultimate capacity

parameter, $C_{S,ult}$ in m/s, is independent of the superficial liquid velocity, L_S in m/s, below a critical value; but above that value it decreases with increasing L_S . It is given by the smaller of C_1 and C_2 , both in m/s, where

$$C_1 = 0.445(1 - F) \left(\frac{\sigma}{\rho_L - \rho_V} \right)^{0.25} - 1.4L_S \quad (6-46)$$

$$C_2 = 0.356(1 - F) \left(\frac{\sigma}{\rho_L - \rho_V} \right)^{0.25} \quad (6-47)$$

where

$$F = \frac{1}{\left[1 + 1.4 \left(\frac{\rho_L - \rho_V}{\rho_V} \right)^{1/2} \right]} \quad (6-48)$$

ρ is in kg/m^3 and σ is the surface tension in dynes/cm.

EXAMPLE 6.5

(a) Estimate the required tray diameter for the absorber of Example 6.1, assuming a tray spacing of 24 in., a foaming factor of $F_F = 0.90$, a fraction flooding of $f = 0.80$, and a surface tension of $\sigma = 70$ dynes/cm. (b) Estimate the ultimate superficial vapor velocity.

SOLUTION

Because tower conditions are almost the same at the top and bottom, the calculation of column diameter is made only at the bottom, where the gas rate is highest. From Example 6.1,

$$T = 30^\circ\text{C} \quad P = 110 \text{ kPa}$$

$$V = 180 \text{ kmol/h}, \quad L = 151.5 + 3.5 = 155.0 \text{ kmol/h}$$

$$M_V = 0.98(44) + 0.02(46) = 44.0,$$

$$M_L = \frac{151.5(18) + 3.5(46)}{155} = 18.6$$

$$\rho_V = \frac{PM}{RT} = \frac{(110)(44)}{(8.314)(303)} = 1.92 \text{ kg/m}^3,$$

$$\rho_L = (0.986)(1,000) = 986 \text{ kg/m}^3$$

$$F_{LV} = \frac{(155)(18.6)}{(180)(44.0)} \left(\frac{1.92}{986} \right)^{0.5} = 0.016$$

(a) For tray spacing = 24 in., from Figure 6.24, $C_F = 0.39$ ft/s,

$$F_{ST} = \left(\frac{\sigma}{20} \right)^{0.2} = \left(\frac{70}{20} \right)^{0.2} = 1.285, \quad F_F = 0.90$$

Because $F_{LV} < 0.1$, $A_d/A = 0.1$ and $F_{HA} = 1.0$. Then, from (6-42),

$$C = 1.285(0.90)(1.0)(0.39) = 0.45 \text{ ft/s}$$

From (6-40),

$$U_f = 0.45 \left(\frac{986 - 1.92}{1.92} \right)^{0.5} = 10.2 \text{ ft/s}$$

From (6-44), using SI units and time in seconds,

$$D_T = \left[\frac{4(180/3,600)(44.0)}{(0.80)(10.2/3.28)(3.14)(1-0.1)(1.92)} \right]^{0.5} \\ = 0.81 \text{ m} = 2.65 \text{ ft}$$

(b) From (6-48),

$$F = \frac{1}{\left[1 + 1.4 \left(\frac{986 - 1.92}{1.92} \right)^{1/2} \right]} = 0.0306$$

From (6-47),

$$C_2 = 0.356(1 - 0.0306) \left(\frac{70}{986 - 1.92} \right)^{0.25} \\ = 0.178 \text{ m/s} = 0.584 \text{ ft/s}$$

If C_2 is the smaller value of C_1 and C_2 , then from (6-45),

$$U_{S,ult} = 0.178 \left(\frac{986 - 1.92}{1.92} \right)^{1/2} = 4.03 \text{ m/s} = 13.22 \text{ ft/s}$$

To apply (6-46) to compute C_1 , the value of L_S is required. This value is related as follows to the value of the superficial vapor velocity, U_S .

$$L_S = U_S \frac{\rho_V L M_L}{\rho_L V M_V} = U_S \frac{(1.92)(155)(18.6)}{(986)(180)(44.0)} = 0.000709 U_S$$

With this expression for L_S , (6-46) becomes

$$C_1 = 0.445(1 - 0.0306) \left(\frac{70}{986 - 1.92} \right)^{0.25} - 1.4(0.000709) U_S \\ = 0.223 - 0.000993 U_S, \text{ m/s}$$

If C_1 is the smaller, then, using (6-45),

$$U_{S,ult} = (0.223 - 0.000993 U_{S,ult}) \left(\frac{986 - 1.92}{1.92} \right)^{1/2} \\ = 5.05 - 0.0225 U_{S,ult}$$

Solving, $U_{S,ult} = 4.94 \text{ m/s}$, which gives $C_1 = 0.223 - 0.000993(4.94) = 0.218 \text{ m/s}$. Thus, C_2 is the smaller value and $U_{S,ult} = 4.03 \text{ m/s} = 13.22 \text{ ft/s}$. This ultimate velocity is 30% higher than the flooding velocity computed in part (a).

Tray Vapor Pressure Drop

Typical tray pressure drop for flow of vapor in a tower is from 0.05 to 0.15 psi/tray. Referring to Figure 6.3, pressure drop (head loss) for a sieve tray is due to friction for vapor flow through the tray perforations, holdup of the liquid on the tray, and a loss due to surface tension:

$$h_t = h_d + h_l + h_\sigma \quad (6-49)$$

where

h_t = total pressure drop/tray, in. of liquid

h_d = dry tray pressure drop, in. of liquid

h_l = equivalent head of clear liquid on tray, in. of liquid

h_σ = pressure drop due to surface tension, in. of liquid

The dry sieve-tray pressure drop is given by a modified orifice equation, applied to the holes in the tray,

$$h_d = 0.186 \left(\frac{u_o^2}{C_o^2} \right) \left(\frac{\rho_V}{\rho_L} \right) \quad (6-50)$$

where u_o = hole velocity (ft/s) and C_o depends on the percent hole area and the ratio of tray thickness to hole diameter. For a typical 0.078-in.-thick tray with $\frac{3}{16}$ -in.-diameter holes and a percent hole area (based on the cross-sectional area of the tower) of 10%, C_o may be taken as 0.73. Otherwise, C_o lies between about 0.65 and 0.85.

The equivalent height of clear liquid holdup on a tray depends on weir height, liquid and vapor densities and flow rates, and downcomer weir length, as given by the following empirical expression developed from experimental data by Bennett, Agrawal, and Cook [30]:

$$h_l = \phi_e \left[h_w + C_l \left(\frac{q_L}{L_w \phi_e} \right)^{2/3} \right] \quad (6-51)$$

where

h_w = weir height, in.

ϕ_e = effective relative froth density (height of clear liquid/froth height) (6-52)

$$= \exp(-4.257 K_s^{0.91})$$

K_s = capacity parameter, ft/s = $U_a \left(\frac{\rho_V}{\rho_L - \rho_V} \right)^{1/2}$ (6-53)

U_a = superficial vapor velocity based on active bubbling area,

$A_a = (A - 2A_d)$, of the tray, ft/s,

L_w = weir length, in.

q_L = liquid flow rate across tray, gal/min

$$C_l = 0.362 + 0.317 \exp(-3.5h_w) \quad (6-54)$$

The second term in (6-51) is related to the Francis weir equation for a straight segmental weir, taking into account the froth nature of the liquid flow over the weir. For $A_d/A = 0.1$, $L_w = 73\%$ of the tower diameter.

As the gas emerges from the tray perforations, the bubbles must overcome surface tension. The pressure drop due to surface tension is given by the difference between the pressure inside the bubble and that of the liquid, according to the theoretical relation

$$h_\sigma = \frac{6\sigma}{g\rho_L D_{B(\max)}} \quad (6-55)$$

where, except for tray perforations much smaller than $\frac{3}{16}$ -in. in diameter, $D_{B(\max)}$, the maximum bubble size, may be taken as the perforation diameter, D_H .

Methods for estimating vapor pressure drop for bubble-cap trays and valve trays are given by Smith [31] and Klein [32], respectively, and are discussed by Kister [33] and Lockett [34].

EXAMPLE 6.6

Estimate the tray vapor pressure drop for the absorber of Example 6.1, assuming use of sieve trays with a tray diameter of 1 m, a weir height of 2 in., and a hole diameter of $\frac{3}{16}$ in.

SOLUTION

From Example 6.5,

$$\rho_V = 1.92 \text{ kg/m}^3 \quad \rho_L = 986 \text{ kg/m}^3$$

At the bottom of the tower, vapor velocity based on the total cross-sectional area of the tower is

$$\frac{(180/3,600)(44)}{(1.92)[3.14(1)^2/4]} = 1.46 \text{ m/s}$$

For a 10% hole area, based on the total cross-sectional area of the tower,

$$u_o = \frac{1.46}{0.10} = 14.6 \text{ m/s} \quad \text{or} \quad 47.9 \text{ ft/s}$$

Using the above densities, (6-50) gives

$$h_d = 0.186 \left(\frac{47.9^2}{0.732} \right) \left(\frac{1.92}{986} \right) = 1.56 \text{ in. of liquid}$$

Take weir length as 73% of tower diameter, with $A_d/A = 0.10$. Then

$$L_w = 0.73(1) = 0.73 \text{ m} \quad \text{or} \quad 28.7 \text{ in.}$$

$$\text{Liquid flow rate in gpm} = \frac{(155/60)(18.6)}{986(0.003785)} = 12.9 \text{ gpm}$$

$$\text{with} \quad A_d/A = 0.1 \quad A_a/A = (A - 2A_d)/A = 0.8$$

$$\text{Therefore, } U_a = 1.46/0.8 = 1.83 \text{ m/s} = 5.99 \text{ ft/s}$$

From (6-53),

$$K_s = 5.99[1.92/(986 - 1.92)]^{0.5} = 0.265 \text{ ft/s}$$

From (6-52),

$$\phi_e = \exp[-4.257(0.265)^{0.91}] = 0.28$$

From (6-54),

$$C_l = 0.362 + 0.317 \exp[-3.5(2)] = 0.362$$

From (6-51),

$$\begin{aligned} h_l &= 0.28[2 + 0.362(12.9/28.7/0.28)^{2/3}] \\ &= 0.28(2 + 0.50) = 0.70 \text{ in.} \end{aligned}$$

From (6-55), in metric units, using $D_{B(\max)} = D_H = \frac{3}{16}$ in. = 0.00476 m,

$$\sigma = 70 \text{ dynes/cm} = 0.07 \text{ N/m} = 0.07 \text{ kg/s}^2,$$

$$g = 9.8 \text{ m/s}^2, \text{ and } \rho_L = 986 \text{ kg/m}^3$$

$$h_\sigma = \frac{6(0.07)}{9.8(986)(0.00476)} = 0.00913 \text{ m} = 0.36 \text{ in.}$$

From (6-45), the total tray head loss is $h_t = 1.56 + 0.70 + 0.36 = 2.62 \text{ in.}$

$$\text{For } \rho_L = 986 \text{ kg/m}^3 = 0.0356 \text{ lb/in}^3,$$

$$\text{tray vapor pressure drop} = h_t \rho_L = 2.62(0.0356) = 0.093 \text{ psi/tray}$$

Mass-Transfer Coefficients and Transfer Units

Following the determination of tower diameter and major details of the tray layout, an estimate of the Murphree vapor point efficiency, defined by (6-30), can be made using empirical correlations for mass-transfer coefficients, based on experimental data. For a vertical path for vapor flow up through the froth from a point on the bubbling area of the tray, (6-29) applies to the Murphree vapor-point efficiency:

$$N_{OG} = -\ln(1 - E_{OV}) \quad (6-56)$$

where

$$N_{OG} = \frac{K_G a P Z_f}{(V/A_b)} \quad (6-57)$$

The overall, volumetric mass-transfer coefficient, $K_G a$, is related to the individual volumetric mass-transfer coefficients by the sum of the mass-transfer resistances, which from equations in Section 3.7 can be shown to be

$$\frac{1}{K_G a} = \frac{1}{k_G a} + \frac{(K P M_L / \rho_L)}{k_L a} \quad (6-58)$$

where the two terms on the right-hand side are the gas- and liquid-phase resistances, respectively, and the symbols k_p for the gas and k_c for the liquid used in Chapter 3 have been replaced by k_G and k_L , respectively. In terms of individual transfer units, defined by

$$N_G = \frac{k_G a P Z_f}{(V/A_b)} \quad (6-59)$$

and

$$N_L = \frac{k_L a \rho_L Z_f}{M_L (L/A_b)} \quad (6-60)$$

we obtain from (6-57) and (6-58)

$$\frac{1}{N_{OG}} = \frac{1}{N_G} + \frac{(K V / L)}{N_L} \quad (6-61)$$

Important empirical mass-transfer correlations have been published by the AIChE [35] for bubble-cap trays, Chan and Fair [36, 37] for sieve trays, and Scheffe and Weiland [38] for one type of valve tray (Glitsch V-1). These correlations have been developed in terms of N_L , N_G , k_L , k_G , a , and N_{Sh} for either the gas or liquid phase. In this section, we present only correlations for sieve trays, as given for binary systems by Chan and Fair [36], who used a correlation for the liquid phase based on the work of Foss and Gerster [39] as reported by the AIChE [40], and who developed a correlation for the vapor phase from a fairly extensive experimental data bank of 143 points for towers 2.5 to 4.0 ft in diameter, operating at pressures from 100 mmHg to 400 psia.

Experimental data for sieve trays have validated the assumed direct dependence of mass transfer on the interfacial area between the gas and liquid phases, and on the residence times in the froth of the gas and liquid phases. Accordingly,

Chan and Fair give the following modifications of (6-59) and (6-60):

$$N_G = k_G \bar{a} \bar{t}_G \quad (6-62)$$

$$N_L = k_L \bar{a} \bar{t}_L \quad (6-63)$$

where $\bar{a}$ is the interfacial area per unit volume of equivalent clear liquid, $\bar{t}_G$ is the average residence time of the gas in the froth, and $\bar{t}_L$ is the average residence time of the liquid in the froth.

Average residence times are estimated from the following dimensionally consistent, theoretical continuity equations, using (6-51) for the equivalent head of clear liquid on the tray and (6-52) for the effective relative density of the froth:

$$\bar{t}_L = \frac{h_l A_a}{q_L} \quad (6-64)$$

and

$$\bar{t}_G = \frac{(1 - \phi_e) h_l}{(\phi_e U_a)} \quad (6-65)$$

where $(1 - \phi_e) h_l / \phi_e$ is the equivalent height of vapor holdup in the froth, and the residence times are usually computed in seconds.

Empirical expressions for $k_G \bar{a}$ and $k_L \bar{a}$ in units of s^{-1} are

$$k_G \bar{a} = \frac{1,030 D_V^{0.5} (f - 0.842 f^2)}{(h_l)^{0.5}} \quad (6-66)$$

and

$$k_L \bar{a} = 78.8 D_L^{0.5} (F + 0.425) \quad (6-67)$$

where the variables and their units are

D_V, D_L = diffusion coefficients, cm^2/s

h_l = clear liquid height, cm

$f = U_a/U_f$, fractional approach to flooding

$F = F\text{-factor} = U_a \rho_V^{0.5}$, $(kg/m)^{0.5}/s$

From (6-66), it is seen that an important factor influencing the value of $k_G \bar{a}$ is the fractional approach to flooding, $f = U_a/U_f$. This effect is shown in Figure 6.26, where (6-66) is compared to experimental data. At gas rates corresponding to a fractional approach to flooding of greater than 0.60, the mass-transfer factor decreases with increasing value of f . This may be due to entrainment, which is discussed in the next sub-section. On an entrainment-free basis, the curve in Figure 6.26 might be expected to at least remain at its peak value for conditions above $f = 0.60$.

From (6-67), it is seen that the F -factor is an important consideration for liquid-phase mass transfer. Experimental data that support this are shown in Figure 6.27, where $k_L \bar{a}$ depends strongly on F but is almost independent of liquid flow rate and weir height. The Murphree vapor-point efficiency model of (6-56), (6-61), (6-64), (6-65), (6-66), and (6-67) correlates the 143 points of the Chan and Fair [36] data bank with an average absolute deviation of 6.27%. Lockett [34] pointed out that (6-67) implies that $k_L \bar{a}$ depends on tray spacing, which seems unreasonable.

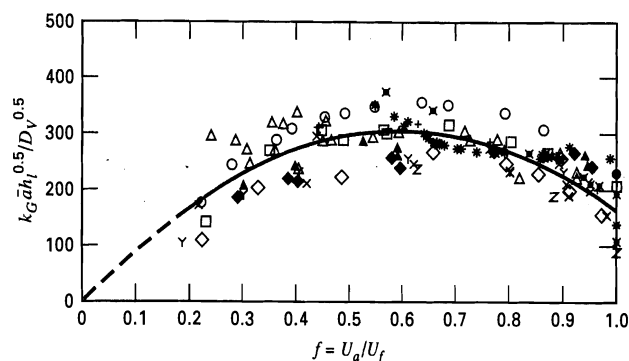


Figure 6.26 Comparison of experimental data to the correlation of Chan and Fair for gas-phase mass transfer.

[From H. Chan and J.R. Fair, *Ind. Eng. Chem. Process Des. Dev.*, **23**, 817 (1984) with permission.]

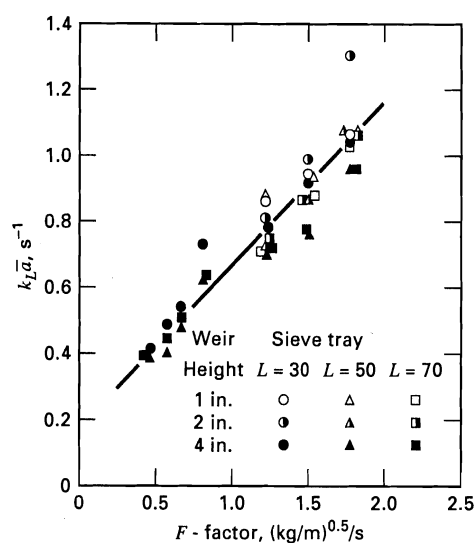


Figure 6.27 Effect of the F -factor on the liquid-phase volumetric mass-transfer coefficient for desorption of oxygen from water with air at 1 atm. and 25°C, where $L = gal/(min)/(ft \text{ of average flow width})$.

However, the data bank did include data for tray spacings from 6 to 24 in.

EXAMPLE 6.7

Estimate the Murphree vapor-point efficiency for the absorber of Example 6.1, using results from Examples 6.5 and 6.6, for the tray of Example 6.6. In addition, determine the controlling resistance to mass transfer.

SOLUTION

Pertinent data for the two phases are as follows.

	Gas	Liquid
Molar flow rate, kmol/h	180.0	155.0
Molecular weight	44.0	18.6
Density, kg/m^3	1.92	986
Ethanol diffusivity, cm^2/s	7.86×10^{-2}	1.81×10^{-5}

Pertinent tray dimensions from Example 6.6 are $D_T = 1$ m, and $A = 0.785$ m²; $A_a = 0.80$, $A = 0.628$ m² = 6,280 cm²; $L_w = 28.7$ in. = 0.73 m.

From Example 6.6,

$$\phi_e = 0.28; \quad h_l = 0.70 \text{ in.} = 1.78 \text{ cm}; \\ U_a = 5.99 \text{ ft/s} = 183 \text{ cm/s} = 1.83 \text{ m/s}$$

From Example 6.5,

$$U_f = 10.2 \text{ ft/s}; \quad f = U_a/U_f = 5.99/10.2 = 0.59 \\ F = 1.83(1.92)^{0.5} = 2.54 \text{ (kg/m)}^{0.5}/\text{s} \\ q_L = \frac{(155.0)(18.6)}{986} \left(\frac{10^6}{3,600} \right) = 812 \text{ cm}^3/\text{s}$$

From (6-64),

$$\bar{t}_L = (1.78)(6,280)/812 = 13.8 \text{ s}$$

From (6-65),

$$\bar{t}_G = (1 - 0.28)(1.78)/[(0.28)(183)] = 0.025 \text{ s}$$

From (6-67),

$$k_L \bar{a} = 78.8(1.81 \times 10^{-5})^{0.5}(2.54 + 0.425) = 0.99 \text{ s}^{-1}$$

From (6-66),

$$k_G \bar{a} = 1,030(7.86 \times 10^{-2})^{0.5}[0.59 - 0.842(0.59)^2]/(1.78)^{0.5} \\ = 64.3 \text{ s}^{-1}$$

From (6-63),

$$N_L = (0.99)(13.8) = 13.7$$

From (6-62),

$$N_G = (64.3)(0.025) = 1.61$$

From Example 6.1, $K = 0.57$. Therefore, $KV/L = (0.57)(180)/155 = 0.662$.

From (6-61),

$$N_{OG} = \frac{1}{(1/1.61) + (0.662/13.7)} = \frac{1}{0.621 + 0.048} = 1.49$$

and the mass transfer of ethanol is seen to be controlled by the vapor-phase resistance. From (6-56), solving for E_{OV} ,

$$E_{OV} = 1 - \exp(-N_{OG}) = 1 - \exp(-1.49) = 0.77 = 77\%$$

Weeping, Entrainment, and Downcomer Backup

For a tray to operate at high efficiency, (1) weeping of liquid through the tray perforations must be small compared to flow over the outlet weir and into the downcomer, (2) entrainment of liquid by the gas must not be excessive, and (3) to prevent downcomer flooding, froth height in the downcomer must not approach tray spacing. The tray must operate in the stable region shown schematically in Figure 6.22. Weeping is associated with the lower limit of gas velocity, while entrainment flooding is associated with the upper limit.

Weeping occurs at low vapor velocities and/or high liquid rates when the clear liquid height on the tray exceeds the sum of the dry (no liquid flow) tray pressure drop, due to vapor flow, and the surface tension effect. Thus, to prevent weeping, it is necessary that

$$h_d + h_\sigma > h_l \quad (6-68)$$

everywhere on the active area of the tray. If weeping occurs uniformly over the tray active area or mainly near the downcomer, a ratio of weep rate to downcomer liquid rate as high as 0.1 may not cause an unacceptable decrease in tray efficiency. Methods for estimating weep rates are discussed by Kister [33].

The prediction of fractional liquid entrainment by the vapor, defined as $\psi = e/(L + e)$, can be made by the correlation of Fair [41], given in Figure 6.28. As shown, entrainment becomes excessive at high values of fraction of flooding, $f = U_a/U_f$, particularly for small values of the kinetic-energy ratio, F_{LV} . The effect of entrainment on the Murphree vapor efficiency can be estimated by the following relation derived by Colburn [42], where E_{MV} is the usual "dry" efficiency and $E_{MV, \text{wet}}$ is the "wet" efficiency:

$$\frac{E_{MV, \text{wet}}}{E_{MV}} = \frac{1}{1 + eE_{MV}/L} \\ = \frac{1}{1 + E_{MV}[\psi/(1 - \psi)]} \quad (6-69)$$

Equation (6-69) assumes that $\lambda = KV/L = 1$ and that the liquid is well mixed on the tray such that the composition of the entrained liquid is that of the liquid flowing to the tray below. For a given value of the entrainment ratio, ψ , the larger the value of E_{MV} , the greater is the effect of entrainment. For $E_{MV} = 1.0$ and $\psi = 0.10$, the "wet" efficiency is 0.90. An equation similar to (6-69) for the effect of weeping

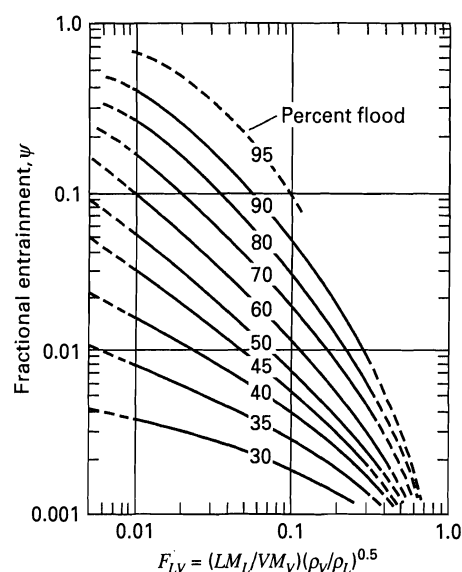


Figure 6.28 Correlation of Fair for fractional entrainment for sieve trays.

[Reproduced by permission from B.D. Smith, *Design of Equilibrium Stage Processes*, McGraw-Hill, New York (1963).]

is not available, because this effect depends greatly on the degree of liquid mixing on the tray and on the distribution of weeping over the active area of the tray. If weeping occurs only in the vicinity of the downcomer, no decrease in the value of E_{MV} is observed.

The height of clear liquid in the downcomer, h_{dc} , is always greater than the height of clear liquid on the tray because, by reference to Figure 6.3, the pressure difference across the froth in the downcomer is equal to the total pressure drop across the tray from which liquid enters the downcomer, plus the height of clear liquid on the tray below to which the liquid flows, and plus the head loss for liquid flow under the downcomer apron. Thus, the clear liquid head in the downcomer is

$$h_{dc} = h_t + h_l + h_{da} \quad (6-70)$$

where h_t is given by (6-49) and h_l by (6-51), and the hydraulic gradient is assumed to be negligible. The head loss for liquid flow under the downcomer, h_{da} , in inches of liquid can be estimated from an empirical orifice-type equation:

$$h_{da} = 0.03 \left(\frac{q_L}{100 A_{da}} \right)^2 \quad (6-71)$$

where q_L is the liquid flow in gpm and A_{da} is the area in ft^2 for liquid flow under the downcomer apron. If the height of the opening under the apron (typically 0.5 in. less than h_w) is h_a , then $A_{da} = L_w h_a$. The height of the froth in the downcomer is

$$h_{df} = h_{dc} / \phi_{df} \quad (6-72)$$

where the froth density, ϕ_{df} , can be taken conservatively as 0.5.

EXAMPLE 6.8

Using data from Examples 6.5, 6.6, and 6.7, estimate the entrainment rate, the froth height in the downcomer, and whether weeping occurs.

SOLUTION

Weeping criterion: From Example 6.6,

$$h_d = 1.56 \text{ in.} \quad h_\sigma = 0.36 \text{ in.} \quad h_l = 0.70 \text{ in.}$$

$$\text{From (6-68), } 1.56 + 0.36 > 0.70$$

Therefore, if the liquid level is uniform across the active area, no weeping occurs.

Entrainment: From Example 6.5,

$$F_{LV} = 0.016 \quad \text{From Example 6.7, } f = 0.59$$

From Figure 6.28, $\psi = 0.06$. Therefore, for $L = 155$ from Example 6.7, the entrainment rate is $0.06(155) = 9.3 \text{ kmol/h}$. Assuming that (6-69) is reasonably accurate for $\lambda = 0.662$ from Example 6.7, and that $E_{MV} = 0.78$, the effect of ψ on E_M is given by

$$\frac{E_{MV, \text{wet}}}{E_{MV}} = \frac{1}{1 + 0.78(0.06/0.94)} = 0.95$$

$$\text{or} \quad E_{MV} = 0.95(0.78) = 0.74$$

Downcomer backup:

From Example 6.6, $h_t = 2.62 \text{ in.}$

From Example 6.7, $L_w = 28.7 \text{ in.}$

From Example 6.6, $h_w = 2.0 \text{ in.}$

Assume that $h_a = 2.0 - 0.5 = 1.5 \text{ in.}$ Then

$$A_{da} = L_w h_a = 28.7(1.5) = 43.1 \text{ in.}^2 = 0.299 \text{ ft}^2$$

From Example 6.6, $q_L = 12.9 \text{ gpm}$

$$\text{From (6-71), } h_{da} = 0.03 \left[\frac{12.9}{(100)(0.299)} \right]^2 = 0.006 \text{ in.}$$

From (6-70), $h_{dc} = 2.62 + 0.70 + 0.006 = 3.33 \text{ in.}$ of clear liquid backup

From (6-72), $h_{df} = \frac{3.33}{0.5} = 6.66 \text{ in.}$ of froth in the downcomer

Based on these results, neither weeping nor downcomer backup appear to be problems. An estimated 5% loss in tray efficiency occurs due to entrainment.

6.7 RATE-BASED METHOD FOR PACKED COLUMNS

Absorption and stripping are frequently conducted in packed columns, particularly when (1) the required column diameter is 2 ft or less; (2) the pressure drop must be low, as for a vacuum service; (3) corrosion considerations favor the use of ceramic or polymeric materials; and/or (4) low liquid holdup is desirable. Structured packing is often favored over random packing for revamps to overcome capacity limitations of trayed towers.

Packed columns are continuous, differential-contacting devices that do not have the physically distinguishable stages found in trayed towers. Thus, packed columns are best analyzed by mass-transfer considerations rather than by the equilibrium-stage concept described in earlier sections of this chapter for trayed towers. Nevertheless, in practice, packed-tower performance is often analyzed on the basis of equivalent equilibrium stages using a packed height equivalent to a theoretical (equilibrium) plate (stage), called the HETP or HETS and defined by the equation

$$\text{HETP} = \frac{\text{packed height}}{\text{number of equivalent equilibrium stages}} = \frac{l_T}{N_t} \quad (6-73)$$

The HETP concept, unfortunately, has no theoretical basis. Accordingly, although HETP values can be related to mass-transfer coefficients, such values are best obtained by back-calculation from (6-73) using experimental data from laboratory or commercial-size columns. To illustrate the application of the HETP concept, consider Example 6.1, which involves the recovery of ethyl alcohol from a CO_2 -rich vapor by absorption with water. The required number of equilibrium stages is found to be just slightly more than 6, say, 6.1. Suppose that experience shows that if 1.5-in. metal Pall rings are used in a packed tower, an average HETP of

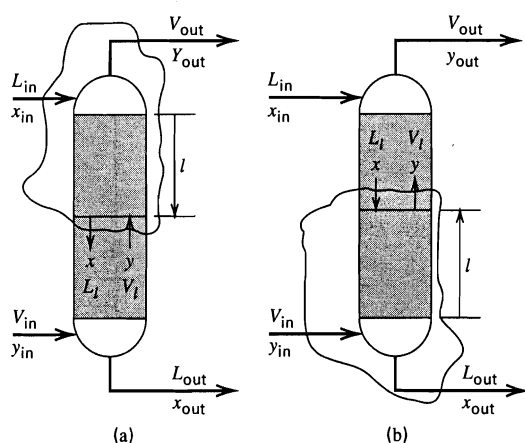


Figure 6.29 Packed columns with countercurrent flow: (a) absorber; (b) stripper.

2.25 ft can be achieved. From (6-73), the required packed height, l_T , is $l_T = (\text{HETP})N_i = 2.25(6.1) = 13.7$ ft. With metal Intalox IMTP #40 random packing, the HETP might be 2.0 ft, giving $l_T = 12.3$ ft. With Mellapak 250Y corrugated, sheet-metal structured packing, the HETP might be only 1.2 ft, giving $l_T = 7.3$ ft.

For packed columns, it is preferable to determine packed height from a more theoretically based method involving mass-transfer coefficients for the liquid and vapor phases. As with cascades of equilibrium stages, countercurrent flow of vapor and liquid is generally preferred over cocurrent flow. Consider the countercurrent-flow packed columns of packed height l_T , shown in Figure 6.29, which is analogous to Figure 6.8 for trayed towers. For packed absorbers and strippers, operating-line equations, that are analogous to those of Section 6.3 can be derived in terms of mole fractions and total molar flow rates. Thus, for the absorber in Figure 6.29a, a material balance around the upper envelope, for the solute, gives

$$x_{in}L_{in} + yV_l = xL_l + y_{out}V_{out} \quad (6-74)$$

or solving for y , assuming dilute solutions such that $V_l = V_{in} = V_{out} = V$ and $L_l = L_{in} = L_{out} = L$

$$y = x \left(\frac{L}{V} \right) + y_{out} - x_{in} \left(\frac{L}{V} \right) \quad (6-75)$$

Similarly for the stripper in Figure 6.29b,

$$y = x \left(\frac{L}{V} \right) + y_{in} - x_{out} \left(\frac{L}{V} \right) \quad (6-76)$$

In Equations (6-74) to (6-76), mole fractions y and x represent, respectively, bulk compositions of the gas and liquid streams in contact with each other at any elevation of the packed part of the column. For the case of absorption, with mass transfer of the solute from the gas stream to the liquid stream, the two-film theory, developed in Section 3.7, can be applied as illustrated in Figure 6.30. A concentration gradient exists in each film. At the interface between the two phases, physical equilibrium is assumed to exist. Thus, as with trayed towers, an operating line and an equilibrium line are of great importance in a packed column. For a given

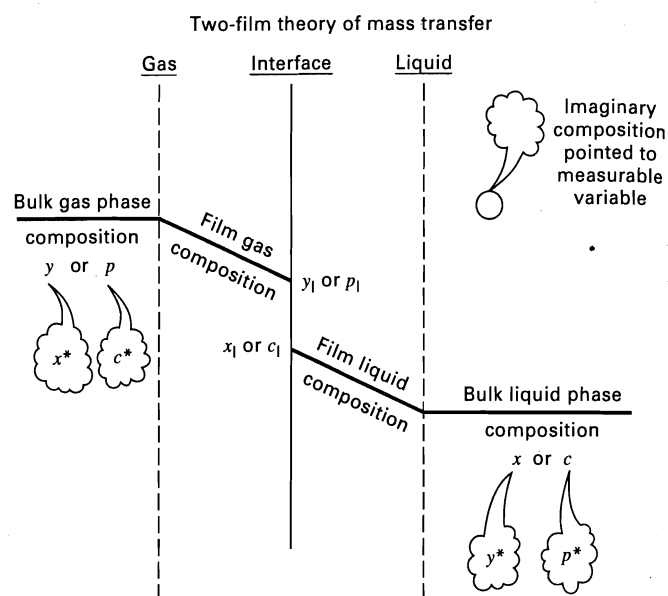


Figure 6.30 Interface properties in terms of bulk properties.

problem specification, the location of the two lines is independent of whether the tower is trayed or packed. Thus, the method for determining the minimum absorbent liquid or stripping vapor flow rates in a packed column is identical to the method for trayed towers, as presented in Section 6.3 and illustrated in Figure 6.9.

The rate of mass transfer for absorption or stripping in a packed column can be expressed in terms of mass-transfer coefficients for each phase. Coefficients, k , based on a unit area for mass transfer could be used, but the area for mass transfer in a packed bed is difficult to determine. Accordingly, as with mass transfer in the froth of a trayed tower, it is more common to use volumetric mass-transfer coefficients, ka , where the quantity a represents the area for mass transfer per unit volume of packed bed. Thus, ka is based on a unit volume of packed bed. At steady state in an absorber, in the absence of chemical reactions, and since species moles are conserved, the rate of solute mass transfer across the gas-phase film must equal the rate across the liquid-phase film. If the system is dilute with respect to the solute, unimolecular diffusion (UMD) may be approximated by the simpler equations for equimolar counterdiffusion (EMD) discussed in Chapter 3. The rate of mass transfer per unit volume of packed bed, r , may be written in terms of mole-fraction driving forces in each of the two phases or in terms of a partial-pressure driving force in the gas phase and a concentration driving force in the liquid phase, as indicated in Figure 6.30. Using the former, for absorption, with the subscript I to denote the interface:

$$r = k_y a (y - y_I) = k_x a (x_I - x) \quad (6-77)$$

The composition at the interface depends on the ratio, $k_x a / k_y a$, of the volumetric mass-transfer coefficients, because (6-77) can be rearranged to

$$\frac{y - y_I}{x - x_I} = -\frac{k_x a}{k_y a} \quad (6-78)$$

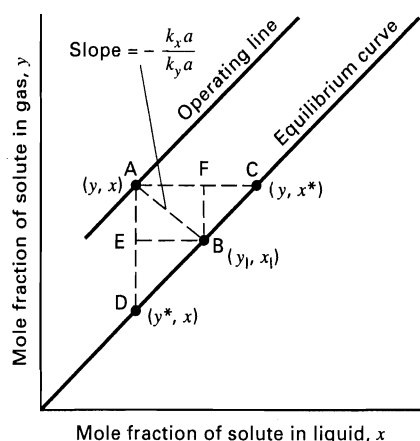


Figure 6.31¹ Interface composition in terms of the ratio of mass-transfer coefficients.

Thus, a straight line of slope $-k_x a / k_y a$ drawn from the operating line at point (y, x) intersects the equilibrium curve at (y_1, x_1) . This result is shown graphically in Figure 6.31.

The slope $-k_x a / k_y a$ determines the relative resistances of the two phases to mass transfer. In Figure 6.31 the distance AE is the gas-phase driving force $(y - y_1)$, while AF is the liquid-phase driving force $(x_1 - x)$. If the mass-transfer resistance in the gas phase is very low, y_1 is approximately equal to y . Then, the resistance resides entirely in the liquid phase. This situation occurs in the absorption of a solute that is only slightly soluble in the liquid phase (i.e., a solute with a high K -value) and is referred to as a liquid-film resistance-controlling process. Alternatively, if the resistance in the liquid phase is very low, x_1 is approximately equal to x . This situation occurs in the absorption of a solute that is very soluble in the liquid phase (i.e., a solute with a low K -value) and is referred to as a gas-film resistance-controlling process. It is important to know if one of the two resistances is controlling. If so, the rate of mass transfer can be increased by promoting turbulence in and/or increasing the dispersion of the controlling phase.

To avoid the need to determine the composition at the interface between the two phases, overall, volumetric mass-transfer coefficients can be defined in terms of overall driving forces for either the gas phase or the liquid phase. Thus, for mole-fraction driving forces,

$$r = K_y a (y - y^*) = K_x a (x^* - x) \quad (6-79)$$

where, as shown in Figure 6.31, y^* is the fictitious vapor mole fraction that is in equilibrium with the mole fraction, x , in the bulk liquid; and x^* is the fictitious liquid mole fraction that is in equilibrium with the mole fraction, y , in the bulk vapor. By combining (6-77) to (6-79), the overall coefficients can be expressed in terms of the separate coefficients for the two phases. Thus,

$$\frac{1}{K_y a} = \frac{1}{k_y a} + \frac{1}{k_x a} \left(\frac{y_1 - y^*}{x_1 - x} \right) \quad (6-80)$$

and

$$\frac{1}{K_x a} = \frac{1}{k_x a} + \frac{1}{k_y a} \left(\frac{x^* - x_1}{y - y_1} \right) \quad (6-81)$$

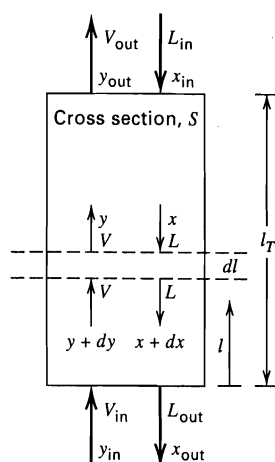


Figure 6.32 Differential contact in a countercurrent-flow, packed absorption column.

However, from Figure 6.31, for dilute solutions when the equilibrium curve is approximately a straight line through the origin,

$$\frac{y_1 - y^*}{x_1 - x} = \frac{\overline{ED}}{\overline{BE}} = K \quad (6-82)$$

and

$$\frac{x^* - x_1}{y - y_1} = \frac{\overline{CF}}{\overline{FB}} = \frac{1}{K} \quad (6-83)$$

where K is the K -value for the solute. Combining (6-80) with (6-82) and (6-81) with (6-83),

$$\frac{1}{K_y a} = \frac{1}{k_y a} + \frac{K}{k_x a} \quad (6-84)$$

and

$$\frac{1}{K_x a} = \frac{1}{k_x a} + \frac{1}{K k_y a} \quad (6-85)$$

Determination of the packed height of a column most commonly involves the overall gas-phase coefficient, $K_y a$, because the liquid usually has a strong affinity for the solute so that resistance to mass transfer is mostly in the gas. This is analogous to a trayed tower, where the tray efficiency from mass transfer considerations is commonly based on $K_{OG} a$ or N_{OG} . Consider the countercurrent-flow absorption column in Figure 6.32. For a dilute system, a differential material balance for a solute being absorbed over a differential height of packing dl , gives:

$$-V dy = K_y a (y - y^*) S dl \quad (6-86)$$

where S is the inside cross-sectional area of the tower. In integral form, with nearly constant terms placed outside the integral, (6-86) becomes

$$\frac{K_y a S}{V} \int_0^{l_T} dl = \frac{K_y a S l_T}{V} = \int_{y_{out}}^{y_{in}} \frac{dy}{y - y^*} \quad (6-87)$$

Solving for the packed height gives

$$l_T = \frac{V}{K_y a S} \int_{y_{out}}^{y_{in}} \frac{dy}{y - y^*} \quad (6-88)$$

Chilton and Colburn [43] suggested that the right-hand side of (6-88) be written as the product of two terms:

$$l_T = H_{OG} N_{OG} \quad (6-89)$$

where

$$H_{OG} = \frac{V}{K_y a S} \quad (6-90)$$

and

$$N_{OG} = \int_{y_{out}}^{y_{in}} \frac{dy}{y - y^*} \quad (6-91)$$

If (6-89) is compared to (6-73), it is seen that H_{OG} is analogous to HETP, as is N_{OG} to N_t .

The term H_{OG} is called the *overall height of a transfer unit* (HTU) based on the gas phase. Experimental data show that the HTU varies less with V than $K_y a$. The smaller the HTU, the more efficient is the contacting. The term N_{OG} is called the *overall number of transfer units* (NTU) based on the gas phase. It represents the overall change in solute mole fraction divided by the average mole-fraction driving force. The larger the NTU, the greater is the extent of contacting required.

Equation (6-91) was first integrated by Colburn [44]. By using the linear equilibrium condition $y^* = Kx$ to eliminate y^* and using the linear, solute material-balance operating line, (6-75), to eliminate x , the result is

$$\int_{y_{out}}^{y_{in}} \frac{dy}{y - y^*} = \int_{y_{out}}^{y_{in}} \frac{dy}{(1 - KV/L)y + y_{out}(KV/L) - Kx_{in}} \quad (6-92)$$

Letting $L/(KV) = A$, the absorption factor, and integrating (6-88), gives

$$N_{OG} = \frac{\ln\{[(A - 1)/A][(y_{in} - Kx_{in})/(y_{out} - Kx_{in})] + (1/A)\}}{(A - 1)/A} \quad (6-93)$$

By applying (6-93) and (6-90), the required packed height, l_T , can be determined from (6-89). However, (6-93) is very sensitive when $A < 0.9$.

The NTU (e.g., N_{OG}) and the HTU (e.g., H_{OG}) should not be confused with the number of equilibrium (theoretical) stages, N_t , and the HETP, respectively. However, when the operating and equilibrium lines are not only straight but also parallel, $NTU = N_t$ and $HTU = HETP$. Otherwise, the NTU is greater than or less than N_t as shown in Figure 6.33 for the case of absorption. When the operating and equilibrium lines are straight but not parallel, then

$$HETP = H_{OG} \frac{\ln(1/A)}{(1 - A)/A} \quad (6-94)$$

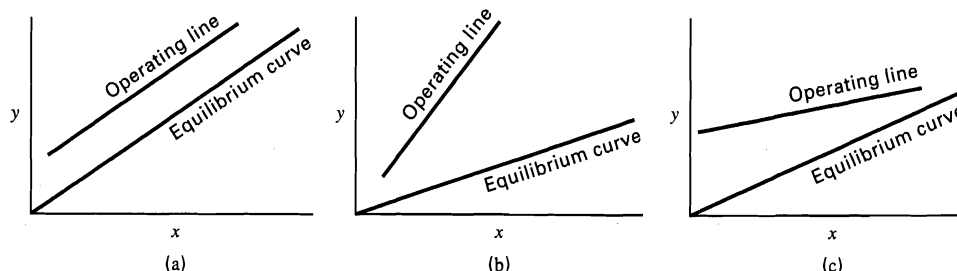


Figure 6.33 Relationship between the NTU and the number of theoretical stages N_t : (a) $NTU = N_t$; (b) $NTU > N_t$; (c) $NTU < N_t$.

and

$$N_{OG} = N_t \frac{\ln(1/A)}{(1 - A)/A} \quad (6-95)$$

Although the most common applications of the HTU and NTU are based on (6-89) to (6-91) and (6-93), a number of alternative groupings have been used, depending on the selected driving force for mass transfer and whether the overall basis is the gas phase, as above, or the liquid phase, where H_{OL} and N_{OL} apply. These groupings are summarized in Table 6.7. Included are driving forces based on partial pressures, p ; mole ratios, X , Y ; and concentrations, c ; as well as mole fractions, x , y . Also included in Table 6.7 for later reference in the last section of this chapter are groupings for unimolecular diffusion (UMD) when solute concentration is not dilute. It is frequently necessary to convert a mass-transfer coefficient based on one type of driving force to another coefficient based on a different type of driving force. Table 3.15 gives the relationships among the different mass-transfer coefficients. The relationships include coefficients based on a concentration and mole-fraction driving force. In addition, a partial-pressure driving force is included for the gas phase.

EXAMPLE 6.9

Repeat Example 6.1 for absorption in a tower packed with 1.5-in. metal Pall rings. If $H_{OG} = 2.0$ ft, compute the required packed height.

SOLUTION

From Example 6.1, $V = 180$ kmol/h, $L = 151.5$ kmol/h, $y_{in} = 0.020$, $x_{in} = 0.0$, and $K = 0.57$. For 97% recovery of ethyl alcohol, by material balance,

$$y_{out} = \frac{(0.03)(0.02)(180)}{180 - (0.97)(0.02)(180)} = 0.000612$$

$$A = \frac{L}{KV} = \frac{151.5}{(0.57)(180)} = 1.477$$

$$\frac{y_{in}}{y_{out}} = \frac{0.020}{0.000612} = 32.68$$

From (6-93),

$$N_{OG} = \frac{\ln\{[(1.477 - 1)/1.477](32.68) + (1/1.477)\}}{(1.477 - 1)/1.477} = 7.5 \text{ transfer units}$$

Table 6.7 Alternative Mass-Transfer Coefficient Groupings

Driving Force	Symbol	Height of a Transfer Unit, HTU		Symbol	Number of Transfer Units, NTU	
		EM Diffusion or Dilute UM Diffusion	UM Diffusion		EM Diffusion ^a or Dilute UM Diffusion	UM Diffusion
1. $(y - y^*)$	H_{OG}	$\frac{V}{K_y a S}$	$\frac{V}{K'_y a (1 - y)_{LM} S}$	N_{OG}	$\int \frac{dy}{(y - y^*)}$	$\int \frac{(1 - y)_{LM} dy}{(1 - y)(y - y^*)}$
2. $(p - p^*)$	H_{OG}	$\frac{V}{K_G a P S}$	$\frac{V}{K'_G a (1 - y)_{LM} P S}$	N_{OG}	$\int \frac{dp}{(p - p^*)}$	$\int \frac{(P - p)_{LM} dp}{(P - p)(p - p^*)}$
3. $(Y - Y^*)$	H_{OG}	$\frac{V'}{K_Y a S}$	$\frac{V'}{K'_Y a S}$	N_{OG}	$\int \frac{dY}{(Y - Y^*)}$	$\int \frac{dY}{(Y - Y^*)}$
4. $(y - y_1)$	H_G	$\frac{V}{k_y a S}$	$\frac{V}{k'_y a (1 - y)_{LM} S}$	N_G	$\int \frac{dy}{(y - y_1)}$	$\int \frac{(1 - y)_{LM} dy}{(1 - y)(y - y_1)}$
5. $(p - p_1)$	H_G	$\frac{V}{k_p a P S}$	$\frac{V}{k'_p a (P - p)_{LM} S}$	N_G	$\int \frac{dp}{(p - p_1)}$	$\int \frac{(P - p)_{LM} dp}{(P - p)(p - p_1)}$
6. $(x^* - x)$	H_{OL}	$\frac{L}{K_x a S}$	$\frac{L}{K'_x a (1 - x)_{LM} S}$	N_{OL}	$\int \frac{dx}{(x^* - x)}$	$\int \frac{(1 - x)_{LM} dx}{(1 - x)(x^* - x)}$
7. $(c^* - c)$	H_{OL}	$\frac{L}{K_L a (\rho_L / M_L) S}$	$\frac{L}{K'_L a (\rho_L / M_L - c)_{LM} S}$	N_{OL}	$\int \frac{dc}{(c^* - c)}$	$\int \frac{(\rho_L / M_L - c)_{LM} dc}{(\rho_L / M_L - c)(c^* - c)}$
8. $(X^* - X)$	H_{OL}	$\frac{L'}{K_X a S}$	$\frac{L'}{K'_X a S}$	N_{OL}	$\int \frac{dX}{(X^* - X)}$	$\int \frac{dX}{(X^* - X)}$
9. $(x_1 - x)$	H_L	$\frac{L}{k_x a S}$	$\frac{L}{k'_x a (1 - x)_{LM} S}$	N_L	$\int \frac{dx}{(x_1 - x)}$	$\int \frac{(1 - x)_{LM} dx}{(1 - x)(x_1 - x)}$
10. $(c_1 - c)$	H_L	$\frac{L}{k_L a (\rho_L / M_L) S}$	$\frac{L}{k'_L a (\rho_L / M_L - c)_{LM} S}$	N_L	$\int \frac{dc}{(c_1 - c)}$	$\int \frac{(\rho_L / M_L - c)_{LM} dc}{(\rho_L / M_L - c)(c_1 - c)}$

^aThe substitution $K_y = K'_y y_{B,LM}$ or its equivalent can be made.

The packed height, from (6-89), is

$$l_T = 2.0(7.5) = 15 \text{ ft}$$

Note that N_L for this example was determined in Example 6.1 to be about 6.1. The value of 7.5 for N_{OG} is greater than N_L because the slope of the operating line, L/G , is greater than the slope of the equilibrium line, K , so Figure 6.33b applies.

EXAMPLE 6.10

Experimental data have been obtained for air containing 1.6% by volume SO_2 being scrubbed with pure water in a packed column of 1.5 m^2 in cross-sectional area and 3.5 m in packed height. Entering gas and liquid flow rates are 0.062 and 2.2 kmol/s, respectively. If the outlet mole fraction of SO_2 in the gas is 0.004 and column temperature is near-ambient with $K_{\text{SO}_2} = 40$, calculate from the data:

- The N_{OG} for absorption of SO_2
- The H_{OG} in meters

- The volumetric, overall mass-transfer coefficient, $K_y a$ for SO_2 in $\text{kmol/m}^3\text{-s-(}\Delta y\text{)}$.

SOLUTION

- Assume a straight operating line because the system is dilute in SO_2 .

$$A = \frac{L}{KV} = \frac{2.2}{(40)(0.062)} = 0.89, \quad y_{\text{in}} = 0.016, \\ y_{\text{out}} = 0.004, \quad x_{\text{in}} = 0.0$$

From (6-93),

$$N_{OG} = \frac{\ln\{[(0.89 - 1)/0.89](0.016/0.004) + (1/0.89)\}}{(0.89 - 1)/0.89} \\ = 3.75$$

- $l_T = 3.5 \text{ m}$. From (6-89), $H_{OG} = l_T/N_{OG} = 3.5/3.75 = 0.93 \text{ m}$
- $V = 0.062 \text{ kmol/s}$, $S = 1.5 \text{ m}^2$.

From (6-90), $K_y a = V/H_{OG} S = 0.062/[(0.93)(1.5)] = 0.044 \text{ kmol/m}^3\text{-s-(}\Delta y\text{)}$

EXAMPLE 6.11

A gaseous reactor effluent consisting of 2 mol% ethylene oxide in an inert gas is scrubbed with water at 30°C and 20 atm. The total gas feed rate is 2,500 lbmol/h, and the water rate entering the scrubber is 3,500 lbmol/h. The column, with a diameter of 4 ft, is packed in two 12-ft-high sections with 1.5-in. metal Pall rings. A liquid redistributor is located between the two packed sections. Under the operating conditions for the scrubber, the K -value for ethylene oxide is 0.85 and estimated values of $k_y a$ and $k_x a$ are 200 lbmol/h-ft³-Δy and 165 lbmol/h-ft³-Δx, respectively.

Calculate: (a) $K_y a$ and (b) H_{OG} .

SOLUTION

(a) From (6-84),

$$K_y a = \frac{1}{(1/k_y a) + (K/k_x a)} = \frac{1}{(1/200) + (0.85/165)} = 98.5 \text{ lbmol/h-ft}^2 \Delta y$$

(b) $S = 3.14(4)^2/4 = 12.6 \text{ ft}^2$

From (6-90), $H_{OG} = V/K_y a S = 2,500/[(98.5)(12.6)] = 2.02 \text{ ft}$.

Note that in this example, both gas-phase and liquid-phase resistances are important.

The value of H_{OG} can also be computed from values of H_G and H_L using equations in Table 6.7:

$$H_G = V/k_y a S = 2,500/[(200)(12.6)] = 1.0 \text{ ft}$$

$$H_L = L/k_x a S = 3,500/[(165)(12.6)] = 1.68 \text{ ft}$$

Substituting these two expressions and (6-90) into (6-84) gives the following relationship for H_{OG} in terms of H_G and H_L :

$$\begin{aligned} H_{OG} &= H_G + H_L/A \\ A &= L/KV = 3,500/[(0.85)(2,500)] = 1.65 \quad (6-96) \\ H_{OG} &= 1.0 + 1.68/1.65 = 2.02 \text{ ft} \end{aligned}$$

6.8 PACKED-COLUMN EFFICIENCY, CAPACITY, AND PRESSURE DROP

Values of volumetric mass-transfer coefficients and corresponding HTUs depend on gas and/or liquid flow rates per unit inside cross-sectional area of the packed column. Therefore, column diameter must be estimated before determining required height of packing. The estimation of a suitable column diameter for a given system, packing, and operating conditions requires consideration of liquid holdup, flooding, and pressure drop.

Liquid Holdup

Typical experimental curves, taken from Billet [45] and shown also by Stichlmair, Bravo, and Fair [46], for specific pressure drop in meters of water head per meter of packed height, and specific liquid holdup in cubic meters per cubic meter of packed bed as a function of superficial gas velocity for different values of superficial water velocity are shown in Figures 6.34 and 6.35, respectively, for a 0.15-m-diameter column packed with 1-in. metal Bialecki rings to a height of 1.5 m and operated at 25°C and 1 bar. In Figure 6.34, the

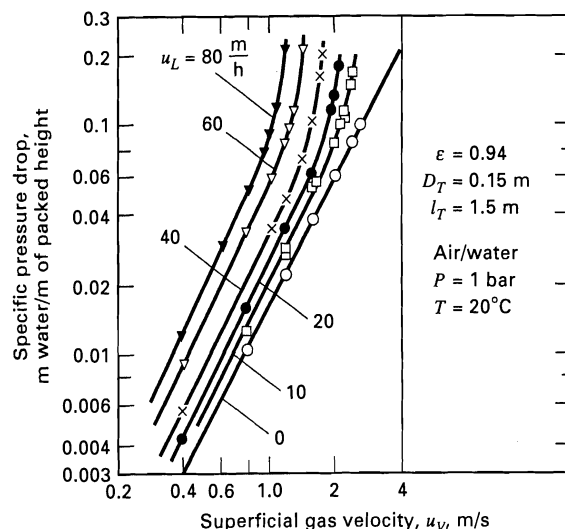


Figure 6.34 Specific pressure drop for dry and irrigated 25-mm metal Bialecki rings.

[From R. Billet, *Packed Column Analysis and Design*, Ruhr-University Bochum (1989) with permission.]

lowest curve corresponds to zero liquid flow, that is, the dry pressure drop. Over an almost 10-fold range of superficial air velocity (the velocity the air would have in the absence of packing), the pressure drop for air flowing up through the packing is proportional to air velocity to the 1.86 power. As liquid flows down through the packing at an increasing rate, gas-phase pressure drop for a given gas velocity increases. However, below a certain limiting gas velocity, the curve for each liquid velocity is a straight line parallel to the dry-pressure-drop curve. In this region, the liquid holdup for each liquid velocity is constant, as shown in Figure 6.35. Thus, for a liquid velocity of 40 m/h, specific liquid holdup is 0.08 m³/m³ of packed bed until a superficial gas velocity of 1.0 m/h is reached. Instead of a packed-column void fraction, ϵ , of 0.94 for the gas to flow through (corresponding to zero liquid flow), the effective void fraction is reduced by

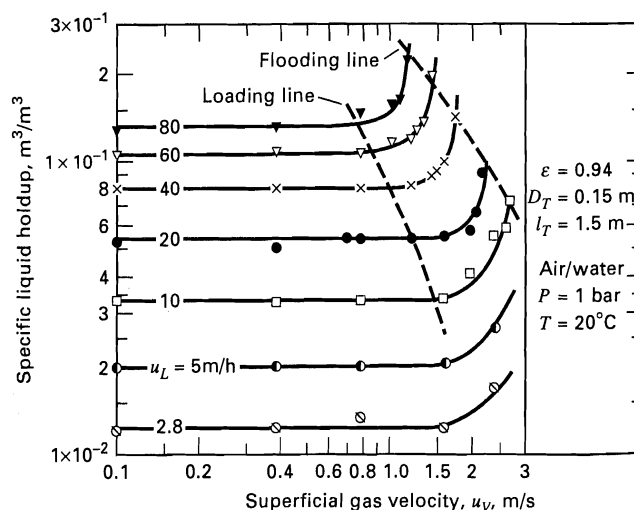


Figure 6.35 Specific liquid holdup for irrigated 25-mm metal Bialecki rings.

[From R. Billet, *Packed Column Analysis and Design*, Ruhr-University Bochum (1989) with permission.]

the liquid holdup to $0.94 - 0.08 = 0.86$, causing an increased pressure drop. For a given liquid velocity, the upper limit to the gas velocity for a constant liquid holdup is termed the *loading point*. Below this point, the gas phase is the continuous phase. Above this point, liquid begins to accumulate or load the bed, replacing gas holdup and causing a sharp increase in pressure drop. Finally, a gas velocity is reached at which the liquid surface is continuous across the top of the packing and the column is flooded. At the *flooding point*, the drag force of the counterflowing gas is sufficient to entrain the entire liquid. Approximate loci of both loading and flooding points are included in Figure 6.35.

The region between the loading point and the flooding point is the *loading region*, where significant liquid entrainment is observed, liquid holdup increases sharply, mass-transfer efficiency decreases, and column operation is unstable. Typically, according to Billet [45], the superficial gas velocity at the loading point is approximately 70% of that at the flooding point. Although a packed column can operate in the loading region, most packed columns are designed to operate at or below the loading point, in the *preloading region*.

The specific liquid holdup in the preloading region has been found, from extensive experiments by Billet and Schultes [47, 69] for a wide variety of random and structured packings and, for a number of gas-liquid systems, to depend on packing characteristics, and the viscosity, density, and superficial velocity of the liquid, u_L , according to the dimensionless expression

$$h_L = \left(12 \frac{N_{FrL}}{N_{ReL}} \right)^{1/3} \left(\frac{a_h}{a} \right)^{2/3} \quad (6-97)$$

where

$$N_{ReL} = \text{liquid Reynolds number} = \frac{\text{inertial force}}{\text{viscous force}} = \frac{u_L \rho_L}{a \mu_L} = \frac{u_L}{a \nu_L} \quad (6-98)$$

where ν_L is the kinematic viscosity.

$$N_{FrL} = \text{liquid Froude number} = \frac{\text{inertial force}}{\text{gravitational force}} = \frac{u_L^2 a}{g} \quad (6-99)$$

and the ratio of specific hydraulic area of packing, a_h , to specific surface area of packing, a , is given by

$$a_h/a = C_h N_{ReL}^{0.15} N_{FrL}^{0.1} \quad \text{for } N_{ReL} < 5 \quad (6-100)$$

$$a_h/a = 0.85 C_h N_{ReL}^{0.25} N_{FrL}^{0.1} \quad \text{for } N_{ReL} \geq 5 \quad (6-101)$$

Values of $a_h/a > 1$ are reasonable because of the creation of droplets and jet flow beside the rivulets that cover the packing surface [70].

Values of a and C_h are characteristic of the particular type and size of packing, as listed, together with packing void fraction, ϵ , and other packing constants in Table 6.8. Because the specific liquid holdup is constant in the preloading region, as seen in Figure 6.35, (6-97) does not involve gas-phase properties or gas velocity.

At low liquid velocities, liquid holdup can become so small that the packing is no longer completely wetted. When this occurs, packing efficiency decreases dramatically, particularly for aqueous systems of high surface tension. To ensure adequate wetting of packing, proven liquid distributors and redistributors should be used and superficial liquid velocities should exceed the following values:

Type of Packing Material	u_{Lmin} , m/s
Ceramic	0.00015
Oxidized or etched metal	0.0003
Bright metal	0.0009
Plastic	0.0012

EXAMPLE 6.12

An absorption column is to be designed using oil absorbent with a kinematic viscosity of three times that of water at 20°C. The superficial liquid velocity will be 0.01 m/s, which is safely above the minimum value for good wetting. The superficial gas velocity will be such that operation will be in the preloading region. Two packing materials are being considered: (1) randomly packed 50-mm metal Hiflow rings and (2) metal Montz B1-200 structured packing. Estimate the specific liquid holdup for each of these two packings.

SOLUTION

From Table 6.8,

Packing	a , m ² /m ³	ϵ	C_h
50-mm metal Hiflow rings	92.3	0.977	0.876
Montz metal B1-200	200.0	0.979	0.547

At 20°C for water, kinematic viscosity, $\nu = \mu/\rho = 1 \times 10^{-6}$ m²/s. Therefore, for the oil, $\mu/\rho = 3 \times 10^{-6}$ m²/s. From (6-98) and (6-99),

$$N_{ReL} = \frac{0.01}{3 \times 10^{-6} a} \quad N_{FrL} = \frac{(0.01)^2 a}{9.8}$$

Therefore,

Packing	N_{ReL}	N_{FrL}
Hiflow	36.1	0.000942
Montz	16.67	0.00204

From (6-101), since $N_{ReL} > 5$, for the Hiflow packing, $a_h/a = (0.85)(0.876)(36.1)^{0.25}(0.000942)^{0.1} = 0.909$. For the Montz packing, $a_h/a = 0.85(0.547)(16.67)^{0.25}(0.00204)^{0.10} = 0.506$.

From (6-97), for the Hiflow packing,

$$h_L = \left[\frac{12(0.000942)}{36.1} \right]^{1/3} (0.909)^{2/3} = 0.0637 \text{ m}^3/\text{m}^3$$

For the Montz packing,

$$h_L = \left[\frac{12(0.00204)}{16.67} \right]^{1/3} (0.506)^{2/3} = 0.0722 \text{ m}^3/\text{m}^3$$

Note that for the Hiflow packing, the void fraction available for gas flow is reduced by the liquid flow from $\epsilon = 0.977$ (Table 6.8) to $0.977 - 0.064 = 0.913$ m³/m³. For the Montz packing, the reduction is from 0.979 to 0.907 m³/m³.

Table 6.8 Characteristics of Packings

Packing	Characteristics from Billet										
	Material	Size	F_P , ft ² /ft ³	a , m ² /m ³	ϵ , m ³ /m ³	C_h	C_p	C_L	C_V	C_s	C_{FI}
Random Packings											
Berl saddles	Ceramic	25 mm	110	260.0	0.680	0.620		1.246	0.387		
Berl saddles	Ceramic	13 mm	240	545.0	0.650	0.833		1.364	0.232		
Bialecki rings	Metal	50 mm		121.0	0.966	0.798	0.719	1.721	0.302	2.916	1.896
Bialecki rings	Metal	35 mm		155.0	0.967	0.787	1.011	1.412	0.390	2.753	1.885
Bialecki rings	Metal	25 mm		210.0	0.956	0.692	0.891	1.461	0.331	2.521	1.856
DIN-PAK rings	Plastic	70 mm		110.7	0.938	0.991	0.378	1.527	0.326	2.970	1.912
DIN-PAK rings	Plastic	47 mm		131.2	0.923	1.173	0.514	1.690	0.354	2.929	1.991
Envi Pac rings	Plastic	80 mm, no. 3		60.0	0.955	0.641	0.358	1.603	0.257	2.846	1.522
Envi Pac rings	Plastic	60 mm, no. 2		98.4	0.961	0.794	0.338	1.522	0.296	2.987	1.864
Envi Pac rings	Plastic	32 mm, no. 1		138.9	0.936	1.039	0.549	1.517	0.459	2.944	2.012
Glitsch rings	Metal	30 PMK		180.5	0.975	0.930	0.851	1.920	0.450	2.694	1.900
Glitsch rings	Metal	30 P		164.0	0.959	0.851	1.056	1.577	0.398	2.564	1.760
Glitsch CMR rings	Metal	1.5"		174.9	0.974	0.935	0.632			2.697	1.841
Glitsch CMR rings	Metal	1.5", T		188.0	0.972	0.870	0.627			2.790	1.870
Glitsch CMR rings	Metal	1.0"		232.5	0.971	1.040	0.641			2.703	1.996
Glitsch CMR rings	Metal	0.5"		356.0	0.952		0.882	2.038	0.495	2.644	2.178
Cascade minirings	Metal	30 PMK		180.2	0.975	0.930					
Cascade minirings	Metal	30 P		168.9	0.958	0.851					
Cascade minirings	Metal	1.5" CMR, T		188.0	0.972	0.870					
Cascade minirings	Metal	1.5" CMR	29	174.9	0.974	0.935					
Cascade minirings	Metal	1.0" CMR	40	232.5	0.971	1.040					
Cascade minirings	Metal	0.5" CMR		356.0	0.955	1.338					
Hackettes	Plastic	45 mm		139.5	0.928	0.643	0.399			2.832	1.966
Hiflow rings	Ceramic	75 mm	15	54.1	0.868		0.435				
Hiflow rings	Ceramic	50 mm	29	89.7	0.809		0.538	1.377	0.379	2.819	1.694
Hiflow rings	Ceramic	38 mm	37	111.8	0.788		0.621	1.659	0.464	2.840	1.930
Hiflow rings	Ceramic	20 mm, 6 stg.		265.8	0.776	0.958					
Hiflow rings	Ceramic	20 mm, 4 stg.		261.2	0.779	1.167	0.628	1.744	0.465		
Hiflow rings	Metal	50 mm	16	92.3	0.977	0.876	0.421	1.168	0.408	2.702	1.626
Hiflow rings	Metal	25 mm	42	202.9	0.962	0.799	0.689	1.641	0.402	2.918	2.177
Hiflow rings	Plastic	90 mm	9	69.7	0.968		0.276				
Hiflow rings	Plastic	50 mm, hydr.		118.4	0.925		0.311	1.553	0.369	2.894	1.871
Hiflow rings	Plastic	50 mm	20	117.1	0.924	1.038	0.327	1.487	0.345		

Hiflow rings	Plastic	25 mm		194.5	0.918		0.741	1.577	0.390	2.841	1.989
Hiflow rings, super	Plastic	50 mm, S		82.0	0.942		0.414	1.219	0.342	2.866	1.702
Hiflow saddles	Plastic	50 mm		86.4	0.938		0.454				
Intalox saddles	Ceramic	50 mm	40	114.6	0.761		0.747	-			
Intalox saddles	Plastic	50 mm	28	122.1	0.908		0.758				
NORPAC rings	Plastic	50 mm	14	86.8	0.947	0.651	0.350	1.080	0.322	2.959	1.786
NORPAC rings	Plastic	35 mm	21	141.8	0.944	0.587	0.371	0.756	0.425	3.179	2.242
NORPAC rings	Plastic	25 mm, type B		202.0	0.953	0.601	0.397	0.883	0.366	3.277	2.472
NORPAC rings	Plastic	25 mm, 10 stg.		197.9	0.920		0.383	0.976	0.410	2.865	2.083
NORPAC rings	Plastic	25 mm	31	180.0	0.927	0.601					
NORPAC rings	Plastic	22 mm		249.0	0.913		0.397				
NORPAC rings	Plastic	15 mm		311.4	0.918	0.343	0.365				
Pall rings	Ceramic	50 mm	43	155.2	0.754	1.066	0.233	1.278	0.333	3.793	3.024
Pall rings	Metal	50 mm	27	112.6	0.951	0.784	0.763	1.192	0.410	2.725	1.580
Pall rings	Metal	35 mm	40	139.4	0.965	0.644	0.967	1.012	0.341	2.629	1.679
Pall rings	Metal	25 mm	56	223.5	0.954	0.719	0.957	1.440	0.336	2.627	2.083
Pall rings	Metal	15 mm	70	368.4	0.933	0.590	0.990				
Pall rings	Plastic	50 mm	26	111.1	0.919	0.593	0.698	1.239	0.368	2.816	1.757
Pall rings	Plastic	35 mm	40	151.1	0.906	0.718	0.927	0.856	0.380	2.654	1.742
Pall rings	Plastic	25 mm	55	225.0	0.887	0.528	0.865	0.905	0.446	2.696	2.064
Raflux rings	Plastic	15 mm		307.9	0.894	0.491	0.595	1.913	0.370	2.825	2.400
Ralu flow	Plastic	1		165	0.940	0.640	0.485	1.486	0.360	3.612	2.401
Ralu flow	Plastic	2		100	0.945	0.640	0.350	1.270	0.320	3.412	2.174
Ralu rings	Plastic	50 mm, hydr.		94.3	0.939	0.439		1.481	0.341		
Ralu rings	Plastic	50 mm		95.2	0.983	0.640	0.468	1.520	0.303	2.843	1.812
Ralu rings	Plastic	38 mm		150	0.930	0.640	0.672	1.320	0.333	2.843	1.812
Ralu rings	Plastic	25 mm		190	0.940	0.719	0.800	1.320	0.333	2.841	1.989
Ralu rings	Metal	50 mm		105	0.975	0.784	0.763	1.192	0.345	2.725	1.580
Ralu rings	Metal	38 mm		135	0.965	0.644	1.003	1.277	0.341	2.629	1.679
Ralu rings	Metal	25 mm		215	0.960	0.714	0.957	1.440	0.336	2.627	2.083
Raschig rings	Carbon	25 mm		202.2	0.720	0.623		1.379	0.471		
Raschig rings	Ceramic	25 mm	179	190.0	0.680	0.577	1.329	1.361	0.412		
Raschig rings	Ceramic	15 mm	380	312.0	0.690	0.648		1.276	0.401		
Raschig rings	Ceramic	10 mm	1,000	440.0	0.650	0.791		1.303	0.272		
Raschig rings	Ceramic	6 mm	1,600	771.9	0.620	1.094		1.130			
Raschig rings	Metal	15 mm	170	378.4	0.917	0.455					
Raschig rings	Ceramic	25		190.0	0.680	0.577	1.329	1.361	0.412	2.454	1.899
Raschig Super-rings	Metal	0.3		315	0.960	0.750	0.760	1.500	0.450	3.560	2.340

(Continued)

Table 6.8 (Continued)

Packing	Material	Size	Characteristics from Billet								
			F_P , ft ² /ft ³	a , m ² /m ³	ϵ , m ³ /m ³	C_h	C_p	C_L	C_V	C_s	C_{FI}
Raschig Super-rings	Metal	0.5		250	0.975	0.620	0.780	1.450	0.430	3.350	2.200
Raschig Super-rings	Metal	1		160	0.980	0.750	0.500	1.290	0.440	3.491	2.200
Raschig Super-rings	Metal	2		97.6	0.985	0.720	0.464	1.323	0.400	3.326	2.096
Raschig Super-rings	Metal	3		80	0.982	0.620	0.430	0.850	0.300	3.260	2.100
Raschig Super-rings	Plastic	2		100	0.960	0.720	0.377	1.250	0.337	3.326	2.096
Tellerettes	Plastic	25 mm	40	190.0	0.930	0.588	0.538	0.899		2.913	2.132
Top-Pak rings	Aluminum	50 mm		105.5	0.956	0.881	0.604	1.326	0.389	2.528	1.579
VSP rings	Metal	50 mm, no. 2		104.6	0.980	1.135	0.773	1.222	0.420	2.806	1.689
VSP rings	Metal	25 mm, no. 1		199.6	0.975	1.369	0.782	1.376	0.405	2.755	1.970
Structured Packings											
Euroform	Plastic	PN-110		110.0	0.936	0.511	0.250	0.973	0.167	3.075	1.975
Gempak	Metal	A2 T-304		202.0	0.977	0.678	0.344			2.986	2.099
Impulse	Ceramic	100		91.4	0.838	1.900	0.417	1.317	0.327	2.664	1.655
Impulse	Metal	250		250.0	0.975	0.431	0.262	0.983	0.270	2.610	1.996
Koch-Sulzer	Metal	CY	70								
Koch-Sulzer	Metal	BX	21								
Mellapak	Plastic	250 Y	22	250.0	0.970	0.554	0.292			3.157	2.464
Montz	Metal	B1-100		100.0	0.987	0.626					
Montz	Metal	B1-200		200.0	0.979	0.547	0.355	0.971	0.390	3.116	2.339
Montz	Metal	B1-300	33	300.0	0.930	0.482	0.295	1.165	0.422	3.098	2.464
Montz	Plastic	C1-200		200.0	0.954		0.453	1.006	0.412		
Montz	Plastic	C2-200		200.0	0.900		0.481	0.739		2.653	1.973
Ralu Pak	Metal	YC-250		250.0	0.945	0.650	0.191	1.334	0.385	3.178	2.558

Column Diameter and Pressure Drop

Most packed columns consist of cylindrical vertical vessels. The column diameter is determined so as to safely avoid flooding and operate in the preloading region with a pressure drop of no greater than 1.5 in. of water head per foot of packed height (equivalent to 0.054 psi/ft of packing). In addition, for random packings, a nominal packing diameter not greater than one-eighth of the diameter of the column is selected; otherwise, poor distribution of liquid and vapor flow over the cross-sectional area of the column can occur, with liquid tending to migrate to the wall of the column.

Flooding data for packed columns with countercurrent flow of liquid and gas were first correlated successfully by Sherwood et al. [26], who used the same liquid-to-gas kinetic energy ratio, $F_{LV} = (LM_L/VM_V)(\rho_V/\rho_L)^{0.5}$, already discussed for the correlation of flooding and entrainment in trayed towers, as shown in Figures 6.24 and 6.28, respectively. The superficial gas velocity, u_V , was embedded in the dimensionless term $u_V^2 a/g\epsilon^3$, which was arrived at by considering the square of the actual gas velocity, u_V^2/ϵ^2 , the hydraulic radius, $r_H = \epsilon/a$, which is the volume available for flow divided by the wetted surface area of the packing, and the gravitational acceleration, g , to give the dimensionless expression, $u_V^2 a/g\epsilon^3 = u_V^2 F_P/g$. The ratio, a/ϵ^3 , is a function of the packing only, and is known as the packing factor, F_P . Values of a , ϵ , and F_P are included in Table 6.8. In some cases, F_P is a modified packing factor, treated as an empirical constant, backed out from experimental data so as to fit a generalized correlation. Additional factors were added by Sherwood et al. to account for liquid density and viscosity, and gas density.

In 1954, Leva [48] used experimental data on ring and saddle packings to extend the Sherwood et al. [26] flooding correlation to include lines of constant pressure drop, with the resulting chart becoming known as the generalized pressure-drop correlation (GPDC).

A modern version of the GPDC chart is that of Leva [49], as shown in Figure 6.36a. The abscissa is the same F_{LV} parameter, but the ordinate is given by

$$Y = \frac{u_V^2 F_P}{g} \left(\frac{\rho_V}{\rho_{H_2O(L)}} \right) f\{\rho_L\} f\{\mu_L\} \quad (6-102)$$

where the density of H_2O is taken as 62.4 lb/ft³ with ρ_V in the same units. The functions $f\{\rho_L\}$ and $f\{\mu_L\}$ are corrections for liquid properties as given by Figures 6.36b and 6.36c, respectively.

For given fluid flow rates and properties, and a given packing material, the GPDC chart is used to compute u_{Vf} , the superficial gas velocity at flooding. Then a fraction of flooding, f , is selected (usually from 0.5 to 0.7), followed by calculation of the tower diameter from an equation similar to (6-44):

$$D_T = \left(\frac{4VM_V}{fu_{Vf}\pi\rho_V} \right)^{0.5} \quad (6-103)$$

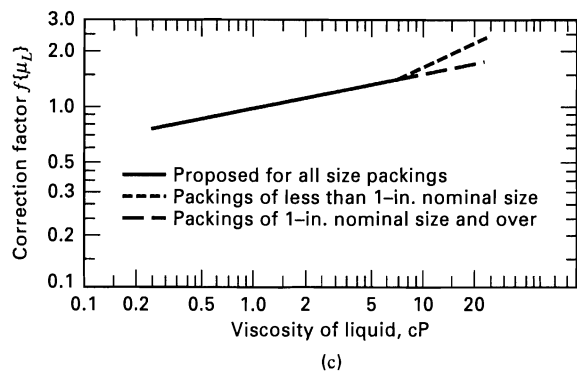
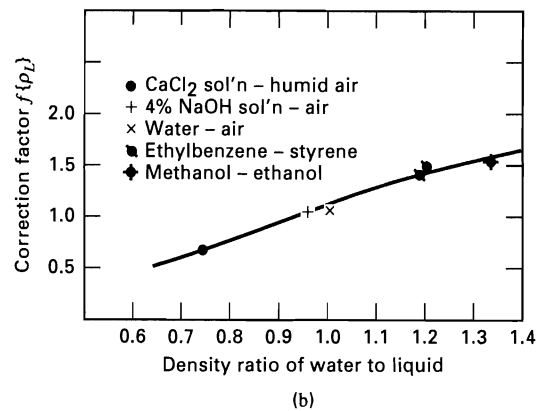
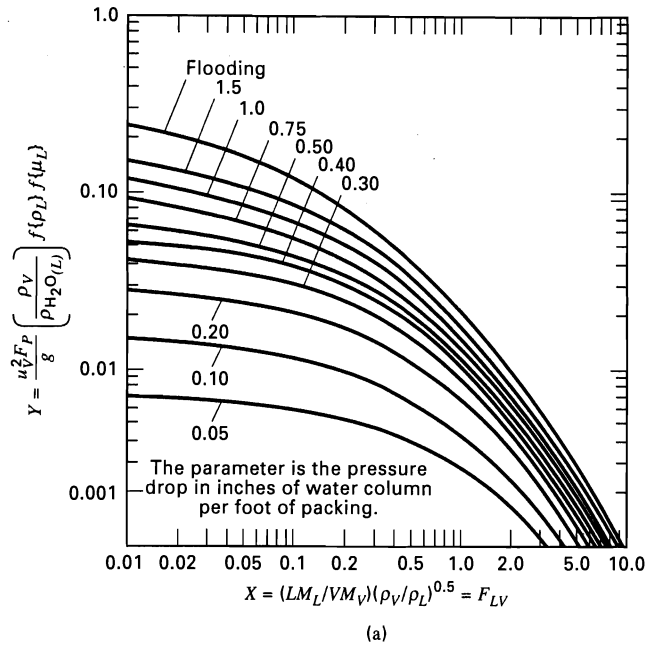


Figure 6.36 (a) Generalized pressure-drop correlation of Leva for packed columns. (b) Correction factor for liquid density. (c) Correction factor for liquid viscosity.

[From M. Leva, *Chem. Eng. Prog.*, **88** (1), 65–72 (1992) with permission.]

EXAMPLE 6.13

Air containing 5 mol% NH_3 at a total flow rate of 40 lbmol/h enters a packed column operating at 20°C and 1 atm, where 90% of the ammonia is scrubbed by a countercurrent flow of 3,000 lb/h of water. Use the GPDC chart of Figure 6.36 to estimate the superficial, gas-flooding velocity, the column inside diameter for

operation at 70% of flooding, and the pressure drop per foot of packing for two packing materials:

(a) One-inch ceramic Raschig rings ($F_P = 179 \text{ ft}^2/\text{ft}^3$)

(b) One-inch metal IMTP packing ($F_P = 41 \text{ ft}^2/\text{ft}^3$)

SOLUTION

Because the superficial gas velocity is highest at the bottom of the column, calculations are made for conditions there.

Inlet gas:

$$M_V = 0.95(29) + 0.05(17) = 28.4, \quad V = 40 \text{ lbmol/h}$$

$$\rho_V = PM_V/RT = (1)(28.4)/[(0.730)(293)(1.8)] \\ = 0.0738 \text{ lb/ft}^3$$

Exiting liquid:

$$\text{Ammonia absorbed} = 0.90(0.05)(40)(17) = 30.6 \text{ lb/h or } 1.8 \text{ lbmol/h}$$

$$\text{Water rate (neglecting any stripping by the gas)} = 3,000 \text{ lb/h or } 166.7 \text{ lbmol/h}$$

$$\text{Mole fraction of ammonia} = 1.8/(166.7 + 1.8) = 0.0107$$

$$M_L = 0.0107(17) + (0.9893)(18) = 17.9,$$

$$L = 1.8 + 166.7 = 168.5 \text{ lbmol/h}$$

$$\text{Take: } \rho_L = 62.4 \text{ lb/ft}^3 \text{ and } \mu_L = 1.0 \text{ cP}$$

Now,

$$X = F_{LV}(\text{abscissa in Figure 6.36a}) \\ = \frac{(168.5)(17.9)}{(40)(28.4)} \left(\frac{0.0738}{62.4} \right)^{0.5} = 0.092$$

From Figure 6.36a, $Y = 0.125$ at flooding.

From Figure 6.36b, $f\{\rho_L\} = 1.14$.

From Figure 6.36c, $f\{\mu_L\} = 1.0$.

From (6-102),

$$u_V^2 = 0.125 \left(\frac{g}{F_P} \right) \frac{62.4}{(0.0738)(1.14)(1.0)} = 92.7 \text{ g}/F_P.$$

Using $g = 32.2 \text{ ft/s}^2$,

Packing Material	$F_P, \text{ft}^2/\text{ft}^3$	$u_o, \text{ft/s}$
Raschig rings	179	4.1
IMTP packing	41	8.5

For $f = 0.70$, using (6-103),

Packing Material	$fu_{V,f}, \text{ft/s}$	$D_T, \text{in.}$
Raschig rings	2.87	16.5
IMTP packing	5.95	11.5

From Figure 6.36a, for $F_{LV} = 0.092$ and $Y = 0.70^2(0.125) = 0.0613$ at 70% of flooding, the pressure drop is 0.88 in. of water head per foot of packed height for both packings.

Based on these results, the IMTP packing has a much greater capacity than the Raschig rings, since the required column cross-sectional area is reduced by about 50%.

Experimental flooding-point data for a variety of packing materials are in reasonable agreement with the upper curve of the GPDC chart of Figure 6.36. Unfortunately, such good agreement is not always the case for pressure drop, particularly for operation at superficial vapor velocities

above 50% of flooding, where pressure drop is greater than 0.5 in. of water head per foot of packed height. Reasons for the difficulty of achieving a simple generalization of pressure drop measurements are discussed in detail by Kister [33]. As an example of the possible magnitude of the disparity, the predicted pressure drop of 0.88 in. of water per foot in Example 6.13 for operation with IMTP packing at 70% of flooding is in poor agreement with the value of 0.63 in. of water head per foot determined from data supplied by the packing manufacturer.

If Figure 6.36a is crossplotted as pressure drop versus Y for constant values of F_{LV} , it is found that a pressure drop of from 2.5 to 3 in. of water head per foot is predicted at the flooding condition for all packings. However, studies by Kister and Gill [33, 50] for both random and structured packings show that the pressure drop at flooding is strongly dependent on the packing factor, F_P , by the empirical expression

$$\Delta P_{\text{flood}} = 0.115 F_P^{0.7} \quad (6-104)$$

where ΔP_{flood} has units of inches of water head per foot of packed height and F_P has units of ft^2/ft^3 . As seen in Table 6.8, the range of F_P is from about 10 to 100. Thus, (6-103) predicts pressure drops at flooding from as low as 0.6 to as high as 3 in. of water head per foot of packed height. Kister and Gill also give an interpolation procedure for estimating pressure drop, which utilizes experimental data in conjunction with a GPDC-type plot.

Theoretically based models for predicting pressure drop in packed beds with countercurrent gas-liquid flows have been presented by Stichlmair et al. [46], who use a particle model, and Billet and Schultes [51, 69], who use a channel model. Both models extend well-accepted equations for dry-bed pressure drop to account for the effect of liquid holdup. Billet and Schultes [69] include a semitheoretical model for predicting the superficial vapor velocity at the loading point, $u_{V,l}$, which provides an alternative, perhaps more accurate, method for estimating column diameter. Their model, which is based on a liquid velocity of zero at the phase boundary at the loading point, gives

$$u_{V,l} = \left(\frac{g}{\Psi_l} \right)^{1/2} \left[\frac{\epsilon}{a^{1/6}} - a^{1/2} \xi_l^{1/3} \right] \xi_l^{1/6} \left(\frac{\rho_L}{\rho_V} \right)^{1/2} \quad (6-105)$$

where $u_{V,l}$ is in m/s,

$$g = \text{gravitational acceleration} = 9.807 \text{ m/s}^2$$

$$\Psi_l = \frac{g}{C^2} \left[F_{LV} \left(\frac{\mu_L}{\mu_V} \right)^{0.4} \right]^{-2n_s} \quad (6-106)$$

ϵ and a are obtained from Table 6.8,

$$F_{LV} = \text{kinetic energy ratio of Figures 6.24 and 6.36a,}$$

$$\xi_l = \left(12 \frac{\mu_L}{g \rho_L} u_{L,l} \right) \quad (6-107)$$

μ_L and μ_V are in kg/m-s

ρ_L and ρ_V are in kg/m³

$u_{L,l}$ = superficial liquid velocity at loading point

$$= u_{V,l} \frac{\rho_V L M_L}{\rho_L V M_V} \text{ in m/s}$$

The values for n_s and C depend on the value of the kinetic energy ratio as follows:

If $F_{LV} \leq 0.4$, the liquid trickles downward over the packing as a disperse phase and $n_s = -0.326$, while $C = C_s$ from Table 6.8.

If $F_{LV} > 0.4$, the column holdup reaches such a large value that the empty spaces within the bed close up and the liquid flows downward as a continuous phase while the gas rises in the form of bubbles, with $n_s = -0.723$ and

$$C = 0.695 \left(\frac{\mu_L}{\mu_V} \right)^{0.1588} C_s \quad (\text{from Table 6.8}) \quad (6-108)$$

Billet and Schultes [69] also present a model for predicting the superficial vapor velocity at the flooding point, $u_{V,f}$, that involves the flooding constant, C_{Ff} , in Table 6.8, but a suitable expression is

$$u_{V,f} = \frac{u_{V,l}}{0.7} \quad (6-109)$$

When a gas flows through a packed column under conditions of no liquid flow, a correlation for the pressure drop can be obtained in a manner similar to that for flow through an empty, straight pipe, by plotting a modified friction factor against a modified Reynolds number as shown in Figure 6.37 from the widely used study by Ergun [52]. In this plot, in which D_P is an effective packing material diameter, it can be seen that at low, superficial gas velocities (modified $N_{Re} < 10$), typical of laminar flow, the pressure drop per unit height is proportional to the superficial vapor velocity, u_V . At high gas velocities, typical of turbulent flow, the pressure drop per unit height approaches a dependency of the square of

the gas velocity. Most packed columns used for separations operate in the turbulent region (modified $N_{Re} > 1,000$). Thus, dry pressure-drop data shown in Figure 6.34 for Bialecki rings show an exponential dependency on gas velocity of about 1.86. Also, as shown in Figure 6.34, when liquid flows countercurrent to the gas in the preloading region, this same dependency continues, but at a higher pressure drop because the volume for gas flow decreases due to liquid holdup.

Based on extensive experimental studies using more than 50 different packing materials, including structured packings, Billet and Schultes [51, 69] developed a correlation for dry-gas pressure drop, ΔP_o , similar in form to that of Figure 6.37. Their dimensionally consistent correlating equation is

$$\frac{\Delta P_o}{l_T} = \Psi_o \frac{a}{\epsilon^3} \frac{u_V^2 \rho_V}{2} \frac{1}{K_W} \quad (6-110)$$

where

l_T = height of packing

K_W = a wall factor

K_W can be important for columns with an inadequate ratio of effective packing diameter to inside column diameter, and is given by

$$\frac{1}{K_W} = 1 + \frac{2}{3} \left(\frac{1}{1 - \epsilon} \right) \frac{D_P}{D_T} \quad (6-111)$$

where the effective packing diameter, D_P , is determined from

$$D_P = 6 \left(\frac{1 - \epsilon}{a} \right) \quad (6-112)$$

The dry-packing resistance coefficient (a modified friction factor), Ψ_o , is given by the empirical expression

$$\Psi_o = C_p \left(\frac{64}{N_{ReV}} + \frac{1.8}{N_{ReV}^{0.08}} \right) \quad (6-113)$$

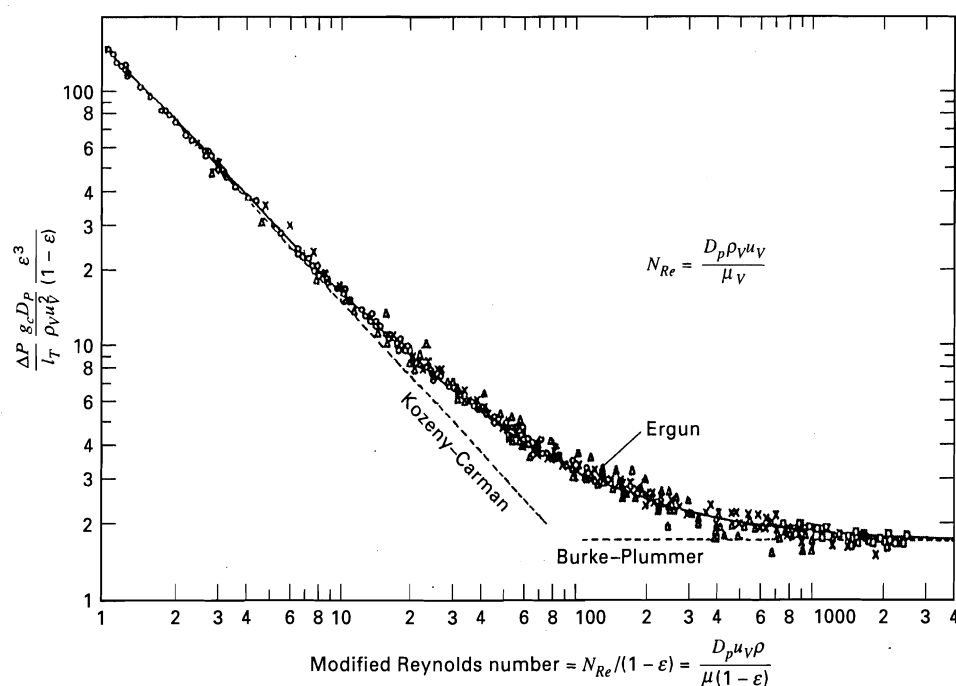


Figure 6.37 Ergun correlation for dry-bed pressure drop.

[From S. Ergun, *Chem. Eng. Prog.* 48 (2), 89-94 (1952) with permission.]

where

$$N_{ReV} = \frac{u_V D_P \rho_V}{(1 - \epsilon) \mu_V} K_W \quad (6-114)$$

and C_p is a packing constant, determined from experimental data, and tabulated for a number of packings in Table 6.8. In (6-113), the laminar-flow region is characterized by the term $64/N_{ReV}$, while the next term characterizes the more common turbulent-flow regime.

When a packed tower is irrigated with a downward-flowing liquid, the cross-sectional area for gas flow is reduced by the liquid holdup and the surface structure exposed to the gas is changed as a result of the coating of the packing with a liquid film. The pressure drop now becomes dependent on the holdup and a two-phase flow resistance, and was found by Billet and Schultes [69] to depend on the liquid-flow Froude number as follows for flow rates up to the loading point:

$$\frac{\Delta P}{\Delta P_o} = \left(\frac{\epsilon}{\epsilon - h_L} \right)^{3/2} \exp \left[\frac{13300}{a^{3/2}} (N_{FrL})^{1/2} \right] \quad (6-115)$$

where

h_L is given by (6-97) and is in m^2/m^3 ,

ϵ and a are given in Table 6.8, where a in (6-115) must be in m^2/m^3 , and

N_{FrL} is given by (6-99).

EXAMPLE 6.14

A column packed with 25-mm metal Bialecki rings is to be designed for the following vapor and liquid conditions:

	Vapor	Liquid
Mass flow rate, kg/h	515	1,361
Density, kg/m ³	1.182	1,000
Viscosity, kg/m-s	1.78×10^{-5}	1.00×10^{-3}
Molecular weight	28.4	18.02
Surface tension, kg/s ²		2.401×10^{-2}

Using the equations of Billet and Schultes, determine the vapor and liquid superficial velocities at the loading and flooding points, the specific liquid holdup at the loading point, the specific pressure drop at the loading point, and the column diameter for operation at the loading point.

SOLUTION

From Table 6.8, the following constants apply to the Bialecki rings:

$$a = 210 \text{ m}^2/\text{m}^3$$

$$\epsilon = 0.956$$

$$C_h = 0.692$$

$$C_p = 0.891$$

$$C_s = 2.521$$

First, compute the superficial vapor velocity at the loading point. From the abscissa label of Figure 6.36a,

$$F_{LV} = \frac{1,361}{515} \left(\frac{1.182}{1,000} \right)^{1/2} = 0.0908$$

Because $F_{LV} < 0.4$, $n_s = -0.326$ and C in (6-106) = $C_s = 2.521$.

From (6-106),

$$\Psi_l = \frac{9.807}{2.521^2} \left[0.0908 \left(\frac{0.001}{0.0000178} \right)^{0.4} \right]^{-2(-0.326)} = 0.923$$

$$u_{L,l} = u_{V,l} \frac{\rho_V L M_L}{\rho_L V M_V} = u_{V,l} \frac{(1.182)(1,361)}{(1,000)(515)} = 0.00312 u_{V,l}$$

From (6-107),

$$\xi_l = \left(12 \frac{(0.001)}{(9.807)(1,000)} (0.00312) u_{V,l} \right) = 3.82 \times 10^{-9} u_{V,l}$$

From (6-105),

$$\begin{aligned} u_{V,l} &= \left(\frac{9.807}{0.923} \right)^{1/2} \left[\frac{0.956}{210^{1/6}} - 210^{1/2} (3.82 \times 10^{-9} u_{V,l})^{1/3} \right] \\ &\quad \times (3.82 \times 10^{-9} u_{V,l})^{1/6} \left(\frac{1,000}{1.182} \right)^{1/2} \\ &= 3.26 \left[0.392 - 0.0227 u_{V,l}^{1/3} \right] 1.15 u_{V,l}^{1/6} \\ &= (1.47 - 0.0851 u_{V,l}^{1/3}) u_{V,l}^{1/6} \end{aligned}$$

Solving this nonlinear equation in $u_{V,l}$ gives $u_{V,l}$ = superficial vapor velocity at the loading point = 1.46 m/s.

The corresponding superficial liquid velocity = $u_{L,l}$
= $0.00312 u_{V,l} = 0.00312(1.46) = 0.00457$ m/s.

The superficial vapor flooding velocity = $u_{V,f} = \frac{u_{V,l}}{0.7} = \frac{1.46}{0.7} = 2.09$ m/s.

The corresponding superficial liquid velocity =

$$u_{L,f} = \frac{0.00457}{0.7} = 0.00653 \text{ m/s.}$$

Next, compute the specific liquid holdup at the loading point.

From (6-98) and (6-99),

$$N_{ReL} = \frac{(0.00457)(1,000)}{(210)(0.001)} = 21.8$$

$$\text{and } N_{FrL} = \frac{(0.00457)^2(210)}{9.807} = 0.000447$$

Because $N_{ReL} > 5$, (6-101) applies:

$$a_h/a = 0.85(0.692)(21.8)^{0.25}(0.000447)^{0.1} = 0.588$$

From (6-97), the specific liquid holdup at the loading point is

$$h_L = \left(12 \frac{0.000447}{21.8} \right)^{1/3} 0.588^{2/3} = 0.0440 \text{ m}^3/\text{m}^3$$

Before computing the specific pressure drop at the loading point, we must compute the column diameter for operation at the loading point.

Applying (6-103),

$$D_T = \left[\frac{4(515/3600)}{(1.46)(3.14)(1.182)} \right]^{1/2} = 0.325 \text{ m}$$

From (6-112),

$$D_P = 6 \left(\frac{1 - 0.956}{210} \right) = 0.00126 \text{ m}$$

From (6-111),

$$\frac{1}{K_W} = 1 + \frac{2}{3} \left(\frac{1}{1 - 0.956} \right) \frac{0.00126}{0.325} = 1.059 \text{ and } K_W = 0.944$$

From (6-114),

$$N_{Rev} = \frac{(1.46)(0.00126)(1.182)}{(1 - 0.956)(0.0000178)} (0.944) = 2,621$$

From (6-113),

$$\Psi_o = 0.891 \left(\frac{64}{2,621} + \frac{1.8}{2,621^{0.08}} \right) = 0.876$$

From (6-110), the specific dry-gas pressure drop is

$$\frac{\Delta P_o}{l_T} = 0.876 \frac{(210)(1.46)^2(1.182)}{(0.956)^3(2)} (1.059) \\ = 281 \text{ kg/m}^2\text{-s}^2 = \text{Pa/m}$$

From (6-115), the specific pressure drop at the loading point is

$$\frac{\Delta P}{l_T} = 281 \left(\frac{0.956}{0.956 - 0.0440} \right)^{3/2} \exp \left[\frac{13300}{210^{3/2}} (0.000447)^{1/2} \right] \\ = 331 \text{ kg/m}^2\text{-s}^2$$

or 0.406 in. of water/ft

Mass-Transfer Efficiency

The mass-transfer efficiency of a packed column is incorporated in the HETP or the more theoretically based HTUs and volumetric mass-transfer coefficients. Although the HETP concept lacks a sound theoretical basis, its simplicity, coupled with the relative ease with which equilibrium-stage calculations can be made with computer-aided simulation programs, has made it a widely used method for estimating packing height. In the preloading region and where good distribution of vapor and liquid is initiated and maintained, values of the HETP depend mainly on packing type and size, liquid viscosity, and surface tension. For rough estimates the following relations, taken from Kister [33], can be used.

1. Pall rings and similar high-efficiency random packings with low-viscosity liquids:

$$\text{HETP, ft} = 1.5D_P, \text{ in.} \quad (6-116)$$

2. Structured packings at low-to-moderate pressure with low-viscosity liquids:

$$\text{HETP, ft} = 100/a, \text{ ft}^2/\text{ft}^3 + 4/12 \quad (6-117)$$

3. Absorption with viscous liquid:

$$\text{HETP} = 5 \text{ to } 6 \text{ ft}$$

4. Vacuum service:

$$\text{HETP, ft} = 1.5D_P, \text{ in.} + 0.5 \quad (6-118)$$

5. High-pressure service (> 200 psia):

HETP for structured packings may be greater than predicted by (6-117)

6. Small-diameter columns, $D_T < 2$ ft:

$$\text{HETP, ft} = D_T, \text{ ft, but not less than } 1 \text{ ft}$$

In general, lower values of HETP are achieved with smaller-size random packings, particularly in small-diameter columns, and with structured packings, particularly those with large values of a , the packing surface area per packed volume. The experimental data of Figure 6.38 for no. 2 (2-in.-diameter) Nutter rings from Kunesh [53] show that in the preloading

region, the HETP is relatively independent of the vapor-flow F -factor:

$$F = u_V(\rho_V)^{0.5} \quad (6-119)$$

provided that the ratio L/V is maintained constant as the superficial gas velocity, u_V , is increased. Beyond the loading point, and as the flooding point is approached, the HETP can increase dramatically like the pressure drop and liquid holdup.

Experimental mass-transfer data for packed columns are usually correlated in terms of volumetric mass-transfer coefficients and/or HTUs, rather than in terms of HETPs. The data are obtained from experiments in which either the liquid-phase or the gas-phase mass-transfer resistance is negligible, so that the other resistance can be studied and correlated independently. For applications where both resistances may be important, the two resistances are added together according to the two-film theory of Whitman [54], as discussed in Chapter 3, to obtain the overall resistance. This theory assumes the absence of any mass-transfer resistance at the interface between the gas and liquid phases. Thus, the two phases are in equilibrium at the interface.

The two-film theory defines an overall coefficient in terms of the individual volumetric mass-transfer coefficients discussed in Section 6.7. Most commonly, reference is made to the overall gas-phase resistance, (6-84),

$$\frac{1}{K_y a} = \frac{1}{k_y a} + \frac{K}{k_x a}$$

for mass-transfer rates expressed in terms of mole-fraction driving forces by (6-77),

$$r = k_y a(y - y_1) = k_x a(x_1 - x) = K_y a(y - y^*)$$

where K is the vapor-liquid equilibrium ratio.

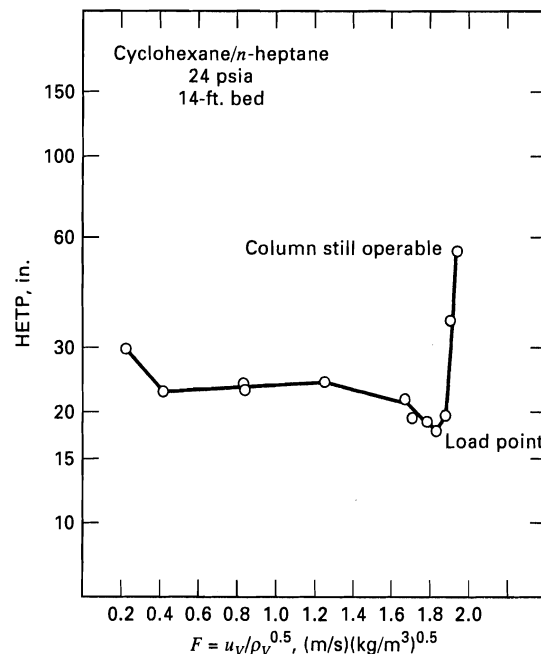


Figure 6.38 Effect of F -factor on HETP.

Alternatively, as summarized in Table 6.7, mass-transfer rates can be expressed in terms of liquid-phase concentrations and gas-phase partial pressure

$$r = k_p a(p - p_1) = k_L a(c_1 - c) = K_G a(p - p^*) \quad (6-120)$$

If we define a Henry's-law constant at the equilibrium interface between the two phases by

$$p_1 = H' c_1 \quad (6-121)$$

and let

$$p^* = H' c \quad (6-122)$$

then

$$\frac{1}{K_G a} = \frac{1}{k_p a} + \frac{H'}{k_L a} \quad (6-123)$$

Alternatively, expressions can be derived for $K_x a$ and $K_L a$.

It should be noted that the units of various mass transfer coefficients differ:

	SI Units	American Engineering Units
r	mol/m ³ -s	lbmol/ft ³ -h
$k_y a, k_x a, K_x a, K_y a$	mol/m ³ -s	lbmol/ft ³ -h
$k_p a, K_G a$	mol/m ³ -s-kPa	lbmol/ft ³ -h-atm
$k_L a, k_G a, k_c a$	s ⁻¹	h ⁻¹
k_L, k_G, k_c	m/s	ft/h

Instead of using mass-transfer coefficients directly for column design, the transfer-unit concept of Chilton and Colburn [43,44] is often employed because HTUs: (1) have only one dimension (length), (2) generally vary with column conditions less than mass-transfer coefficients, and (3) are related to an easily understood geometrical quantity, namely, height per theoretical stage. Definitions of individual and overall HTUs are included in Table 6.7 for the dilute case. By substituting these definitions into (6-84),

$$H_{OG} = H_G + (KV/L)H_L \quad (6-124)$$

Alternatively, an expression can be derived for H_{OL} . In the absorption or stripping of very insoluble gases, the solute K -value or Henry's law constant, H' in (6-112), is very large, making the last terms in (6-84), (6-123), and (6-124) large such that the resistance of the gas phase is negligible and the rate of mass transfer is controlled by the liquid phase. Such data can then be used to study the effect of the variables on the volumetric, liquid-phase mass-transfer coefficient and HTU. Typical data are shown in Figure 6.39 for three different-size Berl-saddle packings for the stripping of oxygen from water by air, in a 20-in.-I.D. column operated at near-ambient temperature and pressure in the preloading region, as reported in an early study by Sherwood and Holloway [55]. The effect of liquid velocity on $k_L a$ is seen to be quite pronounced, with $k_L a$ increasing at about the 0.75 power of the liquid mass velocity. Gas velocity was observed to have no effect on $k_L a$ in the preloading region. Also included in Figure 6.39 are the data plotted in terms of H_L , where

$$H_L = \frac{M_L L}{\rho_L k_L a S} \quad (6-125)$$

As seen, H_L does not depend as strongly as $k_L a$ on liquid mass velocity, $M_L L/S$.

Another system for which the rate of mass transfer is controlled by the liquid phase is CO₂-air-H₂O, where CO₂ can be either absorbed or stripped. Measurements on this system for a variety of modern metal, ceramic, and plastic packings are reported by Billet [45]. Data on the effect of liquid loading on $k_L a$ in the preloading region for two different-size ceramic Hiflow ring packings are shown in Figure 6.40. The effect of gas velocity on $k_L a$ in terms of the F -factor at a constant liquid rate is shown in Figure 6.41 for the same system, but with 50-mm plastic Pall rings and Hiflow rings. Up to an F -factor value of about 1.8 m^{-1/2}-s⁻¹-kg^{1/2}, which is in the preloading region, no effect of gas velocity is

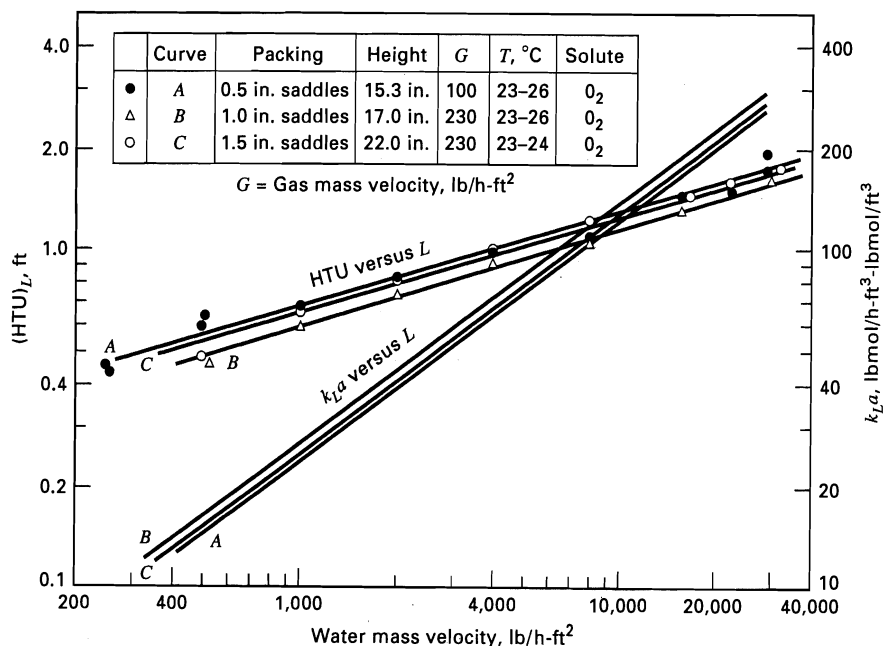


Figure 6.39 Effect of liquid rate on liquid-phase mass transfer of O₂. [From T.K. Sherwood and F.A.L. Holloway, *Trans. AIChE.*, 36, 39-70 (1940) with permission.]

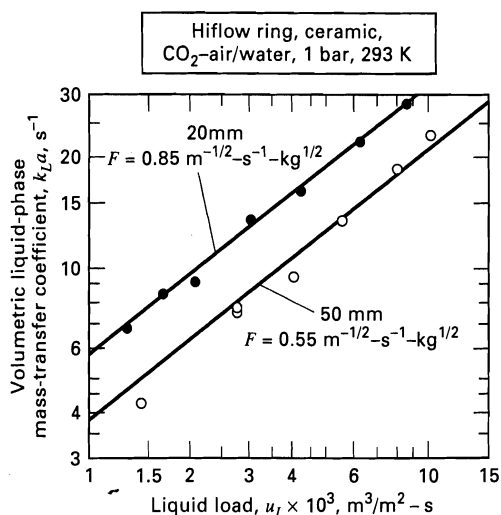


Figure 6.40 Effect of liquid load on liquid-phase mass transfer of CO₂.

[From R. Billet, *Packed Column Analysis and Design*, Ruhr-University Bochum (1989) with permission.]

observed. Above the loading limit, $k_L a$ increases with increasing gas velocity because of increased liquid holdup, which increases interfacial surface area for mass transfer. Although it is not illustrated in Figures 6.39 to 6.41, another major factor that influences the rate of mass transfer in the liquid phase is the solute molecular diffusivity in the solvent. For a given packing, experimental data on different systems in the preloading region can usually be correlated satisfactorily by the following empirical expression, which includes only the liquid velocity and liquid diffusivity:

$$k_L a = C_1 D_L^{1/2} u_L^n \quad (6-126)$$

where n has been observed by different investigators to vary from about 0.6 to 0.95, with 0.75 being a typical value. The exponent on the diffusivity is consistent with the penetration theory discussed in Chapter 3.

A convenient system for studying gas-phase mass transfer is NH₃-air-H₂O. The high solubility of NH₃ in H₂O corresponds to a relatively low K -value. Accordingly, the last terms in (6-84), (6-123), and (6-124) may be negligible so that the

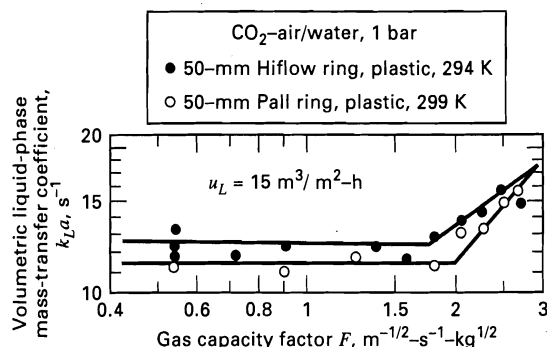


Figure 6.41 Effect of gas rate on liquid-phase mass transfer of CO₂.

[From R. Billet, *Packed Column Analysis and Design*, Ruhr-University Bochum (1989) with permission.]

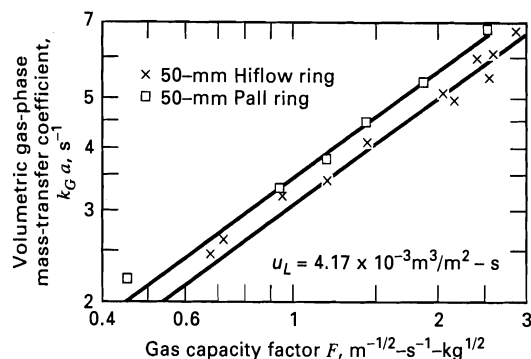


Figure 6.42 Effect of gas rate on gas-phase mass transfer of NH₃.

[From R. Billet, *Packed Column Analysis and Design*, Ruhr-University Bochum (1989) with permission.]

gas-phase resistance controls the rate of mass transfer. The small effect of the liquid-phase resistance can be backed out using a correlation such as (6-126). The typical effect of superficial vapor velocity, expressed in terms of the F -factor of (6-119), on the volumetric, gas-phase mass-transfer coefficient in the preloading region is shown in Figure 6.42 for two different plastic packings at the same liquid velocity. The coefficients are proportional to about the 0.75 power of F . Figure 6.43 shows that the liquid velocity also affects the gas-phase mass-transfer coefficient, probably because as the liquid rate is increased, the holdup increases and more interfacial surface is created.

The volumetric, gas-phase mass-transfer coefficients, $k_G a$, plotted in Figures 6.42 and 6.43, are based on gas-phase molar concentrations. Thus, they have the same units as $k_L a$. For a given packing, experimental data on $k_p a$ or $k_G a$ for different systems in the preloading region can usually be correlated satisfactorily with empirical correlations of the form

$$k_p a = C_2 D_G^{0.67} F^{m'} u_L^{n'} \quad (6-127)$$

where D_G is the gas diffusivity of the solute and m' and n' have been observed by different investigators to vary from 0.65 to 0.85 and from 0.25 to 0.5, respectively, a typical value for m' being 0.8.

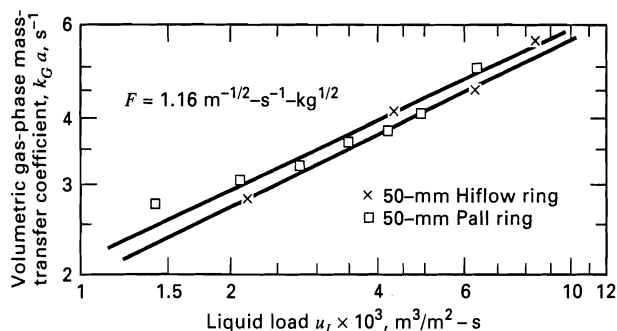


Figure 6.43 Effect of liquid rate on gas-phase mass transfer of NH₃.

[From R. Billet, *Packed Column Analysis and Design*, Ruhr-University Bochum (1989) with permission.]

Table 6.9 Generalized Correlations for Mass Transfer in Packed Columns

Investigator	Year	Ref. No.	Type of Correlations	Packings
Shulman et al.	1955	64	k_p, k_L, a	Raschig rings, Berl saddles
Cornell et al.	1960	56, 57	H_G, H_L	Raschig rings, Berl saddles
Onda et al.	1968	65	k_p, k_L, a	Raschig rings, Berl saddles
Bolles and Fair	1979, 1982	58, 59	H_G, H_L	Raschig rings, Berl saddles, Pall rings
Bravo and Fair	1982	60	a	Raschig rings, Berl saddles, Pall rings
Bravo et al.	1985	61	k_G, k_L	Sulzer
Fair and Bravo	1987	62	k_G, k_L, a	Sulzer, Gempak, Mellapak, Montz, Ralu Pak
Fair and Bravo	1991	63	k_G, k_L, a	Flexipac, Gempak, Intalox 2T, Montz, Mellapak, Sulzer
Billet and Schultes	1991	67	$k_G a, k_L a$	14 random packings and 4 structured packings
Billet and Schultes	1999	69	$k_G a, k_L a$	19 random packings and 6 structured packings

The development of separate generalized correlations for gas- and liquid-phase mass-transfer coefficients and/or HTUs, which began with the study of Sherwood and Holloway [55] on the liquid phase, has led to a significant number of empirical and semitheoretical equations, most of which are based on the application of the two-film theory by Fair and co-workers [56–63] and others [64, 65]. In some cases, values of k_G and k_L are correlated separately from a ; in others, the combinations $k_G a$ and $k_L a$ are used. Important features of some of these correlations are summarized in Table 6.9. The development of such correlations from experimental data is difficult because, as shown by Billet [66] in a comprehensive study with metal Pall rings, values of the mass-transfer coefficients are significantly affected by the technique used to pack the column and the number of liquid feed-distribution points per unit of column cross section, when this number is less than 10 points/ft². When 25 points/ft² are used and $D_T/D_P > 10$, column diameter has little, if any, effect on mass-transfer coefficients for packed heights up to 20 ft.

In an extensive investigation, Billet and Schultes [67] measured and correlated volumetric mass-transfer coefficients and HTUs for 31 different binary and ternary chemical systems with 67 different types and sizes of packings in columns of diameter ranging from 2.4 in. to 4.6 ft. Additional data are reported by Billet and Schultes [69], particularly for Hiflow rings and Raschig Super-rings. The systems include some for which mass-transfer resistance resides mainly in the liquid phase and others for which resistance in the gas phase is predominant. They assume uniform distribution of gas and liquid over the cross-sectional area of the column and apply the two-film theory of mass transfer discussed in Chapter 3. For the liquid-phase resistance, they assume that the liquid flows in a thin film through the irregular channels of the packing, with continual remixing of the liquid at points of contact with the packing such that Higbie's penetration theory of diffusion [68], as developed in Chapter 3, can be applied. Thus, for the diffusing component, in terms of concentration units, the volumetric mass-transfer coefficient is defined by

$$r = (k_L a_{Ph})(c_{L1} - c_L) \quad (6-128)$$

From the penetration theory of Higbie, (3-194),

$$k_L = 2(D_L/\pi t_L)^{0.5} \quad (6-129)$$

where t_L = time of exposure of the liquid film before remixing. Billet and Schultes assume that this time is governed by a length of travel equal to the hydraulic diameter of the packing:

$$t_L = h_L d_H / u_L \quad (6-130)$$

where d_H , the hydraulic diameter, is equal to $4r_H$ or $4\epsilon/a$. Thus, in terms of the height of a liquid transfer unit, (6-129) and (6-130) give

$$H_L = \frac{u_L}{k_L a_{Ph}} = \frac{\sqrt{\pi}}{2} \left(\frac{4h_L \epsilon}{D_L a u_L} \right)^{1/2} \frac{u_L}{a_{Ph}} \quad (6-131)$$

Equation (6-131) was modified to include an empirical constant, C_L , which is back-calculated for each packing to fit the experimental data. The final predictive equation given by Billet and Schultes is

$$H_L = \frac{1}{C_L} \left(\frac{1}{12} \right)^{1/6} \left(\frac{4h_L \epsilon}{D_L a u_L} \right)^{1/2} \frac{u_L}{a} \left(\frac{a}{a_{Ph}} \right) \quad (6-132)$$

where values of C_L are included in Table 6.8.

A similar development was made by Billet and Schultes for the gas-phase resistance, except that the time of exposure of the gas between periods of mixing was determined empirically, to give

$$H_G = \frac{1}{C_V} (\epsilon - h_L)^{1/2} \left(\frac{4\epsilon}{a^4} \right)^{1/2} (N_{Re_V})^{-3/4} (N_{Sc_V})^{-1/3} \left(\frac{u_V a}{D_G a_{Ph}} \right) \quad (6-133)$$

where C_V is included in Table 6.8 and

$$N_{Re_V} = \frac{u_V \rho_V}{a \mu_V} \quad (6-134)$$

$$N_{Sc_V} = \frac{\mu_V}{\rho_V D_V} \quad (6-135)$$

Equations (6-132) and (6-133) contain an area ratio, a_{Ph}/a , which is the ratio of the phase interface area to the packing surface area, which from Billet and Schultes [69] is not the same as the hydraulic area ratio, a_h/a , given by (6-100) and

(6-101). Instead, they give the following correlation:

$$\frac{a_{Ph}}{a} = 1.5(ad_h)^{-1/2}(N_{ReL,h})^{-0.2}(N_{WeL,h})^{0.75}(N_{FrL,h})^{-0.45} \quad (6-136)$$

where

$$d_h = \text{packing hydraulic diameter} = 4\frac{\epsilon}{a} \quad (6-137)$$

and the following liquid-phase dimensionless groups use the packing hydraulic diameter as the characteristic length:

$$\text{Reynolds number} = N_{ReL,h} = \frac{u_L d_h \rho_L}{\mu_L} \quad (6-138)$$

$$\text{Weber number} = N_{WeL,h} = \frac{u_L^2 \rho_L d_h}{\sigma} \quad (6-139)$$

$$\text{Froude number} = N_{FrL,h} = \frac{u_L^2}{g d_h} \quad (6-140)$$

Following the estimation of H_L and H_G from (6-132) and (6-133), respectively, the overall HTU value can be determined from (6-124), followed by the determination of packed height from

$$l_T = H_{OG} N_{OG} \quad (6-141)$$

where the determination of N_{OG} is discussed in Section 6.7.

EXAMPLE 6.15

For the absorption of ethyl alcohol from CO_2 with water, as considered in Example 6.1, a 2.5-ft-I.D. tower, packed with 1.5-in. metal Pall-like rings, is to be used. It is estimated that the tower will operate in the preloading region with a pressure drop of approximately 1.5 in. of water head per foot of packed height. From Example 6.9, the required number of overall transfer units based on the gas phase is 7.5. Estimate H_G , H_L , H_{OG} , HETP, and the required packed height in feet using the following estimates of flow conditions and physical properties at the bottom of the packing:

	Vapor	Liquid
Flow rate, lb/h	17,480	6,140
Molecular weight	44.05	18.7
Density, lb/ft ³	0.121	61.5
Viscosity, cP	0.0145	0.63
Surface tension, dynes/cm	—	101
Diffusivity of ethanol, m ² /s	7.75×10^{-6}	1.82×10^{-9}
Kinematic viscosity, m ² /s	0.75×10^{-5}	0.64×10^{-6}

SOLUTION

Cross-sectional area of tower = $(3.14)(2.5)^2/4 = 4.91 \text{ ft}^2$.

Volumetric liquid flow rate = $6,140/61.5 = 99.8 \text{ ft}^3/\text{h}$.

u_L = superficial liquid velocity = $99.8/[(4.91)(3,600)] = 0.0056 \text{ ft/s}$ or 0.0017 m/s.

From Section 6.8, $u_L > u_{L,\min}$, but the velocity is on the low side.

u_V = superficial gas velocity = $17,480/[(0.121)(4.91)(3,600)] = 8.17 \text{ ft/s} = 2.49 \text{ m/s}$.

Let the packing characteristics for the 1.5-inch metal Pall-like rings be as follows (somewhat different from values for Pall rings

in Table 6.8):

$$a = 149.6 \text{ m}^2/\text{m}^3, \quad \epsilon = 0.952$$

$$C_h = \text{approximately } 0.7, \quad C_L = 1.227, \quad C_V = 0.341$$

Estimation of specific liquid holdup, h_L :

From (6-98),

$$N_{ReL} = \frac{0.0017}{(0.64 \times 10^{-6})(149.6)} = 17.8.$$

From (6-99),

$$N_{FrL} = \frac{(0.0017)^2(149.6)}{9.8} = 4.41 \times 10^{-5}$$

From (6-101),

$$\frac{a_h}{a} = 0.85(0.7)(17.8)^{0.25}(4.41 \times 10^{-5})^{0.10} = 0.045$$

$$a_h = 0.045(149.6) = 6.73 \text{ m}^2/\text{m}^3$$

From (6-97),

$$h_L = \left[\frac{12(4.41 \times 10^{-5})}{17.8} \right]^{1/3} (0.045)^{2/3} = 0.0128 \text{ m}^3/\text{m}^3$$

Estimation of H_L :

First compute a_{Ph} , the ratio of phase interface area to packing surface area.

From (6-137),

$$d_h = 4\frac{0.952}{149.6} = 0.0255 \text{ m}$$

From (6-138),

$$N_{ReL,h} = \frac{(0.0017)(0.0255)}{(0.64 \times 10^{-6})} = 67.7$$

From (6-139),

$$N_{WeL,h} = \frac{(0.0017)^2[(61.5)(16.02)](0.0255)}{[(101)(0.001)]} = 0.000719$$

From (6-140),

$$N_{FrL,h} = \frac{(0.0017)^2}{(9.807)(0.0255)} = 1.156 \times 10^{-5}$$

From (6-136),

$$\frac{a_{Ph}}{a} = 1.5(149.6)^{-1/2}(0.0255)^{-1/2}(67.7)^{-0.2} \times (0.000719)^{0.75}(1.156 \times 10^{-5})^{-0.45} = 0.242$$

Estimation of H_L :

From (6-132), using consistent SI units:

$$H_L = \frac{1}{1.227} \left(\frac{1}{12} \right)^{1/6} \left[\frac{(4)(0.0128)(0.952)}{(1.82 \times 10^{-9})(149.6)(0.0017)} \right]^{1/2} \times \left(\frac{0.0017}{149.6} \right) \left(\frac{1}{0.242} \right) = 0.26 \text{ m} = 0.85 \text{ ft}$$

Estimation of H_G :

From (6-134),

$$N_{ReV} = 2.49/[(149.6)(0.75 \times 10^{-5})] = 2,220$$

From (6-135),

$$N_{ScV} = 0.75 \times 10^{-5}/7.75 \times 10^{-6} = 0.968$$

From (6-133), using consistent SI units,

$$H_G = \frac{1}{0.341} (0.952 - 0.0128)^{1/2} \left[\frac{(4)(0.952)}{(149.6)^4} \right]^{1/2} \\ \times (2220)^{-3/4} (0.968)^{-1/3} \left[\frac{(2.49)}{7.75 \times 10^{-6} (0.242)} \right] \\ = 1.03 \text{ m or } 3.37 \text{ ft}$$

Estimation of H_{OG} :

From Example 6.1, the K -value for ethyl alcohol = 0.57,

$$V = 17,480/44.05 = 397 \text{ lbmol/h,}$$

$$L = 6,140/18.7 = 328 \text{ lbmol/h,}$$

and $1/A = KV/L = (0.57)(397)/328 = 0.69$

From (6-124),

$$H_{OG} = 3.37 + 0.69(0.85) = 3.96 \text{ ft}$$

The mass-transfer resistance in the gas phase is much larger than that in the liquid phase.

Estimation of Packed Height:

From (6-141),

$$l_T = 3.96(7.5) = 29.7 \text{ ft}$$

Estimation of HETP:

From (6-94), for straight operating and equilibrium lines, with $A = 1/0.69 = 1.45$,

$$\text{HETP} = 3.96 \left[\frac{\ln(0.69)}{(1 - 1.45)/1.45} \right] = 4.73 \text{ ft}$$

6.9 CONCENTRATED SOLUTIONS IN PACKED COLUMNS

When the solute concentration in the gas and/or liquid is concentrated so that the operating line and/or equilibrium line are noticeably curved, then the procedure given in Section 6.7 for determining N_{OG} and l_T cannot be used because (6-91) cannot be analytically integrated to give (6-93). Instead, alternative methods can be employed or the computer-aided methods discussed in Chapters 10 and 11 can be applied.

For concentrated solutions, the two columns in Table 6.7 labeled UM (unimolecular) diffusion apply. To obtain these columns from the two columns labeled EM (equimolar) diffusion, we let

$$L' = L(1 - x) \quad \text{and} \quad V' = V(1 - y)$$

where L' and V' are the constant flow rates of the inert (solvent) liquid and (carrier) gas, respectively on a solute-free basis. Then

$$d(Vy) = V'd \left(\frac{y}{1 - y} \right) = V' \frac{dy}{(1 - y)^2} = V \frac{dy}{(1 - y)} \quad (6-141)$$

$$d(Lx) = L'd \left(\frac{x}{1 - x} \right) = L' \frac{dx}{(1 - x)^2} = L \frac{dx}{(1 - x)} \quad (6-142)$$

Equation (6-88) now becomes

$$l_T = \int_{y_2}^{y_1} \left(\frac{V}{K'_y a S} \right) \frac{dy}{(1 - y)(y - y^*)} \\ = \frac{V}{K'_y a S} \int_{y_2}^{y_1} \frac{dy}{(1 - y)(y - y^*)} \quad (6-143)$$

where 1 refers to inlet and 2 refers to outlet conditions. Based on the liquid phase,

$$l_T = \int_{x_1}^{x_2} \left(\frac{L}{K'_x a S} \right) \frac{dx}{(1 - x)(x^* - x)} \\ = \frac{L}{K'_x a S} \int_{x_1}^{x_2} \frac{dx}{(1 - x)(x^* - x)} \quad (6-144)$$

where the overall mass-transfer coefficients are primed to signify UM diffusion.

If the numerators and denominators of (6-143) and (6-144) are multiplied by $(1 - y)_{LM}$ and $(1 - x)_{LM}$, respectively, where $(1 - y)_{LM}$ is the log mean of $(1 - y)$ and $(1 - y^*)$, and $(1 - x)_{LM}$ is the log mean of $(1 - x)$ and $(1 - x^*)$, we obtain the expressions in rows 1 and 6 of columns 4 and 7 in Table 6.7:

$$l_T = \int_{y_2}^{y_1} \left[\frac{V}{K'_y a (1 - y)_{LM} S} \right] \frac{(1 - y)_{LM} dy}{(1 - y)(y - y^*)} \\ = \frac{V}{K'_y a (1 - y)_{LM} S} \int_{y_2}^{y_1} \frac{(1 - y)_{LM} dy}{(1 - y)(y - y^*)} \quad (6-145)$$

$$l_T = \int_{x_1}^{x_2} \left[\frac{L}{K'_x a (1 - x)_{LM} S} \right] \frac{(1 - x)_{LM} dx}{(1 - x)(x^* - x)} \\ = \frac{L}{K'_x a (1 - x)_{LM} S} \int_{x_1}^{x_2} \frac{(1 - x)_{LM} dx}{(1 - x)(x^* - x)} \quad (6-146)$$

In these equations $K'_y(1 - y)_{LM}$ is equal to the concentration-independent K_y , and $K'_x(1 - x)_{LM}$ is equal to the concentration-independent K_x . If there is appreciable absorption, vapor flow rate V decreases from the bottom to the top of the absorber. However, the values of Ka are also a function of flow rate, such that the ratio V/Ka is approximately constant and HTU groupings, $[L/K'_x a (1 - x)_{LM} S]$ and $[V/K'_y a (1 - y)_{LM} S]$, can often be taken out of the integral sign without incurring errors larger than those inherent in experimental measurements of Ka . Usually, average values of V , L , and $(1 - y)_{LM}$ are used.

Another approach is to leave all of the terms in (6-145) or (6-146) under the integral sign and evaluate l_T by a stepwise or graphical integration. In either case, to obtain the terms $(y - y^*)$ or $(x^* - x)$, the equilibrium and operating lines must be established. The equilibrium curve is determined from appropriate thermodynamic data or correlations. To establish the operating line, which will not be straight if the solutions are concentrated, the appropriate material-balance equations must be developed. With reference to Figure 6.29, an overall balance around the upper part of the absorber gives

$$V + L_{in} = V_{out} + L \quad (6-147)$$

Similarly a balance around the upper part of the absorber for the component being absorbed, assuming a pure-liquid absorbent, gives:

$$Vy = V_{\text{out}}y_{\text{out}} + Lx \quad (6-148)$$

An absorbent balance around the upper part of the absorber is:

$$L_{\text{in}} = L(1 - x) \quad (6-149)$$

Combining (6-147) to (6-149) to eliminate V and L gives

$$y = \frac{V_{\text{out}}y_{\text{out}} + [L_{\text{in}}x/(1 - x)]}{V_{\text{out}} + [L_{\text{in}}x/(1 - x)]} \quad (6-150)$$

Equation (6-150) allows the y - x operating line to be calculated from a knowledge of terminal conditions only.

A simpler approach to the problem of concentrated gas or liquid mixtures is to linearize the operating line by expressing all concentrations in mole ratios, and the gas and liquid flows on a solute-free basis, that is, $V' = (1 - y)V$, $L' = (1 - x)L$. Then, in place of (6-145) and (6-146), we have

$$l_T = \int_{Y_2}^{Y_1} \left(\frac{V'}{K_Y a S} \right) \frac{dY}{(Y - Y^*)} = \frac{V'}{K_Y a S} \int_{Y_2}^{Y_1} \frac{dY}{(Y - Y^*)} \quad (6-151)$$

$$l_T = \int_{X_1}^{X_2} \left(\frac{L'}{K_X a S} \right) \frac{dX}{(X^* - X)} = \frac{L'}{K_X a S} \int_{X_1}^{X_2} \frac{dX}{(X^* - X)} \quad (6-152)$$

This set of equations is listed in rows 3 and 8 of Table 6.7.

EXAMPLE 6.16

To remove 95% of the ammonia from an air stream containing 40% ammonia by volume, 488 lbmol/h of an absorbent per 100 lbmol/h of entering gas are to be used, which is greater than the minimum requirement.

Equilibrium data are given in Figure 6.44. Pressure is 1 atm and temperature is assumed constant at 298 K. Calculate the number of transfer units by:

- Equation (6-145) using a curved operating line determined from (6-150)
- Equation (6-151) using mole ratios.

SOLUTION

(a) Take as a basis $L_{\text{in}} = 488$ lbmol/h. Then $V_{\text{out}} = 100 - (40)(0.95) = 62$ lbmol/h, and $y_{\text{out}} = (0.05)(40)/62 = 0.0323$. From (6-150), it is possible to construct the curved operating line of Figure 6.44. For example, if $x = 0.04$,

$$y = \frac{(62)(0.0323) + [(488)(0.04)/(1 - 0.04)]}{62 + [(488)(0.04)/(1 - 0.04)]} = 0.27$$

It is now possible to calculate the following values of y , y^* , $(1 - y)_{\text{LM}} = [(1 - y) - (1 - y^*)]/\ln[(1 - y)/(1 - y^*)]$, and $(1 - y)_{\text{LM}}/[(1 - y)(y - y^*)]$ for use in (6-145). For example, in Figure 6.44, for $x = 0.044$, y (on the operating line) = 0.30, and y^*

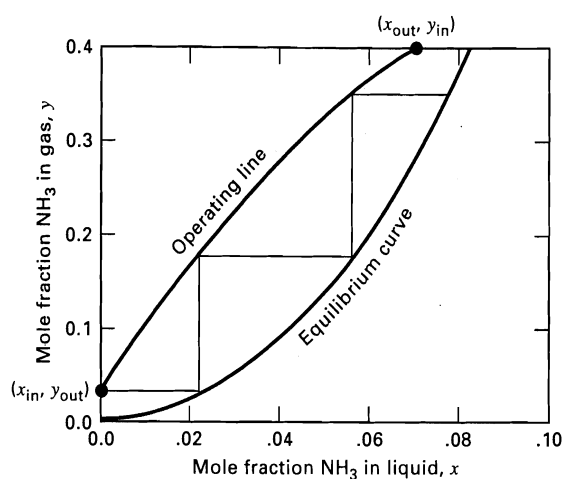


Figure 6.44 Determination of the number of theoretical stages for Example 6.15.

(on the equilibrium curve) = 0.12, from which the other four quantities in the following table follow.

y	y^*	$(y - y^*)$	$(1 - y)$	$(1 - y)_{\text{LM}}$	$\frac{(1 - y)_{\text{LM}}}{(1 - y)(y - y^*)}$
0.03	0.002	0.028	0.97	0.99	36.47
0.05	0.005	0.045	0.95	0.97	22.68
0.10	0.01	0.09	0.90	0.94	11.60
0.15	0.025	0.125	0.85	0.91	8.56
0.20	0.04	0.16	0.80	0.89	6.95
0.25	0.08	0.17	0.75	0.85	6.66
0.30	0.12	0.18	0.70	0.82	6.51
0.35	0.17	0.18	0.65	0.73	6.24
0.40	0.26	0.14	0.60	0.67	7.97

Note that since $(1 - y) \approx (1 - y)_{\text{LM}}$, these two terms frequently cancel out of the NTU equations, particularly when y is small.

Figure 6.45 is a plot of $(1 - y)_{\text{LM}}/[(1 - y)(y - y^*)]$ versus y to determine N_{OG} . The integral on the right-hand side of (6-145), between $y = 0.4$ and $y = 0.0322$, is $3.44 = N_{\text{OG}}$. This is approximately 1 more than the number of equilibrium stages of 2.6, as seen in the steps of Figure 6.44.

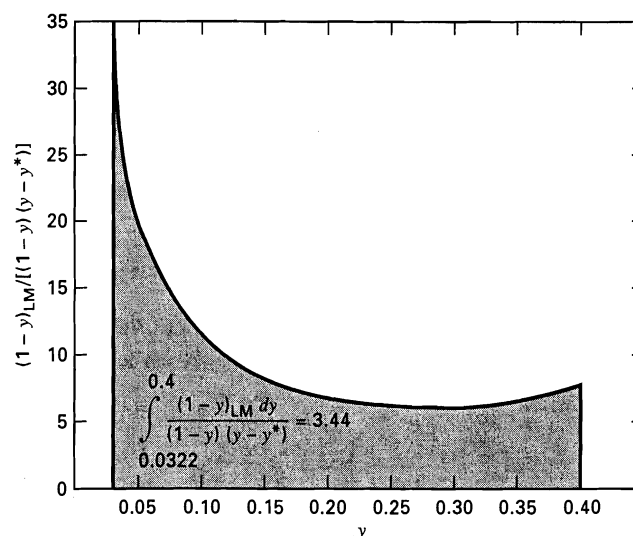


Figure 6.45 Determination of the number of transfer units for Examples 6.15.

(b) It is a simple matter to obtain the following values for $Y = y/(1 - y)$, $Y^* = y^*/(1 - y^*)$, $(Y - Y^*)$, and $(Y - Y^*)^{-1}$.

y	Y	y^*	Y^*	$(Y - Y^*)^{-1}$
0.03	0.031	0.002	0.002	34.48
0.05	0.053	0.005	0.005	20.83
0.1	0.111	0.01	0.010	9.9
0.15	0.176	0.025	0.026	6.66
0.20	0.250	0.04	0.042	4.8
0.25	0.333	0.08	0.087	4.06
0.30	0.43	0.12	0.136	3.40
0.35	0.54	0.17	0.205	2.98
0.40	0.67	0.26	0.310	2.78

SUMMARY

1. A liquid can be used to selectively absorb one or more components from a gas mixture. A gas can be used to selectively desorb or strip one or more components from a liquid mixture.
2. The fraction of a component that can be absorbed or stripped in a countercurrent cascade depends on the number of equilibrium stages and the absorption factor, $A = L/(KV)$, or the stripping factor, $S = KV/L$, respectively.
3. Absorption and stripping are most commonly conducted in trayed towers equipped with sieve or valve trays, or in towers packed with random or structured packings.
4. Absorbers are most effectively operated at high pressure and low temperature. The reverse is true for stripping. However, high costs of gas compression, refrigeration, and vacuum often preclude operation at the most thermodynamically favorable conditions.
5. For a given gas flow rate and composition, a desired degree of absorption of one or more components, a choice of absorbent, and an operating temperature and pressure, there is a minimum absorbent flow rate, given by (6-9) to (6-11), that corresponds to the use of an infinite number of equilibrium stages. For the use of a finite and reasonable number of stages, an absorbent rate of 1.5 times the minimum is typical. A similar criterion, (6-12), holds for a stripper.
6. The number of equilibrium stages required for a selected absorbent or stripping agent flow rate for the absorption or stripping of a dilute solution can be determined from the equilibrium line, (6-1), and an operating line, (6-3) or (6-5), using graphical, algebraic, or numerical methods. Graphical methods, such as Figure 6.11, offer considerable visual insight into stage-by-stage changes in compositions of the gas and liquid streams.
7. Rough estimates of overall stage efficiency, defined by (6-21), can be made with the correlations of Drickamer and Bradford (6-22), O'Connell (6-23), and Figure 6.15. More accurate and reliable procedures involve the use of a small Oldershaw column or semitheoretical equations, e.g., of Chan and Fair, based on mass-transfer considerations, to determine a Murphree vapor-point

Graphical integration of the right-hand-side integral of (6-151) is carried out by determining the area under the curve of Y versus $(Y - Y^*)^{-1}$ between $Y = 0.67$ and $Y = 0.033$. The result is $N_{OG} = 3.46$. Alternatively, the numerical integration can be performed on a computer with a spreadsheet.

It must be pointed out that for concentrated solutions, the assumption of constant temperature may not be valid and can result in a large error. If an overall energy balance indicates a temperature change that alters the equilibrium curve significantly, it is best to use a computer-aided method that includes the energy balance. Such methods are presented in Chapter 10.

efficiency, (6-30), from which a Murphree vapor tray efficiency can be estimated from (6-31) to (6-34), which can then be related to the overall efficiency using (6-37).

8. Tray diameter can be determined from (6-44) based on entrainment flooding considerations using Figure 6.24. Tray vapor pressure drop, the weeping constraint, entrainment, and downcomer backup can be estimated from (6-49), (6-68), (6-69), and (6-70), respectively.

9. Packed-column height can be estimated using the HETP, (6-73), or HTU/NTU, (6-89), concepts, with the latter having a more fundamental theoretical basis in the two-film theory of mass transfer. For straight equilibrium and operating lines, HETP is related to the HTU by (6-94), and the number of equilibrium stages is related to the NTU by (6-95).

10. Below a so-called loading point, in a preloading region, the liquid holdup in a packed column is independent of the vapor velocity. The loading point is typically about 70% of the flooding point, and most packed columns are designed to operate in the preloading region at from 50% to 70% of flooding. From the GPDC chart of Figure 6.36, the flooding point can be estimated, from which the column diameter can be determined with (6-102). The loading point can be estimated from (6-105).

11. One significant advantage of a packed column is its relatively low pressure drop per unit of packed height, as compared to a trayed tower. Packed-column pressure drop can be roughly estimated from Figure 6.36 or more accurately from (6-106) or (6-115).

12. Numerous rules of thumb are available for estimating the HETP of packed columns. However, the preferred approach is to estimate H_{OG} from separate semitheoretical mass-transfer correlations for the liquid and gas phases, such as those of (6-132) and (6-133) based on the extensive experimental work of Billet and Schultes.

13. Determination of theoretical stages for concentrated solutions involves numerical integration because of curved equilibrium and/or operating lines.

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EXERCISES

Section 6.1

6.1 In any absorption operation, the absorbent is stripped to some extent depending on the K -value of the absorbent. In any stripping operation, the stripping agent is absorbed to some extent depending on its K -value. In Figure 6.1, it is seen that both absorption and stripping occur. Which occurs to the greatest extent in terms of kilomoles per hour? Should the operation be called an absorber or a stripper? Why?

6.2 Prior to 1950, only two types of commercial random packings were in common use: Raschig rings and Berl saddles. Starting in the 1950s, a wide variety of commercial random packings began to appear. What advantages do these newer packings have? By what advances in packing design and fabrication techniques were these advantages achieved? Why were structured packings introduced?

6.3 Bubble-cap trays were widely used in the design of trayed towers prior to the 1960s. Today sieve and valve trays are favored. However, bubble-cap trays are still occasionally specified, especially for operations that require very high turndown ratios or appreciable liquid residence time. What characteristics of bubble-cap trays make it possible for them to operate satisfactorily at low vapor and liquid rates?

Section 6.2

6.4 In Example 6.3, a lean oil of 250 MW is used as the absorbent. Consideration is being given to the selection of a new absorbent. Available streams are:

	Rate, gpm	Density, lb/gal	MW
C ₅ s	115	5.24	72
Light oil	36	6.0	130
Medium oil	215	6.2	180

Which stream would you choose? Why? Which streams, if any, are unacceptable?

6.5 Volatile organic compounds (VOCs) can be removed from water effluents by stripping in packed towers. Possible stripping agents are steam and air. Alternatively, the VOCs can be removed by carbon adsorption. The U.S. Environmental Protection Agency (EPA) has identified air stripping as the best available technology from an economic standpoint. What are the advantages and disadvantages of air compared to steam?

6.6 Prove by equations why, in general, absorbers should be operated at high pressure and low temperature, while strippers should be operated at low pressure and high temperature. Also prove, by equations, why a tradeoff exists between number of stages and flow rate of the separating agent.

Section 6.3

6.7 The exit gas from an alcohol fermenter consists of an air-CO₂ mixture containing 10 mol% CO₂ that is to be absorbed in a 5.0-N solution of triethanolamine, containing 0.04 mol of carbon dioxide per mole of amine solution. If the column operates isothermally at 25°C, if the exit liquid contains 78.4% of the CO₂ in the feed gas to the absorber, and if absorption is carried out in a six-theoretical-plate column, calculate:

- Moles of amine solution required per mole of feed gas.
- Exit gas composition.

Equilibrium Data

Y	0.003	0.008	0.015	0.023	0.032	-0.043
X	0.01	0.02	0.03	0.04	0.05	0.06
Y	0.055	0.068	0.083	0.099	0.12	
X	0.07	0.08	0.09	0.10	0.11	

Y = moles CO₂/mole air; X = moles CO₂/mole amine solution

6.8 Ninety-five percent of the acetone vapor in an 85 vol% air stream is to be absorbed by countercurrent contact with pure water in a valve-tray column with an expected overall tray efficiency of 50%. The column will operate essentially at 20°C and 101 kPa pressure. Equilibrium data for acetone-water at these conditions are:

Mole percent acetone in water	3.30	7.20	11.7	17.1
Acetone partial pressure in air, torr	30.00	62.80	85.4	103.0

Calculate:

- The minimum value of L'/V' , the ratio of moles of water per mole of air.
- The number of equilibrium stages required using a value of L'/V' of 1.25 times the minimum.
- The concentration of acetone in the exit water.

From Table 5.2 for N connected equilibrium stages, there are $2N + 2C + 5$ degrees of freedom. Specified in this problem are

Stage pressures (101 kPa)	N
Stage temperatures (20°C)	N
Feed stream composition	$C-1$
Water stream composition	$C-1$
Feed stream T, P	2
Water stream, T, P	2
Acetone recovery	1
L/V	1
	$2N + 2C + 4$

The remaining specification is the feed flow rate, which can be taken on a basis of 100 kmol/h.

6.9 A solvent-recovery plant consists of a plate-column absorber and a plate-column stripper. Ninety percent of the benzene (B) in the gas stream is recovered in the absorption column. Concentration of benzene in the inlet gas is 0.06 mol B/mol B-free gas. The oil entering the top of the absorber contains 0.01 mol B/mol pure oil. In the leaving liquid, $X = 0.19$ mol B/mol pure oil. Operating temperature is 77°F (25°C).

Open, superheated steam is used to strip benzene out of the benzene-rich oil at 110°C. Concentration of benzene in the oil = 0.19 and 0.01 (mole ratios) at inlet and outlet, respectively. Oil (pure)-to-steam (benzene-free) flow rate ratio = 2.0. Vapors are condensed, separated, and removed.

MW oil = 200 MW benzene = 78 MW gas = 32

Equilibrium Data at Column Pressures

X in Oil	Y in Gas, 25°C	Y in Steam, 110°C
0	0	0
0.04	0.011	0.1
0.08	0.0215	0.21
0.12	0.032	0.33
0.16	0.042	0.47
0.20	0.0515	0.62
0.24	0.060	0.795
0.28	0.068	1.05

Calculate:

- (a) The molar flow rate ratio of B-free oil to B-free gas in the absorber; (b) The number of theoretical plates in the absorber; and (c) The minimum steam flow rate required to remove the benzene from 1 mol of oil under given terminal conditions, assuming an infinite-plates column.

6.10 A straw oil used to absorb benzene from coke-oven gas is to be steam-stripped in a sieve-plate column at atmospheric pressure to recover the dissolved benzene. Equilibrium conditions at the operating temperature are approximated by Henry's law such that, when the oil phase contains 10 mol% C_6H_6 , the C_6H_6 partial pressure above the oil is 5.07 kPa. The oil may be considered non-volatile. The oil enters containing 8 mol% benzene, 75% of which is to be recovered. The steam leaving contains 3 mol% C_6H_6 . (a) How many theoretical stages are required? (b) How many moles of steam are required per 100 mol of oil-benzene mixture? (c) If 85% of the benzene is to be recovered with the same oil and steam rates, how many theoretical stages are required?

Section 6.4

6.11 Groundwater at a flow rate of 1,500 gpm, containing three volatile organic compounds (VOCs), is to be stripped in a trayed tower with air to produce drinking water that will meet EPA standards. Relevant data are given below. Determine the maximum air flow rate in scfm (60F, 1 atm) and the number of equilibrium stages required if an air flow rate of twice the minimum is used and the tower operates at 25°C and 1 atm. Also determine the composition in parts per million for each VOC in the resulting drinking water.

Component	K-value	Concentration, ppm	
		Ground water	Max. for Drinking water
1,2-Dichloroethane (DCA)	60	85	0.005
Trichloroethylene (TCE)	650	120	0.005
1,1,1-Trichloroethane (TCA)	275	145	0.200

Note: ppm = parts per million by weight.

6.12 Sulfur dioxide and butadienes (B3 and B2) are to be stripped with nitrogen from the liquid stream as shown in Figure 6.46 so that

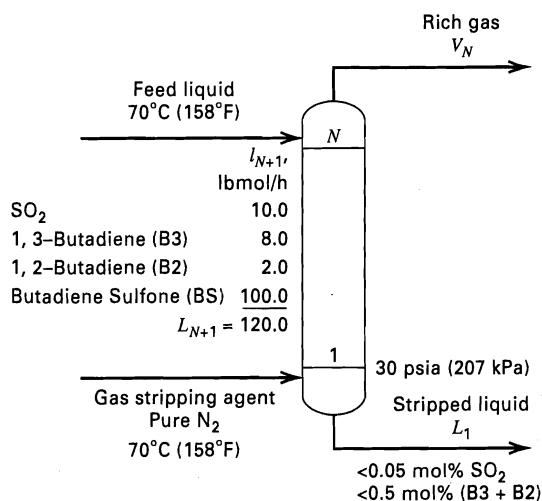


Figure 6.46 Data for Exercise 6.12.

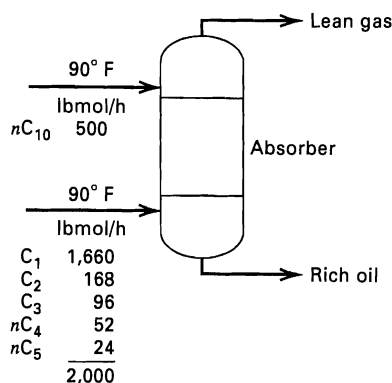


Figure 6.47 Data for Exercise 6.13.

butadiene sulfone (BS) product will contain less than 0.05 mol% SO_2 and less than 0.5 mol% butadienes. Estimate the flow rate of nitrogen, N_2 , and the number of equilibrium stages required. At 70°C, K -values for SO_2 , B2, B3, and BS are, respectively, 6.95, 3.01, 4.53, and 0.016.

6.13 Determine by the Kremser method the separation that can be achieved for the absorption operation indicated in Figure 6.47 for the following combinations of conditions: (a) Six equilibrium stages and 75 psia operating pressure, (b) Three equilibrium stages and 150 psia operating pressure, (c) Six equilibrium stages and 150 psia operating pressure. At 90°F and 75 psia, the K -value of $nC_{10} = 0.0011$.

6.14 One thousand kilomoles per hour of rich gas at 70°F with 25% C_1 , 15% C_2 , 25% C_3 , 20% nC_4 , and 15% nC_5 by moles is to be absorbed by 500 kmol/h of nC_{10} at 90°F in an absorber operating at 4 atm. Calculate by the Kremser method the percent absorption of each component for 4, 10, and 30 theoretical stages. What do you conclude from the results? (Note: The K -value of nC_{10} at 80°F and 4 atm is 0.0014.)

Section 6.5

6.15 Using the performance data of Example 6.3, back-calculate the overall stage efficiency for propane and compare the result with estimates from the Drickamer-Bradford and O'Connell correlations.

6.16 Several hydrogenation processes are being considered that will require hydrogen of 95% purity. A refinery stream of 800,000 scfm (at 32°F, 1 atm), currently being used for fuel and containing 72.5% H_2 , 25% CH_4 , and 2.5% C_2H_6 is available. To convert this gas to the required purity, oil absorption, activated charcoal adsorption, and membrane separation are being considered. For oil absorption, an available n -octane stream can be used as the absorbent. Because the 95% H_2 must be delivered to a hydrogenation process at not less than 375 psia, it is proposed to operate the absorber at 400 psia and 100°F. If at least 80% of the hydrogen fed to the absorber is to leave in the exit gas, determine:

- The minimum absorbent rate in gallons per minute.
- The actual absorbent rate if 1.5 times the minimum amount is used.
- The number of theoretical stages.
- The stage efficiency for each of the three species in the feed gas, using the O'Connell correlation.
- The number of trays actually required.

(f) The composition of the exit gas, taking into account the stripping of octane.

(g) If the octane lost to the exit gas is not recovered, estimate the annual cost of this lost oil if the process operates 7,900 h/year and the octane is valued at \$1.00/gal.

6.17 The absorption operation of Examples 6.1 and 6.4 is being scaled up by a factor of 15, such that a column with an 11.5-ft diameter will be needed. In addition, because of the low efficiency of 30% for the original operation, a new tray design has been developed and tested in an Oldershaw-type column. The resulting Murphree vapor-point efficiency, E_{OV} , for the new tray design for the system of interest is estimated to be 55%. Estimate E_{MV} and E_o . (To estimate the length of the liquid flow path, Z_L , use Figure 6.16. Also, assume that $u/D_E = 6 \text{ ft}^{-1}$.)

Section 6.6

6.18 Conditions at the bottom tray of a reboiled stripper are as shown in Figure 6.48. If valve trays are used with a 24-in. tray spacing, estimate the required column diameter for operation at 80% of flooding.

6.19 Determine the flooding velocity and column diameter for the following conditions at the top tray of a hydrocarbon absorber equipped with valve trays:

Pressure	400 psia
Temperature	128°F
Vapor rate	530 lbmol/h
Vapor MW	26.6
Vapor density	1.924 lb/ft ³
Liquid rate	889 lbmol/h
Liquid MW	109
Liquid density	41.1 lb/ft ³
Liquid surface tension	18.4 dynes/cm
Foaming factor	0.75
Tray spacing	24 in.
Fraction flooding	0.85
Valve trays	

6.20 For Exercise 6.16, if a flow rate of 40,000 gpm of octane is used to carry out the absorption in a sieve-tray column using 24-in. tray spacing, a weir height of 2.5 in., and holes of $\frac{1}{4}$ -in. diameter, determine for a foaming factor of 0.80 and a fraction flooding of 0.70:

- The column diameter based on conditions near the bottom of the column.
- The vapor pressure drop per tray.
- Whether weeping will occur.
- The entrainment rate.
- The fractional decrease in E_{MV} due to entrainment.
- The froth height in the downcomer.

6.21 Repeat the calculations of Examples 6.5, 6.6, and 6.7 for a column diameter corresponding to 40% of flooding.

6.22 For the acetone absorber of Figure 6.1, assuming the use of sieve trays with a 10% hole area and $\frac{3}{16}$ -in. holes with an 18-in. tray spacing, estimate:

- The column diameter for a foaming factor of 0.85 and a fraction of flooding of 0.75.
- The vapor pressure drop per tray.

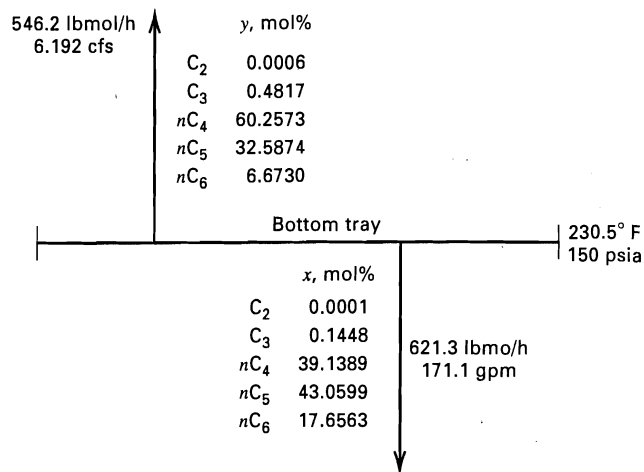


Figure 6.48 Data for Exercise 6.18.

(c) The number of transfer units, N_G and N_L , from (6-62) and (6-63), respectively.

(d) N_{OG} from (6-61).

(e) The controlling resistance to mass transfer.

(f) E_{OV} from (6-56).

From your results, determine if 30 actual trays are adequate.

6.23 Design a VOC stripper for the flow conditions and separation of Example 6.2 except that the wastewater and air flow rates are twice as high. To develop the design, determine:

- The number of equilibrium stages required.
- The column diameter for sieve trays.
- The vapor pressure drop per tray.
- Murphree vapor-point efficiencies using the Chan and Fair method.
- The number of trays actually required.

Section 6.7

6.24 Air containing 1.6 vol% sulfur dioxide is scrubbed with pure water in a packed column of 1.5-m² cross-sectional area and 3.5-m height packed with no. 2 plastic Super Intalox saddles, at a pressure of 1 atm. Total gas flow rate is 0.062 kmol/s, the liquid flow rate is 2.2 kmol/s, and the outlet gas SO₂ concentration is $y = 0.004$. At the column temperature, the equilibrium relationship is given by $y^* = 40x$.

- What is $L/L_{\min}$?
- Calculate N_{OG} and compare your answer to that for the number of theoretical stages required.
- Determine H_{OG} and the HETP from the operating data.
- Calculate $K_G a$ from the data, based on a partial-pressure driving force as in item 2 of Table 6.7.

6.25 An SO₂-air mixture is being scrubbed with water in a countercurrent-flow packed tower operating at 20°C and 1 atm. Solute-free water enters the top of the tower at a constant rate of 1,000 lb/h and is well distributed over the packing. The liquor leaving contains 0.6 lb SO₂/100 lb of solute-free water. The partial pressure of SO₂ in the spent gas leaving the top of the tower is 23 torr. The mole ratio of water to air is 25. The necessary equilibrium data are given below.

- What percent of the SO₂ in the entering gases is absorbed in the tower?

(b) In operating the tower it was found that the rate coefficients k_p and k_L remained substantially constant throughout the tower at the following values:

$$k_L = 1.3 \text{ ft/h}$$

$$k_p = 0.195 \text{ lbmol/h-ft}^2\text{-atm}$$

At a point in the tower where the liquid concentration is 0.001 lbmol SO_2 per lbmol of water, what is the liquid concentration at the gas-liquid interface in lbmol/ft³? Assume that the solution has the same density as H_2O .

Solubility of SO_2 in H_2O at 20°C

lb SO_2 100 lb H_2O	Partial Pressure of SO_2 in Air, torr
0.02	0.5
0.05	1.2
0.10	3.2
0.15	5.8
0.20	8.5
0.30	14.1
0.50	26.0
0.70	39.0
1.0	59

6.26 A wastewater stream of 600 gpm, containing 10 ppm (by weight) of benzene, is to be stripped with air in a packed column operating at 25°C and 2 atm to produce water containing 0.005 ppm of benzene. The packing is 2-in. Flexirings made of polypropylene. The vapor pressure of benzene at 25°C is 95.2 torr. The solubility of benzene in water at 25°C is 0.180 g/100 g. An expert in VOC stripping with air has suggested use of 1,000 scfm of air (60°F, 1 atm), at which condition one should achieve for the mass transfer of benzene:

$$k_L a = 0.067 \text{ s}^{-1} \quad \text{and} \quad k_G a = 0.80 \text{ s}^{-1}$$

Determine:

- The minimum air stripping rate in scfm. Is it less than the rate suggested by the expert? If not, use 1.4 times your minimum value.
- The stripping factor based on the air rate suggested by the expert.
- The number of transfer units, N_{OG} , required.
- The overall mass-transfer coefficient, $K_G a$, in units of mol/m³-s-kPa and s⁻¹. Which phase controls mass transfer?
- The volume of packing in cubic meters.

Section 6.8

6.27 Germanium tetrachloride (GeCl_4) and silicon tetrachloride (SiCl_4) are used in the production of optical fibers. Both chlorides are oxidized at high temperature and converted to glasslike particles. However, the GeCl_4 oxidation is quite incomplete and it is necessary to scrub the unreacted GeCl_4 from its air carrier in a packed column operating at 25°C and 1 atm with a dilute caustic solution. At these conditions, the dissolved GeCl_4 has no vapor pressure and mass transfer is controlled by the gas phase. Thus, the equilibrium curve is a straight line of zero slope. Why? The entering gas is 23,850 kg/day of air containing 288 kg/day of GeCl_4 . The air also contains 540 kg/day of Cl_2 , which, when dissolved, also will have no vapor pressure. The two

liquid-phase reactions are



It is desired to absorb 99% of both GeCl_4 and Cl_2 in an existing 2-ft-diameter column that is packed to a height of 10 ft with $\frac{1}{2}$ -in. ceramic Raschig rings. The liquid rate should be set so that the column operates at 75% of flooding. For the packing: $\epsilon = 0.63$, $F_P = 580 \text{ ft}^{-1}$, and $D_P = 0.01774 \text{ m}$.

Gas-phase mass-transfer coefficients for GeCl_4 and Cl_2 can be estimated from the following empirical equations developed from experimental studies, where μ , ρ , and D_i are gas-phase properties:

$$K_y a = k_y a$$

$$\frac{k_y}{(V/S)} = 1.195 \left[\frac{D_P V'}{\mu(1 - \epsilon_o)} \right]^{-0.36} (N_{Sc})^{-2/3}$$

$$\epsilon_o = \epsilon - h_L$$

$$h_L = 0.03591(L')^{0.331}$$

$$a = \frac{14.69(808 V'/\rho^{1/2})^n}{(L')^{0.111}}$$

$$n = 0.01114L' + 0.148$$

where

S = column cross sectional area, m²

k_y = kmol/m²-s

V = molar gas rate, kmol/s

D_P = equivalent packing diameter, m

μ = gas viscosity, kg/m-s

ρ = gas density, kg/m³

N_{Sc} = Schmidt number = $\mu/\rho D_i$

D_i = molecular diffusivity of component i in the gas, m²/s

a = interfacial area for mass transfer, m²/m³ of packing

L' = liquid mass velocity, kg/m²-s

V' = gas mass velocity, kg/m²-s

For the two diffusing species, take

$$D_{\text{GeCl}_4} = 0.000006 \text{ m}^2/\text{s}$$

$$D_{\text{Cl}_2} = 0.000013 \text{ m}^2/\text{s}$$

Determine:

- The dilute caustic flow rate in kilograms per second.
- The required packed height in feet based on the controlling species (GeCl_4 or Cl_2). Is the 10 ft of packing adequate?
- The percent absorption of GeCl_4 and Cl_2 based on the available 10 ft of packing. If the 10 ft of packing is not sufficient, select an alternative packing that is adequate.

6.28 For the VOC stripping task of Exercise 6.26, the expert has suggested that we use a tower diameter of 0.80 m for which we can expect a pressure drop of 500 N/m²-m of packed height (0.612 in. $\text{H}_2\text{O}/\text{ft}$). Verify the information from the expert by estimating:

- The fraction of flooding using the GPDC chart of Figure 6.36 with $F_P = 24 \text{ ft}^2/\text{ft}^3$.
- The pressure drop at flooding.
- The pressure drop at the operating conditions of Exercise 6.26 using the GPDC chart.

(d) The pressure drop at operating conditions using the correlation of Billet and Schultes by assuming that 2-in. plastic Flexiring packing has the same characteristics as 2-in. plastic Pall rings.

6.29 For the VOC stripping task of Exercise 6.26, the expert suggested certain mass-transfer coefficients. Check this information by estimating the coefficients from the correlations of Billet and Schultes by assuming that 2-in. plastic Flexiring packing has the same characteristics as 2-in. plastic Pall rings.

6.30 A 2 mol% NH_3 -in-air mixture at 68°F and 1 atm is to be scrubbed with water in a tower packed with 1.5-in. ceramic Berl saddles. The inlet water mass velocity will be 2400 lb/h-ft^2 , and the inlet gas mass velocity 240 lb/h-ft^2 . Assume that the tower temperature remains constant at 68°F , at which the gas solubility relationship follows Henry's law, $p = Hx$, where p is the partial pressure of ammonia over the solution, x is the mole fraction of ammonia in the liquid, and H is the Henry's law constant, equal to $2.7 \text{ atm/mole fraction}$.

(a) Calculate the required packed height for absorption of 90% of the NH_3 .

(b) Calculate the minimum water mass velocity in lb/h-ft^2 for absorbing 98% of the NH_3 .

(c) The use of 1.5-in. ceramic Hiflow rings rather than the Berl saddles has been suggested. What changes would this cause in K_Ga , pressure drop, maximum liquid rate, K_La , column height, column diameter, H_{OG} , and N_{OG} ?

6.31 You are to design a packed column to absorb CO_2 from air into fresh, dilute-caustic solution. The entering air contains 3 mol% CO_2 , and a 97% recovery of CO_2 is desired. The gas flow rate is $5,000 \text{ ft}^3/\text{min}$ at 60°F , 1 atm. It may be assumed that in the range of operation, the equilibrium curve is $Y^* = 1.75X$, where Y and X are mole ratios of CO_2 to carrier gas and liquid, respectively. A column diameter of 30 in. with 2-in. Intalox saddle packing can be assumed for the initial design estimates. Assume the caustic solution has the properties of water. Calculate:

- The minimum caustic solution-to-air molar flow rate ratio.
- The maximum possible concentration of CO_2 in the caustic solution.
- The number of theoretical stages at $L/V = 1.4$ times minimum.
- The caustic solution rate.
- The pressure drop per foot of column height. What does this result suggest?
- The overall number of gas transfer units, N_{OG} .
- The height of packing, using a K_Ga of $2.5 \text{ lbmol/h-ft}^3\text{-atm}$.

Section 6.9

6.32 At a point in an ammonia absorber using water as the absorbent and operating at 101.3 kPa and 20°C , the bulk gas phase contains 10 vol% NH_3 . At the interface, the partial pressure of NH_3 is 2.26 kPa . The concentration of the ammonia in the body of the liquid is 1 wt%. The rate of ammonia absorption at this point is 0.05 kmol/h-m^2 .

- Given this information and the equilibrium curve in Figure 6.49, calculate X , Y , Y_1 , X_1 , X^* , Y^* , K_Y , K_X , k_Y , and k_X .
- What percent of the mass-transfer resistance is in each phase?
- Verify for these data that $1/K_Y = 1/k_Y + H'/k_X$.

6.33 One thousand cubic feet per hour of a 10 mol% NH_3 in air mixture is required to produce nitrogen oxides. This mixture is to be obtained by desorbing an aqueous 20 wt% NH_3 solution with air at 20°C . The spent solution should not contain more than 1 wt% NH_3 .

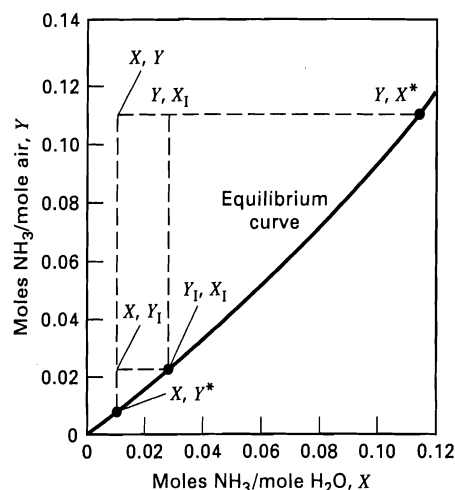


Figure 6.49 Data for Exercise 6.32.

Calculate the volume of packing required for the desorption column. Vapor-liquid equilibrium data for Exercise 6.32 can be used and $K_Ga = 4 \text{ lbmol/h-ft}^3\text{-atm}$ partial pressure.

6.34 Ammonia, present at a partial pressure of 12 torr in an air stream saturated with water vapor at 68°F and 1 atm, must be removed to the extent of 99.6% by water absorption at the same temperature and pressure. Two thousand pounds of dry air per hour are to be handled.

- Calculate the minimum amount of water necessary using the equilibrium data for Exercise 6.32 in Figure 6.49.
- Assuming an operation at 2 times the minimum water flow and at one-half the flooding gas velocity, compute the dimensions of a column packed with 38-mm ceramic Berl Saddles.
- Repeat part (b) for 50-mm Pall rings.
- Which of the two packings would you recommend?

6.35 Exit gas from a chlorinator consists of a mixture of 20 mol% chlorine in air. This concentration is to be reduced to 1% chlorine by water absorption in a packed column to operate isothermally at 20°C and atmospheric pressure. Using the following equilibrium x - y data, calculate for 100 kmol/h of feed gas:

- The minimum water rate in kilograms per hour.
- N_{OG} for twice the minimum water rate.

Data for x - y at 20°C (in chlorine mole fractions):

x	0.0001	0.00015	0.0002	0.00025	0.0003
y	0.006	0.012	0.024	0.04	0.06

6.36 Calculate the diameter and height for the column of Example 6.15 if the tower is packed with 1.5-in. metal Pall rings. Assume that the absorbing solution has the properties of water and use conditions at the bottom of the tower, where flow rates are highest.

6.37 You are asked to design a packed column to recover acetone from air continuously, by absorption with water at 60°F . The air contains 3 mol% acetone, and a 97% recovery is desired. The gas flow rate is $50 \text{ ft}^3/\text{min}$ at 60°F , 1 atm. The maximum-allowable gas superficial velocity in the column is 2.4 ft/s . It may be assumed that in the range of operation, $Y^* = 1.75X$, where Y and X are mole ratios for acetone.

Calculate:

- The minimum water-to-air molar flow rate ratio.
- The maximum acetone concentration possible in the aqueous solution.

(c) The number of theoretical stages for a flow rate ratio of 1.4 times the minimum.

(d) The corresponding number of overall gas transfer units.

(e) The height of packing, assuming $K_{ya} = 12.0 \text{ lbmol/h-ft}^3$ -molar ratio difference.

(f) The height of packing as a function of the molar flow rate ratio, assuming that V and HTU remain constant.

6.38 Determine the diameter and packed height of a countercurrently operated packed tower required to recover 99% of the ammonia from a gas mixture that contains 6 mol% NH_3 in air. The tower, packed with 1-in. metal Pall rings, must handle $2,000 \text{ ft}^3/\text{min}$ of gas as measured at 68°F and 1 atm. The entering water-absorbent rate will be twice the theoretical minimum, and the gas velocity will be such that it is 50% of the flooding velocity. Assume isothermal operation at 68°F and 1 atm. Equilibrium data are given in Figure 6.49.

6.39 A tower, packed with Montz B1-200 metal structured packing, is to be designed to absorb SO_2 from air by scrubbing with water. The entering gas, at an SO_2 -free flow rate of $6.90 \text{ lbmol/h-ft}^2$

of bed cross section, contains 80 mol% air and 20 mol% SO_2 . Water enters at a flow rate of 364 lbmol/h-ft^2 of bed cross section. The exiting gas is to contain only 0.5 mol% SO_2 . Assume that neither air nor water will be transferred between phases and that the tower operates at 2 atm and 30°C . Equilibrium data in mole fractions for SO_2 solubility in water at 30°C and 2 atm (*Perry's Chemical Engineers' Handbook*, 4th ed., Table 14.31, p. 14-6) have been fitted by a least-squares method to the equation

$$y = 12.697x + 3148.0x^2 - 4.724 \times 10^5 x^3 + 3.001 \times 10^7 x^4 - 6.524 \times 10^8 x^5$$

(a) Derive the following molar material balance operating line for SO_2 mole fractions:

$$x = 0.0189 \left(\frac{y}{1-y} \right) - 0.00010$$

(b) Write a computer program or use a spreadsheet program to calculate the number of required transfer units based on the overall gas-phase resistance.

Chapter 7

Distillation of Binary Mixtures

In *distillation* (*fractionation*), a feed mixture of two or more components is separated into two or more products, including, and often limited to, an overhead distillate and a bottoms, whose compositions differ from that of the feed. Most often, the feed is a liquid or a vapor–liquid mixture. The bottoms product is almost always a liquid, but the distillate may be a liquid or a vapor or both. The separation requires that (1) a second phase be formed so that both liquid and vapor phases are present and can contact each other on each stage within a separation column, (2) the components have different volatilities so that they will partition between the two phases to different extents, and (3) the two phases can be separated by gravity or other mechanical means. Distillation differs from absorption and stripping in that the second fluid phase is usually created by thermal means (vaporization and condensation) rather than by the introduction of a second phase that may contain an additional component or components not present in the feed mixture.

According to Forbes [1], the art of distillation dates back to at least the first century A.D. By the eleventh century, distillation was being used in Italy to produce alcoholic beverages. At that time, distillation was probably a batch process based on the use of just a single stage, the boiler. The feed to be separated, a liquid, was placed in a vessel to which heat was applied, causing part of the liquid to evaporate. The vapor passed out of the heating vessel and was cooled in

another chamber by transfer of heat through the wall of the chamber to water, producing condensate that dripped into a product receiver. The word *distillation* is derived from the Latin word *destillare*, which means dripping or trickling down. By at least the sixteenth century, it was known that the extent of separation could be improved by providing multiple vapor–liquid contacts (stages) in a so-called Rectificatorium. The term *rectification* is derived from the Latin words *recte facere*, meaning to improve. Modern distillation derives its ability to produce almost pure products from the use of multistage contacting.

Throughout the twentieth century, multistage distillation was by far the most widely used industrial method for separating liquid mixtures of chemical components. Unfortunately, distillation is a very energy-intensive technique especially when the relative volatility, α , of the component being separated is low (< 1.50). Mix et al. [2] report that the energy consumption for distillation in the United States for 1976 totaled 2×10^{15} Btu (2 quads), which was nearly 3% of the entire national energy consumption. Approximately two thirds of the distillation energy was consumed by petroleum refining, where distillation is widely used to separate crude oil into petroleum fractions, light hydrocarbons (C_2 's to C_5 's), and aromatic chemicals. The separation of other organic chemicals, often in the presence of water, is widely practiced in the chemical industry.

7.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Explain the difference between distillation and absorption or stripping.
- Explain the need in distillation for a condenser to produce reflux and a reboiler to produce boilup.
- Enumerate factors that influence design of a distillation column.
- Distinguish between required specifications and results that can be obtained from the McCabe–Thiele method for continuous binary distillation.
- Determine the five construction lines used in the McCabe–Thiele method using component material balances and vapor–liquid equilibrium relations.
- Explain the concept of “constant molar overflow,” assumptions required for its validity, and why it eliminates the need for energy balances around stages.
- Distinguish among five possible phase conditions of the feed.
- Apply the McCabe–Thiele method for determining minimum reflux ratio, minimum number of equilibrium stages, number of equilibrium stages for a specified reflux ratio greater than minimum, and optimal feed-stage location, given the required split between the two feed components.

- Select an appropriate condenser and a suitable operating pressure for a distillation separation.
- Explain differences among the three most common types of reboilers.
- Calculate condenser and reboiler heat duties and consider use of a feed preheater.
- Determine the optimal reflux ratio.
- Use the Murphree vapor stage efficiency to determine actual number of stages (plates) from the number of equilibrium stages.
- Extend the McCabe–Thiele method to multiple feeds, side streams, and open steam (in place of a reboiler).
- Estimate overall stage efficiency for binary distillation from correlations and laboratory column data.
- Determine the diameter of a trayed tower and size of the reflux drum.
- Determine packed height and diameter of a packed column.
- Use an enthalpy-concentration diagram when assumption of “constant molar overflow” is not valid.

Industrial Example

The fundamentals of distillation are best understood by the study of binary distillation, the separation of a two-component mixture, which is the subject of this chapter. The more general and much more difficult case of a multicomponent mixture is covered in Chapters 10 and 11. A representative binary distillation operation is shown in Figure 7.1 for the separation of 620 lbmol/h (0.0781 kmol/s) of a binary mixture of 46 mol% benzene (the more volatile component) and 54 mol% toluene. The purpose of the 25-sieve-tray (equivalent to 20 theoretical stages plus a partial reboiler that acts as an additional theoretical stage)

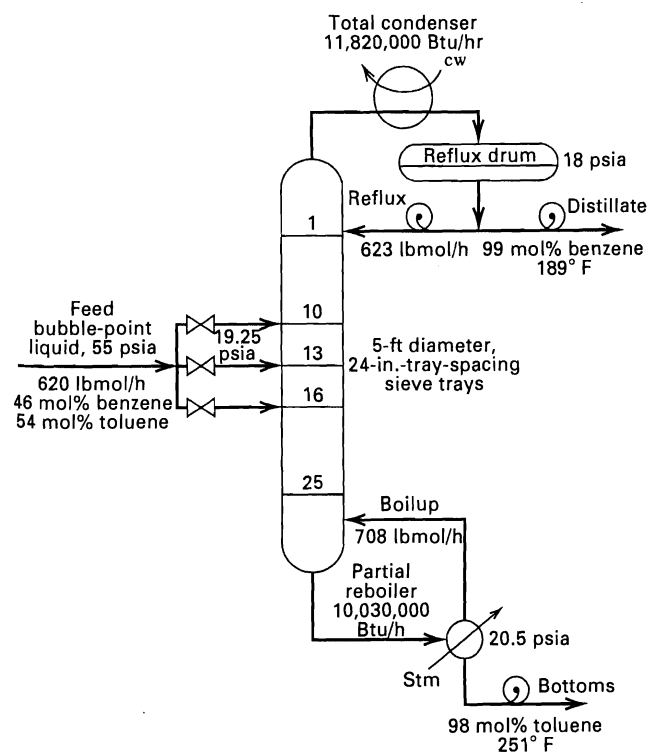


Figure 7.1 Distillation of a binary mixture of benzene and toluene.

distillation column is to separate the feed into a liquid distillate of 99 mol% benzene and a liquid bottoms product of 98 mol% toluene. The column operates at a pressure in the reflux drum of 18 psia (124 kPa), just slightly above ambient pressure. For a negligible pressure drop across the condenser and a vapor pressure drop of 0.1 psi/tray (0.69 kPa/tray), the pressure in the reboiler is $18 + 0.1(25) = 20.5$ psia (141 kPa). In this range of pressure, benzene and toluene form nearly ideal mixtures with a relative volatility of from 2.26 at conditions of the bottom tray to 2.52 at the top tray, as determined from Raoult's law by (2-44). The reflux ratio (reflux rate to distillate rate) is 2.215. If an infinite number of stages were used, the required reflux ratio would be a minimum value of 1.708. Thus, the ratio of reflux rate to minimum reflux rate for this example is 1.297. Most distillation columns are designed to operate with optimal-reflux-to-minimum-reflux ratios of 1.1 to 1.5. If an infinite ratio of reflux to minimum reflux were used, only 10.7 theoretical stages would be required. Thus, the ratio of theoretical stages to minimum theoretical stages for this example is $21/10.7 = 1.96$. For most distillation columns, this ratio is approximately 2. The stage efficiency is 20/25 or 80%. This is close to the average efficiency observed for distillation.

The feed to the separation operation of Figure 7.1 is a saturated liquid at 55 psia (379 kPa). A bubble-point calculation gives a temperature of 294°F (419 K). When this feed is flashed adiabatically across the feed valve to the feed tray pressure of 19.25 psia (133 kPa), the feed temperature drops to 220°F (378 K), causing 23.4 mol% of the feed to be vaporized. A total condenser is used to obtain saturated liquid reflux and liquid distillate at a bubble-point temperature of 189°F (360 K) at 18 psia (124 kPa). The duty of the condenser is 11,820,000 Btu/h (3.46 MW). At the bottom of the column, a partial reboiler is used to produce vapor boilup and a liquid bottoms product. Assuming that the boilup and bottoms are in physical equilibrium, the

partial reboiler functions as an additional theoretical stage, giving a total of 21 theoretical stages. Because the bottoms product is a saturated liquid, its temperature of 251°F (395 K) corresponds to the bubble point of the bottoms at 20.5 psia (141 kPa). The duty of the reboiler is 10,030,000 Btu/h (2.94 MW), which is within 15% of the condenser duty.

The inside diameter of the distillation column in Figure 7.1 is a constant 5 ft (1.53 m). At the top tray this diameter corresponds to 84% of flooding, while at the bottom tray the percent flooding is 81%. As shown, the column can be fed at any one of three trays. For the design conditions, the optimal feed entry is between trays 12 and 13. However, should the feed composition or product specifications change, one of the other two feed trays could become optimal.

Distillation columns similar to that of Figure 7.1 have been built for diameters up to at least 30 ft (9.14 m). With a 24-in. (0.61-m) tray spacing, the maximum number of trays included in a single column is usually no greater than 150. In general, for the sharp separation of a binary mixture with a relative volatility less than 1.05, distillation can require many hundreds of trays, so a more efficient separation

technique should be sought. Even when distillation is the most economical separation technique, its second-law efficiency, using the calculational procedure developed in Chapter 2, can be less than 10%.

Technically, distillation is the most mature separation operation. Design and operation procedures are well established; for example, see Kister [3, 4]. Only when vapor-liquid equilibrium or other data are uncertain is a laboratory and/or pilot-plant study necessary prior to the design of a commercial unit. Table 7.1, taken partially from the study of Mix et al. [2], lists just some of the more common commercial binary distillation operations in decreasing order of difficulty of separation. Included are representative nominal values of relative volatility, number of trays, column operating pressure, and reflux-to-minimum-reflux ratio. Although the data in Table 7.1 refer to trayed towers, distillation can also be carried out in packed columns. More and more frequently, additional distillation capacity is being achieved with existing trayed towers by replacing all or some of the trays with sections of random or structured packing.

Table 7.1 Representative Commercial Binary Distillation Operations [2]

Binary Mixture	Average Relative Volatility	Number of Trays	Typical Operating Pressure, psia	Reflux-to- Minimum- Reflux Ratio
1,3-Butadiene/vinyl acetylene	1.16	130	75	1.70
Vinyl acetate/ethyl acetate	1.16	90	15	1.15
<i>o</i> -Xylene/ <i>m</i> -xylene	1.17	130	15	1.12
Isopentane/ <i>n</i> -pentane	1.30	120	30	1.20
Isobutane/ <i>n</i> -butane	1.35	100	100	1.15
Ethylbenzene/styrene	1.38	34	1	1.71
Propylene/propane	1.40	138	280	1.06
Methanol/ethanol	1.44	75	15	1.20
Water/acetic acid	1.83	40	15	1.35
Ethylene/ethane	1.87	73	230	1.07
Acetic acid/acetic anhydride	2.02	50	15	1.13
Toluene/ethylbenzene	2.15	28	15	1.20
Propane/1,3-butadiene	2.18	40	120	1.13
Ethanol azeotrope/water	2.21	60	15	1.35
Isopropanol/water	2.23	12	15	1.28
Benzene/toluene	3.09	34	15	1.15
Methanol/water	3.27	60	45	1.31
Cumene/phenol	3.76	38	1	1.21
Benzene/ethylbenzene	6.79	20	15	1.14
HCN/water	11.20	15	50	1.36
Ethylene oxide/water	12.68	50	50	1.19
Formaldehyde/methanol	16.70	23	50	1.17
Water/ethylene glycol	81.20	16	4	1.20

In this chapter, equipment for conducting distillation operations is discussed and fundamental equilibrium-based and rate-based calculational procedures are developed for binary mixtures. Trayed and packed distillation columns are

identical in most respects to the absorption and stripping columns discussed in the previous chapter. Therefore, where appropriate, reference is made to Chapter 6 and only important differences are discussed in this chapter.

7.1 EQUIPMENT AND DESIGN CONSIDERATIONS

Industrial distillation operations are most commonly conducted in trays towers, but packed columns are finding increasing use. Occasionally, distillation columns contain both trays and packing. Types of trays and packings are identical to those used for absorption and stripping, as described in Section 6.1, shown in Figures 6.2 to 6.7, and compared in Tables 6.2 and 6.3.

Factors that influence the design or analysis of a binary-distillation operation include:

1. Feed flow rate, composition, temperature, pressure, and phase condition
2. Desired degree of separation between two components
3. Operating pressure (which must be below the critical pressure of the mixture)
4. Vapor pressure drop, particularly for vacuum operation
5. Minimum reflux ratio and actual reflux ratio
6. Minimum number of equilibrium stages and actual number of equilibrium stages (stage efficiency)
7. Type of condenser (total, partial, or mixed)
8. Degrees of subcooling, if any, of the liquid reflux
9. Type of reboiler (partial or total)
10. Type of contacting (trays or packing or both)
11. Height of the column
12. Feed-entry stage
13. Diameter of the column
14. Column internals.

The phase condition (also called thermal condition) of the feed is determined at the feed-tray pressure by an adiabatic-flash calculation across the feed valve. As the molar fraction of vapor in the feed increases, the required reflux ratio (L/D) increases, but the corresponding boilup ratio (V/B) decreases. The column operating pressure in the reflux drum should correspond to a distillate temperature somewhat higher (e.g., 10 to 50°F or 6 to 28°C) than the supply temperature of the cooling water used as the coolant in the overhead condenser. However, if this pressure approaches the critical pressure of the more volatile component, then a lower operating pressure must be used and a refrigerant is required as coolant. For example, in Table 7.1, the separation of ethylene/ethane is

conducted at 230 psia (1,585 kPa), giving a column top temperature of -40°F (233 K), which requires a refrigerant. Water at 80°F (300 K) cannot be used in the condenser because the critical temperature of ethylene is 48.6°F (282 K). If the estimated pressure is less than atmospheric pressure, the operating pressure at the top of the column is often set just above atmospheric pressure to avoid vacuum operation, unless the temperature at the bottom of the column is found to exceed a bottoms temperature limited by decomposition, polymerization, excessive corrosion, or other chemical reaction. In that case, vacuum operation is necessary. In Table 7.1, vacuum operation is required for the separation of ethylbenzene from styrene to maintain a bottoms temperature sufficiently low to prevent polymerization of styrene.

For given (1) feed, (2) desired degree of separation, and (3) operating pressure, a minimum reflux ratio exists that corresponds to an infinite number of theoretical stages; and a minimum number of theoretical stages exists that corresponds to an infinite reflux ratio. A design trade-off is usually made between the number of stages and the reflux ratio. A graphical method for determining the data needed to make this trade-off and to determine the optimal feed-stage location is developed in the next section.

7.2 MCCABE-THIELE GRAPHICAL EQUILIBRIUM-STAGE METHOD FOR TRAYED TOWERS

Consider the general countercurrent-flow, multistage, binary-distillation operation shown in Figure 7.2. The operation consists of a column containing the equivalent of N theoretical stages; a total condenser in which the overhead vapor leaving the top stage is totally condensed to give a liquid distillate product and liquid reflux that is returned to the top stage; a partial reboiler in which liquid from the bottom stage is partially vaporized to give a liquid bottoms product and vapor boilup that is returned to the bottom stage, and an intermediate feed stage. By means of multiple, countercurrent contacting stages arranged in a two-section cascade with reflux and boilup, as discussed in Section 5.4, it is possible to achieve a sharp separation between the two components in the feed unless an azeotrope is formed, in which case one of the two products will approach the azeotropic composition.

The feed, which contains a more-volatile (light) component (the *light key*, LK), and a less-volatile (heavy)

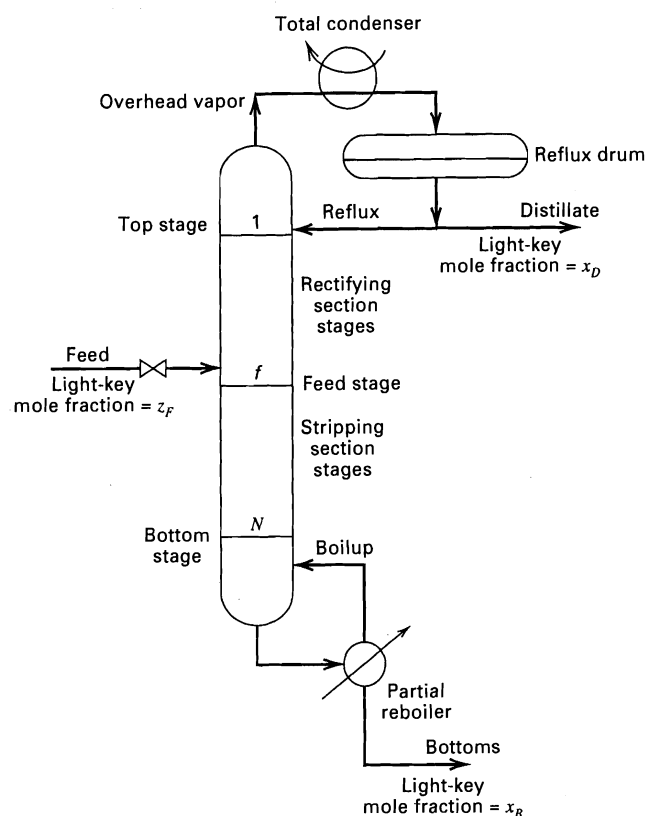


Figure 7.2 Distillation operation using a total condenser and partial reboiler.

component (the *heavy key*, HK), enters the column at a feed stage, f . At the feed-stage pressure, the feed may be liquid, vapor, or a mixture of liquid and vapor, with its overall mole-fraction composition with respect to the light component denoted by z_F . The mole fraction of the light key in the distillate is x_D , while the mole fraction of the light key in the bottoms product is x_B . Corresponding compositions with respect to the heavy key are $1 - z_F$, $1 - x_D$, and $1 - x_B$.

The goal of distillation is to produce from the feed a distillate, rich in the light key (i.e., x_D approaching 1.0), and a bottoms product, rich in the heavy key (i.e., x_B approaching 0.0). The ease or difficulty with which the separation can be achieved depends on the relative volatility, α , of the two components (LK = 1 and HK = 2), where

$$\alpha_{1,2} = K_1/K_2 \quad (7-1)$$

Methods for estimating K -values are discussed in Chapter 2.

If the two components form ideal solutions and follow the ideal-gas law in the vapor phase, Raoult's law applies to give

$$K_1 = P_1^s/P \quad \text{and} \quad K_2 = P_2^s/P$$

and from (7-1), the relative volatility is given simply by the ratio of vapor pressures, $\alpha_{1,2} = P_1^s/P_2^s$ and thus is a function only of temperature. As discussed in Section 4.2, as the temperature (and therefore the pressure) increases, $\alpha_{1,2}$

decreases. At the convergence pressure of the mixture, $\alpha_{1,2} = 1.0$ and a separation cannot be achieved at this or any higher pressure.

The relative volatility can be expressed in terms of equilibrium vapor and liquid compositions from the definition of the K -value as $K_i = y_i/x_i$. For a binary mixture,

$$\alpha_{1,2} = \frac{y_1/x_1}{y_2/x_2} = \frac{y_1(1-x_1)}{x_1(1-y_1)} \quad (7-2)$$

Solving (7-2) for y_1 ,

$$y_1 = \frac{\alpha_{1,2}x_1}{1 + x_1(\alpha_{1,2} - 1)} \quad (7-3)$$

For ideal binary mixtures of components with close boiling points, the temperature change over the column is small and $\alpha_{1,2}$ is almost constant. In any case, for a given pressure P and liquid-phase composition x_1 , the Gibbs phase rule, discussed in Chapter 4, fixes the temperature and equilibrium-vapor composition. An equilibrium curve for the benzene-toluene system is shown in Figure 7.3, where y and x correspond to the light key, benzene, and the pressure is 1 atm, at which pure benzene and pure toluene boil at 176 and 231°F, respectively. Thus, these two components are not close-boiling. Using (7-3) with this curve, α varies from about 2.6 at the bottom of the curve to about 2.35 at the top of the curve. Representative equilibrium curves for some average values of α are shown in Figure 4.5. The higher the average value of α , the easier it is to achieve the desired separation. Average values of α for the distillation operations in Table 7.1 range from 1.16 to 81.2.

In 1925, McCabe and Thiele [5] published an approximate graphical method for combining the equilibrium curve of Figure 7.3 with operating-line curves to estimate, for a given binary-feed mixture and column operating pressure, the number of equilibrium stages and the amount of reflux required for a desired degree of separation of the feed.

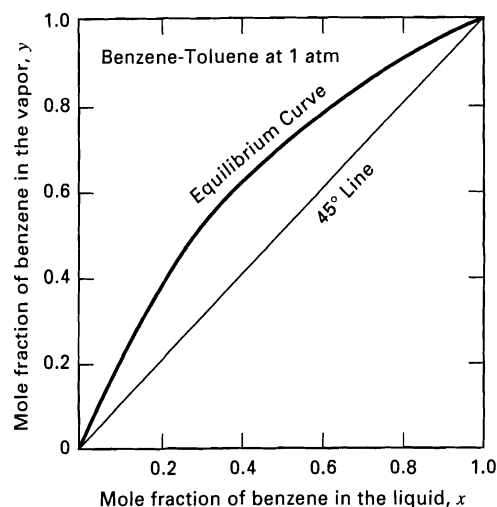


Figure 7.3 Equilibrium curve for benzene-toluene at 1 atm.

Although computer-aided methods, discussed later in Chapter 10, are more accurate and easier to apply, the graphical construction of the McCabe–Thiele method greatly facilitates the visualization of many of the important aspects of multistage distillation, and therefore the effort required to learn the method is well justified.

Typical problem specifications for and results from the McCabe–Thiele method are summarized in Table 7.2. This table applies to a simple, binary-distillation operation, like that in Figure 7.2, for a single feed and two products. The distillate can be a liquid from a total condenser, as shown in Figure 7.2, or a vapor from a partial condenser. The feed phase condition must be known at the column pressure, which is assumed to be uniform throughout the column for the McCabe–Thiele method. The type of condenser and reboiler must be specified, as well as the ratio of reflux to minimum reflux. From the specification of x_D and x_B for the light key, the distillate and bottoms flow rates, D and B , are fixed by material balance, since

$$Fz_F = x_D D + x_B B$$

But, $B = F - D$ and therefore

$$Fz_F = x_D D + x_B (F - D)$$

Table 7.2 Specifications for and Results from the McCabe–Thiele Method for Binary Distillation

Specifications	
F	Total feed rate
z_F	Mole-fraction composition of the feed
P	Column operating pressure (assumed uniform throughout the column)
	Phase condition of the feed at column pressure
	Vapor–liquid equilibrium curve for the binary mixture at column pressure
	Type of overhead condenser (total or partial)
	Type of reboiler (usually partial)
x_D	Mole-fraction composition of the distillate
x_B	Mole-fraction composition of the bottoms
$R/R_{\min}$	Ratio of reflux to minimum reflux
Results	
D	Distillate flow rate
B	Bottoms flow rate
$N_{\min}$	Minimum number of equilibrium stages
$R_{\min}$	Minimum reflux ratio, $L_{\min}/D$
R	Reflux ratio, L/D
V_B	Boilup ratio, $\bar{V}/B$
N	Number of equilibrium stages
	Optimal feed-stage location
	Stage vapor and liquid compositions

All mole fraction compositions are for the light key.

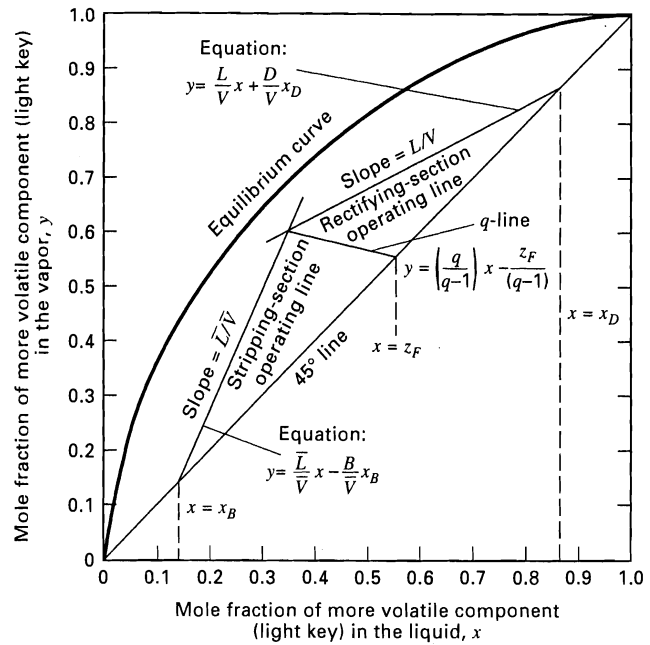


Figure 7.4 Construction lines for McCabe–Thiele method.

or

$$D = F \left(\frac{z_F - x_B}{x_D - x_B} \right)$$

This result requires $x_B < z_F < x_D$.

The McCabe–Thiele method determines not only N , the number of equilibrium stages, but also $N_{\min}$, $R_{\min}$, and the optimal stage for feed entry. Following the application of the McCabe–Thiele method, energy balances are applied to estimate condenser and reboiler heat duties.

Besides the equilibrium curve, the McCabe–Thiele method involves a 45° reference line, separate operating lines for the upper *rectifying* (enriching) section of the column and the lower *stripping* (exhausting) section of the column, and a fifth line (the *q*-line or feed line) for the phase or thermal condition of the feed. A typical set of these lines is shown in Figure 7.4. Equations for these lines are derived in the following subsection.

Rectifying Section

As shown in Figure 7.2, the rectifying section of equilibrium stages extends from the top stage, 1, to just above the feed stage, f . Consider a top portion of the rectifying stages, including the total condenser. A material balance for the light key over the envelope shown in Figure 7.5a for the total condenser and stages 1 to n is as follows, where y and x refer to vapor and liquid mole fractions, respectively, for the light key:

$$V_{n+1}y_{n+1} = L_n x_n + D x_D \quad (7-4)$$

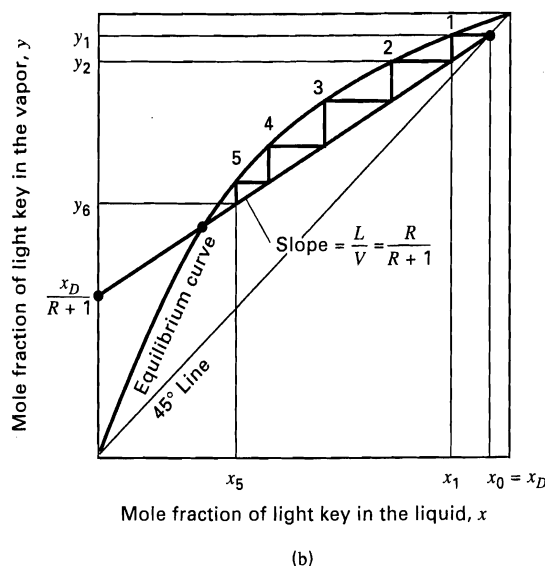
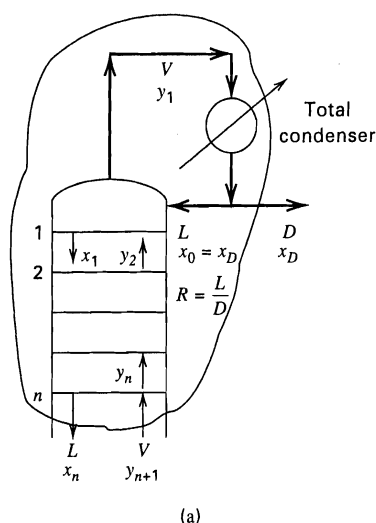


Figure 7.5 McCabe-Thiele operating line for the rectifying section.

Solving for y_{n+1} :

$$y_{n+1} = \frac{L_n}{V_{n+1}}x_n + \frac{D}{V_{n+1}}x_D \quad (7-5)$$

Equation (7-5) relates the compositions, y_{n+1} and x_n , of two passing streams, V_{n+1} and L_n , respectively. For (7-5) to plot as a straight line of the form $y = mx + b$, which is the locus of compositions of all passing streams in the rectifying section, total molar flow rates L and V must not vary from stage to stage. This is the case if:

1. The two components have equal and constant molar enthalpies of vaporization (latent heats).
2. Component sensible-enthalpy changes ($C_P \Delta T$) and heat of mixing are negligible compared to latent heat changes.
3. The column is well insulated so that heat loss is negligible.
4. The pressure is uniform throughout the column (no pressure drop).

These assumptions are referred to as the *McCabe-Thiele assumptions* leading to the condition of *constant molar overflow* in the rectifying section, which refers to a molar liquid flow rate that remains constant as the liquid overflows each weir from one stage to the next. Since a total material balance for the rectifying-section envelope in Figure 7.5a gives $V_{n+1} = L_n + D$, if L is constant, then V is also constant for a particular value of D . Thus, (7-5) can be rewritten as

$$y = \frac{L}{V}x + \frac{D}{V}x_D \quad (7-6)$$

as shown in Figure 7.4. Thus, the slope of the operating line is L/V , which is constant. Because $V > L$, $L/V < 1$ in the rectifying section, as seen in Figure 7.5b.

For constant molar overflow, it is not necessary to consider energy balances in either the rectifying or stripping

sections; only material balances and a vapor-liquid equilibrium curve are required. However, energy balances are needed to determine condenser and reboiler duties, as discussed later.

The liquid entering the top stage is the external reflux rate, L_0 , and its ratio to the distillate rate, L_0/D , is the reflux ratio, R . Because of the assumption of constant molar overflow, R is a constant in the rectifying section, equal to L/D . Since $V = L + D$, the slope of the operating line is readily related to the reflux ratio:

$$\frac{L}{V} = \frac{L}{L + D} = \frac{L/D}{L/D + D/D} = \frac{R}{R + 1} \quad (7-7)$$

Similarly,

$$\frac{D}{V} = \frac{D}{L + D} = \frac{1}{R + 1} \quad (7-8)$$

Combining (7-6), (7-7), and (7-8) produces the most useful form of the operating line for the rectifying section:

$$y = \left(\frac{R}{R + 1} \right)x + \left(\frac{1}{R + 1} \right)x_D \quad (7-9)$$

If values of R and x_D are specified, (7-9) plots as a straight line with an intersection at $y = x_D$ on the 45° line, a slope of $L/V = R/(R + 1)$, and an intersection at $y = x_D/(R + 1)$ for $x = 0$, as shown in Figure 7.5b, which also contains a 45° line and an equilibrium curve. The equilibrium stages are stepped off in the manner described in Section 6.3 for absorption. Starting from the point ($y_1 = x_D$, $x_0 = x_D$) on the operating line and the 45° line, a horizontal line is drawn to the left until it intersects the equilibrium curve at (y_1 , x_1), that is, the compositions of the *equilibrium phases* leaving the top equilibrium stage. A vertical line is now dropped until it intersects the operating line at the point (y_2 , x_1), the compositions of the two *phases passing* each other between stages 1 and 2. The horizontal- and vertical-line constructions are

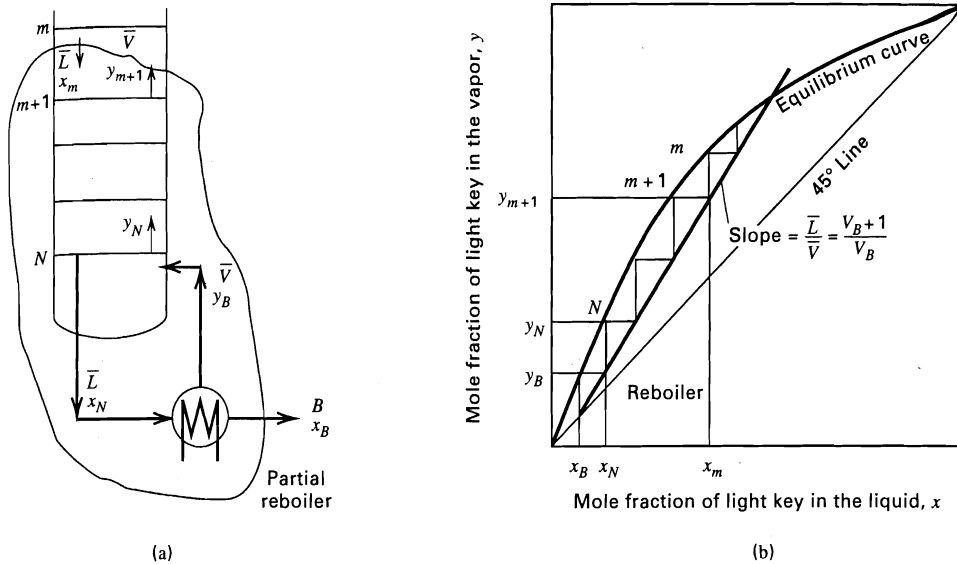


Figure 7.6 McCabe–Thiele operating line for the stripping section.

continued down the rectifying section in the manner shown in Figure 7.5b to give the staircase construction shown, which is arbitrarily terminated at stage 5. The optimal stage for termination is considered later.

Stripping Section

As shown in Figure 7.2, the stripping section of equilibrium stages extends from the feed to the bottom stage. In Figure 7.6a, consider a bottom portion of the stripping stages, including the partial reboiler and extending up from stage N to stage $m + 1$, located somewhere below the feed. A material balance for the light key over the envelope shown in Figure 7.6a results in

$$\bar{L}x_m = \bar{V}y_{m+1} + Bx_B \quad (7-10)$$

Solving for y_{m+1} :

$$y_{m+1} = \frac{\bar{L}}{\bar{V}}x_m - \frac{B}{\bar{V}}x_B$$

or

$$y = \frac{\bar{L}}{\bar{V}}x - \frac{B}{\bar{V}}x_B \quad (7-11)$$

where $\bar{L}$ and $\bar{V}$ are the total molar flows, which by the constant-molar-overflow assumption remain constant from stage to stage. The slope of this operating line for the compositions of passing steams in the stripping section is seen to be $\bar{L}/\bar{V}$. Because $\bar{L} > \bar{V}$, $\bar{L}/\bar{V} > 1$, as seen in Figure 7.6b. This is the inverse of conditions in the rectifying section.

The vapor leaving the partial reboiler is assumed to be in equilibrium with the liquid bottoms product. Thus, the partial reboiler acts as an additional equilibrium stage. The vapor rate leaving it is called the *boilup*, $\bar{V}_{N+1}$, and its ratio to the bottoms product rate, $V_B = \bar{V}_{N+1}/B$, is the *boilup ratio*. Because of the constant-molar-overflow assumption, V_B is constant in the stripping section. Since $\bar{L} = \bar{V} + B$,

$$\frac{\bar{L}}{\bar{V}} = \frac{\bar{V} + B}{\bar{V}} = \frac{V_B + 1}{V_B} \quad (7-12)$$

Similarly,

$$\frac{B}{\bar{V}} = \frac{1}{V_B} \quad (7-13)$$

Combining (7-11), (7-12), and (7-13), the operating-line equation for the stripping section becomes

$$y = \left(\frac{V_B + 1}{V_B} \right) x - \left(\frac{1}{V_B} \right) x_B \quad (7-14)$$

If values of V_B and x_B are known, (7-14) can be plotted, together with the equilibrium curve and a 45° line, as a straight line with an intersection at $y = x_B$ on the 45° line and a slope of $\bar{L}/\bar{V} = (V_B + 1)/V_B$, as shown in Figure 7.6b. The equilibrium stages are stepped off, in a manner similar to that described for the rectifying section, starting from the point ($y = x_B$, $x = x_B$) on the operating and 45° lines and moving upward on a vertical line until the equilibrium curve is intersected at ($y = y_B$, $x = x_B$), which represents the equilibrium mole fractions in the vapor and liquid leaving the partial reboiler. From that point, the staircase is constructed by drawing horizontal and then vertical lines, moving back and forth between the operating line and equilibrium curve, as observed in Figure 7.6b, where the staircase is arbitrarily terminated at stage m . Next, we determine where to terminate the two operating lines.

Feed-Stage Considerations

Thus far, the McCabe–Thiele construction has not considered the feed to the column. In determining the operating lines for the rectifying and stripping sections, it is very important to note that although x_D and x_B can be selected independently, R and V_B are related by the feed phase condition.

Consider the five possible feed conditions shown in Figure 7.7, which assumes that the feed has been flashed adiabatically to the feed-stage pressure. If the feed is a bubble-point liquid, it adds to the reflux, L , coming from the stage above to give $\bar{L} = L + F$. If the feed is a dew-point

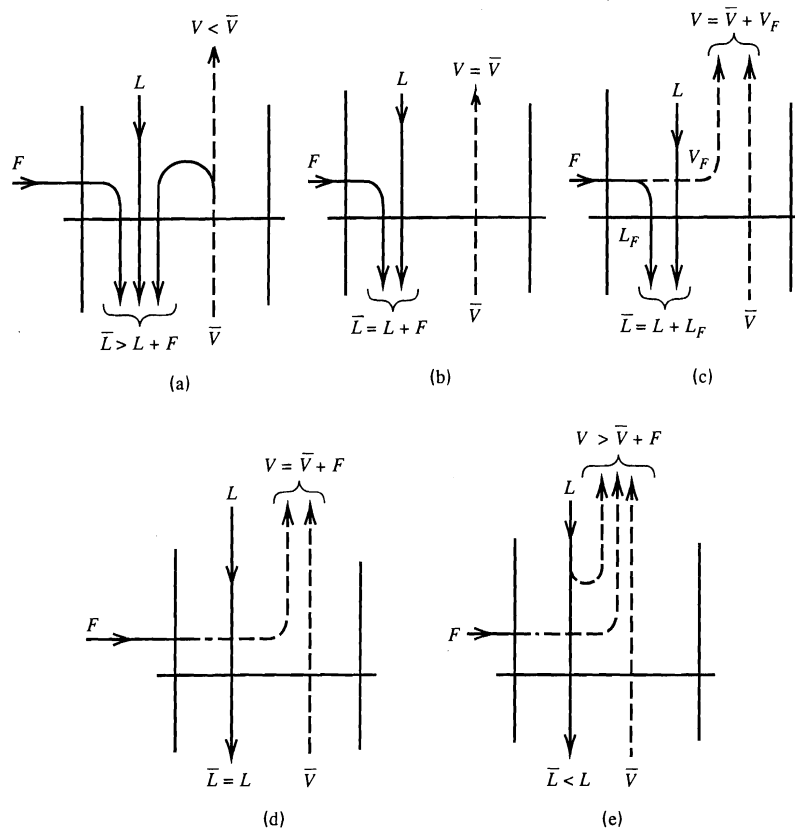


Figure 7.7 Possible feed conditions: (a) subcooled-liquid feed; (b) bubble-point liquid feed; (c) partially vaporized feed; (d) dew-point vapor feed; (e) superheated-vapor feed.

[Adapted from W.L. McCabe, J.C. Smith, and P. Harriott, *Unit Operations of Chemical Engineering*, 5th ed., McGraw-Hill, New York (1993).]

vapor, it adds to the boilup vapor, $\bar{V}$, coming from the stage below to give $V = \bar{V} + F$. For a partially vaporized feed, as shown in Figure 7.7c, $F = L_F + V_F$ and $\bar{L} = L + L_F$ and $V = \bar{V} + V_F$. If the feed is a subcooled liquid, it will cause a portion of the boilup, $\bar{V}$, to condense, giving $\bar{L} > L + F$ and $V < \bar{V}$. If the feed is a superheated vapor, it will cause a portion of the reflux, L , to vaporize, giving $\bar{L} < L$ and $V > \bar{V} + F$.

For cases (b), (c), and (d) of Figure 7.7, covering a range of feed conditions from a saturated liquid to a saturated vapor, the boilup $\bar{V}$ is related to the reflux L by the material balance:

$$\bar{V} = L + D - V_F \quad (7-15)$$

and the boilup ratio, $V_B = \bar{V}/B$, is

$$V_B = \frac{L + D - V_F}{B} \quad (7-16)$$

Alternatively, the reflux can be determined from the boilup by

$$L = \bar{V} + B - L_F \quad (7-17)$$

Although distillation operations can be specified by either the reflux ratio R or the boilup ratio V_B , by tradition R or $R/R_{\min}$ is used because the distillate product is most often the more important product.

For the other two cases, (a) and (e) of Figure 7.7, V_B and R cannot be related by simple material balances alone. It is necessary to consider an energy balance to convert sensible enthalpy into latent enthalpy of phase change. This is most

conveniently done by defining a parameter, q , as the ratio of the increase in molar reflux rate across the feed stage to the molar feed rate,

$$q = \frac{\bar{L} - L}{F} \quad (7-18)$$

or by material balance around the feed stage,

$$q = 1 + \frac{\bar{V} - V}{F} \quad (7-19)$$

Values of q for the five feed conditions are

Feed condition	q
Subcooled liquid	> 1
Bubble-point liquid	1
Partially vaporized	$L_F/F = 1 - \text{molar fraction vaporized}$
Dew-point vapor	0
Superheated vapor	< 0

To determine values of q for subcooled liquid and superheated vapor, a more general definition of q is applied:

q = enthalpy change to bring the feed to a dew-point vapor divided by enthalpy of vaporization of the feed (dew-point vapor minus bubble-point liquid), that is,

$$q = \frac{(h_F)_{\text{sat'd vapor temperature}} - (h_F)_{\text{feed temperature}}}{(h_F)_{\text{sat'd vapor temperature}} - (h_F)_{\text{sat'd liquid temperature}}} \quad (7-20)$$

For a subcooled liquid feed, (7-20) becomes

$$q = \frac{\Delta H^{\text{vap}} + C_{P_L}(T_b - T_F)}{\Delta H^{\text{vap}}} \quad (7-21)$$

For a superheated vapor, (7-20) becomes

$$q = \frac{C_{P_V}(T_d - T_F)}{\Delta H^{\text{vap}}} \quad (7-22)$$

where C_{P_L} and C_{P_V} are the liquid and vapor molar heat capacities, respectively, ΔH^{vap} is the molar enthalpy change from the bubble point to the dew point, and T_F , T_d , and T_b are the feed, dew-point, and bubble-point temperatures, respectively, of the feed at the column operating pressure.

Instead of using (7-14) to locate the stripping operating line on the McCabe–Thiele diagram, it is more common to use an alternative method that involves a q -line (feed line), which is included in Figure 7.4. The q -line, one point of which is the intersection of the rectifying and stripping operating lines, is derived in the following manner. Combining (7-11) with (7-6) gives

$$y(V - \bar{V}) = (L - \bar{L})x + Dx_D + Bx_B \quad (7-23)$$

But

$$Dx_D + Bx_B = Fz_F \quad (7-24)$$

and a material balance around the feed stage gives

$$F + \bar{V} + L = V + \bar{L} \quad (7-25)$$

Combining (7-23) to (7-25) with (7-18) gives

$$y = \left(\frac{q}{q-1} \right) x - \left(\frac{z_F}{q-1} \right) \quad (7-26)$$

which is the equation for the q -line. This line is located on the McCabe–Thiele diagram by noting that when $x = z_F$, (7-26) reduces to the point $y = z_F = x$, which lies on the 45° line. From (7-26), the slope of the line is $q/(q-1)$. This construction is shown in Figure 7.4 for a partially vaporized feed, for which $0 < q < 1$ and $-\infty < [q/(q-1)] < 0$. Following the placement of the rectifying-section operating line and the q -line, the stripping-section operating line is located by drawing a straight line from the point $(y = x_B, x = x_B)$ on the 45° line to and through the point of intersection of the q -line and the rectifying-section operating line as shown in Figure 7.4. The point of intersection must lie somewhere between the equilibrium curve and the 45° line.

As q changes from a value greater than 1 (subcooled liquid) to a value less than 0 (superheated vapor), the slope of the q -line, $q/(q-1)$, changes from a positive value to a negative value and back to a positive value, as shown in Figure 7.8. For a saturated liquid feed, the q -line is vertical; for a saturated vapor, the q -line is horizontal.

Determination of Number of Equilibrium Stages and Feed-Stage Location

Following the construction of the five lines shown in Figure 7.4, the number of equilibrium stages required for the entire column, as well as the location of the feed stage, are determined by stepping off stages by any of several ways. The stages can be stepped off first from the top down and

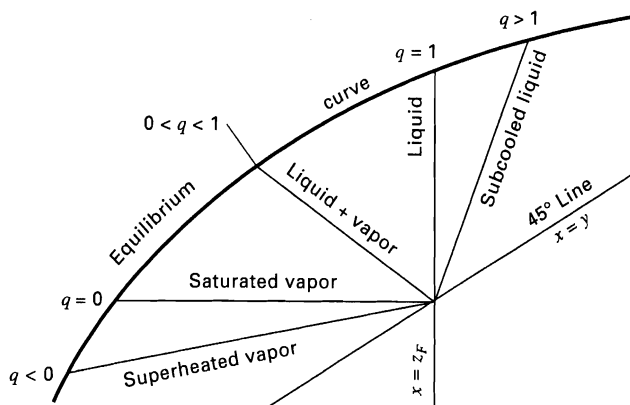


Figure 7.8 Effect of thermal condition of feed on slope of q -line.

then from the bottom up, as described above, until a point of merger is found for the feed stage. Alternatively, the stages can be stepped off from the bottom all the way to the top, or vice versa. Hardly ever will an integer number of stages result, but rather a fractional stage will appear near the middle, at the top, or at the bottom. Usually the staircase is stepped off from the top and continued all the way to the bottom, starting from the point $(y = x_D, x = x_D)$ on the 45° line, as shown in Figure 7.9 for the case of a partially vaporized feed. In that figure, point P is the intersection of the q -line with the two operating lines. The transfer point for stepping off stages between the rectifying-section operating line and the equilibrium curve to stepping off stages between the stripping-section operating line and the equilibrium curve occurs at the feed stage. In Figure 7.9a, the feed stage is stage 3 from the top and a fortuitous total of exactly five stages is required, where the last stage is the partial reboiler. In Figure 7.9b the feed stage is stage 5 and a total of about 6.4 stages is required. In Figure 7.9c, the feed stage is stage 2 and a total of about 5.9 stages is required. In Figure 7.9b, the stepping off of stages in the rectifying section can be continued indefinitely, finally approaching, but never reaching, point K. In Figure 7.9c, if the stepping off of stages had started from the partial reboiler at the point $(y = x_B, x = x_B)$ and proceeded upward, the staircase in the stripping section could have been continued indefinitely, finally approaching, but never reaching, point R. In Figure 7.9, it is seen that the smallest number of total stages occurs when the transfer is made at the first opportunity after a horizontal line of the staircase passes over point P, as in Figure 7.9a. This feed-stage location is optimal.

Limiting Conditions

For a given specification (Table 7.2), a reflux ratio can be selected anywhere from the minimum, $R_{\min}$, to an infinite value (total reflux) where all of the overhead vapor is condensed and returned to the top stage (thus, no distillate is withdrawn). As shown in Figure 7.10b, the minimum reflux corresponds to the need for an infinite number of stages, while in Figure 7.10a the infinite reflux ratio corresponds to

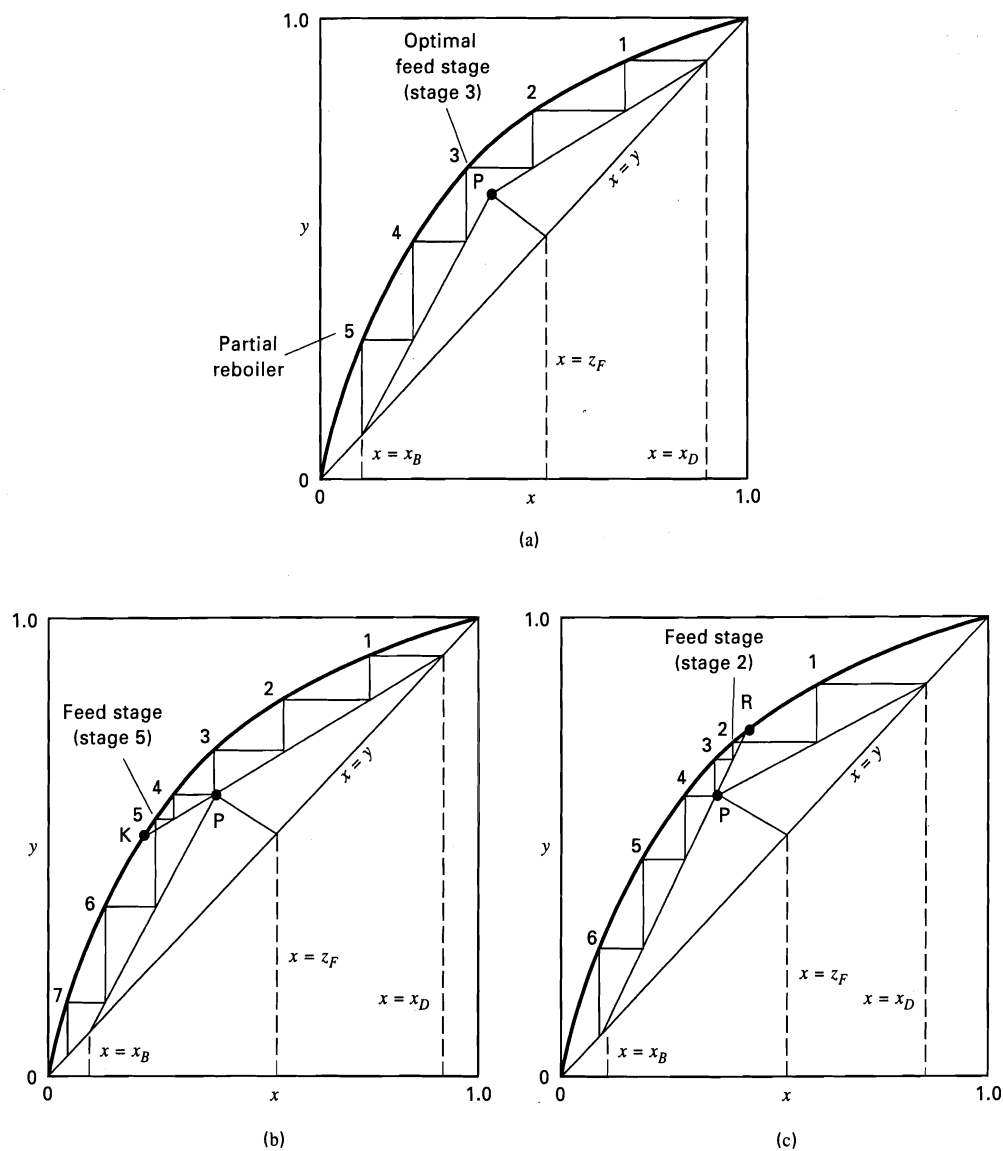


Figure 7.9 Optimal and nonoptimal locations of feed stage: (a) optimal feed-stage location; (b) feed-stage location below optimal stage; (c) feed-stage location above optimal stage.

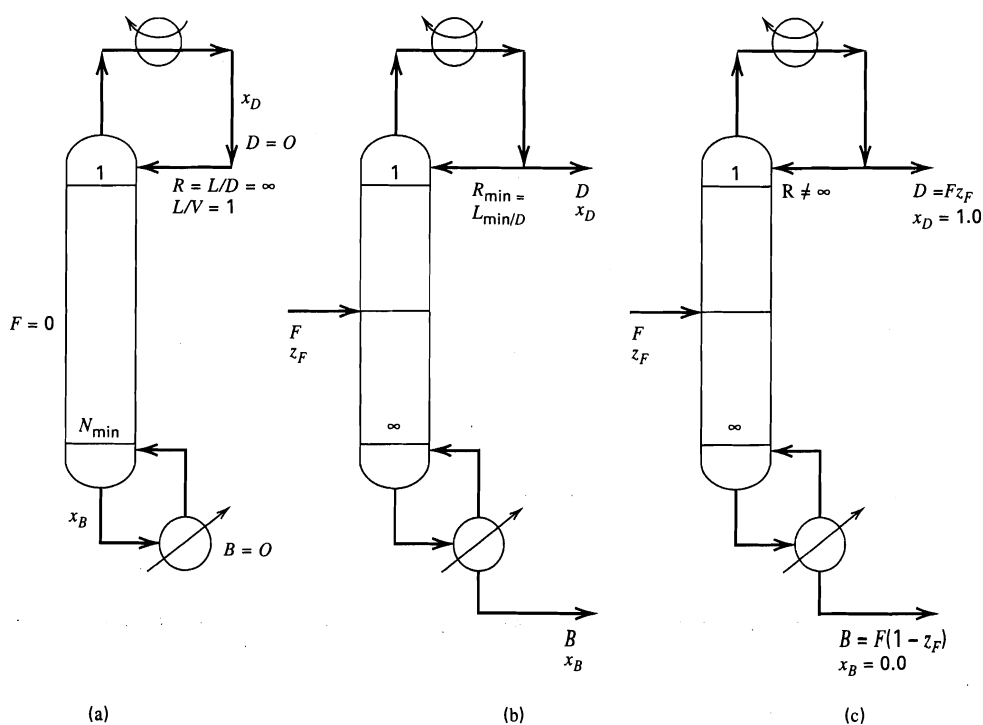


Figure 7.10 Limiting conditions for distillation: (a) total reflux, minimum stages; (b) minimum reflux, infinite stages; (c) perfect separation for nonazeotropic system.

the minimum number of equilibrium stages. The McCabe–Thiele graphical method can quickly determine the two limits, $N_{\min}$ and $R_{\min}$. Then, for a practical operation, $N_{\min} < N < \infty$ and $R_{\min} < R < \infty$.

Minimum Number of Equilibrium Stages

As the reflux ratio is increased, the slope of the rectifying-section operating line, given by (7-7), increases from $L/V < 1$ to a limiting value of $L/V = 1$. Correspondingly, the boilup ratio increases and the slope of the stripping section operating line, given by (7-12), decreases from $\bar{L}/\bar{V} > 1$ to a limiting value of $\bar{L}/\bar{V} = 1$. Thus, at this limiting condition, both the rectifying and stripping operating lines coincide with the 45° line and neither the feed composition, z_F , nor the q -line influences the staircase construction. This is total reflux because when $L = V$, $D = B = 0$, and the total condensed overhead is returned to the column as reflux. Furthermore, all liquid leaving the bottom stage is vaporized and returned as boilup to the column. If both distillate and bottoms flow rates are zero, the feed to the column is also zero, which is consistent with the lack of influence of the feed condition. It is possible to operate a column at total reflux, and such an operation is convenient for the experimental measurement of tray efficiency because a steady-state operating condition is readily achieved.

A simple example of the McCabe–Thiele construction for this limiting condition is shown in Figure 7.11 for two equilibrium stages. Because the operating lines are located as far away as possible from the equilibrium curve, a minimum number of stages is required.

Minimum Reflux Ratio

As the reflux ratio decreases from the limiting case of infinity (i.e., total reflux), the intersection of the two operating lines and the q -line moves away from the 45° line toward the equilibrium curve. The number of equilibrium stages required increases because the operating lines move closer and closer to the equilibrium curve, thus requiring more and

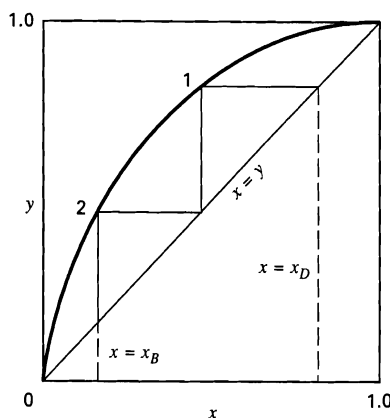


Figure 7.11 Construction for minimum stages at total reflux.

more stairs to move from the top of the column to the bottom. Finally a limiting condition is reached when the point of intersection is on the equilibrium curve, as shown in Figure 7.12. For binary mixtures that are not highly nonideal, the typical case is shown in Figure 7.12a, where the intersection, P, is at the feed stage. To reach that stage from either the rectifying section or the stripping section, an infinite number of stages is required. The point P is called a *pinch point* because the two operating lines each pinch the equilibrium curve.

For a highly nonideal binary system, the pinch point may occur at a stage above or below the feed stage. The former case is illustrated in Figure 7.12b, where the operating line for the rectifying section intersects the equilibrium curve before the feed stage is reached. The slope of this operating line cannot be reduced further because it would then cross over the equilibrium curve and thereby violate the second law of thermodynamics because of a reversal in the direction of mass transfer. This would require spontaneous mass transfer from a region of low concentration to a region of high concentration. This is similar to a second-law violation by a temperature crossover in a heat exchanger. Now, the pinch point occurs entirely in the rectifying section, where an infinite number of stages exists; the stripping section contains a finite number of stages.

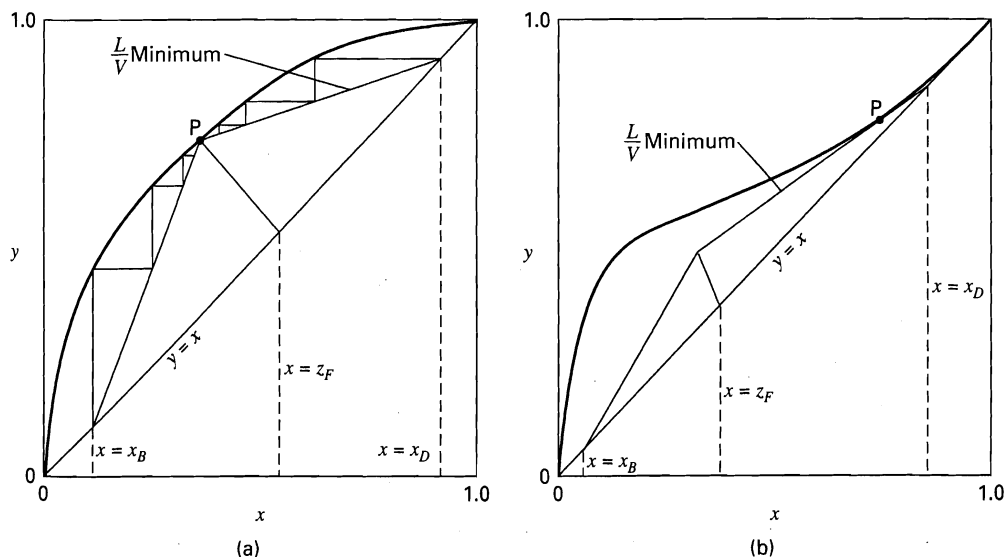


Figure 7.12 Construction for minimum reflux at infinite stages: (a) typical ideal or near-ideal system, pinch point at the feed stage; (b) typical nonideal system, pinch point above the feed stage.

From the slope of the limiting operating line for the rectifying section, the minimum reflux ratio can be determined. From (7-7), the minimum feasible slope is

$$(L/V)_{\min} = R_{\min}/(R_{\min} + 1)$$

or

$$R_{\min} = (L/V)_{\min}/[1 - (L/V)_{\min}] \quad (7-27)$$

Alternatively, the limiting condition of infinite stages corresponds to a minimum boilup ratio for $(\bar{L}/\bar{V})_{\max}$. From (7-12),

$$(V_B)_{\min} = 1/[(\bar{L}/\bar{V})_{\max} - 1] \quad (7-28)$$

Perfect Separation

A third limiting condition of interest involves the degree of separation. As a perfect split ($x_D = 1$, $x_B = 0$) is approached, for a reflux ratio at or greater than the minimum value, the number of stages required near the top and near the bottom

of the column increases rapidly and without limit until pinches are encountered at $x_D = 1$ and $x_B = 0$. Thus, a perfect separation of a binary mixture that does not form an azeotrope requires an infinite number of stages in both sections of the column. However, this is not the case for the reflux ratio. In Figure 7.12a, as x_D is moved from, say, 0.90 toward 1.0, the slope of the operating line at first increases, but in the range of x_D from 0.99 to 1.0 the slope changes only slightly. Furthermore, the value of the slope, and therefore the value of R , is finite for a perfect separation. For example, if the feed is a saturated liquid, application of (7-4) and (7-7) gives the following equation for the minimum reflux of a perfect binary separation:

$$R_{\min} = \frac{1}{z_F(\alpha - 1)} \quad (7-29)$$

where the relative volatility, α , is evaluated at the feed condition.

EXAMPLE 7.1

A trayed tower is to be designed to continuously distill 450 lbmol/h (204 kmol/h) of a binary mixture of 60 mol% benzene and 40 mol% toluene. A liquid distillate and a liquid bottoms product of 95 mol% and 5 mol% benzene, respectively, are to be produced. The feed is preheated so that it enters the column with a molar percent vaporization equal to the distillate-to-feed ratio. Use the McCabe–Thiele method to compute the following, assuming a uniform pressure of 1 atm (101.3 kPa) throughout the column: (a) Minimum number of theoretical stages, $N_{\min}$; (b) Minimum reflux ratio, $R_{\min}$; and (c) Number of equilibrium stages N , for a reflux-to-minimum reflux ratio, $R/R_{\min}$, of 1.3 and the optimal location of the feed stage.

SOLUTION

Calculate D and B . An overall material balance on benzene gives

$$0.60(450) = 0.95D + 0.05B \quad (1)$$

$$\text{A total balance gives } 450 = D + B \quad (2)$$

Combining (1) and (2) to eliminate B , followed by solving the resulting equation for D and (2) for B gives $D = 275$ lbmol/h, $B = 175$ lbmol/h, and $D/F = 0.611$

Calculate the slope of the q -line:

$V_F/F = D/F$ for this example = 0.611 and q for a partially vaporized feed is

$$\frac{L_F}{F} = \frac{(F - V_F)}{F} = 1 - \frac{V_F}{F} = 0.389$$

From (7-26),

$$\text{the slope of the } q\text{-line is } \frac{q}{q - 1} = \frac{0.389}{0.389 - 1} = -0.637$$

(a) In Figure 7.13, where y and x refer to benzene, the more-volatile component, with $x_D = 0.95$ and $x_B = 0.05$, the number of minimum equilibrium stages is stepped off from the top between the equilibrium curve and the 45° line, giving $N_{\min} = 6.7$.

(b) In Figure 7.14, a q -line is drawn that has a slope of -0.637 and passes through the feed composition ($z_F = 0.60$) on the 45° line. For the minimum-reflux condition, an operating line for the rectifying section passes through the point $x = x_D = 0.95$ on

the 45° line and through the point of intersection of the q -line and the equilibrium curve ($y = 0.684$, $x = 0.465$). The slope of this operating line is 0.55, which from (7-9) equals $R/(R + 1)$. Therefore, $R_{\min} = 1.22$.

(c) The operating reflux ratio is $1.3R_{\min} = 1.3(1.22) = 1.59$

From (7-9), the slope of the operating line for the rectifying section is

$$\frac{R}{R + 1} = \frac{1.59}{1.59 + 1} = 0.614$$

The construction for the resulting two operating lines, together with the q -line, is shown in Figure 7.15, where the operating line for the stripping section is drawn to pass through the point $x = x_B = 0.05$ on the 45° line and the point of intersection of the q -line and the operating line for the stripping section. The number of equilibrium stages is stepped off between, first, the rectifying-section operating line and the equilibrium curve and then the stripping-section operating line and the equilibrium curve, starting from point A

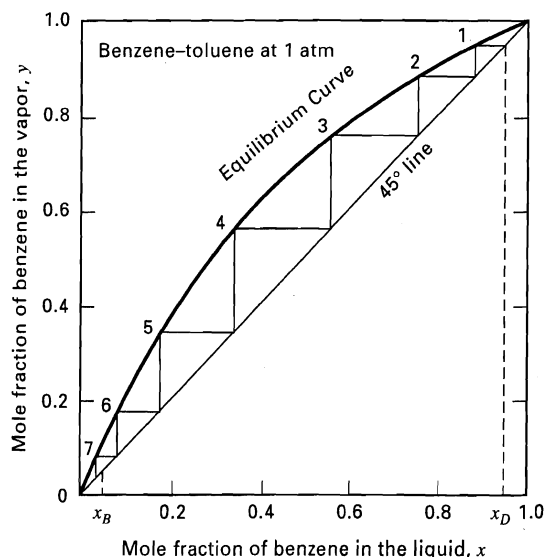


Figure 7.13 Determination of minimum stages for Example 7.1.

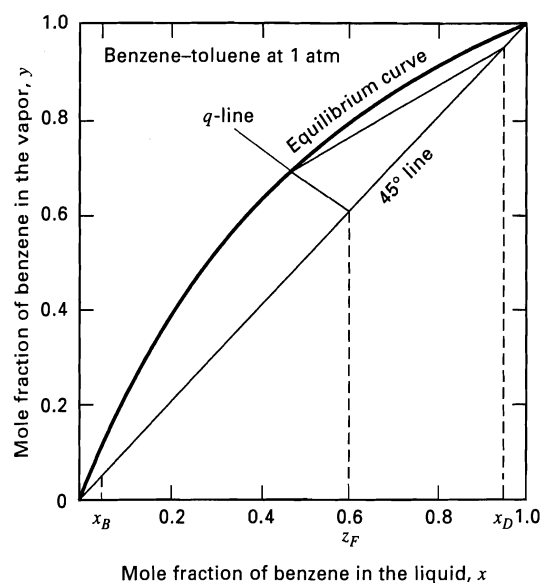


Figure 7.14 Determination of minimum reflux for Example 7.1.

(at $x = x_D = 0.95$) and finishing at point B (to the left of $x = x_B = 0.05$). For the optimal feed-stage location, the transfer from the rectifying-section operating line to the stripping-section operating line takes place at point P. The result is $N = 13.2$ equilibrium

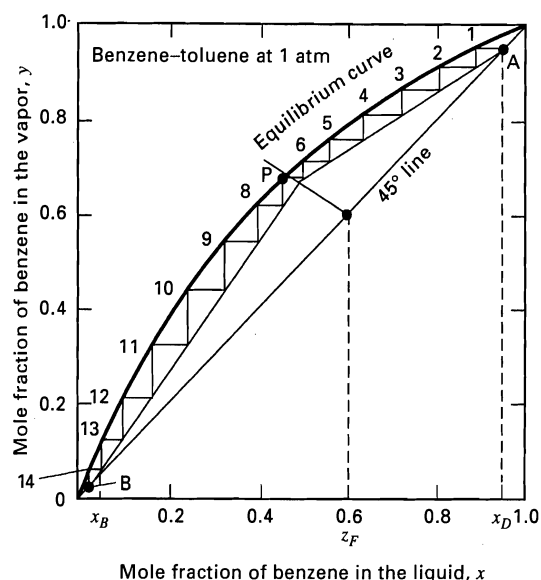


Figure 7.15 Determination of number of equilibrium stages and feed-stage location for Example 7.1.

stages, with stage 7 from the top being the feed stage. Thus, for this example, $N/N_{\min} = 13.2/6.7 = 1.97$. The bottom stage is the partial reboiler, leaving 12.2 equilibrium stages contained in the column. If the plate efficiency were 0.8, 16 trays would be needed.

Column Operating Pressure and Condenser Type

For preliminary design, column operating pressure and condenser type are established by the procedure shown in Figure 7.16, which is formulated to achieve, if possible, a reflux-drum pressure, P_D , between 0 and 415 psia (2.86 MPa) at a minimum temperature of 120°F (49°C) (corresponding to the use of water as the coolant in the condenser). The pressure and temperature limits are representative only and depend on economic factors. Columns can operate at pressures higher than 415 psia if the critical or convergence

pressure of the mixture is not approached. A condenser pressure drop of 0 to 2 psi (0 to 14 kPa) and an overall, column pressure drop of 5 psi (35 kPa) may be assumed. However, when column tray requirements are known, more refined computations should result in approximately 0.1 psi/tray (0.7 kPa/tray) pressure drop for atmospheric and superatmospheric pressure operation and 0.05 psi/tray (0.35 kPa/tray) pressure drop for vacuum-column operation. Column bottom temperature must not result in bottoms decomposition or correspond to a near-critical condition. Therefore, after

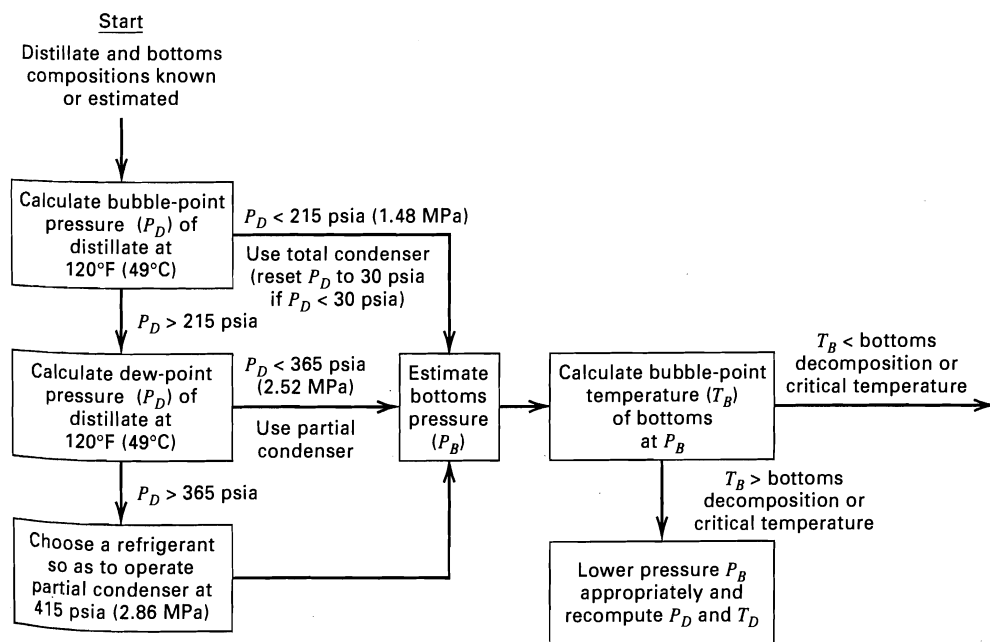


Figure 7.16 Algorithm for establishing distillation-column pressure and condenser type.

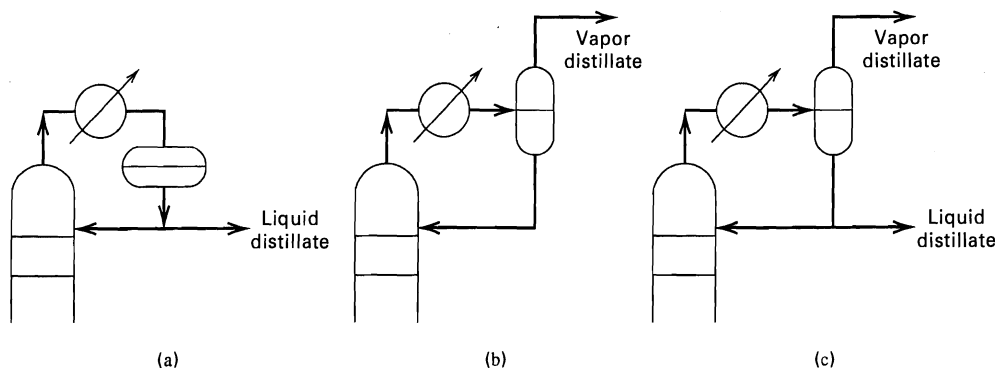


Figure 7.17 Condenser types: (a) total condenser; (b) partial condenser; (c) mixed condenser.

the bottoms pressure is estimated from the pressure in the reflux drum, a bubble-point temperature of the bottoms is computed at the bottoms pressure. If that temperature exceeds the bottoms decomposition or critical temperature, then the bottoms pressure is recomputed at or below the bubble-point decomposition or critical temperature. The pressure in the reflux drum will then be lower and must be recomputed, together with the distillate temperature, from the assumed column and condenser pressure drops. This will result often in vacuum operation. If the recomputed distillate temperature is less than 120°F (49°C), a refrigerant, rather than cooling water, is used for the condenser.

A total condenser is recommended for reflux drum pressures to 215 psia (1.48 MPa). A partial condenser is appropriate from 215 psia to 365 psia (2.52 MPa). However, a partial condenser can be used below 215 psia when a vapor distillate is desired. A mixed condenser can provide both vapor and liquid distillates. The three types of condenser configurations are shown in Figure 7.17. A refrigerant is often used as condenser coolant if pressure tends to exceed 365 psia.

When a partial condenser is specified, the McCabe–Thiele staircase construction for the case of a total condenser must be modified, as will be illustrated in Example 7.2, to account for the fact that the first equilibrium stage, counted down from the top, is now the partial condenser. This is based on the assumption that the liquid reflux leaving the reflux drum is in equilibrium with the vapor distillate.

Subcooled Reflux

Although most distillation columns are designed so that the reflux is a saturated (bubble-point) liquid, such is not always the case for operating columns. If the condenser type is partial or mixed, the reflux is a saturated liquid unless heat losses cause its temperature to decrease. For a total condenser, however, the operating reflux is often a subcooled liquid at column pressure, particularly if the condenser is not tightly designed and the distillate bubble-point temperature is significantly higher than the inlet cooling-water temperature. If the condenser outlet pressure is lower than the top-tray pressure of the column, the reflux is subcooled for any of the three types of condensers.

When subcooled reflux enters the top tray, its temperature rises and causes vapor entering the tray to condense. The

latent enthalpy of condensation of the vapor provides the sensible enthalpy to heat the subcooled reflux to the bubble point. In that event, the internal reflux ratio within the rectifying section of the column is higher than the external reflux ratio from the reflux drum. The McCabe–Thiele construction should be based on the internal reflux ratio, which can be estimated by the following equation derived from an approximate energy balance around the top tray:

$$R_{\text{internal}} = R \left(1 + \frac{C_{P_L} \Delta T_{\text{subcooling}}}{\Delta H^{\text{vap}}} \right) \quad (7-30)$$

where C_{P_L} and ΔH^{vap} are per mole and $\Delta T_{\text{subcooling}}$ is the degrees of subcooling. The internal reflux ratio replaces R , the external reflux ratio, in (7-9). If a correction is not made for subcooled reflux, the calculated number of equilibrium stages is somewhat more than required.

EXAMPLE 7.2

One thousand kmol/h of a feed containing 30 mol% *n*-hexane and 70% *n*-octane is to be distilled in a column consisting of a partial reboiler, one equilibrium (theoretical) plate, and a partial condenser, all operating at 1 atm (101.3 kPa). Thus, hexane is the light key and octane is the heavy key. The feed, a bubble-point liquid, is fed to the reboiler, from which a liquid bottoms product is continuously withdrawn. Bubble-point reflux is returned from the partial condenser to the plate. The vapor distillate, in equilibrium with the reflux, contains 80 mol% hexane, and the reflux ratio, L/D , is 2. Assume that the partial reboiler, plate, and partial condenser each function as equilibrium stages.

- Using the McCabe–Thiele method, calculate the bottoms composition and kmol/h of distillate produced.
- If the relative volatility α is assumed constant at a value of 5 over the composition range (the relative volatility actually varies from approximately 4.3 at the reboiler to 6.0 at the condenser), calculate the bottoms composition analytically.

SOLUTION

First determine whether the problem is completely specified. From Table 5.4c, we have $N_D = C + 2N + 6$ degrees of freedom, where N includes the partial reboiler and the stages in the column, but not the partial condenser. With $N = 2$ and $C = 2$, $N_D = 12$. Specified in

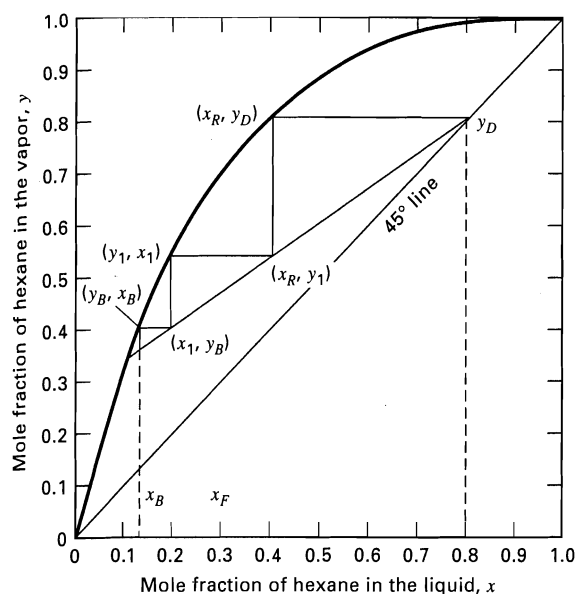


Figure 7.18 Solution to Example 7.2.

this problem are

Feed-stream variables	4
Plate and reboiler pressures	2
Condenser pressure	1
$Q (=0)$ for plate	1
Number of stages	1
Feed-stage location	1
Reflux ratio, L/D	1
Distillate composition	$\frac{1}{1}$
Total	12

Thus, the problem is fully specified and can be solved.

(a) *Graphical solution.* A diagram of the separator is given in Figure 7.18 as is the McCabe–Thiele graphical solution, which is constructed in the following manner.

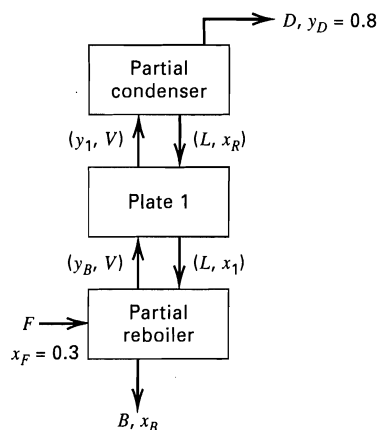
1. The point $y_D = 0.8$ at the partial condenser is located on the $x = y$ line.
2. Conditions in the condenser are fixed because x_R (reflux composition) is in equilibrium with y_D . Hence, the point (x_R, y_D) is located on the equilibrium curve.
3. Noting that $(L/V) = 1 - 1/[1 + (L/D)] = 2/3$, the operating line with slope $L/V = 2/3$ is drawn through the point $y_D = 0.8$ on the 45° line until it intersects the equilibrium curve. Because the feed is introduced into the partial reboiler, there is no stripping section.
4. Three theoretical stages (partial condenser, plate 1, and partial reboiler) are stepped off and the bottoms composition $x_B = 0.135$ is read.

The amount of distillate is determined from overall material balances. For hexane, $z_F F = y_D D + x_B B$. Therefore, $(0.3)(1,000) = (0.8)D + (0.135)B$. For the total flow, $B = 1,000 - D$. Solving these two equations simultaneously, $D = 248$ kmol/h.

(b) *Analytical solution.* For constant α , equilibrium liquid compositions for the light key, in terms of α and y are given by a rearrangement of (7-3):

$$x = \frac{y}{y + \alpha(1 - y)} \quad (1)$$

where α is assumed constant at a value of 5.



The steps in the solution are as follows:

1. The liquid leaving the partial condenser at x_R is calculated from (1), for $y = y_D = 0.8$:

$$x_R = \frac{0.8}{0.8 + 5(1 - 0.8)} = 0.44$$

2. Then y_1 is determined by a material balance about the partial condenser:

$$V y_1 = D y_D + L x_R \quad \text{with } D/V = 1/3 \quad \text{and } L/V = 2/3$$

$$y_1 = (1/3)(0.8) + (2/3)(0.44) = 0.56$$

3. From (1), for plate 1, $x_1 = \frac{0.56}{0.56 + 5(1 - 0.56)} = 0.203$

4. By material balance around plate 1 and the partial condenser,

$$V y_B = D y_D + L x_1$$

$$\text{and } y_B = (1/3)(0.8) + (2/3)(0.203) = 0.402$$

5. From (1), for the partial reboiler,

$$x_B = \frac{0.402}{0.402 + 5(1 - 0.402)} = 0.119.$$

By approximating the equilibrium curve with $\alpha = 5$, an answer of 0.119 is obtained rather than 0.135 for x_B obtained in part (a). Note that for a larger number of theoretical plates, part (b) can be readily computed with a spreadsheet program.

EXAMPLE 7.3

Consider Example 7.2. (a) Solve it graphically, assuming that the feed is introduced on plate 1, rather than into the reboiler. (b) Determine the minimum number of stages required to carry out the separation. (c) Determine the minimum reflux ratio.

SOLUTION

(a) The flowsheet and solution given in Figure 7.19 are obtained as follows.

1. The point x_R, y_D is located on the equilibrium line.
2. The operating line for the enriching section is drawn through the point $y = x = 0.8$, with a slope of $L/V = 2/3$.

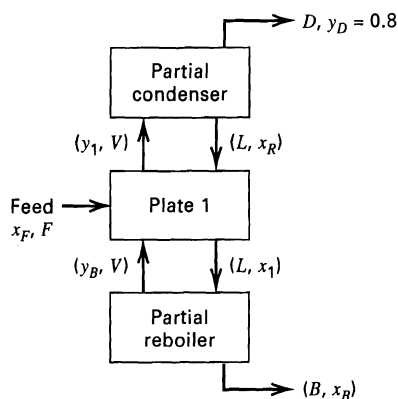
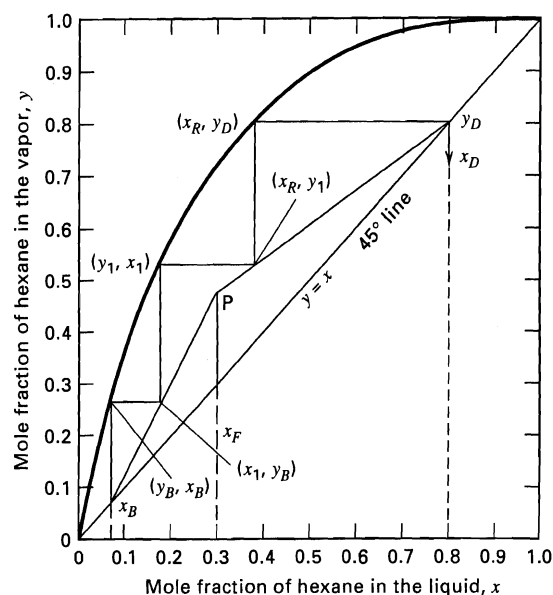


Figure 7.19 Solution to Example 7.3.

3. The intersection of the q -line, $x_F = 0.3$ (which, for a saturated liquid, is a vertical line), with the enriching-section operating line is located at point P. The stripping-section operating line must also pass through this point, but its slope and the point x_B are not known initially.
4. The slope of the stripping-section operating line is found by trial and error to give three equilibrium contacts in the column, with the middle stage involved in the switch from one operating line to the other. If the middle stage is the optimal feed-stage location, the result is $x_B = 0.07$, as shown in Figure 7.19. The amount of distillate is obtained from the combined total and hexane overall material balances to give $(0.3)(1,000) = (0.8D) + 0.07(1,000 - D)$. Solving, $D = 315$ kmol/h.

Comparing this result to that obtained in Example 7.2, we find that the bottoms purity and distillate yield are improved by introduction of the feed to plate 1, rather than to the reboiler. This improvement could have been anticipated if the q -line had been constructed in Figure 7.18. That is, the partial reboiler is not the optimal feed-stage location.

- (b) The construction corresponding to total reflux ($L/V = 1$, no products, no feed, minimum equilibrium stages) is shown

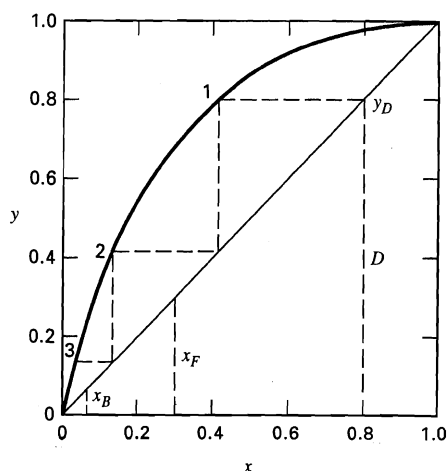


Figure 7.20 Solution for total reflux in Example 7.3.

in Figure 7.20. Slightly more than two stages are required for an x_B of 0.07, compared to the three stages previously required.

- (c) To determine the minimum-reflux ratio, the vertical q -line in Figure 7.19 is extended from point P until the equilibrium curve is intersected, which is determined to be the point (0.71, 0.3). The slope, $(L/V)_{\min}$ of the operating line for the rectifying section, which connects this point to the point (0.8, 0.8) on the 45° line is 0.18. Thus $(L/D)_{\min} = (L/V_{\min}) / [1 - (L/V_{\min})] = 0.22$. This is considerably less than the $L/D = 2$ specified.

Reboiler Type

Different types of reboilers are used to provide boilup vapor to the stripping section of a distillation column. For small laboratory and pilot-plant-size columns, the reboiler consists of a reservoir of liquid located just below the bottom plate to which heat is supplied from (1) a jacket or mantle that is heated by an electrical current or by condensing steam, or (2) tubes that pass through the liquid reservoir carrying condensing steam. Both of these types of reboilers have limited heat-transfer surface and are not suitable for industrial applications.

For plant-size distillation columns, the reboiler is usually an external heat exchanger, as shown in Figure 7.21, of either the kettle or vertical thermosyphon type. Both can provide the amount of heat-transfer surface required for large installations. In the former case, liquid leaving the sump (reservoir) at the bottom of the column enters the kettle, where it is partially vaporized by the transfer of heat from tubes carrying condensing steam or some other heating medium. The bottoms product liquid leaving the reboiler is assumed to be in equilibrium with the vapor returning to the bottom tray of the column. Thus the kettle reboiler is a partial reboiler equivalent to one equilibrium stage. The kettle

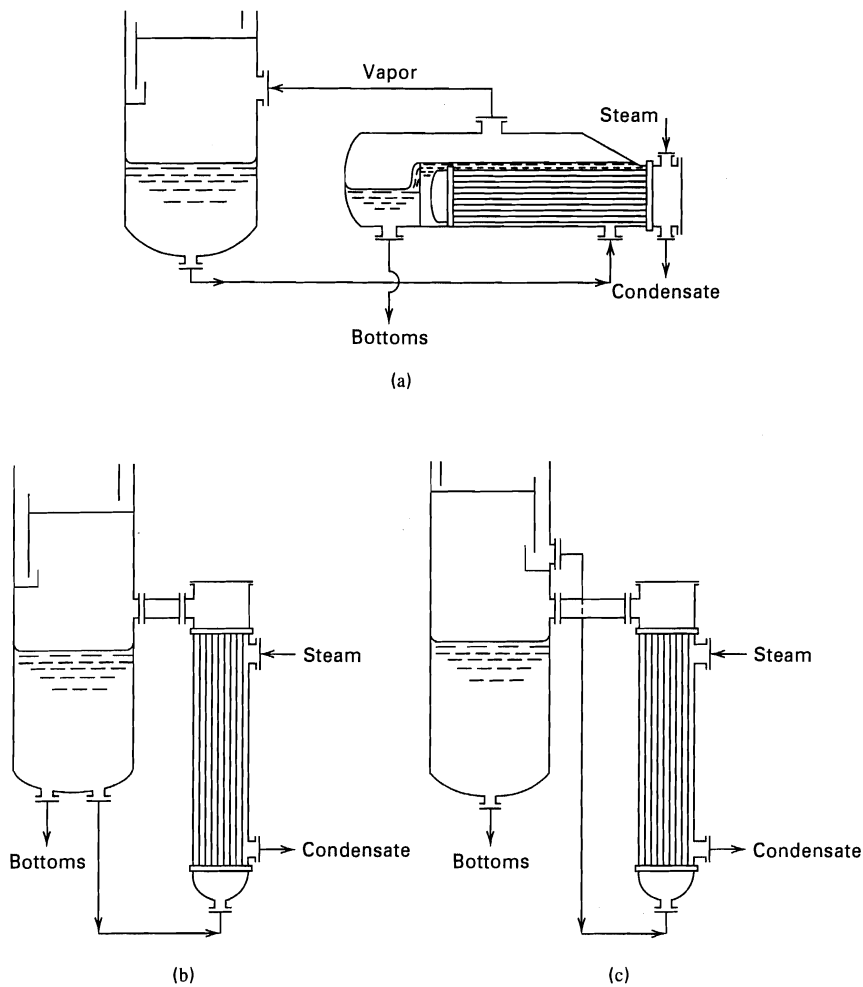


Figure 7.21 Reboilers for plant-size distillation columns: (a) kettle-type reboiler; (b) vertical thermosyphon-type reboiler, reboiler liquid withdrawn from bottom sump; (c) vertical thermosyphon-type reboiler, reboiler liquid withdrawn from bottom-tray downcomer.

reboiler is sometimes located in the bottom of the column to avoid piping.

The vertical thermosyphon reboiler may be of the type shown in Figure 7.21b or 7.21c. In the former, both the bottoms product and the reboiler feed are withdrawn from the column bottom sump. Circulation through the tubes of the reboiler occurs because of the difference in static heads of the supply liquid and the column of partially vaporized fluid flowing through the reboiler tubes. The partial vaporization provides enrichment of the exiting vapor in the more volatile component. However, the exiting liquid is then mixed with liquid leaving the bottom tray, which contains a higher percentage of the more volatile component. The result is that this type of reboiler arrangement provides only a fraction of an equilibrium stage and it is best to take no credit for it.

A more complex and less-common vertical thermosyphon reboiler is that of Figure 7.21c, where the reboiler liquid is withdrawn from the downcomer of the bottom tray. Partially vaporized liquid is returned to the column, where the bottoms product from the bottom sump is withdrawn. This type of reboiler does function as an equilibrium stage.

Kettle reboilers are common, but thermosyphon reboilers are favored when (1) the bottoms product contains thermally sensitive compounds, (2) bottoms pressure is high, (3) only a small ΔT is available for heat transfer, and (4) heavy

fouling occurs. Horizontal thermosyphon reboilers are sometimes used in place of the vertical types when only small static heads are needed for circulation, surface-area requirement is very large, and/or when frequent cleaning of the tubes is anticipated. A pump may be added for either thermosyphon type to improve circulation. Liquid residence time in the column bottom sump should be at least 1 minute and perhaps as much as 5 minutes or more.

Condenser and Reboiler Duties

Following the determination of the feed condition, reflux ratio, and number of theoretical stages by the McCabe–Thiele method, estimates of the heat duties of the condenser and reboiler are made. An energy balance for the entire column gives

$$Fh_F + Q_R = Dh_D + Bh_B + Q_C + Q_{\text{loss}} \quad (7-31)$$

Except for small and/or uninsulated distillation equipment, Q_{loss} is negligible and can be ignored. We can approximate the energy balance of (7-31) by applying the assumptions of the McCabe–Thiele method. An energy balance for a total condenser is

$$Q_C = D(R + 1) \Delta H^{\text{vap}} \quad (7-32)$$

where ΔH^{vap} = average molar heat of vaporization of the two components being separated. For a partial condenser,

$$Q_C = DR \Delta H^{\text{vap}} \quad (7-33)$$

For a partial reboiler,

$$Q_R = BV_B \Delta H^{\text{vap}} \quad (7-34)$$

When the feed is at the bubble point and a total condenser is used, (7-16) can be arranged to:

$$BV_B = L + D = D(R + 1) \quad (7-35)$$

Comparing this to (7-34) and (7-32), note that $Q_R = Q_C$. When the feed is partially vaporized and a total condenser is used, the heat required by the reboiler is less than the condenser duty and is given by

$$Q_R = Q_C \left[1 - \frac{V_F}{D(R + 1)} \right] \quad (7-36)$$

If saturated steam is the heating medium for the reboiler, the steam rate required is given by an energy balance:

$$m_s = \frac{M_s Q_R}{\Delta H_s^{\text{vap}}} \quad (7-37)$$

where

m_s = mass flow rate of steam

Q_R = reboiler duty (rate of heat transfer)

M_s = molecular weight of steam

ΔH_s^{vap} = molar enthalpy of vaporization of steam

The cooling water rate for the condenser is

$$m_{\text{cw}} = \frac{Q_C}{C_{P_{\text{H}_2\text{O}}}(T_{\text{out}} - T_{\text{in}})} \quad (7-38)$$

where

m_{cw} = mass flow rate of cooling water

Q_C = condenser duty (rate of heat transfer)

$C_{P_{\text{H}_2\text{O}}}$ = specific heat of water

$T_{\text{out}}, T_{\text{in}}$ = temperature of cooling water out of and into the condenser, respectively

Because the annual cost of reboiler steam can be an order of magnitude higher than the annual cost of cooling water, the feed to a distillation column is frequently preheated and partially vaporized to reduce Q_R , in comparison to Q_C , as indicated by (7-36).

Feed Preheat

The feed to a distillation column is usually a process feed, an effluent from a reactor, or a liquid product from another separator. The feed pressure must be greater than the pressure in the column at the feed-tray location. If so, any excess feed pressure is dropped across a valve, which may cause the feed to partially vaporize before entering the column; if not, additional pressure is added with a pump.

The temperature of the feed as it enters the column does not necessarily equal the temperature in the column at the feed-tray location. However, such equality will increase second-law efficiency. It is usually best to avoid a subcooled liquid or superheated vapor feed and supply a partially vaporized feed. This is achieved by preheating the feed in a heat exchanger with the bottoms product or some other process stream that possesses a suitably high temperature, to ensure a reasonable ΔT driving force for heat transfer, and a sufficient available enthalpy.

Optimal Reflux Ratio

An industrial distillation column must be operated between the two limiting conditions of minimum reflux and total reflux. As shown in Table 7.3, for a typical case adapted from Peters and Timmerhaus [6], as the reflux ratio is increased from the minimum value, the number of plates decreases, the column diameter increases, and the reboiler steam and condenser cooling-water requirements increase. When the annualized fixed investment costs for the column, condenser, reflux drum, reflux pump, and reboiler are added to the annual cost of steam and cooling water, an optimal reflux ratio is established, as shown, for the conditions of Table 7.3, in Figure 7.22. For this example the optimal R/R_{min} is 1.1.

Table 7.3 Effect of Reflux Ratio on Annualized Cost of a Distillation Operation

R/R_{min}	Actual N	Diam., ft	Reboiler Duty, Btu/h	Condenser Duty, Btu/h	Annualized Cost, \$/yr			Total Annualized Cost, \$/yr
					Equipment	Cooling Water	Steam	
1.00	Infinite	6.7	9,510,160	9,416,000	Infinite	17,340	132,900	Infinite
1.05	29	6.8	9,776,800	9,680,000	44,640	17,820	136,500	198,960
1.14	21	7.0	10,221,200	10,120,000	38,100	18,600	142,500	199,200
1.23	18	7.1	10,665,600	10,560,000	36,480	19,410	148,800	204,690
1.32	16	7.3	11,110,000	11,000,000	35,640	20,220	155,100	210,960
1.49	14	7.7	11,998,800	11,880,000	35,940	21,870	167,100	224,910
1.75	13	8.0	13,332,000	13,200,000	36,870	24,300	185,400	246,570

(Adapted from an example by Peters and Timmerhaus [6].)

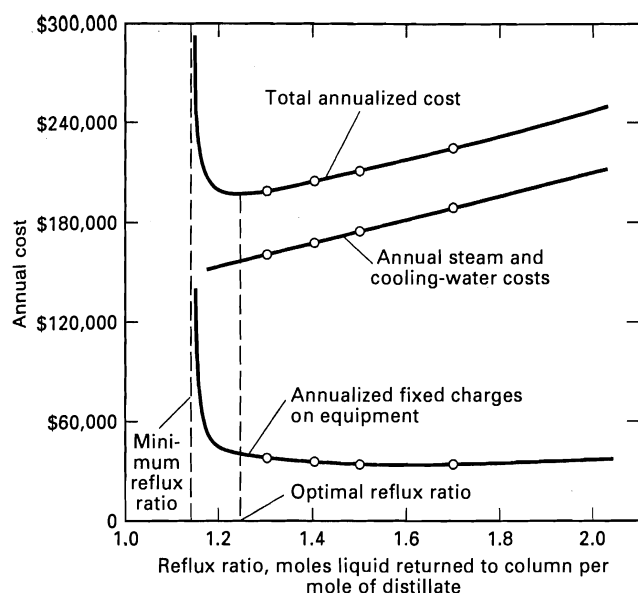


Figure 7.22 Optimal reflux ratio for a representative distillation operation.

[Adapted from M.S. Peters and K.D. Timmerhaus, *Plant Design and Economics for Chemical Engineers*, 4th ed., McGraw-Hill, New York (1991).]

The data in Table 7.3 show that although the condenser and reboiler duties are almost identical for a given reflux ratio, the annual cost of steam for the reboiler is almost eight times that of the cost of condenser cooling water. The total annual cost is dominated by the cost of steam except at the minimum-reflux condition. At the optimal reflux ratio, the cost of steam is 70% of the total annualized cost. Because the cost of steam is dominant, the optimal reflux ratio is sensitive to the steam cost. For example, at the extreme of zero cost for steam, the optimal $R/R_{\min}$ for this example is shifted from 1.1 to 1.32. This example assumes that the heat removed by cooling water in the condenser has no value.

The range of optimal ratio of reflux to minimum reflux often is from 1.05 to 1.50, with the lower value applying to a difficult separation (e.g., $\alpha = 1.2$) and the higher value applying to an easy separation (e.g., $\alpha = 5$). However, as seen

in Figure 7.22, the optimal reflux ratio is not sharply defined. Accordingly, to achieve greater operating flexibility, columns are often designed for reflux ratios greater than the optimum.

Large Number of Stages

The McCabe–Thiele graphical construction is difficult to apply when conditions of relative volatility and/or product purities are such that a large number of stages must be stepped off. In that event, one of the following techniques can be used to determine the stage requirements.

1. Separate plots of expanded scales and/or larger dimensions are used for stepping off stages at the ends of the y - x diagram. For example, the additional plots might cover just the regions (1) 0.95 to 1.0 and (2) 0 to 0.05.
2. As described by Horvath and Schubert [7] and shown in Figure 7.23, a plot based on logarithmic coordinates is used for the low (bottoms) end of the y - x diagram, while for the high (distillate) end, the log-log graph is turned upside down and rotated 90°. Unfortunately, as seen in Figure 7.23, the operating lines become curved, but they can be plotted from a few points computed from (7-9) and (7-14). The 45° line remains straight and the normally curved equilibrium curve becomes nearly straight at the two ends.
3. The stages at the two ends are computed algebraically in the manner of part (b) of Example 7.2. This is readily done with a spreadsheet computer program.
4. If the equilibrium data are given in analytical form, commercially available McCabe–Thiele computer programs can be used.
5. The stages are determined by combining the McCabe–Thiele graphical construction, for a suitable region in the middle, with the Kremser equations of Section 5.4 for the low and/or high ends, where absorption and stripping factors are almost constant. This technique, which is often preferred, is illustrated in the following example.

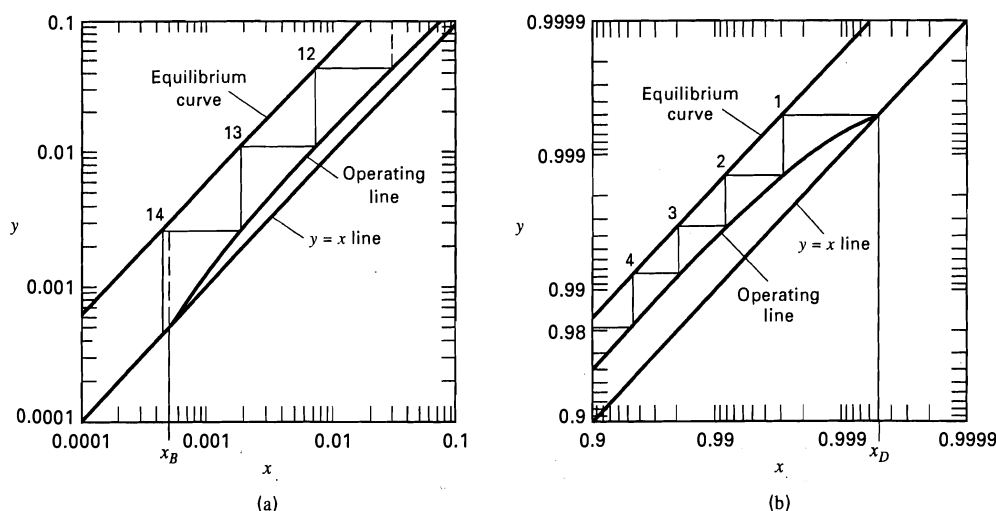


Figure 7.23 Use of log-log coordinates for McCabe–Thiele construction: (a) bottoms end of column; (b) distillate end of column.

EXAMPLE 7.4

Repeat part (c) of Example 7.1 for benzene distillate and bottoms purities of 99.9 and 0.1 mol%, respectively, using a reflux ratio of 1.88, which is about 30% higher than the minimum reflux of 1.44 for these new purities. At the top of the column, $\alpha = 2.52$; at the bottom, $\alpha = 2.26$.

SOLUTION

Figure 7.24 shows the McCabe–Thiele construction for the region of x from 0.028 to 0.956, where the stages have been stepped off in two directions, starting from the feed stage. In this middle region, seven stages are stepped off above the feed stage and eight below the feed stage, for a total of 16 stages, including the feed stage. The Kremser equations can now be applied to determine the remaining stages needed to achieve the desired high purities for the distillate and bottoms.

Additional stages for the rectifying section. With respect to Figure 5.8a, counting stages from the top down, from Figure 7.24:

From (7-3), for $(x_N)_{\text{benzene}} = 0.956$,

$$(y_{N+1})_{\text{benzene}} = 0.982 \quad \text{and} \quad (y_{N+1})_{\text{toluene}} = 0.018$$

Also, $(x_0)_{\text{benzene}} = (y_1)_{\text{benzene}} = 0.999$ and $(x_0)_{\text{toluene}} = (y_1)_{\text{toluene}} = 0.001$

Combining the Kremser equations (5.55), (5-34), (5-35), (5-48), and (5-50) and performing a number of algebraic manipulations:

$$N_R = \frac{\log \left[\frac{1}{A} + \left(1 - \frac{1}{A} \right) \left(\frac{y_{N+1} - x_0 K}{y_1 - x_0 K} \right) \right]}{\log A} \quad (7-39)$$

where N_R = additional equilibrium stages for the rectifying section. For that section, which is like an absorption section, it is best to apply (7-39) to toluene, the heavy key. Because $\alpha = 2.52$ at the top of the column, where K_{benzene} is close to one, take $K_{\text{toluene}} = 1/2.52 = 0.397$. Since $R = 1.88$, $L/V = R/(R+1) = 0.653$.

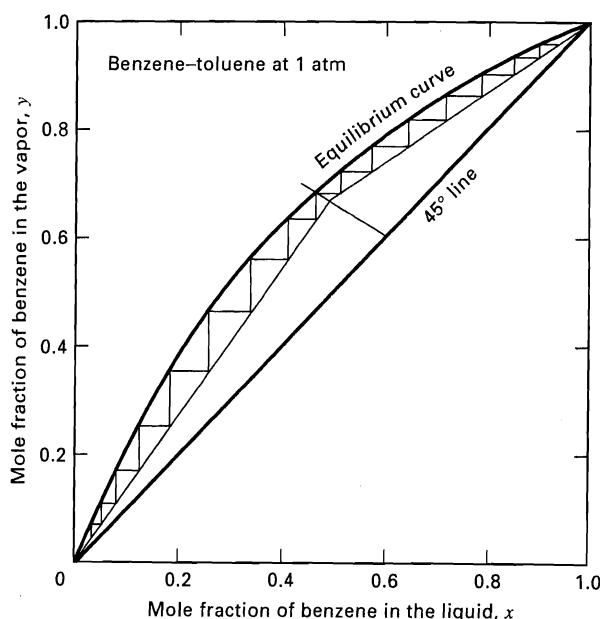


Figure 7.24 McCabe–Thiele construction for Example 7.4 from $x = 0.028$ to $x = 0.956$.

Therefore, the absorption factor for toluene is $A_{\text{toluene}} = L/(VK_{\text{toluene}}) = 0.653/0.397 = 1.64$ which is assumed to remain constant in the uppermost part of the rectifying section. Therefore, from (7-39) for toluene,

$$N_R = \frac{\log \left[\frac{1}{1.64} + \left(1 - \frac{1}{1.64} \right) \left(\frac{0.018 - 0.001(0.397)}{0.001 - 0.001(0.397)} \right) \right]}{\log 1.64} = 5.0$$

Additional stages for the stripping section. With respect to Figure 5.8b, counting stages from the bottom up, we have from Figure 7.24: $(x_{N+1})_{\text{benzene}} = 0.048$. Also, $(x_1)_{\text{benzene}} = (x_B)_{\text{benzene}} = 0.001$. Combining the Kremser equations for a stripping section gives

$$N_S = \frac{\log \left[\bar{A} + (1 - \bar{A}) \left(\frac{x_{N+1} - x_1/K}{x_1 - x_1/K} \right) \right]}{\log(1/\bar{A})} \quad (7-40)$$

where

N_S = additional equilibrium stages for the stripping section

$\bar{A}$ = absorption factor in the stripping section $= \bar{L}/K\bar{V}$

Because benzene is being stripped in the stripping section, it is best to apply (7-40) to the benzene. At the bottom of the column, where K_{toluene} is approximately 1.0, $\alpha = 2.26$, and therefore $K_{\text{benzene}} = 2.26$. By material balance, with flows in lbmol/h, $D = 270.1$. For $R = 1.88$, $L = 507.8$, and $V = 270.1 + 507.8 = 777.9$. From Example 7.1, $V_F = D = 270.1$ and $L_F = 450 - 270.1 = 179.9$. Therefore, $\bar{L} = L + L_F = 507.8 + 179.9 = 687.7$ lbmol/h and $\bar{V} = V - V_F = 777.9 - 270.1 = 507.8$ lbmol/h.

$$\bar{L}/\bar{V} = 687.7/507.8 = 1.354;$$

$$\bar{A}_{\text{benzene}} = \bar{L}/K\bar{V} = 1.354/2.26 = 0.599$$

Substitution into (7-40) gives

$$N_S = \frac{\log \left[0.599 + (1 - 0.599) \left(\frac{0.028 - 0.001/2.26}{0.001 - 0.001/2.26} \right) \right]}{\log(1/0.599)} = 5.9$$

This value includes the partial reboiler. Accordingly, the total number of equilibrium stages starting from the bottom is: partial reboiler + 5.9 + 8 + feed stage + 7 + 5.0 = 26.9.

Use of Murphree Efficiency

The McCabe–Thiele method assumes that the two phases leaving each stage are in thermodynamic equilibrium. In industrial, countercurrent, multistage equipment, it is not always practical to provide the combination of residence time and intimacy of contact required to approach equilibrium closely. Hence, concentration changes for a given stage are usually less than predicted by equilibrium.

As discussed in Section 6.5, a stage efficiency frequently used to describe individual tray performance for individual components is the Murphree plate efficiency. This efficiency can be defined on the basis of either phase and, for a given component, is equal to the change in actual composition in

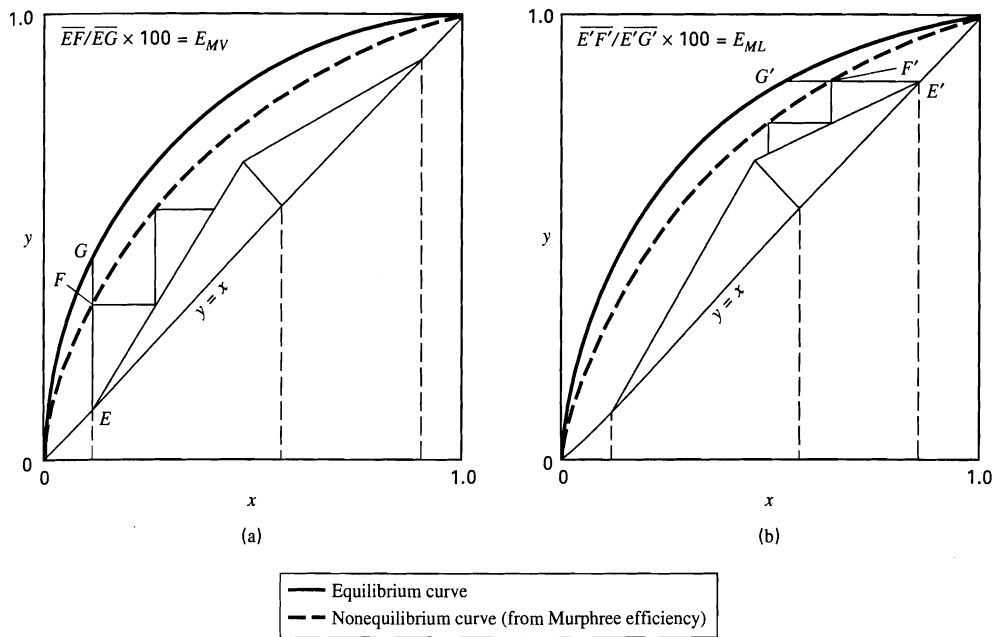


Figure 7.25 Use of Murphree plate efficiencies in McCabe–Thiele construction.

the phase, divided by the change predicted by equilibrium. This definition applied to the vapor phase can be expressed in a manner similar to (6-28):

$$E_{MV} = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}} \quad (7-41)$$

Where E_{MV} is the Murphree vapor efficiency for stage n , where $n + 1$ is the stage below and y_n^* is the composition in the hypothetical vapor phase in equilibrium with the liquid composition leaving stage n . Values of E_{MV} can be less than or somewhat more than 100%. The component subscript in (7-41) is dropped because values of E_{MV} are equal for the two components of a binary mixture.

In stepping off stages, the Murphree vapor efficiency, if known, can be used to dictate the percentage of the distance taken from the operating line to the equilibrium line; only E_{MV} of the total vertical path is traveled. This is shown in Figure 7.25a for the case of Murphree efficiencies based on the vapor phase. Figure 7.25b shows the case when the Murphree tray efficiency is based on the liquid. In effect, the dashed curve for actual exit-phase composition replaces the thermodynamic equilibrium curve for a particular set of operating lines. In Figure 7.25a, $E_{MV} = \overline{EF}/\overline{EG} = 0.7$ for the bottom stage.

Multiple Feeds, Side Streams, and Open Steam

The McCabe–Thiele method for a single feed and two products is readily extended to the case of multiple feeds and/or side streams by adding one additional operating line for each additional feed or side stream. A multiple-feed arrangement is shown in Figure 7.26. In the absence of side stream L_S , this arrangement has no effect on the material balance associated with the rectifying section of the column above the upper-feed point, F_1 . The section of column between the upper-feed point

and the lower-feed point F_2 (in the absence of feed F) is represented by an operating line of slope L'/V' , this line intersecting the rectifying-section operating line. A similar argument holds for the stripping section of the column. Hence it is possible to apply the McCabe–Thiele graphical construction shown in Figure 7.27a, where feed F_1 is a dew-point vapor, while feed F_2 is a bubble-point liquid. Feed F and side stream L_S of Figure 7.26 are not present. Thus, between the two feed points for this example, the molar vapor flow rate is $V' = V - F_1$ and $\bar{L} = L' + F_2 = L + F_2$. For given x_B, z_{F_2} ,

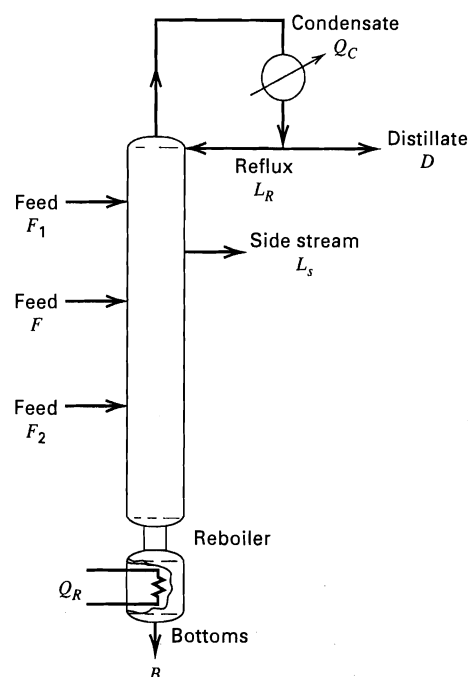


Figure 7.26 Complex distillation column with multiple feeds and side stream.

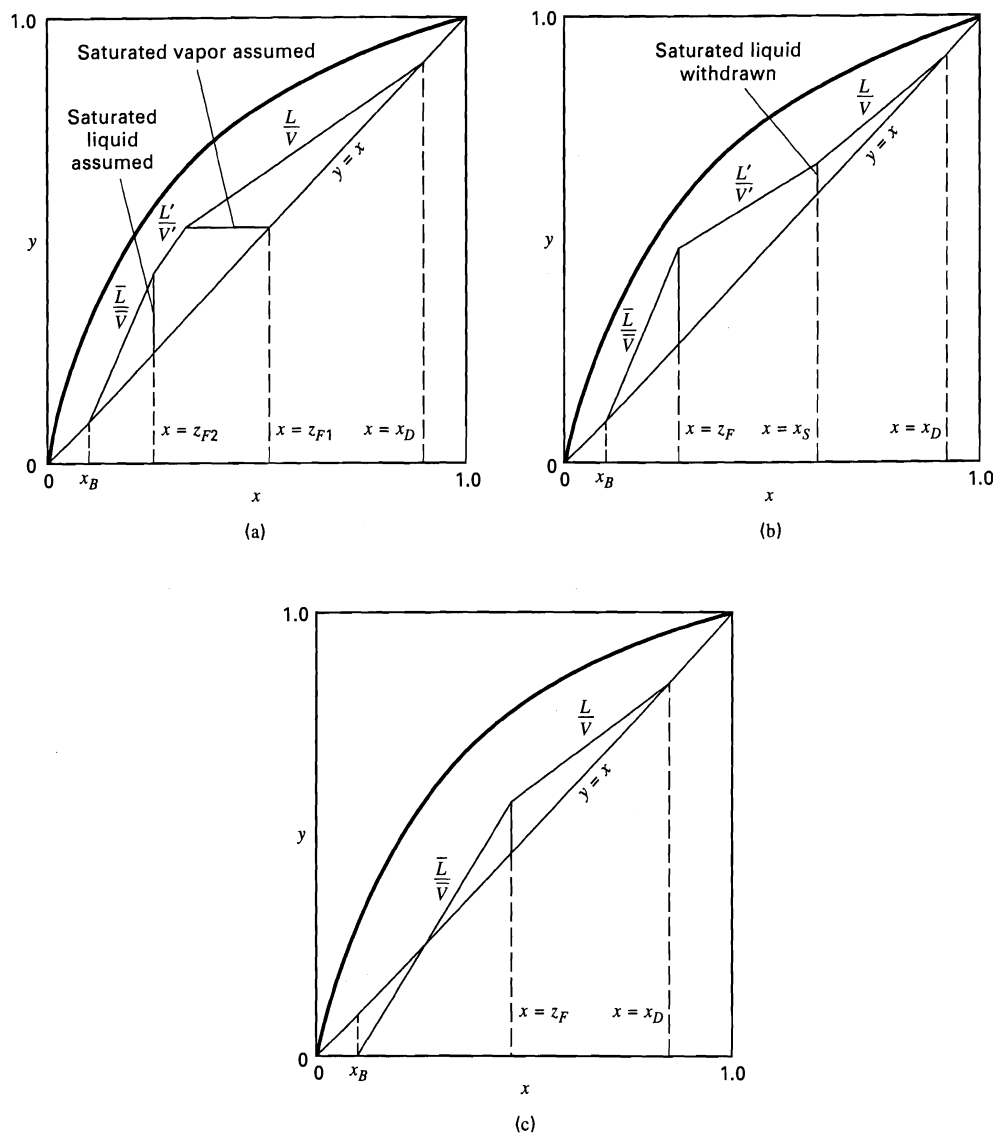


Figure 7.27 McCabe-Thiele construction for complex columns: (a) two feeds (saturated liquid and saturated vapor); (b) one feed, one side stream (saturated liquid); (c) use of open steam.

z_{F1} , x_D , and L/D , the three operating lines in Figure 7.27a are readily constructed.

A side stream may be withdrawn from the rectifying section, the stripping section, or between multiple feed points, as a saturated vapor or saturated liquid. Within material-balance constraints, L_S and x_S can both be specified. In Figure 7.27b, a saturated-liquid side stream of composition x_S and molar flow rate L_S is withdrawn from the rectifying section above feed F . In the section of stages between the side stream-withdrawal stage and the feed stage, $L' = L - L_S$, while $V' = V$. The McCabe-Thiele constructions determine the location of the side stream stage. However, if it is not located directly above x_S , the reflux ratio must be varied until it does.

For certain types of distillation, an inert hot gas is introduced directly into the base of the column. Open steam, for example, can be used if one of the components in the mixture is water, or if water can form a second liquid phase, thereby reducing the boiling point, as in the steam distillation of fats, where heat is supplied by live, superheated steam and no reboiler is used. Most commonly, the feed contains water,

which is removed as bottoms. In that application, Q_R of Figure 7.26 is replaced by a stream of composition $y = 0$ (pure steam) which, with $x = x_B$, becomes a point on the operating line, since the passing streams at this point actually exist at the end of the column. With open steam, the bottoms flow rate is increased by the flow rate of the open steam. The use of open steam rather than a reboiler for the operating condition $F_1 = F_2 = L_S = 0$ is represented graphically in Figure 7.27c.

EXAMPLE 7.5

A complex distillation column, equipped with a partial reboiler and total condenser, and operating at steady state with a saturated-liquid feed, has a liquid side stream draw-off in the enriching (rectifying) section. Making the usual simplifying assumptions of the McCabe-Thiele method: (a) Derive an equation for the two operating lines in the enriching section. (b) Find the point of intersection of these operating lines. (c) Find the intersection of the operating line between F and L_S with the diagonal. (d) Show the construction on a y - x diagram.

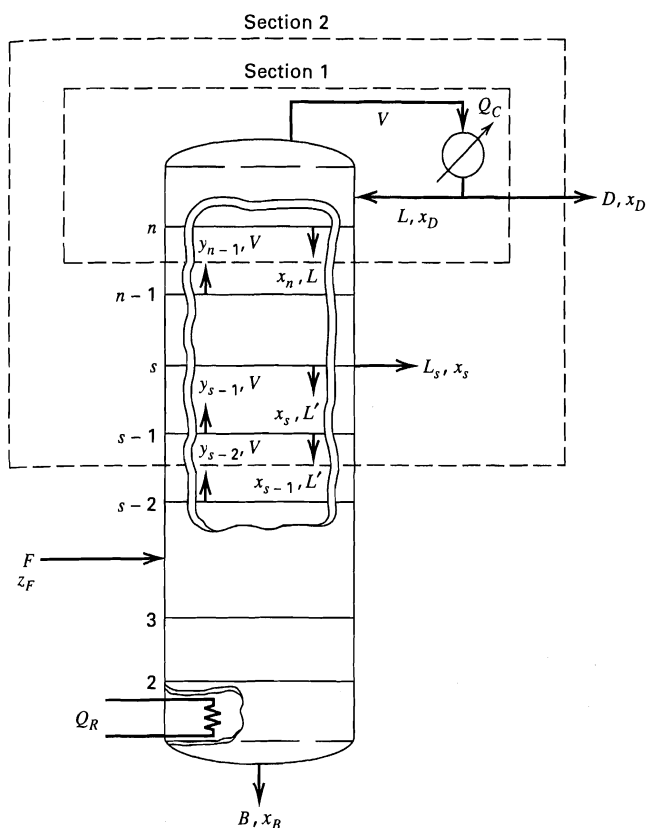


Figure 7.28 Distillation column with side stream for Example 7.5.

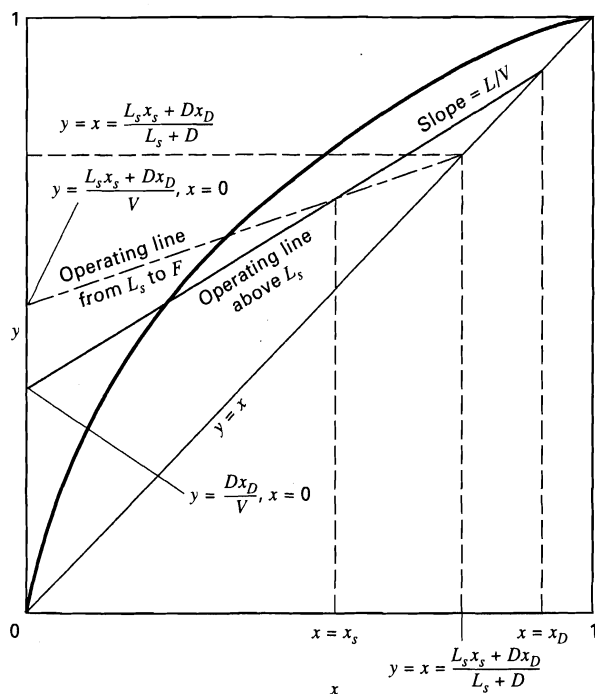


Figure 7.29 McCabe-Thiele diagram for Example 7.5.

SOLUTION

- (a) By material balance over section 1 in Figure 7.28, $V_{n-1}y_{n-1} = L_n x_n + Dx_D$. About section 2, $V_{s-2}y_{s-2} = L'_{s-1}x_{s-1} + L_s x_s + Dx_D$. The two operating lines for conditions of

constant molar overflow become:

$$y = \frac{L}{V}x + \frac{D}{V}x_D \quad \text{and} \quad y = \frac{L'}{V}x + \frac{L_s x_s + Dx_D}{V}$$

- (b) Equating the two operating lines, the intersection occurs at $(L - L')x = L_s x_s$ and since $L - L' = L_s$, the point of intersection becomes $x = x_s$.

- (c) The intersection of the lines

$$y = \frac{L'}{V}x + \frac{L_s x_s + Dx_D}{V}$$

and $y = x$ occurs at $x = \frac{L_s x_s + Dx_D}{L_s + D}$

- (d) The y - x diagram is shown in Figure 7.29.

7.3 ESTIMATION OF STAGE EFFICIENCY

Methods for estimating the stage efficiency for binary distillation are analogous to those for absorption and stripping, presented in Section 6.5. The efficiency is a complex function of tray design, fluid properties, and flow patterns. However, in hydrocarbon absorption and stripping, the liquid phase is often rich in heavy components so that liquid viscosity is high and mass-transfer rates are relatively low. This leads to low stage efficiencies, usually less than 50%. In contrast, for binary distillation, particularly of close-boiling mixtures, liquid viscosity is low, with the result that stage efficiencies, for well-designed trays and optimal operating conditions, are often higher than 70% and can be even higher than 100% for large-diameter columns where a cross-flow effect is present.

Performance Data

As discussed in *AIChE Equipment Testing Procedure* [8], performance data for an industrial distillation column are best obtained at conditions of total reflux (no feed or products) so as to avoid possible column-feed fluctuations, simplify location of the operating line, and avoid discrepancies between feed and feed-tray compositions. However, as shown by Williams, Stigger, and Nichols [9], efficiency measured at total reflux can differ markedly from that at design reflux ratio. Ideally, the column is operated in the range of 50% to 85% of flooding. If liquid samples are taken from the top and bottom of the column, the overall plate efficiency, E_o , can be determined from (6-21), where the number of theoretical stages required is determined by applying the McCabe-Thiele method at total reflux, as in Figure 7.11. If liquid samples are taken from the downcomers of intermediate trays, Murphree vapor efficiencies, E_{MV} , can be determined using (6-28). If liquid samples are withdrawn from different points on one tray, (6-30) can be applied to obtain point efficiencies, E_{OV} . Reliable values for these efficiencies require the availability of accurate vapor-liquid equilibrium data. For that reason, efficiency data for binary mixtures that form ideal solutions are preferred.

Table 7.4 Performance Data for the Distillation of a Mixture of Methylene Chloride and Ethylene Chloride

Company	Eastman Kodak
Location	Rochester, New York
Column diameter	5.5 ft (65.5 in. I.D.)
No. of trays	60
Tray spacing	18 in.
Type tray	10 rows of 3-in.-diameter bubble caps on 4-7/8-in. triangular centers. 115 caps/tray
Bubbling area	20 ft ²
Length of liquid travel	49 in.
Outlet-weir height	2.25 in.
Downcomer clearance	1.5 in.
Liquid rate	24.5 gal/min-ft = 1,115.9 lb/min
Vapor <i>F</i> -factor	1.31 ft/s (lb/ft ³) ^{0.5}
Percent of flooding	85
Pressure, top tray	33.8 psia
Pressure, bottom tray	42.0 psia
Liquid composition, mole % methylene chloride:	
From tray 33	89.8
From tray 32	72.6
From tray 29	4.64

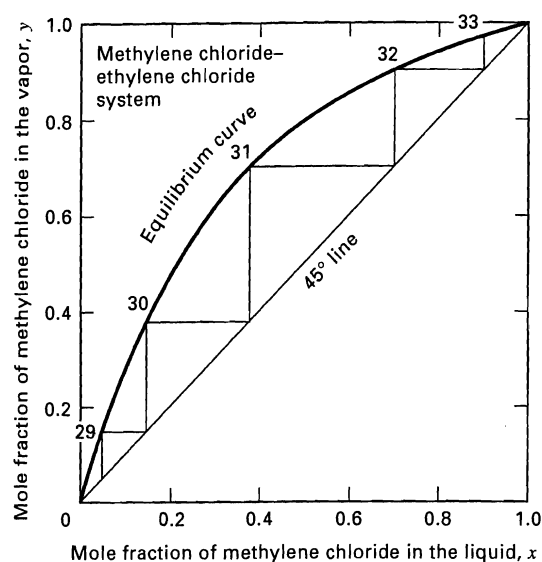
Source: J.A. Gerster, A.B. Hill, N.H. Hochgrof, and D.B. Robinson, *Tray Efficiencies in Distillation Columns, Final Report from the University of Delaware*, AIChE, New York (1958).

Table 7.4, from Gerster et al. [10], lists plant data, obtained from Eastman Kodak Company in Rochester, New York, for the distillation at total reflux of a methylene chloride (MC)–ethylene chloride (EC) mixture in a 5.5-ft-diameter column containing 60 bubble-cap trays on 18-in. tray spacing and operating at 85% of flooding at total reflux. MC is the light key.

EXAMPLE 7.6

Using the performance data of Table 7.4, estimate: (a) the overall tray efficiency for the section of trays from 35 to 29 and (b) the Murphree vapor efficiency for tray 32. Assume the following values for relative volatility:

x_{MC}	$\alpha_{MC,EC}$	y_{MC} from (7-3)
0.00	3.55	0.00
0.10	3.61	0.286
0.20	3.70	0.481
0.30	3.76	0.617
0.40	3.83	0.719
0.50	3.91	0.796
0.60	4.00	0.857
0.70	4.03	0.904
0.80	4.09	0.942
0.90	4.17	0.974
1.00	4.25	1.00

**Figure 7.30** McCabe–Thiele diagram for Example 7.6.

SOLUTION

- (a) The above x – α – y data are plotted in Figure 7.30. Four theoretical stages are stepped off from $x_{33} = 0.898$ to $x_{29} = 0.0464$ for total reflux. Since the actual number of stages is also 4, the overall stage efficiency from (6-21) is 100%.
- (b) At total reflux conditions, passing vapor and liquid streams have the same composition. That is, the operating line is the 45° line. Using this together with the above performance data and the equilibrium curve in Figure 7.30, we obtain for methylene chloride, with trays counted from the bottom up:

$$y_{32} = x_{33} = 0.898 \quad \text{and} \quad y_{31} = x_{32} = 0.726$$

From (6-28),

$$(E_{MV})_{32} = \frac{y_{32} - y_{31}}{y_{32}^* - y_{31}}$$

From Figure 7.30, for $x_{32} = 0.726$, $y_{32}^* = 0.917$,

$$(E_{MV})_{32} = \frac{0.898 - 0.726}{0.917 - 0.726} = 0.90 \quad \text{or} \quad 90\%$$

Empirical Correlations

Based on 41 sets of performance data for bubble-cap-tray and sieve-tray columns, distilling mainly hydrocarbon mixtures and a few water and miscible organic mixtures, Drickamer and Bradford [11] correlated the overall stage efficiency for the separation of the two key components in terms of the molar-average liquid viscosity of the tower feed at the average tower temperature. The data covered average temperatures from 157 to 420°F, pressures from 14.7 to 366 psia, feed liquid viscosities from 0.066 to 0.355 cP, and overall tray efficiencies from 41% to 88%. The empirical equation

$$E_o = 13.3 - 66.8 \log \mu \quad (7-42)$$

where E_o is in percent and μ is in centipoise, fits the data with average and maximum percent deviations of 5.0% and 13.0%, respectively. A plot of the Drickamer and Bradford

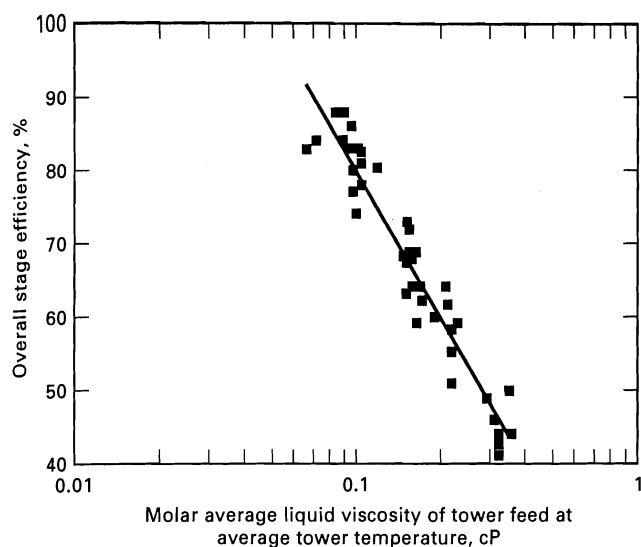


Figure 7.31 Drickamer and Bradford correlation for plate efficiency of distillation columns.

correlation, compared to performance data for distillation, is given in Figure 7.31. Equation (7-42) is restricted to the range of the data and is intended mainly for hydrocarbon mixtures.

Mass-transfer theory, discussed in Section 6.5, indicates that, when the relative volatility covers a wide range, the relative importance of liquid-phase and gas-phase mass-transfer resistances can shift. Thus, as might be expected, O'Connell [12] found that the Drickamer-Bradford correlation correlates data inadequately for fractionators operating on key components with large relative volatilities. Separate correlations in terms of a viscosity-volatility product were developed for fractionators and for absorbers and strippers by O'Connell. However, as shown in Figure 7.32, Lockhart and Leggett [13] were able to obtain a single correlation by using the product of liquid viscosity and an appropriate volatility as the correlating variable. For fractionators, the relative volatility of the key components was used; for hydrocarbon absorbers, the volatility was taken as 10 times the K -value of a selected key component, which must be one

that is reasonably distributed between top and bottom products. The data used by O'Connell cover a range of relative volatility from 1.16 to 20.5. A comprehensive study of the effect on E_o of the ratio of liquid-to-vapor molar flow rates, L/V , for eight different binary systems in a 10-in.-diameter column with bubble-cap trays was reported by Williams, et al. [9]. The systems included water, hydrocarbons, and other organic compounds. While L/V did have an effect, it could not be correlated. For fractionation with L/V nearly equal to 1.0 (i.e., total reflux), their distillation data, which are included in Figure 7.32, are in reasonable agreement with the O'Connell correlation. For the distillation of hydrocarbons in a column having a diameter of 0.45 m, Zuiderweg, Verburg, and Gilissen [14] found differences in E_o among bubble-cap, sieve, and valve trays to be insignificant at 85% of flooding. Accordingly, Figure 7.32 is assumed to be applicable to all three tray types, but may be somewhat conservative for well-designed trays. For example, data of Fractionation Research Incorporated (FRI) for valve trays operating with the cyclohexane/*n*-hexane and isobutane/*n*-butane systems are also included in Figure 7.32 and show efficiencies 10% to 20% higher than the correlation.

For the distillation data plotted in Figure 7.32, which cover a viscosity-relative volatility range for distillation of from 0.1 to 10 cP, the O'Connell correlation fits the empirical equation

$$E_o = 50.3(\alpha\mu)^{-0.226} \quad (7-43)$$

where E_o is in percent and μ is in centipoise. The relative volatility is determined for the two key components at average column conditions.

Most of the data for developing the correlation of Figure 7.32 are for columns having a liquid flow path across the active tray area of from 2 to 3 ft. Gautreaux and O'Connell [15], using theory and experimental data, showed that higher efficiencies are achieved for longer flow paths. For short liquid flow paths, the liquid flowing across the tray is usually mixed completely. For longer flow paths, the equivalent of two or more completely mixed, successive liquid zones may

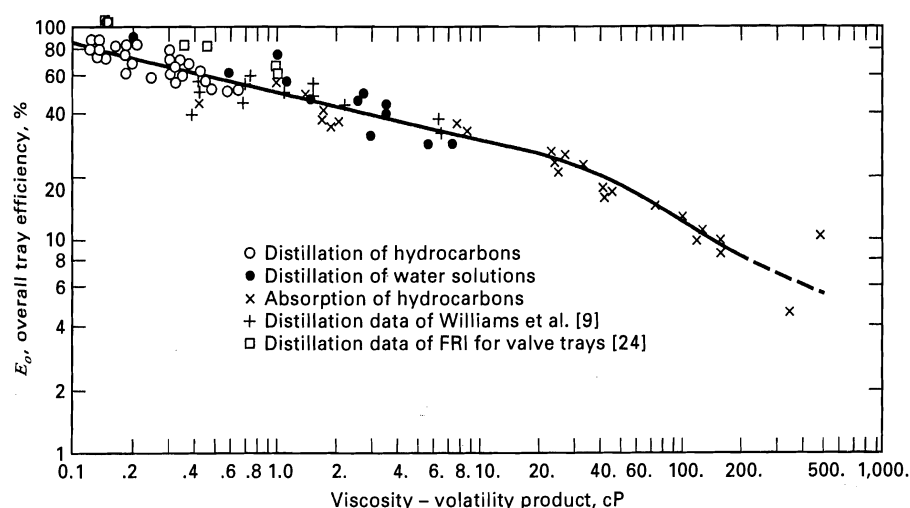


Figure 7.32 Lockhart and Leggett version of the O'Connell correlation for overall tray efficiency of fractionators, absorbers, and strippers.

[Adapted from F.J. Lockhart and C.W. Leggett, in *Advances in Petroleum Chemistry and Refining*, Vol. 1, Eds., K.A. Kobe and John J. McKetta, Jr., Interscience, New York, pp. 323-326 (1958).]

Table 7.5 Correction to Overall Tray Efficiency for Length of Liquid Flow Path ($0.1 \leq \mu\alpha \leq 1.0$)

Length of Liquid Flow Path, ft	Value to Be Added to E_o from Figure 7.32, %
3	0
4	10
5	15
6	20
8	23
10	25
15	27

Source: F.J. Lockhart and C.W. Leggett, in K.A. Kobe and J.J. McKetta, Jr., Eds., *Advances in Petroleum Chemistry and Refining*, Vol. 1, Interscience, New York, pp. 323–326 (1958).

be present. The result is a greater average driving force for mass transfer, and, thus, a higher efficiency—sometimes even greater than 100%. Provided that the viscosity–volatility product lies between 0.1 and 1.0, Lockhart and Leggett [13] recommend addition of the increments in Table 7.5 to the value of E_o from Figure 7.32 when the liquid flow path is greater than 3 ft. However, at large liquid rates, long liquid-path lengths are undesirable because they lead to excessive liquid gradients, causing maldistribution of vapor flow. The use of multipass trays, shown in Figure 6.16, to prevent excessive liquid gradients is discussed in Section 6.5.

EXAMPLE 7.7

For the benzene–toluene distillation of Figure 7.1, use the Drickamer–Bradford and O’Connell correlations to estimate the overall stage efficiency and number of actual plates required. Calculate the height of the tower assuming 24-in. tray spacing, with 4 ft above the top tray for removal of entrained liquid and 10 ft below the bottom tray for bottoms surge capacity. The separation requires 20 equilibrium stages plus a partial reboiler that acts as an equilibrium stage.

SOLUTION

For estimating overall stage efficiency, the liquid viscosity is determined at the feed-stage condition of 220°F, assuming a liquid composition of 50 mol% benzene.

$$\begin{aligned}\mu \text{ of benzene} &= 0.10 \text{ cP}; & \mu \text{ of toluene} &= 0.12 \text{ cP}; \\ \text{Average } \mu &= 0.11 \text{ cP}.\end{aligned}$$

From Figure 7.3, take the average relative volatility as

$$\text{Average } \alpha = \frac{\alpha_{\text{top}} + \alpha_{\text{bottom}}}{2} = \frac{2.52 + 2.26}{2} = 2.39$$

From the Drickamer–Bradford correlation (7-42), $E_o = 13.3 - 66.8 \log(0.11) = 77\%$

This is close to the value given in the description of this problem. Therefore, 26 actual trays are required and column height = $4 + 2(26 - 1) + 10 = 64$ ft.

From the O’Connell correlation (7-43),

$$E_o = 50.3[(2.39)(0.11)]^{-0.226} = 68\%.$$

For a 5-ft-diameter column, the length of the liquid flow path is about 3 ft for a single-pass tray and even less for a two-pass tray. From Table 7.5, the efficiency correction is zero. Therefore, the actual number of trays required is $20/0.68 = 29.4$, or call it 30 trays. Column height = $4 + 2(30 - 1) + 10 = 72$ ft.

Semi-Theoretical Models

In Section 6.5, semi-theoretical tray models based on the Murphree vapor efficiency and the Murphree vapor-point efficiency are applied to absorption and stripping. These same relationships are valid for distillation. However, because the equilibrium line is curved for distillation, λ must be taken as mV/L (not $KV/L = 1/A$), where m = local slope of the equilibrium curve = dy/dx . In Section 6.6, the method of Chan and Fair [16] is used for estimating the Murphree vapor-point efficiency from mass-transfer considerations. The Murphree vapor efficiency can then be estimated. The Chan and Fair correlation is specifically applicable to binary distillation because it was developed from experimental data that includes six different binary systems.

Scale-up from Laboratory Data

When binary mixtures form ideal or nearly ideal solutions, it is rarely necessary to obtain laboratory distillation data. Where nonideal solutions are formed and/or the possibility of azeotrope formation exists, use of a small laboratory Oldershaw column, of the type discussed in Section 6.5, should be used to verify the desired degree of separation and to obtain an estimate of the Murphree vapor-point efficiency. The ability to predict the efficiency of an industrial-size sieve-tray column from measurements with 1-in. glass and 2-in. metal diameter Oldershaw columns is shown in Figure 7.33, from the work of Fair, Null, and Bolles [17]. The measurements were made for the cyclohexane/*n*-heptane system at vacuum conditions (Figure 7.33a) and at near-atmospheric conditions (Figure 7.33b) and for the isobutane/*n*-butane system at 11.2 atm (Figure 7.33c). The Oldershaw data are correlated by the solid lines. Data for the 4-ft-diameter column with sieve trays of 8.3% and 13.7% open area were obtained by Sakata and Yanagi [18] and Yanagi and Sakata [19], respectively, of FRI. The Oldershaw column is assumed to measure point efficiency. The FRI column measured overall efficiency, but the relations of Section 6.5 were used to convert the FRI data to the point efficiencies shown in Figure 7.33. The data cover a percent of flooding ranging from about 10% to 95%. Data from the Oldershaw column are in reasonable agreement with the FRI data for 14% open area, except at the lower part of the flooding range. In Figures 7.33b and 7.33c, the FRI data for 8% open area show efficiencies as much as 10 percentage points higher.

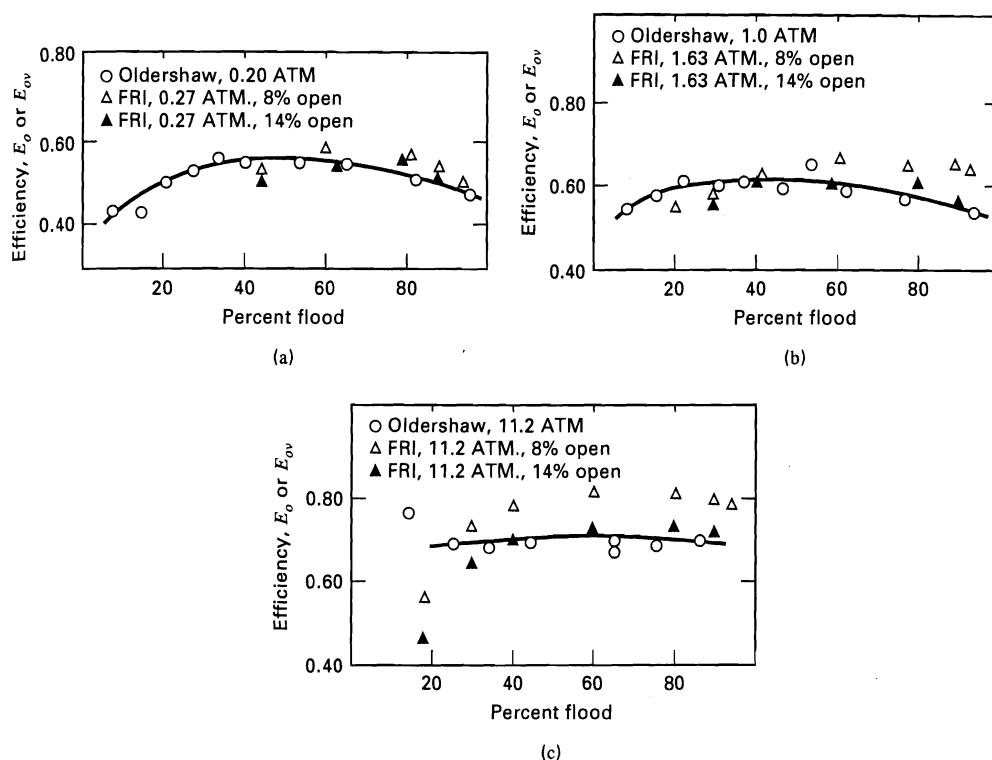


Figure 7.33 Comparison of Oldershaw column efficiency with point-efficiency in 4-ft-diameter FRI column with sieve trays: (a) cyclohexane/*n*-heptane system; (b) cyclohexane/*n*-heptane systems; (c) isobutane/*n*-butane system.

7.4 DIAMETER OF TRAYED TOWERS AND REFLUX DRUMS

In Section 6.6, methods for estimating tray diameter and pressure drop for absorbers and strippers are presented. These same methods apply to distillation columns. Calculations of column diameter are usually made for conditions at the top and bottom trays of the tower. If the diameters differ by 1 ft or less, the larger diameter is used for the entire column. If the diameters differ by more than 1 ft, it is often more economical to swage the column, using the different diameters computed for the sections above and below the feed.

Reflux Drums

Almost all commercial towers are provided with a cylindrical reflux drum, as shown in Figure 7.1. This drum is usually located near ground level, necessitating a pump to lift the reflux to the top of the column. If a partial condenser is used, the drum is often oriented vertically to facilitate the separation of vapor from liquid—in effect, acting as a flash drum. Vertical reflux and flash drums are sized by calculating a minimum drum diameter, D_T , to prevent liquid carryover by entrainment, using (6-44) in conjunction with the curve for 24-in. tray spacing in Figure 6.24 and a value of $F_{HA} = 1.0$ in (6-42). Also, $f = 0.85$ and $A_d = 0$ are used. To absorb process upsets and fluctuations, and otherwise facilitate control, vessel volume, V_V , is determined on the basis of liquid residence time, t , which should be at least 5 min, with the

vessel half full of liquid [20]:

$$V_V = \frac{2LM_L t}{\rho_L} \quad (7-44)$$

where L is the molar liquid flow rate leaving the vessel. Assuming a vertical, cylindrical vessel and neglecting the volume associated with the heads, the height H of the vessel is

$$H = \frac{4V_V}{\pi D_T^2} \quad (7-45)$$

However, if $H > 4D_T$, it is generally preferable to increase D_T and decrease H to give $H = 4D_T$. Then

$$D_T = \frac{H}{4} = \left(\frac{V_V}{\pi} \right)^{1/3} \quad (7-46)$$

A height above the liquid level of at least 4 ft is necessary for feed entry and disengagement of liquid droplets from the vapor. Within this space, it is common to install a wire mesh pad, which serves as a mist eliminator.

When vapor is totally condensed, a cylindrical, horizontal reflux drum is commonly employed to receive the condensate. Equations (7-44) and (7-46) permit estimates of the drum diameter, D_T , and length, H , by assuming a near-optimal value for H/D_T of 4, and the same liquid residence time suggested for a vertical drum. A horizontal drum is also used following a partial condenser when the liquid flow rate is appreciably greater than the vapor flow rate.

EXAMPLE 7.8

Equilibrium vapor and liquid streams leaving a flash drum, supplied by a partial condenser, are as follows:

Component	Vapor	Liquid
Pound-moles per hour:		
HCl	49.2	0.8
Benzene	118.5	81.4
Monochlorobenzene	71.5	178.5
Total	239.2	260.7
Pounds per hour	19,110	26,480
T , °F	270	270
P , psia	35	35
Density, lb/ft ³	0.371	57.08

Determine the dimensions of the flash drum.

SOLUTION

Using Figure 6.24,

$$F_{LV} = \frac{26,480}{19,110} \left(\frac{0.371}{57.08} \right)^{0.5} = 0.112$$

C_F at a 24-in. tray spacing is 0.34. Assume, in (6-24), that $C = C_F$. From (6-40),

$$U_f = 0.34 \left(\frac{57.08 - 0.371}{0.371} \right)^{0.5} = 4.2 \text{ ft/s} = 15,120 \text{ ft/h}$$

From (6-44) with $A_d/A = 0$,

$$D_T = \left[\frac{(4)(19,110)}{(0.85)(15,120)(3.14)(1)(0.371)} \right]^{0.5} = 2.26 \text{ ft}$$

From (7-44), with $t = 5 \text{ min} = 0.0833 \text{ h}$,

$$V_V = \frac{(2)(26,480)(0.0833)}{(57.08)} = 77.3 \text{ ft}^3$$

From (7-43),

$$H = \frac{(4)(77.3)}{(3.14)(2.26)^2} = 19.3 \text{ ft}$$

However, $H/D_T = 19.3/2.26 = 8.54 > 4$. Therefore, redimension V_V for $H/D_T = 4$.

From (7-46),

$$D_T = \left(\frac{77.3}{3.14} \right)^{1/3} = 2.91 \text{ ft and } H = 4D_T = (4)(2.91) = 11.64 \text{ ft}$$

Height above the liquid level is $11.64/2 = 5.82 \text{ ft}$, which is adequate.

Alternatively, with a height of twice the minimum disengagement height, $H = 8 \text{ ft}$ and $D_T = 3.5 \text{ ft}$.

7.5 RATE-BASED METHOD FOR PACKED COLUMNS

With the availability of more efficient liquid distributors and economical and efficient packings, packed towers are finding increasing use in new distillation processes and for

Table 7.6 Modified Efficiency and Mass-Transfer Equations for Binary Distillation

$$\lambda = mV/L \quad (7-47)$$

$$m = dy/dx = \text{local slope of equilibrium curve}$$

Efficiency:

Equations (6.31) to (6.37) hold if λ is defined by (7.47)

Mass transfer:

$$\frac{1}{N_{OG}} = \frac{1}{N_G} + \frac{\lambda}{N_L} \quad (7-48)$$

$$\frac{1}{K_{OG}} = \frac{1}{k_{Ga}} + \frac{mPM_L/\rho_L}{k_L a} \quad (7-49)$$

$$\frac{1}{K_y a} = \frac{1}{k_y a} + \frac{m}{k_x a} \quad (7-50)$$

$$\frac{1}{K_x a} = \frac{1}{k_x a} + \frac{1}{mk_y a} \quad (7-51)$$

$$H_{OG} = H_G + \lambda H_L \quad (7-52)$$

$$\text{HETP} = H_{OG} \ln \lambda / (\lambda - 1) \quad (7-53)$$

retrofitting existing trayed towers. Methods in Section 6.8 for estimating packed-column efficiency, diameter, and pressure drop for absorbers are applicable to distillation. Methods for determining packed height are similar to those presented in Section 6.7 and are extended here for use in conjunction with the McCabe–Thiele diagram. Both the HETP and the HTU methods are discussed and illustrated. Unlike the case of absorption or stripping of dilute solutions, where values of HETP and HTU may be constant throughout the packed height, values of HETP and HTU can vary over the packed height of a distillation column, especially across the feed entry, where appreciable changes in vapor and liquid traffic occur. Also, because the equilibrium line for distillation is curved rather than straight, the mass-transfer equations of Section 6.8 must be modified by replacing $\lambda = KV/L = 1/A$ with

$$\lambda = \frac{mV}{L} = \frac{\text{slope of equilibrium curve}}{\text{slope of operating line}}$$

where $m = dy/dx$ varies with location in the tower. The modified efficiency and mass-transfer relationships are summarized in Table 7.6.

HETP Method

In the HETP method, the equilibrium stages are first stepped off on a McCabe–Thiele diagram. The case of equimolar counterdiffusion (EMD) applies to distillation. At each stage, the temperature, pressure, phase-flow ratio, and phase compositions are noted. A suitable packing material is selected and the column diameter is estimated for operation at, say, 70% of flooding by one of the methods of Section 6.8. Mass-transfer coefficients for the individual phases are estimated for the conditions at each stage from correlations also

discussed in Section 6.8. From these coefficients, values of H_{OG} and HETP are estimated for each stage. The latter values are then summed to obtain the separate packed heights of the rectifying and stripping sections. If experimental values of HETP are available, they are used directly. In computing values of H_{OG} from H_G and H_L , or K_y from k_y and k_x , (6-92) and (6-80) must be modified because for binary distillation where the mole fraction of the light key may range from almost 0 at the bottom of the column to almost 1 at the top of the column, the ratio $(y_1 - y^*)/(x_1 - x)$ in (6-76) is no longer a constant equal to the K -value, but is dy/dx equal to the slope, m , of the equilibrium curve. The modified equations are included in Table 7.6.

EXAMPLE 7.9

For the benzene–toluene distillation of Example 7.1, determine packed heights of the rectifying and stripping sections based on a column diameter and packing material with the following values for the individual HTUs. Included are the L/V values for each section from Example 7.1.

	H_G , ft	H_L , ft	L/V
Rectifying section	1.16	0.48	0.62
Stripping section	0.90	0.53	1.40

SOLUTION

Slopes dy/dx of the equilibrium curve are obtained from Figure 7.15 and values of λ from (7-47). H_{OG} for each stage is determined from (7-52) in Table 7.6. HETP for each stage is determined from (7-53) in Table 7.6. The results are given in Table 7.7, where only 0.2 of stage 13 is needed and stage 14 is the partial reboiler.

Based on the results in Table 7.7, 10 ft of packing should be used in each of the two sections.

Table 7.7 Results for Example 7.9

Stage	m	$\lambda = \frac{mV}{L} \text{ or } m \frac{\bar{V}}{\bar{L}}$	H_{OG} , ft	HETP, ft
1	0.47	0.76	1.52	1.74
2	0.53	0.85	1.56	1.70
3	0.61	0.98	1.62	1.64
4	0.67	1.08	1.68	1.62
5	0.72	1.16	1.71	1.59
6	0.80	1.29	1.77	1.56
Total for rectifying section:				9.85
7	0.90	0.64	1.32	1.64
8	0.98	0.70	1.28	1.52
9	1.15	0.82	1.34	1.47
10	1.40	1.00	1.43	1.43
11	1.70	1.21	1.53	1.40
12	1.90	1.36	1.62	1.38
13	2.20	1.57	1.73	1.37(0.2) = 0.27
Total for stripping section:				9.11
Total packed height:				18.96

HTU Method

In the HTU method, equilibrium stages are not stepped off on a McCabe–Thiele diagram. Instead, the diagram provides data to perform an integration over the packed height of each section using either mass-transfer coefficients or transfer units.

Consider the schematic diagram of a packed distillation column and its accompanying McCabe–Thiele diagram in Figure 7.34. Assume that V , L , $\bar{V}$, and $\bar{L}$ are constant in their respective sections. For equimolar countercurrent diffusion (EMD), the rate of mass transfer of the light-key component from the liquid phase to the vapor phase is

$$n = k_x a(x - x_1) = k_y a(y_1 - y) \quad (7-54)$$

Rearranging:

$$\frac{k_x a}{k_y a} = \frac{y_1 - y}{x_1 - x} \quad (7-55)$$

Thus, as shown in Figure 7.34b, for any point (x, y) on the operating line, the corresponding interfacial point (x_1, y_1) on the equilibrium curve is obtained by drawing a line of slope $(-k_x a/k_y a)$ from the point (x, y) to the point where it intersects the equilibrium curve.

By material balance over an incremental section of packed height, assuming constant molar overflow,

$$V dy = k_y a(y_1 - y) S dl \quad (7-56)$$

$$L dx = k_x a(x - x_1) S dl \quad (7-57)$$

where S is the cross-sectional area of the packed section. Integrating over the rectifying section,

$$(l_T)_R = \int_0^{(l_T)_R} dl = \int_{y_F}^{y_2} \frac{V dy}{k_y a S (y_1 - y)} = \int_{x_F}^{x_D} \frac{L dx}{k_x a S (x - x_1)} \quad (7-58)$$

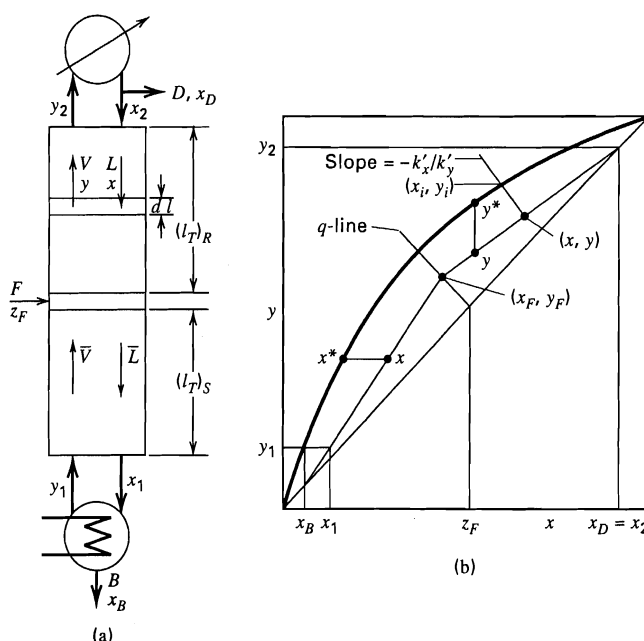


Figure 7.34 Distillation in a packed column.

or

$$(l_T)_R = \int_{y_F}^{y_2} \frac{H_G dy}{(y_1 - y)} = \int_{x_F}^{x_D} \frac{H_L dx}{(x - x_1)} \quad (7-59)$$

Integrating over the stripping section,

$$(l_T)_S = \int_0^{(l_T)_S} dl = \int_{y_1}^{y_F} \frac{V dy}{k_y a S (y_1 - y)} = \frac{L dx}{k_x a S (x - x_1)} \quad (7-60)$$

or

$$(l_T)_S = \int_{y_1}^{y_F} \frac{H_G dy}{(y_1 - y)} = \int_{x_1}^{x_F} \frac{H_L dx}{(x - x_1)} \quad (7-61)$$

In general, values of k_y and k_x vary over the packed height, causing the slope $(-k_x a / k_y a)$ to vary. If $k_x a > k_y a$, the main resistance to mass transfer resides in the vapor and it is most accurate to evaluate the integrals in y . For $k_y a > k_x a$, the integrals in x are used. Usually, it is sufficient to evaluate k_y and k_x at just three points in each section, from which their variation with x can be determined. Then by computing and plotting their ratios from (7-55), a locus of points P can be found, from which values of $(y_1 - y)$ for any value of y , or $(x - x_1)$ for any value of x can be read for use in integrals (7-58) to (7-61). These integrals can be evaluated either graphically or numerically to determine the packed heights.

EXAMPLE 7.10

Suppose that 250 kmol/h of a mixture of 40 mol% isopropyl ether in isopropanol is distilled in a packed column operating at 1 atm to obtain a distillate of 75 mol% isopropyl ether and a bottoms of 95 mol% isopropanol. At the feed entry, the mixture is a saturated liquid. A reflux ratio of 1.5 times minimum is used and the column is equipped with a total condenser and a partial reboiler. For the packing and column diameter, mass-transfer coefficients given below have been estimated from empirical correlations of the type discussed in Section 6.8. Compute the required packed heights of the rectifying and stripping sections.

SOLUTION

The distillate and bottoms rates are computed by an overall material balance on isopropyl ether:

$$0.40(250) = 0.75D + 0.05(250 - D)$$

$$\text{Solving, } D = 125 \text{ kmol/h and } B = 250 - 125 = 125 \text{ kmol/h}$$

The equilibrium curve for this mixture at 1 atm is shown in Figure 7.35, where it is noted that isopropyl ether is the light key and an azeotrope is formed at 78 mol% isopropyl ether. The distillate composition of 75 mol% is safely below the azeotropic composition. Also shown in Figure 7.35 are the q -line and the rectification-section operating line for the condition of minimum reflux. The slope of the latter line is measured to be $(L/V)_{\min} = 0.39$. From (7-27),

$$R_{\min} = 0.39 / (1 - 0.39) = 0.64 \quad \text{and} \quad R = 1.5 R_{\min} = 0.96$$

$$L = RD = 0.96(125) = 120 \text{ kmol/h}$$

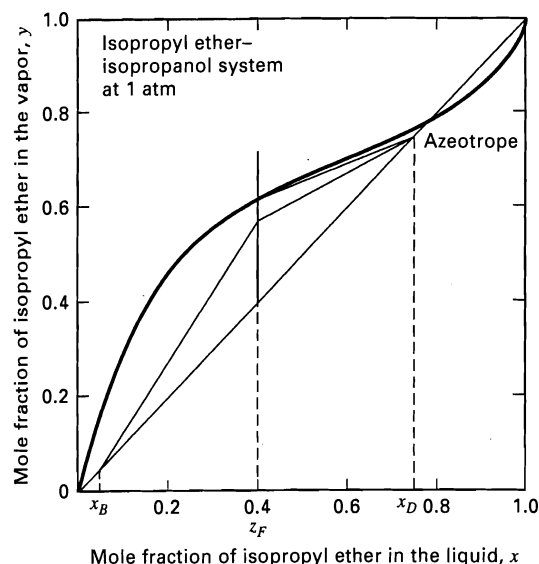


Figure 7.35 Operating lines and minimum-reflux line for Example 7.10.

$$\text{and } V = L + D = 120 + 125 = 245 \text{ kmol/h}$$

$$\bar{L} = L + L_F = 120 + 250 = 370 \text{ kmol/h}$$

$$\text{and } \bar{V} = V - V_F = 245 - 0 = 245 \text{ kmol/h}$$

$$\text{Slope of rectification-section operating line} = L/V = 120/245 = 0.49$$

This line and the stripping-section operating line are plotted in Figure 7.35. The partial reboiler, R , is stepped off in Figure 7.36 to give the following end points for determining the packed heights of the two sections, where the symbols refer to Figure 7.34a:

	Stripping Section	Rectifying Section
Top	$(x_F = 0.40, y_F = 0.577)$	$(x_2 = 0.75, y_2 = 0.75)$
Bottom	$(x_1 = 0.135, y_1 = 0.18)$	$(x_F = 0.40, y_F = 0.577)$

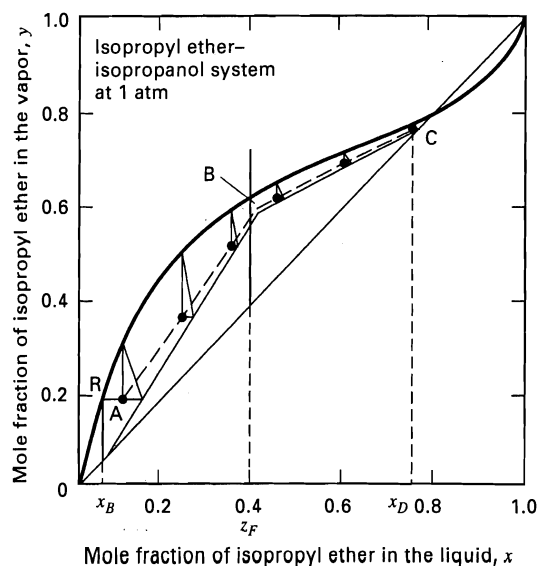


Figure 7.36 Mass-transfer driving forces for Example 7.10.

Mass-transfer coefficients at three values of x in each section are as follows:

x	$k_y a$ kmol/m ³ -h-(mole fraction)	$k_x a$ kmol/m ³ -h-(mole fraction)
Stripping section:		
0.15	305	1,680
0.25	300	1,760
0.35	335	1,960
Rectifying section:		
0.45	185	610
0.60	180	670
0.75	165	765

Slopes of the above mass-transfer coefficients are computed, for each point, x , on the operating line using (7-55), and drawn from the operating line to the equilibrium line, as shown in Figure 7.36. These lines are often referred to as tie lines because they tie the operating line to the equilibrium line. Using the tie lines as hypotenuses, right triangles are drawn, as shown in Figure 7.36. Dashed locus lines, AB and BC, are then drawn through the points at the 90° corners of the triangles. Using these locus lines, additional tie lines can quickly be added to the three plotted in each section, as needed, to give sufficient accuracy. From the tie lines, values of $(y_1 - y)$ can be tabulated for values of y on the operating lines. Since the diameter of the column is not given, the packed volumes are determined from the following rearrangements of (7-58) and (7-60), where $V = Sl_T$:

$$V_R = \int_{y_F}^{y_2} \frac{V dy}{k_y a (y_1 - y)} \quad (7-62)$$

$$V_S = \int_{y_1}^{y_F} \frac{V dy}{k_y a (y_1 - y)} \quad (7-63)$$

Values of $k_y a$ are interpolated as necessary. Results are given in the following table.

y	$(y_1 - y)$	$k_y a$	$\frac{V(\text{or } \bar{V})}{k_y a (y_1 - y)}, \text{m}^3$
Stripping section:			
0.18	0.145	307	5.5
0.25	0.150	303	5.4
0.35	0.143	300	5.7
0.45	0.103	320	7.4
0.577	0.030	350	23.3
Rectifying section:			
0.577	0.030	187	43.7
0.60	0.033	185	40.1
0.65	0.027	182	49.9
0.70	0.017	175	82.3
0.75	0.010	165	148.5

By numerical integration, $V_S = 3.6 \text{ m}^3$ and $V_R = 12.3 \text{ m}^3$.

7.6 PONCHON–SAVARIT GRAPHICAL EQUILIBRIUM-STAGE METHOD FOR TRAYED TOWERS

The McCabe–Thiele method, in Section 7.2 for binary distillation, assumes that molar vapor and liquid flow rates are constant in each section of the column. This assumption

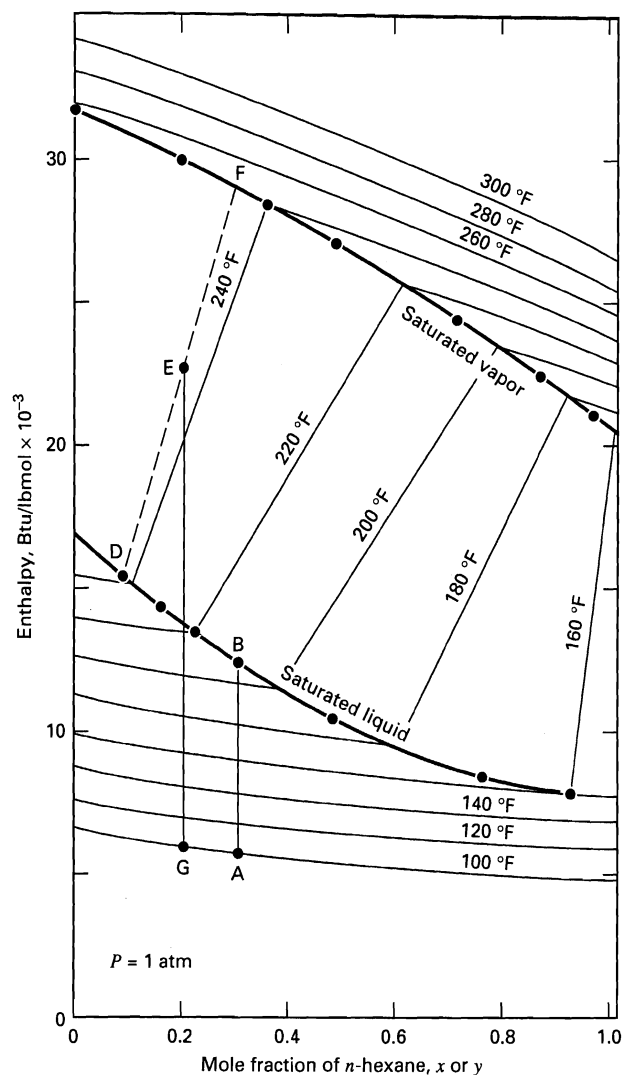


Figure 7.37 Enthalpy–concentration diagram for *n*-hexane/*n*-octane.

(constant molar overflow) eliminates the need to make an energy balance around each stage. For nonideal binary mixtures, such an assumption may not be valid and the McCabe–Thiele method may not be accurate. A graphical method that includes energy balances as well as material balances and phase equilibrium relations is the Ponchon–Savarit method [21, 22], which utilizes an enthalpy–composition diagram of the type shown in Figure 7.37 for the *n*-hexane/*n*-octane system at 1 atm. This diagram includes curves for the enthalpies of saturated vapor and liquid mixtures. Terminal points of tie lines connecting these two curves represent the equilibrium vapor and liquid compositions, together with vapor and liquid enthalpies, for the given temperature. Isotherms above the saturated vapor curve represent enthalpies of the superheated vapor, while isotherms below the saturated liquid curve represent the subcooled liquid. In Figure 7.37, a mixture of 30 mol% hexane and 70 mol% octane at 100°F (Point A) is a subcooled liquid. By heating it to Point B at 204°F, it becomes a liquid at its bubble point (Point B). When a mixture of 20 mol% hexane and 80 mol% octane at 100°F (Point G) is heated to 243°F (Point E), at equilibrium, it splits into a vapor phase at Point F and a liquid phase at Point D. The

liquid phase contains 7 mol% hexane, while the vapor contains 29 mol% hexane.

The application of the enthalpy–concentration diagram to equilibrium-stage calculations may be illustrated by considering a single equilibrium stage, $n - 1$, where vapor from stage $n - 2$ below is mixed adiabatically with liquid from stage n above to give an overall mixture, denoted by mole-fraction z , and then brought to equilibrium. The process is represented schematically in two steps, mixing followed by equilibration, at the top of Figure 7.38. The energy-balance equations for stage $n - 1$ are

Mixing:

$$V_{n-2}H_{n-2} + L_n h_n = (V_{n-2} + L_n)h_z \quad (7-64)$$

Equilibration:

$$(V_{n-2} + L_n)h_z = V_{n-1}H_{n-1} + L_{n-1}h_{n-1} \quad (7-65)$$

where H and h are vapor and liquid molar enthalpies, respectively. The governing material-balance equations for the light component are

Mixing:

$$y_{n-2}V_{n-2} + x_n L_n = z(V_{n-2} + L_n) \quad (7-66)$$

Equilibration:

$$z(V_{n-2} + L_n) = y_{n-1}V_{n-1} + x_{n-1}L_{n-1} \quad (7-67)$$

Simultaneous solution of (7-64) and (7-66) gives

$$\frac{H_{n-2} - h_z}{y_{n-2} - z} = \frac{h_z - h_n}{z - x_n} \quad (7-68)$$

which is the three-point form of a straight line plotted in Figure 7.38. Similarly, the simultaneous solution of (7-65) and (7-67) gives

$$\frac{H_{n-1} - h_z}{y_{n-1} - z} = \frac{h_z - h_{n-1}}{z - x_{n-1}} \quad (7-69)$$

which is also the equation for a straight line. However, in this case y_{n-1} and x_{n-1} are in equilibrium and, therefore, the points (H_{n-1}, y_{n-1}) and (h_{n-1}, x_{n-1}) must lie on the

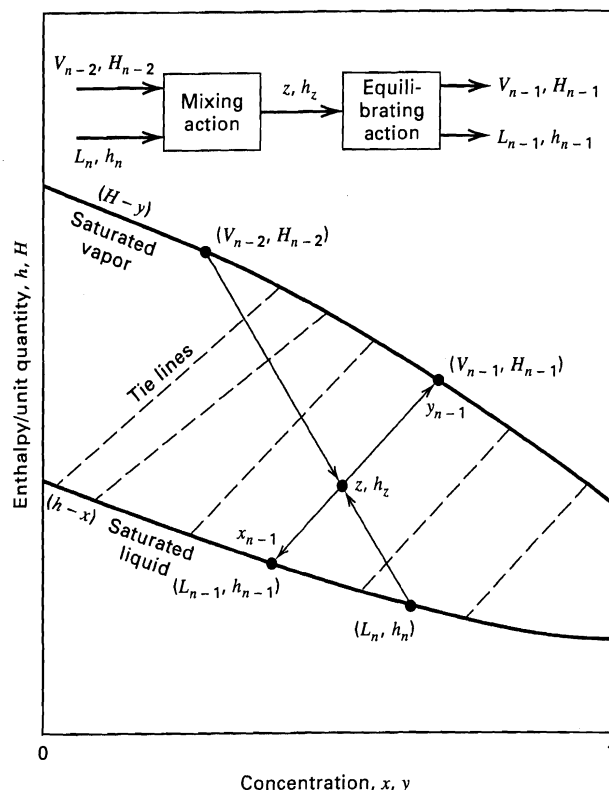


Figure 7.38 Two-phase mixing and equilibration on an enthalpy–concentration diagram.

opposite ends of the tie line that passes through the mixing point (h_z, z) , as shown in Figure 7.38.

The Ponchon–Savarit method for binary distillation is an extension of the construction in Figure 7.38 to countercurrent cascades above and below the feed stage, with consideration of the condenser and reboiler. A detailed description of the method is not given here because the method has been largely superseded by the rigorous computer-aided calculation procedures, discussed in Chapter 10, which include energy balances and can be applied to multicomponent as well as binary mixtures. A detailed presentation of the Ponchon–Savarit method for binary distillation is given by Henley and Seader [23].

SUMMARY

1. A binary-liquid and/or binary-vapor mixture can be separated economically into two nearly pure products (distillate and bottoms) by distillation, provided that the value of the relative volatility of the two components is high enough, usually greater than 1.05.
2. Distillation is the most mature and widely used separation operation, with design procedures and operation practices well established.
3. The purities of the products from distillation depend on the number of equilibrium stages in the rectifying section above the feed entry and in the stripping section below the feed entry, and on the reflux ratio. Both the number of stages and the reflux ratio must be greater than the minimum values corresponding to total reflux and infinite stages, respectively. The optimal reflux-to-minimum-reflux ratio is usually in the range of 1.10 to 1.50.

4. Distillation is most commonly conducted in trayed towers equipped with sieve or valve trays, or in columns packed with random or structured packings. Many older towers are equipped with bubble-cap trays.
5. Most distillation towers are equipped with a condenser, cooled with cooling water, to provide reflux, and a reboiler, heated with steam, to provide boilup.
6. When the assumption of constant molar overflow is valid in each of the two sections of the distillation tower, the McCabe–Thiele graphical method is convenient for determining stage and reflux requirements. This method facilitates the visualization of many aspects of distillation and provides a procedure for locating the optimal feed-stage location.

7. Miscellaneous considerations involved in the design of a distillation tower include selection of operating pressure, type of condenser, degree of reflux subcooling, type of reboiler, and extent of feed preheat.
8. The McCabe–Thiele method can be extended to handle Murphree stage efficiency, multiple feeds, side streams, open steam, and use of interreboilers and intercondensers.
9. Rough estimates of overall stage efficiency, defined by (6-21), can be made with the Drickamer and Bradford, (7-42), or O'Connell, (7-43), correlations. More accurate and reliable procedures use data from a small Oldershaw column or the same semi-theoretical equations for mass transfer in Chapter 6 that are used for absorption and stripping.
10. Tray diameter, pressure drop, weeping, entrainment, and downcomer backup can all be estimated by the procedures in Chapter 6.

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EXERCISES

Unless otherwise stated, the usual simplifying assumptions of saturated-liquid reflux, optimal feed-stage location, no heat losses, steady state, and constant molar liquid and vapor flows apply to each of the following problems.

Section 7.1

- 7.1 List as many differences between absorption and distillation as you can. List as many differences between stripping and distillation as you can.
- 7.2 Prior to the 1980s, packed columns were rarely used for distillation unless column diameter was less than 2.5 ft. Explain why, in recent years, some existing trayed towers are being retrofitted with packing and some new large-diameter columns are being designed for packing rather than trays.

11. Reflux and flash drums are sized by a procedure based on avoidance of entrainment and provision for adequate liquid residence time.
12. Packed-column diameter and pressure drop are determined by the same procedures presented in Chapter 6 for absorption and stripping.
13. The height of a packed column may be determined by the HETP method, or preferably from the HTU method. Application of the latter method is similar to that of Chapter 6 for absorbers and strippers, but differs in the manner in which the curved equilibrium line must be handled, as given by (7-47).
14. The Ponchon–Savarit graphical method removes the assumption of constant molar overflow in the McCabe–Thiele method by employing energy balances with an enthalpy-concentration diagram. However, use of the Ponchon–Savarit method has largely been supplanted by rigorous computer-aided methods.

7.3 A mixture of methane and ethane is to be separated by distillation. Explain why water cannot be used as the coolant in the condenser. What would you choose as the coolant?

7.4 A mixture of ethylene and ethane is to be separated by distillation. Determine the maximum operating pressure of the column. What operating pressure would you suggest? Why?

7.5 Under what circumstances would it be advisable to conduct laboratory or pilot-plant tests of a proposed distillation separation?

7.6 Explain why an economic tradeoff exists between the number of trays and the reflux ratio.

Section 7.2

7.7 Following the development by Sorel in 1894 of a mathematical model for continuous, steady-state, equilibrium-stage

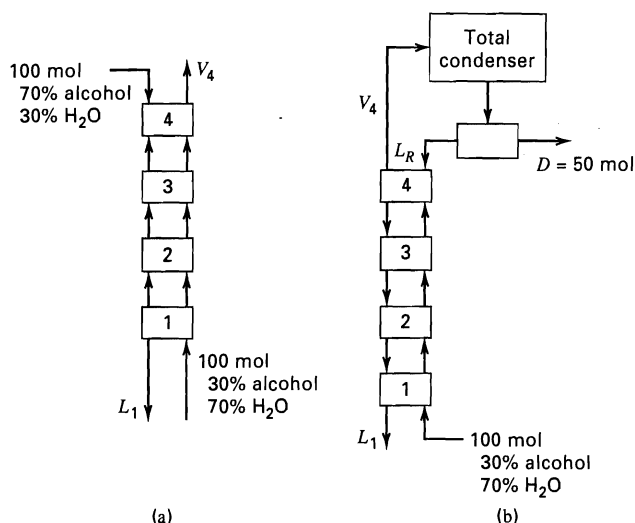


Figure 7.39 Data for Exercise 7.8.

distillation, a number of methods were proposed for solving the equations graphically or algebraically during an 18-year period from 1920 to 1938, prior to the availability of digital computers. Today, the only method from that era that remains in widespread use is the McCabe–Thiele method. What are the attributes of this method that are responsible for its continuing popularity?

7.8 (a) For the cascade shown in Figure 7.39a, calculate the compositions of streams V_4 and L_1 . Assume atmospheric pressure, saturated liquid and vapor feeds, and the vapor–liquid equilibrium data given below. Compositions are in mole percent.

(b) Given the feed compositions in cascade (a), how many equilibrium stages are required to produce a V_4 containing 85 mol% alcohol?

(c) For the cascade configuration shown in Figure 7.39b, with $D = 50$ mol, what are the compositions of D and L_1 ?

(d) For the configuration of cascade (b), how many equilibrium stages are required to produce a D of 50 mol% alcohol?

EQUILIBRIUM DATA, MOLE-FRACTION ALCOHOL

x	0.1	0.3	0.5	0.7	0.9
y	0.2	0.5	0.68	0.82	0.94

7.9 Liquid air is fed to the top of a perforated-tray reboiled stripper operated at substantially atmospheric pressure. Sixty percent of the oxygen in the feed is to be drawn off in the bottoms vapor product from the still. This product is to contain 0.2 mol% nitrogen. Based on the assumptions and data given below, calculate:

- The mole percent of nitrogen in the vapor leaving the top plate.
- The moles of vapor generated in the still per 100 mol of feed.
- The number of theoretical plates required.

Notes: To simplify the problem, assume constant molar overflow equal to the moles of feed. Liquid air contains 20.9 mol% of oxygen and 79.1 mol% of nitrogen. The equilibrium data [*Chem. Met. Eng.*, 35, 622 (1928)] at atmospheric pressure are

Temperature, K	Mole-Percent N_2 in Liquid	Mole-Percent N_2 in Vapor
77.35	100.00	100.00
77.98	90.00	97.17
78.73	79.00	93.62

Temperature, K	Mole-Percent N_2 in Liquid	Mole-Percent N_2 in Vapor
79.44	70.00	90.31
80.33	60.00	85.91
81.35	50.00	80.46
82.54	40.00	73.50
83.94	30.00	64.05
85.62	20.00	50.81
87.67	10.00	31.00
90.17	0.00	0.00

7.10 A mixture of A (more volatile) and B is being separated in a plate distillation column. In two separate tests run with a saturated-liquid feed of 40 mol% A, the following compositions, in mol% A, were obtained for samples of liquid and vapor streams from three consecutive stages between the feed and total condenser at the top:

Stage	Mol % A			
	Test 1		Test 2	
	Vapor	Liquid	Vapor	Liquid
$M + 2$	79.5	68.0	75.0	68.0
$M + 1$	74.0	60.0	68.0	60.5
M	67.9	51.0	60.5	53.0

Determine the reflux ratio and overhead composition in each case, assuming that the column has more than three stages.

7.11 A saturated-liquid mixture containing 70 mol% benzene and 30 mol% toluene is to be distilled at atmospheric pressure to produce a distillate of 80 mol% benzene. Five procedures, described below, are under consideration. For each of the procedures, calculate and tabulate:

- Moles of distillate per 100 moles of feed,
- Moles of total vapor generated per mole of distillate,
- Mole percent benzene in the residue, and
- For each part, construct a y - x diagram. On this, indicate the compositions of the overhead product, the reflux, and the composition of the residue.
- If the objective is to maximize total benzene recovery, which, if any, of these procedures is preferred?

Note: Assume that the relative volatility equals 2.5.

The procedures are as follows:

- Continuous distillation followed by partial condensation. The feed is sent to the direct-heated still pot, from which the residue is continuously withdrawn. The vapors enter the top of a helically coiled partial condenser that discharges into a trap. The liquid is returned (refluxed) to the still, while the residual vapor is condensed as a product containing 80 mol% benzene. The molar ratio of reflux to product is 0.5.
- Continuous distillation in a column containing one equilibrium plate. The feed is sent to the direct-heated still, from which residue is withdrawn continuously. The vapors from the plate enter the top of a helically coiled partial condenser that discharges into a trap. The liquid from the trap is returned to the plate, while the uncondensed vapor is condensed to form a distillate containing 80 mol% benzene. The molar ratio of reflux to product is 0.5.
- Continuous distillation in a column containing the equivalent of two equilibrium plates. The feed is sent to the direct-heated still, from which residue is withdrawn continuously. The

vapors from the top plate enter the top of a helically coiled partial condenser that discharges into a trap. The liquid from the trap is returned to the top plate (refluxed) while the uncondensed vapor is condensed to form a distillate containing 80 mol% benzene. The molar ratio of reflux to product is 0.5.

4. The operation is the same as that described for Procedure 3 with the exception that the liquid from the trap is returned to the bottom plate.
5. Continuous distillation in a column containing the equivalent of one equilibrium plate. The feed at its boiling point is introduced on the plate. The residue is withdrawn continuously from the direct-heated still pot. The vapors from the plate enter the top of a helically coiled partial condenser that discharges into a trap. The liquid from the trap is returned to the plate while the uncondensed vapor is condensed to form a distillate containing 80 mol% benzene. The molar ratio of reflux to product is 0.5.

7.12 A saturated-liquid mixture of benzene and toluene containing 50 mol% benzene is distilled in an apparatus consisting of a still pot, one theoretical plate, and a total condenser. The still pot is equivalent to one equilibrium stage, and the pressure is 101 kPa.

The still is supposed to produce a distillate containing 75 mol% benzene. For each of the following procedures, calculate, if possible, the number of moles of distillate per 100 moles of feed. Assume a relative volatility of 2.5.

- (a) No reflux with feed to the still pot.
- (b) Feed to the still pot, reflux ratio $L/D = 3$.
- (c) Feed to the plate with a reflux ratio of 3.
- (d) Feed to the plate with a reflux ratio of 3. However, in this case, a partial condenser is employed.
- (e) Part (b) using minimum reflux.
- (f) Part (b) using total reflux.

7.13 A fractionation column operating at 101 kPa is to separate 30 kg/h of a solution of benzene and toluene containing 0.6 mass-fraction toluene into an overhead product containing 0.97 mass-fraction benzene and a bottoms product containing 0.98 mass-fraction toluene. A reflux ratio of 3.5 is to be used. The feed is liquid at its boiling point, feed is to the optimal tray, and the reflux is at saturation temperature.

- (a) Determine the quantity of top and bottom products.
- (b) Determine the number of stages required.

EQUILIBRIUM DATA IN MOLE-FRACTION BENZENE, 101 KPA

y	0.21	0.37	0.51	0.64	0.72	0.79	0.86	0.91	0.96	0.98
x	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	0.95

7.14 A mixture of 54.5 mol% benzene in chlorobenzene at its bubble point is fed continuously to the bottom plate of a column containing two theoretical plates. The column is equipped with a partial reboiler and a total condenser. Sufficient heat is supplied to the reboiler to give $\bar{V}/F = 0.855$, and the reflux ratio L/V in the top of the column is kept constant at 0.50. Under these conditions, what quality of product and bottoms (x_D , x_B) can be expected?

EQUILIBRIUM DATA AT COLUMN PRESSURE, MOLE-FRACTION BENZENE

x	0.100	0.200	0.300	0.400	0.500	0.600	0.700	0.800
y	0.314	0.508	0.640	0.734	0.806	0.862	0.905	0.943

7.15 A continuous distillation operation with a reflux ratio (L/D) of 3.5 yields a distillate containing 97 wt% B (benzene) and bottoms containing 98 wt% T (toluene). Due to weld failures, the 10 plates in the bottom section of the column are ruined, but the 14 upper plates are intact. It is suggested that the column still be used, with the feed (F) as saturated vapor at the dew point, with $F = 13,600$ kg/h containing 40 wt% B and 60 wt% T. Assuming that the plate efficiency remains unchanged at 50%: (a) Can this column still yield a distillate containing 97 wt% B, (b) How much distillate can we get, and (c) What will the composition of the residue be in mole percent?

For vapor-liquid equilibrium data, see Exercise 7.13.

7.16 A distillation column having eight theoretical stages (seven in the column + partial reboiler + total condenser) is being used to separate 100 kmol/h of a saturated-liquid feed containing 50 mol% A into a product stream containing 90 mol% A. The liquid-to-vapor molar ratio at the top plate is 0.75. The saturated-liquid feed is introduced on plate 5 from the top. Determine: (a) The composition of the bottoms, (b) The $\bar{L}/\bar{V}$ ratio in the stripping section, and (c) The moles of bottoms per hour.

Unbeknown to the operators, the bolts holding plates 5, 6, and 7 rust through, and the plates fall into the still pot. If no adjustments are made, what is the new bottoms composition?

It is suggested that, instead of returning reflux to the top plate, an equivalent amount of liquid product from another column be used as reflux. If this product contains 80 mol% A, what now is the composition of: (a) The distillate, and (b) The bottoms.

EQUILIBRIUM DATA, MOLE FRACTION OF A

y	0.19	0.37	0.5	0.62	0.71	0.78	0.84	0.9	0.96
x	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9

7.17 A distillation unit consists of a partial reboiler, a column with seven equilibrium plates, and a total condenser. The feed consists of a 50 mol% mixture of benzene in toluene. It is desired to produce a distillate containing 96 mol% benzene, when operating at 101 kPa.

- (a) With saturated-liquid feed fed to the fifth plate from the top, calculate: (1) Minimum reflux ratio $(L_R/D)_{\min}$, (2) The bottoms composition, using a reflux ratio (L_R/D) of twice the minimum, and (3) Moles of product per 100 moles of feed.
- (b) Repeat part (a) for a saturated-vapor feed fed to the fifth plate from the top.
- (c) With saturated-vapor feed fed to the reboiler and a reflux ratio (L/V) of 0.9, calculate: (1) Bottoms composition, (2) Moles of product per 100 mole of feed.

Equilibrium data are given in Exercise 7.13.

7.18 A valve-tray fractionating column containing eight theoretical plates, a partial reboiler equivalent to one theoretical plate, and a total condenser is in operation separating a benzene-toluene mixture containing 36 mol% benzene at 101 kPa. Under normal operating conditions, the reboiler generates 100 kmol of vapor per hour. A request has been made for very pure toluene, and it is proposed to operate this column as a stripper, introducing the feed on the top plate as a saturated liquid, employing the same boilup at the still, and returning no reflux to the column. Equilibrium data are given in Exercise 7.13.

- (a) What is the minimum feed rate under the proposed conditions, and what is the corresponding composition of the liquid in the reboiler at the minimum feed?

(b) At a feed rate 25% above the minimum, what is the rate of production of toluene, and what are the compositions in mole percent of the product and distillate?

7.19 A solution of methanol and water at 101 kPa containing 50 mol% methanol is continuously rectified in a seven-theoretical-plate, perforated-tray column, equipped with a total condenser and a partial reboiler heated by steam.

During normal operation, 100 kmol/h of feed is introduced on the third plate from the bottom. The overhead product contains 90 mol% methanol, and the bottoms product contains 5 mol% methanol. One mole of liquid reflux is returned to the column for each mole of overhead product.

Recently it has been impossible to maintain the product purity in spite of an increase in the reflux ratio. The following test data were obtained:

Stream	kmol/h	mol% alcohol
Feed	100	51
Waste	62	12
Product	53	80
Reflux	94	—

What is the most probable cause of this poor performance? What further tests would you make to establish definitely the reason for the trouble? Could some 90% product be obtained by further increasing the reflux ratio, while keeping the vapor rate constant?

Vapor-liquid equilibrium data at 1 atm [Chem. Eng. Prog. 48, 192 (1952)] in mole-fraction methanol are

x	0.0321	0.0523	0.075	0.154	0.225	0.349	0.813	0.918
y	0.1900	0.2940	0.352	0.516	0.593	0.703	0.918	0.963

7.20 A fractionating column equipped with a partial reboiler heated with steam, as shown in Figure 7.40, and with a total condenser, is operated continuously to separate a mixture of 50 mol% A and 50 mol% B into an overhead product containing 90 mol% A and a bottoms product containing 20 mol% A. The column has three theoretical plates, and the reboiler is equivalent to one theoretical plate. When the system is operated at an $L/V = 0.75$ with

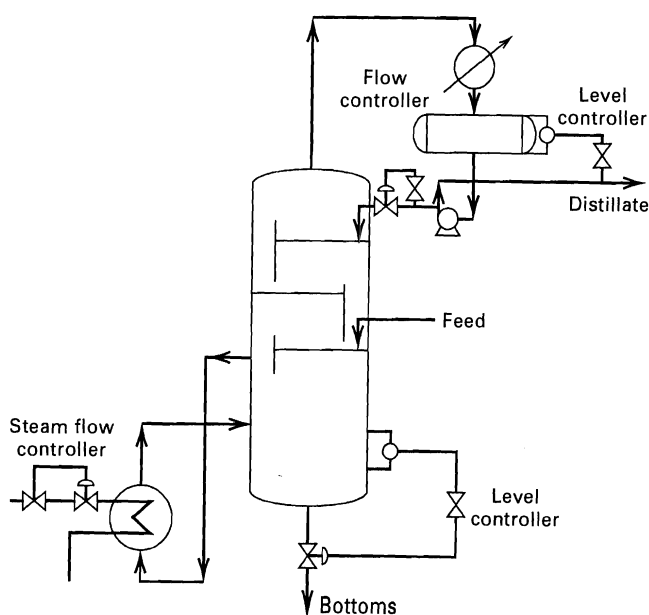


Figure 7.40 Data for Exercise 7.20.

the feed as a saturated liquid to the bottom plate of the column, the desired products can be obtained. The system is instrumented as shown. The steam to the reboiler is controlled by a flow controller so that it remains constant. The reflux to the column is also on a flow controller so that the quantity of reflux is constant. The feed to the column is normally 100 kmol/h, but it was inadvertently cut back to 25 kmol/h. What would be the composition of the reflux, and what would be the composition of the vapor leaving the reboiler under these new conditions? Assume that the vapor leaving the reboiler is not superheated. Relative volatility for the system is 3.0.

7.21 A saturated-vapor mixture of maleic anhydride and benzoic acid containing 10 mol% acid is a by-product of the manufacture of phthalic anhydride. This mixture is distilled continuously at 13.3 kPa to give a product of 99.5 mol% maleic anhydride and a bottoms of 0.5 mol% anhydride. Using the data below, calculate the number of theoretical plates needed using an L/D of 1.6 times the minimum.

VAPOR PRESSURE, TORR:

Temperature, °C:	10	50	100	200	400
Maleic anhydride	78.7	116.8	135.8	155.9	179.5
Benzoic acid	131.6	167.8	185.0	205.8	227

7.22 A bubble-point binary mixture containing 5 mol% A in B is to be distilled to give a distillate containing 35 mol% A and a bottoms product containing 0.2 mol% A. If the relative volatility is constant at a value of 6, calculate the following algebraically, assuming that the column will be equipped with a partial reboiler and a partial condenser.

- The minimum number of equilibrium stages
- The minimum boilup ratio $\bar{V}/B$ leaving the reboiler
- The actual number of equilibrium stages for a boilup ratio equal to 1.2 times the minimum value

7.23 Methanol (M) is to be separated from water (W) by distillation as shown in Figure 7.41. The feed is subcooled such that $q = 1.12$. Determine the feed-stage location and the number of theoretical stages required. Vapor-liquid equilibrium data are given in Exercise 7.19.

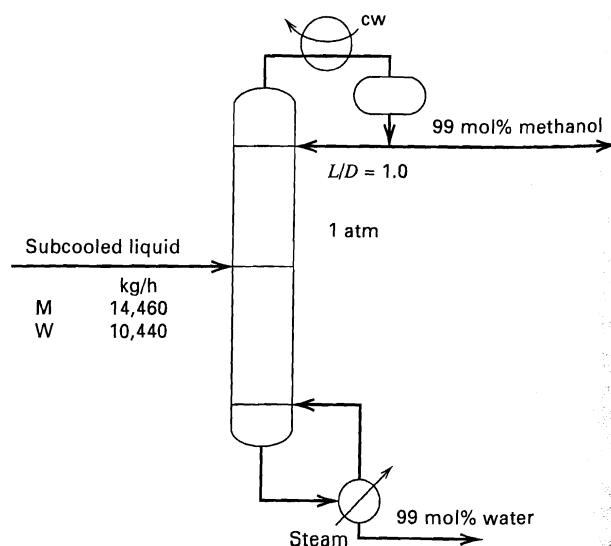


Figure 7.41 Data for Exercise 7.23.

7.24 A saturated-liquid mixture of 69.4 mol% benzene (B) in toluene (T) is to be continuously distilled at atmospheric pressure to produce a distillate containing 90 mol% benzene, with a yield of 25 moles of distillate per 100 moles of feed. The feed is sent to a steam-heated still (reboiler), where residue is to be withdrawn continuously. The vapors from the still pass directly to a partial condenser. From a liquid separator following the condenser, reflux is returned to the still. Vapors from the separator, which are in equilibrium with the liquid reflux, are sent to a total condenser and are continuously withdrawn as distillate. At equilibrium the mole ratio of B to T in the vapor may be taken as 2.5 times the mole ratio of B to T in the liquid. Calculate analytically and graphically the total moles of vapor generated in the still per 100 mol of feed.

7.25 A plant has a batch of 100 kmol of a liquid mixture containing 20 mol% benzene and 80 mol% chlorobenzene. It is desired to rectify this mixture at 1 atm to obtain bottoms containing only 0.1 mol% benzene. The relative volatility may be assumed constant at 4.13. There are available a suitable still to vaporize the feed, a column containing the equivalent of four theoretical plates, a total condenser, and a reflux drum to collect the condensed overhead. The run is to be made at total reflux. While the steady state is being approached, a finite amount of distillate is held in a reflux trap. When the steady state is reached, the bottoms contain 0.1 mol% benzene. With this apparatus, what yield of bottoms can be obtained? The liquid holdup in the column is negligible compared to that in the still and in the reflux drum.

7.26 A mixture of acetone and isopropanol containing 50 mol% acetone is to be distilled continuously to produce an overhead product containing 80 mol% acetone and a bottoms containing 25 mol% acetone. If a saturated-liquid feed is employed, if the column is operated with a reflux ratio of 0.5, and if the Murphree vapor efficiency is 50%, how many trays will be required? Assume a total condenser, partial reboiler, saturated-liquid reflux, and optimal feed stage. The vapor-liquid equilibrium data for this system are

EQUILIBRIUM DATA, MOLE-PERCENT ACETONE

Liquid	0	2.6	5.4	11.7	20.7	29.7	34.1	44.0	52.0
Vapor	0	8.9	17.4	31.5	45.6	55.7	60.1	68.7	74.3
Liquid	63.9	74.6	80.3	86.5	90.2	92.5	95.7	100.0	
Vapor	81.5	87.0	89.4	92.3	94.2	95.5	97.4	100.0	

7.27 A mixture of 40 mol% carbon disulfide (CS_2) in carbon tetrachloride (CCl_4) is continuously distilled. The feed is 50% vaporized ($q = 0.5$). The top product from a total condenser is 95 mol% CS_2 , and the bottoms product from a partial reboiler is a liquid of 5 mol% CS_2 .

The column operates with a reflux ratio, L/D , of 4 to 1. The Murphree vapor efficiency is 80%.

(a) Calculate graphically the minimum reflux, the minimum boilup ratio from the reboiler, $\bar{V}/B$, and the minimum number of stages (including reboiler).

(b) How many trays are required for the actual column at 80% efficiency by the McCabe-Thiele method.

The vapor-liquid equilibrium data at column pressure for this mixture in terms of CS_2 mole fraction are

x	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
y	0.135	0.245	0.42	0.545	0.64	0.725	0.79	0.85	0.905	0.955

7.28 A distillation unit consists of a partial reboiler, a bubble-cap column, and a total condenser. The overall plate efficiency is 65%.

The feed is a liquid mixture, at its bubble point, consisting of 50 mol% benzene in toluene. This liquid is fed to the optimal plate. The column is to produce a distillate containing 95 mol% benzene and a bottoms of 95 mol% toluene. Calculate for an operating pressure of 1 atm: (a) Minimum reflux ratio $(L/D)_{\min}$, (b) Minimum number of actual plates to carry out the desired separation, (c) Using a reflux ratio (L/D) of 50% more than the minimum, the number of actual plates needed, (d) The kilograms per hour of product and residue, if the feed is 907.3 kg/h, (e) The saturated steam at 273.7 kPa required in kilograms per hour for heat to the reboiler using enthalpy data below and any assumptions necessary, and (f) A rigorous enthalpy balance on the reboiler, using the enthalpy data, tabulated below and assuming ideal solutions. Enthalpies in Btu/lbmol at reboiler temperature:

	h_L	h_V
benzene	4,900	18,130
toluene	8,080	21,830

Vapor-liquid equilibrium data are given in Exercise 7.13.

7.29 A continuous distillation unit, consisting of a perforated-tray column together with a partial reboiler and a total condenser, is to be designed to operate at atmospheric pressure to separate ethanol and water. The feed, which is introduced into the column as liquid at its bubble point, contains 20 mol% alcohol. The distillate is to contain 85 mol% alcohol, and the alcohol recovery is to be 97%.

(a) What is the molar concentration of the bottoms?

(b) What is the minimum value of:

(1) The reflux ratio L/V ?

(2) The reflux ratio L/D ?

(3) The boilup ratio $\bar{V}/B$ from the reboiler?

(c) What is the minimum number of theoretical stages and the corresponding number of actual plates if the overall plate efficiency is 55%?

(d) If the reflux ratio L/V used is 0.80, how many actual plates will be required?

Vapor-liquid equilibrium for ethanol-water at 1 atm in terms of mole fractions of ethanol are [*Ind. Eng. Chem.*, **24**, 881 (1932)]:

x	y	$T, ^\circ\text{C}$	x	y	$T, ^\circ\text{C}$
0.0190	0.1700	95.50	0.3273	0.5826	81.50
0.0721	0.3891	89.00	0.3965	0.6122	80.70
0.0966	0.4375	86.70	0.5079	0.6564	79.80
0.1238	0.4704	85.30	0.5198	0.6599	79.70
0.1661	0.5089	84.10	0.5732	0.6841	79.30
0.2337	0.5445	82.70	0.6763	0.7385	78.74
0.2608	0.5580	82.30	0.7472	0.7815	78.41
			0.8943	0.8943	78.15

7.30 A solvent A is to be recovered by distillation from its water solution. It is necessary to produce an overhead product containing 95 mol% A and to recover 95% of the A in the feed. The feed is available at the plant site in two streams, one containing 40 mol% A and the other 60 mol% A. Each stream will provide 50 kmol/h of component A, and each will be fed into the column as saturated liquid. Since the less volatile component is water, it has been proposed to supply the necessary heat in the form of open steam. For the preliminary design, it has been suggested that the operating reflux ratio, L/D , be 1.33 times the minimum value. A total condenser will be employed. For this system, it is estimated that the overall plate

efficiency will be 70%. How many plates will be required, and what will be the bottoms composition? The relative volatility may be assumed to be constant at 3.0. Determine analytically the points necessary to locate the operating lines. Each feed should enter the column at its optimal location.

7.31 A saturated-liquid feed stream containing 40 mol% *n*-hexane (H) and 60 mol% *n*-octane is fed to a plate column. A reflux ratio L/D equal to 0.5 is maintained at the top of the column. An overhead product of 0.95 mole fraction H is required, and the column bottoms is to be 0.05 mole fraction H. A cooling coil submerged in the liquid of the second plate from the top removes sufficient heat to condense 50 mol% of the vapor rising from the third plate down from the top.

- Derive the equations needed to locate the operating line.
- Locate the operating lines and determine the required number of theoretical plates if the optimal feed plate location is used.

7.32 One hundred kilogram-moles per hour of a saturated liquid mixture of 12 mol% ethyl alcohol in water is distilled continuously by direct steam at 1 atm introduced directly to the bottom plate. The distillate required is 85 mol% alcohol, representing 90% recovery of the alcohol in the feed. The reflux is saturated liquid with $L/D = 3$. Feed is on the optimal stage. Vapor-liquid equilibrium data are given in Exercise 7.29. Calculate:

- Steam requirement, kmol/h
- Number of theoretical stages
- The feed stage (optimal)
- Minimum reflux ratio, $(L/D)_{\min}$

7.33 A water-isopropanol mixture at its bubble point containing 10 mol% isopropanol is to be continuously rectified at atmospheric pressure to produce a distillate containing 67.5 mol% isopropanol. Ninety-eight percent of the isopropanol in the feed must be recovered. If a reflux ratio L/D of 1.5 times the minimum is used, how many theoretical stages will be required: (a) If a partial reboiler is used? (b) If no reboiler is used and saturated steam at 101 kPa is introduced below the bottom plate? (c) How many stages are required at total reflux?

Vapor-liquid equilibrium data in mole fraction of isopropanol at 101 kPa are

$T, ^\circ\text{C}$	93.00	84.02	82.12	81.25	80.62	80.16	80.28	81.51
y	0.2195	0.4620	0.5242	0.5686	0.5926	0.6821	0.7421	0.9160
x	0.0118	0.0841	0.1978	0.3496	0.4525	0.6794	0.7693	0.9442

Notes: Composition of the azeotrope is $x = y = 0.6854$. Boiling point of azeotrope = 80.22°C

7.34 An aqueous solution containing 10 mol% isopropanol is fed at its bubble point to the top of a continuous stripping column, operated at atmospheric pressure, to produce a vapor containing 40 mol% isopropanol. Two procedures are under consideration, both involving the same heat expenditure with V/F (moles of vapor generated/mole of feed) = 0.246 in each case. Scheme 1 uses a partial reboiler at the bottom of a plate-type stripping column, generating vapor by the use of steam condensing inside a closed coil. In Scheme 2, the reboiler is omitted and live steam is injected directly below the bottom plate. Determine the number of stages required in each case.

Equilibrium data for the system isopropanol-water are given in Exercise 7.33. The usual simplifying assumptions may be made.

7.35 Determine the optimal-stage location for each feed and the number of theoretical stages required for the distillation separation

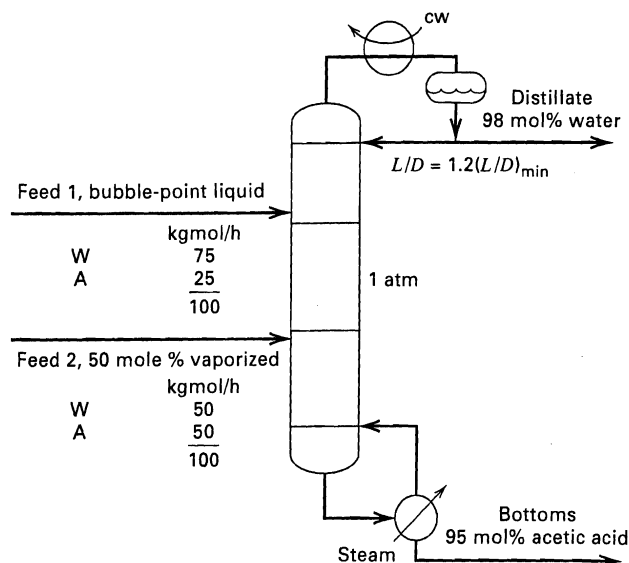


Figure 7.42 Data for Exercise 7.35.

shown in Figure 7.42 using the following equilibrium data in mole fractions.

WATER (W)/ACETIC ACID (A), 1 ATM

x_W	0.0055	0.053	0.125	0.206	0.297	0.510	0.649	0.803	0.9594
y_W	0.0112	0.133	0.240	0.338	0.437	0.630	0.751	0.866	0.9725

7.36 Determine the number of theoretical stages required and the optimal-stage locations for the feed and liquid side stream for the distillation process shown in Figure 7.43 assuming that methanol (M) and ethanol (E) form an ideal solution.

7.37 A mixture of *n*-heptane (H) and toluene (T) is separated by extractive distillation with phenol (P). Distillation is then used to recover the phenol for recycle as shown in Figure 7.44a, where the small amount of *n*-heptane in the feed is ignored. For the conditions shown in Figure 7.44a, determine the number of theoretical stages

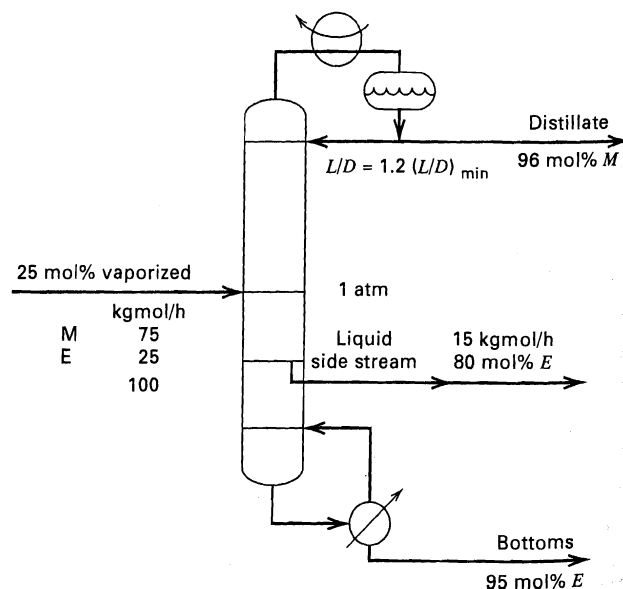


Figure 7.43 Data for Exercise 7.36.

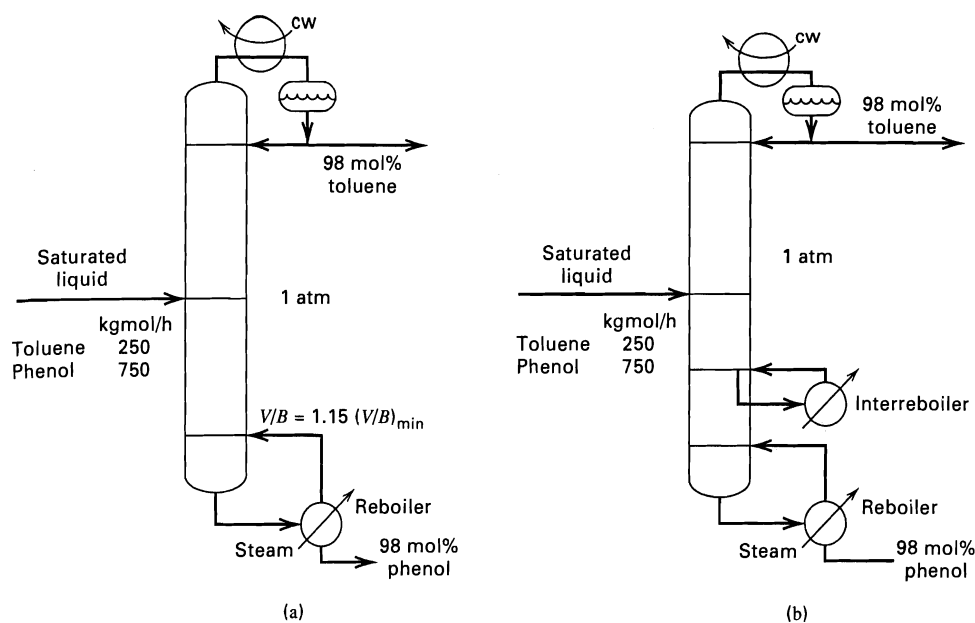


Figure 7.44 Data for Exercise 7.37.

required. Note that heat will have to be supplied to the reboiler at a high temperature because of the high boiling point of phenol.

Therefore, consider the alternative scheme in Figure 7.44b, where an interreboiler, located midway between the bottom plate and the feed stage, is used to provide 50% of the boilup used in Figure 7.44a. The remainder of the boilup is provided by the reboiler. Determine the number of theoretical stages required for the case with the interreboiler and the temperature of the interreboiler stage. Unsmoothed vapor-liquid equilibrium data at 1 atm are [Trans. AICHE, 41, 555 (1945)]:

x_T	y_T	$T, ^\circ\text{C}$	x_T	y_T	$T, ^\circ\text{C}$
0.0435	0.3410	172.70	0.6512	0.9260	120.00
0.0872	0.5120	159.40	0.7400	0.9463	119.70
0.1186	0.6210	153.80	0.7730	0.9536	119.40
0.1248	0.6250	149.40	0.8012	0.9545	115.60
0.2190	0.7850	142.20	0.8840	0.9750	112.70
0.2750	0.8070	133.80	0.9108	0.9796	112.20
0.4080	0.8725	128.30	0.9394	0.9861	113.30
0.4800	0.8901	126.70	0.9770	0.9948	111.10
0.5898	0.9159	122.20	0.9910	0.9980	111.10
0.6348	0.9280	120.20	0.9939	0.9986	110.50
			0.9973	0.9993	110.50

7.38 A distillation column for the separation of *n*-butane from *n*-pentane was recently put into operation in a petroleum refinery. Apparently, an error was made in the design because the column fails to make the desired separation as shown in the following table [Chem. Eng. Prog., 61 (8), 79 (1965)]:

	Design Specification	Actual Operation
Mol% $n\text{C}_5$ in distillate	0.26	13.49
Mol% $n\text{C}_4$ in bottoms	0.16	4.28

In order to correct the situation, it is proposed to add an intercondenser in the rectifying section to generate more reflux and an interreboiler in the stripping section to produce additional boilup. Show by use of a McCabe-Thiele diagram how such a proposed change can improve the operation.

7.39 In the production of chlorobenzenes by chlorination of benzene, two close-boiling isomers, *para*-dichlorobenzene (P) and *ortho*-dichlorobenzene (O), are separated by distillation. The feed to the column consists of 62 mol% of the *para* isomer and 38 mol% of the *ortho* isomer. Assume that the pressures at the bottom and top of the column are 20 psia (137.9 kPa) and 15 psia (103.4 kPa), respectively. The distillate is a liquid containing 98 mol% *para* isomer. The bottoms product is to contain 96 mol% *ortho* isomer. At column pressure, the feed is slightly vaporized with $q = 0.9$. Calculate the number of theoretical stages required for a reflux ratio equal to 1.15 times the minimum-reflux ratio. Base your calculations on a constant relative volatility obtained as the arithmetic average between the column top and column bottom using appropriate vapor pressure data and the assumption of Raoult's and Dalton's laws. The McCabe-Thiele construction should be supplemented at the two ends by use of the Kremser equations as illustrated in Example 7.4.

7.40 Relatively pure oxygen and nitrogen can be obtained by the distillation of air using the Linde double column, which, as shown in Figure 7.45, consists of a lower column operating at elevated pressure surmounted by an atmospheric-pressure column. The boiler of the upper column is at the same time the reflux condenser for both columns. Gaseous air plus enough liquid to take care of heat leak into the column (more liquid, of course, if liquid-oxygen product is withdrawn) enters the exchanger at the base of the lower column and condenses, giving up heat to the boiling liquid and thus supplying the vapor flow for this column. The liquid air enters an intermediate point in this column, as shown in Figure 7.45. The vapors rising in this column are partially condensed to form the reflux, and the uncondensed vapor passes to an outer row of tubes and is totally condensed, the liquid nitrogen collecting in an annulus, as shown. By operating this column at 4 to 5 atm, the liquid oxygen boiling at 1 atm is cold enough to condense pure nitrogen. The liquid that collects in the bottom of the lower column contains about 45 mol% O_2 and forms the feed for the upper column. Such a double column can produce very pure oxygen with high oxygen recovery, and relatively pure nitrogen. On a single McCabe-Thiele diagram—using equilibrium lines, operating lines, q -lines, 45° line, stepped-off stages, and other illustrative aids—show qualitatively how the stage requirements of the double column can be computed.

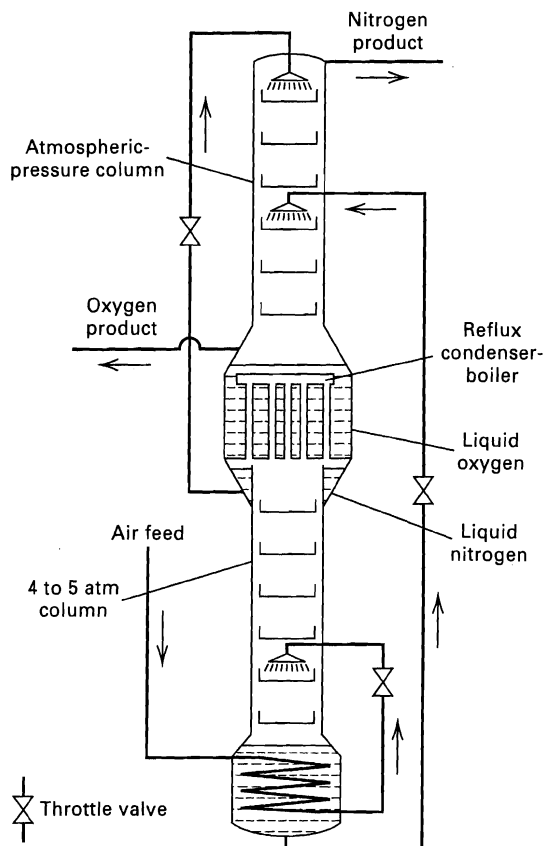


Figure 7.45 Data for Exercise 7.40.

Section 7.3

7.41 The following performance data have been obtained for a distillation tower separating a 50/50 by weight percent mixture of methanol and water:

Feed rate = 45,438 lb/h; Feed condition = bubble-point liquid at feed-tray pressure, Wt% methanol in distillate = 95.04, and Wt% methanol in bottoms = 1.00

Reflux ratio = 0.947; Reflux condition = saturated liquid

Boilup ratio = 1.138; Pressure in reflux drum = 14.7 psia

Type condenser = total; Type reboiler = partial

Condenser pressure drop = 0.0 psi; Tower pressure drop = 0.8 psi

Trays above feed tray = 5; Trays below feed tray = 6

Total trays = 12; Tray diameter = 6 ft

Type tray = single-pass sieve tray; Flow path length = 50.5 in.

Weir length = 42.5 in.; Hole area = 10%; Hole size = 3/16 in.

Weir height = 2 in.; Tray spacing = 24 in.

Viscosity of feed = 0.34 cP

Surface tension of distillate = 20 dyne/cm; Surface tension of bottoms = 58 dyne/cm

Temperature of top tray = 154°F; Temperature of bottom tray = 207°F

Vapor-liquid equilibrium data at column pressure in mole fraction of methanol are

y	0.0412	0.156	0.379	0.578	0.675	0.729	0.792	0.915
x	0.00565	0.0246	0.0854	0.205	0.315	0.398	0.518	0.793

Based on the above data:

(a) Determine the overall tray efficiency from the data, assuming that the reboiler is the equivalent of a theoretical stage.

(b) Estimate the overall tray efficiency from the Drickamer-Bradford correlation.

(c) Estimate the overall tray efficiency from the O'Connell correlation, accounting for length of flow path.

(d) Estimate the Murphree vapor tray efficiency by the method of Chan and Fair.

7.42 For the conditions of Exercise 7.41, a laboratory Oldershaw column measures an average Murphree vapor-point efficiency of 65%. Estimate E_{MV} and E_o .

Section 7.4

7.43 Conditions for the top tray of a distillation column are as shown in Figure 7.46. Determine the column diameter corresponding to 85% of flooding if a valve tray is used. Make whatever assumptions necessary.

7.44 A separation of propylene from propane is achieved by distillation as shown in Figure 7.47, where two columns in series are used because a single column would be too tall. The tray numbers refer to equilibrium stages. Determine the column diameters, tray efficiency using the O'Connell correlation, number of actual trays, and column heights if perforated trays are used.

7.45 Determine the height and diameter of a vertical flash drum for the conditions shown in Figure 7.48.

7.46 Determine the length and diameter of a horizontal reflux drum for the conditions shown in Figure 7.49.

7.47 Results of design calculations for a methanol-water distillation operation are given in Figure 7.50.

(a) Calculate the column diameter at the top tray and at the bottom tray for sieve trays. Should the column be swaged?

(b) Calculate the length and diameter of the horizontal reflux drum.

7.48 For the conditions given in Exercise 7.41, estimate for the top tray and the bottom tray: (a) Percent of flooding, (b) Tray pressure drop in psi, (c) Whether weeping will occur, (d) Entrainment rate, and (e) Froth height in the downcomer.

7.49 If the feed rate to the tower of Exercise 7.41 is increased by 30% with all other conditions except for tower pressure drop remaining the same, estimate for the top and bottom trays: (a) Per-

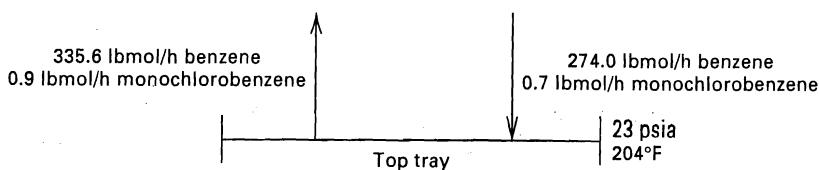


Figure 7.46 Data for Exercise 7.43.

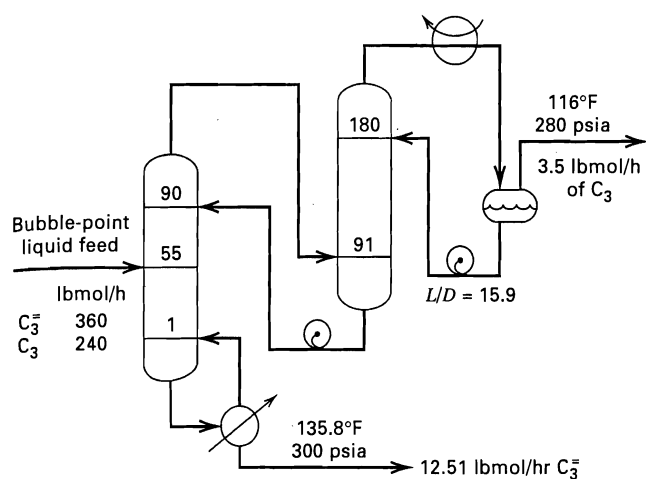


Figure 7.47 Data for Exercise 7.44.

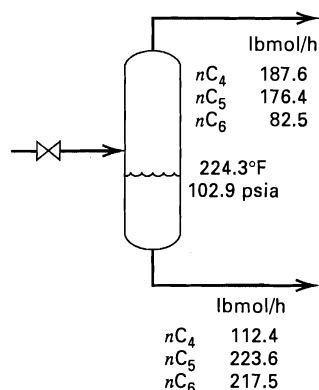


Figure 7.48 Data for Exercise 7.45.

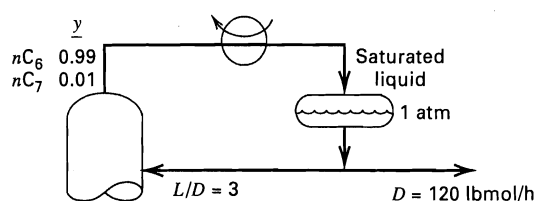


Figure 7.49 Data for Exercise 7.46.

cent of flooding, (b) Tray pressure drop in psi, (c) Entrainment rate, (d) Froth height in the downcomer. Will the new operation be acceptable? If not, should you consider a retrofit with packing? If so, should both sections of the column be packed or could just one section be packed to achieve an acceptable operation?

Section 7.5

7.50 A mixture of benzene and dichloroethane is used to test the efficiency of a packed column that contains 10 ft of packing and operates adiabatically at atmospheric pressure. The liquid is charged to the reboiler, and the column is operated at total reflux until equilibrium is established. At equilibrium, liquid samples from the distillate and reboiler, as analyzed by refractive index, give the following compositions for benzene: $x_D = 0.653$, $x_B = 0.298$.

Calculate the value of HETP in inches for this packing. What are the limitations on using this calculated value for design?

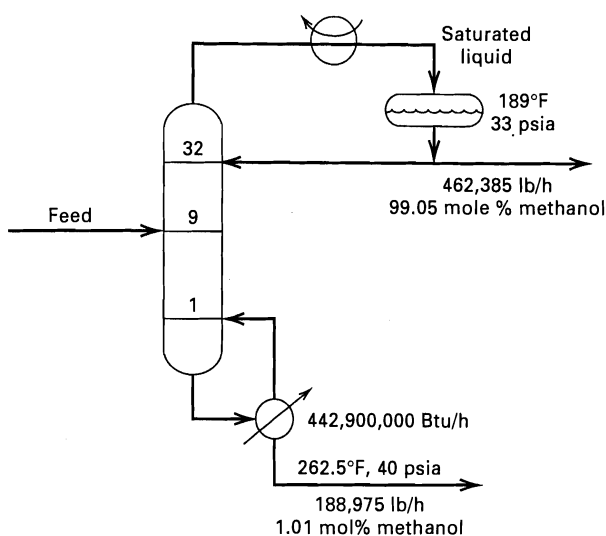


Figure 7.50 Data for Exercise 7.47.

Data for x - y at 1 atm (in benzene mole fractions) are

x	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
y	0.11	0.22	0.325	0.426	0.526	0.625	0.720	0.815	0.91

7.51 Consider a distillation column for separating ethanol from water at 1 atm. The following specifications are set:

Feed: 10 mol% ethanol (bubble-point liquid)

Bottoms: 1 mol% ethanol

Distillate: 80 mol% ethanol (saturated liquid)

Reflux ratio: 1.5 times the minimum

Constant molar overflow may be assumed and vapor-liquid equilibrium data are given in Exercise 7.29.

(a) How many theoretical plates are required above and below the feed if a plate column is used?

(b) How many transfer units are required above and below the feed if a packed column is used?

(c) Assuming that the plate efficiency is approximately 80% and the plate spacing is 18 in., how high is the plate column?

(d) Using an H_{OG} value of 1.2 ft., how high is the packed column?

(e) Assuming that you had HTU data available only on the benzene-toluene system, how would you go about applying the data to obtain the HTU for the ethanol-water system?

7.52 Plant capacity for the methanol-water distillation of Exercise 7.41 is to be doubled. Rather than installing a second trayed tower identical to the one in operation, a packed column is to be considered for the new installation. This column will have a feed location identical to the present trayed tower and will be expected to achieve the same product purities with the same top pressure and reflux ratio. Two packings are being considered:

1. 50-mm plastic NOR PAC rings (a random packing)
2. Montz metal B1-300 (a structured packing)

For each of these two packings, design a packed column to operate at 70% of flooding by calculating for each section: (a) Liquid holdup, (b) Column diameter, (c) H_{OG} , (d) Packed height, (e) Pressure drop. What are the advantages, if any, of each of the packed-column designs over a second trayed tower? Which packing, if either, is preferable?

Table 7.8 Methanol–Water Vapor–Liquid Equilibrium and Enthalpy Data for 1 atm (MeOH = Methyl Alcohol)

Mol% MeOH y or x	Enthalpy above 0°C, Btu/lbmol Solution				Vapor Liquid Equilibrium Data		
	Saturated Vapor		Saturated Liquid		Mol% MeOH in		Temperature, °C
	$T, ^\circ\text{C}$	h_V	$T, ^\circ\text{C}$	h_L	Liquid	Vapor	
0	100	20,720	100	3,240	0	0	100
5	98.9	20,520	92.8	3,070	2.0	13.4	96.4
10	97.7	20,340	87.7	2,950	4.0	23.0	93.5
15	96.2	20,160	84.4	2,850	6.0	30.4	91.2
20	94.8	20,000	81.7	2,760	8.0	36.5	89.3
30	91.6	19,640	78.0	2,620	10.0	41.8	87.7
40	88.2	19,310	75.3	2,540	15.0	51.7	84.4
50	84.9	18,970	73.1	2,470	20.0	57.9	81.7
60	80.9	18,650	71.2	2,410	30.0	66.5	78.0
70	76.6	18,310	69.3	2,370	40.0	72.9	75.3
80	72.2	17,980	67.6	2,330	50.0	77.9	73.1
90	68.1	17,680	66.0	2,290	60.0	82.5	71.2
100	64.5	17,390	64.5	2,250	70.0	87.0	69.3
					80.0	91.5	67.6
					90.0	95.8	66.0
					95.0	97.9	65.0
					100.0	100.0	64.5

Source: J.G. Dunlop, "Vapor–Liquid Equilibrium Data," M.S. thesis, Brooklyn Polytechnic Institute, Brooklyn, NY (1948).

7.53 For the specifications of Example 7.1, design a packed column using 50-mm metal Hiflow rings and operating at 70% of flooding by calculating for each section: (a) Liquid holdup, (b) Column diameter, (c) H_{OG} , (d) Packed height, and (e) Pressure drop.

What are the advantages and disadvantages of a packed column as compared to a trayed tower for this service?

Section 7.6

7.54 An enthalpy–concentration diagram is given in Figure 7.37 for a mixture of *n*-hexane (H), and *n*-octane (O) at 101 kPa. Using this diagram, determine the following:

- The mole-fraction composition of the vapor when a liquid containing 30 mol% H is heated from point A to the bubble-point temperature at point B.
- The energy required to vaporize 60 mol% of a mixture initially at 100°F and containing 20 mol% H (point G).
- The compositions of the equilibrium vapor and liquid resulting from part (b).

7.55 Using the enthalpy–concentration diagram of Figure 7.37, determine the following for a mixture of *n*-hexane (H) and *n*-octane (O) at 1 atm:

- The temperature and compositions of equilibrium liquid and vapor resulting from adiabatic mixing of 950 lb/h of a mixture of

30 mol% H in O at 180°F with 1,125 lb/h of a mixture of 80 mol% H in O at 240°F.

(b) The energy required to partially condense, by cooling, a mixture of 60 mol% H in O from an initial temperature of 260°F to 200°F. What are the compositions and amounts of the resulting vapor and liquid phases per pound-mole of original mixture?

(c) If the equilibrium vapor from part (b) is further cooled to 180°F, determine the compositions and relative amounts of the resulting vapor and liquid.

7.56 One hundred pound-moles per hour of a mixture of 60 mol% methanol in water at 30°C and 1 atm is to be separated by distillation at the same pressure into a liquid distillate containing 98 mol% methanol and a bottoms liquid product containing 96 mol% water. Enthalpy and equilibrium data for the mixture at 1 atm are given in Table 7.8. The enthalpy of the feed mixture is 765 Btu/lbmol.

- Using the given data, plot an enthalpy–concentration diagram.
- Devise a procedure to determine, from the diagram of part (a), the minimum number of equilibrium stages for the condition of total reflux and the required separation.
- From the procedure developed in part (b), determine $N_{\min}$. Why is the value independent of the feed condition?

(d) What are the temperatures of the distillate and the bottoms?

Liquid-Liquid Extraction with Ternary Systems

In *liquid-liquid extraction*, a liquid feed of two or more components to be separated is contacted with a second liquid phase, called the *solvent*, which is immiscible or only partly miscible with one or more components of the liquid feed and completely or partially miscible with one or more of the other components of the liquid feed. Thus, the solvent, which is a single chemical species or a mixture, partially dissolves certain components of the liquid feed, effecting at least a partial separation of the feed. Liquid-liquid extraction is sometimes called *extraction*, *solvent extraction*, or *liquid extraction*. These, as well as the term *solid-liquid extraction*, are also applied to the recovery of substances from a solid by contact with a liquid solvent, such as the recovery of oil from seeds by an organic solvent. Solid-liquid extraction (leaching) is covered in Chapter 16.

According to Derry and Williams [1], liquid extraction has been practiced since at least the time of the Romans, who separated gold and silver from molten copper by extraction using molten lead as a solvent. This was followed by the discovery that sulfur could selectively dissolve silver from an alloy with gold. However, it was not until the early 1930s that the first large-scale liquid-liquid extraction

process began operation. In that industrial process, named after its inventor L. Edeleanu, aromatic and sulfur compounds were selectively removed from liquid kerosene by liquid-liquid extraction with liquid sulfur dioxide at 10 to 20°F. Removal of aromatic compounds resulted in a cleaner-burning kerosene. Liquid-liquid extraction has grown in importance in recent years because of the growing demand for temperature-sensitive products, higher-purity requirements, more efficient equipment, and availability of solvents with higher selectivity.

The simplest liquid-liquid extraction involves only a ternary system. The feed consists of two miscible components, the *carrier*, *C*, and the *solute*, *A*. Solvent, *S*, is a pure compound. Components *C* and *S* are at most only partially soluble in each other. Solute *A* is soluble in *C* and completely or partially soluble in *S*. During the extraction process, mass transfer of *A* from the feed to the solvent occurs, with less transfer of *C* to the solvent, or *S* to the feed. However, complete or nearly complete transfer of *A* to the solvent is seldom achieved in just one stage, as discussed in Chapter 4. In practice, a number of stages are used in one- or two-section, countercurrent cascades, as discussed in Chapter 5.

8.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Explain differences among liquid-liquid extraction, stripping, and distillation.
- List situations where liquid-liquid extraction might be preferred to distillation.
- Explain why so many different types of equipment are used for liquid-liquid extraction.
- List major types of equipment used for liquid-liquid extraction and compare their advantages and disadvantages.
- List major factors involved in the selection of extraction equipment.
- List factors that influence liquid-liquid extraction.
- List characteristics of an ideal solvent.
- Define the distribution coefficient and show its relationship to activity coefficients and relative selectivity of a solute between carrier and solvent.
- Make a preliminary selection of a solvent using group-interaction rules.
- Distinguish, for ternary mixtures, between Type I and Type II systems.
- For a specified recovery of a solute, calculate with the Hunter and Nash method, using a triangular diagram, minimum solvent requirement and number of equilibrium stages for ternary liquid-liquid extraction in a countercurrent cascade.

- Determine usefulness of extract reflux and carry out calculations with the Maloney and Schubert graphical method for a two-section extraction cascade that uses extract reflux.
- Design a cascade of mixer-settler units based on mass-transfer considerations.
- Determine the size of multicompartment extraction columns, including consideration of the effect of axial dispersion.

Industrial Example

Acetic acid is produced by methanol carbonylation or oxidation of acetaldehyde, or as a by-product of cellulose-acetate manufacture. In all three cases, a mixture of acetic acid (normal b.p. = 118.1°C) and water (normal b.p. = 100°C) must be separated to give glacial acetic acid (99.8 wt% min). When the mixture contains less than 50% acetic acid, separation by distillation is expensive because of the need to vaporize large amounts of the more volatile water, with its very high heat of vaporization. Accordingly, an alternative

liquid-liquid extraction process is often considered. A typical implementation is shown in Figure 8.1. In this process, it is important to note that two additional distillation separation steps are required to recover the solvent for recycle to the extractor. These additional separation steps are common to almost all extraction processes.

In the process of Figure 8.1, a feed of 30,260 lb/h, of 22 wt% acetic acid in water, is sent to a single-section extraction column, operating at near-ambient conditions, where the feed is countercurrently contacted with 71,100 lb/h of

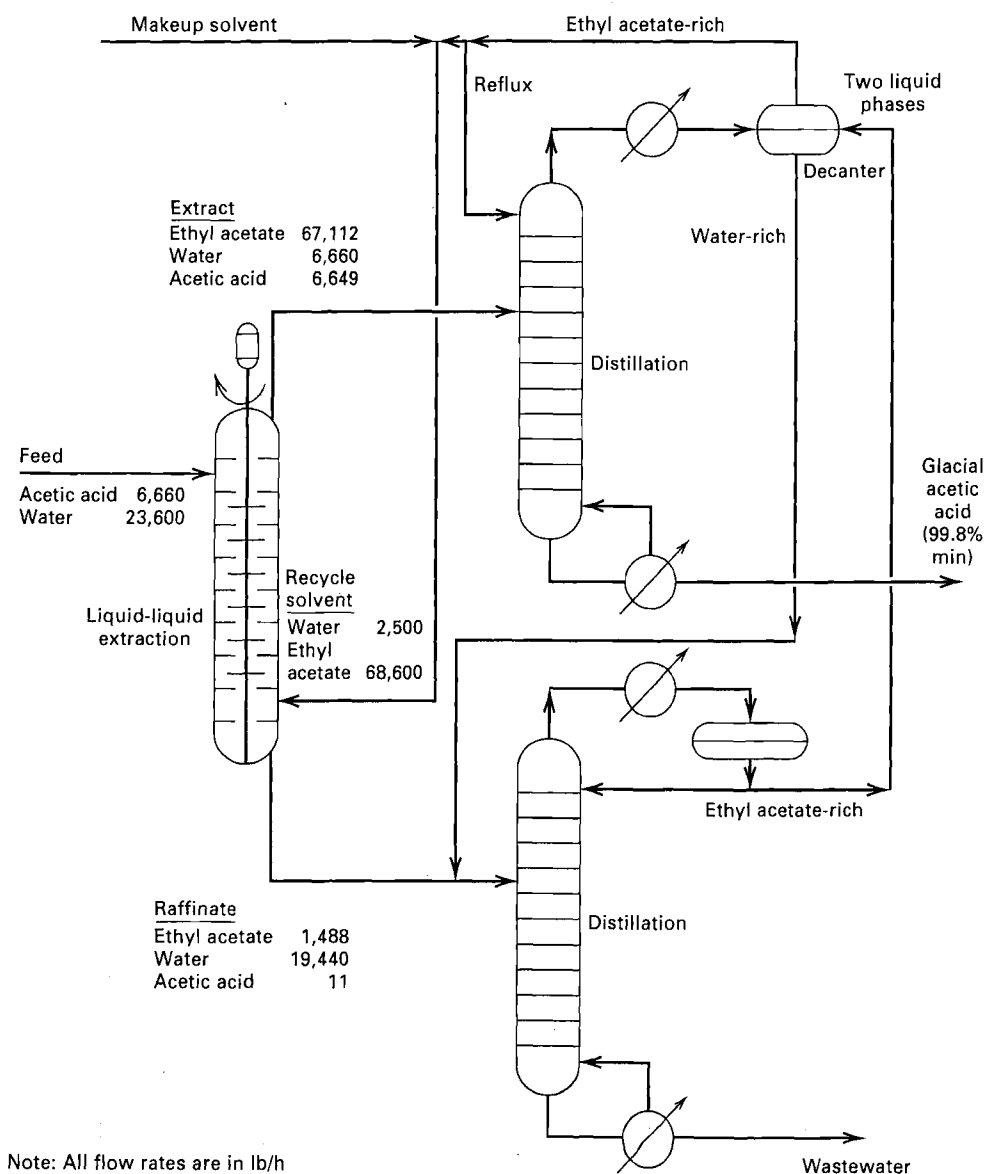


Figure 8.1 Typical liquid-liquid extraction process.

ethyl-acetate solvent (normal b.p. = 77.1°C), saturated with water. The extract (solvent-rich product), being the low-density liquid phase, exits from the top of the extractor with 99.8% of the acetic acid originally contained in the feed. The raffinate (carrier-rich product), being the high-density liquid phase, exits from the bottom of the extractor and contains only 0.05 wt% acetic acid. The extract is sent to a distillation column, where glacial acetic acid is the bottoms product. The overhead vapor, which is rich in ethyl acetate but which also contains appreciable water vapor, splits into two liquid phases upon condensation. The two phases are separated by gravity in the decanter. The lighter ethyl-acetate-rich phase is divided into two streams. One is used for reflux for the distillation operation and the other is used for solvent recycle to the extractor.

The water-rich phase from the decanter is sent, together with the raffinate from the extractor, to a second distillation column, where wastewater is removed from the bottom and the ethyl-acetate-rich overhead distillate is recycled to the decanter. Makeup ethyl-acetate solvent is provided for solvent losses to the glacial acetic acid and wastewater products.

At an average extraction temperature of 100°F, six equilibrium stages are required to transfer 99.8% of the acetic acid from the feed to the extract using a solvent-to-feed ratio of 2.35 on a weight basis, where the recycled solvent is saturated with water. For six theoretical stages, a mechanically assisted extractor is preferred and a rotating-disk contactor (RDC), in a column configuration, is shown in Figure 8.1. The organic-rich phase is dispersed into droplets by rotating disks, while the water-rich phase is a continuous phase throughout the column. Dispersion and subsequent coalescence and settling takes place easily because at extractor operating conditions, liquid-phase viscosities are less than 1 cP, the phase-density difference is more than 0.08 g/cm³, and the interfacial tension between the two phases is appreciable, at more than 30 dyne/cm.

The column has an inside diameter of 5.5 ft and a total height from the tangent of the top head to the tangent of the bottom head of 28 ft. The column is divided into 40 compartments, each 7.5 in. high and each containing a 40-in.-diameter rotor disk located between a pair of stator (donut) rings of 46-in. inside diameter. Above the top stator ring and below the bottom stator ring are settling zones. Because the light liquid phase is dispersed, the liquid-liquid interface is maintained near the top of the column. The rotors are mounted on a centrally located single shaft driven at a nominal 60 rpm by a 5-hp motor, equipped with a speed changer, the optimal disk speed being determined during plant operation. The HETP for the extractor is 50 in., equivalent to 6.67 compartments per theoretical stage. The HETP would be only 33 in. if axial (longitudinal) mixing did not occur.

Because of the corrosive nature of aqueous acetic acid solutions, the extractor is constructed of stainless steel. Since 1948, hundreds of extraction columns similar to that of Figure 8.1, with diameters ranging up to at least 25 ft, have been built. As discussed in Section 8.1, a number of other extraction devices are suitable for the process in Figure 8.1.

Liquid-liquid extraction is a reasonably mature separation operation, although not as mature or as widely applied as distillation, absorption, and stripping. Since the 1930s, more than 1,000 laboratory, pilot-plant, and industrial extractors have been installed. Procedures for determining the number of theoretical stages to achieve a desired solute recovery are well established. However, in the thermodynamics of liquid-liquid extraction, no simple limiting theory, such as that of ideal solutions for vapor-liquid equilibrium, exists. In many cases, experimental equilibrium data are preferred over predictions based on activity-coefficient correlations. However, such data can often be correlated well by semi-theoretical activity-coefficient equations such as the NRTL or UNIQUAC equations discussed in Chapter 2. Also, considerable laboratory effort may be required just to find an acceptable and efficient solvent. Furthermore, as will be discussed in the next section, a wide variety of industrial extraction equipment is available, making it necessary to consider many alternatives before making a final selection. Unfortunately, no generalized capacity and efficiency correlations are available for all equipment types. Often, equipment vendors must be relied upon to determine equipment size, or pilot-plant tests must be performed, followed by application of scale-up procedures recommended by the vendor or taken from sources such as this textbook.

Since the introduction of industrial liquid-liquid extraction processes, a large number of applications have been proposed and developed. The petroleum industry represents the largest-volume application for liquid-liquid extraction. By the late 1960s, more than 100,000 m³/day of liquid feedstocks were being processed with physically selective solvents [2]. Extraction processes are well suited to the petroleum industry because of the need to separate heat-sensitive liquid feeds according to chemical type (e.g., aliphatic, aromatic, naphthenic) rather than by molecular weight or vapor pressure. Table 8.1 shows some representative, industrial extraction processes. Other major applications exist in the biochemical industry, where emphasis is on the separation of antibiotics and protein recovery from natural substrates; in the recovery of metals, such as copper from ammoniacal leach liquors, and in separations involving rare metals and radioactive isotopes from spent-fuel elements; and in the inorganic chemical industry, where high-boiling constituents

Table 8.1 Representative Industrial Liquid-Liquid Extraction Processes

Solute	Carrier	Solvent
Acetic acid	Water	Ethyl acetate
Acetic acid	Water	Isopropyl acetate
Aconitic acid	Molasses	Methyl ethyl ketone
Ammonia	Butenes	Water
Aromatics	Paraffins	Diethylene glycol
Aromatics	Paraffins	Furfural
Aromatics	Kerosene	Sulfur dioxide
Aromatics	Paraffins	Sulfur dioxide
Asphaltenes	Hydrocarbon oil	Furfural
Benzoic acid	Water	Benzene
Butadiene	1-Butene	aq. Cuprammonium acetate
Ethylene cyanohydrin	Methyl ethyl ketone	Brine liquor
Fatty acids	Oil	Propane
Formaldehyde	Water	Isopropyl ether
Formic acid	Water	Tetrahydrofuran
Glycerol	Water	High alcohols
Hydrogen peroxide	Anthrahydroquinone	Water
Methyl ethyl ketone	Water	Trichloroethane
Methyl borate	Methanol	Hydrocarbons
Naphthenes	Distillate oil	Nitrobenzene
Naphthenes/aromatics	Distillate oil	Phenol
Phenol	Water	Benzene
Phenol	Water	Chlorobenzene
Penicillin	Broth	Butyl acetate
Sodium chloride	aq. Sodium hydroxide	Ammonia
Vanilla	Oxidized liquors	Toluene
Vitamin A	Fish-liver oil	Propane
Vitamin E	Vegetable oil	Propane
Water	Methyl ethyl ketone	aq. Calcium chloride

such as phosphoric acid, boric acid, and sodium hydroxide need to be recovered from aqueous solutions.

In general, extraction is preferred to distillation for the following applications:

1. In the case of dissolved or complexed inorganic substances in organic or aqueous solutions.

8.1 EQUIPMENT

Given the wide diversity of applications, one might expect a correspondingly large variety of liquid-liquid extraction devices. Indeed, such is the case. Equipment similar to that used for absorption, stripping, and distillation is sometimes used, but such devices are inefficient unless liquid viscosities are low and the difference in phase density is high. For that reason, centrifugal and mechanically agitated devices are often preferred. Regardless of the type of equipment, the

2. The removal of a component present in small concentrations, such as a color former in tallow or hormones in animal oil.
3. When a high-boiling component is present in relatively small quantities in an aqueous waste stream, as in the recovery of acetic acid from cellulose acetate. Extraction becomes competitive with distillation because of the expense of evaporating large quantities of water with its very high heat of vaporization.
4. The recovery of heat-sensitive materials, where extraction may be less expensive than vacuum distillation.
5. The separation of a mixture according to chemical type rather than relative volatility.
6. The separation of close-melting or close-boiling liquids, where solubility differences can be exploited.
7. Mixtures that form azeotropes.

The key to an effective extraction process is the discovery of a suitable solvent. In addition to being stable, nontoxic, inexpensive, and easily recoverable, a good solvent should be relatively immiscible with feed components(s) other than the solute and have a different density from the feed to facilitate phase separation. Also, it must have a very high affinity for the solute, from which it should be easily separated by distillation, crystallization, or other means. Ideally, the distribution coefficient for the solute between the two liquid phases should be greater than 1; otherwise a large solvent-to-feed ratio is required. When the degree of solute extraction is not particularly high and/or when a large extraction factor can be achieved, an extractor will not require many stages. This is fortunate because mass-transfer resistance in liquid-liquid systems is often high and stage efficiency is low in commercial contacting devices, unless mechanical agitation is provided.

In this chapter, equipment for conducting liquid-liquid extraction operations is discussed and fundamental equilibrium-based and rate-based calculation procedures are presented mainly for extraction in ternary systems. The use of graphical methods is emphasized. Except for systems dilute in solute(s), calculations for higher-order multicomponent systems are best conducted with computer-aided methods discussed in Chapter 10.

necessary number of theoretical stages is computed. Then the size of the device for a continuous, countercurrent process is obtained from experimental HETP or mass-transfer-performance-data characteristic of the particular piece of equipment. In extraction, some authors use the acronym HETS, height equivalent to a theoretical stage, rather than HETP. Also, the dispersed phase is sometimes referred to as the *discontinuous phase*, the other phase being the *continuous phase*.

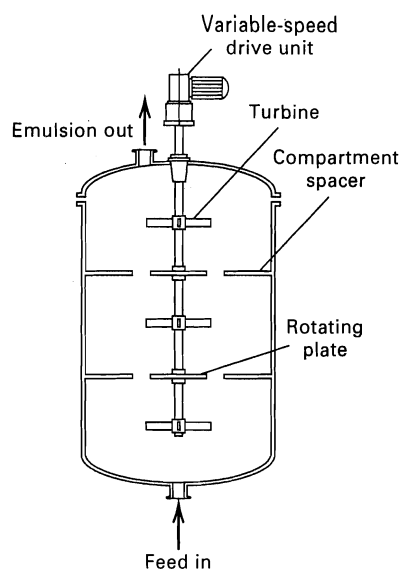


Figure 8.2 Compartmented mixing vessel with variable-speed turbine agitators.

[Adapted from R.E. Treybal, *Mass Transfer*, 3rd ed., McGraw-Hill, New York (1980).]

Mixer-Settlers

In mixer-settlers, the two liquid phases are first mixed (Figure 8.2) and then separated by settling (Figure 8.4). Any number of mixer-settler units may be connected together to form a multistage, countercurrent cascade. During mixing, one of the liquids is dispersed in the form of small droplets into the other liquid phase. The dispersed phase may be either the heavier or the lighter of the two phases. The mixing

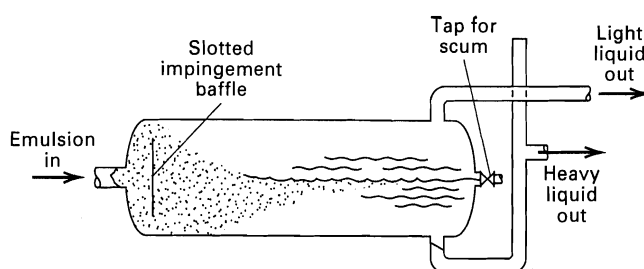


Figure 8.4 Horizontal gravity-settling vessel.

[Adapted from R.E. Treybal, *Liquid Extraction*, 2nd ed., McGraw-Hill, New York (1963) with permission.]

step is commonly conducted in an agitated vessel, with sufficient agitation and residence time so that a reasonable approach to equilibrium (e.g., 80% to 90% of a theoretical stage) is attained. The vessel may be compartmented as shown in Figure 8.2, and is usually agitated by means of impellers of the type shown in Figure 8.3. If dispersion is easily achieved and equilibrium is rapidly approached, as with liquids of low interfacial tension and viscosity, the mixing step can be carried out by impingement in a jet mixer; by turbulence in a nozzle mixer, orifice mixer, or other in-line mixing device; by shearing action if both phases are fed simultaneously into a centrifugal pump; or by injectors, wherein the flow of one liquid is induced by another.

The settling step is by gravity in a second vessel called a settler or decanter. In the configuration shown in Figure 8.4, a horizontal vessel, with an impingement baffle to prevent the jet of the entering two-phase dispersion (emulsion) from disturbing the gravity-settling process, is used. Vertical and inclined vessels are also used. A major problem in settlers is the emulsification in the mixing vessel, which may occur if the agitation is so intense that the dispersed droplet size falls below 1 to 1.5 micrometers. When this happens, coalescers, separator membranes, meshes, electrostatic forces, ultrasound, chemical treatment, or other ploys are required to speed settling. The rate of settling can also be increased by substituting centrifugal for gravitational force. This may be necessary if the phase-density difference is small.

A large number of commercial single- and multi-stage mixer-settler units are available, many of which are described by Bailes, Hanson, and Hughes [3] and by Lo, Baird, and Hanson [4]. Particularly worthy of mention is the Lurgi extraction tower [4], which was originally developed for extracting aromatics from hydrocarbon mixtures. In this device, the phases are mixed by centrifugal mixers stacked vertically outside the column and driven from a single shaft. Settling takes place in the column, with phases flowing interstage-wise, guided by a complex baffle design located within the settling zones.

Spray Columns

The simplest and one of the oldest extraction devices is the spray column. Either the heavy phase or the light phase can be dispersed, as shown in Figure 8.5. The droplets of the

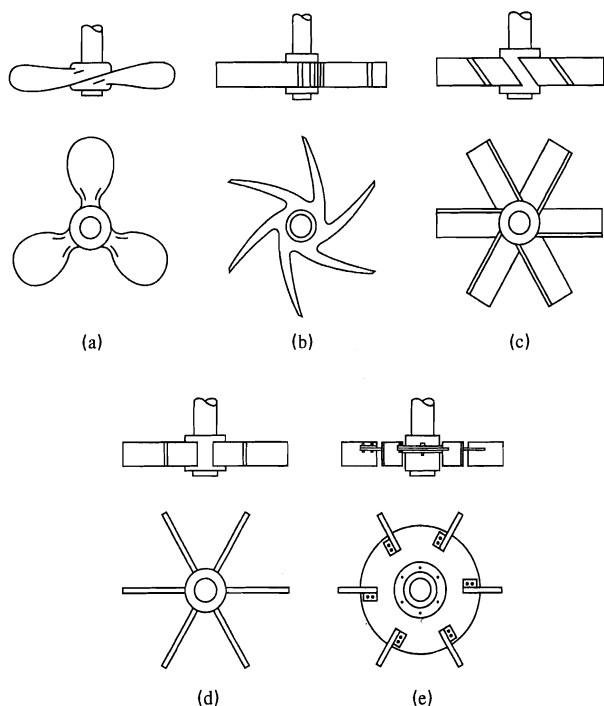


Figure 8.3 Some common types of mixing impellers: (a) marine-type propeller; (b) centrifugal turbine; (c) pitched-blade turbine; (d) flat-blade paddle; (e) flat-blade turbine.

[From R.E. Treybal, *Mass Transfer*, 3rd ed., McGraw-Hill, New York (1980) with permission.]

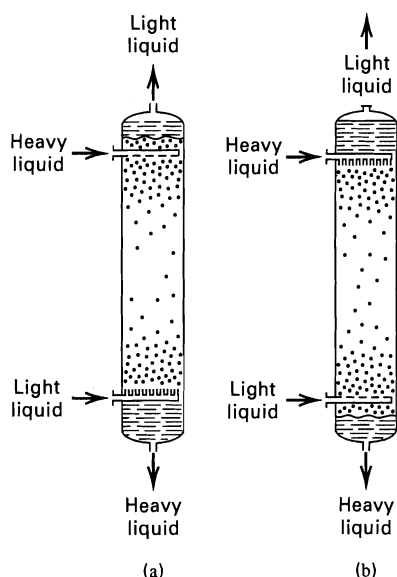


Figure 8.5 Spray columns: (a) light liquid dispersed, heavy liquid continuous; (b) heavy liquid dispersed, light liquid continuous.

dispersed phase are generated only at the inlet, usually by spray nozzles. Because of lack of column internals, throughputs are large, depending upon phase-density difference and phase viscosities. As in gas absorption, *axial dispersion (backmixing)* in the continuous phase limits these devices to applications where only one or two stages are required. Axial dispersion is so serious for columns with large diameter-to-length ratio that the continuous phase may be completely mixed. Therefore, spray columns are rarely used, despite their very low cost.

Packed Columns

Axial mixing in a spray column can be substantially reduced, but not eliminated, by packing the column. The packing also improves mass transfer by breaking up large drops to increase interfacial area and promotes mixing in drops by distorting droplet shape. With the exception of Raschig rings [5], the same packings used in distillation and absorption are

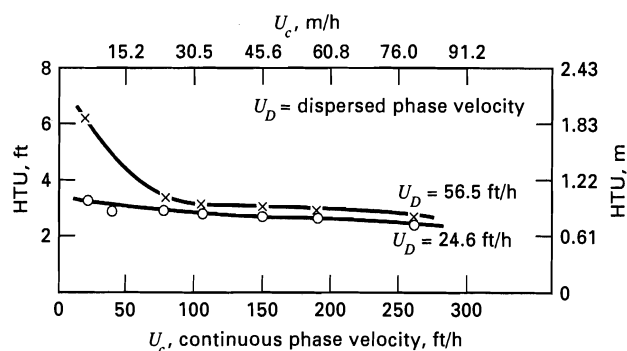


Figure 8.6 Efficiency of 1-in. Intalox saddles in a column 60 in. high with MEK-water-kerosene.

[From R.R. Neumatis, J.S. Eckert, E.H. Foote, and L.R. Rollinson, *Chem. Eng. Progr.*, 67(1), 60 (1971) with permission.]

employed for liquid-liquid extraction. The choice of packing material, however, is somewhat more critical. A material preferentially wetted by the continuous phase is preferred. Figure 8.6 shows performance data, in terms of HTU, for Intalox saddles in an extraction service as a function of continuous, U_c , and discontinuous, U_D , phase superficial velocities. Because of backmixing, the HETP is generally larger than for staged devices. For that reason, packed columns are used only where few stages are needed.

Plate Columns

Sieve plates in a column also reduce axial mixing and achieve a more stagewise type of contact. The dispersed phase may be the light or the heavy phase. In the former case, the dispersed phase, analogous to vapor bubbles in distillation, flows vertically up the column, with redispersion at each tray. The heavy phase is the continuous phase, flowing at each stage through a *downcomer* and then across the tray the way a liquid does in a trayed distillation tower. If the heavy phase is dispersed, *upcomers* are used for the light phase. Columns have been built and successfully operated for diameters larger than 4.5 m. Holes from 0.64 to 0.32 cm in diameter and 1.25 to 1.91 cm apart are commonly used. Tray spacings are much closer than in distillation—10 to 15 cm in most applications involving low-interfacial-tension liquids. Plates are usually built without outlet weirs on the downspouts. A variation of the simple sieve column is the Koch Cascade Tower, where perforated plates are set in vertical arrays of complex designs.

If operated in the proper hydrodynamic flow regime, extraction rates in sieve-plate columns are high because the dispersed-phase droplets coalesce and re-form on each stage. This helps destroy concentration gradients, which develop if a droplet passes through the entire column without disturbance. Sieve-plate columns in extraction service are subject to the same limitations as distillation columns: flooding, entrainment, and, to a lesser extent, weeping. Additional problems, such as scum formation at interfaces due to small amounts of impurities, are frequently encountered in all types of extraction devices.

Columns with Mechanically Assisted Agitation

If the surface tension is high, and/or the density difference between the two liquid phases is low, and/or liquid viscosities are high, gravitational forces are inadequate for proper phase dispersal and the creation of turbulence. In that case, some type of mechanical agitation is necessary to increase interfacial area per unit volume and/or decrease mass-transfer resistance. For packed and plate columns, agitation is provided by an oscillating pulse to the liquid, either by mechanical or pneumatic means. Pulsed, perforated-plate columns found considerable application in the nuclear

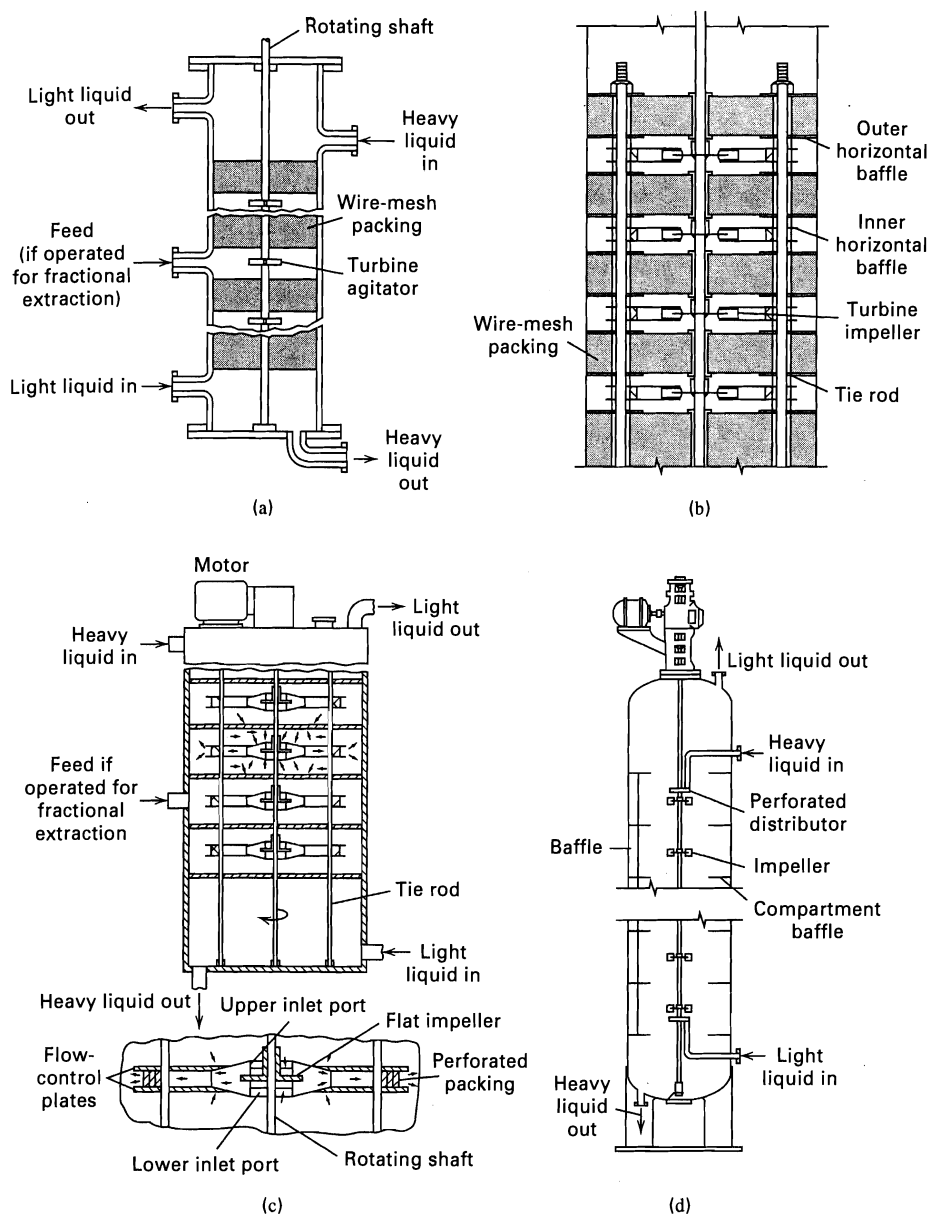


Figure 8.7 Commercial extractors with mechanically assisted agitation: (a) Scheibel column—first design; (b) Scheibel column—second design; (c) Scheibel column—third design; (d) Oldshue-Rushton (Mixco) column; (continued)

industry in the 1950s, but their popularity declined because of mechanical problems and the difficulty of propagating a pulse through a large volume [6]. The most important mechanically agitated columns are those that employ rotating agitators, driven by a shaft that extends axially through the column. The agitators create shear mixing zones, which alternate with settling zones in the column. Differences among the various agitated columns lie primarily in the mixers and settling chambers used. Nine of the more popular arrangements are shown in Figure 8.7. Agitation can also be induced in a column by moving the plates back and forth in a reciprocating motion (Figure 8.7j) or in a novel horizontal contactor (Figure 8.7k). Such devices are also included in Figure 8.7. These devices answer the plea of Fenske, Carlson, and Quiggle [7] in 1947 for equipment that can efficiently provide large numbers of equilibrium stages in a compact device without large numbers of pumps and

motors, and extensive piping. They stated, "Despite . . . advantages of liquid-liquid separational processes, the problems of accumulating twenty or more theoretical stages in a small compact and relatively simple countercurrent operation have not yet been fully solved." Indeed, in 1946 it was considered impractical to design for more than seven theoretical stages, which represented the number of mixer-settler units in the only large-scale, commercial, liquid-extraction process in use at that time.

Perhaps the first mechanically agitated column of importance was the Scheibel column [8] (Figure 8.7a), in which countercurrent liquid phases are contacted at fixed intervals by unbaffled, flat-bladed, turbine-type agitators (Figure 8.3) mounted on a vertical shaft. In the unbaffled separation or calming zones, located between the mixing zones, knitted wire-mesh packing is installed to prevent backmixing between mixing zones and to induce coalescence and

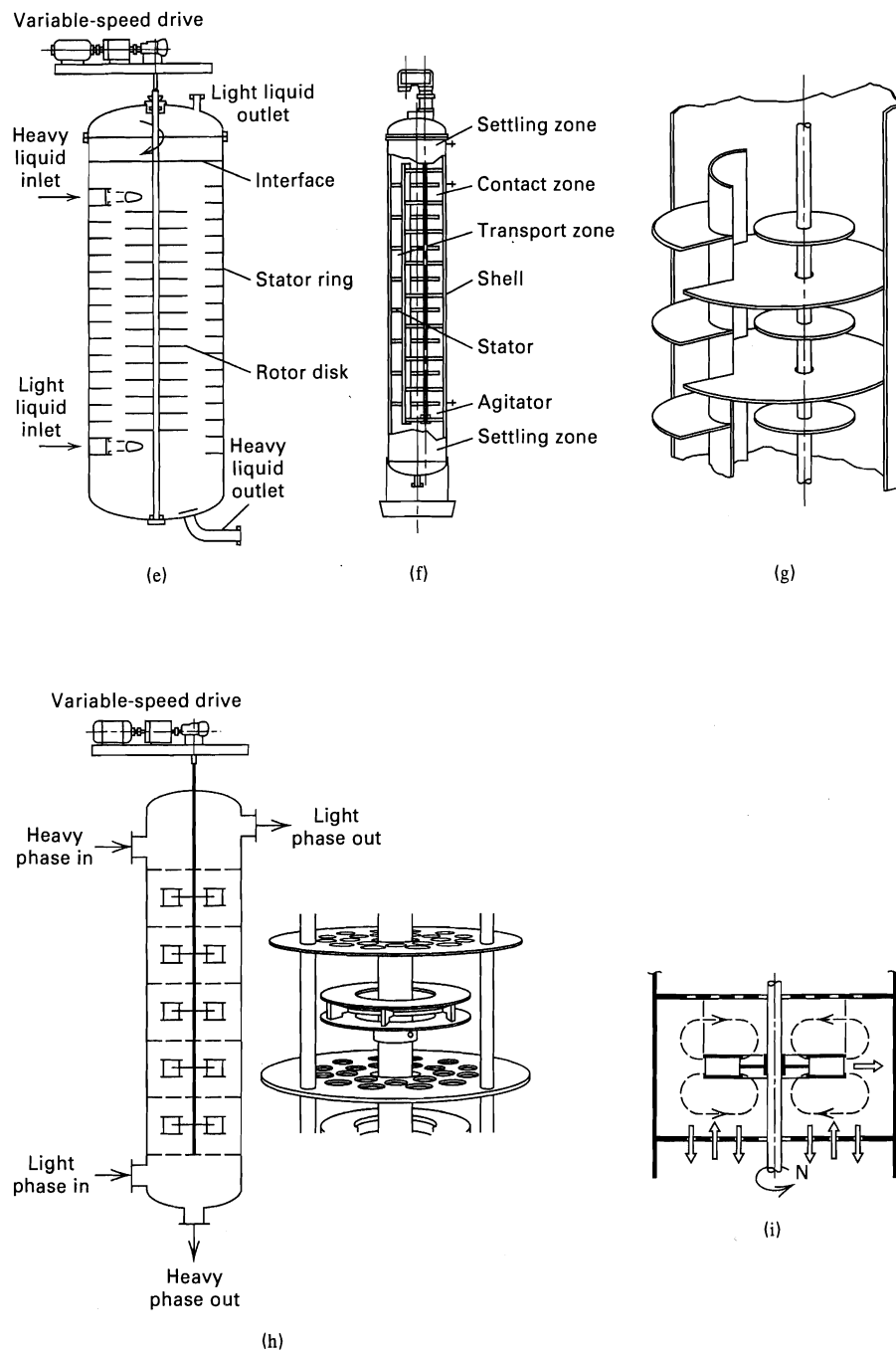


Figure 8.7 (Continued) (e) rotating-disk-contactor (RDC); (f) asymmetric rotating-disk contactor (ARD); (g) section of ARD contactor; (h) Kuhni column; (i) flow pattern in Kuhni column.

settling of drops. The mesh material must be wetted by the dispersed phase. For more economical designs for larger-diameter installations (>1 m), Scheibel [9] (Figure 8.7b) added outer and inner horizontal annular baffles to divert the vertical flow of the phases in the mixing zone and to ensure complete mixing. For systems with high interfacial surface tension and viscosities, the wire mesh is removed. The first two Scheibel designs did not permit removal of the agitator shaft for inspection and maintenance. Instead, the entire internal assembly (called the cartridge) had to be removed. To permit removal of just the agitator assembly shaft, especially for large-diameter columns (e.g., >1.5 m), and allow an access way through the column for any necessary

inspection, cleaning, and repair, Scheibel [10] offered a third design, shown in Figure 8.7c. Here the agitator assembly shaft can be removed because it has a smaller diameter than the opening in the inner baffle.

The Oldshue-Rushton extractor [11] (Figure 8.7d) consists of a column with a series of compartments separated by annular outer stator-ring baffles, each with four vertical baffles attached to the wall. The centrally mounted vertical shaft drives a flat-bladed turbine impeller in each compartment.

A third type of column with rotating agitators that appeared about the same time as the Scheibel and Oldshue-Rushton columns is the rotating-disk contactor (RDC)

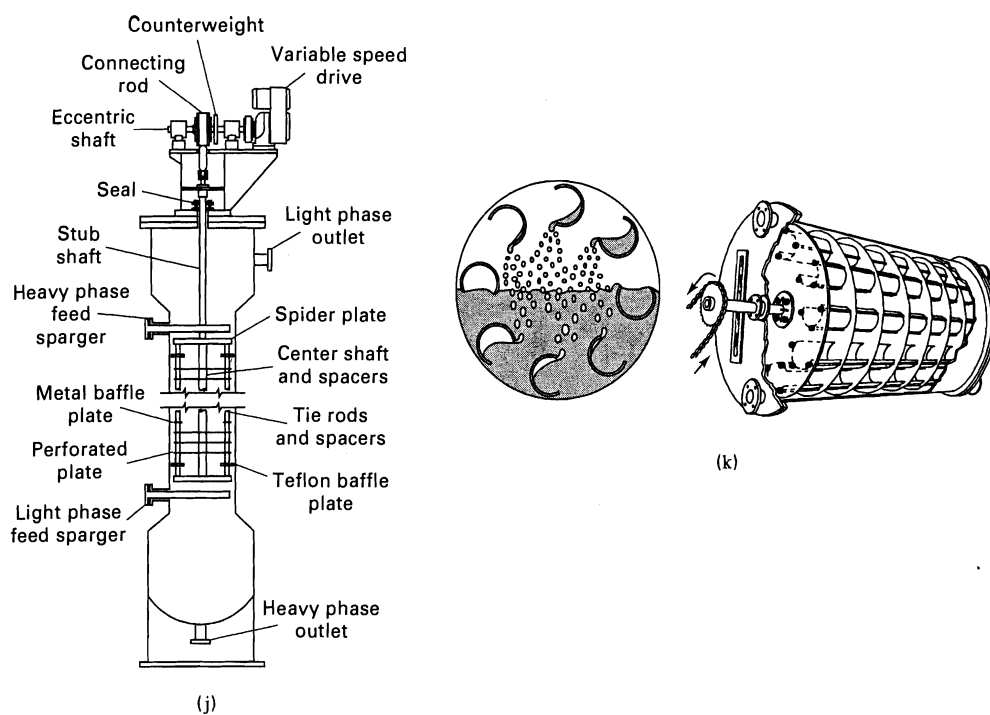


Figure 8.7 (Continued)
(j) Karr reciprocating-plate column (RPC); (k) Graesser raining-bucket (RTL) extractor.

[12, 13] (Figure 8.7e), an example of which is described at the beginning of this chapter and shown in Figure 8.1. On a worldwide basis, it is probably the most extensively used liquid-liquid extraction device, with hundreds of units in use by 1983 [4]. Horizontal disks, mounted on a centrally located rotating shaft, are the agitation elements. Mounted at the column wall are annular stator rings with an opening larger than the agitator-disk diameter. Thus, the agitator assembly shaft is easily removed from the column. Because the rotational speed of the rotor controls the drop size, the rotor speed can be continuously varied over a wide range.

A modification of the RDC concept is the asymmetric rotating-disk contactor (ARD) [14], which has been in industrial use since 1965. As shown in Figure 8.7f, the contactor consists of a column, a baffled stator, and an offset multi-stage agitator fitted with disks. The asymmetric arrangement, shown in more detail in Figure 8.7g, provides contact and transport zones that are separated by a vertical baffle, to which is attached a series of horizontal baffles. Compared to the RDC, this design retains the efficient shearing action, but reduces backmixing because of the separate mixing and settling compartments.

Another extractor based on the Scheibel concept is the Kuhni extraction column [15]. As shown in Figure 8.7h, the column is compartmented by a series of stator disks made of perforated plates. On a centrally positioned shaft is mounted a series of double-entry, radial-flow, shrouded-turbine mixers, which promote, in each compartment, the circulation action shown in Figure 8.7i. For columns of diameter greater than 3 m, three turbine-mixer shafts on parallel axes are normally provided to preserve scale-up.

Three hundred of these extractors were in use, mainly in Europe, by 1983 [4].

Rather than provide agitation by rotating impellers on a vertical shaft, or by pulsing the liquid phases, Karr [16, 17] devised a reciprocating, perforated-plate extractor, also called the Karr column, in which the plates move up and down approximately two times per second with a stroke length of 0.75 inch. As shown in Figure 8.7j, annular baffle plates are provided periodically in the plate stack to minimize axial mixing. The perforated plates use large holes (typically 9/16-in. diameter) and a high hole area (typically 58%). The central shaft, which supports both sets of plates, is reciprocated by a drive mechanism located at the top of the column. A modification of the Karr column is the vibrating-plate extractor (VPE) of Prochazka et al. [18], which uses perforated plates of smaller hole size and smaller percent hole area than the Karr column. The small holes provide passage for the dispersed phase, while one or more large holes on each plate provide passage for the continuous phase. Some VPE columns operate like the Karr column with uniform motion of all plates; others are provided with two shafts to obtain countermotion of alternate plates.

Another novel device for providing agitation is the Graesser raining-bucket contactor (RTL), which was developed in the late 1950s [4], primarily for extraction processes involving liquids of small density difference, low interfacial tension, and a tendency to form emulsions. As shown in Figure 8.7k, a series of disks is mounted inside a shell on a central, horizontal, rotating shaft, with a series of horizontal, C-shaped buckets fitted between and around the periphery of the disks. An annular gap between the disks and the inside periphery of the shell allows countercurrent, longitudinal

flow of the phases. Dispersing action is very gentle, with each phase cascading through the other in opposite directions toward the two-phase interface, which is maintained close to the equatorial position.

A number of industrial centrifugal extractors have been available since 1944, when the Podbielniak (POD) extractor, with its short residence time, was successfully applied to penicillin extraction [19]. In the POD, several concentric sieve trays are arranged around a horizontal axis through which the two liquid phases flow countercurrently. Liquid inlet pressures of 4 to 7 atm are required to overcome pressure drop and centrifugal force. As many as five theoretical stages can be achieved in one unit.

Many of the commercial extractors described above have seen numerous industrial applications. Maximum loadings and sizes for column-type equipment, as given by Reissinger and Schroeter [5, 20] and Lo et al. [4], are listed in Table 8.2. As seen, the Lurgi tower, RDC, and Graesser extractors have been built in very large sizes. Throughputs per unit cross-sectional area are highest for the Karr extractor and lowest for the Graesser extractor.

The selection of an appropriate extractor is based on a large number of factors. Table 8.3 lists the advantages and disadvantages of the various types of extractors. Figure 8.8 shows a selection scheme for commercial extractors. For example, if only a small number of stages is required, a

Table 8.2 Maximum Size and Loading for Commercial Liquid-Liquid Extraction Columns

Column Type	Approximate Maximum Liquid Throughput, $\text{m}^3/\text{m}^2\text{-h}$	Maximum Column Diameter, m
Lurgi tower	30	8.0
Pulsed packed	40	3.0
Pulsed sieve tray	60	3.0
Scheibel	40	3.0
RDC	40	8.0
ARD	25	5.0
Kuhni	50	3.0
Karr	100	1.5
Graesser	<10	7.0

Above data apply to systems of:

1. High interfacial surface tension (30 to 40 dyne/cm).
2. Viscosity of approximately 1 cP.
3. Volumetric phase ratio of 1:1.
4. Phase-density difference of approximately 0.6 g/cm³.

mixer-settler unit might be selected. If more than five theoretical stages, a high throughput, and a large load range ($\text{m}^3/\text{m}^2\text{-h}$) are needed, and floor space is limited, an RDC or ARD contactor should be considered.

Table 8.3 Advantages and Disadvantages of Different Extraction Equipment

Class of Equipment	Advantages	Disadvantages
Mixer-settlers	Good contacting Handles wide flow ratio Low headroom High efficiency Many stages available Reliable scale-up	Large holdup High power costs High investment Large floor space Interstage pumping may be required
Continuous, counterflow contactors (no mechanical drive)	Low initial cost Low operating cost Simplest construction	Limited throughput with small density difference Cannot handle high flow ratio High headroom Sometimes low efficiency Difficult scale-up
Continuous, counterflow contactors (mechanical agitation)	Good dispersion Reasonable cost Many stages possible Relatively easy scale-up	Limited throughput with small density difference Cannot handle emulsifying systems Cannot handle high flow ratio
Centrifugal extractors	Handles low-density difference between phases Low holdup volume Short holdup time Low space requirements Small inventory of solvent	High initial costs High operating cost High maintenance cost Limited number of stages in single unit

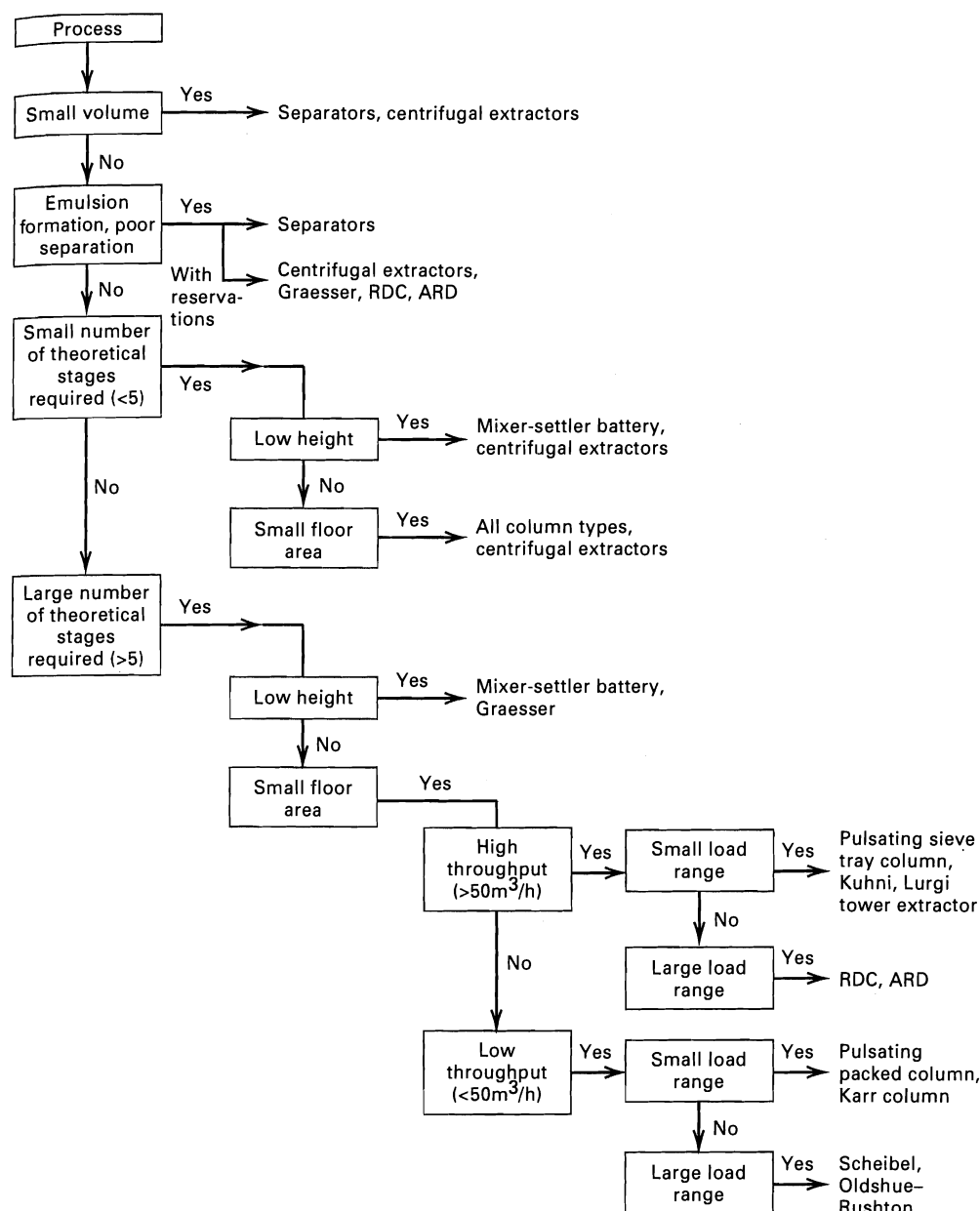


Figure 8.8 Scheme for selecting extractors.

[From K.-H. Reissinger and J. Schroeter, *I. Chem. E. Symp. Ser. No. 54*, 33-48 (1978).]

8.2 GENERAL DESIGN CONSIDERATIONS

The design and analysis of a liquid-liquid extractor involves more factors than for vapor-liquid operations because of complications introduced by the two liquid phases. One of the three different cascade arrangements in Figure 8.9, or a more complex arrangement, must be selected. The single-section cascade of Figure 8.9a is similar to that used for absorption and stripping. It is designed to transfer to the solvent a certain percentage of the solute in the feed. The two-section cascade of Figure 8.9b is similar to distillation. Solvent enters at one end and reflux, derived from the extract, enters at the other end. The feed enters between the two sections. With two sections, depending on solubility considerations, it is sometimes possible to achieve a reasonably sharp separation between components of the feed; if not, a dual-solvent arrangement with two sections, as in Figure 8.9c, with or without reflux at the two ends, may be advantageous.

For the latter configuration, which involves a minimum of four components (two in the feed and two solvents), computer-aided calculations are preferred, as discussed in Chapter 10. Although the configurations in Figure 8.9 are shown with packed sections, any of the extractors discussed in Section 8.1 may be considered. The factors influencing extraction include:

1. Entering feed flow rate, composition, temperature, and pressure
2. Type of stage configuration (one-section or two-section)
3. Desired degree of recovery of one or more solutes for one-section cascades
4. Degree of separation of the feed for two-section cascades
5. Choice of liquid solvent(s)

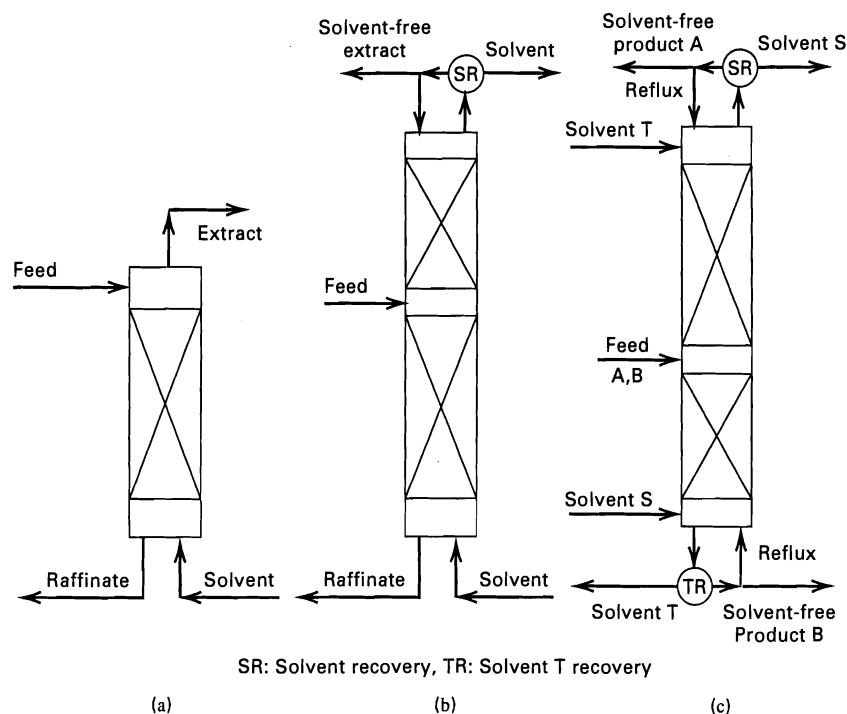


Figure 8.9 Common liquid-liquid extraction cascade configurations: (a) single-section cascade; (b) two-section cascade; (c) dual solvent with two-section cascade.

6. Operating temperature
7. Operating pressure (greater than the bubble point of the system)
8. Minimum-solvent flow rate and actual solvent flow rate as a multiple of the minimum rate for one-section cascades or reflux rate and minimum reflux ratio for two-section cascades
9. Number of equilibrium stages
10. Emulsification and scum-formation tendency
11. Interfacial tension
12. Phase-density difference
13. Type of extractor
14. Extractor size and horsepower requirement

The ideal solvent has:

1. A high selectivity for the solute relative to the carrier, so as to minimize the need to recover carrier from the solvent
2. A high capacity for dissolving the solute, so as to minimize the solvent-to-feed ratio
3. A minimal solubility in the carrier
4. A volatility sufficiently different from the solute that recovery of the solvent can be achieved by distillation, but the vapor pressure should not be so high that a high extractor pressure is needed or so low that a high temperature is needed if the solvent is recovered by distillation
5. Stability to maximize the solvent life and minimize the solvent make-up requirement
6. Inertness to permit use of common materials of construction

7. A low viscosity to promote phase separation, minimize pressure drop, and provide a high solute mass-transfer rate
8. Nontoxic and nonflammable characteristics to facilitate its safe use
9. Availability at a relatively low cost
10. A moderate interfacial tension to balance the ease of dispersion and the promotion of phase separation
11. A large difference in density relative to the carrier to achieve a high capacity in the extractor
12. Compatibility with the solute and carrier to avoid contamination
13. A lack of tendency to form a stable rag or scum layer at the phase interface
14. Desirable wetting characteristics with respect to extractor internals

Solvent selection is frequently a compromise among all the properties listed above. However, initial consideration is usually given first to selectivity and environmental concerns, and second to capacity. From (2-20) in Chapter 2, the distribution coefficient for solute A between solvent S and carrier C is given by:

$$(K_A)_D = (x_A)^{II}/(x_A)^I = (\gamma_A)^I/(\gamma_A)^{II} \quad (8-1)$$

where II is the extract phase, rich in S, and I is the raffinate phase, rich in C. Similarly, for the carrier and the solvent, respectively,

$$(K_C)_D = (x_C)^{II}/(x_C)^I = (\gamma_C)^I/(\gamma_C)^{II} \quad (8-2)$$

$$(K_S)_D = (x_S)^{II}/(x_S)^I = (\gamma_S)^I/(\gamma_S)^{II} \quad (8-3)$$

From (2-22), the relative selectivity of the solute with respect to the carrier is obtained by taking the ratio of (8-1)

Table 8.4 Group Interactions for Solvent Selection

Group	Solute	Solvent								
		1	2	3	4	5	6	7	8	9
1	Acid, aromatic OH (phenol)	0	—	—	—	—	0	+	+	+
2	Paraffinic OH (alcohol), water, imide or amide with active H	—	0	+	+	+	+	+	+	+
3	Ketone, aromatic nitrate, tertiary amine, pyridine, sulfone, trialkyl phosphate, or phosphine oxide	—	+	0	+	+	—	0	+	+
4	Ester, aldehyde, carbonate, phosphate, nitrite or nitrate, amide without active H; intramolecular bonding, e.g., <i>o</i> -nitrophenol	—	+	+	0	+	—	+	+	+
5	Ether, oxide, sulfide, sulfoxide, primary and secondary amine or imine	—	+	+	+	0	—	0	+	+
6	Multihaloparaffin with active H	0	+	—	—	—	0	0	+	0
7	Aromatic, halogenated aromatic, olefin	+	+	0	+	0	0	0	0	0
8	Paraffin	+	+	+	+	+	+	0	0	0
9	Monohaloparaffin or olefin	+	+	+	+	+	0	0	+	0

(+) Plus sign means that compounds in the column group tend to raise activity coefficients of compounds in the row group.

(—) Minus sign means a lowering of activity coefficients.

(0) Zero means no effect.

Choose a solvent that lowers the activity coefficient.

Source: Cusack, R.W., P. Fremaux, and D. Glate, *Chem Eng.*, 98(2), 66–76 (1991).

to (8-2), giving

$$\beta_{AC} = (K_A)_D / (K_C)_D = \frac{(x_A)^{II} / (x_A)^I}{(x_C)^{II} / (x_C)^I} = \frac{(\gamma_A)^I / (\gamma_A)^{II}}{(\gamma_C)^I / (\gamma_C)^{II}} \quad (8-4)$$

For high selectivity, the value of β_{AC} should be high, that is, at equilibrium there should be a high concentration of A and a low concentration of C in the solvent. A first estimate of β_{AC} is made from available values or predictions of the activity coefficients $(\gamma_A)^I$, $(\gamma_A)^{II}$, and $(\gamma_C)^{II}$, at infinite dilution where $(\gamma_C)^I = 1$, or by using liquid–liquid equilibrium data for the lowest tie line on a triangular diagram of the type discussed in Chapter 4. If A and C form a nearly ideal solution, the value of $(\gamma_A)^I$ in (8-4) can also be taken as 1.

For high solvent capacity, the value of $(K_A)_D$ should be high. From (8-2) it is seen that this is difficult to achieve if A and C form nearly ideal solutions, such that $(\gamma_A)^I = 1.0$, unless A and S have a great affinity for each other, which would result in a negative deviation from Raoult's law to give $(\gamma_A)^{II} < 1$. Unfortunately, such systems are rare.

For ease in solvent recovery, $(K_S)_D$ should be as large as possible and $(K_C)_D$ as small as possible to minimize the presence of solvent in the raffinate and carrier in the extract. This will generally be the case if activity coefficients $(\gamma_S)^I$ and $(\gamma_C)^{II}$ at infinite dilution are large.

If a water-rich feed is to be separated, it is common to select an organic solvent; for an organic-rich feed, an aqueous solvent is often selected. In either case, it is desirable

to select a solvent that lowers the activity coefficient of the solute. Consideration of molecule group interactions can help narrow the search for such a solvent before activity coefficients are estimated or liquid–liquid equilibrium data are sought. A table of interactions for solvent-screening purposes, as given by Cusack et al. [21], based on a modification of the work of Robbins [22], is shown as Table 8.4, where the solute and solvent each belong to any of nine different chemical groups. In this table, a minus (—) sign for a given solute–solvent pair means that the solvent will desirably lower the value of the activity coefficient of the solute relative to its value in the feed solution. For example, suppose it is desired to extract acetone from water. Acetone, the solute, is a ketone. Thus, in Table 8.4, group 3 applies for the solute, and desirable solvents are of the type given in groups 1 and 6. In particular, trichloroethane, a group 6 compound, is known to be a highly selective solvent with high capacity for acetone over a wide range of feed compositions. However, if the compound is environmentally objectionable, it must be rejected. A more sophisticated solvent-selection method, based on the UNIFAC group-contribution method for estimating activity coefficients and utilizing a computer-aided constrained optimization approach, has been developed by Naser and Fournier [23].

In Chapter 4, ternary diagrams were introduced for representing liquid–liquid equilibrium data for three-component systems at constant temperature. Such diagrams are available for a large number of systems, as discussed by

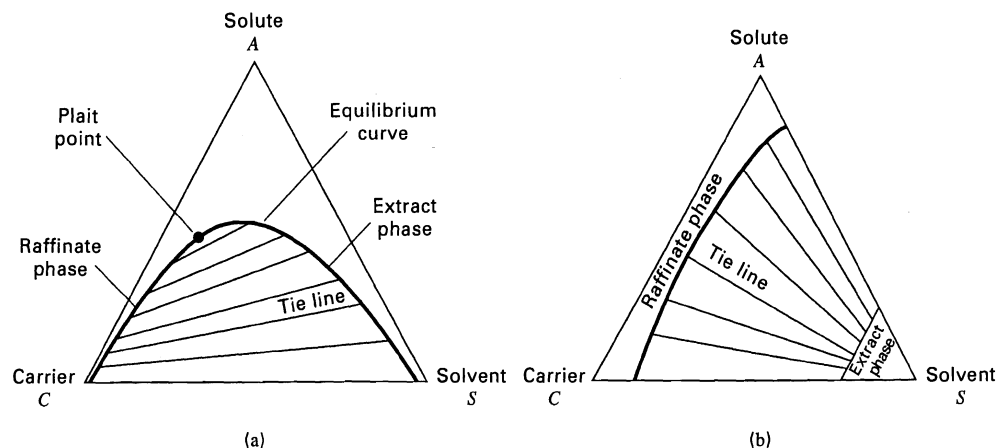


Figure 8.10 Most common classes of ternary systems: (a) type I, one immiscible pair; (b) type II, two immiscible pairs.

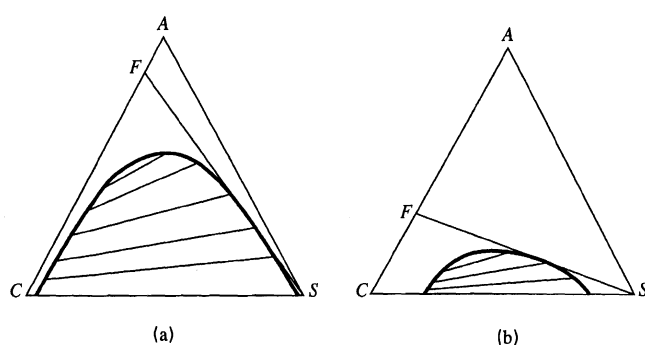


Figure 8.11 Effect of solubility on range of feed composition that can be extracted.

Humphrey et al. [6]. For liquid-liquid extraction with ternary systems, the most common diagram is Type I, shown in Figure 8.10a; much less common is Type II, shown in Figure 8.10b. Examples of Type II systems are (1) *n*-heptane/aniline/methyl cyclohexane, (2) styrene/ethylbenzene/diethylene glycol, and (3) chlorobenzene/water/methylethyl ketone. For Type I, the solute and solvent are miscible in all proportions, while in Type II they are not. For Type I systems, the greater the two-phase region on line $\overline{CS}$, the greater will be the immiscibility of carrier and solvent. The closer the top of the two-phase region is to apex A, the greater will be the range of feed composition, along line $\overline{AC}$, that can be separated with solvent S. In Figure 8.11, it is possible to separate feed solutions only in the composition range from C to F because, regardless of the amount of solvent added, two liquid phases are not formed in the feed composition range of $\overline{FA}$ (i.e., $\overline{FS}$ does not pass through the two-phase region).

The system in Figure 8.11a has a wider range of feed composition than the system in Figure 8.11b. For Type II systems, a high degree of insolubility of S in C and C in S will produce a desirable high relative selectivity, but at the expense of solvent capacity. Thus, solvents that result in Type I systems are more desirable.

Whether a ternary system is of Type I or Type II often depends on the temperature. For example, data of Darwent and Winkler [24] for the ternary system *n*-hexane (H)/methylcyclopentane (M)/aniline (A) for temperatures of 25, 34.5, and 45°C are shown in Figure 8.12. At the lowest temperature, 25°C, we have a Type II system because both H and M are only partially miscible in the aniline solvent. As the temperature increases, the solubility of M in aniline increases more rapidly than the solubility of H in aniline until at 34.5°C, the critical solution temperature for M in aniline is reached. At this temperature, the system is at the border of Type II and Type I. At 45°C, the system is clearly of type I, with aniline more selective for M than H. Type I systems have a plait point (P in Figure 8.10a); type II systems do not.

Except in the near-critical region, pressure has little if any effect on liquid-phase activity coefficients and, therefore, on liquid-liquid equilibrium. It is only necessary to select an operating pressure of at least ambient, and greater than the bubble-point pressure of the two-liquid-phase mixture at any location in the extractor. Most extractors operate at near-ambient temperature. If feed and solvent enter the extractor at the same temperature, the operation will be nearly isothermal because the only thermal effect is the heat of mixing, which is usually small.

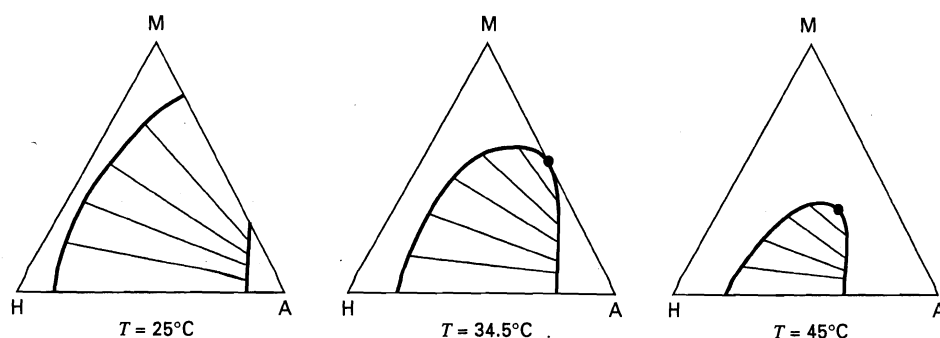


Figure 8.12 Effect of temperature on solubility for the system *n*-hexane (H)/methylcyclopentane (M)/aniline (A).

Laboratory or pilot-plant work, using actual or expected plant feed and solvent, is almost always necessary to ascertain dispersion and coalescence properties of the liquid-liquid system. Although rapid coalescence of drops is desirable, this reduces interfacial area, leading to reduced mass-transfer rates. Thus, compromises are necessary. Coalescence is enhanced when the solvent phase is continuous and mass transfer of solute is from the droplets. This phenomenon, called the *Marangoni effect*, is due to a lowering of interfacial tension by a significant presence of the solute in the interfacial film. When the solvent is the dispersed phase, the interfacial film is depleted of solute, causing an increase in interfacial tension and inhibition of coalescence.

For a given (1) feed liquid, (2) degree of solute extraction, (3) operating pressure and temperature, and (4) choice of solvent for a single-section cascade, a minimum solvent-to-feed flow-rate ratio exists that corresponds to an infinite number of countercurrent, equilibrium contacts. As with absorption and stripping, a trade-off then exists between the number of equilibrium stages and the solvent-to-feed flow-rate ratio. For a two-section cascade, as for distillation, the trade-off involves the reflux ratio and the number of stages. Algebraic methods, similar to those for absorption and stripping described in Chapter 6, for computing the minimum ratios and the trade-off are rapid, but are useful only for very dilute solutions, where values of the solute activity coefficients are essentially those at infinite dilution. When the carrier and the solvent are mutually insoluble, the algebraic method of Sections 5.3 and 5.4 can be used. For more general applications, use of the graphical methods described in this chapter is preferred for ternary systems. Computer-aided methods discussed in Chapter 10 are necessary for higher-order multicomponent systems.

8.3 HUNTER-NASH GRAPHICAL EQUILIBRIUM-STAGE METHOD

Stagewise extraction calculations for ternary systems of Type I and Type II (Figure 8.10) are most conveniently carried out with equilibrium diagrams [25]. In this section, procedures are developed and illustrated, using mainly triangular diagrams. The use of other diagrams is covered in the next section.

Consider a countercurrent-flow, N -equilibrium-stage contactor for liquid-liquid extraction of a ternary system operating under isothermal, continuous, steady-state flow conditions at a pressure sufficient to prevent vaporization, as

shown in Figure 8.13. Stages are numbered from the feed end. Thus, the final extract is E_1 and the final raffinate is R_N . Equilibrium is assumed to be achieved at each stage, so that for any stage, n , the extract, E_n , and the raffinate, R_n , are in equilibrium with all three components. Mass transfer of all components occurs at each stage. The feed, F , contains the carrier, C , and the solute, A , and can also contain solvent, S , up to the solubility limit. The entering solvent, S , can contain C and A , but preferably contains little of either, if any. Because most liquid-liquid equilibrium data are given in mass rather than mole concentrations, let:

F = mass flow rate of feed to the cascade

S = mass flow rate of solvent to the cascade

E_n = mass flow rate of extract leaving stage n

R_n = mass flow rate of raffinate leaving stage n

$(y_i)_n$ = mass fraction of species i in extract leaving stage n

$(x_i)_n$ = mass fraction of species i in raffinate leaving stage n

Although Figure 8.13 might seem to imply that the extract is the light phase, either phase can be the light phase. Phase equilibrium may be represented, as discussed in Chapter 4, on an equilateral-triangle diagram, as proposed by Hunter and Nash [26], or on a right-triangle diagram as proposed by Kinney [27]. Assume, for illustration purposes, that the ternary system is A (solute), C (carrier), and S (solvent) at a particular temperature, T , such that the liquid-liquid equilibrium data are represented on the equilateral-triangle diagram of Figure 8.14, where the bold line is the equilibrium curve and the dashed lines are the tie lines that connect equilibrium phases of the equilibrium curve (also called the *binodal curve* because the plait point separates the curve into an extract and a raffinate). The equilibrium tie lines slope upward from the C side of the diagram toward the S side. Therefore, at equilibrium A has a concentration higher in S than in C . Thus, S is an effective solvent for extracting A from a mixture with C . On the other hand, because the tie lines slope downward from the S side toward the C side, C is not an effective solvent for extracting A from S .

Some ternary systems, such as isopropanol-water-benzene, exhibit a phenomenon called *solutropy*, wherein moving from the plait point, the tie lines first slope in one direction. However, the slope diminishes until an intermediate tie line becomes horizontal. Below that tie line, the remaining tie lines slope in the other direction. Sometimes, the solutropy phenomenon disappears if mole-fraction coordinates, rather than mass-fraction coordinates, are used.

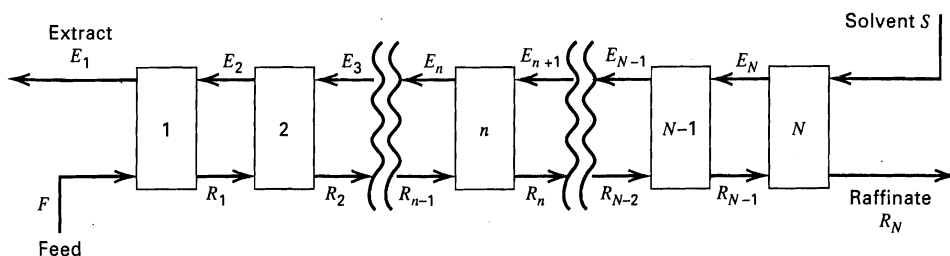


Figure 8.13 Countercurrent-flow, N -equilibrium-stage liquid-liquid extraction cascade.

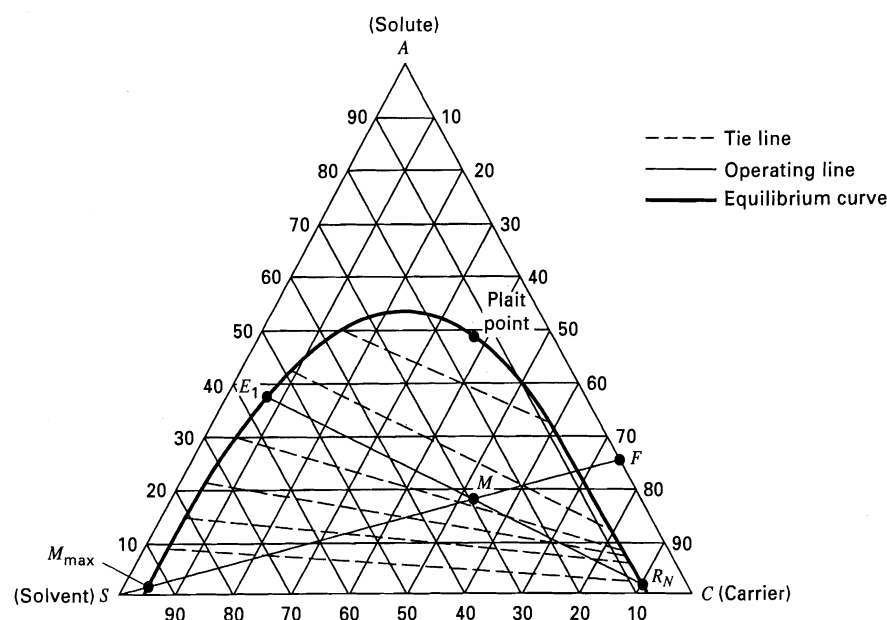


Figure 8.14 Construction 1: Location of product points.

Number of Equilibrium Stages

From the degrees-of-freedom discussion in Chapters 4 and 5, the following sets of specifications, for the cascade of Figure 8.13 with a ternary system, can be made, where all sets include the specification of F , $(x_i)_F$, $(y_i)_S$, and T :

- | | |
|--|------------------------------|
| Set 1. S and $(x_i)_{R_N}$ | Set 4. N and $(x_i)_{R_N}$ |
| Set 2. S and $(y_i)_{E_1}$ | Set 5. N and $(y_i)_{E_1}$ |
| Set 3. $(x_i)_{R_N}$ and $(y_i)_{E_1}$ | Set 6. S and N |

where values of $(x_i)_{R_N}$ and $(y_i)_{E_1}$ and all exiting phases must lie on the equilibrium curve.

Calculations for sets 1 to 3, which involve the determination of N , are made directly using the triangular diagram. Sets 4 to 6, which involve a specified N , require an iterative procedure. We first consider the calculational procedure for Set 1, with the procedures for Sets 2 and 3 being just minor modifications. The procedure, sometimes referred to as the *Hunter-Nash method* [26], involves three kinds of construction on the triangular diagram and is somewhat more difficult than the McCabe-Thiele staircase-type construction for distillation. Although the procedure is illustrated only for the Type I system, the principles are readily extended to a Type II system. The constructions are shown in Figure 8.14, where A is the solute, C is the carrier, and S is the solvent. On the binodal curve, all extract compositions lie on the equilibrium curve to the left of the plait point, while all raffinate compositions lie to the right. To determine the number of stages, given the flow rates and compositions of the feed and solvent, and the desired raffinate composition, the constructions are as follows:

Construction 1 (Product Points)

First, on Figure 8.14, we locate mixing point M , which represents the overall composition of the combination of feed, F , and entering solvent, S . Assume the following feed and

solvent specifications, as plotted in Figure 8.14, where pure S is the solvent:

Feed	Solvent
$F = 250$ kg	$S = 100$ kg
$(x_A)_F = 0.24$	$(x_A)_S = 0.00$
$(x_C)_F = 0.76$	$(x_C)_S = 0.00$
$(x_S)_F = 0.00$	$(x_S)_S = 1.00$

The overall composition M for combined F and S is obtained from the following material balances:

$$\begin{aligned}
 M &= F + S = 250 + 100 = 350 \text{ kg} \\
 (x_A)_M M &= (x_A)_F F + (x_A)_S S \\
 &= 0.24(250) + 0(100) = 60 \text{ kg} \\
 (x_A)_M &= 60/350 = 0.171 \\
 (x_C)_M M &= (x_C)_F F + (x_C)_S S \\
 &= 0.76(250) + 0(100) = 190 \text{ kg} \\
 (x_C)_M &= 190/350 = 0.543 \\
 (x_S)_M M &= (x_S)_F F + (x_S)_S S \\
 &= 0(250) + 1(100) = 100 \text{ kg} \\
 (x_S)_M &= 100/350 = 0.286
 \end{aligned}$$

From any two of these $(x_i)_M$ values, point M is located, as shown, in Figure 8.14. Based on the properties of the triangular diagram, presented in Chapter 4, point M must be located somewhere on the straight line connecting F and S . Therefore, M can be located knowing just one value of $(x_i)_M$, say $(x_S)_M$. Also, the ratio S/F is given by the inverse-lever-arm rule as

$$S/F = \overline{MF}/\overline{MS} = 100/250 = 0.400$$

or

$$S/M = \overline{MF}/\overline{SF} = 100/350 = 0.286$$

Thus, point M can be located by two points or by measurement, employing either of these ratios.

With point M located, the composition of extract, E_1 , exiting from a countercurrent, multistage extractor, can be determined from overall material balances:

$$M = R_N + E_1 = 350 \text{ kg}$$

$$(x_A)_M M = 60 = (x_A)_{R_N} R_N + (x_A)_{E_1} E_1$$

$$(x_C)_M M = 190 = (x_C)_{R_N} R_N + (x_C)_{E_1} E_1$$

$$(x_S)_M M = 100 = (x_S)_{R_N} R_N + (x_S)_{E_1} E_1$$

Because the raffinate, R_N , is assumed to be at equilibrium, its composition must lie on the equilibrium curve of Figure 8.14. Therefore, if we specify the value $(x_A)_{R_N} = 0.025$, we can locate the point R_N , and the values of $(x_C)_{R_N}$ and $(x_S)_{R_N}$ can be read from Figure 8.14. A straight line drawn from R_N through M will locate E_1 at the intersection of the equilibrium curve, from which, in Figure 8.14, the composition of E_1 can be read. Values of the flow rates R_N and E_1 can then be determined from the overall material balances above or from Figure 8.14 by the inverse-lever-arm rule:

$$E_1/M = \overline{MR_N}/\overline{E_1R_N}$$

$$R_N/M = \overline{ME_1}/\overline{E_1R_N}$$

with $M = 350 \text{ kg}$ for this illustration. By either method, we find:

Raffinate Product	Extract Product
$R_N = 198.6 \text{ kg}$	$E_1 = 151.4 \text{ kg}$
$(x_A)_{R_N} = 0.025$	$(x_A)_{E_1} = 0.364$
$(x_C)_{R_N} = 0.90$	$(x_C)_{E_1} = 0.075$
$(x_S)_{R_N} = 0.075$	$(x_S)_{E_1} = 0.561$

Also included in Figure 8.14 is the point $M_{\max}$, which lies on the equilibrium curve along the straight line connecting F to S . $M_{\max}$ corresponds to the maximum possible solvent addition. If more solvent were added, two liquid phases could not exist.

Construction 2 (Operating Point and Lines)

In Chapters 6 and 7, we learned that an operating line is the locus of passing streams in a cascade. Referring to Figure 8.13, material balances around groups of stages from the

feed end are

$$F - E_1 = \cdots = R_{n-1} - E_n = \cdots = R_N - S = P \quad (8-5)$$

Because the passing streams are differenced, P defines a *difference point* rather than a *mixing point*. From the same geometric considerations as apply to a mixing point, a difference point also lies on a straight line drawn through the points involved. However, while the mixing point always lies inside the triangular diagram and between the two end points, the difference point usually lies outside the triangular diagram along an extrapolation of the line through two points such as F and E_1 , R_N and S , and so on.

To locate the difference point, two straight lines are drawn, respectively, through the point pairs (E_1, F) and (S, R_N) , which are established by Construction 1 and shown in Figure 8.15. These lines are extrapolated until they intersect at difference point P . Figure 8.15 shows these lines and the difference point, P . From (8-5), straight lines drawn through points on the triangular diagram for any other pair of passing streams, such as (E_n, R_{n-1}) , must also pass through point P . Thus, we refer to the difference point as an *operating point*, and the lines drawn through pairs of points for passing streams and extrapolated to point P as *operating lines*.

The difference point has properties similar to those of the mixing point. If $F - E_1 = P$ is rewritten as $F = E_1 + P$, we see that F can be interpreted as the mixing point for P and E_1 . Therefore, by the inverse-lever-arm rule, the length of line $\overline{E_1P}$ relative to the length of the line $\overline{FP}$ is given by

$$\frac{\overline{E_1P}}{\overline{FP}} = \frac{E_1 + P}{E_1} = \frac{F}{E_1} \quad (8-6)$$

Thus, point P can be located, if desired, by measurement with a ruler using either pair of feed-product passing streams.

The operating point, P , lies on the feed or raffinate side of the triangular diagram in the illustration of Figure 8.15. Depending on the relative amounts of feed and solvent and the slope of the tie lines, point P may be located on the solvent or feed side of the diagram, and inside or outside the diagram.

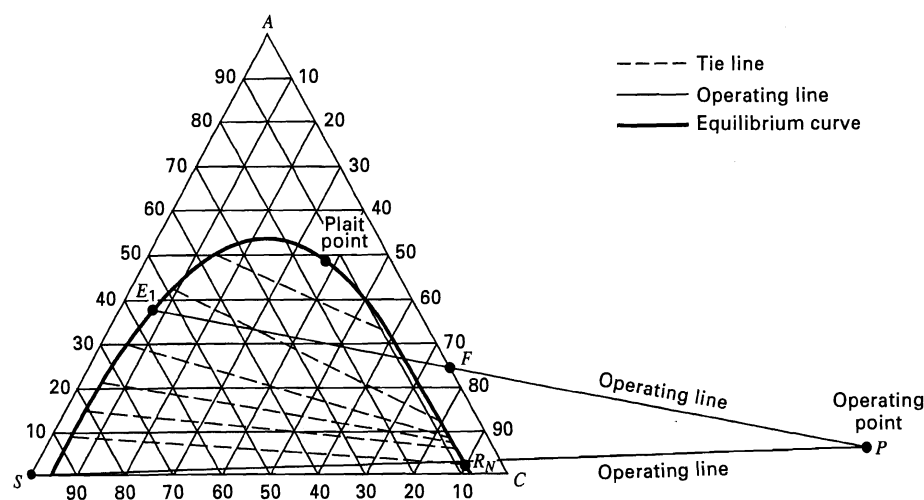


Figure 8.15 Construction 2: Location of operating point.

Construction 3 (Equilibrium Lines)

The third type of construction involves the dashed *tie lines* that connect opposite sides of the equilibrium curve, which is divided into the two sides by the plait point, which for type I diagrams is the point where the two equilibrium phases become one phase. A material balance around any stage n for any of the three components is

$$(x_i)_{n-1}R_{n-1} + (y_i)_{n+1}E_{n+1} = (x_i)_nR_n + (y_i)_nE_n \quad (8-7)$$

Because R_n and E_n are in equilibrium, their composition points are on the triangular diagram at the two ends of a tie line. Typically a diagram will not contain all the tie lines needed. Tie lines may be added by centering them between existing experimental or predicted tie lines, or by using either of two interpolation procedures illustrated in Figure 8.16. In Figure 8.16a, the conjugate line from the plait point to J is determined from four tie lines and the plait point. From tie line $\overline{DE}$, lines $\overline{DG}$ and $\overline{EF}$ are drawn parallel to triangle sides $\overline{CB}$ and $\overline{AC}$, respectively. The intersection at point H gives a second point on the conjugate curve. Subsequent intersections, using the other tie lines, establish additional points from which the conjugate curve is drawn. Then, using the curve, additional tie lines are drawn by reversing the procedure. If it is desired to keep the conjugate curve inside the two liquid-phase region of the triangular diagram, the procedure illustrated in Figure 8.16b is used, where lines are drawn parallel to triangle sides $\overline{AB}$ and $\overline{AC}$.

Stepping off Stages

Equilibrium stages are stepped off on the triangular diagram by alternating the use of tie lines and operating lines, as shown in Figure 8.17, where Constructions 1 and 2 have already been employed to locate the five points F , E , S , R_1 , and P . We start at the feed end from point E_1 . Referring to Figure 8.13, we see that R_1 is in equilibrium with E_1 . Therefore, by Construction 3, R_1 in Figure 8.17 must be at the opposite end of a tie line (shown as a dashed line) connecting to E_1 . From Figure 8.13, R_1 passes E_2 . Therefore, by Construction 2, E_2 must lie at the

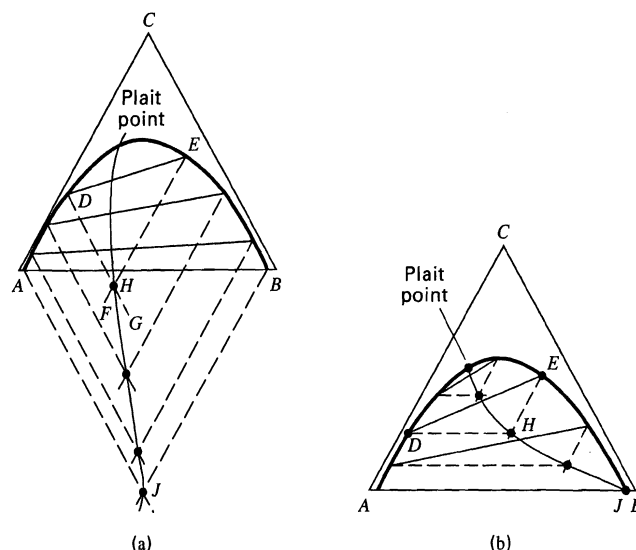


Figure 8.16 Use of conjugate curves to interpolate tie lines: (a) method of *International Critical Tables*, Vol. III, McGraw-Hill, New York, p. 393 (1928); (b) method of T.K. Sherwood, *Absorption and Extraction*, McGraw-Hill, New York, p. 242 (1937). [From R.E. Treybal, *Liquid Extraction*, 2nd ed., McGraw-Hill, New York (1963) with permission.]

intersection of the straight operating line, drawn through points R_1 and P , and back to the extract side of the equilibrium curve. From E_2 , we locate R_2 with a tie line by Construction 3; from R_2 , we locate E_3 by Construction 2. Continuing in this fashion by alternating between equilibrium tie lines and operating lines, we finally reach or pass the specified point R_N . If the latter, a fraction of the last stage is taken. In Figure 8.17 approximately 2.8 equilibrium stages are required, where stages are counted by the number of tie lines used.

Procedures for problem specification sets 2 and 3 are very similar to that for set 1. Sets 4 and 5 can be handled by iteration on assumed values for S and following the above procedure for set 1. Set 6 can also use the procedure of set 1 by iterating on E_1 .

From (8-6), we see that if the ratio F/E_1 approaches a value of 1, the operating point, P , will be located at a large distance

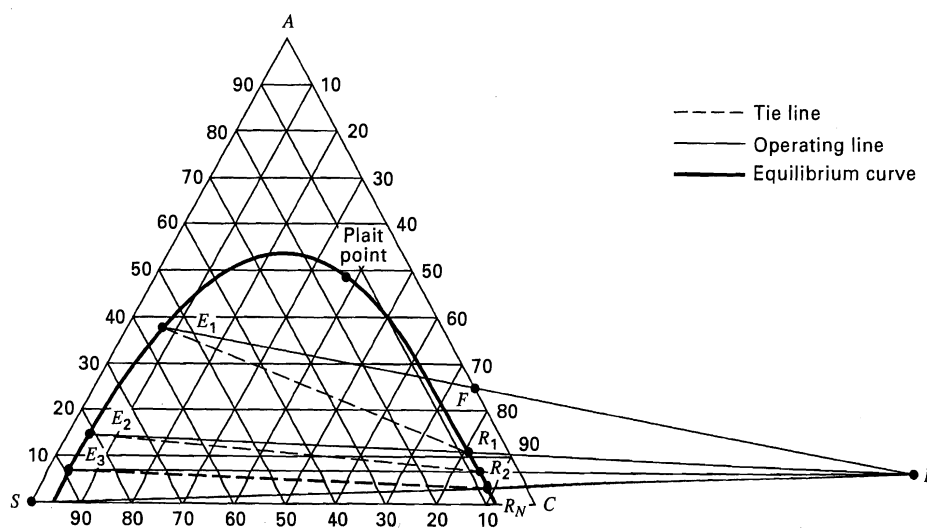


Figure 8.17 Determination of the number of equilibrium stages.

from the triangular diagram. In that case, using an arbitrary rectangular-coordinate system superimposed over the triangular diagram, the coordinates of P can be calculated from (8-6) using the equations for the two straight lines established in Construction 2. Operating lines for intermediate stages can then be located on the triangular diagram so as to pass through P . Details of this procedure are given by Treybal [25].

Minimum and Maximum Solvent-to-Feed Flow-Rate Ratios

The graphical procedure just described for determining the number of equilibrium stages to achieve a desired solute extraction for a given solvent-to-feed flow-rate ratio presupposes that this ratio is greater than the minimum ratio corresponding to the need for an infinite number of stages, but less than the maximum ratio that would prevent the formation of the required second liquid phase. In practice, one usually determines the minimum ratio before solving specification sets 1 or 2. In essence, we must solve set 4 with $N = \infty$, where, as in distillation, absorption, and stripping, the infinity of stages occurs at a pinch point of the equilibrium curve and the operating line(s). In ternary systems, the pinch point occurs when a tie line is coincident with an operating line. The calculation is somewhat involved because the location of the pinch point is not always at the feed end of the cascade. Consider the previous A-C-S system, as shown in Figure 8.18. The points F , S , and R_N are specified, but E_1 is not because the solvent rate has not yet been specified. The operating line $\overline{OL}$ is drawn through the points S and R_N and extended to the left and right of the diagram. This line is the locus of all possible material balances determined by adding S to R_N . Each tie line is then assumed to be a pinch point by extending each tie line until it intersects the line $\overline{OL}$. In this manner, a sequence of intersections, P_1 , P_2 , P_3 , and so on, is found. If these points lie on the raffinate side of the diagram, as in Figure 8.18, the pinch point corresponds to the point $P_{\min}$ located at the greatest distance from R_N . If the triangular diagram does not have a sufficient number of tie lines to determine that point accurately, additional tie lines are introduced by a

method described previously and illustrated in Figure 8.16. If we assume in Figure 8.18 that no other tie line gives a point P_i farther away from R_N than P_1 , then $P_1 = P_{\min}$.

With $P_{\min}$ known, an operating line can be drawn through point F and extended to E_1 at an intersection with the extract side of the equilibrium curve. From the compositions of the four points, S , R_N , F , and E_1 , the mixing point M can be found and the following material balances can then be used to solve for $S_{\min}/F$:

$$F + S_{\min} = R_N + E_1 = M \quad (8-8)$$

$$(x_A)_F F + (x_A)_S S_{\min} = (x_A)_M M \quad (8-9)$$

from which

$$\frac{S_{\min}}{F} = \frac{(x_A)_F - (x_A)_M}{(x_A)_M - (x_A)_S} \quad (8-10)$$

A solvent flow rate greater than $S_{\min}$ must be selected for the extraction to be conducted in a finite number of stages. In Figure 8.18, such a solvent rate results in an operating point P to the right of $P_{\min}$, that is, at a location farther away from R_N . A reasonable value for S might be $1.5 S_{\min}$. From Figure 8.18, we find $(x_A)_M = 0.185$, from which, by (8-10), $S_{\min}/F = 0.30$. In our example of Figure 8.17, we used $S/F = 0.40$, giving $S/S_{\min} = 1.33$.

In Figure 8.18 the tie lines slope downward toward the raffinate side of the diagram. If the tie lines slope downward toward the extract side of the diagram, the above procedure for finding $S_{\min}/F$ must be modified. The sequence of points P_1 , P_2 , P_3 , and so on, is now found on the other side of the diagram. However, the pinch point now corresponds to that point, $P_{\min}$, that is closest to point S and an operating point, P , must be chosen between points $P_{\min}$ and S . For a system that exhibits solutropy, intersections P_1 , P_2 , and so on, will be found on both sides of the diagram. Those on the extract side will determine the minimum solvent-to-feed ratio.

In Figure 8.14, the mixing point M must lie in the two-phase region. As this point is moved along the line $\overline{SF}$ toward S , the ratio S/F increases according to the inverse-lever-arm rule. In the limit, a maximum S/F ratio is reached when $M = M_{\max}$ arrives at the equilibrium curve on the

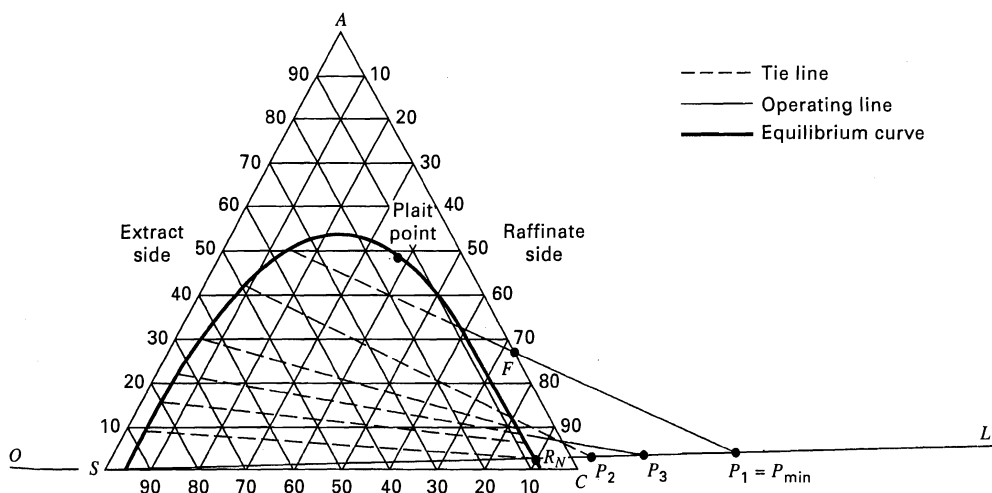


Figure 8.18 Determination of minimum solvent-to-feed ratio.

extract side. At this point, all of the feed is dissolved in the solvent, no raffinate is obtained, and only one stage is required. To avoid this impractical condition, as well as the other extreme of infinite stages, we must select a solvent ratio, S/F , such that $(S/F)_{\min} < (S/F) < (S/F)_{\max}$. In Figure 8.14, the mixing point $M_{\max}$ is located as shown, from which $(S/F)_{\max}$ is determined to be about 16.

EXAMPLE 8.1

Acetone is to be extracted from a feed mixture of 30 wt% acetone (A) and 70 wt% ethyl acetate (C) at 30°C by using pure water (S) as the solvent by the cascade shown at the bottom of Figure 8.19. The final raffinate is to contain 5 wt% acetone on a water-free basis. Determine the minimum and maximum solvent-to-feed ratios and the number of equilibrium stages required for two intermediate S/F ratios. The equilibrium data, which are shown in Figure 8.19 and are taken from Venkataranam and Rao [28], correspond to a type I system, but with tie lines sloping downward toward the extract side of the diagram. Thus, although water is a convenient solvent, it does not have a high capacity, relative to ethyl acetate, for dissolving acetone. The flow diagram of the cascade in Figure 8.19 shows the nomenclature to be used for this example. Also, determine, for the feed, the maximum weight percent acetone that can enter the extractor. This example, as well as Example 8.2 later, are taken largely from an analysis by Sawistowski and Smith [29].

SOLUTION

Point B represents the solvent-free final raffinate. By drawing a straight line from B to S , the intersection with the equilibrium curve on the raffinate side, B' , is the actual raffinate composition leaving stage N .

Minimum S/F . Because the tie lines slope downward toward the extract side of the diagram, we seek the extrapolated tie line that intersects the extrapolated line $\overline{SB}$ closest to S . This tie line, leading to $P_{\min}$, is shown in Figure 8.20. The intersection is not shown because it occurs far to the left of the triangular diagram. Because this tie line is at the feed end of the extractor, the location of the extract composition, $D'_{\min}$, is determined as shown in Figure 8.20. The mixing point, $M_{\min}$, for $(S/F)_{\min}$ is the intersection of lines $\overline{B'D'_{\min}}$ and $\overline{SF}$. By the inverse-lever-arm rule, $(S/F)_{\min} = \overline{FM_{\min}}/\overline{SM_{\min}} = 0.60$.

Maximum S/F . If M in Figure 8.20 is moved along line $\overline{FS}$ toward S , the intersection for $(S/F)_{\max}$ occurs at the point shown on the extract side of the binodal curve. By the inverse-lever-arm rule, using line $\overline{FS}$, $(S/F)_{\max} = \overline{FM_{\max}}/\overline{SM_{\max}} = 12$.

Equilibrium stages for other S/F ratios. First consider $S/F = 1.75$. In Figure 8.19, the composition of the saturated extract D' is obtained from a material balance about the extractor,

$$S + F = D' + B' = M$$

For $S/F = 1.75$, point M can be located such that $\overline{FM}/\overline{MS} = 1.75$. A straight line must also pass through D' , B' , and M . Therefore, D' can be located by extending $\overline{B'M}$ to the extract envelope.

The flow difference point P is located to the left of the triangular diagram. Therefore, $P = S - B' = D' - F$. It is located at the intersection of extensions of lines $\overline{FD'}$ and $\overline{B'S}$.

Stepping off stages poses no problem. Starting at D' , we follow a tie line to L_1 . Then V_2 is located by noting the intersection of the operating line $\overline{L_1P}$ with the phase envelope. Additional stages are stepped off in the same manner by alternating between the tie lines and operating lines. For the sake of clarity, only the first stage is shown; four are required.

For $S/F = 5(S/F)_{\min} = 3.0$, M is determined and the stages are stepped off in a similar manner to give two equilibrium stages.

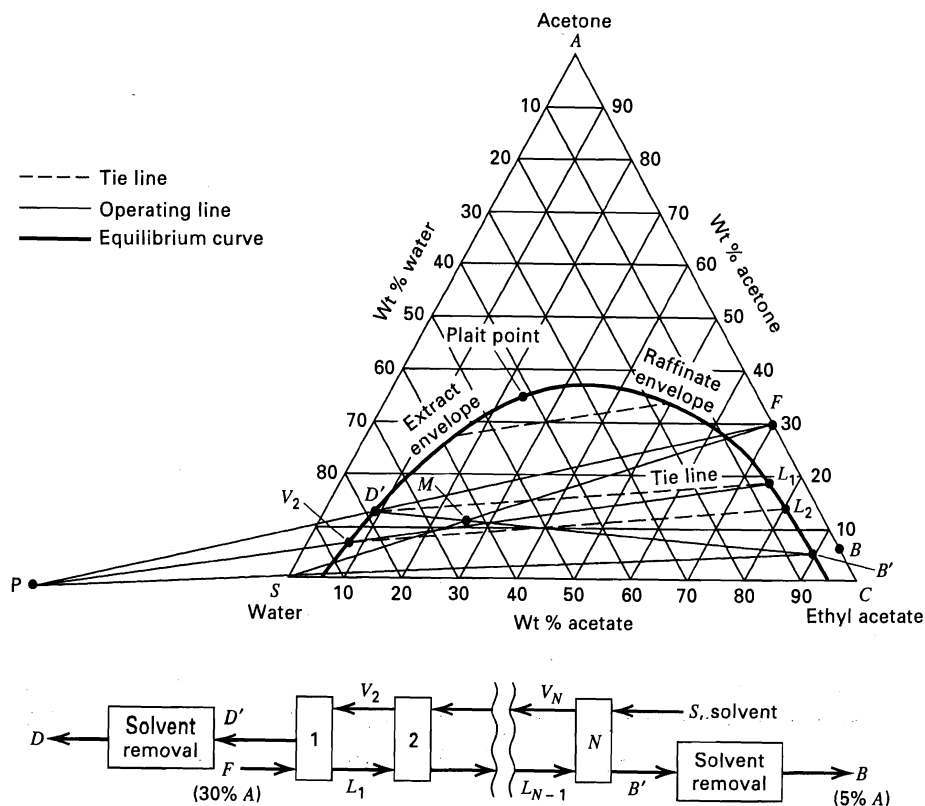
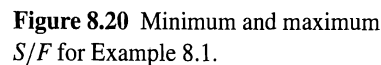


Figure 8.19 Determination of stages for Example 8.1 with $S/F = 1.75$.



S/F (solvent/feed ratio)	0.60	1.75	3	12
N (equilibrium stages)	∞	4	2	1
x_D (wt% acetone, solvent free)	64	62	50	30

be extracted because two liquid phases in equilibrium will not form (no phase splitting). This feed composition is determined by extending a line from S , tangent to the equilibrium curve, until it intersects $\overline{AC}$. This is shown as point D in Figure 8.20. The feed composition is 64 wt% acetone. Feed mixtures with a higher acetone content cannot be extracted with water.

As discussed in Chapter 4, diagrams other than the equilateral-triangle diagram are used for calculations involving ternary liquid-liquid systems. Ternary, countercurrent extraction calculations can also be made on a right-triangle diagram as shown by Kinney [27]; no new principles are involved. The disadvantage is that mass-percent compositions of only two of the components are plotted; the third being determined, when needed, by difference from 100%. The advantage of right-triangle diagrams is that ordinary rectangular-coordinates graph paper can be used and either one of the coordinates can be expanded, if necessary, to increase the accuracy of the constructions.

Consider the right-triangle diagram in Figure 8.22 for the A-C-S system of Figure 8.14. The compositions of S (the solvent) and A (the solute) are plotted in weight (mass) fractions, x_i . For example, point M represents a liquid mixture of overall composition ($x_A = 0.43$, $x_S = 0.28$). By difference, x_C , the carrier, which is not shown on the diagram, is $1 - 0.43 - 0.28 = 0.29$. Although lines of constant x_C

Because point M falls within the phase envelope, the mixture separates into two liquid phases, whose compositions are given by points A' and A'' at the ends of the tie line that passes through point M . In this case, the extract at A'' is richer in the solute (A) and the solvent (S) than the raffinate at A' .

Figure 8.23 represents the portion of an n -stage, countercurrent-flow cascade, where x and y are weight fractions of solute, A, in the raffinate and extract, respectively, and L and V are total amounts of raffinate and extract, respectively. The feed to stage N is $L_{N+1} = 180$ kg of 35 wt% A in a saturated mixture with C and S ($x_{N+1} = 0.35$), and

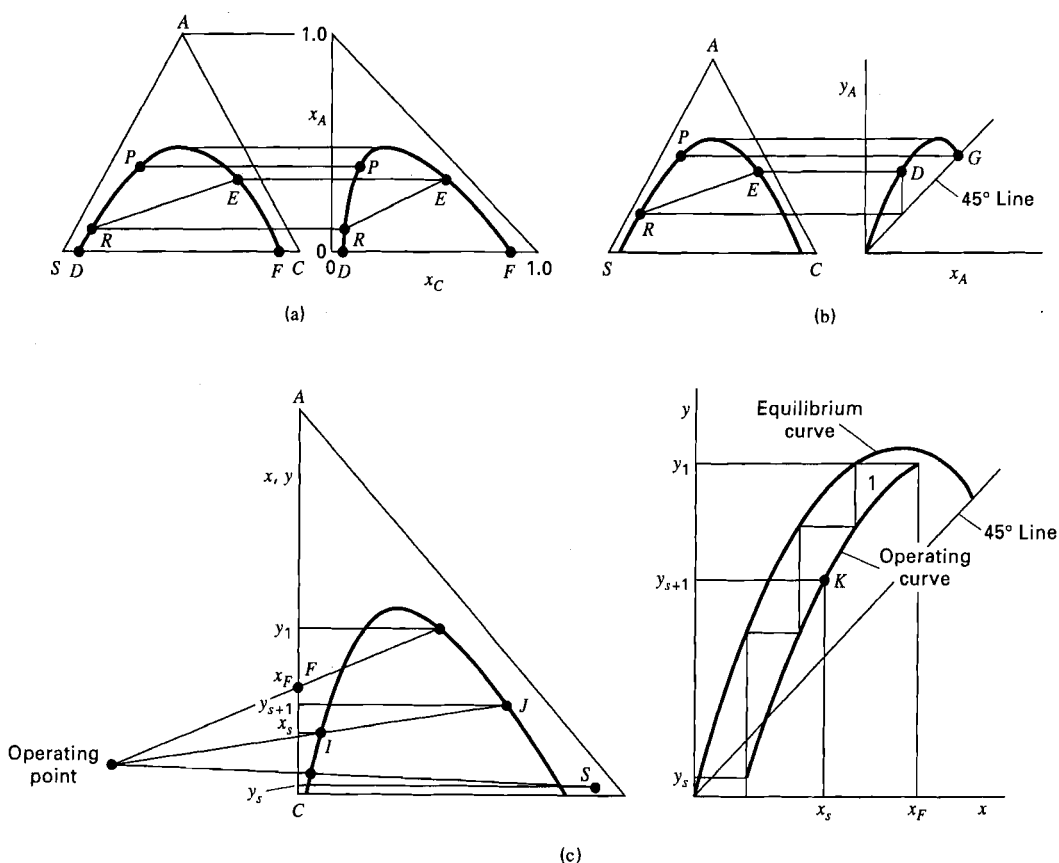


Figure 8.21 Development of other coordinate systems from the equilateral-triangle diagram: (a) to right-triangle diagram; (b) to auxiliary distribution curve. (c) Location of operating point on auxiliary McCabe-Thiele diagram. [From R.E. Treybal, *Liquid Extraction*, 2nd ed., McGraw-Hill, New York (1963) with permission.]

the solvent to stage 1 is $V_W = 100$ kg of pure S ($y_W = 0.0$). Thus, the solvent-to-feed ratio is $100/180 = 0.556$. These two points are shown on the right-triangle diagram of Figure 8.24. The mixing point for L_{N+1} and V_W is shown as point M_1 , as determined by the inverse lever-arm rule. Suppose

that the final raffinate, L_W , leaving stage 1 is to contain 0.05 weight-fraction glycol ($x_W = 0.05$). By an overall balance,

$$M_1 = V_W + L_{N+1} = V_N + L_W \quad (8-11)$$

Applying the mixing rule, since V_W , L_{N+1} , and M_1 lie on a straight line, V_N , L_W , and M_1 must also lie on a straight line. Furthermore, because V_N leaves stage N at equilibrium and L_W leaves stage 1 at equilibrium, these two streams must lie on the extract and raffinate sides, respectively, of the equilibrium curve. The resulting points are shown in Figure 8.24, where it is seen that the weight fraction of glycol in the final extract is $y_N = 0.34$.

Figures 8.23b and 8.23c, and 8.24 include two additional cases of solvent-to-feed ratio, each with the same compositions for the solvent and the feed and the same

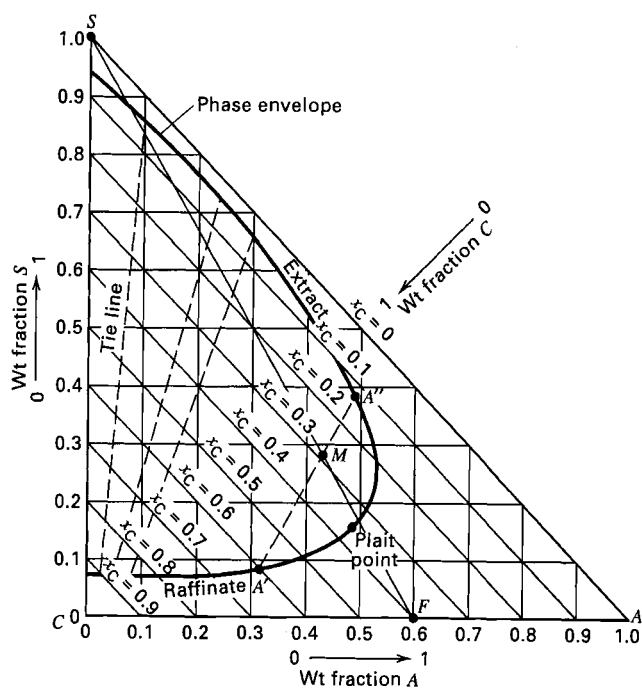


Figure 8.22 Right-triangle diagram for the ternary system of Figure 8.14.

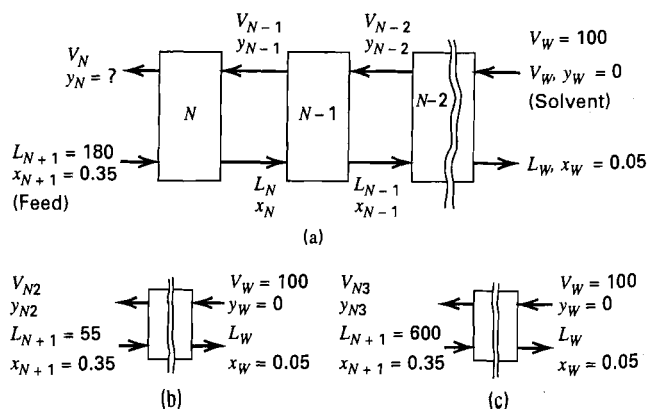


Figure 8.23 Multistage countercurrent contactors.

If an operating line is added to the equilibrium curve in Figure 8.21b, a staircase construction of the type used in the McCabe-Thiele method of Chapter 7 can rapidly determine the number of equilibrium stages. However, unlike distillation, where the operating line is straight because of the assumption of constant molar overflow, the operating line for liquid-liquid extraction in a ternary system will always be curved except in the low-solute-concentration region. Fortunately, the curved operating line is quite readily drawn using the following technique of Varteressian and Fenske [30]. In Figure 8.19 for the equilateral-triangle diagram, or in Figure 8.24 for the right-triangle diagram, the intersections of the equilibrium curve with a line drawn through a difference (operating) point represent the compositions of the passing streams. Thus, for each such operating line on the triangular diagram, one point of the operating line for the y - x plot is determined. The operating lines passing through the difference point can be drawn at random; they need not coincide with passing streams of actual equilibrium-stage operating lines. Usually five or six such fictitious operating-line intersections, covering the expected range of compositions in the extraction cascade, are sufficient to establish the curved operating line in the y - x plot. For example, in Figure 8.21c, the arbitrary operating line that intersects the equilibrium curve at I and J in the right-triangle diagram becomes a point K on

the operating line of the y - x diagram. The y - x plot of Figure 8.25 for the A-C-S system includes an operating line established in this manner, based on the data of Figure 8.24, but with a solvent-to-feed ratio of 0.208, that is, $V_W = 100$, $L_{N+1} = 480$ (25% greater than the minimum ratio of 0.167). The stages are stepped off in the McCabe-Thiele manner starting from the feed end. The result is seen to be almost exactly three equilibrium stages.

Extract and Raffinate Reflux

The simple, single-section, countercurrent, equilibrium-stage extraction cascade shown in Figure 8.13 can be refluxed, as in Figure 8.26a, to resemble distillation. In Figure 8.26a, L is used for raffinate flows, V is used for extract flows, and stages are numbered from the solvent end of the process. Extract reflux, L_R , is provided by sending the extract, V_N , to a solvent-recovery step, which removes most of the solvent, to give a solute-rich solution, $L_R + D$, which is divided into extract reflux, L_R , which is returned to stage N , and solute product, D . At the other end of the cascade, a portion, B , of the raffinate, L_1 , is withdrawn in a stream divider and added as raffinate reflux, V_B , to fresh solvent, S . The remaining raffinate, B , is sent to a solvent-removal step (not shown) to produce a carrier-rich raffinate product. When using extract reflux,

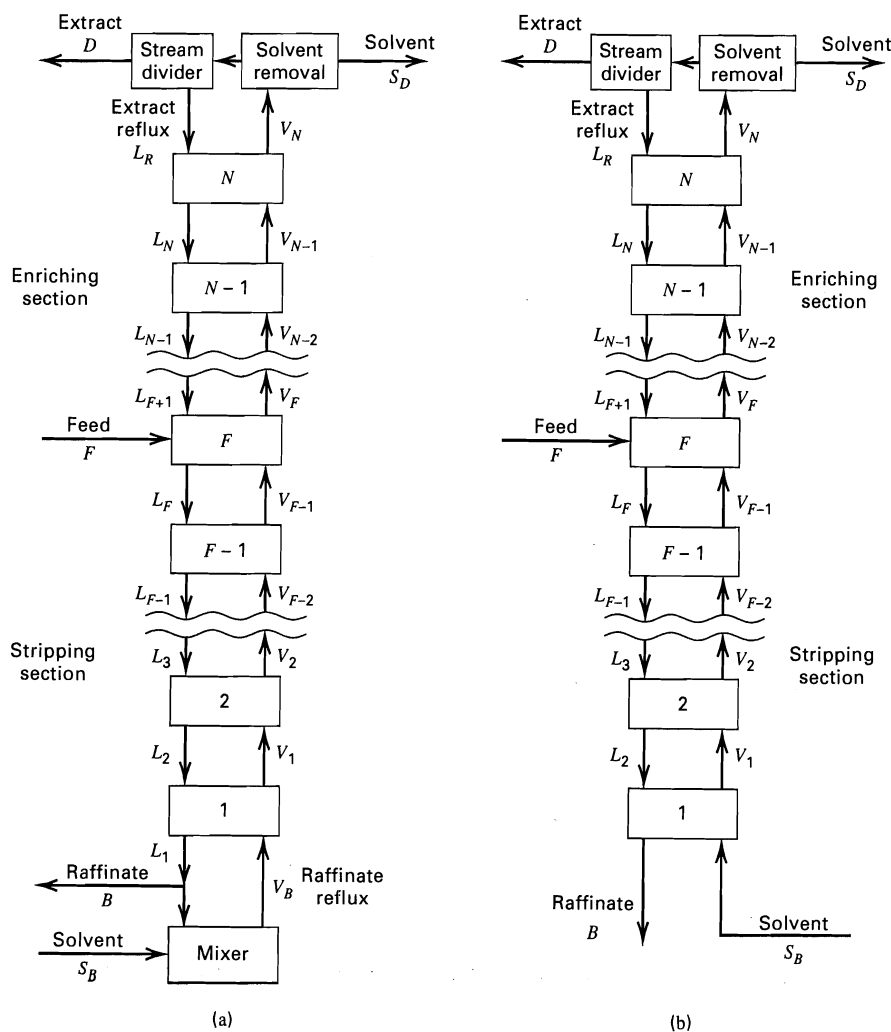


Figure 8.26 Liquid-liquid extraction with reflux: (a) with extract and raffinate reflux; (b) with extract reflux only.

Table 8.5 Analogy between Distillation and Extraction

Distillation	Extraction
Addition of heat	Addition of solvent
Reboiler	Solvent mixer
Removal of heat	Removal of solvent
Condenser	Solvent separator
Vapor at the boiling point	Solvent-rich solution saturated with solvent
Superheated vapor	Solvent-rich solution containing more solvent than that required to saturate it
Liquid below the boiling point	Solvent-lean solution, containing less solvent than that required to saturate it
Liquid at the boiling point	Solvent-lean solution saturated with solvent
Mixture of liquid and vapor	Two-phase liquid mixture
Relative volatility	Relative selectivity
Change of pressure	Change of temperature
D = distillate	D = extract product (solute on a solvent-free basis)
B = bottoms	B = raffinate (solvent-free basis)
L = saturated liquid	L = saturated raffinate (solvent-free)
V = saturated vapor	V = saturated extract (solvent-free)
A = more volatile component	A = solute to be recovered
C = less volatile component	C = carrier from which A is extracted
F = feed	F = feed
x = mole fraction A in liquid	X = mole or weight ratio of A (solvent-free), $A/(A + C)$
y = mole fraction A in vapor	$Y = S/(A + C)$

minimum- and total-reflux conditions, corresponding to infinite and minimum number of stages, bracket the optimal extract reflux ratio. Raffinate reflux is not processed through the solvent-removal unit because fresh solvent is added at this end of the cascade. It is necessary, however, to remove solvent from extract reflux at the enriching end of the cascade.

The analogy between a two-section liquid-liquid extractor with feed entering a middle stage, and distillation, is considered in some detail by Randall and Longtin [32]. Different aspects of the analogy are listed in Table 8.5. The most important analogy is that the solvent (a mass-separating agent) in extraction serves the same purpose as heat (an energy-separating agent) in distillation.

The use of raffinate reflux has been judged to be of little, if any, benefit by Skelland [31], who shows that the amount of raffinate reflux does not affect the number of stages required. Accordingly, we will consider a two-section, countercurrent cascade that includes only extract reflux, as shown in Figure 8.26b.

Analysis of a refluxed extractor, such as that of Figure 8.26b, involves relatively straightforward extensions of the procedures already developed. As will be shown, however, results for a type I system depend critically on the feed composition and the nature of the equilibrium-phase diagram, and it is very difficult to draw any general conclusions with respect to the effect (or even feasibility) of reflux.

For the two-section cascade with extract reflux shown in Figure 8.26b, a degrees-of-freedom analysis can be performed as described in Chapter 5. The result, using as elements

two countercurrent cascades, a feed stage, a splitter, and a divider, is $N_D = 2N + 3C + 13$. All but four of the specifications will usually be

Variable Specification	Number of Variables
Pressure at each stage	N
Temperature for each stage	N
Feed-stream flow rate, composition, temperature, and pressure	$C + 2$
Solvent composition, temperature, and pressure	$C + 1$
Split of each component in the splitter (solvent removal step)	C
Temperature and pressure of the two streams leaving the splitter	4
Pressure and temperature of the divider	2
	$2N + 3C + 9$

The four additional specifications can be taken from one of the following sets:

Set 1	Set 2	Set 3
Solvent rate	Reflux ratio	Solvent rate
Solute concentration in extract (solvent free)	Solute concentration in extract (solvent-free)	Reflux ratio
Solute concentration in raffinate (solvent-free)	Solute concentration in raffinate (solvent-free)	Number of stages
Optimal feed-stage location	Optimal feed-stage location	Feed-stage location

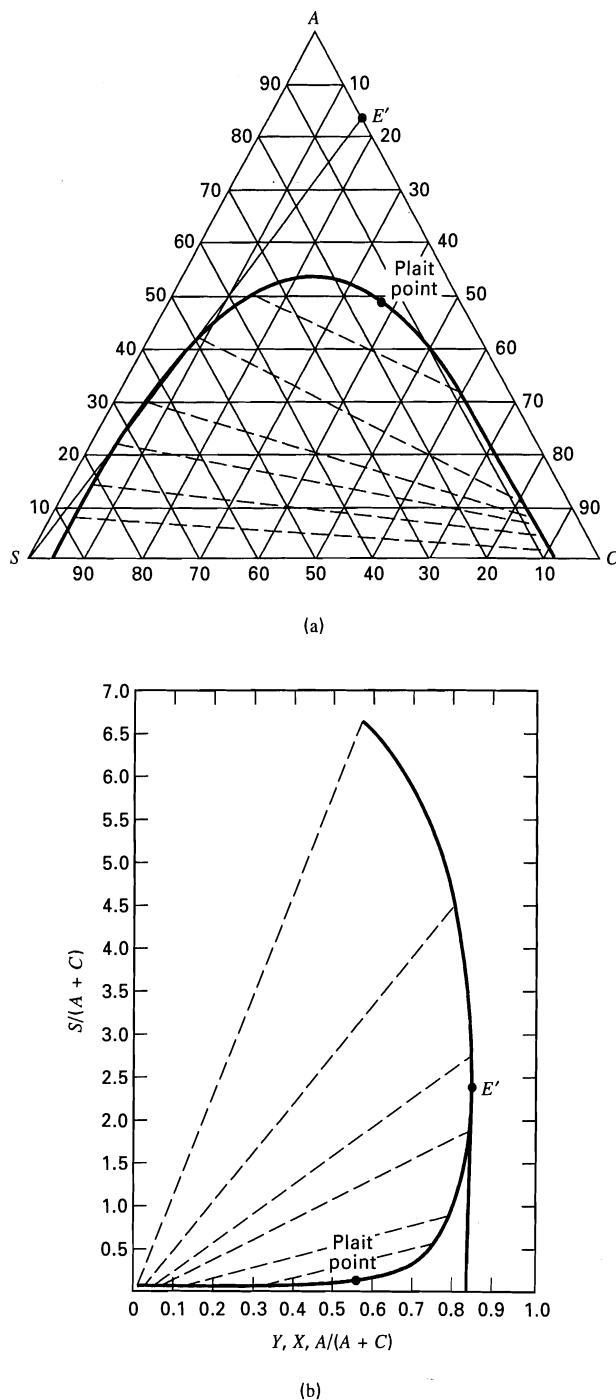


Figure 8.27 Limitation on product purity: (a) using an equilateral-triangle diagram; (b) using a Janecke diagram.

Sets 1 and 2 are of particular interest in the design of a new extractor because two of the specifications deal with the split of the feed into two products of designated purities, on a solvent-free basis. Set 2 is analogous to the design of a binary distillation column using the McCabe–Thiele graphical method, where the purities of the distillate and bottoms, the reflux ratio, and the optimal feed-stage location are specified. For a single-section cascade, it is not feasible to specify the split of the feed with respect to two key components. Instead, as in absorption and stripping, the recovery of just one component in the feed is specified.

It will be recalled for binary distillation that the purity of one of the products may be limited by the formation of an azeotrope. A similar limitation can occur for a type I system when using a two-section cascade with extract reflux, because of the plait point, which separates the two-liquid-phase region from the homogeneous, single-phase region. This limitation can be determined from a triangular diagram, but it is most readily observed on a Janecke diagram, of the type described in Chapter 4 and shown previously in Figure 4.14e. In Figure 8.27, liquid-liquid equilibrium data are repeated for the A–C–S system of Figure 8.14 where A is the solute and S is the solvent. In the triangular representation of Figure 8.27a, the maximum solvent-free solute concentration that can be achieved in the extract by a countercurrent cascade with extract reflux is determined by the intersection of line SE' , drawn tangent to the binodal curve, from the pure solvent point S to the solvent-free composition line AC , giving, in this case, 83 wt% solute. The same value is read from the binodal curve of the Janecke diagram of Figure 8.27b as the value of the abscissa for the point E' farthest to the right of the curve.

Without extract reflux, the maximum solvent-free solute concentration that can be achieved corresponds to an extract that is in equilibrium with the feed, when saturated with the extract. If this maximum value is close to the maximum value determined, as in Figure 8.27, then the use of extract reflux will be of little value. This is often the case for type I systems, as illustrated in the following example.

EXAMPLE 8.2

In Example 8.1, a feed mixture of 30 wt% acetone and 70 wt% ethyl acetate was extracted in a single-section, countercurrent cascade with pure water to obtain a raffinate of 5 wt% acetone on a water (solvent)-free basis. The maximum solvent-free solute concentration in the extract was found to be 64 wt%, as shown in Figure 8.20 at point D, corresponding to the condition of minimum $S/F = 0.60$ at infinite stages. For an actual $S/F = 1.75$ with four equilibrium stages, the extract contains 62 wt% acetone on a solvent-free basis. Thus, use of extract reflux for the purpose of producing a more pure (solvent-free) extract is not very attractive, given the particular phase-equilibrium diagram and feedstock composition. However, to demonstrate the technique, the calculation for extract reflux is carried out nevertheless. Also, the minimum number of equilibrium stages at total reflux and the minimum reflux ratio are determined.

SOLUTION

For the case of the single-section countercurrent cascade, the extract pinch point is at 57 wt% water, 27 wt% acetone, and 16 wt% acetate, as shown in Figure 8.20 at point $D'_{\min}$. If stages are added above the feed point, as in the two-section, refluxed cascade of Figure 8.26b, it is possible, theoretically, to reduce the water content of the extract to about 28 wt%, as shown by point G in Figure 8.20. However, the solvent (water)-free extract would not be as rich in acetone (51 wt%), which is determined from the line drawn through points S' and G and extended to where it intersects the solvent-free line AC .

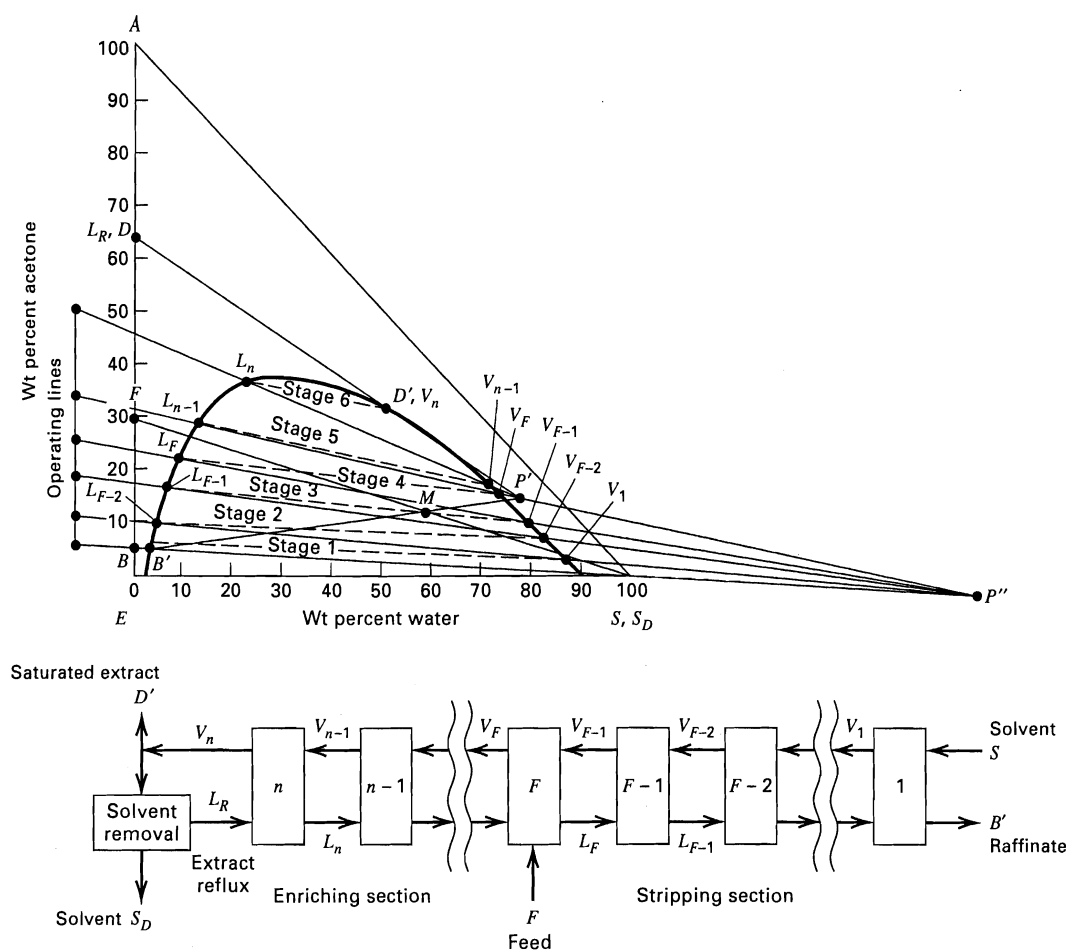


Figure 8.28 Equilibrium stages for Example 8.2.

To make this example more interesting, assume that a saturated extract containing 50 wt% water is required elsewhere in the process. Thus, the extraction cascade is that shown in Figure 8.28, rather than that of Figure 8.26b. The difference lies in the location of the solvent-removal step. This saturated-extract product is shown as point D' in Figure 8.28. Assume the ratio S/F to be 1.43, which is more than twice the minimum ratio found in Example 8.1. The desired raffinate composition is again 5 wt% acetone on a water-free basis (point B in Figure 8.28), which maps to point B' on the raffinate side of the binodal curve on a line connecting points B and S .

As with single-section cascades, the mixing point, M , in Figure 8.28, for the two streams entering the cascade, S and F , is determined by applying the inverse-lever-arm rule, using the S/F ratio, or by computing the overall composition of M , which in this case is 59 wt% water, 29 wt% acetate, and 12 wt% acetone.

The cascade in Figure 8.28 consists of an enriching section to the left of the feed point and a stripping section to the right, where extract is enriched in solute and raffinate is stripped of solute, respectively. A difference or operating point is needed for each section. We will let these be P' and P'' , respectively, for the enriching and stripping sections. In the enriching section, referring to the cascade in Figure 8.28, $P' = V_n - L_R = V_{n-1} - L_n$. But, by material balance, $V_n - L_R = D' + S_D$. Therefore, $P' = D' + S_D$, that is, the total flow leaving the extract end of the cascade. Also, by overall material balance, $M = F + S = B' + D' + S_D = B' + P'$. Thus, P' must lie on a line drawn through points B' and M . To locate the position of P' on that line, we also note that V_n has the same composition as D' , and L_R is simply D' with the solvent

removed (point D). From above, however, $P' = V_n - L_R$ or $V_n = P' + L_R$. Thus, point P' must also lie on a line drawn through points $V_n(D')$ and $L_R(D)$. Thus, point P' is located at the intersection of the extended lines $B'M$ and DD' as shown in Figure 8.28.

The difference point, P'' , for the stripping section is located in a similar manner, if we note that $P'' = B' - S = F - D' - S_D = F - P'$. Thus, P'' must be the intersection of extended lines $F'P'$ and $B'S$ as shown to the right of the triangular diagram of Figure 8.28.

The stages can now be stepped off in the usual manner by starting from V_n and using the difference point P' until an operating line crosses the feed line $F'P'$. From there, the stages are stepped off using the difference point P'' until the raffinate composition is reached or exceeded. In Figure 8.28, it is seen that six equilibrium stages are required, with two in the enriching section and four in the stripping section. The feed enters the third stage from the left.

The reflux ratio, defined for this example as $(V_n - D')/D' = (L_R + S_D)/D'$, can be determined as follows. From above, $P' = D' + S_D$. Therefore, by the mixing rule, $S_D/D' = \overline{D'P'}/\overline{S_DP'}$. By material balance, $V_n - D' = L_R + S_D$. Therefore,

$$\frac{V_n - D'}{S_D} = \frac{L_R + S_D}{S_D} = \frac{\overline{L_RS_D}}{\overline{L_RD'}} \quad \text{and}$$

$$\frac{V_n - D'}{D'} = \left(\frac{S_D}{D'}\right) \left(\frac{V_n - D'}{S_D}\right) = \left(\frac{\overline{D'P'}}{\overline{S_DP'}}\right) \left(\frac{\overline{L_RS_D}}{\overline{L_RD'}}\right)$$

By measurement from Figure 8.28, $(V_n - D')/D' = (1.2)(2.0) = 2.4$. The reflux ratio is valid only for the selected solvent-to-feed ratio of 1.43.

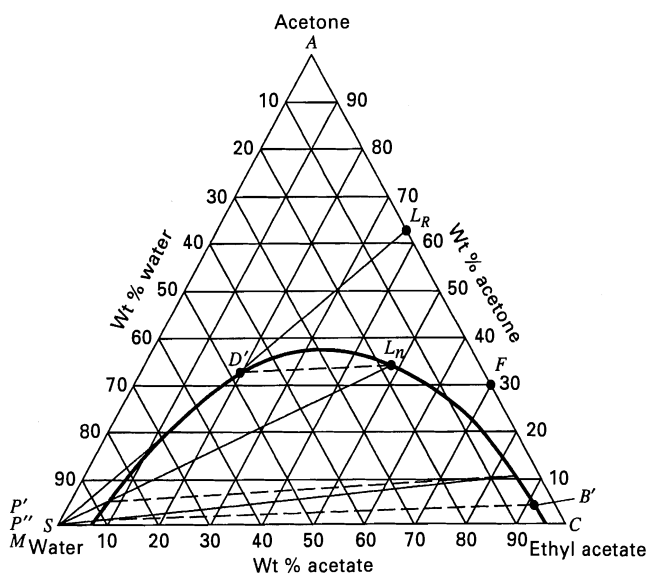


Figure 8.29 Total reflux and minimum stages for Example 8.2.

Next, we consider the case of total reflux, corresponding to the minimum number of stages. With reference to the equilateral-triangle diagram of Figure 8.29, compositions of existing streams are as previously specified or computed. With respect to acetone, we have 30 wt% in F , 4.9 wt% in B' , 33 wt% in D' , and 62 wt% in L_R . As in the case of the single-section cascade of Figure 8.20, as the solvent-to-feed ratio is increased, the mixing point $M = F + S$ moves toward the pure-solvent apex. At the maximum solvent addition, M lies at the intersection of the line through F and S with the extract side of the binodal curve. Difference points P' and P'' also move toward S because $P' = D' + S_D$ approaches S_D at total reflux and $P'' = F - P'$ approaches P' , recalling that at total reflux $F = D' = 0$. As shown in Figure 8.29, the minimum number of equilibrium stages is three, as stepped off from the S apex.

Lastly, we consider the case of minimum reflux ratio at infinite stages, which also corresponds to the minimum solvent ratio. As the solvent ratio is reduced, point M moves toward the feed point, F , and point P'' moves away from the binodal curve. Also, point P' moves toward V_n . Ultimately, a value of the S/F ratio is reached where an operating line in either the enriching section or the stripping section coincides with a tie line, giving a pinch point and, therefore, an infinite number of stages. Often, this occurs for the extended tie line that passes through the feed point. Such is the case here, giving a minimum reflux ratio of about 0.6 and a corresponding minimum S/F ratio of about 0.75.

8.4 MALONEY-SCHUBERT GRAPHICAL EQUILIBRIUM-STAGE METHOD

For type II ternary systems, as shown in Figure 8.10b, use of a two-section cascade with extract reflux is particularly desirable. Without a plait point, the two-phase region extends all the way across the solute composition. Thus, while maximum solvent-free solute concentration in the extract is limited for a type I system as was shown in Figure 8.27, no such limit exists with a type II system. Accordingly, it is possible with extract reflux to achieve as sharp a separation as desired between the solute (A) and carrier (C).

When reflux is used, many stages may be required and the use of triangular diagrams is often not convenient. Instead, use can be made of a McCabe-Thiele-type diagram. Alternatively, a Janecke diagram, of the type shown earlier in Figure 4.14e, often in conjunction with a distribution diagram, has proved to be useful. In Janecke diagrams, which use convenient rectangular coordinates, solvent concentration on a solvent-free basis is plotted as the ordinate against solute concentration on a solvent-free basis as the abscissa, that is, $\%S/(\%A + \%C)$ against $\%A/(\%A + \%C)$, either with mass or mole percents. The Janecke diagram is analogous to an enthalpy-concentration diagram and is consistent with the distillation-extraction analogy of Table 8.5, where enthalpy is replaced by solvent concentration because a mass-separating agent replaces an energy-separating agent. The application of such a diagram to liquid-liquid extraction of a type II system with the use of reflux is considered in detail by Maloney and Schubert [33], who use an auxiliary distribution diagram of the McCabe-Thiele type, but on a solvent-free basis, to facilitate visualization of the stages. This method is also referred to as the *Ponchon-Savarit method for extraction*. Unlike the analogous method for distillation mentioned briefly at the end of Chapter 7 and which requires both enthalpy and vapor-liquid equilibrium data, the method for extraction requires only ternary liquid-liquid solubility data, which are far more common than combined vapor-liquid enthalpy and equilibrium data. Accordingly, despite the development of rigorous computer-aided methods, the Ponchon-Savarit method for extraction has remained useful, while the analogous method for distillation has rapidly declined in popularity. Although the Janecke diagram can also be applied to type I systems, it becomes difficult to use when the carrier and the solvent are highly immiscible, because the resulting values of the ordinate can become very large.

With the Janecke diagram, construction of tie lines, mixing points, operating points, and operating lines are all made in a manner similar to that for a triangular diagram [33]. Consider the case of extraction for a type II system with extract reflux, shown in Figure 8.26b. A representative Janecke diagram is shown in Figure 8.30, where all flow rates are on a solvent-free mass basis and the following solvent-free concentrations for both extract and raffinate phases are based on Janecke coordinates:

$$Y = \frac{\text{mass solvent}}{\text{mass of solvent-free liquid phase}}$$

$$X = \frac{\text{mass solute}}{\text{mass of solvent-free liquid phase}}$$

Values of the ordinate, Y , especially for the saturated-extract phase, can vary over a wide range depending on the solubility of the solute and carrier in the solvent. Values of the abscissa, X , vary from 0 to 1 (pure carrier to pure solute). Equilibrium tie lines relate concentrations in the saturated extract to the saturated raffinate. The Y location of the feed, F , is somewhere between zero and the saturated-raffinate curve. The extract, V_N , leaving stage N and prior to solvent

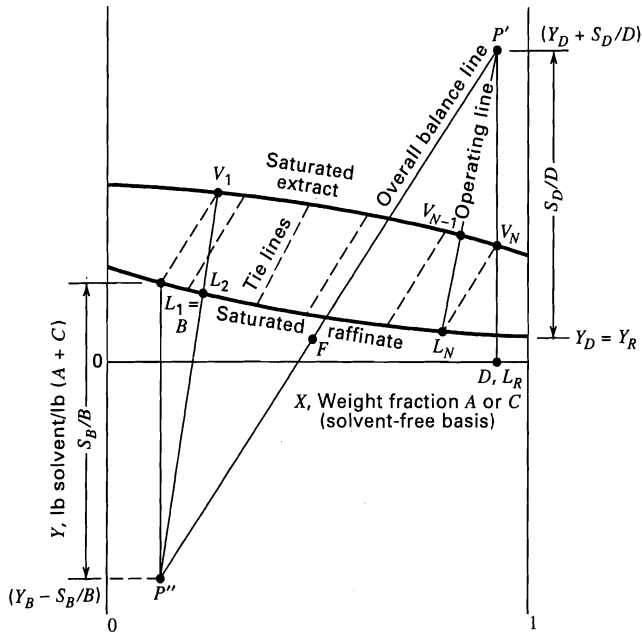


Figure 8.30 Construction of equilibrium stages on a Janecke diagram.

removal, is rich in solute and lies on the saturated extract curve. Upon solvent removal, the extract, D , and extract reflux, L_R , with identical compositions, lie on the $Y = 0$ horizontal line. The solvent removed from the cascade, S_D , is assumed to be pure. The raffinate, L_1 , leaving stage 1 is rich in the carrier and lies on the saturated-raffinate line. Solvent S_B , fed to the cascade, is assumed to be pure.

Points P' and P'' are difference or operating points for the enriching and stripping sections, respectively, that are used to draw operating lines. The locations of P' and P'' are derived as follows: Referring to Figure 8.26b, solvent-free and solvent material balances around the solvent-recovery step, in terms of passing streams, are, respectively,

$$V_N - L_R = D \quad (8-13)$$

$$Y_{V_N} V_N - Y_D L_R = S_D + Y_D D \quad (8-14)$$

For a solvent difference balance around a section of top stages down to stage n , located above stage F in Figure 8.26b, we obtain

$$Y_{V_n} V_n - Y_{L_{n+1}} L_{n+1} = S_D + Y_D D \quad (8-15)$$

Thus, any solvent flow difference between passing streams in the enriching section above the feed stage is given by $S_D + Y_D D$. If (8-13) and (8-14) are combined to eliminate V_N , we obtain

$$\frac{L_R}{D} = \frac{(Y_D + S_D/D) - Y_{V_N}}{Y_{V_N} - Y_D} \quad (8-16)$$

In Figure 8.30, $(Y_{V_N} - Y_D)$ is the vertical distance between the points V_N and (D, L_R) . The difference point, P' , in Figure 8.30 becomes $(S_D + Y_D D)$ divided by D to give $(Y_D + S_D/D)$. That is,

$$P' = Y_D + S_D/D \quad (8-17)$$

$$\text{and} \quad L_R/D = \overline{P'V_N}/\overline{V_N L_R} \quad (8-18)$$

Similarly, for the stripping section, it can be shown that,

$$P'' = Y_B - S_B/B \quad (8-19)$$

$$D/B = \overline{F P''}/\overline{P' F} \quad (8-20)$$

Stages are stepped off in a manner analogous to that for the triangular diagram, starting from either the extract, D , or the raffinate, B , alternating between operating lines and tie lines. For example, in Figure 8.30, we can start from the top of the cascade at the extract D and step off stages in the enriching section. The solute compositions of D , L_R , and V_N on a solvent-free basis are identical. Thus, the operating line for passing streams L_R and V_N is a vertical line passing through the difference point P' . From point V_N on the saturated extract curve, a tie line is followed down to the equilibrium raffinate phase, L_N . An operating line connecting points P' and L_N intersects the extract curve at the passing stream V_{N-1} . Subsequent stages in the enriching section are stepped off in a similar manner until the feed stage is reached. The optimal location of this stage is determined in a manner analogous to the intersection of the rectification and stripping operating lines in the McCabe-Thiele method. On the Janecke diagram, this intersection is the line $\overline{P' P''}$, which passes through the feed point, as shown in Figure 8.30. Thus, the transition from the enriching section (where the difference point P' is used) to the stripping section (where P'' is used) is made when an equilibrium tie line for a stage crosses the line $\overline{P' P''}$. Following the location of the feed stage, the remaining stripping-section stages are stepped off until the desired product raffinate solvent-free concentration is reached or crossed over.

The Janecke diagram can also be used to determine the two limiting conditions of total reflux (minimum stages) and minimum reflux (infinite stages). For total reflux, the difference points P' and P'' lie at $Y = +\infty$ and $-\infty$, respectively, because $F = B = D = 0$. Thus, all operating lines become vertical lines and the minimum number of stages are stepped off in the manner illustrated in Figure 8.31a.

For the condition of minimum reflux, a pinch condition is sought either at the feed stage or some other stage location. In Figure 8.31b, where the pinch is assumed to be at the feed stage, an operating line is drawn coincident with a tie line and the feed point, F , to determine points P' and P'' . To determine if the pinch does occur at the feed stage, tie lines to the right of the feed-stage tie line are extended to an intersection with the vertical line through D . If a higher intersection occurs, then that P' may be the correct $P'_{\min}$ difference point for minimum reflux. In a similar manner, tie lines to the left of the feed-stage tie line are extended to an intersection with the vertical line through B . If a lower intersection occurs, then that P'' may be the factor that determines the minimum reflux. The former case is shown in Figure 8.31c, where P' is higher than P'_1 . Thus, $P' = P'_{\min}$. In any case, once the controlling P' or P'' is determined, a line through F determines the other difference point and the minimum reflux ratio is computed from (8-18) using $P'_{\min}$.

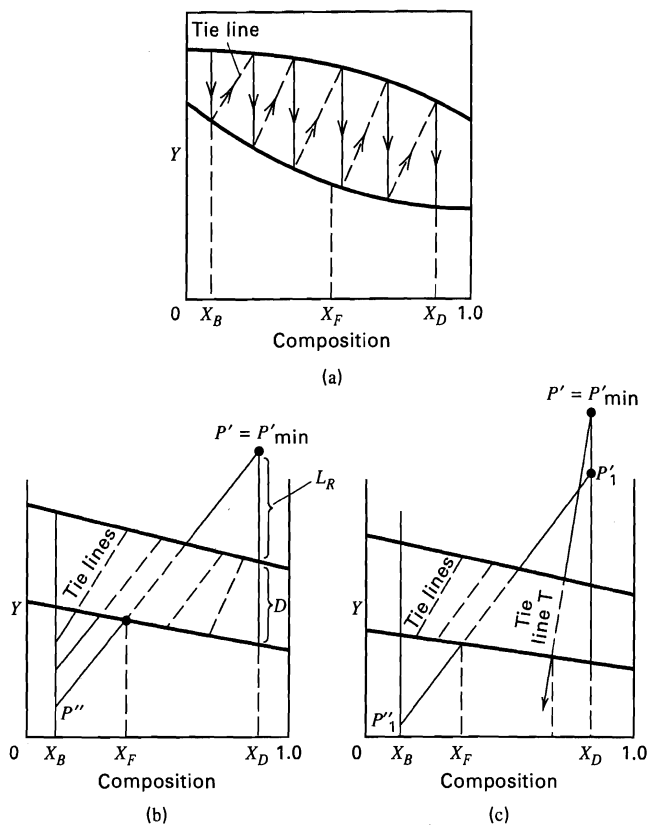


Figure 8.31 Limiting conditions on a Janecke diagram: (a) minimum number of stages at total reflux; (b) minimum reflux determined by a tie line through the feed point; (c) minimum reflux determined by a tie line to the right of the feed point.

EXAMPLE 8.3

As shown in Figure 8.32, a countercurrent, extraction cascade equipped with a perfect solvent separator to provide extract reflux is used to separate methylcyclopentane (A) and *n*-hexane (C) into a final extract and raffinate containing, on a solvent-free basis, 95 wt% and 5 wt% A, respectively, using aniline (S) as the solvent. The feed rate is 1,000 kg/h with 55 wt% A, and the mass ratio of solvent to feed is 4.0. The feed contains no aniline and the fresh solvent is pure. Recycle solvent is also assumed to be pure. Equilibrium curves and tie lines are given in Figure 8.33.

- (a) Determine the reflux ratio and number of stages. Equilibrium data at extractor temperature and pressure are shown for mass units in the Janecke diagram of Figure 8.33. Feed is to enter at the optimal stage.

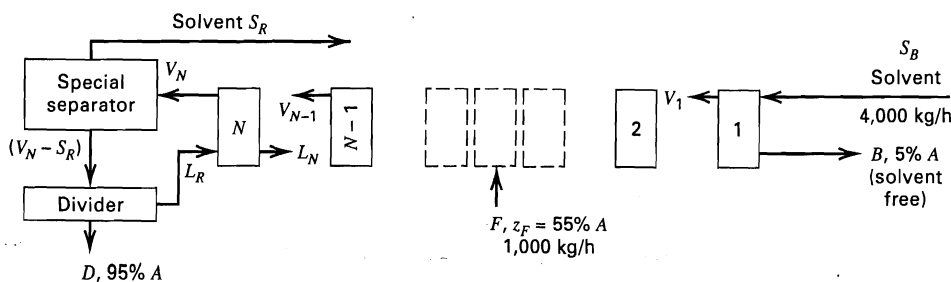


Figure 8.32 Countercurrent extraction cascade with extract reflux for Example 8.3.

- (b) Determine the minimum number of stages for the specified solvent-free extract and raffinate compositions.
(c) Determine the minimum reflux ratio for the specified feed and product compositions.

SOLUTION

- (a) First, determine all product rates by material-balance calculations. An overall balance on solute plus carrier gives $D + B = 1,000$ kg/h

$$\begin{aligned} \text{A solute balance gives } 0.95D + 0.05B &= (0.55)(1,000) \\ &= 550 \text{ kg/h} \end{aligned}$$

Solving these two equations simultaneously gives $D = 556$ kg/h and $B = 444$ kg/h. Since $S_B = 4,000$ kg/h, $S_B/B = 9.0$. In Figure 8.30, point P'' is located at a distance of S_B/B below the raffinate composition, X_B , at point B. Since Y at point B, from Figure 8.33, is approximately 0.3, point P'' is located at $0.3 - 9.0 = -8.7$.

A line drawn through P'' and F , extended to the intersection with the vertical line through D , gives $P' = 6.7$. By measurement from Figure 8.33, using (8-16), $L_R/D = 3.7$.

In Figure 8.33, stages are stepped off starting from point D . At the third stage ($N - 2$), the tie line crosses line $P''FP'$. Thus, this is the optimal feed stage. Three more stages are required to reach B , giving a total of six equilibrium stages.

- (b) If the construction for minimum stages, shown in Figure 8.31a, is used in Figure 8.33, just less than five stages are determined.

- (c) If the construction for minimum reflux, shown in Figure 8.31b for a pinch at the feed stage, is used in Figure 8.33, a value of $P' = 2.90$ is obtained. No other tie line in either section gives a larger value. Therefore, $P_{\min} = 2.9$. By measurement, using (8-18), $(L_R/D)_{\min} = 0.83$. Using the construction indicated in Figure 8.30, the corresponding $(S_B/B)_{\min}$ is found to be 4.2. Thus, $(S_B)_{\min} = 4.2(444) = 1,865$ kg/h or $(S_B/F)_{\min} = 1,865/1,000 = 1.865$.

For this example, a relatively high reflux ratio and corresponding solvent-to-feed ratio is employed to keep the required number of equilibrium stages small. When the number of equilibrium stages is large, the Janecke diagram becomes cluttered with operating lines and tie lines. In that case, an auxiliary McCabe-Thiele-type plot of solute mass fraction in the extract layer versus solute mass fraction in the raffinate layer, both on a solvent-free basis, as in the Janecke diagram, can be drawn, with points on the enriching and stripping operating lines determined, as discussed above, from arbitrary operating lines on the Janecke diagram. Stages are

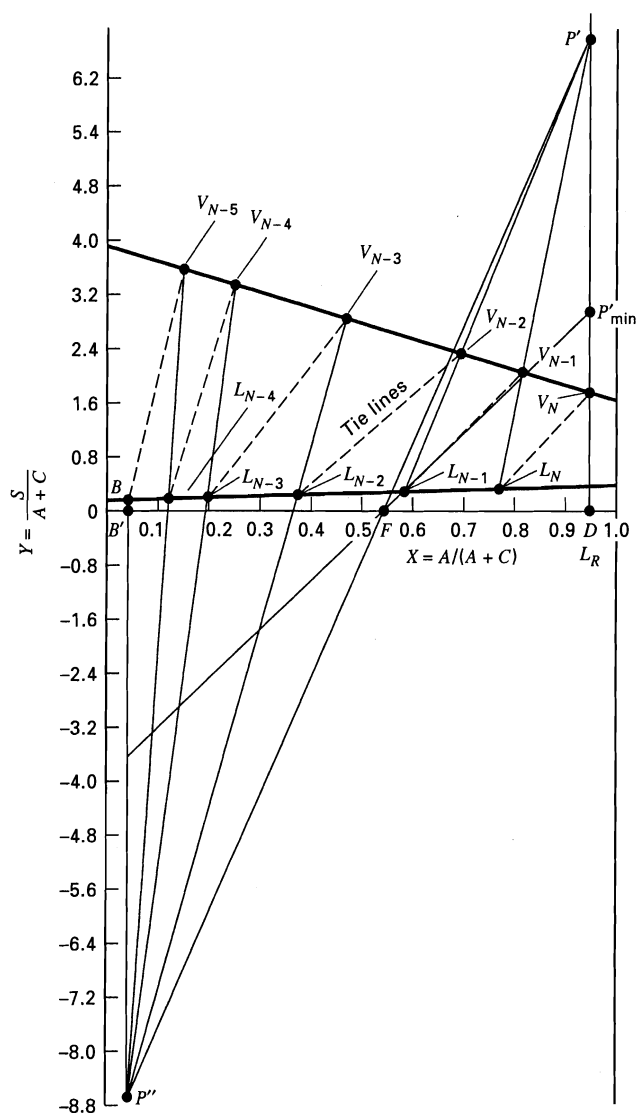


Figure 8.33 Maloney-Schubert constructions on Janecke diagram for Example 8.3.

then stepped off in the McCabe-Thiele manner. An example of the Janecke diagram with such an auxiliary McCabe-Thiele diagram is shown in Figure 8.34, taken from Maloney and Schubert [33].

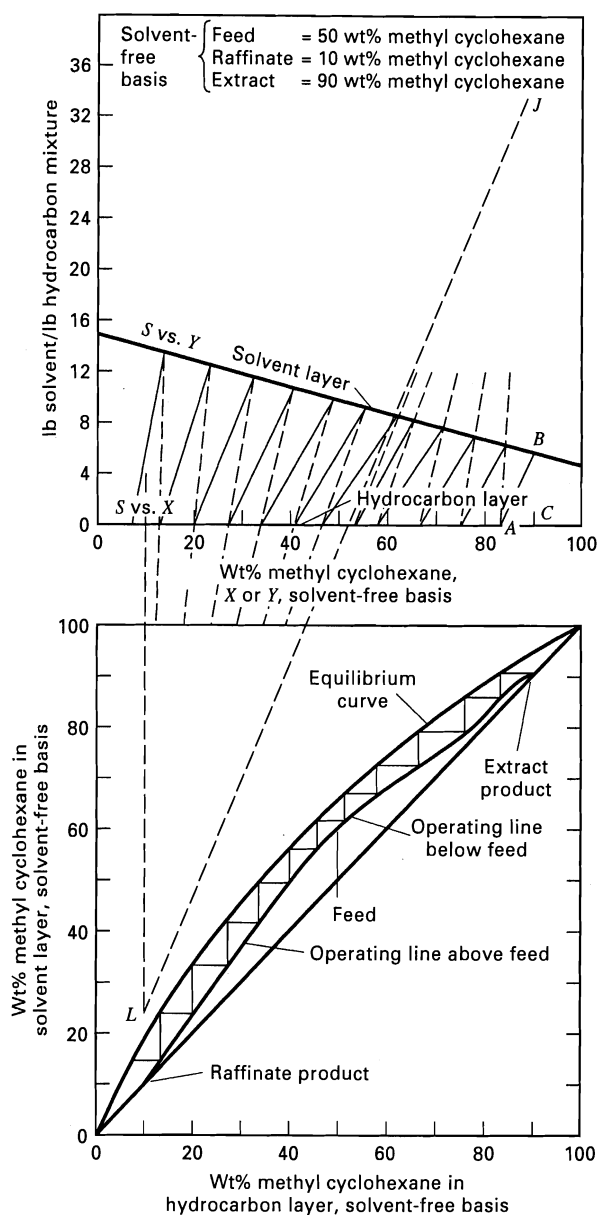


Figure 8.34 Use of Janecke diagram with auxiliary distribution diagram.

[From *Chemical Engineers' Handbook*, 5th ed., R.H. Perry and C.H. Chilton, Eds., McGraw-Hill, New York (1973).]

8.5 THEORY AND SCALE-UP OF EXTRACTOR PERFORMANCE

Following the estimation, by methods described in Sections 8.3 and 8.4, of the number of equilibrium stages, suitable extraction equipment can be selected using the scheme of Figure 8.8. Often, the choice is between a cascade of mixer-settler units or a multicompartiment, column-type extractor with mechanical agitation, the main considerations being the number of stages required, and the floor space and head room available. Methods for estimating size and power requirements of these two general types of extractors are presented next. Column devices with no mechanical agitation are also considered.

Mixer-Settler Units

Sizing of mixer-settler units is done most accurately by scale-up from batch or continuous runs in laboratory or pilot-plant equipment. However, preliminary-sizing calculations can be made using available theory and empirical correlations. Experimental data of Flynn and Treybal [34] show that when liquid-phase viscosities are less than 5 cP and the specific-gravity difference between the two liquid phases is greater than about 0.10, the average residence time required of the two liquid phases in the mixing vessel to achieve at least 90% stage efficiency may be as low as 30 s and is usually not more than 5 min, when an agitator-power input per mixer volume of 1,000 ft-lbf/min-ft³ (4 hp/1,000 gal) is used.

Based on experiments reported by Ryan, Daley, and Lowrie [35], the capacity of a settler vessel can be expressed in terms of C gal/min of combined extract and raffinate per square foot of phase-disengaging area. For a horizontal, cylindrical vessel of length L and diameter D_T , the economic ratio of L to D_T is approximately 4. Thus, if the phase interface is located at the middle of the vessel, the disengaging area is $D_T L$ or $4D_T^2$. A typical value of C given by Happel and Jordan [36] is about 5. Frequently, the settling vessel will be larger than the mixing vessel, as is the case in the following example.

EXAMPLE 8.4

Benzoic acid is to be continuously extracted from a dilute solution in water with a solvent of toluene in a series of discrete mixer-settler vessels operated in countercurrent flow. The flow rates of the feed and solvent are 500 and 750 gal/min, respectively. Assuming a residence time, t_{res} , of 2 min in each mixer and a settling vessel capacity of 5 gal/min-ft², estimate:

- Diameter and height of a mixing vessel, assuming $H/D_T = 1$
- Agitator horsepower for a mixing vessel
- Diameter and length of a settling vessel, assuming $L/D_T = 4$
- Residence time in a settling vessel in minutes

SOLUTION

- $Q = \text{total flow rate} = 500 + 750 = 1,250 \text{ gal/min}$
 $V = \text{volume} = Qt_{\text{res}} = 1,250(2) = 2,500 \text{ gal or } 2,500/7.48 = 334 \text{ ft}^3$

$$V = \pi D_T^2 H/4, H = D_T, \text{ and } V = \pi D_T^3/4$$

$$D_T = (4V/\pi)^{1/3} = [(4)(334)/3.14]^{1/3} = 7.52 \text{ ft and } H = 7.52 \text{ ft}$$

- Horsepower = $4(2,500/1,000) = 10 \text{ hp}$

- $D_T L = 1,250/5 = 250 \text{ ft}^2$; $D_T^2 = 250/4 = 62.5 \text{ ft}^2$

$$D_T = 7.9 \text{ ft}; L = 4D_T = 4(7.9) = 31.6 \text{ ft}$$

- Volume of settler = $\pi D_T^2 L/4 = 3.14(7.9)^2(31.6)/4 = 1,548 \text{ ft}^3$
or $1,548(7.48) = 11,580 \text{ gal}$

$$t_{\text{res}} = V/Q = 11,580/1,250 = 9.3 \text{ min}$$

A typical single-compartment mixing tank for liquid-liquid extraction is shown in Figure 8.35. The vessel is closed with the two liquid phases entering at the bottom and the effluent, in the form of a two-phase emulsion, leaving at the top. Although flat tank heads are shown in Figure 8.35, rounded heads of the type in Figure 8.2 are preferred to eliminate stagnant fluid regions. Air or other gases must be evacuated from the vessel so no gas-liquid interface exists.

Mixing is accomplished by an appropriate, centrally located impeller selected from the many types available, some of which are shown in Figure 8.3. For example, a flat-blade turbine might be chosen as in Figure 8.35. A single turbine is adequate unless the vessel height is greater than the vessel diameter, in which case a compartmented vessel with two or

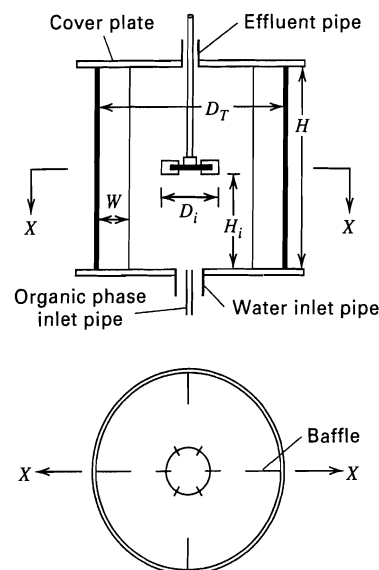


Figure 8.35 Agitated vessel with flat-blade turbine and baffles.

more impellers might be employed. When the vessel is open, vertical side baffles are mandatory to prevent vortex formation at the gas-liquid interface. For closed vessels that run full of liquid, vortexing will not occur. Nevertheless, it is common to install baffles, even in closed tanks, to minimize swirling and improve circulation patterns. Although no standards exist for vessel and turbine geometry, the following, with reference to Figure 8.35, give good dispersion performance in liquid-liquid agitation:

$$\text{Number of turbine blades} = 6;$$

$$\text{Number of vertical baffles} = 4$$

$$H/D_T = 1; D_i/D_T = 1/3;$$

$$W/D_T = 1/12 \text{ and } H_i/H = 1/2$$

To achieve a high stage efficiency for extraction in a mixing vessel—say, between 90 and 100%—it is necessary to provide fairly vigorous agitation. For a given type of impeller and vessel-impeller geometry, the agitator power, P , can be estimated from an empirical correlation in terms of a power number, N_{Po} , which depends on an impeller Reynolds number, N_{Re} , where

$$N_{Po} = \frac{Pg_c}{N^3 D_i^5 \rho_M} \quad (8-21)$$

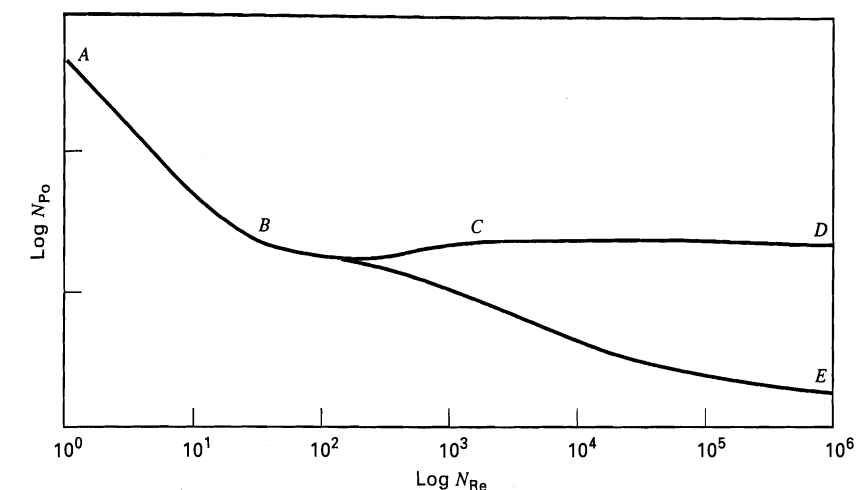
$$N_{Re} = \frac{D_i^2 N \rho_M}{\mu_M} \quad (8-22)$$

The impeller Reynolds number is the ratio of the inertial force to the viscous force:

$$\text{Inertial force} \propto (ND_i)^2 \rho_M D_i^2$$

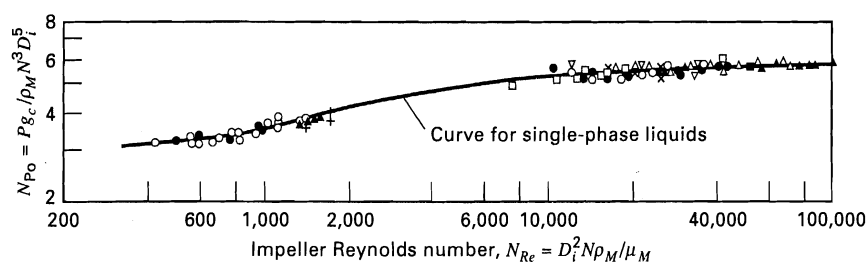
$$\text{Viscous force} \propto \frac{\mu_M (ND_i) D_i^2}{D_i}$$

where N = rate of impeller rotation. Thus, the characteristic length in the impeller Reynolds number is the impeller diameter and the characteristic velocity is ND_i = impeller peripheral velocity.



For curve ABCD, no vortex present
For curve BE, vortex present

(a)



(b)

Figure 8.36 Power consumption of agitated vessels. (a) Typical power characteristics.

[From J.H. Rushton and J.Y. Oldshue, *Chem. Eng. Prog.*, **49**, 161–168 (1953).]

(b) Power correlation for six-bladed, flat-blade turbines with no vortex.

[From D.S. Laity and R.E. Treybal, *AIChE J.*, **3**, 176–180 (1957).]

The agitator power is proportional to the product of the volumetric liquid flow produced by the impeller and the applied kinetic energy per unit volume of fluid. The result is

$$P \propto (ND_i^3)[\rho_M(ND_i)^2/2g_c]$$

which can be rewritten as (8-21), where the constant of proportionality is $2N_{Po}$. Both the impeller Reynolds number and the power number (also called the Newton number) are dimensionless groups. Thus, any consistent set of units can be used. The power number for an agitated vessel serves the same purpose as the friction factor for the flow of a fluid through a pipe. This is illustrated, over a wide range of impeller Reynolds number, for a typical mixing impeller in Figure 8.36a, taken from the work of Rushton and Oldshue [37]. The upper curve, ABCD, pertains to a vessel with baffles, while the lower curve, ABE, pertains to the same tank with no baffles. In the low-Reynolds-number region, AB, viscous forces dominate and the impeller power is proportional to $\mu_M N^2 D_i^3$. Somewhere beyond a Reynolds number of about 200, a vortex appears if no baffles are present and the power-number relation is given by curve BE. In this region, the Froude number, $N_{Fr} = N^2 D_i / g$, which is the ratio of inertial to gravitational forces, also becomes a factor. With baffles present, and the Reynolds number greater than about 1,000, a region, CD, is reached where fully developed turbulent flow exists. Now, inertial forces dominate and the power is proportional to $\rho_M N^3 D_i^5$. It is clear that the addition of baffles greatly increases power requirements in the turbulent flow region.

A correlation of experimental data for liquid–liquid mixing in baffled vessels with six-bladed, flat-blade turbines is shown in Figure 8.36b, from a study by Laity and Treybal [38]. The range of impeller Reynolds number covers only the turbulent–flow region, where efficient liquid–liquid mixing is achieved. The solid line represents batch mixing of single-phase liquids. The data points represent liquid–liquid mixing, where agreement is achieved with the single-phase curve by computing two-phase mixture properties from

$$\rho_M = \rho_C \phi_C + \rho_D \phi_D \quad (8-23)$$

$$\mu_M = \frac{\mu_C}{\phi_C} \left(1 + \frac{1.5 \mu_D \phi_D}{\mu_C + \mu_D} \right) \quad (8-24)$$

where ϕ is the volume fraction of holdup in the tank, with subscripts C for the continuous phase and D the dispersed phase, such that $\phi_D + \phi_C = 1$. When measurements were made for continuous flow from inlets at the bottom of the vessel to an outlet for the emulsion from the top of the vessel and with the impeller located at a position above the liquid–liquid interface when at rest, the data were correlated with the curve of Figure 8.36b.

With fully developed turbulent flow, the volume fraction of dispersed phase in the vessel closely approximates that in the feed to the vessel; otherwise the volume fraction may be different from that in the total feed to the vessel. That is, the residence times of the two phases in the vessel may not be the same. At best, spheres of uniform size can pack tightly to

give a void fraction of 0.26. Therefore, $\phi_C > 0.26$ and $\phi_D < 0.74$ is sometimes quoted. However, some experiments have shown a 0.20–0.80 range. For continuous flow, the vessel is first filled with the phase to be continuous. Following initiation of agitation, the two-feed liquids are then introduced into the vessel in their desired volume ratio.

Based on the work of Skelland and Ramsay [39] and Skelland and Lee [40], a minimum impeller rate of rotation is required for complete and uniform dispersion of one liquid into another. For a flat-blade turbine in a baffled vessel of the type discussed above, this minimum rotation rate can be estimated from

$$\frac{N_{\min}^2 \rho_M D_i}{g \Delta \rho} = 1.03 \left(\frac{D_T}{D_i} \right)^{2.76} \phi_D^{0.106} \left(\frac{\mu_M^2 \sigma}{D_i^5 \rho_M g^2 (\Delta \rho)^2} \right)^{0.084} \quad (8-25)$$

where $\Delta \rho$ is the absolute value of the difference in density and σ is the interfacial tension between the two liquid phases. The dimensionless group on the left-hand side of (8-25) is the two-phase Froude number. The dimensionless group at the far right of (8-25) is a ratio of forces:

$$\frac{(\text{viscous})^2 (\text{interfacial tension})}{(\text{inertial})(\text{gravitational})^2}$$

EXAMPLE 8.5

Furfural is to be continuously extracted from a dilute solution in water by toluene at 25°C in an agitated vessel of the type shown in Figure 8.35. The feed enters at a flow rate of 20,400 lb/h, while the solvent enters at 11,200 lb/h. For a residence time in the vessel of 2 min, estimate for either phase as the dispersed phase:

- The dimensions of the mixing vessel and the diameter of the flat-blade turbine impeller
- The minimum rate of rotation of the impeller for complete and uniform dispersion
- The power requirement of the agitator at the minimum rotation rate

SOLUTION

Mass flow rate of feed = 20,400 lb/h; feed density = 62.3 lb/ft³

Volumetric flow rate of feed = $Q_F = 20,400/62.3 = 327 \text{ ft}^3/\text{h}$

Mass flow rate of solvent = 11,200 lb/h

Solvent density = 54.2 lb/ft³; volumetric flow rate of solvent = $Q_S = 11,200/54.2 = 207 \text{ ft}^3/\text{h}$

Because of the dilute concentration of solute in the feed and sufficient agitation to achieve complete and uniform dispersion, assume fractional volumetric holdups of raffinate and extract in the vessel are equal to the corresponding volume fractions in the combined feed (raffinate, *R*) and solvent (extract, *E*) entering the mixer:

$$\phi_R = 327/(327 + 207) = 0.612; \quad \phi_E = 1 - 0.612 = 0.388$$

- Mixer volume = $(Q_F + Q_S)t_{\text{res}} = V = (327 + 207)(2/60) = 17.8 \text{ ft}^3$. Assume a cylindrical vessel with $D_T = H$ and neglect the volume of the bottom and top heads and the volume

occupied by the agitator and the baffles. Then

$$V = (\pi D_T^2/4) H = \pi D_T^3/4$$

$$D_T = [(4/\pi)V]^{1/3} = [(4/3.14)17.8]^{1/3} = 2.83 \text{ ft}$$

$$H = D_T = 2.83 \text{ ft}$$

Make the vessel 3 ft in diameter by 3 ft high, giving a volume $V = 21.2 \text{ ft}^3 = 159 \text{ gal}$. Assume that

$$D_i/D_T = 1/3; D_i = D_T/3 = 3/3 = 1 \text{ ft.}$$

- Case 1—Raffinate phase dispersed:

$$\phi_D = \phi_R = 0.612; \phi_C = \phi_E = 0.388$$

$$\rho_D = \rho_R = 62.3 \text{ lb/ft}^3; \rho_C = \rho_E = 54.2 \text{ lb/ft}^3$$

$$\mu_D = \mu_R = 0.89 \text{ cP} = 2.16 \text{ lb/h-ft};$$

$$\mu_C = \mu_E = 0.59 \text{ cP} = 1.43 \text{ lb/h-ft}$$

$$\Delta \rho = 62.3 - 54.2 = 8.1 \text{ lb/ft}^3;$$

$$\sigma = 25 \text{ dyne/cm} = 719,000 \text{ lb/h}^2$$

From (8-23),

$$\rho_M = (54.2)(0.388) + (62.3)(0.612) = 59.2 \text{ lb/ft}^3$$

From (8-24),

$$\mu_M = \frac{1.43}{0.388} \left[1 + \frac{1.5(2.16)(0.612)}{1.43 + 2.16} \right] = 5.72 \text{ lb/h-ft}$$

From (8-25), using American engineering units, with $g = 4.17 \times 10^8 \text{ ft/h}^2$,

$$\frac{\mu_M^2 \sigma}{D_i^5 \rho_M g^2 (\Delta \rho)^2} = \frac{(5.72)^2 (719,000)}{(1)^5 (59.2) (4.17 \times 10^8)^2 (8.1)^2} = 3.47 \times 10^{-14}$$

$$\begin{aligned} N_{\min}^2 &= 1.03 \left(\frac{g \Delta \rho}{\rho_M D_i} \right) \left(\frac{D_T}{D_i} \right)^{2.76} \phi_D^{0.106} (3.47 \times 10^{-14})^{0.084} \\ &= 1.03 \left[\frac{(4.17 \times 10^8)(8.1)}{(59.2)(1)} \right] \left(\frac{3}{1} \right)^{2.76} (0.612)^{0.106} (0.0740) \\ &= 8.56 \times 10^7 (\text{rph})^2 \end{aligned}$$

$$N_{\min} = 9,250 \text{ rph} = 155 \text{ rpm}$$

Case 2—Extract phase dispersed: Calculations similar to case 1 result in $N_{\min} = 8,820 \text{ rph} = 147 \text{ rpm}$

- Case 1—Raffinate phase dispersed:

$$\text{From (8-22), } N_{\text{Re}} = \frac{(1)^2 (9,250)(59.2)}{(5.72)} = 9.57 \times 10^4$$

From Figure 8.36b, it is seen that a fully turbulent flow exists, with the power number given by its asymptotic value of $N_{\text{Po}} = 5.7$.

From (8-21),

$$\begin{aligned} P &= N_{\text{Po}} N^3 D_i^5 \rho_M / g_c \\ &= (5.7)(9,250)^3 (1)^5 (59.2) / (4.17 \times 10^8) \\ &= 640,000 \text{ ft-lbf/h} = 0.323 \text{ hp} \end{aligned}$$

$$P/V = 0.323(1000)/159 = 2.0 \text{ hp/1,000 gal}$$

Case 2—Extract phase dispersed:

Calculations similar to case 1 result in $P = 423,000 \text{ ft-lbf/h} = 0.214 \text{ hp}$.

$$P/V = 0.214(1000)/159 = 1.4 \text{ hp/1,000 gal}$$

Mass-Transfer Efficiency

When dispersion is complete and uniform, the contents of the vessel are perfectly mixed with respect to both phases. In that case, the concentration of the solute in each of the two phases in the mixing vessel is uniform and equal to the concentrations in the two-phase emulsion leaving the mixing vessel. This is the so-called ideal CFSTR or CSTR (continuous-flow stirred-tank reactor) model, sometimes called the completely back-mixed or perfectly mixed model, first discussed by MacMullin and Weber [41] and widely applied to reactor design. The Murphree dispersed-phase efficiency for liquid-liquid extraction, based on the raffinate as the dispersed phase, can be expressed as the fractional approach to equilibrium. In terms of bulk molar concentrations of the solute,

$$E_{MD} = \frac{c_{D,in} - c_{D,out}}{c_{D,in} - c_D^*} \quad (8-26)$$

where c_D^* is the solute concentration in equilibrium with the bulk solute concentration in the exiting continuous phase, $c_{C,out}$. The molar rate of mass transfer of the solute, n , from the dispersed phase to the continuous phase can be expressed as

$$n = K_{OD}a(c_{D,out} - c_D^*)V \quad (8-27)$$

where the concentration driving force for mass transfer is uniform throughout the well-mixed vessel and is equal to the driving force based on the exit concentrations, a is the interfacial area for mass transfer per unit volume of liquid phases, V is the total volume of liquid phases in the vessel, and K_{OD} is the overall mass-transfer coefficient based on the dispersed phase, which is given in terms of the separate resistances of the dispersed and continuous phases by

$$\frac{1}{K_{OD}} = \frac{1}{k_D} + \frac{1}{mk_C} \quad (8-28)$$

where equilibrium is assumed at the interface between the two phases and m = the slope of the equilibrium curve for the solute plotted as c_C versus c_D :

$$m = dc_C/dc_D \quad (8-29)$$

For dilute solutions, changes in volumetric flow rates of the raffinate and extract are small, and thus the rate of mass transfer based on the change in solute concentration in the dispersed phase is given by material balance:

$$n = Q_D(c_{D,in} - c_{D,out}) \quad (8-30)$$

where Q_D is the volumetric flow rate of the dispersed phase.

To obtain an expression for E_{MD} in terms of $K_{OD}a$, (8-26), (8-27), and (8-30) are combined in the following manner. From (8-26),

$$\frac{E_{MD}}{1 - E_{MD}} = \frac{c_{D,in} - c_{D,out}}{c_{D,out} - c_D^*} \quad (8-31)$$

Equating (8-27) and (8-30), and noting that the right-hand side of (8-31) is the number of dispersed-phase transfer units

for a perfectly mixed vessel with $c_D = c_{D,out}$,

$$N_{OD} = \int_{c_{D,out}}^{c_{D,in}} \frac{dc_D}{c_D - c_D^*} = \frac{c_{D,in} - c_{D,out}}{c_{D,out} - c_D^*} = \frac{K_{OD}aV}{Q_D} \quad (8-32)$$

Combining (8-31) and (8-32) and solving for E_{MD} ,

$$E_{MD} = \frac{K_{OD}aV/Q_D}{1 + K_{OD}aV/Q_D} = \frac{N_{OD}}{1 + N_{OD}} \quad (8-33)$$

When $N_{OD} = (K_{OD}aV/Q_D) \gg 1$, $E_{MD} = 1$.

Drop Size and Interfacial Area

From (8-33) and (8-28), it is seen that an estimate of E_{MD} requires generalized correlations of experimental data for the interfacial area for mass transfer, a , and the dispersed- and continuous-phase mass-transfer coefficients, k_D and k_C , respectively. The population of dispersed-phase droplets in an agitated vessel will cover a range of sizes and shapes. For each droplet, it is useful to define d_e , the equivalent diameter of a spherical drop, using the method of Lewis, Jones, and Pratt [42],

$$d_e = (d_1^2 d_2)^{1/3} \quad (8-34)$$

where d_1 and d_2 are the major and minor axes, respectively, of an ellipsoidal-drop image. For a spherical drop, d_e is simply the diameter of the drop. For the population of drops, it is useful to define an average or mean drop diameter. A number of different definitions are available depending on whether weight-mean, mean-volume, surface-mean, mean-surface, length-mean, or mean-length diameter is appropriate [43]. For mass-transfer calculations, the surface-mean diameter, d_{vs} (also called the *Sauter mean diameter*), is most appropriate because it is the mean drop diameter that gives the same interfacial surface area as the entire population of drops for the same mass of drops. It is determined from experimental drop-size distribution data for N drops by the definition:

$$\frac{\pi d_{vs}^2}{(\pi/6)d_{vs}^3} = \frac{\pi \sum_N d_e^2}{(\pi/6) \sum_N d_e^3}$$

which, when solved for d_{vs} , gives

$$d_{vs} = \frac{\sum_N d_e^3}{\sum_N d_e^2} \quad (8-35)$$

With this definition, the interfacial surface area per unit volume of a two-phase mixture is

$$a = \frac{\pi N d_{vs}^2 \phi_D}{\pi N d_{vs}^3 / 6} = \frac{6\phi_D}{d_{vs}} \quad (8-36)$$

Equation (8-36) is used to estimate the interfacial area, a , from a measurement of d_{vs} or vice versa. Early experimental investigations, such as those of Vermeulen, Williams, and

Langlois [44], found that d_{vs} is dependent on a Weber number:

$$N_{We} = \frac{(\text{inertial force})}{(\text{interfacial tension force})} = \frac{D_i^3 N^2 \rho_C}{\sigma} \quad (8-37)$$

High Weber numbers give small droplets and high interfacial areas. Gnanasundaram, Degaleesan, and Laddha [45] correlated d_{vs} over a wide range of N_{We} . Below a critical value of $N_{We} = 10,000$, d_{vs} is dependent on dispersed-phase holdup, ϕ_D , because of coalescence effects. For $N_{We} > 10,000$, inertial forces dominate so that coalescence effects are much less prominent and d_{vs} is almost independent of holdup up to $\phi_D = 0.5$. The recommended correlations are

$$\frac{d_{vs}}{D_i} = 0.052(N_{We})^{-0.6} e^{4\phi_D}, \quad N_{We} < 10,000 \quad (8-38)$$

$$\frac{d_{vs}}{D_i} = 0.39(N_{We})^{-0.6}, \quad N_{We} > 10,000 \quad (8-39)$$

Typical values of N_{We} for industrial extractors are less than 10,000, so (8-38) applies. Values of d_{vs}/D_i are frequently in the range of 0.0005 to 0.01.

Experimental studies, for example, those of Chen and Middleman [46] and Sprow [47], show that the dispersion produced in an agitated vessel is a dynamic phenomenon. Droplet breakup by turbulent pressure fluctuations dominates in the vicinity of the impeller blades, while for reasonable dispersed-phase holdup, coalescence of drops by collisions dominates away from the impeller. Thus, a distribution of drop sizes is found in the vessel, with smaller drops in the vicinity of the impeller blades and larger drops elsewhere. Typically, when both drop breakup and coalescence occur, the drop-size distribution is such that $d_{\min} \approx d_{vs}/3$ and $d_{\max} \approx 3d_{vs}$. Thus, the drop size varies over about a 10-fold range, and the distribution approximates a normal Gaussian distribution.

EXAMPLE 8.6

For the conditions and results of Example 8.5, with the extract phase as the dispersed phase, estimate the Sauter mean drop diameter, the range of drop sizes, and the interfacial area.

SOLUTION

$$D_i = 1 \text{ ft}; \quad N = 147 \text{ rpm} = 8,820 \text{ rph}$$

$$\rho_C = 62.3 \text{ lb/ft}^3; \quad \sigma = 718,800 \text{ lb/h}^2$$

From (8-37),

$$N_{We} = (1)^3(8,820)^2(62.3)/718,800 = 6,742; \quad \phi_D = 0.388$$

From (8-38),

$$d_{vs} = (1)(0.052)(6,742)^{-0.6} \exp[4(0.388)] = 0.00124 \text{ ft}$$

or

$$(0.00124)(12)(25.4) = 0.38 \text{ mm}$$

$$d_{\min} = d_{vs}/3 = 0.126 \text{ mm}; \quad d_{\max} = 3d_{vs} = 1.134 \text{ mm}$$

$$\text{From (8-36),} \quad a = 6(0.388)/0.00124 = 1,880 \text{ ft}^2/\text{ft}^3$$

Mass-Transfer Coefficients

Experimental studies, conducted since the early 1940s, show that mass transfer in mechanically agitated liquid-liquid systems is very complex. This is true for mass transfer in (1) the dispersed-phase droplets, (2) the continuous phase, and (3) at the interface. The reasons for this complexity are many. The magnitude of k_D depends on drop diameter, solute diffusivity, and fluid motion within the drop. When drop diameter is small (less than 1 mm according to Davies [48]), interfacial tension is high (say > 15 dyne/cm), and trace amounts of surface-active agents are present, droplets are rigid (internally stagnant), and they behave like solids. As droplets become larger, interfacial tension decreases, surface-active agents become relatively ineffective, and internal toroidal fluid circulation patterns, caused by viscous drag of the continuous phase, appear within the drops. For larger-diameter drops, the shape of the drop may oscillate between spheroid and ellipsoid or other shapes.

Mass-transfer coefficients, k_C , in the continuous phase depend on the relative motion between the droplets and the continuous phase, and whether the drops are forming or breaking, or are coalescing. Interfacial movements or turbulence, called *Marangoni effects*, occur due to interfacial-tension gradients. Such effects can induce substantial increases in mass-transfer rates.

A relatively conservative estimate of the overall mass-transfer coefficient, K_{OD} , in (8-28), can be made from estimates of k_D and k_C , by assuming rigid drops, the absence of Marangoni effects, and a stable drop size (i.e., no drop forming, breaking, or coalescing). For k_D , the asymptotic steady-state solution for mass transfer in a rigid sphere with negligible resistance of the surroundings is given by Treybal [25] as

$$(N_{Sh})_D = \frac{k_D d_{vs}}{D_D} = \frac{2}{3} \pi^2 = 6.6 \quad (8-40)$$

where D_D is the diffusivity of the solute in the droplet. N_{Sh} is the Sherwood number. Exercise 3.31 in Chapter 3 for diffusion from the surface of a sphere into an infinite, quiescent fluid gives the following result for the continuous-phase Sherwood number:

$$(N_{Sh})_C = \frac{k_C d_{vs}}{D_C} = 2 \quad (8-41)$$

where D_C is the diffusivity of the solute in the continuous phase. However, if other spheres of equal diameter are located near the sphere of interest, $(N_{Sh})_C$ may decrease to a value as low as 1.386, according to Cornish [49]. In an agitated vessel, the continuous-phase Sherwood number will usually be much greater than 1.386. A reasonable estimate can be made with the semi-theoretical correlation of Skelland and Moeti [50]. They fitted 180 data points for three different solutes, three different dispersed organic solvents, and water as the continuous phase. Mass transfer was from the dispersed phase to the continuous phase, but only

for $\phi_D = 0.01$. Skelland and Moeti assumed an equation of the form

$$(N_{Sh})_C \propto (N_{Re})_C^y (N_{Sc})_C^x \quad (8-42)$$

where

$$(N_{Sh})_C = k_C d_{vs} / D_C \quad (8-43)$$

$$(N_{Sc})_C = \mu_C / \rho_C D_C \quad (8-44)$$

For the Reynolds number, they assumed that the characteristic velocity is the square root of the mean-square, local fluctuating velocity in the vicinity of the droplet, based on the theory of local isotropic turbulence of Batchelor [51]:

$$\bar{u}^2 \propto \left(\frac{P g_c}{V} \right)^{2/3} \left(\frac{d_{vs}}{\rho_C} \right)^{2/3} \quad (8-45)$$

Thus,

$$(N_{Re})_C = \frac{(\bar{u}^2)^{1/2} d_{vs} \rho_C}{\mu_C} \quad (8-46)$$

Combining (8-45) and (8-46), with omission of the proportionality constant:

$$(N_{Re})_C = \frac{d_{vs}^{4/3} \rho_C^{2/3} (P g_c / V)^{1/3}}{\mu_C} \quad (8-47)$$

As discussed previously in conjunction with Figure 8.36, in the turbulent-flow region,

$$P g_c \propto \rho_M N^3 D_i^5 \quad \text{or} \quad \text{for low } \phi_D, P g_c / V \propto \rho_C N^3 D_i^5 / D_T^3$$

Thus,

$$(N_{Re})_C = \frac{d_{vs}^{4/3} \rho_C N D_i^{5/3}}{\mu_C D_T} \quad (8-48)$$

Skelland and Moeti correlated their mass-transfer coefficient data with

$$k_C \propto D_C^{2/3} \mu_C^{-1/3} N^{3/2} d_{vs}^0$$

The exponents in this proportionality are used to determine the exponents y and x in (8-42) as $\frac{2}{3}$ and $\frac{1}{3}$, respectively. In addition, based on the work of previous investigators, a droplet Eotvos number,

$$N_{Eo} = \rho_D d_{vs}^2 g / \sigma \quad (8-49)$$

where N_{Eo} = (gravitational force)/(surface tension force) and the dispersed-phase holdup, ϕ_D , are incorporated into the following final correlation, which predicts 180 experimental data points to an average absolute deviation of 19.71%:

$$\begin{aligned} (N_{Sh})_C &= \frac{k_C d_{vs}}{D_C} = 1.237 \times 10^{-5} \left(\frac{\mu_C}{\rho_C D_C} \right)^{1/3} \\ &\times \left(\frac{D_i^2 N \rho_C}{\mu_C} \right)^{2/3} \phi_D^{-1/2} \left(\frac{D_i N^2}{g} \right)^{5/12} \\ &\times \left(\frac{D_i}{d_{vs}} \right)^2 \left(\frac{d_{vs}}{D_T} \right)^{1/2} \left(\frac{\rho_D d_{vs}^2 g}{\sigma} \right)^{5/4} \end{aligned} \quad (8-50)$$

EXAMPLE 8.7

For the system, conditions, and results of Examples 8.5 and 8.6, with the extract as the dispersed phase, estimate:

- The dispersed-phase mass-transfer coefficient, k_D
- The continuous-phase mass-transfer coefficient, k_C
- The Murphree dispersed-phase efficiency, E_{MD}
- The fractional extraction of furfural

The molecular diffusivities of furfural in toluene (dispersed) and water (continuous) at dilute conditions are, respectively,

$$D_D = 8.32 \times 10^{-5} \text{ ft}^2/\text{h} \quad \text{and} \quad D_C = 4.47 \times 10^{-5} \text{ ft}^2/\text{h}$$

The distribution coefficient for dilute conditions is $m = d_{cC}/d_{cD} = 0.0985$.

SOLUTION

(a) From (8-40), $k_D = 6.6(D_D)/d_{vs} = 6.6(8.32 \times 10^{-5})/0.00124 = 0.44 \text{ ft/h}$

(b) To apply (8-50) to the estimation of k_C , first compute each of the dimensionless groups in that equation:

$$N_{Sc} = \mu_C / \rho_C D_C = 2.165 / [(62.3)(4.47 \times 10^{-5})] = 777$$

$$N_{Re} = D_i^2 N \rho_C / \mu_C = (1)^2 (8,820)(62.3) / 2.165 = 254,000$$

$$N_{Fr} = D_i N^2 / g = (1)(8,820)^2 / (4.17 \times 10^8) = 0.187$$

$$D_i / d_{vs} = 1 / 0.00124 = 806; \quad d_{vs} / D_T = 0.00124 / 3 = 0.000413$$

$$N_{Eo} = \rho_D d_{vs}^2 g / \sigma = (54.2)(0.00124)^2 (4.17 \times 10^8) / 718,800 = 0.0483$$

From (8-50),

$$N_{Sh} = 1.237 \times 10^{-5} (777)^{1/3} (254,000)^{2/3} (0.388)^{-1/2} (0.187)^{5/12} \times (806)^2 (0.000413)^{1/2} (0.0483)^{5/4} = 109$$

which is much greater than the value of 2 in a quiescent fluid.

$$k_C = N_{Sh} D_C / d_{vs} = (109)(4.47 \times 10^{-5}) / 0.00124 = 3.93 \text{ ft/h}$$

(c) From (8-28) and the results of Example 8.6,

$$K_{ODa} = \left\{ \frac{1}{1/0.44 + 1/[(0.0985)(3.93)]} \right\} 1,880 = 387 \text{ h}^{-1}$$

From (8-32), with $V = \pi D_T^2 H / 4 = (3.14)(3)^2(3) / 4 = 21.2 \text{ ft}^3$

$$N_{OD} = K_{ODa} V / Q_D = 387(21.2) / 207 = 39.6$$

From (8-33),

$$E_{MD} = (N_{OD} / (1 + N_{OD})) = 39.6 / (1 + 39.6) = 0.975 = 97.5\%$$

(d) By material balance,

$$Q_C(c_{C,in} - c_{C,out}) = Q_D c_{D,out} \quad (1)$$

From (8-26),

$$E_{MD} = c_{D,out} / c_D^* = m c_{D,out} / c_{C,out} \quad (2)$$

Combining (1) and (2) to eliminate $c_{D,out}$ gives

$$\frac{c_{C,out}}{c_{C,in}} = \frac{1}{1 + Q_D E_{MD} / (Q_C m)} \quad (3)$$

and

$$f_{\text{Extracted}} = \frac{c_{C,\text{in}} - c_{C,\text{out}}}{c_{C,\text{in}}} = 1 - \frac{c_{C,\text{out}}}{c_{C,\text{in}}} = \frac{Q_D E_{MD} / (Q_C m)}{1 + Q_D E_{MD} / (Q_C m)}$$

$$\frac{Q_D}{Q_C} \frac{E_{MD}}{m} = \frac{(207)(0.975)}{(327)(0.0985)} = 6.27$$

Thus,

$$f_{\text{Extracted}} = \frac{6.27}{1 + 6.27} = 0.862 \quad \text{or} \quad 86.2\%$$

Multicompartment Columns

Sizing extraction columns, which may or may not include mechanical agitation, involves the determination of column diameter and column height. The diameter must be sufficiently large to permit the two phases to flow countercurrently through the column without flooding. The column height must be sufficient to achieve the number of equilibrium stages corresponding to the desired degree of extraction.

For small-diameter columns, rough estimates of the diameter and height can be made using the results of a study by Stichlmair [52] with the toluene–acetone–water system for $Q_D/Q_C = 1.5$. Typical ranges of $1/\text{HETS}$ and the sum of the superficial phase velocities for a number of extractor types are given in Table 8.6.

Because of the large number of important variables, an accurate estimation of column diameter for liquid–liquid contacting devices is far more complex and more uncertain than for vapor–liquid contactors. These variables include individual phase flow rates, density difference between the two phases, interfacial tension, direction of mass-transfer, viscosity and density of the continuous phase, rotating or reciprocating speed, and geometry of internals. Column diameter is best determined by scale-up from tests run in standard laboratory or pilot-plant test units with a diameter of 1 in. or larger. The sum of the measured superficial velocities of the two liquid phases in the test unit can then be assumed to hold for larger commercial units. This sum is often expressed in total gallons per hour per square foot of empty column cross-section area.

In the absence of laboratory data, preliminary estimates of diameter for some columns can be made by a simplification of the theory of Logsdaile, Thornton, and Pratt [53], which is

Table 8.6 Performance of Several Types of Column Extractors

Extractor Type	$1/\text{HETS}, \text{m}^{-1}$	$U_D + U_C, \text{m/h}$
Packed column	1.5–2.5	12–30
Pulsed packed column	3.5–6	17–23
Sieve-plate column	0.8–1.2	27–60
Pulsed-plate column	0.8–1.2	25–35
Scheibel column	5–9	10–14
RDC	2.5–3.5	15–30
Kuhni column	5–8	8–12
Karr column	3.5–7	30–40
RTL contactor	6–12	1–2

Source: J. Stichlmair, *Chemie-Ingenieur-Technik*, 52, 253 (1980).

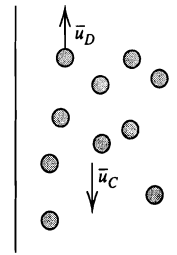


Figure 8.37 Countercurrent flows of dispersed and continuous liquid phases in a column.

compared to other procedures by Landau and Houlihan [54] in the case of the rotating-disk contactor. Because the relative motion between a dispersed droplet phase and a continuous phase is involved, this theory is based on a concept that is similar to that developed in Chapter 6 for liquid droplets dispersed in a vapor phase.

Consider the case of liquid droplets of the lower-density phase rising through the denser, downward-flowing, continuous liquid phase, as shown in Figure 8.37. If the average superficial velocities of the discontinuous (droplet) phase and the continuous phase are U_D in the upward direction and U_C in the downward direction (i.e., both of these velocities are positive), respectively, the corresponding average actual velocities relative to the column wall are

$$\bar{u}_D = \frac{U_D}{\phi_D} \quad (8-51)$$

and

$$\bar{u}_C = \frac{U_C}{1 - \phi_D} \quad (8-52)$$

The average droplet rise velocity relative to the continuous phase is the sum of (8-51) and (8-52):

$$\bar{u}_r = \frac{U_D}{\phi_D} + \frac{U_C}{1 - \phi_D} \quad (8-53)$$

This relative velocity (also called *slip velocity*) can be expressed in terms of a modified form of (6-40) where the continuous-phase density in the buoyancy term is replaced by the density of the two-phase mixture, ρ_M . Thus, after noting for the case here that the drag force, F_d , and gravitational force, F_g , act downward while buoyancy, F_b , acts upward, we obtain

$$\bar{u}_r = C \left(\frac{\rho_M - \rho_D}{\rho_C} \right)^{1/2} f\{1 - \phi_D\} \quad (8-54)$$

where C is the same parameter as in (6-41) and $f\{1 - \phi_D\}$ is a factor that allows for the hindered rising effect of neighboring droplets. The density ρ_M is a volumetric mean given by

$$\rho_M = \phi_D \rho_D + (1 - \phi_D) \rho_C \quad (8-55)$$

$$\rho_M - \rho_D = (1 - \phi_D)(\rho_C - \rho_D) \quad (8-56)$$

Substitution of (8-56) into (8-54) yields

$$\bar{u}_r = C \left(\frac{\rho_C - \rho_D}{\rho_C} \right)^{1/2} (1 - \phi_D)^{1/2} f\{1 - \phi_D\} \quad (8-57)$$

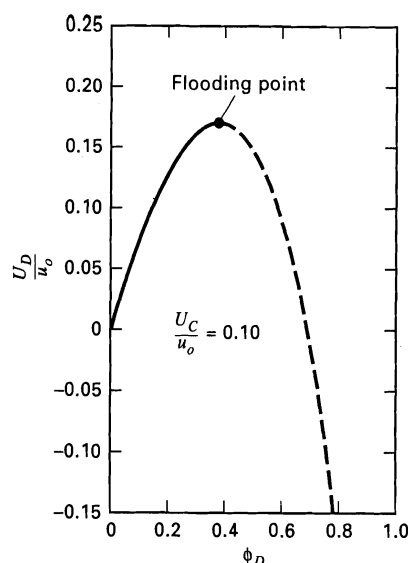


Figure 8.38 Typical holdup curve for liquid-liquid extraction column.

From experimental data, Gayler, Roberts, and Pratt [55] found that, for a given liquid-liquid system, the right-hand side of (8-57) can be expressed empirically as

$$\bar{u}_r = u_0(1 - \phi_D) \quad (8-58)$$

where u_0 is a characteristic rise velocity for a single droplet, which depends on all the variables discussed above, except those on the right-hand side of (8-53). Thus, for a given liquid-liquid system, column design, and operating conditions, the combination of (8-53) and (8-58) gives

$$\frac{U_D}{\phi_D} + \frac{U_C}{1 - \phi_D} = u_0(1 - \phi_D) \quad (8-59)$$

where u_0 is a constant. Equation (8-59) is cubic in ϕ_D , with a typical solution shown in Figure 8.38 for $U_C/u_0 = 0.1$. Thornton [56] argues that, with U_C fixed, an increase in U_D results in an increased value of the holdup ϕ_D , until the flooding point is reached, at which $(\partial U_D / \partial \phi_D)_{U_C} = 0$. Thus, in Figure 8.38, only that portion of the curve for $\phi_D = 0$ to $(\phi_D)_f$, the holdup at the flooding point, is realized in practice. Alternatively, with U_D fixed, $(\partial U_C / \partial \phi_D)_{U_D} = 0$ at the flooding point. If these two derivatives are applied to (8-59), we obtain, respectively,

$$U_C = u_0[1 - 2(\phi_D)_f][1 - (\phi_D)_f]^2 \quad (8-60)$$

$$U_D = 2u_0[1 - (\phi_D)_f](\phi_D)_f^2 \quad (8-61)$$

where the subscript f denotes flooding. Combining (8-60) and (8-61) to eliminate u_0 gives the following expression for $(\phi_D)_f$:

$$(\phi_D)_f = \frac{[1 + 8(U_C/U_D)]^{0.5} - 3}{4[(U_C/U_D) - 1]} \quad (8-62)$$

This equation predicts values of $(\phi_D)_f$ ranging from zero at $U_D/U_C = 0$ to 0.5 at $U_C/U_D = 0$. At $U_D/U_C = 1$, $(\phi_D)_f = \frac{1}{3}$. The simultaneous solution of (8-59) and (8-62) results in Figure 8.39 for the variation of total capacity as a

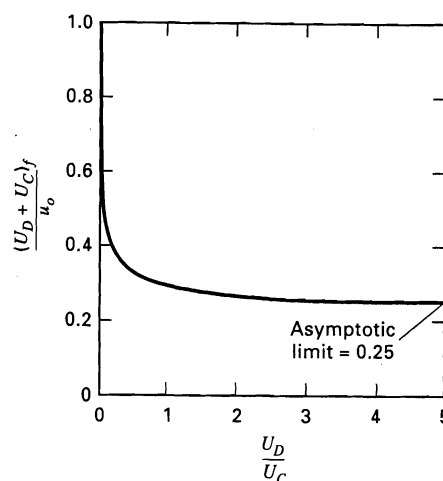


Figure 8.39 Effect of phase ratio on total capacity of liquid-liquid extraction column.

function of phase flow ratio. The largest total capacities are achieved, as might be expected, at the smallest ratios of dispersed-phase flow rate to continuous-phase flow rate.

For fixed values of column geometry and rotor speed, experimental data of Logsdail et al. [53] for a laboratory-scale RDC indicate that the dimensionless group $(u_0 \mu_C \rho_C / \sigma \Delta \rho)$ is approximately constant. Data of Reman and Olney [57] and Strand, Olney, and Ackerman [58] for well-designed and efficiently operated commercial RDC columns ranging from 8 to 42 in. in diameter indicate that this dimensionless group has a value of roughly 0.01 for systems involving water as either the continuous or dispersed phase. This value is suitable for preliminary calculations of RDC and Karr column diameters, when the sum of the actual superficial phase velocities is taken as 50% of the estimated sum at flooding conditions.

EXAMPLE 8.8

Estimate the diameter of an RDC to extract acetone from a dilute toluene-acetone solution into water at 20°C. The flow rates for the dispersed organic and continuous aqueous phases are 27,000 and 25,000 lb/h, respectively.

SOLUTION

The necessary physical properties are

$$\mu_C = 1.0 \text{ cP } (0.000021 \text{ lbf-s/ft}^2) \text{ and } \rho_C = 1.0 \text{ g/cm}^3$$

$$\Delta \rho = 0.14 \text{ g/cm}^3 \text{ and } \sigma = 32 \text{ dyne/cm } (0.00219 \text{ lbf/ft})$$

$$\frac{U_D}{U_C} = \left(\frac{27,000}{25,000} \right) \left(\frac{\rho_C}{\rho_D} \right) = \left(\frac{27,000}{25,000} \right) \left(\frac{1.0}{0.86} \right) = 1.26$$

From Figure 8.39, $(U_D + U_C)_f / u_0 = 0.29$.

Assume that $u_0 \mu_C \rho_C / \sigma \Delta \rho = 0.01$.

Therefore,

$$u_0 = \frac{(0.01)(0.00219)(0.14)}{(0.000021)(1.0)} = 0.146 \text{ ft/s}$$

$$(U_D + U_C)_f = 0.29(0.146) = 0.0423 \text{ ft/s}$$

$$(U_D + U_C)_{50\% \text{ of flooding}} = \left(\frac{0.0423}{2} \right) (3,600) = 76.1 \text{ ft/h}$$

$$\text{Total ft}^3/\text{h} = \frac{27,000}{(0.86)(62.4)} + \frac{25,000}{(1.0)(62.4)} = 904 \text{ ft}^3/\text{h}$$

$$\text{Column cross-sectional area} = A_c = \frac{904}{76.1} = 11.88 \text{ ft}^2$$

$$\text{Column diameter} = D_T = \left(\frac{4A_c}{\pi} \right)^{0.5} = \left[\frac{(4)(11.88)}{3.14} \right]^{0.5} = 3.9 \text{ ft}$$

Note that from Table 8.6, a typical $(U_D + U_C)$ for an RDC is 15 to 30 m/h or 49 to 98.4 ft/h.

Despite their compartmentalization, mechanically assisted liquid-liquid extraction columns, such as the RDC and Karr columns, operate more nearly like differential contacting devices than like staged contactors. Therefore, it is more common to consider stage efficiency for such columns in terms of HETS (height equivalent to a theoretical stage) or as some function of mass-transfer parameters, such as HTU (height of a transfer unit). Although it is not on as sound a theoretical basis as the HTU, the HETS is preferred here because it can be applied directly to determine column height from the number of equilibrium stages.

Because of the great complexity of liquid-liquid systems and the large number of variables that influence contacting efficiency, general correlations for HETS have been difficult to develop. However, for well-designed and efficiently operated columns, the available experimental data indicate that the dominant physical properties influencing HETS are the interfacial tension, the phase viscosities, and the density difference between the phases. In addition, it has been observed by Reman [59] for RDC units and by Karr and Lo [60] for Karr columns that HETS increases with increasing column diameter because of axial mixing effects discussed in the next section.

It is preferred to obtain values of HETS by conducting small-scale laboratory experiments with systems of interest. These values are scaled to commercial-size columns by assuming that HETS varies with column diameter D_T , raised to an exponent, which may vary from 0.2 to 0.4 depending on the system.

In the absence of experimental data, the crude correlation of Figure 8.40 can be used for preliminary design if phase viscosities are no greater than 1 cP. The data points correspond to minimum reported HETS values for RDC and Karr units with the exponent on column diameter set arbitrarily to $\frac{1}{3}$. The points represent values of HETS that vary from as low as 6 in. for a 3-in.-diameter, laboratory-size column operating with a low-interfacial-tension/low-viscosity system such as methyl-isobutyl ketone/acetic acid/water, to as high as 25 in. for a 36-in.-diameter commercial column operating with a high-interfacial-tension/low-viscosity system such as xylenes-acetic acid-water. For systems having one phase of high viscosity, values of HETS can be 24 in. or more, even for a small, laboratory-size column.

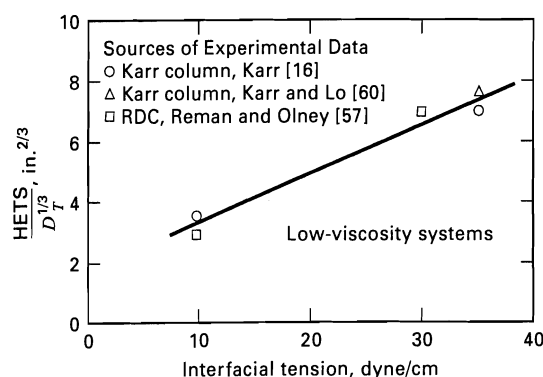


Figure 8.40 Effect of interfacial tension on HETS for RDC and Karr columns.

EXAMPLE 8.9

Estimate HETS for the conditions of Example 8.8.

SOLUTION

Because toluene has a viscosity of approximately 0.6 cP, this is a low-viscosity system. From Example 8.8, the interfacial tension is 32 dyne/cm. From Figure 8.40, $\text{HETS}/D_T^{1/3} = 6.9$. For $D_T = 3.9$ ft, $\text{HETS} = 6.9[(3.9)(12)]^{1/3} = 24.8$ in. Note that from Table 8.6, HETS for an RDC varies from 0.29 to 0.40 m or 11.4 to 15.7 in. for a small column.

More accurate estimates of flooding and HETS are discussed in detail by Lo et al. [4] and by Thornton [61]. Packed column design is considered by Strigle [62].

Axial Dispersion

In this and previous chapters covering liquid-liquid and vapor-liquid countercurrent-flow contactors, plug flow of each phase has been assumed. Each element of a phase is assumed to have the same residence time in the contactor, while each phase may have a different residence time. Because axial concentration gradients in the direction of bulk flow are established in each phase, diffusion of a species is superimposed on the bulk flow of the species in that phase. Axial diffusion degrades the efficiency of multistage separation equipment, and in the limit, a multistage separator behaves like a single well-mixed stage. In Figure 8.41, solute concentration profiles for the extract and raffinate phases of a liquid-liquid extraction column are shown for plug flow (dashed lines) and for flow with significant axial diffusion in each of the two phases (solid lines). The continuous phase is the feed/raffinate (x subscript), which enters the contactor at the top ($z = 0$). The dispersed phase is the solvent/extract (y subscript), which enters the contactor at the bottom ($z = H$). Solute transfer is from the continuous phase to the dispersed phase. Two effects of axial diffusion are seen: (1) The concentration curves in the presence of axial diffusion are closer together than for plug flow and (2) these close proximities are due partially to concentrations at the two ends, which are different from those in the original feed and

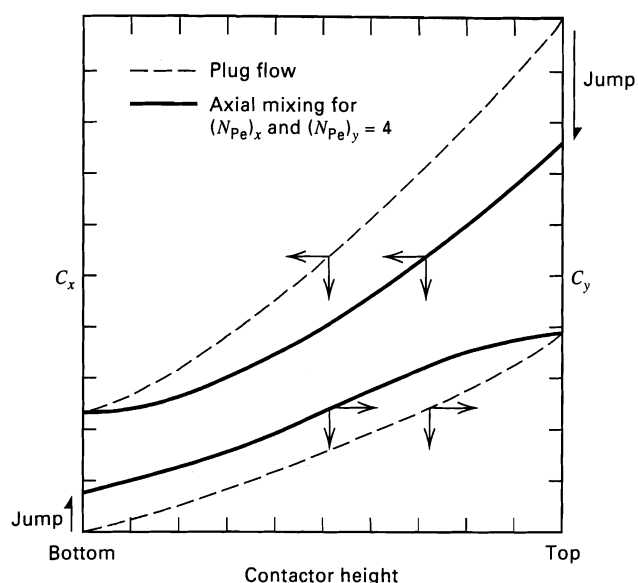


Figure 8.41 Solute concentration profiles for continuous, countercurrent extraction with and without axial mixing.

solvent. These differences are called *jumps* and are due to axial diffusion outside the region in the contactor where the two liquid phases are in contact. The jump at the top is caused by axial diffusion, superimposed on the bulk flow, in the feed liquid before it enters the contactor. This causes the concentration of solute in the feed just as it enters the contactor to be less than its concentration in the original feed liquid. Similarly, diffusion of solute into the incoming solvent causes the concentration of solute in the solvent just entering the bottom of the contactor to be greater than the concentration in the original solvent, which in Figure 8.41 is zero. The overall effect of axial diffusion is a reduction in the average driving force for mass transfer of the solute between the two phases, necessitating a taller column to accomplish the desired separation.

The effects shown in Figure 8.41 are actually due to a number of factors besides diffusion, which are lumped together into one overall effect, commonly referred to as *axial dispersion*, *axial mixing*, *longitudinal dispersion*, or *back-mixing*. These factors include:

1. Molecular and turbulent diffusion of the continuous phase along concentration gradients
2. Circulatory motion of the continuous phase due to the droplets of the dispersed phase
3. Transport and shedding of the continuous phase in the wakes attached to the rear of droplets of the dispersed phase
4. Circulation of continuous and dispersed phases in mechanically agitated columns
5. Channeling and nonuniform velocity profiles leading to distributions of residence times in the two phases

In general, the effect of axial dispersion is most pronounced when (1) a high recovery of solute is necessary, (2) the contactor is short in height, (3) large circulation

patterns occur, (4) a wide range of droplet sizes is present, and/or (5) the feed-to-solvent flow ratio is very small or very large. Although axial-dispersion effects are generally negligible in extractors where phase separation occurs between stages, such as in mixer-settler cascades and sieve-plate columns with downcomers, axial dispersion can be significant in spray columns, packed columns, and RDCs. Although axial dispersion can occur in packed absorbers, packed strippers, and packed distillation columns, it is significant only when operating at very high liquid-to-gas ratios. However, axial dispersion can be significant in spray and bubble columns used for absorption.

Two types of models have been developed for predicting the extent and effect of axial mixing: (1) diffusion models for differential-type contactors, due to Sleicher [63] and Miyauchi and Vermeulen [64]; and (2) backflow models for staged extractors without complete phase separation between stages, due to Sleicher [65] and Miyauchi and Vermeulen [66]. Both types are discussed by Vermeulen et al. [67]. Diffusion models, which have received the most attention and have been most applied more frequently, are convenient for studying the complex nature of axial dispersion.

Consider a differential height, dz , of a differential contactor with countercurrent two-phase flow, as shown in Figure 8.42. Feed enters the top of the column at $z = 0$, while solvent enters the bottom of the column at $z = H$. Assume that: (1) axial dispersion in each phase is characterized by a constant turbulent-diffusion coefficient, E ; (2) phase superficial velocities are each uniform over the cross section and

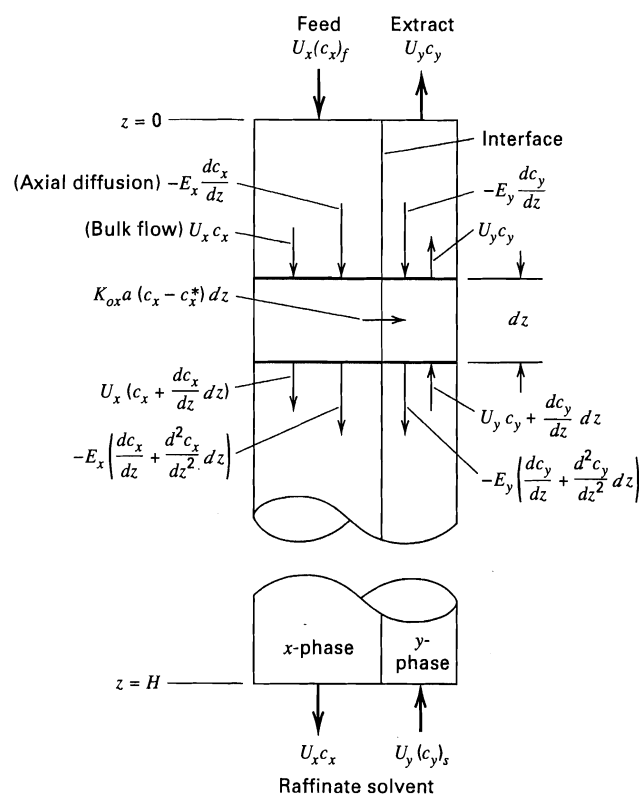


Figure 8.42 Axial dispersion in an extraction column.

[From J.D. Thornton, *Science and Practice of Liquid-Liquid Extraction*, Vol. 1, Clarendon Oxford, (1992) with permission.]

constant in the axial direction; (3) the volumetric, overall mass-transfer coefficients for the solute are constant; (4) only the solute undergoes mass transfer between the two phases; and (5) the phase equilibrium ratio for the solute is constant. Then the solute mass-balance equations for the feed/raffinate (x) and solvent/extract (y) phases, respectively, are

$$E_x \frac{d^2 c_x}{dz^2} - U_x \frac{dc_x}{dz} - K_{Ox} a (c_x - c_x^*) = 0 \quad (8-63)$$

$$E_y \frac{d^2 c_y}{dz^2} + U_y \frac{dc_y}{dz} + K_{Ox} a (c_x - c_x^*) = 0 \quad (8-64)$$

where c_x^* is the concentration of the solute in the raffinate that is equilibrium with the solute concentration in the bulk extract. For these two differential equations, the boundary conditions, which were first proposed by Danckwerts [68] and were further elucidated by Wehner and Wilhelm [69], are

$$\text{at } z = 0, \quad U_x c_{xf} - U_x c_{x0} = -E_x \frac{dc_x}{dz} \quad (8-65)$$

and

$$dc_y/dz = 0 \quad (8-66)$$

at $z = H$,

$$U_y c_{yH} - U_y c_{ys} = E_y \frac{dc_y}{dz} \quad (8-67)$$

and

$$dc_x/dz = 0 \quad (8-68)$$

where:

c_{xf} = concentration of solute in the original feed

c_{x0} = concentration of solute in the feed at $z = 0$

c_{yH} = concentration of solute in the solvent at $z = H$

c_{ys} = concentration of solute in the original solvent

The two terms on the left-hand sides of (8-65) and (8-67) are the jumps shown in Figure 8.41.

It is customary to convert (8-63) and (8-64) to alternative forms in terms of pertinent dimensionless groups. This is readily done by defining

$$Z = z/H \quad (8-69)$$

$$N_{Pe_y} = U_y H / E_y = \text{axial, turbulent column Peclet number for the extract phase} \quad (8-70)$$

$$N_{Pe_x} = U_x H / E_x = \text{axial, turbulent column Peclet number for the raffinate phase} \quad (8-71)$$

$$N_{Ox} = K_{Ox} a H / U_x = K_{Ox} a V / Q_x \quad (8-72)$$

Equations (8-63) and (8-64) then become

$$\frac{d^2 c_x}{dZ^2} - N_{Pe_x} \frac{dc_x}{dZ} - N_{Ox} N_{Pe_x} (c_x - c_x^*) = 0 \quad (8-73)$$

$$\frac{d^2 c_y}{dZ^2} - N_{Pe_y} \frac{dc_y}{dZ} - \left(\frac{U_x}{U_y} \right) N_{Ox} N_{Pe_y} (c_x - c_x^*) = 0 \quad (8-74)$$

The boundary conditions are transformed in a similar way. For a straight equilibrium curve, $c_x^* = m c_y$. Thus, we have a coupled set of ordinary differential equations, whose solutions for c_y and c_x are functions of c_{yf} , c_{ys} , m , N_{Pe_x} , N_{Pe_y} , N_{Ox} , U_x/U_y , and Z .

Further algebraic manipulations involving the substitution of dimensionless solute concentrations can reduce the number of variables from 10 to 7. In either case, the solution of the axial dispersion equations as obtained by Sleicher [65] and Miyauchi and Vermeulen [66] is very difficult to display in tabular or graphical form. However, the possible importance of axial dispersion is most commonly judged by the magnitudes of the Peclet numbers. A Peclet number of 0 corresponds to complete back-mixing, such that at most only one equilibrium stage is achieved; the entire column functions like a single mixer stage. A Peclet number of ∞ corresponds to an absence of axial dispersion. Experimental data on several different types of liquid-liquid extraction columns indicate that N_{Pe} for the dispersed phase is frequently greater than 50, while N_{Pe} for the continuous phase may be in the range of 5 to 30. Thus, as a first approximation, axial dispersion in the dispersed phase can be largely ignored. This effect was observed experimentally by Geankoplis and Hixson [70] in a spray extraction column and by Gier and Hougen [71] in spray and packed extraction columns. They reported end-concentration changes of significant magnitude at the continuous-phase entrance, but not at the dispersed-phase entrance.

A number of approximate solutions to the axial dispersion equations, (8-63) to (8-68), have been published, including one by Sleicher [63]. Alternatively, if the original solvent is free of solute, a rapid and somewhat conservative estimate of the effect of axial dispersion can be made by the method of Watson and Cochran [72] from an empirical relation for the column efficiency:

$$\frac{H_{\text{plug flow}}}{H_{\text{actual}}} = \frac{(HTU_{Ox})(NTU_{Ox})}{H} = 1 - \frac{1}{1 + N_{Pe_x}(HTU_{Ox}/H) - E + (1/NTU_{Ox})} - \frac{E}{N_{Pe_y}(HTU_{Ox}/H) - 1 + E + (1/NTU_{Ox})} \quad (8-75)$$

where

$H_{\text{actual}} = H$ = height of column taking into account axial dispersion

HTU_{Ox} = height of an overall transfer unit based on the raffinate phase for plug flow

NTU_{Ox} = number of overall transfer units based on the raffinate phase for plug flow

E = extraction factor = $m U_x / U_y$

$m = dc_x/dc_y$

The product of HTU_{Ox} and NTU_{Ox} is the column height for plug flow, which is $< H$. Thus, the ratio on the left-hand side

of (8-75) is a column efficiency. The NTU_{Ox} is approximated by:

$$NTU_{Ox} = \ln \left(\frac{X}{XE + 1 - E} \right) / (E - 1) \quad (8-76)$$

$$\text{where } X = (c_x)_{\text{out}} / (c_x)_{\text{in}} \quad (8-77)$$

The HTU_{Ox} is defined by:

$$HTU_{Ox} = U_x / K_{Ox} a \quad (8-78)$$

For given values of HTU_{Ox} , NTU_{Ox} , E , N_{Pe_x} , and N_{Pe_y} , (8-75) is solved for H . Caution must be exercised in using (8-75) because of its empirical nature. The equation is limited to $NTU_{Ox} \geq 2$, $E > 0.25$, $N_{Pe_x}(HTU_{Ox}/H) > 1.5$, and the calculated value of the column efficiency, $H_{\text{plug flow}}/H_{\text{actual}}$, must be ≥ 0.20 . Within these restrictions, an extensive comparison by Watson and Cochran with the exact solution of (8-63) to (8-68) gives conservative efficiency values that deviate by no more than 0.07 (7%), with the highest accuracy for estimated efficiencies greater than 0.5 (50%).

SUMMARY

1. A solvent can be used to selectively extract one or more components from a liquid mixture.
2. Although liquid-liquid extraction is a reasonably mature separation operation, considerable experimental effort is often needed to find a suitable solvent and to determine residence-time requirements or values of HETS, NTU, or mass-transfer coefficients.
3. Compared to vapor-liquid separation operations, extraction has a higher overall mass-transfer resistance. Stage efficiencies in columns are frequently low.
4. A wide variety of commercial extractors are available, as shown in Figures 8.2 to 8.7, ranging from simple columns with no mechanical agitation to centrifugal devices that may spin at several thousand revolutions per minute. A selection scheme, given in Table 8.3, is useful for choosing the most suitable extractors for a given separation.
5. Solvent selection is facilitated by consideration of a number of chemical factors given in Table 8.4 and physical factors discussed in Section 8.2.
6. For liquid-liquid extraction with ternary mixtures, phase equilibrium is conveniently represented on equilateral- or right-triangle diagrams for both type I (solute and solvent completely miscible) and the less common type II (solute and solvent not completely miscible) systems.
7. For determining equilibrium-stage requirements of single-section, countercurrent cascades for ternary systems, the graphical methods of Hunter and Nash (equilateral-triangle diagram), Kinney (right-triangle diagram), or Varteressian and Fenske (distribution diagram of McCabe-Thiele type) can be applied, as described in Section 8.3. These methods can also determine minimum and maximum solvent requirements.
8. A two-section, countercurrent cascade with extract reflux can be employed with a type II ternary system to enable a sharp separa-

EXAMPLE 8.10

Experiments conducted for a dilute system under laboratory conditions approximating plug flow give $HTU_{Ox} = 3$ ft. If a commercial column is to be designed for $NTU_{Ox} = 4$ and N_{Pe_x} and N_{Pe_y} are estimated to be 19 and 50, respectively, determine the necessary column height if $E = 0.5$.

SOLUTION

For plug flow, column height is $(HTU_{Ox})(NTU_{Ox})$. Substitution of the data into (8-75) gives

$$\frac{12}{H} = 1 - \frac{1}{(57/H) + 0.75} - \frac{0.5}{(180/H) - 0.25}$$

This is a nonlinear algebraic equation in H . Solving by an iterative method,

$$H = 17 \text{ ft}$$

$$\text{Efficiency} = (HTU_{Ox})(NTU_{Ox})/H = 12/17 = 0.706 \quad (70.6\%)$$

tion of a binary feed mixture. The calculation of stage requirements of such a two-section cascade is conveniently carried out by the graphical method of Maloney and Schubert using a Janecke equilibrium diagram, as discussed in Section 8.4. The addition of raffinate reflux to such a cascade is of little value. The Maloney-Schubert method can also be applied to single-section cascades.

9. When only a few equilibrium stages are required, a cascade of mixer-settler units may be attractive because each mixer can be designed to closely approach an equilibrium stage. With many ternary and higher-order systems, the residence-time requirement may be only a few minutes for a 90% approach to equilibrium using an agitator input of approximately 4 hp/1,000 gal. Adequate phase-disengaging area for the settlers may be estimated from the rule of 5 gal of combined extract and raffinate per minute per square foot of disengaging area.

10. For mixers utilizing a six-flat-bladed turbine in a closed vessel with side vertical baffles, as shown in Figure 8.35, useful extractor design correlations are available for estimating, for a given extraction, the mixing-vessel dimensions, minimum impeller rotation rate for complete and uniform dispersion, impeller horsepower, mean droplet size, range of droplet sizes, interfacial area per unit volume, dispersed-phase and continuous-phase mass-transfer coefficients, and Murphree efficiency.

11. For column-type extractors, with and without mechanical agitation, correlations for determining column diameter, to avoid flooding, and column height are suitable only for very preliminary sizing calculations. For final extractor selection and design, recommendations of equipment vendors based on experimental data from pilot-size equipment are highly desirable.

12. Sizing of column-type extractors must consider axial dispersion, which can significantly reduce mass-transfer driving forces and thus increase the required column height. Axial dispersion effects are often most significant in the continuous phase.

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EXERCISES

Section 8.1

- 8.1 Explain why it is preferable to separate a dilute mixture of benzoic acid in water by liquid–liquid extraction rather than distillation.
- 8.2 Why is liquid–liquid extraction preferred over distillation for the separation of a mixture of formic acid and water?
- 8.3 Based on the information in Table 8.3 and the selection scheme in Figure 8.8, is the choice of an RDC appropriate for the extraction of acetic acid from water by ethyl acetate in the process described in the introduction to this chapter and shown in Figure 8.1? What other types of extractors might be considered?
- 8.4 What is the major advantage of the ARD over the RDC? What is the disadvantage of the ARD compared to the RDC?
- 8.5 Under what conditions is a cascade of mixer-settler units probably the best choice of extraction equipment?

8.6 A petroleum reformat stream of 4,000 bbl/day is to be contacted with diethylene glycol to extract the aromatics from the paraffins. The ratio of solvent volume to reformat volume is 5. It is estimated that eight theoretical stages will be needed. Using Tables 8.2 and 8.3, and Figure 8.8, which types of extractors would be most suitable?

Section 8.2

- 8.7 Using Table 8.4, select possible liquid–liquid extraction solvents for separating the following mixtures: (a) water–ethyl alcohol, (b) water–aniline, and (c) water–acetic acid. For each case, indicate clearly which of the two components should be the solute.
- 8.8 Using Table 8.4, select possible liquid–liquid extraction solvents for removing the solute from the carrier in the following cases:

	Solute	Carrier
(a)	Acetone	Ethylene glycol
(b)	Toluene	<i>n</i> -Heptane
(c)	Ethyl alcohol	Glycerine

8.9 For the extraction of acetic acid (*A*) from a dilute solution in water (*C*) into ethyl acetate (*S*) at 25°C, estimate or obtain data for (K_A)_D, (K_C)_D, (K_S)_D, and β_{AC} . Does this system exhibit: (a) High selectivity, (b) High solvent capacity and (c) Ease in recovering the solvent? Can you select a solvent that would exhibit better factors than ethyl acetate?

8.10 Interfacial tension can be an important factor in liquid–liquid extraction. Very low values of interfacial tension result in stable emulsions that are difficult to separate, while very high values require large energy inputs to form the dispersed phase. It is best to measure the interfacial tension for the two-phase mixture of interest. However, in the absence of experimental data, propose a method for estimating the interfacial tension of a ternary system

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using only the compositions of the equilibrium phases and the values of surface tension in air for each of the three components.

Section 8.3

8.11 One thousand kilograms per hour of a 45 wt% acetone in-water solution is to be extracted at 25°C in a continuous, counter-current system with pure 1,1,2-trichloroethane to obtain a raffinate containing 10 wt% acetone. Using the following equilibrium data, determine with an equilateral-triangle diagram:

- (a) the minimum flow rate of solvent,
- (b) the number of stages required for a solvent rate equal to 1.5 times the minimum, and
- (c) the flow rate and composition of each stream leaving each stage.

	Acetone, Weight Fraction	Water, Weight Fraction	Trichloroethane, Weight Fraction
Extract	0.60	0.13	0.27
	0.50	0.04	0.46
	0.40	0.03	0.57
	0.30	0.02	0.68
	0.20	0.015	0.785
	0.10	0.01	0.89
Raffinate	0.55	0.35	0.10
	0.50	0.43	0.07
	0.40	0.57	0.03
	0.30	0.68	0.02
	0.20	0.79	0.01
	0.10	0.895	0.005

The tie-line data are:

Raffinate, Weight Fraction Acetone	Extract, Weight Fraction Acetone
0.44	0.56
0.29	0.40
0.12	0.18

8.12 Solve Exercise 8.11 with a right-triangle diagram.

8.13 A distillate containing 45 wt% isopropyl alcohol, 50 wt% diisopropyl ether, and 5 wt% water is obtained from the heads column of an isopropyl alcohol finishing unit. The company desires to recover the ether from this stream by liquid–liquid extraction in a column, with water, as the solvent, entering the top and the feed entering the bottom so as to produce an ether containing no more than 2.5 wt% alcohol and to obtain the extracted alcohol at a concentration of at least 20 wt%. The unit will operate at 25°C and 1 atm. Using the method of Varteressian and Fenske with a McCabe–Thiele diagram, find how many theoretical stages are required.

Is it possible to obtain an extracted alcohol composition of 25 wt%? Equilibrium data are given below.

PHASE EQUILIBRIUM DATA AT 25°C, 1 ATM

Ether Phase			Water Phase		
Wt% Alcohol	Wt% Ether	Wt% Water	Wt% Alcohol	Wt% Ether	Wt% Water
2.4	96.7	0.9	8.1	1.8	90.1
3.2	95.7	1.1	8.6	1.8	89.6
5.0	93.6	1.4	10.2	1.5	88.3
9.3	88.6	2.1	11.7	1.6	86.7
24.9	69.4	5.7	17.5	1.9	80.6
38.0	50.2	11.8	21.7	2.3	76.0
45.2	33.6	21.2	26.8	3.4	69.8

ADDITIONAL POINTS ON PHASE BOUNDARY

Wt% Alcohol	Wt% Ether	Wt% Water
45.37	29.70	24.93
44.55	22.45	33.00
39.57	13.42	47.01
36.23	9.66	54.11
24.74	2.74	72.52
21.33	2.06	76.61
0	0.6	99.4
0	99.5	0.5

8.14 Benzene and trimethylamine (TMA) are to be separated in a three-stage liquid-liquid extraction column using water as the solvent. If the solvent-free extract and raffinate products are to contain, respectively, 70 and 3 wt% TMA, find the original feed composition and the water-to-feed ratio with a right-triangle diagram. There is no reflux and the solvent is pure water. Equilibrium data are as follows:

TRIMETHYLAMINE-WATER-BENZENE COMPOSITIONS ON PHASE BOUNDARY

Extract, wt%			Raffinate, wt%		
TMA	H ₂ O	Benzene	TMA	H ₂ O	Benzene
5.0	94.6	0.4	5.0	0.0	95.0
10.0	89.4	0.6	10.0	0.0	90.0
15.0	84.0	1.0	15.0	1.0	84.0
20.0	78.0	2.0	20.0	2.0	78.0
25.0	72.0	3.0	25.0	4.0	71.0
30.0	66.4	3.6	30.0	7.0	63.0
35.0	58.0	7.0	35.0	15.0	50.0
40.0	47.0	13.0	40.0	34.0	26.0

The tie-line data are:

Extract, wt% TMA	Raffinate, wt% TMA
39.5	31.0
21.5	14.5
13.0	9.0
8.3	6.8
4.0	3.5

8.15 The system docosane-diphenylhexane (DPH)-furfural is representative of more complex systems encountered in the solvent

refining of lubricating oil. Five hundred kilograms per hour of a 40 wt% mixture of DPH in docosane are to be continuously extracted in a countercurrent system with 500 kg/h of a solvent containing 98 wt% furfural and 2 wt% DPH to produce a raffinate that contains only 5 wt% DPH. Calculate with a right-triangle diagram the number of theoretical stages required and the number of kilograms per hour of DPH in the extract at 45°C and at 80°C. Equilibrium data are as follows.

EQUILIBRIUM DATA: BINODAL CURVES IN DOCOSANE-DIPHENYLHEXANE-FURFURAL SYSTEM [IND. ENG. CHEM., 35, 711 (1943)]

Wt% at 45°C			Wt% at 80°C		
Docosane	DPH	Furfural	Docosane	DPH	Furfural
96.0	0.0	4.0	90.3	0.0	9.7
84.0	11.0	5.0	50.5	29.5	20.0
67.0	26.0	7.0	34.2	35.8	30.0
52.5	37.5	10.0	23.8	36.2	40.0
32.6	47.4	20.0	16.2	33.8	50.0
21.3	48.7	30.0	10.7	29.3	60.0
13.2	46.8	40.0	6.9	23.1	70.0
7.7	42.3	50.0	4.6	15.4	80.0
4.4	35.6	60.0	3.0	7.0	90.0
2.6	27.4	70.0	2.2	0.0	97.8
1.5	18.5	80.0			
1.0	9.0	90.0			
0.7	0.0	99.3			

The tie lines in the docosane-diphenylhexane-furfural system are:

Docosane Phase Composition, wt%			Furfural Phase Composition, wt%		
Docosane	DPH	Furfural	Docosane	DPH	Furfural
Temperature, 45°C:					
85.2	10.0	4.8	1.1	9.8	89.1
69.0	24.5	6.5	2.2	24.2	73.6
43.9	42.6	13.3	6.8	40.9	52.3
Temperature, 80°C:					
86.7	3.0	10.3	2.6	3.3	94.1
73.1	13.9	13.0	4.6	15.8	79.6
50.5	29.5	20.2	9.2	27.4	63.4

8.16 For each of the ternary systems shown in Figure 8.43, indicate whether: (a) simple, countercurrent extraction, or (b) countercurrent extraction with extract reflux, or (c) countercurrent extraction with raffinate reflux, or (d) countercurrent extraction with both extract and raffinate reflux would be expected to yield the most economical process.

8.17 Two solutions, feed F at the rate of 7,500 kg/h containing 50 wt% acetone and 50 wt% water, and feed F' at the rate of 7,500 kg/h containing 25 wt% acetone and 75 wt% water, are to be extracted in a countercurrent system with 5,000 kg/h of 1,1,2-trichloroethane at 25°C to give a raffinate containing 10 wt% acetone. Calculate the number of equilibrium stages required and the stage to which each feed should be introduced, using a right-triangle diagram. Equilibrium data are given in Exercise 8.11.

8.18 The three-stage extractor shown in Figure 8.44 is used to extract the amine from a fluid consisting of 40 wt% benzene (B) and 60 wt% trimethylamine (T). The solvent (water) flow to stage 3

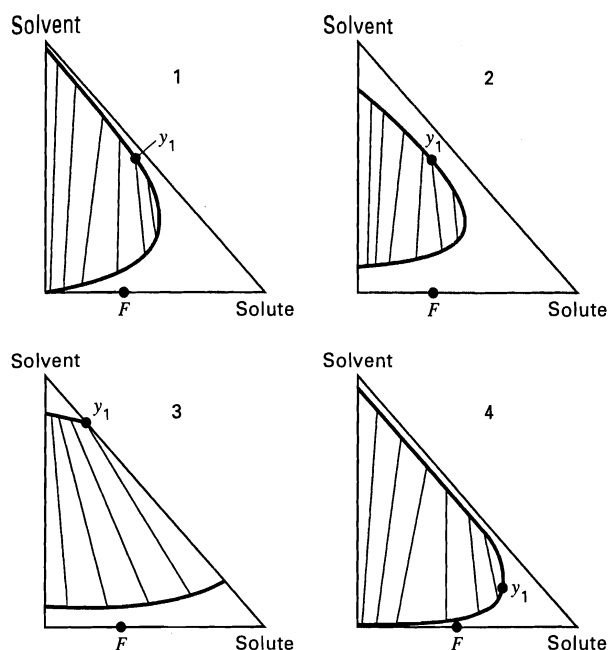


Figure 8.43 Data for Exercise 8.16.

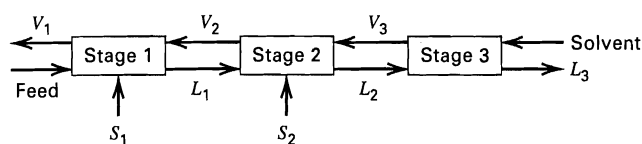


Figure 8.44 Data for Exercise 8.18.

is 5,185 kg/h and the feed flow rate is 10,000 kg/h. On a solvent-free basis V_1 is to contain 76 wt% T and L_3 is to contain 3 wt% T. Determine the required solvent flow rates S_1 and S_2 using an equilateral-triangle diagram. Equilibrium data are given in Exercise 8.14.

8.19 The extraction process shown Figure 8.45 is conducted in a multiple-feed, countercurrent unit without extract or raffinate reflux. Feed F' is composed of solvent and solute, and is an extract-phase feed. Feed F'' is composed of unextracted raffinate and solute and is a raffinate-phase feed. Derive the equations required to establish the three reference points needed to step off the theoretical stages in the extraction column. Show the graphical determination of these points on a right-triangle graph.

8.20 A mixture containing 50 wt% methylcyclohexane (MCH) in *n*-heptane is fed to a countercurrent, stage-type extractor at 25°C. Aniline is used as solvent. Reflux is used on both ends of the column.

An extract containing 95 wt% MCH and a raffinate containing 5 wt% MCH (both on solvent-free basis) are required. The minimum extract reflux ratio is 3.49. Using a right-triangle diagram with the equilibrium data of Exercise 8.22 below, calculate: (a) the raffinate

reflux ratio, (b) the amount of aniline that must be removed at the separator "on top" of the column, and (c) the amount of solvent that must be added to the solvent mixer at the bottom of the column.

8.21 In its natural state, zirconium, which is an important material of construction for nuclear reactors, is associated with hafnium, which has an abnormally high neutron-absorption cross section and must be removed before the zirconium can be used. Refer to Figure 8.46 for a proposed liquid-liquid extraction process wherein tributyl phosphate (TBP) is used as a solvent for the separation of hafnium from zirconium.

One liter per hour of 5.10-N HNO_3 containing 127 g of dissolved Hf and Zr oxides per liter is fed to stage 5 of the 14-stage extraction unit. The feed contains 22,000 g Hf per million g of Zr. Fresh TBP enters stage 14, while scrub water is fed to stage 1. Raffinate is removed at stage 14, while the organic extract phase that is removed at stage 1 goes to a stripping unit. The stripping operation consists of a single contact between fresh water and the organic phase. The following table gives experimental data. (a) Use these data to fashion a complete material balance for the process. (b) Check the data for consistency in as many ways as you can. (c) What is the advantage of running the extractor as shown? Would you recommend that all the stages be used?

STAGewise ANALYSES OF MIXER-SETTLER RUN

Stage	Organic Phase			Aqueous Phase		
	g oxide/ liter	N HNO_3 $\times (100)$	(Hf/Zr) $\times (100)$	g oxide/ liter	N HNO_3 $\times (100)$	(Hf/Zr)
1	22.2	1.95	<0.010	17.5	5.21	<0.010
2	29.3	2.02	<0.010	27.5	5.30	<0.010
3	31.4	2.03	<0.010	33.5	5.46	<0.010
4	31.8	2.03	0.043	34.9	5.46	0.24
5	32.2	2.03	0.11	52.8	5.15	3.6
6	21.1	1.99	0.60	30.8	5.15	6.8
7	13.7	1.93	0.27	19.9	5.05	9.8
8	7.66	1.89	1.9	11.6	4.97	20
9	4.14	1.86	4.8	8.06	4.97	36
10	1.98	1.83	10	5.32	4.75	67
11	1.03	1.77	23	3.71	4.52	110
12	0.66	1.68	32	3.14	4.12	140
13	0.46	1.50	42	2.99	3.49	130
14	0.29	1.18	28	3.54	2.56	72
Stripper		0.65		76.4	3.96	<0.01

[Data From R.P. Cox, H.C. Peterson, and C.H. Beyer, *Ind. Eng. Chem.*, 50 (2), 141 (1958). Exercise adapted from E.J. Henley and H. Bieber, *Chemical Engineering Calculations*, McGraw-Hill, New York, p. 298 (1959).]

8.22 At 45°C, 5,000 kg/h of a mixture of 65 wt% docosane, 7 wt% furfural, and 28 wt% diphenylhexane is to be extracted with pure furfural to obtain a raffinate with 12 wt% diphenylhexane in a continuous, countercurrent, multistage liquid-liquid extraction system. Phase-equilibrium data for this ternary system are given in

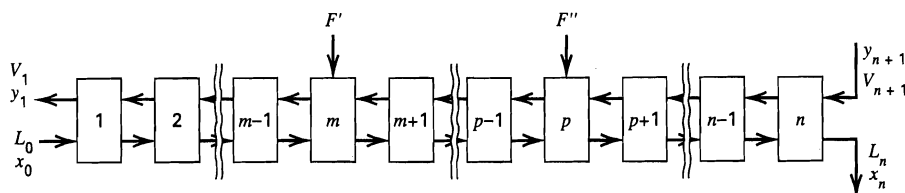


Figure 8.45 Data for Exercise 8.19.

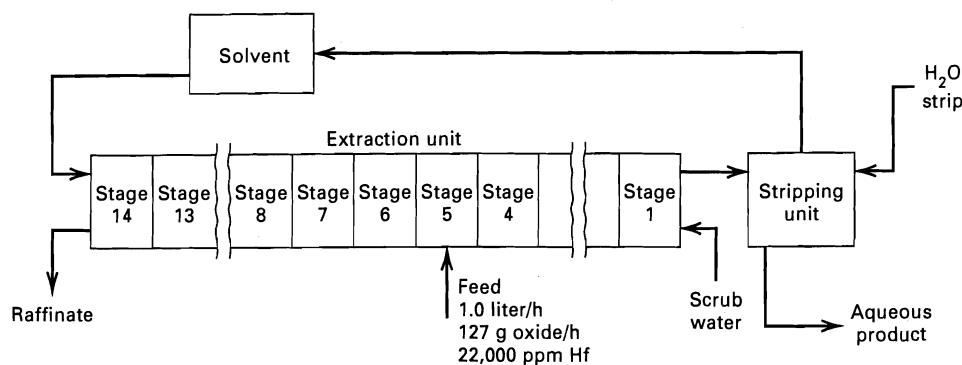


Figure 8.46 Data for Exercise 8.21.

Exercise 8.15. Determine:

- The minimum flow rate of solvent.
- The flow rate and composition of the extract at the minimum solvent flow rate.
- The number of equilibrium stages required if a solvent flow rate of 1.5 times the minimum is used.

8.23 At 45°C, 1,000 kg/h of a mixture of 0.80 mass fraction docosane and 0.20 mass fraction diphenylhexane is to be extracted with pure furfural to remove some of the diphenylhexane from the feed. Phase-equilibrium data for this ternary system are given in Exercise 8.15. Determine:

- The composition and flow rate of the extract and raffinate from a single equilibrium stage for solvent flow rates of 100, 1000, and 10000 kg/h.
- The minimum solvent flow rate to form two liquid phases.
- The maximum solvent flow rate to form two liquid phases.
- The composition and flow rate of the extract and raffinate if a solvent flow rate of 2000 kg/h and two equilibrium stages are used in a countercurrent flow system.

8.24 A liquid mixture of 27 wt% acetone and 73 wt% water is to be separated at 25°C into a raffinate and extract by multistage, steady-state, countercurrent liquid-liquid extraction with a solvent of pure 1,1,2-trichloroethane. Phase equilibrium data are given in Exercise 8.11. Determine:

- The minimum solvent-to-feed ratio to obtain a raffinate that is essentially free of acetone.
- The composition of extract at the minimum solvent-to-feed ratio.
- The composition of the extract stream leaving stage 2 (see Figure 8.13), if a large number of equilibrium stages is used with the minimum solvent rate.

Section 8.4

8.25 A feed mixture containing 50 wt% *n*-heptane and 50 wt% methylcyclohexane (MCH) is to be separated by liquid-liquid extraction into one product containing 92.5 wt% methylcyclohexane and another containing 7.5 wt% methylcyclohexane, both on a solvent-free basis. Aniline will be used as the solvent. Using the equilibrium data given below and the graphical method of Maloney and Schubert: (a) What is the minimum number of theoretical stages necessary to effect this separation? (b) What is the minimum extract reflux ratio? (c) If the reflux ratio is 7.0, how many theoretical contacts are required?

LIQUID-LIQUID EQUILIBRIUM DATA FOR THE SYSTEM *n*-HEPTANE/METHYLCYCLOHEXANE/ANILINE AT 25°C AND AT 1 ATM (101 kPa)

Hydrocarbon Layer		Solvent Layer	
Weight Percent MCH, Solvent-Free Basis	Pounds Aniline/Pound Solvent-Free Mixture	Weight Percent MCH, Solvent-Free Basis	Pounds Aniline/Pound Solvent-Free Mixture
0.0	0.0799	0.0	15.12
9.9	0.0836	11.8	13.72
20.2	0.087	33.8	11.5
23.9	0.0894	37.0	11.34
36.9	0.094	50.6	9.98
44.5	0.0952	60.0	9.0
50.5	0.0989	67.3	8.09
66.0	0.1062	76.7	6.83
74.6	0.1111	84.3	6.45
79.7	0.1135	88.8	6.0
82.1	0.116	90.4	5.9
93.9	0.1272	96.2	5.17
100.0	0.135	100.0	4.92

8.26 Two liquids, A and B, which have nearly identical boiling points, are to be separated by liquid-liquid extraction with solvent C. The following data represent the equilibrium between the two liquid phases at 95°C.

EQUILIBRIUM DATA, WT%

Extract Layer			Raffinate Layer		
A, %	B, %	C, %	A, %	B, %	C, %
0	7.0	93.0	0	92.0	8.0
1.0	6.1	92.9	9.0	81.7	9.3
1.8	5.5	92.7	14.9	75.0	10.1
3.7	4.4	91.9	25.3	63.0	11.7
6.2	3.3	90.5	35.0	51.5	13.5
9.2	2.4	88.4	42.0	41.0	17.0
13.0	1.8	85.2	48.1	29.3	22.6
18.3	1.8	79.9	52.0	20.0	28.0
24.5	3.0	72.5	47.1	12.9	40.0
31.2	5.6	63.2	Plait point		

[Adapted from McCabe and Smith, *Unit Operations of Chemical Engineering*, 4th ed., McGraw-Hill, New York, p. 557 (1985).]

Determine the minimum amount of reflux that must be returned from the extract product to produce an extract containing 83% A and 17% B (on a solvent-free basis) and a raffinate product containing 10% A and 90% B (solvent-free basis). The feed contains 35% A and 65% B on a solvent-free basis and is a saturated raffinate. The raffinate is the heavy liquid. Determine the number of ideal stages on both sides of the feed required to produce the same end products from the same feed when the reflux ratio of the extract, expressed as pounds of extract reflux per pound of extract product (including solvent), is twice the minimum. Calculate the masses of the various streams per 1,000 lb of feed, all on a solvent-free basis. Solve the problem using equilateral-triangle coordinates, right-triangle coordinates, and solvent-free coordinates. Which method is best for this exercise?

8.27 Solve Exercise 8.20 by the graphical method of Maloney and Schubert.

Section 8.5

8.28 Acetic acid is continuously extracted from a 3 wt% dilute solution in water with a solvent of isopropyl ether in a mixer-settler unit. The flow rates of the feed and solvent are 12,400 and 24,000 lb/h, respectively. Assuming a residence time of 1.5 min in the mixer and a settling vessel capacity of 4 gal/min-ft², estimate: (a) Diameter and height of the mixing vessel, assuming $H/D_T = 1$, (b) Agitator horsepower for the mixing vessel, (c) Diameter and length of the settling vessel, assuming $L/D_T = 4$, and (d) Residence time in minutes in the settling vessel.

8.29 A cascade of six mixer-settler units is available, each unit consisting of a 10-ft-diameter by 10-ft-high mixing vessel equipped with a 20-hp agitator, and a 10-ft-diameter by 40-ft-long settling vessel. If this cascade is used for the acetic acid extraction described in the introduction to this chapter, estimate the pounds per hour of feed that could be processed.

8.30 Acetic acid is to be extracted from a dilute aqueous solution with isopropyl ether at 25°C in a countercurrent cascade of mixer-settler units. In one of the units, the following conditions apply:

	Raffinate	Extract
Flow rate, lb/h	21,000	52,000
Density, lb/ft ³	63.5	45.3
Viscosity, cP	3.0	1.0

Interfacial tension = 13.5 dyne/cm. If the raffinate is the dispersed phase and the mixer residence time is 2.5 minutes, estimate for the mixer: (a) The dimensions of a closed, baffled vessel, (b) The diameter of a flat-bladed impeller, (c) The minimum rate of rotation

in revolutions per minute of the impeller for complete and uniform dispersion, and (d) The power requirement of the agitator at the minimum rate of rotation.

8.31 For the conditions of Exercise 8.30, estimate: (a) Sauter mean drop size, (b) Range of drop sizes, and (c) Interfacial area of the two-phase liquid-liquid emulsion.

8.32 For the conditions of Exercises 8.30 and 8.31, and the additional data given below, estimate: (a) The dispersed-phase mass-transfer coefficient, (b) The continuous-phase mass-transfer coefficient, (c) The Murphree dispersed-phase efficiency, and (d) The fraction of acetic acid extracted.

Additional data:

Diffusivity of acetic acid: in the raffinate, 1.3×10^{-9} m²/s and in the extract, 2.0×10^{-9} m²/s.

Distribution coefficient for acetic acid: $c_D/c_C = 2.7$

8.33 For the conditions and results of Example 8.4, involving the extraction of benzoic acid, from a dilute solution in water with toluene, determine the following when using a six-flat-blade turbine impeller in a closed vessel with baffles and with the extract phase dispersed, based on the physical properties given: (a) The minimum rate of rotation of the impeller for complete and uniform dispersion, (b) The power requirement of the agitator at the minimum rotation rate, (c) The Sauter mean droplet diameter, (d) The interfacial area, (e) The overall mass-transfer coefficient, K_{OD} , (f) The number of overall transfer units, N_{OD} , (g) The Murphree efficiency, E_{MD} , and (h) The fractional extraction of benzoic acid. Liquid properties are:

	Raffinate Phase	Extract Phase
Density, g/cm ³	0.995	0.860
Viscosity, cP	0.95	0.59
Diffusivity of benzoic acid, cm ² /s	2.2×10^{-5}	1.5×10^{-5}
Interfacial tension	= 22 dyne/cm	
Distribution coefficient for benzoic acid	$= c_D/c_C = 21$	

8.34 Estimate the diameter of an RDC column to extract acetic acid from water with isopropyl ether for the conditions and data of Exercises 8.28 and 8.30.

8.35 Estimate the diameter of a Karr column to extract benzoic acid from water with toluene for the conditions of Exercise 8.33.

8.36 Estimate the value of HETS for an RDC column operating under the conditions of Exercise 8.34.

8.37 Estimate the value of HETS for a Karr column operating under the conditions of Exercise 8.35.

Chapter 9

Approximate Methods for Multicomponent, Multistage Separations

Although rigorous computer methods, discussed in Chapter 10, are available for solving multicomponent separation problems, approximate methods continue to be used in practice for various purposes, including preliminary design, parametric studies to establish optimal design conditions, process synthesis studies to determine optimal separation sequences, and for obtaining an initial approximation for a rigorous method.

In Section 5.4, the approximate methods of Kremser [1] for absorbers, and Edmister [2] for distillation are discussed. This chapter presents an additional approximate method

that is widely used for making preliminary designs and optimization of simple distillation. The method is commonly referred to as the *Fenske–Underwood–Gilliland* or FUG method. In addition, application of the Kremser method is extended to and illustrated for strippers and liquid–liquid extraction. Although these methods can be applied fairly readily by manual calculation if physical properties are independent of composition, computer calculations are preferred, and FUG models are included in most computer-aided process design programs.

9.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- For multicomponent distillation, select two key components, operating pressure, and type condenser.
- For the specified separation between two key components in a multicomponent distillation column, estimate minimum number of equilibrium stages and distribution of nonkey components by the Fenske equation, minimum reflux ratio by the Underwood method, number of equilibrium stages for a reflux ratio greater than minimum by the Gilliland correlation, and feed stage location.
- Estimate stage requirements for multicomponent absorption, stripping, and liquid–liquid extraction using the Kremser equation.

9.1 FENSKE–UNDERWOOD–GILLILAND METHOD

An algorithm for the empirical Fenske–Underwood–Gilliland method, named after the authors of the three important steps in the procedure, is shown in Figure 9.1 for a simple distillation column of the type shown in Figure 9.3. The column can be equipped with a partial or total condenser. From Table 5.4, the number of degrees of freedom with a total condenser is $2N + C + 9$. In this case, the following variables are generally specified with the partial reboiler counted as a theoretical stage:

	Number of Specifications
Feed flow rate	1
Feed mole fractions	$C - 1$
Feed temperature ¹	1
Feed pressure ¹	1
Adiabatic stages (excluding reboiler)	$N - 1$
Stage pressures (including reboiler)	N

	Number of Specifications
Split of light key component	1
Split of heavy key component	1
Feed-stage location	1
Reflux ratio (as multiple of minimum-reflux ratio)	1
Reflux temperature	1
Adiabatic reflux divider	1
Pressure of total condenser	1
Pressure at reflux divider	1
	$2N + C + 9$

Similar specifications can be written for columns with a partial condenser.

¹ Feed temperature and pressure may correspond to known stream conditions leaving the previous piece of equipment.

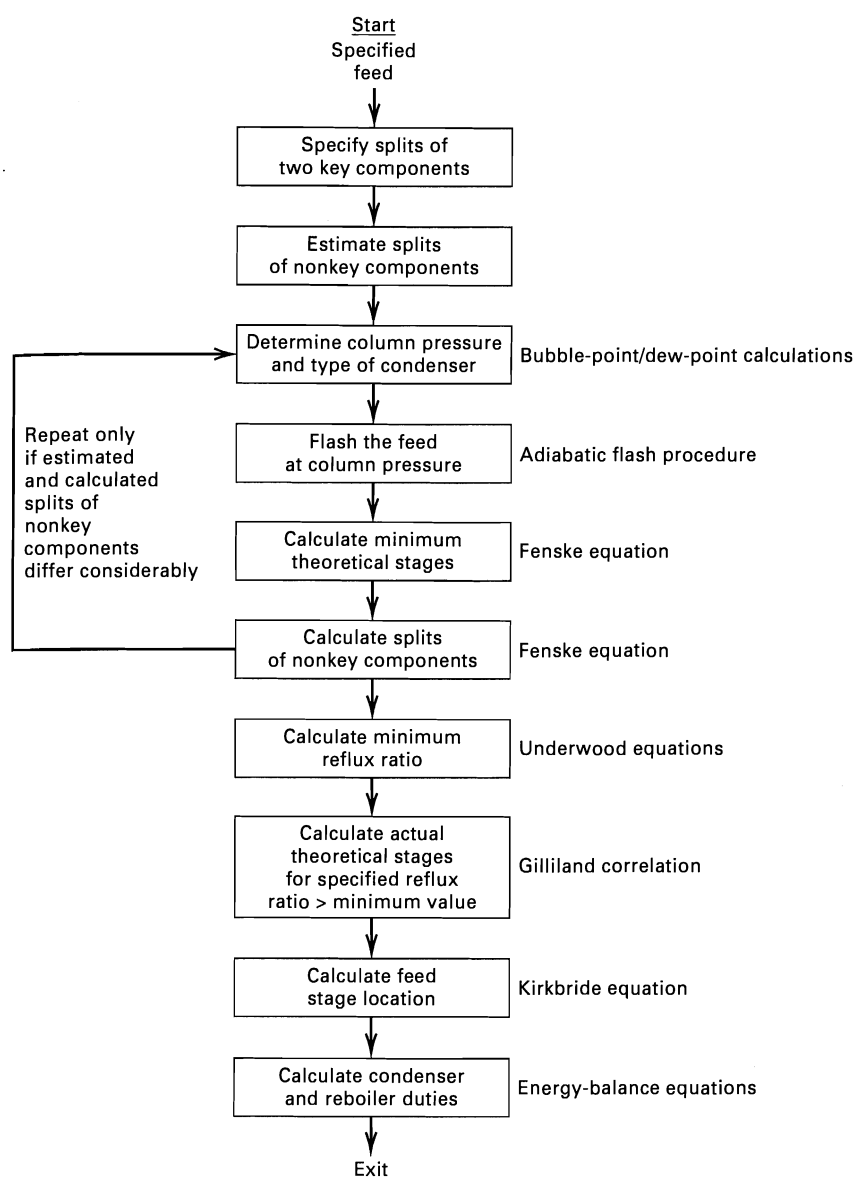
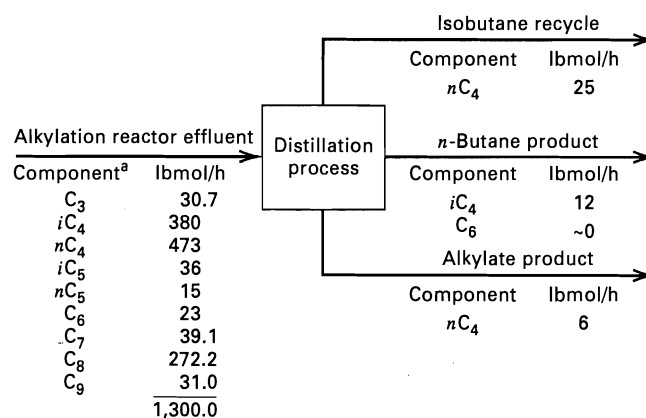


Figure 9.1 Algorithm for multicomponent distillation by FUG method.

Selection of Two Key Components

For multicomponent feeds, specification of two key components and their distribution between distillate and bottoms is accomplished in a variety of ways. Preliminary estimation of the distribution of nonkey components can be sufficiently difficult to require the iterative procedure indicated in Figure 9.1. However, generally only two and seldom more than three iterations are necessary.

Consider the multicomponent hydrocarbon feed in Figure 9.2. This mixture is typical of the feed to the recovery section of an alkylation plant [3]. Components are listed in order of decreasing volatility. A sequence of distillation columns including a deisobutanizer and a debutanizer is to be used to separate this mixture into the three products indicated. In Case 1 of Table 9.1, the deisobutanizer is selected as the first column in the sequence. Since the allowable quantities of *n*-butane in the isobutane recycle, and isobutane in the *n*-butane product, are specified, isobutane is the light key and



^aC₆, C₇, C₈, C₉ are taken as normal paraffins.

Figure 9.2 Separation specifications for alkylation-reactor effluent.

Table 9.1 Specifications of Key Component Splits and Preliminary Estimation of Nonkey Component Splits for Alkylation Reactor Effluent

Component	Feed, lbmol/h	Case 1, Deisobutanizer Column First, lbmol/h		Case 2, Debutanizer Column First (<i>i</i> C ₅ is HK), lbmol/h		Case 3, Debutanizer Column First (C ₆ is HK), lbmol/h	
		Distillate	Bottoms	Distillate	Bottoms	Distillate	Bottoms
C ₃	30.7	(30.7)	(0)	(30.7)	(0)	(30.7)	(0)
<i>i</i> C ₄	380	368 ^a	12 ^b	(380.0)	(0)	(380.0)	(0)
<i>n</i> C ₄	473	25 ^b	448 ^a	467 ^a	6 ^b	467 ^a	6 ^b
<i>i</i> C ₅	36	(0)	(36)	13 ^b	23 ^a	(13)	(23)
<i>n</i> C ₅	15	(0)	(15)	(1)	(14)	(1)	(14)
C ₆	23	(0)	(23)	(0)	(23)	0.01 ^b	22.99 ^a
C ₇	39.1	(0)	(39.1)	(0)	(39.1)	(0)	(39.1)
C ₈	272.2	(0)	(272.2)	(0)	(272.2)	(0)	(272.2)
C ₉	31.0	(0)	(31.0)	(0)	(31.0)	(0)	(31.0)
	1,300.0	423.7	876.3	891.7	408.3	891.71	408.29

^aBy material balance.^bSpecification.

(Preliminary estimate.)

n-butane is the heavy key. These two keys are adjacent in order of volatility. Because a fairly sharp separation between these two keys is indicated and the nonkey components are not close in volatility to the butanes, as a preliminary estimate we can assume the separation of the nonkey components to be perfect.

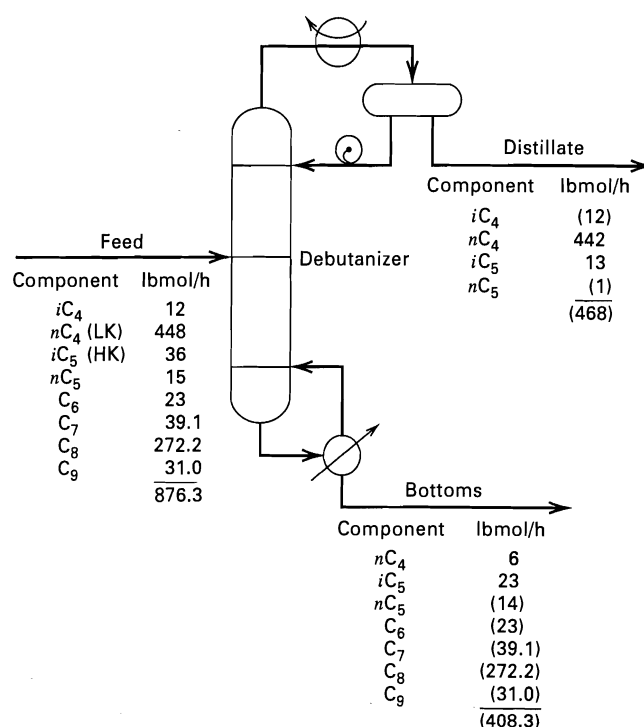
Alternatively, in Case 2, if the debutanizer is placed first in the sequence, specifications in Figure 9.2 require that *n*-butane be selected as the light key. However, selection of the heavy key is uncertain because no recovery or purity is specified for any component less volatile than *n*-butane. Possible heavy-key components for the debutanizer are *i*C₅, *n*C₅, or C₆. The simplest procedure is to select *i*C₅ so that the two keys are again adjacent.

For example, suppose we specify that 13 lbmol/h of *i*C₅ in the feed is allowed to appear in the distillate. Because the split of *i*C₅ is then not sharp and *n*C₅ is close in volatility to *i*C₅, it is probable that the quantity of *n*C₅ in the distillate will not be negligible. A preliminary estimate of the distributions of the nonkey components for Case 2 is given in Table 9.1. Although *i*C₄ may also distribute, a preliminary estimate of zero is made for it in the bottoms.

Finally, in Case 3, we select C₆ as the heavy key for the debutanizer at a specified rate of 0.01 lbmol/h in the distillate, as shown in Table 9.1. Now *i*C₅ and *n*C₅ will distribute between the distillate and bottoms in amounts to be determined; as a preliminary estimate, we assume the same distribution as in Case 2.

In practice, the deisobutanizer is usually placed first in the sequence. In Table 9.1, the bottoms for Case 1 then becomes the feed to the debutanizer, for which, if *n*C₄ and *i*C₅ are selected as the key components, component-separation specifications for the debutanizer are as indicated in Figure 9.3

with preliminary estimates of the separation of nonkey components shown in parentheses. This separation has been treated by Bachelor [4]. Because *n*C₄ and C₈ comprise 82.2 mol% of the feed and differ widely in volatility, the temperature difference between distillate and bottoms is likely to be large. Furthermore, the light key split is rather sharp, but the heavy key split is not. As will be shown later, this case provides a relatively severe test of the empirical design procedure discussed in this section.

**Figure 9.3** Specifications for debutanizer.

Column Operating Pressure

For preliminary design, column operating pressure and type of condenser can be established by the procedure discussed in Section 7.2 and shown in Figure 7.16, as illustrated in the following example. With column operating pressure established, the column feed can be flashed adiabatically at an estimated feed-tray pressure to determine feed-phase condition.

EXAMPLE 9.1

Determine column operating pressures and type of condenser for the debutanizer of Figure 9.3.

SOLUTION

Using the estimated distillate composition in Figure 9.3, we compute the distillate bubble-point pressure at 120°F (48.9°C) iteratively from (4-12) in a manner similar to Example 4.2. This procedure gives 79 psia as the reflux-drum pressure. Thus, a total condenser is indicated. Allowing a 2-psi condenser pressure drop, column top pressure is $(79 + 2) = 81$ psia; and allowing a 5-psi pressure drop through the column, the bottoms pressure is $(81 + 5) = 86$ psia. Assume a feed-tray pressure midway between the column top and bottom pressures or 83.5 psia.

Bachelor [4] sets column pressure at 80 psia throughout. He obtains a distillate temperature of 123°F. A bubble-point calculation for the bottoms composition at 80 psia gives 340°F. This temperature is sufficiently low to prevent decomposition.

Feed to the debutanizer is presumably bottoms from a deisobutanizer operating at a pressure of perhaps 100 psia or more. Results of an adiabatic flash of this feed, by the procedure of Section 4.4, to 80 psia are given by Bachelor [4] as follows.

Component	Pound-Moles per Hour	
	Vapor Feed	Liquid Feed
<i>i</i> C ₄	3.3	8.7
<i>n</i> C ₄	101.5	346.5
<i>i</i> C ₅	4.6	31.4
<i>n</i> C ₅	1.6	13.4
<i>n</i> C ₆	1.3	21.7
<i>n</i> C ₇	1.2	37.9
<i>n</i> C ₈	3.2	269.0
<i>n</i> C ₉	0.2	30.8
	116.9	759.4

The temperature of the flashed feed is 180°F (82.2°C). From above, the feed-mole-fraction vaporized is $(116.9/876.3) = 0.1334$.

Fenske Equation for Minimum Equilibrium Stages

For a specified separation between two key components of a multicomponent mixture, an exact expression is easily developed for the required minimum number of equilibrium stages, which corresponds to total reflux. This condition can be achieved in practice by charging the column with feedstock and operating it with no further input of feed and no withdrawal of distillate or bottoms, as illustrated in Figure 9.4. To

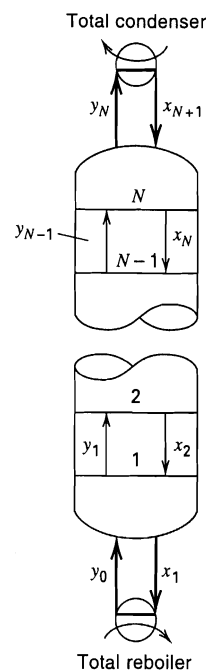


Figure 9.4 Distillation column operation at total reflux.

facilitate derivation of the Fenske equation, stages are numbered from the bottom up. All vapor leaving stage *N* is condensed and returned to stage *N* as reflux. All liquid leaving stage 1 is vaporized and returned to stage 1 as boilup. For steady-state operation within the column, heat input to the reboiler and heat output from the condenser are made equal (assuming no heat losses). Then, by a material balance, vapor and liquid streams passing between any pair of stages have equal flow rates and compositions, for example, $V_{N-1} = L_N$ and $y_{i,N-1} = x_{i,N}$. However, molar vapor and liquid flow rates will change from stage to stage unless the assumption of constant molar overflow is valid.

Derivation of an exact equation for the minimum number of equilibrium stages involves only the definition of the *K*-value and the mole-fraction equality between stages. For component *i* at stage 1 in Figure 9.4,

$$y_{i,1} = K_{i,1}x_{i,1} \quad (9-1)$$

But for passing streams

$$y_{i,1} = x_{i,2} \quad (9-2)$$

Combining these two equations,

$$x_{i,2} = K_{i,1}x_{i,1} \quad (9-3)$$

Similarly, for stage 2,

$$y_{i,2} = K_{i,2}x_{i,2} \quad (9-4)$$

Combining (9-3) and (9-4), we have

$$y_{i,2} = K_{i,2}K_{i,1}x_{i,1} \quad (9-5)$$

Equation (9-5) is readily extended in this fashion to give

$$y_{i,N} = K_{i,N}K_{i,N-1} \cdots K_{i,2}K_{i,1}x_{i,1} \quad (9-6)$$

Similarly, for component j ,

$$y_{j,N} = K_{j,N} K_{j,N-1} \cdots K_{j,2} K_{j,1} x_{j,1} \quad (9-7)$$

Combining (9-6) and (9-7), we find that

$$\frac{y_{i,N}}{y_{j,N}} = \alpha_N \alpha_{N-1} \cdots \alpha_2 \alpha_1 \left(\frac{x_{i,1}}{x_{j,1}} \right) \quad (9-8)$$

or

$$\left(\frac{x_{i,N+1}}{x_{i,1}} \right) \left(\frac{x_{j,1}}{x_{j,N+1}} \right) = \prod_{k=1}^{N_{\min}} \alpha_k \quad (9-9)$$

where $\alpha_k = K_{i,k}/K_{j,k}$, the relative volatility between components i and j . Equation (9-9) relates the relative enrichments of any two components i and j over a cascade of N theoretical stages to the stage relative volatilities between the two components. Although (9-9) is exact, it is rarely used in practice because the conditions of each stage must be known to compute the set of relative volatilities. However, if the relative volatility is assumed constant, (9-9) simplifies to

$$\left(\frac{x_{i,N+1}}{x_{i,1}} \right) \left(\frac{x_{j,1}}{x_{j,N+1}} \right) = \alpha^N \quad (9-10)$$

or

$$N_{\min} = \frac{\log \{ [(x_{i,N+1}/x_{i,1}) (x_{j,1}/x_{j,N+1})] \}}{\log \alpha_{i,j}} \quad (9-11)$$

Equation (9-11) is extremely useful. It is referred to as the *Fenske equation* [5]. When i = the light key (LK) and j = the heavy key (HK), the minimum number of equilibrium stages is influenced by the nonkey components only by their effect (if any) on the value of the relative volatility between the key components.

Equation (9-11) permits a rapid estimation of minimum equilibrium stages. A more convenient form of (9-11) is obtained by replacing the product of the mole-fraction ratios by the equivalent product of mole-distribution ratios in terms of component distillate and bottoms flow rates d and b , respectively,² and by replacing the relative volatility by a geometric mean of the top-stage and bottom-stage values. Thus,

$$N_{\min} = \frac{\log [(d_i/d_j)(b_j/b_i)]}{\log \alpha_m} \quad (9-12)$$

where the mean relative volatility is approximated by

$$\alpha_m = [(\alpha_{i,j})_N (\alpha_{i,j})_1]^{1/2} \quad (9-13)$$

Thus, the minimum number of equilibrium stages depends on the degree of separation of the two key components and their relative volatility, but is independent of feed-phase condition. Equation (9-12) in combination with (9-13) is exact for two minimum stages. For one stage, it is equivalent to the equilibrium-flash equation. In practice, distillation columns are designed for separations corresponding to as many as 150 minimum equilibrium stages.

The Fenske equation is quite reliable except when the relative volatility varies appreciably over the column, and/or when the mixture forms nonideal liquid solutions. In those cases, if the Fenske equation is applied with (9-13), it should be done with great caution, and should be followed by rigorous calculations of the type in Chapter 10.

EXAMPLE 9.2

For the debutanizer shown in Figure 9.3 and considered in Example 9.1, estimate the minimum equilibrium stages by the Fenske equation. Assume uniform operating pressure of 80 psia (552 kPa) throughout and utilize the ideal K -values given by Bachelor [4] as plotted in Figure 9.5.

SOLUTION

The two key components are n -butane and isopentane. Distillate and bottoms conditions based on the estimated product distributions for nonkey components in Figure 9.3 are

Component	$x_{N+1} = x_D$	$x_1 = x_B$
iC_4	0.0256	~0
nC_4 (LK)	0.9445	0.0147
iC_5 (HK)	0.0278	0.0563
nC_5	0.0021	0.0343
nC_6	~0	0.0563
nC_7	~0	0.0958
nC_8	~0	0.6667
nC_9	~0	0.0759
	1.0000	1.0000

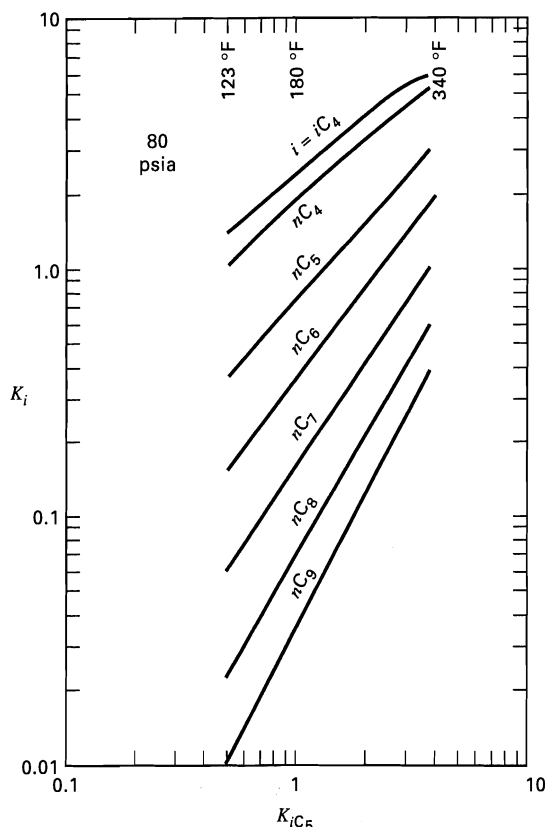


Figure 9.5 Ideal K -values for hydrocarbons at 80 psia.

² This substitution is valid even though no distillate or bottoms products are withdrawn at total reflux.

From Figure 9.5, at 123°F, the assumed top-stage temperature is

$$(\alpha_{nC_4, iC_5})_N = 1.03/0.495 = 2.08$$

At 340°F, the assumed bottom-stage temperature is

$$(\alpha_{nC_4, iC_5})_1 = 5.20/3.60 = 1.44$$

From (9-13),

$$\alpha_m = [(2.08)(1.44)]^{1/2} = 1.73$$

Noting that $(d_i/d_j) = (x_{D_i}/x_{D_j})$ and $(b_i/b_j) = (x_{B_i}/x_{B_j})$, (9-12) becomes

$$N_{\min} = \frac{\log[(0.9445/0.0278)(0.0563/0.0147)]}{\log 1.73} = 8.88 \text{ stages}$$

Distribution of Nonkey Components at Total Reflux

The Fenske equation is not restricted to the two key components. Once $N_{\min}$ is known, (9-12) can be used to calculate molar flow rates d and b for all nonkey components. These values provide a first approximation to the actual product distribution when more than the minimum number of stages is employed.

Let i = a nonkey component and j = the heavy key or reference component denoted by r . Then (9-12) becomes

$$\left(\frac{d_i}{b_i}\right) = \left(\frac{d_r}{b_r}\right) (\alpha_{i,r})_m^{N_{\min}} \quad (9-14)$$

Substituting $f_i = d_i + b_i$ in (9-14) gives

$$b_i = \frac{f_i}{1 + (d_r/b_r)(\alpha_{i,r})_m^{N_{\min}}} \quad (9-15)$$

or

$$d_i = \frac{f_i(d_r/b_r)(\alpha_{i,r})_m^{N_{\min}}}{1 + (d_r/b_r)(\alpha_{i,r})_m^{N_{\min}}} \quad (9-16)$$

Equations (9-15) and (9-16) give the distribution of nonkey components at total reflux as predicted by the Fenske equation.

For accurate calculations, (9-15) and (9-16) should be used to compute the smaller of the two quantities b_i and d_i . The other quantity is best obtained by overall material balance.

EXAMPLE 9.3

Estimate the product distributions for nonkey components by the Fenske equation for the conditions of Example 9.2.

SOLUTION

All nonkey relative volatilities are calculated relative to isopentane using the K -values of Figure 9.5.

Component	α_{i,iC_5}		
	123°F	340°F	Geometric Mean
iC_4	2.81	1.60	2.12
nC_5	0.737	0.819	0.777
nC_6	0.303	0.500	0.389
nC_7	0.123	0.278	0.185
nC_8	0.0454	0.167	0.0870
nC_9	0.0198	0.108	0.0463

Based on $N_{\min} = 8.88$ stages from Example 9.2 and the above geometric-mean relative volatilities, values of $(\alpha_{i,r})_m^{N_{\min}}$ are computed relative to isopentane as tabulated below.

From (9-15), using the feed rate specifications in Figure 9.3 for f_i , the distribution of nonkey iC_4 is

$$b_{iC_4} = \frac{12}{1 + (13/23)790} = 0.0268 \text{ lbmol/h}$$

$$d_{iC_4} = f_{iC_4} - b_{iC_4} = 12 - 0.0268 = 11.9732 \text{ lbmol/h}$$

Results of similar calculations for the other nonkey components are included in the following table.

Component	$(\alpha_{i,iC_5})_m^{N_{\min}}$	d_i	b_i
iC_4	790	11.9732	0.0268
nC_4	130	442.0	6.0
iC_5	1.00	13.0	23.0
nC_5	0.106	0.851	14.149
nC_6	0.000228	0.00297	22.99703
nC_7	3.11×10^{-7}	6.87×10^{-6}	39.1
nC_8	3.83×10^{-10}	5.98×10^{-8}	272.2
nC_9	1.41×10^{-12}	2.48×10^{-11}	31.0
		467.8272	408.4728

Underwood Equations for Minimum Reflux

Minimum reflux is based on the specifications for the degree of separation between two key components. The minimum reflux is finite and feed product withdrawals are permitted. However, a column cannot operate under this condition because of the accompanying requirement of infinite stages. Nevertheless, minimum reflux is a useful limiting condition.

For binary distillation of an ideal mixture at minimum reflux, as shown in Figure 7.12a, most of the stages are crowded into a constant-composition zone that bridges the feed stage. In this zone, all vapor and liquid streams have compositions essentially identical to those of the flashed feed. This zone constitutes a single *pinch point* or *point of infinitude* as shown in Figure 9.6a. If nonideal phase conditions are such as to create a point of tangency between the equilibrium curve and the operating line in the rectifying

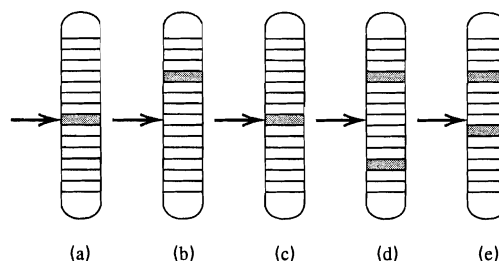


Figure 9.6 Location of pinch-point zones at minimum reflux: (a) binary system; (b) binary system, nonideal conditions giving point of tangency; (c) multicomponent system, all components distributed (Class 1); (d) multicomponent system, not all LLK and HHK distributing (Class 2); (e) multicomponent system, all LLK, if any, distributing, but not all HHK distributing (Class 2). (LLK = lighter than light key; HHK = heavier than heavy key.)

section, as shown in Figure 7.12b, the pinch point will occur in the rectifying section as in Figure 9.6b. Alternatively, the single pinch point can occur in the stripping section.

Shiras, Hanson, and Gibson [6] classified multicomponent systems as having one (Class 1) or two (Class 2) pinch points. For Class 1 separations, all components in the feed distribute to both the distillate and bottoms products. Then the single pinch point bridges the feed stage as shown in Figure 9.6c. Class 1 separations can occur when narrow-boiling-range mixtures are distilled or when the degree of separation between the key components is not sharp.

For Class 2 separations, one or more of the components appear in only one of the products. If neither the distillate nor the bottoms product contains all feed components, two pinch points occur away from the feed stage as shown in Figure 9.6d. Stages between the feed stage and the rectifying-section pinch point remove heavy components that do not appear in the distillate. Light components that do not appear in the bottoms are removed by the stages between the feed stage and the stripping-section pinch point. However, if all feed components appear in the bottoms, the stripping-section pinch point moves to the feed stage as shown in Figure 9.6e.

Consider the general case of a rectifying-section pinch point at or away from the feed stage as shown in Figure 9.7. A component material balance over all stages gives

$$y_{i,\infty} V_{\infty} = x_{i,\infty} L_{\infty} + x_{i,D} D \quad (9-17)$$

A total balance over all stages is

$$V_{\infty} = L_{\infty} + D \quad (9-18)$$

Since phase compositions do not change in the pinch zone, the phase equilibrium relation is

$$y_{i,\infty} = K_{i,\infty} x_{i,\infty} \quad (9-19)$$

Combining (9-17) to (9-19) for components i and j to eliminate $y_{i,\infty}$, $y_{j,\infty}$, and V_{∞} ; solving for the internal reflux ratio at the pinch point; and substituting $(\alpha_{i,j})_{\infty} = K_{i,\infty}/K_{j,\infty}$, we have

$$\frac{L_{\infty}}{D} = \frac{[(x_{i,D}/x_{i,\infty}) - (\alpha_{i,j})_{\infty}(x_{j,D}/x_{j,\infty})]}{(\alpha_{i,j})_{\infty} - 1} \quad (9-20)$$

For Class 1 separations, flashed feed- and pinch-zone compositions are identical.³ Therefore, $x_{i,\infty} = x_{i,F}$ and (9-20) for the light key (LK) and the heavy key (HK) becomes

$$\frac{(L_{\infty})_{\min}}{D} = \frac{(L_F/F)[(Dx_{LK,D})/(L_F x_{LK,F}) - (\alpha_{LK,HK})_F (Dx_{HK,D}/L_F x_{HK,F})]}{(\alpha_{LK,HK})_F - 1} \quad (9-21)$$

This equation is attributed to Underwood [7] and can be applied to subcooled-liquid or superheated-vapor feeds by using fictitious values of L_F and $x_{i,F}$ computed by making a

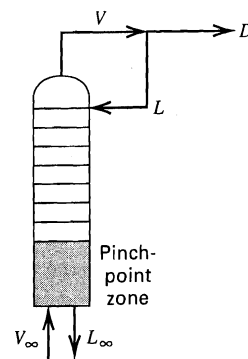


Figure 9.7 Rectifying-section pinch-point zone.

flash calculation outside the two-phase region. As with the Fenske equation, (9-21) applies to components other than the key components. Therefore, for a specified split of two key components, the distribution of nonkey components is obtained by combining (9-21) with the analogous equation for component i in place of the light key to give

$$\frac{Dx_{i,D}}{L_F x_{i,F}} = \left[\frac{(\alpha_{i,HK})_F - 1}{(\alpha_{LK,HK})_F - 1} \right] \left(\frac{Dx_{LK,D}}{L_F x_{LK,F}} \right) + \left[\frac{(\alpha_{LK,HK})_F - (\alpha_{i,HK})_F}{(\alpha_{LK,HK})_F - 1} \right] \left(\frac{Dx_{HK,D}}{L_F x_{HK,F}} \right) \quad (9-22)$$

For a Class 1 separation,

$$0 < \left(\frac{Dx_{i,D}}{L_F x_{i,F}} \right) < 1$$

for all nonkey components. If that is so, the external reflux ratio is obtained from the internal reflux by an enthalpy balance around the rectifying section in the form

$$\begin{aligned} \frac{(L_{\min})_{\text{external}}}{D} &= (R_{\min})_{\text{external}} \\ &= \frac{(L_{\infty})_{\min}(h_{V_{\infty}} - h_{L_{\infty}}) + D(h_{V_{\infty}} - h_V)}{D(h_V - h_L)} \end{aligned} \quad (9-23)$$

where subscripts V and L refer to vapor leaving the top stage and external liquid reflux sent to the top stage, respectively. For conditions of constant molar overflow,

$$(R_{\min})_{\text{external}} = (L_{\infty})_{\min}/D$$

Even when (9-21) is invalid, it is useful because, as shown by Gilliland [8], the minimum-reflux ratio computed by assuming a Class 1 separation is equal to or greater than the true minimum. This is because the presence of distributing nonkey components in the pinch-point zones increases the difficulty of the separation, thus increasing the reflux requirement.

EXAMPLE 9.4

Calculate the minimum internal reflux for the conditions of Example 9.2 assuming a Class 1 separation. Check the validity of this assumption.

³ Assuming the feed is neither subcooled nor superheated.

SOLUTION

From Figure 9.5, the relative volatility between nC_4 (LK) and iC_5 (HK) at the feed temperature of 180°F is 1.93. Feed liquid and distillate quantities are given in Figure 9.3 and Example 9.1. From (9-21),

$$(L_\infty)_{\min} = \frac{759.4[(442/346.5) - 1.93(13/31.4)]}{1.93 - 1} = 389 \text{ lbmol/h}$$

Distribution of nonkey components in the feed is determined by (9-22). The most likely nonkey component to distribute is nC_5 because its volatility is close to that of iC_5 (HK), which does not undergo a sharp separation. For nC_5 , using data for K -values from Figure 9.5, we have

$$\frac{Dx_{nC_5,D}}{L_F x_{nC_5,F}} = \left[\frac{0.765 - 1}{1.93 - 1} \right] \left(\frac{442}{346.5} \right) + \left[\frac{1.93 - 0.765}{1.93 - 1} \right] \left(\frac{13}{31.4} \right) = 0.1963$$

Therefore, $Dx_{nC_5,D} = 0.1963(13.4) = 2.63$ lbmol/h of nC_5 in the distillate. This is less than the quantity of nC_5 in the total feed. Therefore, nC_5 distributes between the distillate and the bottoms. However, similar calculations for the other nonkey components give negative distillate flow rates for the other heavy components and, in the case of iC_4 , a distillate flow rate greater than the feed rate. Thus, the computed reflux rate is not valid. However, as expected, it is greater than the true internal value of 298 lbmol/h reported by Bachelor [4].

For Class 2 separations, (9-17) to (9-20) still apply. However, (9-20) cannot be used directly to compute the internal minimum-reflux ratio because values of $x_{i,\infty}$ are not simply related to feed composition for Class 2 separations. Underwood [9] devised an ingenious algebraic procedure to overcome this difficulty. For the rectifying section, he defined a quantity Φ by

$$\sum \frac{(\alpha_{i,r})_\infty x_{i,D}}{(\alpha_{i,r})_\infty - \Phi} = 1 + (R_\infty)_{\min} \quad (9-24)$$

Similarly, for the stripping section, Underwood defined Φ' by

$$\sum \frac{(\alpha'_{i,r})_\infty x_{i,B}}{(\alpha'_{i,r})_\infty - \Phi'} = 1 - (R'_\infty)_{\min} \quad (9-25)$$

where $R'_\infty = L'_\infty/B$ and the prime refers to conditions in the stripping-section pinch-point zone. In his derivation, Underwood assumed that relative volatilities are constant in the region between the two pinch-point zones and that $(R_\infty)_{\min}$ and $(R'_\infty)_{\min}$ are related by the assumption of constant molar overflow in the region between the feed entry and the rectifying-section pinch point and in the region between the feed entry and the stripping-section pinch point. Hence,

$$(L'_\infty)_{\min} - (L_\infty)_{\min} = qF \quad (9-26)$$

With these two critical assumptions, Underwood showed that at least one common root θ (where $\theta = \Phi = \Phi'$) exists between (9-24) and (9-25).

Equation (9-24) is analogous to the following equation derived from (9-19), and the relation $\alpha_{i,r} = K_i/K_r$,

$$\sum_i \frac{(\alpha_{i,r})_\infty x_{i,D}}{(\alpha_{i,r})_\infty - L_\infty/[V_\infty(K_r)_\infty]} = 1 + (R_\infty)_{\min} \quad (9-27)$$

where $L_\infty/[V_\infty(K_r)_\infty]$ is called the *absorption factor* for a reference component in the rectifying-section pinch-point zone. Although Φ is analogous to the absorption factor, a different root of Φ is used to solve for $(R_\infty)_{\min}$, as discussed by Shiras et al. [6].

The common root θ may be determined by multiplying (9-24) and (9-25) by D and B , respectively, adding the two equations, substituting (9-25) to eliminate $(R'_\infty)_{\min}$ and $(R_\infty)_{\min}$, and utilizing the overall component balance $z_{i,F}F = x_{i,D}D + x_{i,B}B$ to obtain

$$\sum_i \frac{(\alpha_{i,r})_\infty z_{i,F}}{(\alpha_{i,r})_\infty - \theta} = 1 - q \quad (9-28)$$

where q is the thermal condition of the feed from (7-20) and r is conveniently taken as the heavy key, HK. When only the two key components distribute, (9-28) is solved iteratively for a root of θ that satisfies $\alpha_{LK,HK} > \theta > 1$. The following modification of (9-24) is then solved for the internal reflux ratio $(R_\infty)_{\min}$:

$$\sum_i \frac{(\alpha_{i,r})_\infty x_{i,D}}{(\alpha_{i,r})_\infty - \theta} = 1 + (R_\infty)_{\min} \quad (9-29)$$

If any nonkey components are suspected of distributing, estimated values of $x_{i,D}$ cannot be used directly in (9-29). This is particularly true when nonkey components are intermediate in volatility between the two key components. In this case, (9-28) is solved for m roots of θ , where m is one less than the number of distributing components. Furthermore, each root of θ lies between an adjacent pair of relative volatilities of distributing components. For instance, in Example 9.4, it was found the nC_5 distributes at minimum reflux, but nC_6 and heavier do not and iC_4 does not. Therefore, two roots of θ are necessary, where

$$\alpha_{nC_4,iC_5} > \theta_1 > 1.0 > \theta_2 > \alpha_{nC_5,iC_5}$$

With these two roots, (9-29) is written twice and solved simultaneously to yield $(R_\infty)_{\min}$ and the unknown value of $x_{nC_5,D}$. The solution must, of course, satisfy the condition $\sum x_{i,D} = 1.0$.

With the internal reflux ratio $(R_\infty)_{\min}$ known, the external reflux ratio is computed by enthalpy balance with (9-23). This requires a knowledge of the rectifying-section pinch-point compositions. Underwood [9] shows that

$$x_{i,\infty} = \frac{\theta x_{i,D}}{(R_\infty)_{\min}[(\alpha_{i,r})_\infty - \theta]} \quad (9-30)$$

with $y_{i,\infty}$ given by (9-17). The value of θ to be used in (9-30) is the root of (9-29) satisfying the inequality

$$(\alpha_{\text{HNK},r})_\infty > \theta > 0$$

where HNK refers to the heaviest nonkey component in the distillate at minimum reflux. This root is equal to $L_\infty/[V_\infty(K_r)_\infty]$ in (9-27). With wide-boiling feeds, the external reflux can be significantly higher than the internal reflux. Bachelor [4] cites a case where the external reflux rate is 55% greater than the internal reflux.

For the stripping-section pinch-point composition, Underwood obtains

$$x'_{i,\infty} = \frac{\theta x_{i,B}}{[(R'_{\infty})_{\min} + 1][(\alpha_{i,r})_{\infty} - \theta]} \quad (9-31)$$

where, in this case, θ is the root of (9-29) satisfying the inequality

$$(\alpha_{\text{HNK},r})_{\infty} > \theta > 0$$

where HNK refers to the heaviest nonkey in the bottoms product at minimum reflux.

Because of their relative simplicity, the Underwood minimum-reflux equations for Class 2 separations are widely used, but too often without examining the possibility of non-key distribution. In addition, the assumption is frequently made that $(R_{\infty})_{\min}$ equals the external reflux ratio. When the assumptions of constant relative volatility and constant molar overflow in the regions between the two pinch-point zones are not valid, values of the minimum-reflux ratio computed from the Underwood equations for Class 2 separations can be appreciably in error because of the sensitivity of (9-28) to the value of q , as will be shown in Example 9.5. When the Underwood assumptions appear to be valid and a negative minimum-reflux ratio is computed, this may be interpreted to mean that a rectifying section is not required to obtain the specified separation. The Underwood equations show that the minimum reflux depends mainly on the feed condition and relative volatility and, to a lesser extent, on the degree of separation between the two key components. A finite minimum-reflux ratio exists even for a perfect separation.

An extension of the Underwood method for distillation columns with multiple feeds is given by Barnes, Hanson, and King [10]. Exact computer methods for determining minimum reflux are available [11]. For making rigorous distillation calculations at actual reflux conditions by the computer methods of Chapter 10, knowledge of the minimum reflux is not essential, but the minimum number of equilibrium stages is very useful.

EXAMPLE 9.5

Repeat Example 9.4 assuming a Class 2 separation and utilizing the corresponding Underwood equations. Check the validity of the Underwood assumptions. Also calculate the external reflux ratio.

SOLUTION

From the results of Example 9.4, assume that the only distributing nonkey component is n -pentane. Assuming that the feed temperature of 180°F is reasonable for computing relative volatilities in the pinch zone, the following quantities are obtained from Figures 9.3 and 9.5:

Species i	$z_{i,F}$	$(\alpha_{i,\text{HK}})_{\infty}$
$i\text{C}_4$	0.0137	2.43
$n\text{C}_4$ (LK)	0.5113	1.93
$i\text{C}_5$ (HK)	0.0411	1.00

Species i	$z_{i,F}$	$(\alpha_{i,\text{HK}})_{\infty}$
$n\text{C}_5$	0.0171	0.765
$n\text{C}_6$	0.0262	0.362
$n\text{C}_7$	0.0446	0.164
$n\text{C}_8$	0.3106	0.0720
$n\text{C}_9$	0.0354	0.0362
	1.0000	

The q for the feed is assumed to be the mole fraction of liquid in the flashed feed. From Example 9.1, $q = 1 - 0.1334 = 0.8666$. Applying (9-28), we have

$$\begin{aligned} & \frac{2.43(0.0137)}{2.43 - \theta} + \frac{1.93(0.5113)}{1.93 - \theta} + \frac{1.00(0.0411)}{1.00 - \theta} \\ & + \frac{0.765(0.0171)}{0.765 - \theta} + \frac{0.362(0.0262)}{0.362 - \theta} + \frac{0.164(0.0446)}{0.164 - \theta} \\ & + \frac{0.072(0.3106)}{0.072 - \theta} + \frac{0.0362(0.0354)}{0.0362 - \theta} = 1 - 0.8666 \end{aligned}$$

Solving this equation by a bounded-Newton method for two roots of θ that satisfy

$$\alpha_{n\text{C}_4, i\text{C}_5} > \theta_1 > \alpha_{i\text{C}_5, i\text{C}_5} > \theta_2 > \alpha_{n\text{C}_5, i\text{C}_5}$$

or

$$1.93 > \theta_1 > 1.00 > \theta_2 > 0.765$$

$\theta_1 = 1.04504$ and $\theta_2 = 0.78014$. Because distillate rates for $n\text{C}_4$ and $i\text{C}_5$ are specified (442 and 13 lbmol/h, respectively), the following form of (9-29) is preferred:

$$\sum_i \frac{(\alpha_{i,r})_{\infty}(x_{i,D}D)}{(\alpha_{i,r})_{\infty} - \theta} = D + (L_{\infty})_{\min} \quad (9-32)$$

with the restriction that

$$\sum_i (x_{i,D}D) = D \quad (9-33)$$

Assuming that $x_{i,D}D$ equals 0.0 for components heavier than $n\text{C}_5$ and 12.0 lbmol/h for $i\text{C}_4$, we find that these two relations give the following three linear equations:

$$\begin{aligned} D + (L_{\infty})_{\min} &= \frac{2.43(12)}{2.43 - 1.04504} + \frac{1.93(442)}{1.93 - 1.04504} \\ &+ \frac{1.00(13)}{1.00 - 1.04504} + \frac{0.765(x_{n\text{C}_5,D}D)}{0.765 - 1.04504} \\ D + (L_{\infty})_{\min} &= \frac{2.43(12)}{2.43 - 0.78014} + \frac{1.93(442)}{1.93 - 0.78014} \\ &+ \frac{1.00(13)}{1.00 - 0.78014} + \frac{0.765(x_{n\text{C}_5,D}D)}{0.765 - 0.78014} \\ D &= 12 + 442 + 13 + (x_{n\text{C}_5,D}D) \end{aligned}$$

Solving these three equations gives

$$\begin{aligned} x_{n\text{C}_5,D}D &= 2.56 \text{ lbmol/h} \\ D &= 469.56 \text{ lbmol/h} \\ (L_{\infty})_{\min} &= 219.8 \text{ lbmol/h} \end{aligned}$$

The distillate rate for $n\text{C}_5$ is very close to the value of 2.63 computed in Example 9.4, if we assume a Class 1 separation. The internal minimum reflux ratio at the rectifying pinch point is considerably less than the value of 389 computed in Example 9.4 and is also much less than the true internal value of 298 reported by Bachelor [4]. The main reason for the discrepancy between the value of 219.8 and the

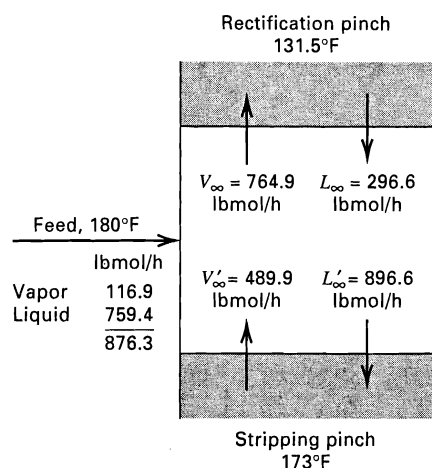


Figure 9.8 Pinch-point region conditions for Example 9.5 from computations by Bachelor.

[From J.B. Bachelor, *Petroleum Refiner*, 36(6), 161–170 (1957).]

true value of 298 is the invalidity of the assumption of constant molar overflow. Bachelor computed the pinch-point region flow rates and temperatures shown in Figure 9.8. The average temperature of the region between the two pinch regions is 152°F (66.7°C), which is appreciably lower than the flashed-feed temperature. The relatively hot feed causes additional vaporization across the feed zone. The effective value of q in the region between the pinch points is obtained from (7-18):

$$q_{\text{eff}} = \frac{L'_{\infty} - L_{\infty}}{F} = \frac{896.6 - 296.6}{876.3} = 0.685$$

This is considerably lower than the value of 0.8666 for q based on the flashed-feed condition. On the other hand, the value of $\alpha_{LK,HK}$ at 152°F (66.7°C) is not much different from the value at 180°F (82.2°C). If this example is repeated using q equal to 0.685, the resulting value of $(L_{\infty})_{\min}$ is 287.3 lbmol/h, which is only 3.6% lower than the true value of 298. Unfortunately, in practice, this corrected procedure cannot be applied because the true value of q cannot be readily determined.

To compute the external-reflux ratio from (9-23), rectifying pinch-point compositions must be calculated from (9-30) and (9-17). The root of θ to be used in (9-30) is obtained from the version of (9-29) used above. Thus,

$$\frac{2.43(12)}{2.43 - \theta} + \frac{1.93(442)}{1.93 - \theta} + \frac{1.00(13)}{1.00 - \theta} + \frac{0.765(2.56)}{0.765 - \theta} = 469.56 + 219.8$$

where $0.765 > \theta > 0$. Solving, $\theta = 0.5803$. Liquid pinch-point compositions are obtained from the following form of (9-30):

$$x_{i,\infty} = \frac{\theta(x_{i,D}D)}{(L_{\infty})_{\min}[(\alpha_{i,r})_{\infty} - \theta]}$$

with $(L_{\infty})_{\min} = 219.8$ lbmol/h.

For iC_4 ,

$$x_{iC_4,\infty} = \frac{0.5803(12)}{219.8(2.43 - 0.5803)} = 0.0171$$

From a combination of (9-17) and (9-18),

$$y_{i,\infty} = \frac{x_{i,\infty}L_{\infty} + x_{i,D}D}{L_{\infty} + D}$$

For iC_4 ,

$$y_{iC_4,\infty} = \frac{0.0171(219.8) + 12}{219.8 + 469.56} = 0.0229$$

Similarly, the mole fractions of the other components appearing in the distillate are

Component	$x_{i,\infty}$	$y_{i,\infty}$
iC_4	0.0171	0.0229
nC_4	0.8645	0.9168
iC_5	0.0818	0.0449
nC_5	0.0366	0.0154
	1.0000	1.0000

The temperature of the rectifying-section pinch point is obtained from either a bubble-point temperature calculation on $x_{i,\infty}$ or a dew-point temperature calculation on $y_{i,\infty}$. The result is 126°F. Similarly, the liquid-distillate temperature (bubble point) and the temperature of the vapor leaving the top stage (dew point) are both computed to be approximately 123°F. Because rectifying-section pinch-point temperature and distillate temperatures are very close, it is expected that $(R_{\infty})_{\min}$ and $(R_{\min})_{\text{external}}$ will be almost identical. Bachelor [4] obtained a value of 292 lbmol/h for the external-reflux rate, compared to 298 lbmol/h for the internal reflux rate.

Gilliland Correlation for Actual Reflux Ratio and Theoretical Stages

To achieve a specified separation between two key components, the reflux ratio and the number of theoretical stages must be greater than their minimum values. The actual reflux ratio is generally established by economic considerations at some multiple of minimum reflux. The corresponding number of theoretical stages is then determined by suitable analytical or graphical methods or, as discussed in this section, by an empirical equation. However, there is no reason why the number of theoretical stages cannot be specified as a multiple of minimum stages and the corresponding actual reflux computed by the same empirical relationship. As shown in Figure 9.9, from studies by Fair and Bolles [12], the optimal value of $R/R_{\min}$ is approximately 1.05. However, near-optimal conditions extend over a relatively broad range of mainly larger values of $R/R_{\min}$. In practice, superfractionators requiring a large number of stages are frequently designed for a value of $R/R_{\min}$ of approximately 1.10, while separations requiring a small number of stages are designed for a value of $R/R_{\min}$ of approximately 1.50. For intermediate cases, a commonly used rule of thumb is $R/R_{\min}$ equal to 1.30.

The number of equilibrium stages required for the separation of a binary mixture assuming constant relative volatility and constant molar overflow depends on $z_{i,F}$, $x_{i,D}$, $x_{i,B}$, q , R , and α . From (9-11), for a binary mixture, $N_{\min}$ depends on $x_{i,D}$, $x_{i,B}$, and α , while $R_{\min}$ depends on $z_{i,F}$, $x_{i,D}$, q , and α . Accordingly, a number of investigators have assumed empirical correlations of the form

$$N = N \{ N_{\min} \{ x_{i,D}, x_{i,B}, \alpha \}, R_{\min} \{ z_{i,F}, x_{i,D}, q, \alpha \}, R \}$$

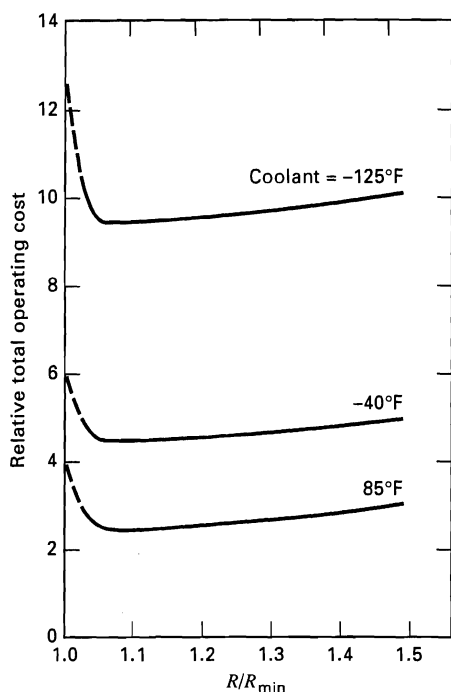


Figure 9.9 Effect of reflux ratio on cost.

[From J.R. Fair and W.L. Bolles, *Chem. Eng.*, 75(9), 156-178 (1968).]

Furthermore, they have assumed that such a correlation might exist for nearly ideal multicomponent systems even though additional feed composition variables and nonkey relative volatilities also influence the value of $R_{\min}$.

The most successful and simplest empirical correlation of this type is the one developed by Gilliland [13] and slightly modified in a later version by Robinson and Gilliland [14]. The correlation is shown in Figure 9.10, where the three sets of data points, which are based on accurate calculations, are

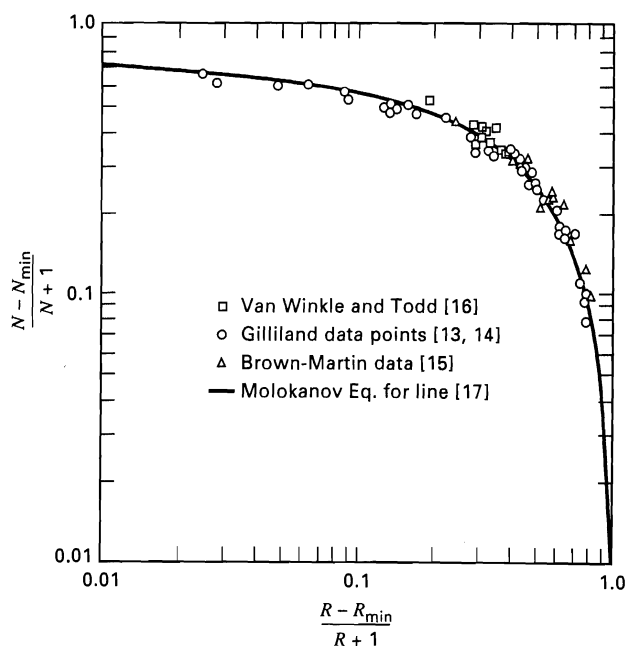


Figure 9.10 Comparison of rigorous calculations with Gilliland correlation.

the original points from Gilliland [13] and the multicomponent data points of Brown and Martin [15] and Van Winkle and Todd [16]. The 61 data points cover the following ranges of conditions:

- | | |
|----------------------------------|------------------------------|
| 1. Number of components: 2 to 11 | 4. α : 1.11 to 4.05 |
| 2. q : 0.28 to 1.42 | 5. $R_{\min}$: 0.53 to 9.09 |
| 3. Pressure: vacuum to 600 psig | 6. $N_{\min}$: 3.4 to 60.3 |

The line drawn through the data represents the equation developed by Molokanov et al. [17]:

$$Y = \frac{N - N_{\min}}{N + 1} = 1 - \exp \left[\left(\frac{1 + 54.4X}{11 + 117.2X} \right) \left(\frac{X - 1}{X^{0.5}} \right) \right] \quad (9-34)$$

where

$$X = \frac{R - R_{\min}}{R + 1}$$

This equation satisfies the end points ($Y = 0, X = 1$) and ($Y = 1, X = 0$). At a value of $R/R_{\min}$ near the optimum of 1.3, Figure 9.10 predicts an optimal ratio for $N/N_{\min}$ of approximately 2. The value of N includes one stage for a partial reboiler and one stage for a partial condenser, if any.

The Gilliland correlation is very useful for preliminary exploration of design variables. Although it was never intended for final design, the Gilliland correlation was used, before the applicability of digital computers, to design many distillation columns for multicomponent separations without benefit of accurate stage-by-stage calculations. In Figure 9.11, a replot of the correlation in linear coordinates shows that a small initial increase in R above $R_{\min}$ causes a large decrease in N , but further changes in R have a much smaller effect on N . The knee in the curve of Figure 9.11 corresponds closely to the optimal value of $R/R_{\min}$ in Figure 9.9.

Robinson and Gilliland [14] state that a more accurate correlation should utilize a parameter involving the feed condition q . This effect is shown in Figure 9.12 using data points for the sharp separation of benzene-toluene mixtures from Guerrieri [18]. The data, which cover feed conditions ranging

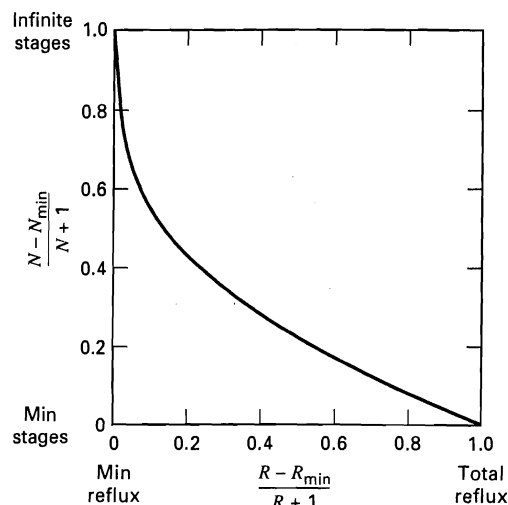


Figure 9.11 Gilliland correlation with linear coordinates.

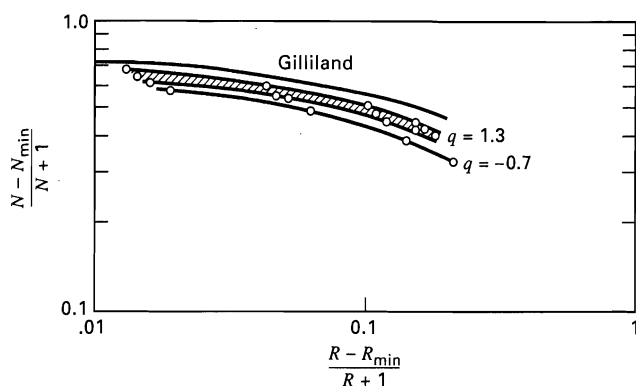


Figure 9.12 Effect of feed condition on Gilliland correlation.

[From G. Guerrieri, *Hydrocarbon Processing*, 48(8), 137–142 (1969).]

from subcooled liquid to superheated vapor (q equals 1.3 to -0.7), show a trend toward decreasing theoretical-stage requirements with increasing feed vaporization. The Gilliland correlation appears to be conservative for feeds having low values of q . Donnell and Cooper [19] state that this effect of q is important only when the α between the key components is high or when the feed is low in volatile components.

A serious problem with the Gilliland correlation can occur when stripping is much more important than rectification. For example, Oliver [20] considers a fictitious binary case with specifications of $z_F = 0.05$, $x_D = 0.40$, $x_B = 0.001$, $q = 1$, $\alpha = 5$, $R/R_{\min} = 1.20$, and constant molar overflow. By exact calculations, $N = 15.7$. From the Fenske equation, $N_{\min} = 4.04$. From the Underwood equation, $R_{\min} = 1.21$. From (9-32) for the Gilliland correlation, $N = 10.3$. This is 34% lower than the exact value. This limitation, which is caused by ignoring boilup, is discussed further by Strangio and Treybal [21], who present a more accurate, but far more tedious, method for such cases.

EXAMPLE 9.6

Use the Gilliland correlation to estimate the theoretical-stage requirements for the debutanizer of Examples 9.1, 9.2, and 9.5 for an external reflux of 379.6 lbmol/h (30% greater than the exact value of the minimum-reflux rate from Bachelor).

SOLUTION

From the examples cited, values of $R_{\min}$ and $[(R - R_{\min})/(R + 1)]$ are obtained using a distillate rate from Example 9.5 of 469.56 lbmol/h. Thus, $R = 379.6/469.56 = 0.808$. With $N_{\min} = 8.88$,

$$R_{\min} = 0.479, \quad \text{and} \quad \frac{R - R_{\min}}{R + 1} = X = 0.182$$

From (9-34),

$$\frac{N - N_{\min}}{N + 1} = 1 - \exp \left[\left(\frac{1 + 54.4(0.182)}{11 + 117.2(0.182)} \right) \left(\frac{0.182 - 1}{0.182^{0.5}} \right) \right]$$

$$= 0.476$$

$$N = \frac{8.88 + 0.476}{1 - 0.476} = 17.85$$

$$N - 1 = 16.85$$

where $N - 1$ corresponds to the equilibrium stages in the tower allowing one theoretical stage for the reboiler, but no stage for the total condenser.

It should be kept in mind that, had the exact value of $R_{\min}$ not been known and a value of R equal to 1.3 times $R_{\min}$ from the Underwood method been used, the value of R would have been 292 lbmol/h. But this, by coincidence, is only the true minimum reflux. Therefore, the desired separation would not be achieved.

Feed-Stage Location

Implicit in the application of the Gilliland correlation is the specification that the theoretical stages be distributed optimally between the rectifying and stripping sections. As suggested by Brown and Martin [15], the optimal feed stage can be located by assuming that the ratio of stages above the feed to stages below the feed is the same as the ratio determined by simply applying the Fenske equation to the separate sections at total reflux conditions to give

$$\frac{N_R}{N_S} \approx \frac{(N_R)_{\min}}{(N_S)_{\min}} = \frac{\log[(x_{LK,D}/z_{LK,F})(z_{HK,F}/x_{HK,D})] \log[(\alpha_B \alpha_F)^{1/2}]}{\log[(z_{LK,F}/x_{LK,B})(x_{HK,B}/z_{HK,F})] \log[(\alpha_D \alpha_F)^{1/2}]} \quad (9-35)$$

Unfortunately, (9-35) is not reliable except for fairly symmetrical feeds and separations.

A reasonably good approximation of optimal feed-stage location can be made by employing the empirical equation of Kirkbride [22]:

$$\frac{N_R}{N_S} = \left[\left(\frac{z_{HK,F}}{z_{LK,F}} \right) \left(\frac{x_{LK,B}}{x_{HK,D}} \right)^2 \left(\frac{B}{D} \right) \right]^{0.206} \quad (9-36)$$

An extreme test of both these equations is provided by the fictitious binary-mixture problem of Oliver [20] cited in the previous section. Exact calculations by Oliver and calculations using (9-35) and (9-36) give the following results:

Method	N_R/N_S
Exact	0.08276
Kirkbride (9-34)	0.1971
Fenske ratio (9-33)	0.6408

Although the result from the Kirkbride equation is not very satisfactory, the result from the Fenske ratio method is much worse.

EXAMPLE 9.7

Use the Kirkbride equation to determine the feed-stage location for the debutanizer of Example 9.1, assuming an equilibrium-stage requirement of 18.27.

SOLUTION

Assume that the product distribution computed in Example 9.3 for total-reflux conditions is a good approximation to the distillate and

bottoms compositions at actual reflux conditions. Then

$$x_{nC_4,B} = \frac{6.0}{408.5} = 0.0147 \quad x_{iC_5,D} = \frac{13}{467.8} = 0.0278$$

$$D = 467.8 \text{ lbmol/h} \quad B = 408.5 \text{ lbmol/h}$$

From Figure 9.3,

$$z_{nC_4,F} = 448/876.3 = 0.5112$$

and

$$z_{iC_5,F} = 36/876.3 = 0.0411$$

From (9-36),

$$\frac{N_R}{N_S} = \left[\left(\frac{0.0411}{0.5112} \right) \left(\frac{0.0147}{0.0278} \right)^2 \left(\frac{408.5}{467.8} \right) \right]^{0.206} = 0.445$$

Therefore, $N_R = (0.445/1.445)(18.27) = 5.63$ stages and $N_S = 18.27 - 5.63 = 12.64$ stages.

Rounding the estimated stage requirements leads to one stage as a partial reboiler, 12 stages below the feed, and six stages above the feed.

Distribution of Nonkey Components at Actual Reflux

For multicomponent mixtures, all components distribute to some extent between distillate and bottoms at total reflux conditions. However, at minimum-reflux conditions, none or only a few of the nonkey components distribute. Distribution ratios for these two limiting conditions are shown in Figure 9.13 for the debutanizer example. For total-reflux conditions, results from the Fenske equation in Example 9.3 plot as a straight line for the log-log coordinates. For minimum reflux, results from the Underwood equation in Example 9.5 are shown as a dashed line.

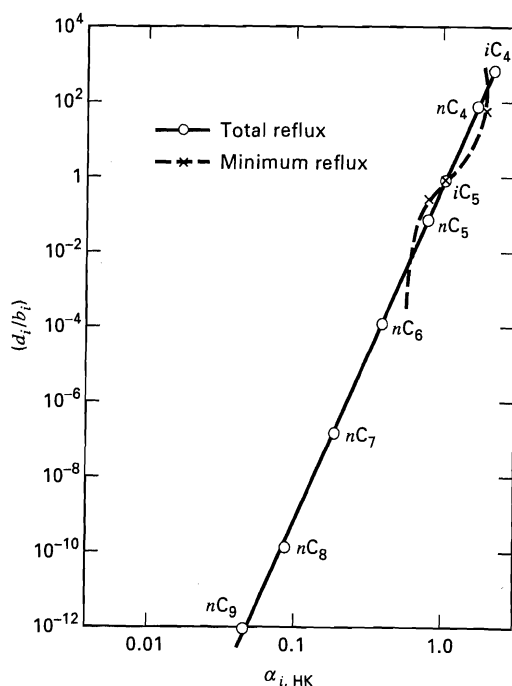


Figure 9.13 Component distribution ratios at extremes of distillation operating conditions.

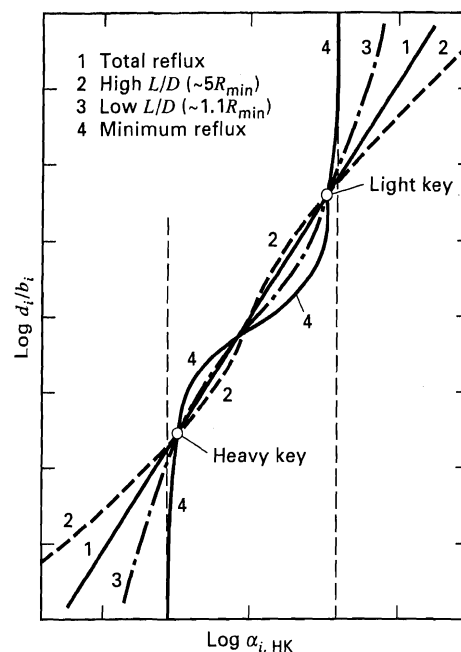


Figure 9.14 Component distribution ratios at various reflux ratios.

It might be expected that a product-distribution curve for actual reflux conditions would lie between the two limiting curves. However, as shown by Stupin and Lockhart [23], product distributions in distillation are complex. A typical result is shown in Figure 9.14. For a reflux ratio near minimum, the product distribution (curve 3) lies between the two limits (curves 1 and 4). However, for a high reflux ratio, the product distribution for a nonkey component (curve 2) may actually lie outside the limits, so that an inferior separation results.

For the behavior of the product distribution in Figure 9.14, Stupin and Lockhart provide an explanation that is consistent with the Gilliland correlation of Figure 9.10. As the reflux ratio is decreased from total reflux while maintaining the specified splits of the two key components, equilibrium-stage requirements increase only slowly at first, but then rapidly as minimum reflux is approached. Initially, large decreases in reflux cannot be adequately compensated for by increasing the number of stages. This causes inferior nonkey component distributions. However, as minimum reflux is approached, comparatively small decreases in reflux are more than compensated for by large increases in equilibrium stages; and the separation of nonkey components becomes superior to that at total reflux. It appears reasonable to assume that, at a near-optimal reflux ratio of 1.3, nonkey-component distribution is close to that estimated by the Fenske equation for total-reflux conditions.

9.2 KREMSEK GROUP METHOD

Many multicomponent separators are cascades of stages where the two contacting phases flow countercurrently. Approximate calculation procedures have been developed to relate compositions of streams entering and exiting cascades

to the number of equilibrium stages required. These approximate procedures are called *group methods* because they provide only an overall treatment of the stages in the cascade without considering detailed changes in temperature, flow rates, and composition in the individual stages. In this section, single cascades used for absorption, stripping, and liquid-liquid extraction are considered.

Kremser [1] originated the group method. He derived overall species material balances for a multistage counter-current absorber. Subsequent articles by Souders and Brown [24] Horton and Franklin [25] and Edmister [26] improved the method. The Kremser equations are derived and applied to absorption in Section 5.4. These equations are illustrated for strippers and extractors here. Another treatment by Smith and Brinkley [27] emphasizes liquid-liquid separations.

Strippers

The vapor entering a stripper is often steam or an inert gas. When the stripping agent contains none of the species in the feed liquid, is not present in the entering liquid, and is not absorbed or condensed in the stripper, the only direction of mass transfer is from the liquid to the gas phase. Then, only values of the effective stripping factor, S_e , as defined by (5-51), are needed to apply the group method via (5-49) and (5-50). The equations for strippers are analogous to those for absorbers.

For optimal stripping, temperatures should be high and pressures low. However, temperatures should not be so high as to cause decomposition, and vacuum should be used only if necessary. The minimum stripping-agent flow rate, for a specified value of ϕ_S for a key component K corresponding to an infinite number of stages, can be estimated from an equation obtained from (5-50) with $N = \infty$,

$$(V_0)_{\min} = \frac{L_{N+1}}{K_K} (1 - \phi_{SK}) \quad (9-37)$$

This equation assumes that $A_K < 1$ and the fraction of liquid feed stripped is small.

EXAMPLE 9.8

Sulfur dioxide and butadienes (B3 and B2) are to be stripped with nitrogen from the liquid stream given in Figure 9.15 so that butadiene sulfone (BS) product will contain less than 0.05 mol% SO_2 and less than 0.5 mol% butadienes. Estimate the flow rate of nitrogen, N_2 , and the number of equilibrium stages required.

SOLUTION

Neglecting stripping of BS, the stripped liquid must have the following component flow rates, and corresponding values for ϕ_S :

Species	l_1 , lbmol/h	$\phi_S = \frac{l_1}{l_{N+1}}$
SO_2	<0.0503	<0.00503
B3 + B2	<0.503	<0.0503
BS	100.0	—

Thermodynamic properties can be computed based on ideal solutions at low pressures. For butadiene sulfone, the vapor pressure is

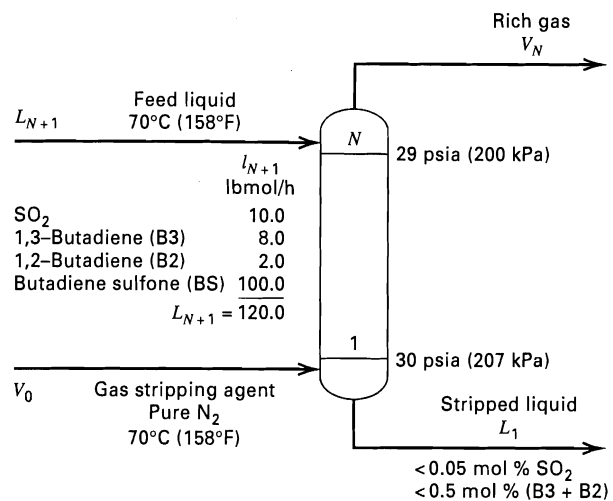


Figure 9.15 Specifications for stripper of Example 9.8.

given by

$$P_{BS}^s = \exp \left(17.30 - \frac{11,142}{T + 459.67} \right)$$

where P_{BS}^s is in pounds force per square inch and T is in degrees Fahrenheit. The liquid enthalpy of BS is

$$(h_L)_{BS} = 50T$$

where $(h_L)_{BS}$ is in British thermal units per pound-mole and T is in degrees Fahrenheit.

The entering flow rate of the stripping agent V_0 is not specified. The minimum rate at infinite stages can be computed from (9-37), provided that a key component is selected. Suppose we choose B2, which is the heaviest component to be stripped to a specified extent. At 70°C, the vapor pressure of B2 is 90.4 psia. From Raoult's law at 30 psia total pressure,

$$K_{B2} = \frac{90.4}{30} = 3.01$$

From (9-37), using $(\phi_S)_{B2} = 0.0503$, we have

$$(V_0)_{\min} = \frac{120}{3.01} (1 - 0.0503) = 37.9 \text{ lbmol/h}$$

For this value of $(V_0)_{\min}$, (9-42) can now be used to determine ϕ_S for B3 and SO_2 . The K -values for these two species are 4.53 and 6.95, respectively. From (9-37), at infinite stages with $V_0 = 37.9 \text{ lbmol/h}$,

$$(\phi_S)_{B3} = 1 - \frac{4.53(37.9)}{120} = -0.43$$

and

$$(\phi_S)_{\text{SO}_2} = 1 - \frac{6.95(37.9)}{120} = -1.19$$

These negative values indicate complete stripping of B3 and SO_2 . Therefore, the total butadienes in the stripped liquid would be only $(0.0503)(2.0) = 0.1006$, compared to the specified value of 0.503. We can obtain a better estimate of $(V_0)_{\min}$ by assuming that all of the butadiene content of the stripped liquid is due to B2. Then $(\phi_S)_{B2} = 0.503/2 = 0.2515$, and $(V_0)_{\min}$ from (9-37) is 29.9 lbmol/h. Values of $(\phi_S)_{B3}$ and $(\phi_S)_{\text{SO}_2}$ are still negative.

The actual entering flow rate for the stripping vapor must be greater than the minimum value. To estimate the effect of V_0 on the theoretical stage requirements and ϕ_S values for the nonkey components, the Kremser approximation is used with K -values at 70°C

and 30 psia, $L = L_{N+1} = 120$ lbmol/h, and $V = V_0$ equal to a series of multiples of 29.9 lbmol/h. The calculations are greatly facilitated if values of N are selected and values of V are determined from (5-51), where S is obtained from Figure 5.9. Because B3 will be found to some extent in the stripped liquid, $(\phi_S)_{B2}$ will be held below 0.2515. By making iterative calculations, one can choose $(\phi_S)_{B2}$ so that $(\phi_S)_{B2+B3}$ satisfies the specification of, say, 0.05. For 10 theoretical stages, assuming essentially complete stripping of B3 such that $(\phi_S)_{B2} \approx 0.25$, $S_{B2} \approx 0.76$ from Figure 5.9. From (5-51),

$$V = V_0 = \frac{(120)(0.76)}{3.01} = 30.3 \text{ lbmol/h}$$

For B3, from (5-51),

$$S_{B3} = \frac{(4.53)(30.3)}{120} = 1.143$$

From Figure 5.9,

$$(\phi_S)_{B3} = 0.04$$

Thus,

$$(\phi_S)_{B2+B3} = \frac{0.25(2) + 0.04(8)}{10} = 0.082$$

This is considerably above the specification of 0.05. Therefore, repeat the calculations with, say, $(\phi_S)_{B2} = 0.09$ and continue to repeat until the specified value of $(\phi_S)_{B2+B3}$ is obtained. In this manner, calculations for various numbers of theoretical stages are carried out with converged results as shown.

N	V_0 , lbmol/h	$V_0/(V_0)_{\min}$	Fraction Not Stripped				
			ϕ_{SO_2}	ϕ_{B3}	ϕ_{B2}	ϕ_{B2+B3}	ϕ_{BS}
∞	29.9	1.00	0.0	0.0	0.2515	0.0503	0.9960
10	33.9	1.134	0.0005	0.017	0.18	0.050	0.9955
5	45.4	1.518	0.0050	0.029	0.117	0.040	0.9940
3	94.3	3.154	0.0050	0.016	0.045	0.022	0.9874

These results show that the specification on SO_2 is also met for all four values of N .

Liquid-Liquid Extraction

A schematic representation of a countercurrent extraction cascade is shown in Figure 9.16, with stages numbered from the top down and solvent V_{N+1} entering at the bottom.⁴ The group method of calculation can be applied, with the equations written by analogy to absorbers. In place of the K -value, the distribution coefficient is used:

$$K_{D_i} = \frac{y_i}{x_i} = \frac{v_i/V}{l_i/L} \quad (9-38)$$

Here, y_i is the mole fraction of i in the solvent or extract phase and x_i is the mole fraction in the feed or raffinate phase. Also, in place of the stripping factor, an *extraction*

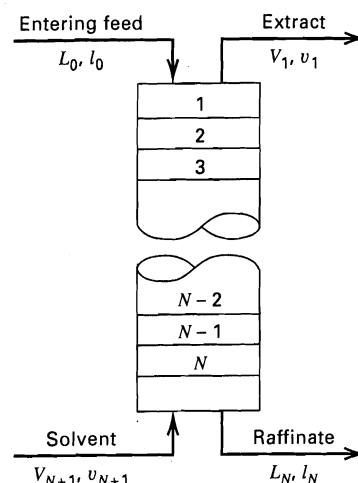


Figure 9.16 Countercurrent, liquid-liquid extraction cascade.

factor, E , is used, where

$$E_i = \frac{K_{D_i} V}{L} \quad (9-39)$$

The reciprocal of E is

$$U_i = \frac{1}{E_i} = \frac{L}{K_{D_i} V} \quad (9-40)$$

The working equations for each component are

$$v_1 = v_{N+1} \phi_U + l_0(1 - \phi_E) \quad (9-41)$$

$$l_N = l_0 + v_{N+1} - v_1 \quad (9-42)$$

where

$$\phi_U = \frac{U_e - 1}{U_e^{N+1} - 1} \quad (9-43)$$

$$\phi_E = \frac{E_e - 1}{E_e^{N+1} - 1} \quad (9-44)$$

For the Kremser approximation, values of E_i and U_i in (9-39) and (9-40) are based on the feed and solvent at entering conditions. However, in liquid-liquid extraction, values of V , L , and K_D may change considerably from stage to stage. Therefore a better approximation is desirable. This is achieved, with reference to Figure 9.16, by the following relations due to Horton and Franklin [25] and Edmister [26], which use average values of E_i and U_i based on estimates of values of V , L , and K_D at each end of the cascade. These equations are applied in Example 9.9.

$$V_2 = V_1 \left(\frac{V_{N+1}}{V_1} \right)^{1/N} \quad (9-45)$$

$$L_1 = L_0 + V_2 - V_1 \quad (9-46)$$

$$V_N = V_{N+1} \left(\frac{V_1}{V_{N+1}} \right)^{1/N} \quad (9-47)$$

$$E_e = [E_1(E_N + 1) + 0.25]^{1/2} - 0.5 \quad (9-48)$$

$$U_e = [U_N(U_1 + 1) + 0.25]^{1/2} - 0.5 \quad (9-49)$$

⁴ In a vertical extractor, solvent would have to enter at the top if of greater density than the feed.

If desired, (9-38) through (9-49) can be applied using mass units rather than mole units. No enthalpy-balance equations are required because ordinarily temperature changes in an adiabatic extractor are not great unless the feed and solvent enter at appreciably different temperatures or the heat of mixing is large. Unfortunately, the group method is not always reliable for liquid-liquid extraction cascades because the distribution coefficient, as discussed in Chapter 2, is a ratio of activity coefficients, which can vary drastically with composition. Accordingly, rigorous methods of Chapter 10 are preferred.

EXAMPLE 9.9

Countercurrent, liquid-liquid extraction with methylene chloride is to be used at 25°C to recover dimethylformamide from an aqueous stream as shown in Figure 9.17. Estimate flow rates and compositions of extract and raffinate streams by the group method using mass units. Distribution coefficients for all components except DMF are essentially constant over the expected composition range and on a mass-fraction basis are

Component	K_{Di}
MC	40.2
FA	0.005
DMA	2.2
W	0.003

The distribution coefficient for DMF depends on concentration in the water-rich phase as shown in Figure 9.18.

SOLUTION

Although the Kremser approximation could be applied for the first trial calculation, the following values will be assumed from guesses based on the magnitudes of the K_D -values.

Pounds per Hour				
Component	Feed, l_0	Solvent, v_{11}	Raffinate, l_{10}	Extract, v_1
FA	20	0	20	0
DMA	20	0	0	20
DMF	400	2	2	400
W	3,560	25	3,560	25
MC	0	9,973	88	9,885
	4,000	10,000	3,670	10,330

From (9-45) through (9-49), we have

$$V_2 = 10,330 \left(\frac{10,000}{10,330} \right)^{1/10} = 10,297 \text{ lb/h;}$$

$$L_1 = 4,000 + 10,297 - 10,330 = 3,967 \text{ lb/h;}$$

$$V_{10} = 10,000 \left(\frac{10,330}{10,000} \right)^{1/10} = 10,033 \text{ lb/h}$$

From (9-39), (9-40), (9-48), and (9-49), assuming a mass fraction of 0.09 for DMF in L_1 in order to obtain $(K_D)_{DMF}$ for stage 1, we have

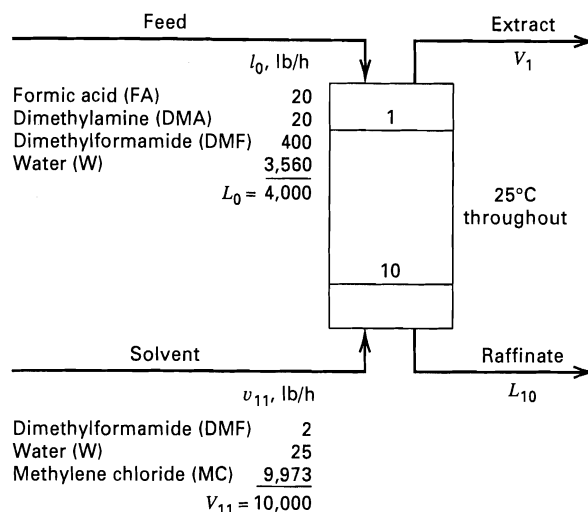


Figure 9.17 Specifications for extractor of Example 9.9.

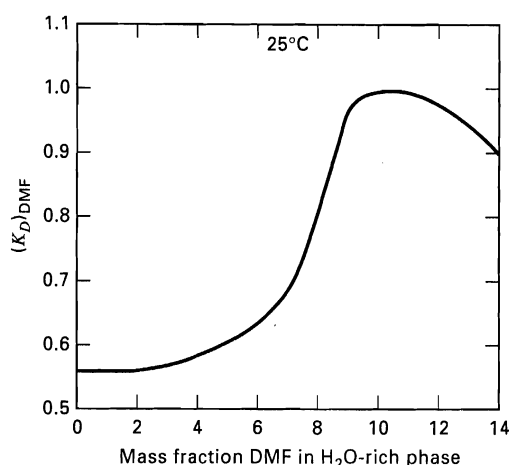


Figure 9.18 Distribution coefficient for dimethylformamide between water and methylene chloride.

Component	E_1	E_{10}	U_1	U_{10}	E_e	U_e
FA	0.013	0.014	—	—	0.013	—
DMA	5.73	6.01	—	—	5.86	—
DMF	2.50	1.53	0.400	0.653	2.06	0.579
W	0.0078	0.0082	128	122	0.0078	125
MC	—	—	0.0096	0.0091	—	0.0091

From (9-44), (9-43), (9-41), and (9-42), we have

Pounds per Hour			
Component	ϕ_E	ϕ_U	
FA	0.9870	—	19.7
DMA	0.0	—	0.0
DMF	0.000374	0.422	1.3
W	0.9922	0.0	3,557.2
MC	—	0.9909	90.8
			3,669.0
			10,331.0

The calculated total flow rates L_{10} and V_1 are almost exactly equal to the assumed rates. Therefore, an additional iteration is not necessary. The degree of extraction of DMF is very high. It would be worthwhile to calculate additional cases with less solvent and/or fewer equilibrium stages.

SUMMARY

1. The Fenske–Underwood–Gilliland (FUG) method for simple distillation of ideal and nearly ideal multicomponent mixtures is useful for making preliminary estimates of stage and reflux requirements.
2. Based on a specified split of two key components in the feed mixture, the theoretical Fenske equation is used to determine the minimum number of equilibrium stages at total reflux. The theoretical Underwood equations are used to determine the minimum-reflux ratio for an infinite number of stages. The empirical Gilliland correlation relates the minimum stages and minimum reflux ratio to the actual reflux ratio and the actual number of equilibrium stages.

3. Estimates of the distribution of nonkey components and the feed-stage location can be made with the Fenske and Kirkbride equations, respectively.
4. The Underwood equations are more restrictive than the Fenske equation and must be used with care and caution.
5. The Kremser group method can be applied to simple strippers and liquid–liquid extractors to make approximate estimates of component recoveries for specified values of entering flow rates and number of equilibrium stages.

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EXERCISES

Section 9.1

9.1 A mixture of propionic and *n*-butyric acids, which can be assumed to form ideal solutions, is to be separated by distillation into a distillate containing 95 mol% propionic acid and a bottoms product containing 98 mol% *n*-butyric acid. Determine the type of condenser to be used and estimate the distillation-column operating pressure.

9.2 A sequence of two distillation columns is to be used to produce the products indicated in Figure 9.19. Establish the type of condenser and an operating pressure for each column for: (a) The direct sequence (C_2/C_3 separation first) and (b) The indirect sequence (C_3/nC_4 separation first). Use K -values from Figures 2.8 and 2.9.

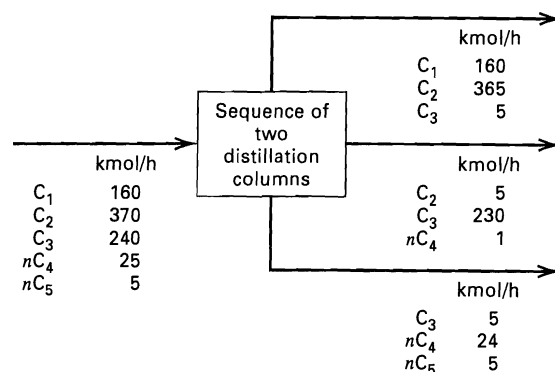


Figure 9.19 Data for Exercise 9.2.

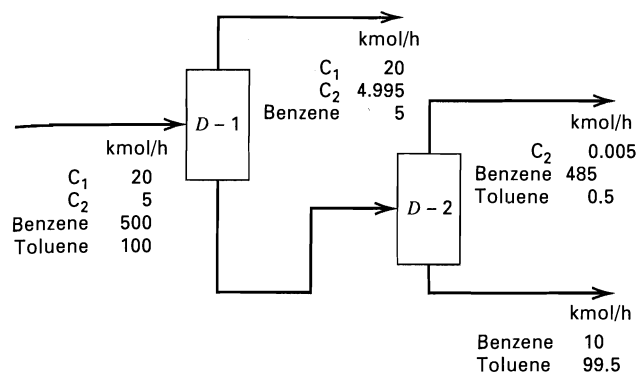


Figure 9.20 Data for Exercise 9.3.

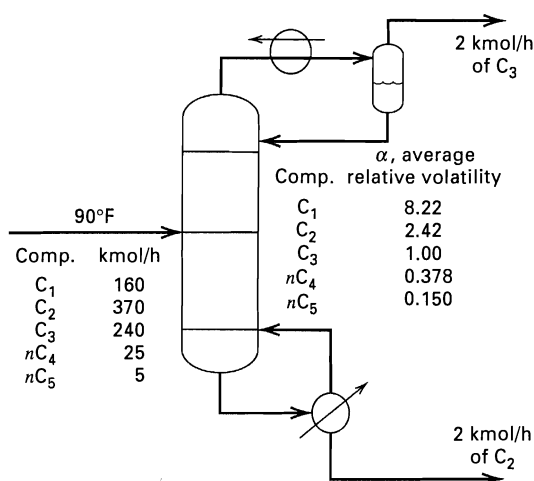


Figure 9.21 Data for Exercise 9.4.

9.3 For each of the two distillation separations (*D-1* and *D-2*) indicated in Figure 9.20, establish the type of condenser and an operating pressure.

9.4 A deethanizer is to be designed for the separation indicated in Figure 9.21. Estimate the number of equilibrium stages required, assuming it is equal to 2.5 times the minimum number of equilibrium stages at total reflux.

9.5 For the complex distillation operation shown in Figure 9.22, use the Fenske equation to determine the minimum number of stages required between: (a) The distillate and feed, (b) The feed and the side stream, and (c) The side stream and bottoms. The *K*-values can be obtained from Raoult's law.

9.6 A 25 mol% mixture of acetone (*A*) in water (*W*) is to be separated by distillation at an average pressure of 130 kPa into a distillate containing 95 mol% acetone and a bottoms containing 2 mol% acetone. The infinite-dilution activity coefficients are

$$\gamma_A^\infty = 8.12 \quad \gamma_W^\infty = 4.13$$

Calculate by the Fenske equation the number of equilibrium stages required. Compare the result to that calculated from the McCabe-Thiele method. Is the Fenske equation reliable for this separation?

9.7 For the distillation operation indicated in Figure 9.23, calculate the minimum number of equilibrium stages and the distribution of the nonkey components by the Fenske equation, using Figures 2.8 and 2.9 for *K*-values.

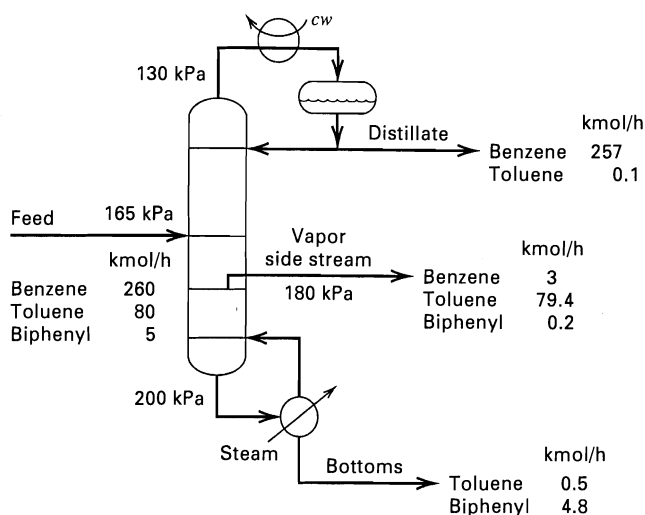


Figure 9.22 Data for Exercise 9.5.

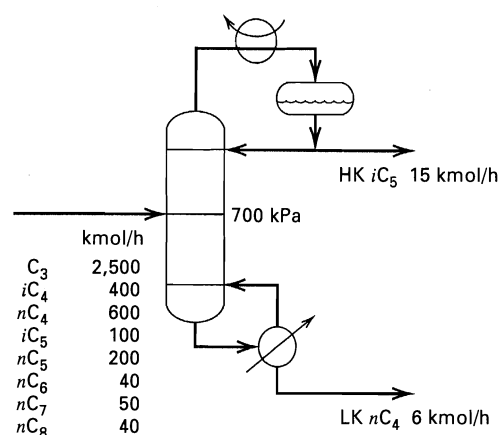


Figure 9.23 Data for Exercise 9.7.

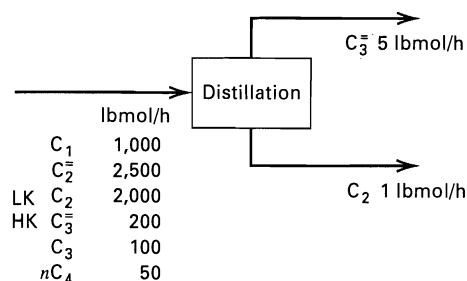


Figure 9.24 Data for Exercise 9.8.

9.8 For the distillation operation shown in Figure 9.24, establish the type of condenser and an operating pressure, calculate the minimum number of equilibrium stages, and estimate the distribution of the nonkey components. Obtain *K*-values from Figures 2.8 and 2.9.

9.9 For 15 minimum equilibrium stages at 250 psia, calculate and plot the percent recovery of *C*₃ in the distillate as a function of distillate flow rate for the distillation of 1,000 lbmol/h of a feed containing 3% *C*₂, 20% *C*₃, 37% *nC*₄, 35% *nC*₅, and 5% *nC*₆ by moles. Obtain *K*-values from Figures 2.8 and 2.9.

9.10 Use the Underwood equations to estimate the minimum external-reflux ratio for the separation by distillation of 30 mol% propane in propylene to obtain 99 mol% propylene and 98 mol% propane, if the feed condition at a column operating pressure of 300 psia is: (a) Bubble-point liquid, (b) Fifty mole percent vaporized, and (c) Dew-point vapor. Use K -values from Figures 2.8 and 2.9.

9.11 For the conditions of Exercise 9.7, compute the minimum-external-reflux rate and the distribution of the nonkey components at minimum reflux by the Underwood equation if the feed is a bubble-point liquid at column pressure.

9.12 Calculate and plot the minimum-external-reflux ratio and the minimum number of equilibrium stages against percent product purity for the separation by distillation of an equimolar bubble-point liquid feed of isobutane/*n*-butane at 100 psia. The distillate is to have the same iC_4 purity as the bottoms is to have nC_4 purity. Consider percent purities from 90% to 99.99%. Discuss the significance of the results.

9.13 Use the Fenske–Underwood–Gilliland shortcut method to determine the reflux ratio required to conduct the distillation operation indicated in Figure 9.25 if $N/N_{\min} = 2.0$, the average relative volatility = 1.11, and the feed is at the bubble-point temperature at column feed-stage pressure. Assume that external reflux equals internal reflux at the upper pinch zone. Assume a total condenser and a partial reboiler.

9.14 A feed consisting of 62 mol% *para*-dichlorobenzene in *ortho*-dichlorobenzene is to be separated by distillation at near atmospheric pressure into a distillate containing 98 mol% *para* isomer and bottoms containing 96 mol% *ortho* isomer.

If a total condenser and partial reboiler are used, $q = 0.9$, average relative volatility = 1.154, and reflux/minimum reflux = 1.15, use the Fenske–Underwood–Gilliland procedure to estimate the number of theoretical stages required.

9.15 Explain why the Gilliland correlation can give erroneous results for cases where the ratio of rectifying to stripping stages is small.

9.16 The hydrocarbon feed to a distillation column is a bubble-point liquid at 300 psia with the mole fraction composition, $C_2 = 0.08$, $C_3 = 0.15$, $nC_4 = 0.20$, $nC_5 = 0.27$, $nC_6 = 0.20$, and $nC_7 = 0.10$.

(a) For a sharp separation between nC_4 and nC_5 , determine the column pressure and type of condenser if condenser outlet temperature is 120°F.

(b) At total reflux, determine the separation for eight theoretical stages overall, specifying 0.01 mole fraction nC_4 in the bottoms product.

(c) Determine the minimum-reflux ratio for the separation in part (b).

(d) Determine the number of theoretical stages at $L/D = 1.5$ times minimum using the Gilliland correlation.

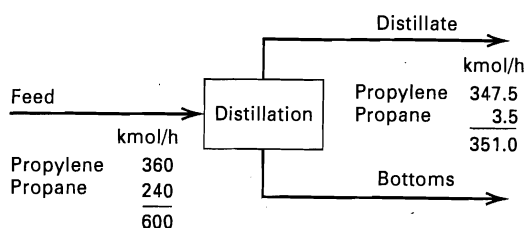


Figure 9.25 Data for Exercise 9.13.

9.17 The following feed mixture is to be separated by ordinary distillation at 120 psia so as to obtain 92.5 mol% of the nC_4 in the liquid distillate and 82.0 mol% of the iC_5 in the bottoms.

Component	lbmol/h
C_3	5
iC_4	15
nC_4	25
iC_5	20
nC_5	35
Total	100

(a) Estimate the minimum number of equilibrium stages required by applying the Fenske equation. Obtain K -values from Figures 2.8 and 2.9.

(b) Use the Fenske equation to determine the distribution of nonkey components between distillate and bottoms.

(c) Assuming that the feed is at its bubble point, use the Underwood method to estimate the minimum-reflux ratio.

(d) Determine the number of theoretical stages required by the Gilliland correlation assuming $L/D = 1.2(L/D)_{\min}$, a partial reboiler, and a total condenser.

(e) Estimate the feed-stage location.

9.18 Consider the separation by distillation of a chlorination effluent to recover C_2H_5Cl . The feed is a bubble-point liquid at the column pressure of 240 psia with the following composition and K -values for the column conditions:

Component	Mole Fraction	K
C_2H_4	0.05	5.1
HCl	0.05	3.8
C_2H_6	0.10	3.4
C_2H_5Cl	0.80	0.15

Specifications are: (x_D/x_B) for $C_2H_5Cl = 0.01$

(x_D/x_B) for $C_2H_6 = 75$

Calculate the product distribution, the minimum theoretical stages, the minimum reflux, and the theoretical stages at 1.5 times minimum L/D and locate the feed stage. The column is to have a partial condenser and a partial reboiler.

9.19 One hundred kilogram-moles per hour of a three-component bubble-point mixture to be separated by distillation has the following composition:

Component	Mole Fraction	Relative Volatility
A	0.4	5
B	0.2	3
C	0.4	1

(a) For a distillate rate of 60 kmol/h, five theoretical stages, and total reflux, calculate the distillate and bottoms compositions by the Fenske equation.

(b) Using the separation in part (a) for components B and C, determine the minimum reflux and minimum boilup ratio by the Underwood equation.

(c) For an operating reflux ratio of 1.2 times the minimum, determine the number of theoretical stages and the feed-stage location.

9.20 For the conditions of Exercise 9.6, determine the ratio of rectifying to stripping equilibrium stages by: (a) Fenske equation, (b) Kirkbride equation, and (c) McCabe–Thiele diagram. Discuss your results.

Section 9.2

9.21 Starting with equations like (5-46) and (5-47), show that for two stages, $S_e = \sqrt{0.25 + S_2(S_1 + 1)} - 0.5$.

9.22 Determine by the Kremser group method the separation that can be achieved for the absorption operation indicated in Figure 9.26 for the following combinations of conditions: (a) Six equilibrium stages and 75-psia operating pressure, (b) Three equilibrium stages and 150-psia operating pressure, and (c) Six equilibrium stages and 150-psia operating pressure.

9.23 One thousand kilogram-moles per hour of rich gas at 70°F with 25% C_1 , 15% C_2 , 25% C_3 , 20% nC_4 , and 15% nC_5 by moles is to be absorbed by 500 kmol/h of nC_{10} at 90°F in an absorber operating at 4 atm. Calculate by the Kremser group method the percent absorption of each component for: (a) Four theoretical stages, (b) Ten theoretical stages, and (c) Thirty theoretical stages. Use Figures 2.8 and 2.9 for K -values.

9.24 For the flashing and stripping operation indicated in Figure 9.27, determine by the Kremser group method the kilogram-moles per hour of steam if the stripper is operated at 2 atm and has five theoretical stages.

9.25 A stripper operating at 50 psia with three equilibrium stages is used to strip 1,000 kmol/h of liquid at 250°F having the following molar composition: 0.03% C_1 , 0.22% C_2 , 1.82% C_3 , 4.47% nC_4 , 8.59% nC_5 , 84.87% nC_{10} . The stripping agent is 100 kmol/h of superheated steam at 300°F and 50 psia. Use the group method to estimate the compositions and flow rates of the stripped liquid and rich gas.

9.26 One hundred kilogram-moles per hour of an equimolar mixture of benzene (B), toluene (T), n -hexane (C_6), and n -heptane (C_7)

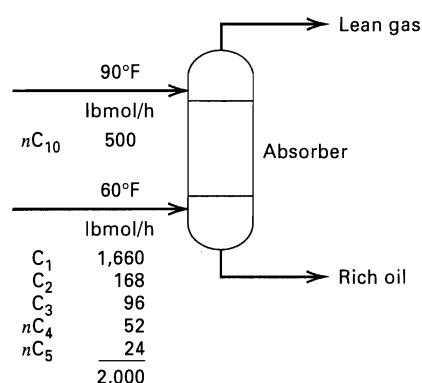


Figure 9.26 Data for Exercise 9.22.

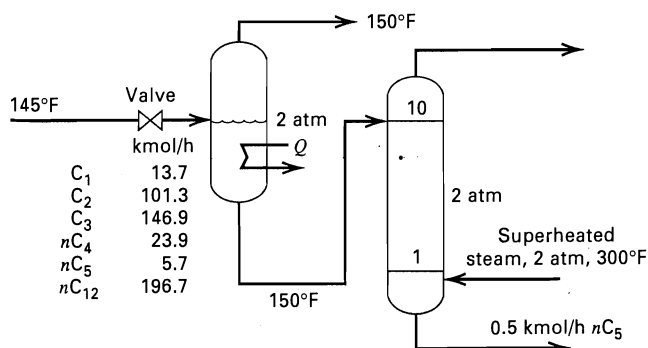


Figure 9.27 Data for Exercise 9.24.

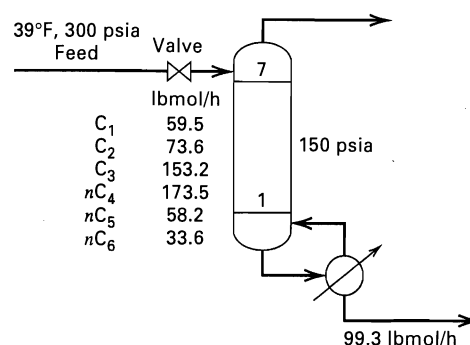


Figure 9.28 Data for Exercise 9.27.

is to be extracted at 150°C by 300 kmol/h of diethylene glycol (DEG) in a countercurrent, liquid-liquid extractor having five equilibrium stages. Estimate the flow rates and compositions of the extract and raffinate streams by the group method. In mole-fraction units, the distribution coefficients for the hydrocarbon can be assumed essentially constant at the following values:

Component	$K_{Di} = y(\text{solvent phase})/x(\text{raffinate phase})$
B	0.33
T	0.29
C_6	0.050
C_7	0.043

For diethylene glycol, assume $K_D = 30$. [E.D. Oliver, *Diffusional Separation Processes*, John Wiley and Sons, New York, p. 432 (1966).]

9.27 A reboiled stripper in a natural-gas plant is to be used to remove mainly propane and lighter components from the feed shown in Figure 9.28. Determine by the group method the compositions of the vapor and liquid products.

9.28 A mixture of ethylbenzene and xylenes is to be distilled as shown in Figure 9.29. Assuming the applicability of Raoult's and Dalton's laws:

(a) Use the Fenske-Underwood-Gilliland method to estimate the number of stages required for a reflux-to-minimum reflux ratio of 1.10. Estimate the feed stage location by the Kirkbride equation.
(b) From the results of part (a) for reflux, stages, and distillate rate, use the Edmister group method of Section 5.4 to predict the compositions of the distillate and bottoms. Compare the results with the specifications.

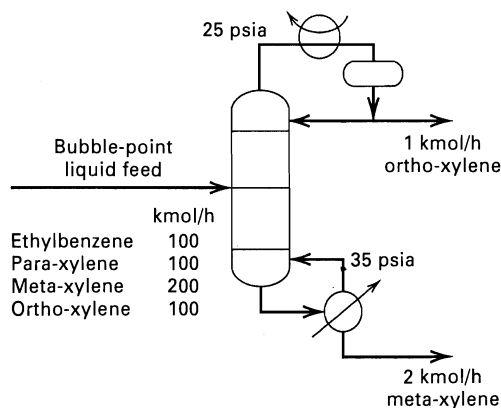


Figure 9.29 Data for Exercise 9.28.

Equilibrium-Based Methods for Multicomponent Absorption, Stripping, Distillation, and Extraction

Previous chapters have considered graphical, empirical, and approximate group methods for the solution of multistage separation problems involving equilibrium stages. Except for simple cases, such as binary distillation, these methods are suitable only for preliminary-design studies. Final design of multistage equipment for conducting multicomponent separations requires rigorous determination of temperatures, pressures, stream flow rates, stream compositions, and heat-transfer rates at each stage. (However, rigorous calculational procedures may not be justified when multicomponent physical properties or stage efficiencies are not well known.) This determination is made by solving material balance, energy balance, and equilibrium relations for each stage. Unfortunately, these relations are nonlinear algebraic equations that interact strongly. Consequently, solution procedures are relatively difficult and tedious. However, once the procedures are programmed for a high-speed digital computer, solutions are achieved fairly rapidly and almost routinely. Such programs are readily available and widely used. This chapter discusses the solution methods used by such programs, with applications to absorption, stripping, distillation, and liquid-liquid extraction. Applications to extractive, azeotropic, and reactive distillation are covered in Chapter 11.

This chapter begins in Section 10.1 with the development of a mathematical model for a general equilibrium stage for vapor-liquid contacting. The resulting equations, when collected together for a countercurrent cascade of stages, are often referred to as the MESH equations. A number of strategies for solving these equations have been proposed, as summarized in Section 10.2, with those most important considered in detail here. All of these methods utilize an algorithm for solving a tridiagonal-matrix equation, described in Section 10.3. When the feed(s) to the cascade contains components of a narrow boiling-point range, the bubble-point (BP) method is very efficient. When the components cover a wide range of volatilities, the sum-rates (SR) method is a better choice. The BP and SR methods are relatively simple, but are restricted to ideal and nearly ideal mixtures, and are limited in allowable specifications. Sections 10.4 and 10.5 present more complex methods, Newton-Raphson (NR) and Inside-Out, respectively, which are required for nonideal systems. These two methods, which are widely available in process simulators such as ASPEN PLUS, CHEMCAD, HYSYS, and PRO/II, also provide many specification options.

10.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Write MESH equations for an equilibrium stage in a multicomponent vapor-liquid cascade.
- Explain how equilibrium stages can be combined to form a countercurrent cascade of N equilibrium stages that can be applied to absorption, stripping, distillation, and extraction.
- Discuss different methods to solve the MESH equations and the use of the tridiagonal-matrix algorithm.
- Solve, rigorously, with a simulation program, countercurrent-flow, multi-equilibrium stage, multicomponent separation problems by the bubble-point, sum rates, Newton-Raphson, and inside-out methods. Select the best method to use for a given problem.

10.1 THEORETICAL MODEL FOR AN EQUILIBRIUM STAGE

Consider a general, continuous, steady-state vapor–liquid or liquid–liquid separator consisting of a number of stages arranged in a countercurrent cascade. Assume that: (1) phase equilibrium is achieved at each stage, (2) no chemical reactions occur, and (3) entrainment of liquid drops in vapor and occlusion of vapor bubbles in liquid are negligible. A general schematic representation of an equilibrium stage j is shown in Figure 10.1 for a vapor–liquid separator, where the stages are numbered down from the top. The same representation applies to a liquid–liquid separator if the higher-density liquid phases are represented by liquid streams and the lower-density liquid phases are represented by vapor streams.

Entering stage j can be one single- or two-phase feed of molar flow rate F_j , with overall composition in mole fractions $z_{i,j}$ of component i , temperature $T_{F,j}$, pressure $P_{F,j}$, and corresponding overall molar enthalpy $h_{F,j}$. Feed pressure is assumed equal to or greater than stage pressure P_j . Any excess feed pressure ($P_F - P_j$) is reduced to zero adiabatically across valve F .

Also entering stage j can be interstage liquid from stage $j - 1$ above, if any, of molar flow rate L_{j-1} , with composition in mole fractions $x_{i,j-1}$, enthalpy $h_{L,j-1}$, temperature T_{j-1} , and pressure P_{j-1} , which is equal to or less than the pressure of stage j . Pressure of liquid from stage $j - 1$ is increased adiabatically by hydrostatic head change across head L .

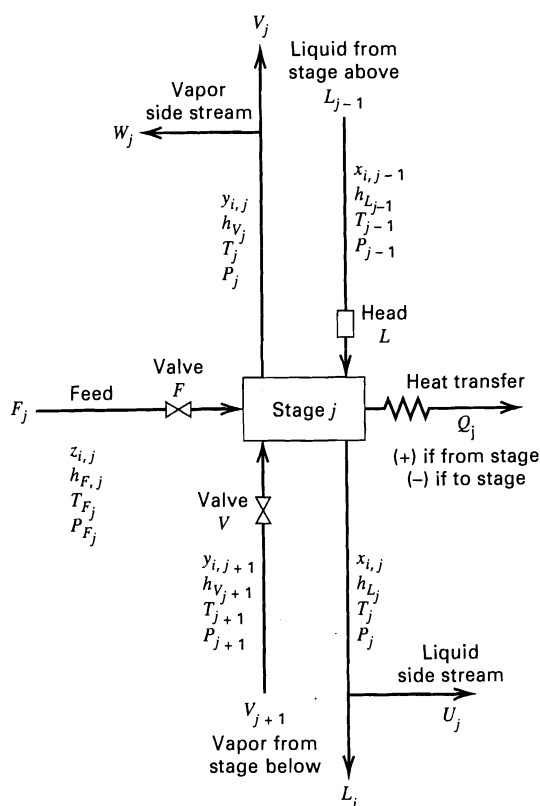


Figure 10.1 General equilibrium stage.

Similarly, from stage $j + 1$ below, interstage vapor of molar flow rate V_{j+1} , with composition in mole fractions $y_{i,j+1}$, enthalpy $h_{V,j+1}$, temperature T_{j+1} , and pressure P_{j+1} can enter stage j . Any excess pressure ($P_{j+1} - P_j$) is reduced to zero adiabatically across valve V .

Leaving stage j is vapor of intensive properties $y_{i,j}$, $h_{V,j}$, T_j , and P_j . This stream can be divided into a vapor side stream of molar flow rate W_j and an interstage stream of molar flow rate V_j to be sent to stage $j - 1$ or, if $j = 1$, to leave the separator as a product. Also leaving stage j is liquid of intensive properties $x_{i,j}$, $h_{L,j}$, T_j , and P_j , which is in equilibrium with vapor ($V_j + W_j$). This liquid can be divided also into a liquid side stream of molar flow rate U_j and an interstage or product stream of molar flow rate L_j to be sent to stage $j + 1$ or, if $j = N$, to leave the multistage separator as a product.

Heat can be transferred at a rate Q_j from (+) or to (–) stage j to simulate stage intercoolers, interheaters, intercondensers, interreboilers, condensers, or reboilers as shown in Figure 1.8. The model in Figure 10.1 does not allow for pumparounds of the type shown in Figure 10.2. Such pumparounds are often used in columns having side streams in order to conserve energy and balance column vapor loads. Some simulator models can handle pumparounds.

Associated with each general theoretical stage are the following indexed equations expressed in terms of the variable set in Figure 10.1. However, variables other than those shown in Figure 10.1 can be used. For example, component flow rates can replace mole fractions, and side-stream flow rates can be expressed as fractions of interstage flow rates.

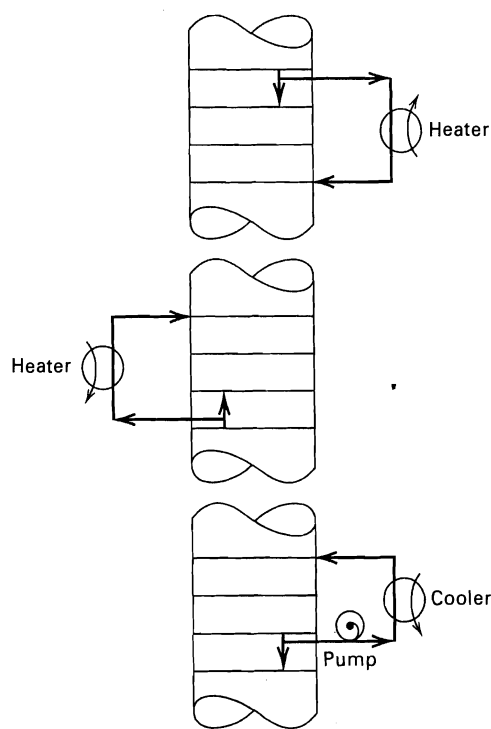


Figure 10.2 Pumparounds.

The equations are similar to those of Section 5.7¹ and are often referred to as the MESH equations after Wang and Henke [1].

1. M equations—Material balance for each component (C equations for each stage).

$$M_{i,j} = L_{j-1}x_{i,j-1} + V_{j+1}y_{i,j+1} + F_j z_{i,j} - (L_j + U_j)x_{i,j} - (V_j + W_j)y_{i,j} = 0 \quad (10-1)$$

2. E equations—phase-Equilibrium relation for each component (C equations for each stage), from (2-19).

$$E_{i,j} = y_{i,j} - K_{i,j}x_{i,j} = 0 \quad (10-2)$$

where $K_{i,j}$ is the phase equilibrium ratio.

3. S equations—mole-fraction Summations (one for each stage),

$$(S_y)_j = \sum_{i=1}^C y_{i,j} - 1.0 = 0 \quad (10-3)$$

$$(S_x)_j = \sum_{i=1}^C x_{i,j} - 1.0 = 0 \quad (10-4)$$

4. H equation—energy balance (one for each stage).

$$H_j = L_{j-1}h_{L,j-1} + V_{j+1}h_{V,j+1} + F_j h_{F_j} - (L_j + U_j)h_{L,j} - (V_j + W_j)h_{V,j} - Q_j = 0 \quad (10-5)$$

where kinetic- and potential-energy changes are ignored.

A total material balance equation can be used in place of (10-3) or (10-4). It is derived by combining these two equations and $\sum_j z_{i,j} = 1.0$ with (10-1) summed over the C components and over stages 1 through j to give

$$L_j = V_{j+1} + \sum_{m=1}^j (F_m - U_m - W_m) - V_1 \quad (10-6)$$

In general, $K_{i,j} = K_{i,j}\{T_j, P_j, \mathbf{x}_j, \mathbf{y}_j\}$, $h_{V_j} = h_{V_j}\{T_j, P_j, \mathbf{y}_j\}$, and $h_{L_j} = h_{L_j}\{T_j, P_j, \mathbf{x}_j\}$, where $\mathbf{x}_j$ and $\mathbf{y}_j$ are vectors of component mole fractions in streams leaving stage j . If these relations are not counted as equations and the three properties are not counted as variables, each equilibrium stage is defined only by the $2C + 3$ MESH equations. A countercurrent cascade of N such stages, as shown in Figure 10.3, is represented by $N(2C + 3)$ such equations in $[N(3C + 10) + 1]$ variables. If N and all $F_j, z_{i,j}, T_{F_j}, P_{F_j}, P_j, U_j, W_j$, and Q_j are specified, the model is represented by $N(2C + 3)$ simultaneous algebraic equations in $N(2C + 3)$ unknown (output) variables comprising all $x_{i,j}, y_{i,j}, L_j, V_j$, and T_j , where the M , E , and H equations are nonlinear. If other variables are

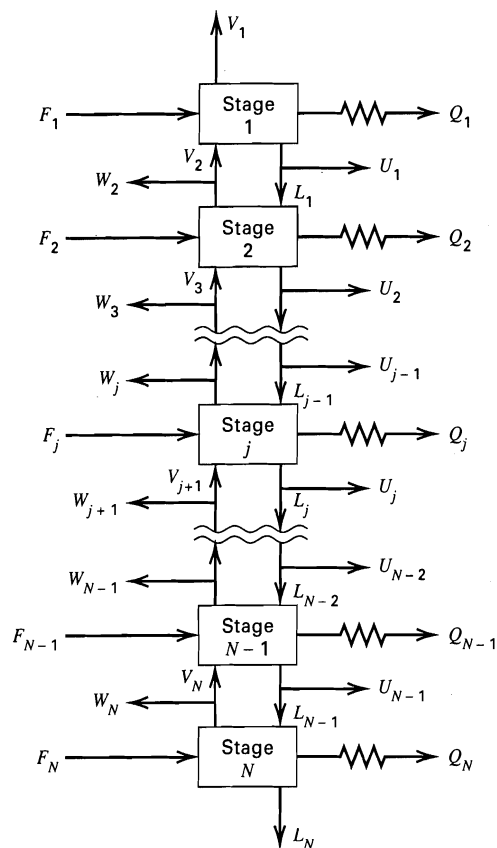


Figure 10.3 General countercurrent cascade of N stages.

specified, as they often are, corresponding substitutions are made to the list of output variables. Regardless of the specifications, the result is a set containing nonlinear equations that must be solved by iterative techniques.

10.2 GENERAL STRATEGY OF MATHEMATICAL SOLUTION

A wide variety of iterative solution procedures for solving nonlinear, algebraic equations has appeared in the literature. In general, these procedures make use of equation partitioning in conjunction with equation tearing and/or linearization by Newton-Raphson techniques, which are described in detail by Myers and Seider [2]. The equation-tearing method was applied in Section 4.4 for computing an adiabatic flash.

Early, pre-computer attempts to solve (10-1) to (10-5) or equivalent forms of these equations resulted in the classical *stage-by-stage, equation-by-equation* calculational procedures of Lewis-Matheson [3] in 1932 and Thiele-Geddes [4] in 1933 based on equation tearing for solving simple fractionators with one feed and two products. Composition-independent K -values and component enthalpies were generally employed. The Thiele-Geddes method was formulated to handle the Case II variable specification in Table 5.4 wherein the number of equilibrium stages above and below the feed, the reflux ratio, and the distillate flow rate are specified, and stage temperatures and interstage vapor (or liquid) flow rates are the iteration (tear) variables. Although widely

¹ Unlike the treatment in Section 5.7, all C component material balances are included here, and the total material balance is omitted. Also, the separate but equal temperature and pressure of the equilibrium phases are replaced by the stage temperature and pressure.

used for hand calculations in the years immediately following its appearance in the literature, the Thiele–Geddes method was found often to be numerically unstable when attempts were made to program it for a digital computer. However, Holland [5] developed an improved Thiele–Geddes procedure called the *theta method*, which in various versions has been applied with considerable success.

The Lewis–Matheson method is also an equation-tearing procedure. It was formulated according to the Case I variable specification in Table 5.4 to determine stage requirements for specifications of the separation of two key components, a reflux ratio, and a feed-stage location criterion. Both outer and inner iterations are required. The outer-loop tear variables are the mole fractions or flow rates of nonkey components in the products. The inner-loop tear variables are the interstage vapor (or liquid) flow rates. The Lewis–Matheson method was widely used for hand calculations, but it also proved often to be numerically unstable when implemented on a digital computer.

Rather than using an equation-by-equation solution procedure, Amundson and Pontinen [6] in a significant development in 1958, showed that (10-1), (10-2), and (10-6) of the MESH equations for a Case II specification could be combined and solved component-by-component from simultaneous-linear-equation sets for all N stages by an equation-tearing procedure using the same tear variables as the Thiele–Geddes method. Although too tedious for hand calculations, such equation sets are readily solved with a digital computer.

In a classic study in 1964, Friday and Smith [7] systematically analyzed a number of tearing techniques for solving the MESH equations. They carefully considered the choice of output variable for each equation. They showed that no one tearing technique could solve all types of problems. For separators where the feed(s) contains only components of similar volatility (narrow-boiling case), a modified Amundson–Pontinen approach termed the bubble-point (BP) method was recommended. For a feed(s) containing components of widely different volatility (wide-boiling case) or solubility, the BP method was shown to be subject to failure and a so-called sum-rates (SR) method was suggested. For intermediate cases, the equation-tearing technique may fail to converge; in that case, Friday and Smith indicated that either a Newton–Raphson method or a combined tearing and Newton–Raphson technique was necessary. Boston and Sullivan [8] in 1974 presented an alternative, robust approach to obtaining a solution to the MESH equations. They defined energy and volatility parameters, which are used as the primary successive-approximation variables. A third parameter, which is a combination of the phase flow rates and temperature at each stage, is employed to iterate on the primary variables; thus the name *inside-out method*. Current practice is based mainly on the BP, SR, Newton–Raphson, and inside-out methods, all of which are treated in this chapter. The latter two methods are the most widely used because they permit considerable flexibility in

the choice of specified variables and generally are capable of solving most problems. However, the first iteration of the BP or SR method is frequently used to initiate the Newton–Raphson (NR) or inside-out method.

10.3 EQUATION-TEARING PROCEDURES

In general, the modern equation-tearing procedures are readily programmed, are rapid, and require a minimum of computer storage. Although they can be applied to a wider variety of problems than the classical Thiele–Geddes tearing procedure, they are usually limited to the same choice of specified variables. Thus, neither product purities, species recoveries, interstage flow rates, nor stage temperatures can be specified.

Tridiagonal Matrix Algorithm

The key to the success of the BP and SR tearing procedures is the tridiagonal matrix that results from a modified form of the M equations (10-1) when they are torn from the other equations by selecting T_j and V_j as the tear variables, which leaves the modified M equations linear in the unknown liquid mole fractions. This set of equations, one for each component, is solved by a highly efficient and reliable modified Gaussian elimination algorithm due to Thomas as applied by Wang and Henke [1]. The modified M equations are obtained by substituting (10-2) into (10-1) to eliminate y and by substituting (10-6) into (10-1) to eliminate L . Thus, equations for calculating y and L are partitioned from the other equations. The result for each component and each stage is as follows, where the i subscripts have been dropped from the B , C , and D terms.

$$A_j x_{i,j-1} + B_j x_{i,j} + C_j x_{i,j+1} = D_j \quad (10-7)$$

where

$$A_j = V_j + \sum_{m=1}^{j-1} (F_m - W_m - U_m) - V_1, \quad 2 \leq j \leq N \quad (10-8)$$

$$B_j = -\left[V_{j+1} + \sum_{m=1}^j (F_m - W_m - U_m) - V_1 + U_j + (V_j + W_j) K_{i,j} \right], \quad 1 \leq j \leq N \quad (10-9)$$

$$C_j = V_{j+1} K_{i,j+1}, \quad 1 \leq j \leq N-1 \quad (10-10)$$

$$D_j = -F_j z_{i,j}, \quad 1 \leq j \leq N \quad (10-11)$$

with $x_{i,0} = 0$, $V_{N+1} = 0$, $W_1 = 0$, and $U_N = 0$, as indicated in Figure 10.3. If the modified M equations are grouped by component, they can be partitioned by writing them as a series of separate tridiagonal matrix equations, one for each of the C components, where the output variable for each matrix equation is x_i over the entire countercurrent cascade of N stages.

$$\begin{bmatrix}
 B_1 & C_1 & 0 & 0 & 0 & \dots & \dots & 0 \\
 A_2 & B_2 & C_2 & 0 & 0 & \dots & \dots & 0 \\
 0 & A_3 & B_3 & C_3 & 0 & \dots & \dots & 0 \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 0 & \dots & \dots & \dots & 0 & A_{N-2} & B_{N-2} & C_{N-2} & 0 \\
 0 & \dots & \dots & \dots & 0 & 0 & A_{N-1} & B_{N-1} & C_{N-1} \\
 0 & \dots & \dots & \dots & 0 & 0 & 0 & A_N & B_N
 \end{bmatrix}
 \begin{bmatrix}
 x_{i,1} \\
 x_{i,2} \\
 x_{i,3} \\
 \dots \\
 \dots \\
 \dots \\
 x_{i,N-2} \\
 x_{i,N-1} \\
 x_{i,N}
 \end{bmatrix}
 =
 \begin{bmatrix}
 D_1 \\
 D_2 \\
 D_3 \\
 \dots \\
 \dots \\
 \dots \\
 D_{N-2} \\
 D_{N-1} \\
 D_N
 \end{bmatrix}
 \quad (10-12)$$

Constants B_j and C_j for each component depend only on tear variables T and V provided that K -values are composition independent. If not, compositions from the previous iteration may be used to estimate the K -values.

The Thomas algorithm for solving the linearized equation set (10-12) is a Gaussian-elimination procedure that involves forward elimination starting from stage 1 and working toward stage N to finally isolate $x_{i,N}$. Other values of $x_{i,j}$ are then obtained starting with $x_{i,N-1}$ by backward substitution. For five stages, the matrix equations at the beginning, middle, and end of the procedure are as shown in Figure 10.4.

The equations used in the Thomas algorithm are as follows:

For stage 1, (10-7) is $B_1 x_{i,1} + C_1 x_{i,2} = D_1$, which can be solved for $x_{i,1}$ in terms of unknown $x_{i,2}$ to give

$$x_{i,1} = \frac{D_1 - C_1 x_{i,2}}{B_1}$$

Let

$$p_1 = \frac{C_1}{B_1} \quad \text{and} \quad q_1 = \frac{D_1}{B_1}$$

Then

$$x_{i,1} = q_1 - p_1 x_{i,2} \quad (10-13)$$

Thus, the coefficients in the matrix become $B_1 \leftarrow 1$, $C_1 \leftarrow p_1$, and $D_1 \leftarrow q_1$, where $\leftarrow$ means "is replaced by." Only values for p_1 and q_1 need be stored.

For stage 2, (10-7) can be combined with (10-13) and solved for $x_{i,2}$ to give

$$x_{i,2} = \frac{D_2 - A_2 q_1}{B_2 - A_2 p_1} - \left(\frac{C_2}{B_2 - A_2 p_1} \right) x_{i,3}$$

$$\begin{bmatrix}
 B_1 & C_1 & 0 & 0 & 0 \\
 A_2 & B_2 & C_2 & 0 & 0 \\
 0 & A_3 & B_3 & C_3 & 0 \\
 0 & 0 & A_4 & B_4 & C_4 \\
 0 & 0 & 0 & A_5 & B_5
 \end{bmatrix}
 \begin{bmatrix}
 x_1 \\
 x_2 \\
 x_3 \\
 x_4 \\
 x_5
 \end{bmatrix}
 =
 \begin{bmatrix}
 D_1 \\
 D_2 \\
 D_3 \\
 D_4 \\
 D_5
 \end{bmatrix}$$

(a)

$$\begin{bmatrix}
 1 & p_1 & 0 & 0 & 0 \\
 0 & 1 & p_2 & 0 & 0 \\
 0 & 0 & 1 & p_3 & 0 \\
 0 & 0 & 0 & 1 & p_4 \\
 0 & 0 & 0 & 0 & 1
 \end{bmatrix}
 \begin{bmatrix}
 x_1 \\
 x_2 \\
 x_3 \\
 x_4 \\
 x_5
 \end{bmatrix}
 =
 \begin{bmatrix}
 q_1 \\
 q_2 \\
 q_3 \\
 q_4 \\
 q_5
 \end{bmatrix}$$

(b)

$$\begin{bmatrix}
 1 & 0 & 0 & 0 & 0 \\
 0 & 1 & 0 & 0 & 0 \\
 0 & 0 & 1 & 0 & 0 \\
 0 & 0 & 0 & 1 & 0 \\
 0 & 0 & 0 & 0 & 1
 \end{bmatrix}
 \begin{bmatrix}
 x_1 \\
 x_2 \\
 x_3 \\
 x_4 \\
 x_5
 \end{bmatrix}
 =
 \begin{bmatrix}
 r_1 \\
 r_2 \\
 r_3 \\
 r_4 \\
 r_5
 \end{bmatrix}$$

(c)

Figure 10.4 The coefficient matrix for the modified M -equations of a given component at various steps in the Thomas algorithm for five equilibrium stages. (Note that the i subscript is deleted from x .) (a) Initial matrix. (b) Matrix after forward elimination. (c) Matrix after backward substitution.

Let

$$q_2 = \frac{D_2 - A_2 q_1}{B_2 - A_2 p_1} \quad \text{and} \quad p_2 = \frac{C_2}{B_2 - A_2 p_1}$$

Then

$$x_{i,2} = q_2 - p_2 x_{i,3}$$

Thus, $A_2 \leftarrow 0$, $B_2 \leftarrow 1$, $C_2 \leftarrow p_2$, and $D_2 \leftarrow q_2$. Only values for p_2 and q_2 need be stored.

In general, we can define

$$p_j = \frac{C_j}{B_j - A_j p_{j-1}} \quad (10-14)$$

$$q_j = \frac{D_j - A_j q_{j-1}}{B_j - A_j p_{j-1}} \quad (10-15)$$

Then

$$x_{i,j} = q_j - p_j x_{i,j+1} \quad (10-16)$$

with $A_j \leftarrow 0$, $B_j \leftarrow 1$, $C_j \leftarrow p_j$, and $D_j \leftarrow q_j$. Only values of p_j and q_j need be stored. Thus, starting with stage 1, values of p_j and q_j are computed recursively in the order $p_1, q_1, p_2, q_2, \dots, p_{N-1}, q_{N-1}, q_N$. For stage N , (10-16) isolates $x_{i,N}$ as

$$x_{i,N} = q_N \quad (10-17)$$

Successive values of x_i are computed recursively by backward substitution from (10-16) in the form

$$x_{i,j-1} = q_{j-1} - p_{j-1} x_{i,j} = r_{j-1} \quad (10-18)$$

Equation (10-18) corresponds to the identity coefficient matrix.